

FREE K.C.S.E REVISION SERIES 2012 to 2015

GEOGRAPHY

TOPICS COVERED

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1. INTRODUCTION TO GEOGRAPHY

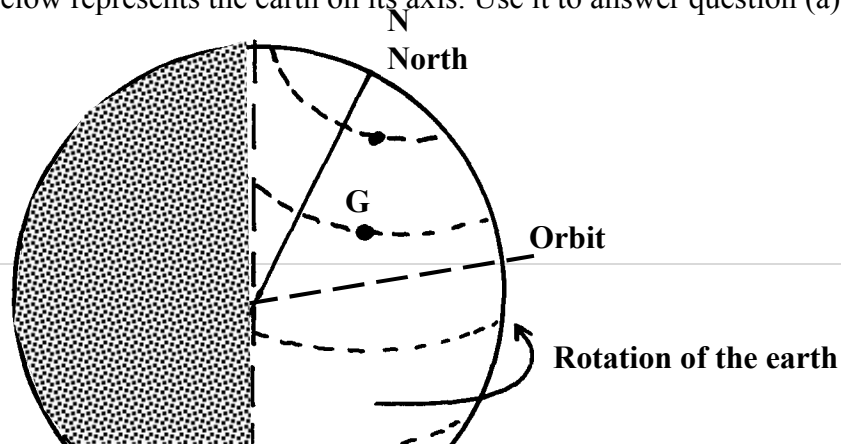
The topic entails:

- (i) Definition of Geography and Environment*
 - (ii) Description of the branches of Geography*
 - (iii) Explaining the importance of studying Geography and its relationship with other disciplines.*
1. Explain why the study of Geography is beneficial in the management and conservation of the environment.

2. THE EARTH AND THE SOLAR SYSTEM

The topic entails:

- (i) Definition of Solar System.*
 - (ii) Explaining the origin of the earth*
 - (iii) Explaining the effects of rotation and revolution of the earth.*
 - (iv) Description of the structure of the earth*
1. The diagram below represents the earth on its axis. Use it to answer question (a)



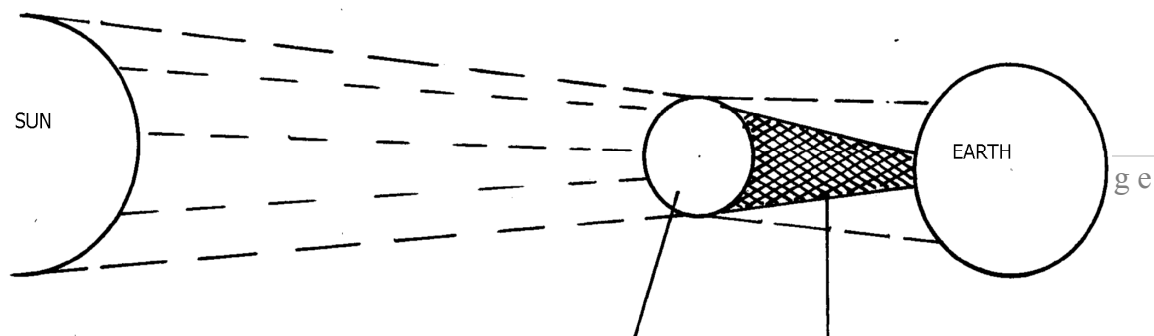
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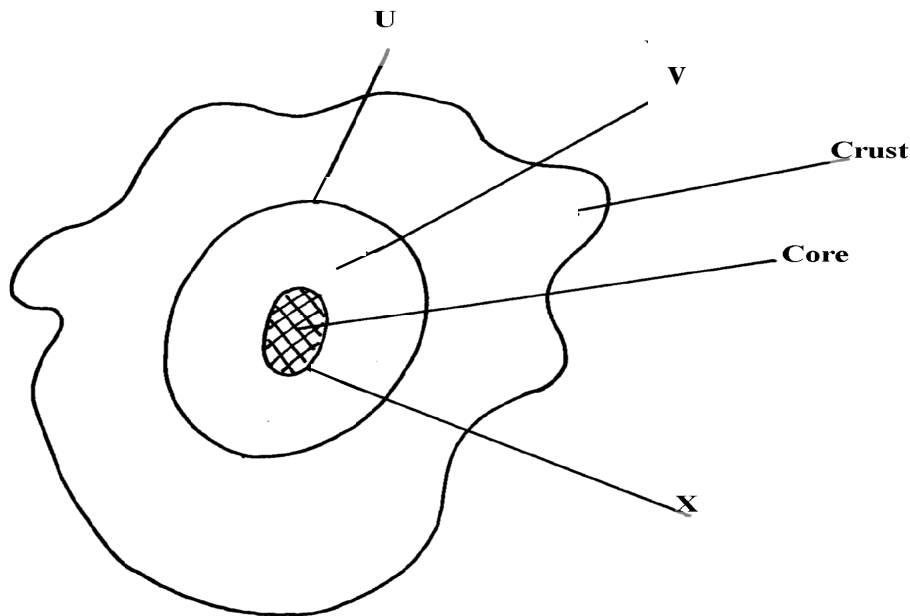
- a)
 - i) Name the latitude marked **G**
 - ii) What is the angle of inclination of the earth's axis from its orbit
 - b)
 - i) State **two** effects of the rotation of the earth
 - ii) When the local time is 2.00 p.m at longitude 45°E, what is the longitude of a place whose local time is 10.30 a.m
 - c) Name **two** local winds experience around lake Victoria region
2. The table below represents rainfall and temperature figure for a town in Kenya .Use it to answer the questions that follow:-

MONTH	J	F	M	A	M	J	J	A	S	O	N	D
Temp (°C)	27	28	28	28	27	25	25	24	25	26	27	26
Rainfall(mm)	25	38	99	140	277	439	277	69	142	201	71	25

- a)
 - i) calculate the annual range of temperature for the town
 - ii) Calculate the total annual rainfall for the town (1mk)
 - b) State **three** characteristics of the climate experience in the town
3. a) What is a solstice
- b) State **three** effects of the revolution of the earth
4. (a) (i) Give **two** theories that explain the evolution of the solar system and the origin of the earth
- (ii) Identify the force that causes the earth to bulge at the equator
- (b) Give **two** reasons that support the belief that the interior of the earth is very hot
5. a) State **two** theories that are used to explain the origin of the earth
- b) What is solar “system”?
6. a). Name **two** planets without natural satellites in the solar system
- (b) (i) What is a time zone?
- (ii) Give the reason why the International Date Line is significant.
- (2mks)
- (c) State any **two** characteristics of Latitudes.
7. (a) What is the solar system
- (b) Give **three** reasons why the interior of the earth is very hot
8. (a) The diagram below represents an eclipse. Use it to answer the following questions:
- (i) Name the type of eclipse



- (ii) Identify the parts marked **Q** and **T**
(b) State **three** effects of the rotation of the earth
9. a) State **three** reasons why the interior of the earth is known to be very hot
b) Give **two** effects of the elliptical shape of the earth
10. (a) (i) What is an equinoxal date?
(ii) Name **two** equinoxal dates
- (iii) State **two** changes caused by the earth's revolution around the sun
11. The diagram below represents the internal structure of the earth. Use it to answer question (a.)



- a) Name the parts named **u**, **v** and **x**
b) Describe the characteristics of
i) Crust ii) Core

3. WEATHER AND CLIMATE

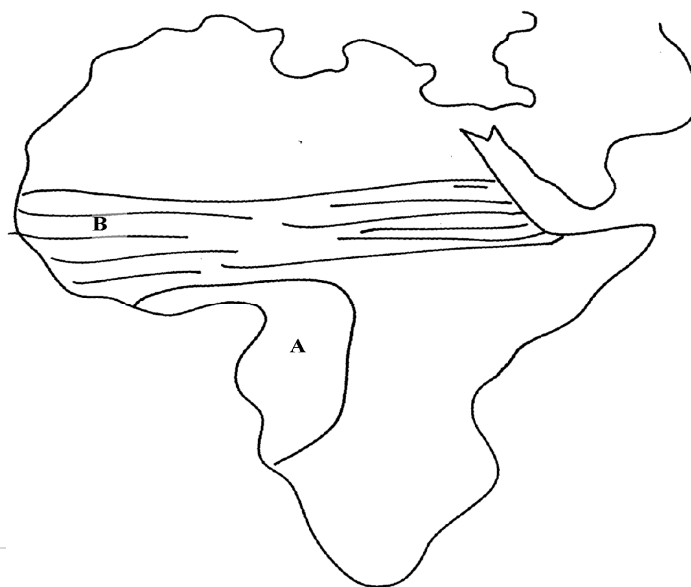
The topic entails:-

- (i) *Defining weather and stating its elements*
(ii) *Explaining conditions necessary for siting a weather station.*

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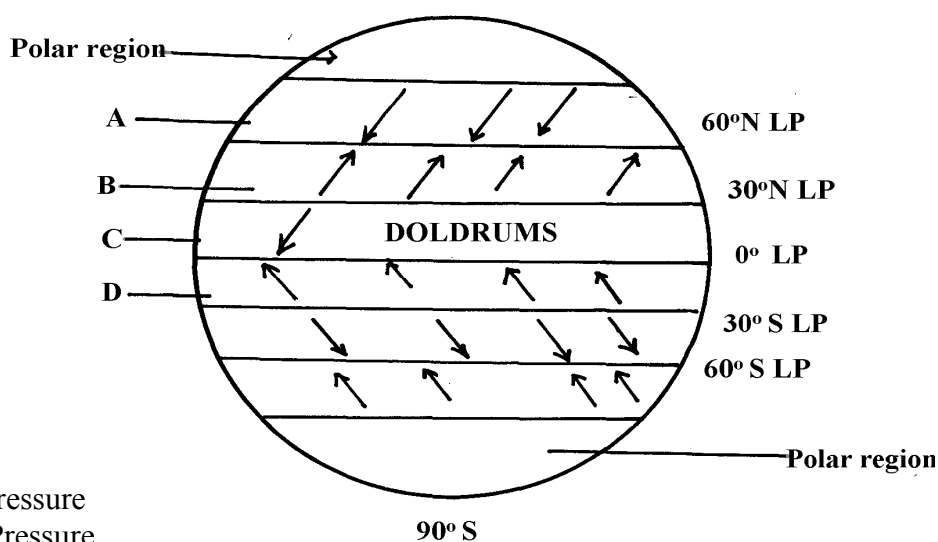
- (iii) *Use instruments to measure elements of weather*
- (iv) *Analyse and interpret data on weather conditions*
- (v) *Description of the structure and composition of the atmosphere*
- (vi) *Explain factors influencing weather*
- (vii) *Carrying out a field study on weather station*
- (viii) *Distinguish between weather and climate*
- (ix) *Explain the factors that influence climate*
- (x) *Description of the characteristics of the climatic regions of Kenya.*
- (xi) *Description of characteristics of major climatic regions of the World.*
- (xii) *Accounting for the causes of aridity and desertification*
- (xiii) *Explaining the effects and possible solutions to aridity and desertification*
- (xiv) *Discussing the causes and impact of climate change on physical and human environment*

1. State **three** causes of desertification.
2. a)i) Differentiate the term aridity and desertification
ii) Differentiate between weather and climate
b) State **two** causes of temperature inversion
c) Explain the occurrence of a land breeze
3. a) Give **three** characteristics of the inter- tropical convergence zone
b) (i) State **three** natural causes of climate change
ii) Explain **four** consequences of climate change on the physical environment
c) Give four reasons why some areas within the equatorial belt do not experience true equatorial climate
4. (a) Describe the climatic conditions experienced in the Kenya highlands
(b) Explain **four** effects of folding to human activities
5. (a) Distinguish between **weather** and **climate**
(b) State **six** characteristics of equatorial climate
6. (a) Explain how the following factors influence climate:
(i) Ocean currents
(ii) Altitude
(b) Study the map of Africa below and answer the following questions:



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- (i) Describe the characteristics of climate marked **A**.
7. (a) State **three** conditions necessary for the formation of dew
(b) State **two** climatic reasons why the government should evict settlers from the Mau forest complex in the Rift valley of Kenya
 8. (a) (i) What is a fog?
(ii) State any **two** conditions necessary for the formation of fog.
(b) Name any **two** isothermic layers of the atmosphere.
 9. (a) What is an air mass?
(b) Give **two** climatic characteristics of the Inter-Tropical Convergence Zone (ITCZ)
 10. (a) Describe **two** characteristics of a mountain climate
(b) (i) State **two** human causes of climate change
(ii) Explain **three** consequences of climate change
(c) Four classes intend to visit a weather station near your school to study the instruments for measuring weather element:-
(i) Describe how you would use a rain gauge to measure rainfall
(ii) Name **two** instruments you would find inside a Stevenson screen
 11. a) What do you understand by:
i) Micro- climate
ii) Green house effect
b) Name **two** weather recording instruments that are placed in a Stevenson's screen
 12. (a) What is an air mass?
(b) What two conditions favour formation of air mass?
 13. (a) What is the Inter-tropical convergence Zone?
(b) Account for any **four** characteristics of tropical rainforests.
 14. (a) What is an air mass?
(b) Study the diagram below and name the air masses marked **A, B, C** and **D**



Key

LP- Low Pressure
HP- High Pressure

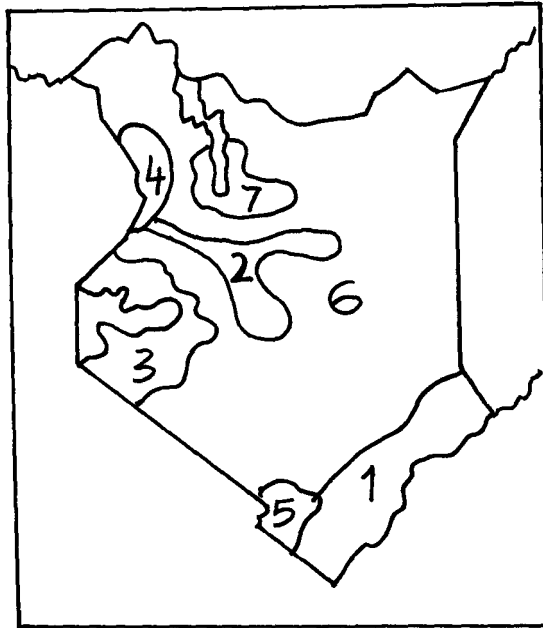
15. (a) (i) What is global warming?

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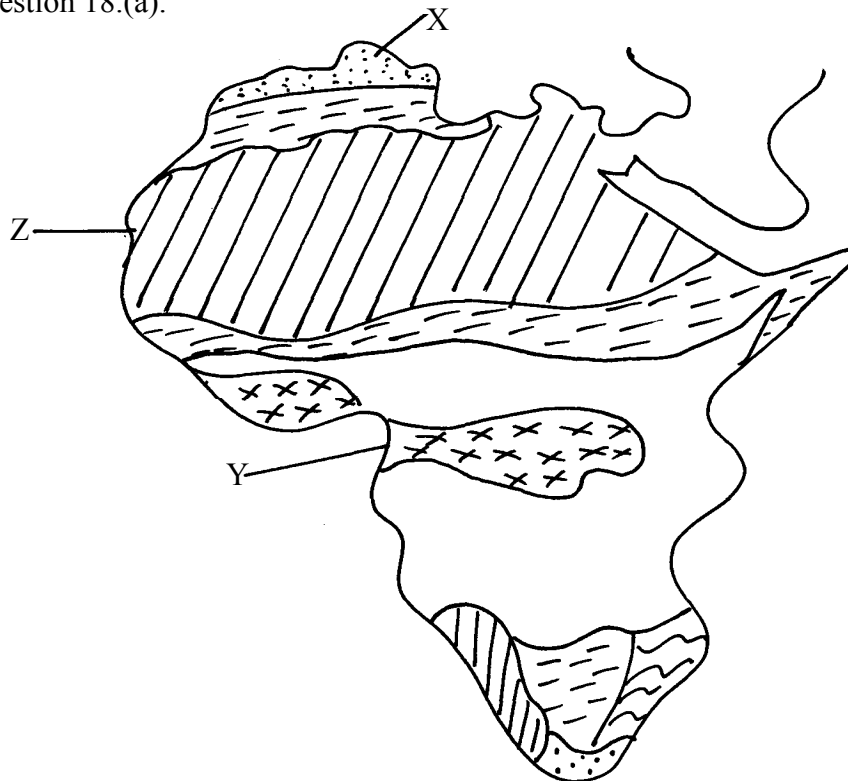
- (ii) Give any **four** causes of climate change.
- (iii) Name **four** Green house gases
- (b)(i) Explain **five** effects of climate change
- (ii) Identify with evidence **two** climatic aspects that could have influenced the distribution of vegetation
- 16. (a) Differentiate between:
 - (i) Aridity and desertification
- (b) Give **three** reasons why the recording of weather data at a weather station may be inaccurate
- (c) State **two** qualities that makes Stevenson screen suitable for its work

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17. The map below shows the climatic regions of Kenya
(a) Use it to answer the question a and b



- (a) Name the climatic region name 2 and 3
(b) State **three** characteristics of climatic region marked 7
18. The map of Africa below shows the different climatic regions of Africa. Use it to answer question 18.(a).



- a) i) Identify the climatic types marked X and Y
ii) Name any **four** characteristics of the climate marked Z
b) i) Briefly explain the green house effect and the global warming

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- ii) Explain **three** effects of climatic change on the physical environment
- c) Explain how the following factors influence climate
 - i) Altitude
 - ii) Continentality
 - iii) Ocean currents

4. STATISTICAL METHODS

The topic entails:-

- (i) *Defining statistics*
- (ii) *Identifying types and sources of statistical data*
- (iii) *Identifying and describing methods of collecting and recording data.*
- (iv) *Analysis, interpretation and presentation of statistical data using appropriate graphical methods.*
- (v) *Explaining the advantages of each method of data presentation.*

1. Study the table below and answer questions that follow:-

CROP	1978	1979	1980	1981	1982
COFFEE	1000	990	870	830	840
TEA	750	700	650	700	600
PYRETHRUM	300	250	350	400	450
MAIZE	500	450	550	600	350

- (a) (i) Using 1cm to represent 500 tons, draw a compound bar graph to represent the data.
- (ii) Give **two** disadvantages of using the method to represent statistical data.

2. The table below shows leading import crops by value (Kshs. Million). Use it to answer questions a – c

	CROP			
Year	Un milled wheat	Maize	Rice	Wheat flour
2000	6,989	4,664	1,968	180
2001	7,515	3,342	2,619	639
2002	5,577	229	2,104	237
2003	6,099	1,417	2,981	168
2004	6,754	4,647	3,659	200

- (a) (i) Using a scale of 1cm represents 100,000 , draw a comparative bar graph to represent the data in the table above
- (ii) Give **three** advantages of using comparative bar graphs

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- (b) Explain **three** reasons why Kenya is a producer of the commodities shown in the table above yet she imports the same

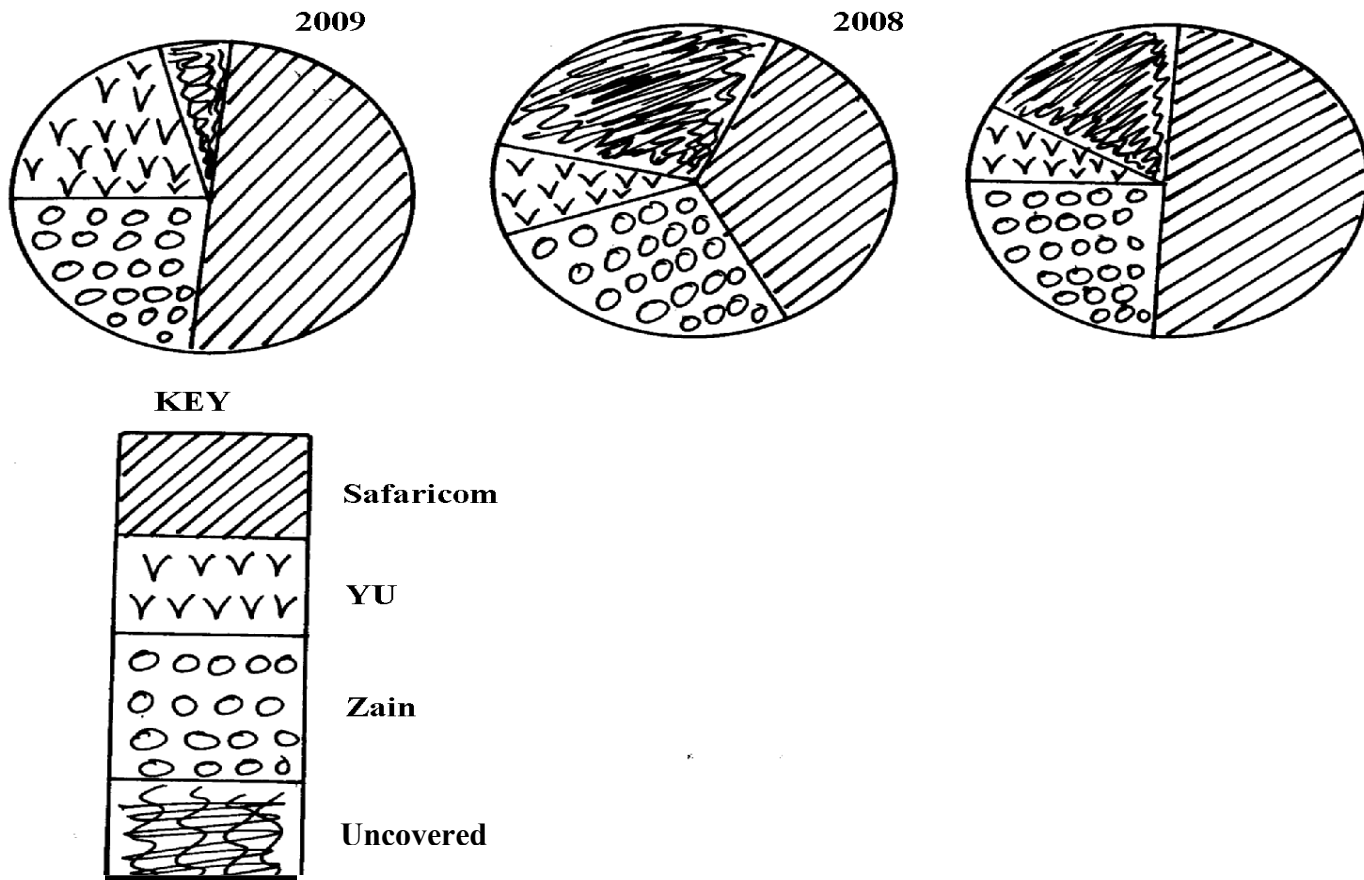
3. The table below shows milk production in '000 units in selected Districts

District	1982	1992	2002
Trans nzoia	24	26	40
Kiambu	23	25	31
Meru	25	27	32
Bungoma	12	14	20

- a) i) Using a vertical scale of 1 centimeter to represent 10,000 units, draw a compound bar graph to represent the above given data

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4. Study the figure below and use it to answer question 6. The figure depicts proportional divided circles showing the extend of network coverage in Kenya between 2007 and 2009



- a) i) State **four** deductions that can be made from the above representation
 ii) State **three** advantages of using proportional circles in representing data
5. The table below shows four principal crops produced in Kenya in the years 2000 and 2001. Use it to answer questions.
- | CROP | AMOUNT IN METRIC TONS | |
|--------|-----------------------|---------|
| YEAR | 2000 | 2001 |
| Wheat | 70,000 | 13,000 |
| Maize | 200,000 | 370,000 |
| Coffee | 98,000 | 55,000 |
| Tea | 240,000 | 295,00 |
- (a) (i) Using a radius of 5 cm, draw a pie chart to represent crop production in the year 2000.
 (ii) State **two** advantages of using pie charts.
 (b) Calculate the percentage increase in wheat production between the years 2000 and 2001.
6. Study the data given and use it to draw a pie chart showing mineral production in Kenya;

Mineral	Amount (000 tonnes)
Gold	26

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Flouspar	14
Soda ash	32
Zink	28

- (a) Using a radius of 5cm, draw a pie chart to represent the above data
(b) List **three** advantages of using a pie chart in representing data

5. MAPS AND MAP WORK

The topic entails:

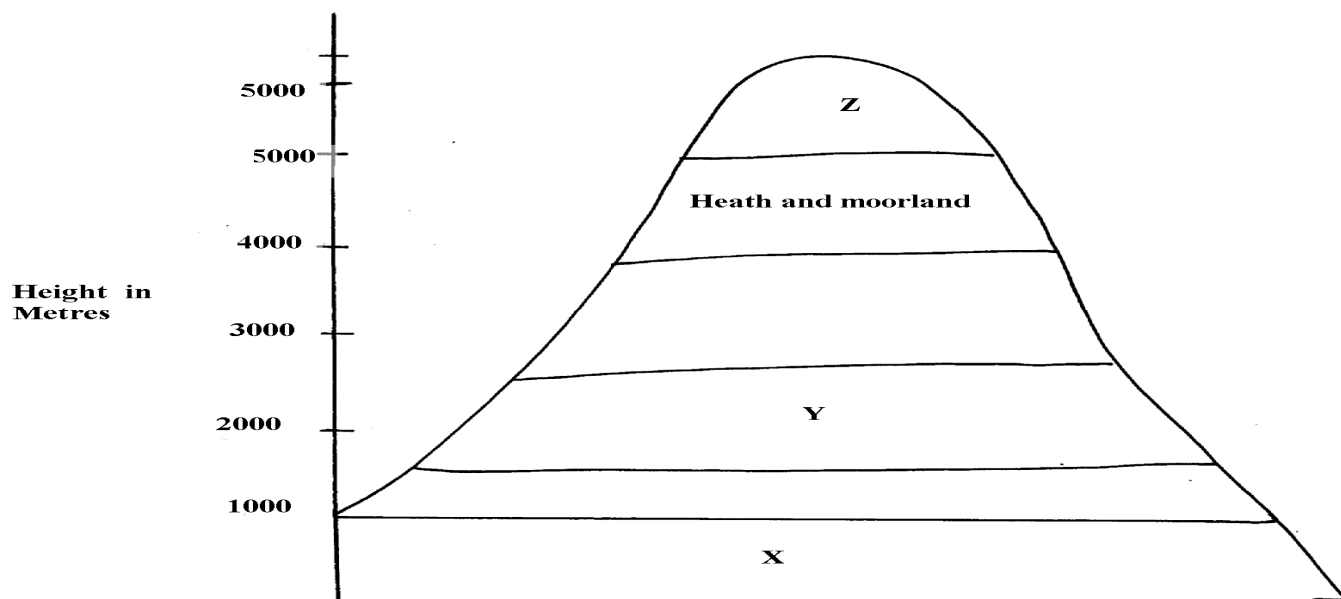
- (i) Definition of pictures, Plans and maps*
- (ii) Explaining the relationship between pictures, plans and maps*
- (iii) Identifying types of maps and stating their uses.*
- (iv) Identifying and use of marginal information on maps*
- (v) Identifying types and uses of scales*
- (vi) Determining distances and areas using scales*
- (vii) Distinguishing direction and bearing*
- (viii) Identifying traditional and modern methods of locating places and features*
- (ix) Locating places and features on maps using various methods.*
- (x) Description of different methods of representing relief on topographical maps.*
- (xi) Enlargement and reduction of topographical maps*
- (xii) Drawing cross-section from topographical maps*
- (xiii) Calculation and interpretation of vertical exaggeration and gradient*
- (xiv) Determination of indivisibility.*

1.
 - (a) Identify **any two** adjoined map sheets to the area covered by the map
 - (b)
 - (i) Give **two** features of the map which shows that the area to the south east of the area covered by the map receives high amount of rainfall(cite evidence)
 - (ii) Name **two** physical features found in grid square 4548
 - (c)
 - (i) Reduce by a half the area covered by easting 23 to 30 and northing 47 to 53
 - (ii) On the reduced area mark and label the following:-
All weather road loose surface
District boundary
 - (d)
 - (i) Describe the drainage of the area covered by the map to the south of northing 50
 - (ii) Citing evidence from the map give **three** economic activities carried out in the area covered by the map
2.
 - (a)
 - (i) What is the **latitudinal** and **longitudinal** location of shopping centre at grid square 5863
 - (ii) Name the continuous man made feature along Road B1
 - (b)
 - (i) Apart from spot height, give **two** other methods which have been used to show relief on the map
 - (ii) Measure the length of the road D313 between Northing 68 and Northing 70.
Give your answer in kilometers
 - (iii) Calculate the bearing of the culvert in grid square 6066 from river confluence in grid square 6269
3.
 - (a)
 - (i) Draw a cross-section along Easting 67 from Northing 67 to Northing 71.
(Use a vertical scale of 1cm rep. 80m)
On the cross section you have drawn, mark and name:
- Hill
- Pass
- Road D313
 - (ii) Calculate the vertical exaggeration of the cross section you have drawn
4. Study the map of Belgut 1:50,000 (sheet 117/3) provided and answer the following questions
a)i) Give the general direction of the flow of river Itare

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- ii) What is the bearing of a trigonometrical station 117S 13 at grid reference 443512 from a tea nursery found at grid reference 443447?
- iii) Name **two** types of trigonometrical station that have been used to show the relief in the area covered by the map
- iv) Calculate the area that is found on the western side of river Sondo. Give answers in meters
- b)i) Using a vertical scale of 1cm to represent 100 meters draw a cross-section from grid reference 260520 to grid reference 340520
- ii) On the cross-section, mark and label the following
 - A hill
 - A provincial boundary
 - All weather road, loose surface
 - Riverine trees
- iii) Calculate the vertical exaggeration of the cross-section
- c) Students from Chemamul School set out to carry out field work in the area of Belgut
 - i) With evidence, name **two** crops they found being grown in the area

Use the diagram below to answer questions 5



5.
 - (a) Name the vegetation types marked **X** and **Y**
 - (b) Give **two** reasons for the absence of vegetation at **Z**
 - (c) Name any **one** part in Kenya represented by this diagram
6. Use the map of Kericho (1:50,000) to answer the questions

Study the map of Belgut 1: 50000 (sheet 117/3) provided and answer the following questions

 - (a) (i) What is the height of the highest contour in the area covered by the map?
 - (ii) Give the longitudinal extent of the area covered by the map.
 - (iii) Calculate the area of the part of Kisii district shown on the map.
 - (iv) What is the six figure grid reference of the junction at Marumbasi?
 - (b) (i) Draw a rectangle 10cm by 14cm to represent the area between easting 30 and 40
 - (ii) On the rectangle mark and name the following:-
 - A plantation

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- River Yurith
 - A seasonal swamp
 - The bridge at Kabirigut
- (iii) Calculate the new scale of your reduction
- (c) Describe **three** ways in which physical factors have influenced the construction of all weather roads in the area.
- (d) (i) Citing evidence from the map, explain **three** factors that favour the growing of tea in Belgut.
- (ii) With evidence from the map name any other crop grown in the area other than tea.
7. Study the map of Belgut: 1:50000 (sheet 117/3) provided and answer the following questions
- a) i) Convert the scale of the map into statement scale
- ii) Give the longitudinal extend of the area covered by the map
- iii) Calculate the bearing of the posho mill at Kiptule from the spot height at Kiptere
- b) Apart from forests name other vegetation types in the area covered by the map
- c) i) Draw a rectangle measuring 15cm by 8cm to represent the area south of northing 50 and West of easting 30. On the rectangle mark
- River Sondo
 - Coffee mill
 - All weather road loose surface
 - Wood land
- ii) Describe the drainage of the area covered by the map
- d) What factors on the map can promote trading activity
8. Study the topographic map of Belgut provided and use it to answer this question
- (a) (i) Name the **two** provinces covered in the area by the map
- (ii) What is the general direction of Ikamu school from Chemamul school?
- (iii) Write the **six** figure grid reference of the Posho Mill at Kiptule
- (b) (i) Measure the distance of all weather road (bound surface) from Kapsuser shops to its ends in the North-East to Kericho. Give your answer in Kilometers
- (ii) Identify **two** methods used in the map to locate places
- (iii) Name **three** types of natural vegetation found in the area
- (iv) Citing evidence from the map, name three agricultural activities carried out in Belgut area
- (c) Citing evidence from the map, explain three physical factors which have influenced settlement in the area
- (d) Describe the drainage of the area covered by the map
9. a) i) What type of map is Belgut sheet?
- ii) Give **two** methods used in representing relief on the map extract.
- b) i) What is the length of the Murram road from Marumbasi to Kiptere Sunchen (Give your answer in Km)
- ii) Name the main crop planted under plantation from the map
- c) i) Explain how relief has influenced settlement in the area covered by the Map
- ii) Citing evidence from the map, give two social – economic activities carried out in the area covered by the map.
- d) i) Using a vertical scale of 1 cm to represent 100 metres, draw a cross section from the East 440000 to easting 500000 on the cross section label:

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- Road
 - River
 - Forest
10. Study the map of BELGUT 1:50,000 provided and answer the following questions
- a) i) Give the longitudinal extent of the area covered by the map
 - ii) Convert the scale of the map into a statement scale
 - iii) Name **two** methods that have been used to represent relief on the map
 - b) i) Draw a cross-section between grid references 29050 and 33050. Use scale of 1cm to represent 20 meters
 - ii) On the cross-section Mark and name:
 - papyrus swamp
 - All weather road loose surface
 - iii) Calculate the vertical exaggeration
11. a) Describe the drainage of the area covered by the map
- b) Citing evidence from the map, give **three** economic activities carried out in the area covered by the map
- c) State **two** functions of the tea factory to the population around.
- d) Give **three** reasons to show the area covered in the map receives high rainfall. The evidence should be deducted from the map
12. Study the map of **Belgut** 1:50,000(sheet 117/3) provided and answer the following questions
- (a) (i) What type of a map is Belgut ?
 - (ii) Give the grid square in which Matongo school is found
 - (iii) Calculate the area enclosed by Kendu-Kisii, all weather roads (bound surface) to the West of the map
 - (iv) Citing evidence from the map, identify **four** social activities taking place in the map
 - (b) (i) Draw a cross-section along Northings 54 between Eastings 26 to 32. Use vertical scale of 1cm to represent 20m. On it mark and name:-
 - (I) River
 - (II) Loose surface road
 - (III) Swamp
 - (ii) Calculate the vertical exaggeration
 - (c) Describe the drainage of the area covered by the map
 - (d) Citing evidence from the map, explain **two** conditions that favour cattle rearing in the area covered by the map
13. Study the map of Kericho (1:50,000) sheet 117/4 provided and answer the following questions
- (a) i) What is the bearing of the secondary trigometrica station 2173 around Kapcheptoror school form the dry weather road junction at Kipchimchim school
 - (ii) Give a six-figure grid reference of the trigometrical station (other) 1811 near Poiywek school
 - (iii) Convert the scale of the map into a statement scale
 - (iv) Using liens of latitudes and longitudes give the position of Keongo school at gird square 5662
 - (v) Give **two** methods used in representing relief in the are covered by the map
 - (b) Citing evidence from the map, state:-

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- (i) Two social functions of Kericho Municipality
 - (ii) Two economic activities carried out in the area covered by the map
 - (c) Using a vertical scale 1cm represents 20m
 - (i) Draw a cross section from grid reference 550640 to 590660
 - (ii) On the cross-section, mark and name the following:-
 - A river
 - Dry weather road
 - A hill
 - (d) Describe the drainage of the area covered by the map
14. Study the map of Belgut (117/3) provided and answer the questions that follow:
- (a) Identify **two** provinces covered by the map of Belgut
 - (b) (i) Using a vertical scale 1cm represents 50m, draw a cross section from grid reference 260590 to 330560. on it mark and name;
 - All weather loose surface roads
 - Marshes
 - River
 - (ii) Calculate the vertical exaggeration of the cross section.
 - (iii) What type of map is Belgut?
15. Study the map of **BELGUT (1:50000 sheet 117/3)** provided and answer the questions below:
- a) i) Give the six figure grid references of the confluence of the river Itare and river Kitoi.
 - ii) Measure the length in kilometers of all weather roads loose surface from the junction at grid square 3957 to the junction at Kipmaso grid square 3751.
 - iii) Give the name to the adjoining sheet found in the North East of Belgut .
 - b) i) What is the longitudinal extent of the area covered by the map?
 - ii) What is the approximate height of the school at Kiptere grid square 3658?
 - c) i) Describe the drainage of the area covered by the map.
 - ii) Using a vertical scale of 1cm to represent 40 metres draw a cross-section along northing 53 from easting 24 to 29.
On it mark and name
 - regional boundary
 - Foot path
 - main track (motorable)
 - d) i) Describe the distribution of the settlement of the area covered by the map.
 - ii) Citing evidence from the map, explain **two** factors that may favour trading activities in the area covered by the map.

6. FIELD WORK

The topic entails:-

- (i) *Definition of field work*
- (ii) *Stating different types of field work*
- (iii) *Explaining the importance of field work*
- (iv) *Explaining the procedure to be followed during field work*
- (v) *Identifying possible problems during field work*
- (vi) *Carrying out field work within the local environment.*

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1. a) Form four students of your school carried out a field study on beef farming in Narok District.
 - (i) State **four** objectives of their study.
 - (ii) Give **four** follow up activities they carried out.
- (b) Students of Kakao secondary school intend to undertake a field study of Olkaria I geothermal power generating project. Answer the following questions;
 - (i) State **three** objectives they would write down for the field study
 - (ii) List **three** preparations they would undertake before the actual field study
- (c) Your class visited a biogas digester near your school;
 - (i) Describe how it was constructed
 - (ii) List **three** raw materials the class may have identified which are used in the production of biogas
2. a) You are planning to carry out a field study on soil in an arid region.
 - i) What are some of the characteristics you would observe?
 - ii) Why would you prepare a working schedule for the study?
3. a)i) Give **three** natural vegetation zones on mount Kenya
 - ii) Name three temperate grasslands found in the world
 - iii) Describe the characteristics of the hot desert vegetation
- b) Explain **three** causes of the decline of the areas under forests in Kenya
- c) You are supposed to carry out a field study of a weather station near your school
 - i) What preparations would you make for the study
 - ii) What instruments are you likely to find within the Stevenson box
4. (a) You are required to carry out a field study on vegetation within the local environment;
 - (i) Apart from identifying different types of plants, state other activities you will carry during the field study
 - (ii) How will you identify the different types of plants
- (b) Form four students from Kisumu west district carried out a field study in an area of soil erosion in Machakos district
 - (i) State **three** causes of soil erosion they could have identified
 - (ii) Name **two** effects of soil erosion they have identified
 - (iii) State any **one** objective for their study
5. (a) Students from Kisumu West secondary school carried out a field study in the area covered by the map
 - (i) What **three** preparations did they make?
 - (ii) State any null hypothesis for the study
- b) Your class intends to carry out a field study on weathering within the vicinity of the school
 - i) State the type of information you are likely to collect
 - ii) State **two** follow up activities you are likely to carry after the study
6. (a) Students from Kericho school set out to conduct a field study on the relationship between climate and vegetation of the area covered by the map.
 - (i) What preparation did they carry out for the study?
 - (ii) State **three** evidences they would identify to support climate change.
 - (iii) State **two** possible alternative hypotheses for the study

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- (b) Students are planning to carry out a field study in the area affected by climate change;
(i) State **three** ways in which observation would be the best method of data collection.
(c) Citing evidence from the map, explain **three** factors that have influenced settlement in the area covered by the map.
7. (a) Students of Chepkosilen school carried a field study on economic activities in the area covered by the map.
(i) Give **two** preparations they made before the study.
(ii) State **two** hypotheses for their study.
(iii) Citing evidence, identify **three** economic activities that they studied.
(iv) What type of map is Belgut?
- (b) Students from your school have conducted a field study on a Lake in Kenya
(i) In their study they identified some of the problems affecting the lake to have been caused by nearby **industries** and **deforestation** in the surrounding areas. Explain how each of the two could have affected the lake.
(ii) Name any **two** methods they might have used to collect the data.
(iii) State any **two** reasons why it would be important to do follow-up after the study
(d) You are required to carry out a field study on soil erosion around your school
(i) State **two** methods you would use to record data
(ii) Give **three** problems you may encounter during the field study
8. a) Students of Masabot School carried out a field study of Changoi tea factory.
i) Name **two** types of roads they used to travel to Changoi tea factory.
ii) What preparations they were likely to make for the study
b)(i) Suppose you were a student in the school at Tegat and you plan to carry out a day's field study of Changoi tea factory. Design a working programme (schedule) you would use during the day of study
ii) Your class is required to carry out a field study of a river. What would be the advantages of dividing the class into groups according to the stages of the long profile of the river?
9. a) You intend to carry out a field study on a desert landscape.
i) Apart from conducting oral interviews, state two other methods you would use to collect information
ii) State **two** problems that you are likely to encounter in the field
b) You are provided to carry out a field study of the vegetation within the local environment;
i) Apart from identifying the different types of plants, state three other activities you will carry out during the field study
ii) How will you identify the different types of plants?
10. a) Your class went for a field study in Samburu.
i) List **three** methods they are likely to have used to present their findings
b) i) State three activities they would be involved in.
ii) Identify **three** problems they are likely to encounter.
11. (a) You are required to carry out a field work on soils around your school:-

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- (i) State **three** objectives for your study
- (ii) State **two** reasons why it would be necessary to carry samples back to school
- (b) You carried out field work on soils around your school:-
- (i) State **three** preparations you made before the actual day of field work
- (ii) State any **three** problems you encountered during the field work
12. (a) You are to carry out a field study on rivers near your school:-
- (i) Name **three** methods you will use to collect your data
- (ii) Why is it important to carry out a pre-visit
- (iii) How will your findings be useful to the local community?
13. (a) A field study was carried out around the rift valley lakes:-
State **two** characteristics of the lakes they would have identified
- b) You intend to go for a field study to a region where folding has occurred.
- i) State **three** reasons why you would conduct a pre-visit.
- ii) Identify **two** methods you would use to record data.
- c) i) State **two** problems they may face during their study
- ii) State **two** follow up activities they may have been involved in after the field study
14. The table below shows the crops produced in Kenya between the years 2000 to 2002

CROPS '000'	2000	2001	2002
WHEAT	22	37	83
MAIZE	131	255	325
BARLEY	12	26	47
TOTAL	165	318	455

- (a) (i) Calculate the percentage of wheat production in the year 2000
- (ii) Using a scale of 1cm rep 200 units, draw proportional circles to show the production of crops each year. Show your calculations
- (b) Explain **three** physical conditions which favour wheat growing in Kenya
- (c) Compare wheat growing in Kenya and Canada under the following:-
- (i) Mechanization
- (ii) Marketing
- (iii) Size
- (d) State **four** uses of wheat
15. (a) Use the following information to answer the questions below:

Type of energy	No. of families using each type	Average monthly income per family (Kshs.)
Fire wood	13,400	900
Kerosene	11,200	1600
Charcoal	9,100	1000
Liquid Petroleum Gas	5,300	3000
Saw dust	4,000	900
Hydroelectricity	2,000	4500
Total	45,000	11,900

- (i) Draw a pie-chart with a radius of 4cm to represent the number of families using each type of energy. Show your calculations

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- (b) Students from Matungu district went to study gold mining in Kakamega South district
- (i) State any **two** main preparations made before field study visit
 - (ii) Give any **two** follow-up activities they engage in after the study
- (c) Suppose you were to conduct a field study in Kakamega forest
- (i) State **three** problems that are likely to hinder your work
 - (ii) How could you determine the following:-
 - Heights of a tree
 - Diameter of stem
 - Tree of the same species
- (d) You have been asked to conduct field study on land pollution in an urban set up;
- (i) State **three** problems that you may encounter;
 - (ii) Your class carried out a field study on forests in your area. List **four** measures you would recommend to conserve forests in the area
-

7. MINERALS AND ROCKS

The topic entails:-

- (i) *Defining minerals and rocks*
 - (ii) *Stating the characteristics of minerals*
 - (iii) *Classification of rocks according to mode of formation*
 - (iv) *Stating the characteristics of rocks*
 - (v) *Accounting for the distribution of major types of rocks in Kenya.*
 - (vi) *Explaining the significance of rocks*
 - (vii) *Identifying major types of rocks and their uses within the local environment.*
1. (a) State **two** formations in which mineral ores occur
- (b) Give **three** negative effects of open cast mining on the environment
2. a) Briefly describe how minerals occur in veins and lodes
- (b) Describe how coral rocks are formed
3. State **three** classification of sedimentary rocks based on their mode of formation
4. a) Apart from Thermal Metamorphism, name two other types of metamorphism
- b) Describe the following types of rocks
- i) Calcareous rocks
 - ii) Carbonaceous rocks.
 - iii) State **three** examples of rocks that are dominant at the coast of Kenya.
- c) i) Explain how sedimentary rocks are formed through the following processes.

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Mechanical Process

ii) Chemical Process

5. (a) Give **three** ways through which rocks may be metamorphosed. (3 mks)
- (b) Name **two** examples of plutonic rocks
- c) State **three** conditions that favour the formation of an artesian basin.

8. MINING

This topic entails

- (i) *Defining mining*
- (ii) *Explaining the factors influencing:-*
 - The occurrence of minerals
 - Mining activities
- (iii) *Describing methods of mining*
- (iv) *Locating major minerals on the map of East Africa*
- (v) *Explaining the significance of minerals in Kenya*
- (vi) *Accounting for the problems facing the mining industry in Kenya.*
- (vii) *Explaining the effects of mining on the environment in Kenya.*
- (viii) *Describing the occurrence of specific minerals and their exploitation in selected countries.*

1. (a) (i) What problem has been brought about by uncontrolled mining of minerals in Kenya?
(ii) Where is fluorspar mined in Kenya?
(iii) Describe how trona is mined and processed in Magadi
- (b) (i) List **two** ways in which mining leads to loss of biodiversity
(ii) List **two** ways in which mining leads to pollution
2. (a) (i) Name **three** methods of mining.
(ii) State **five** factors that influence the exploration of minerals.
(iii) State **three** ways in which mining derelicts can be reclaimed.
3. (a) State **two** ways in which gold in South Africa occurs. (2mks)
(b) Give **three** ways in which abandoned mined sites could be rehabilitated.
4. (a) Name **two** countries in Africa where oil is mined
(b) State **two** by-products of crude oil
5. (a) (i) Define the term '**mining method**'
(ii) Explain how the following factors influence mining:-
 - Technology
 - Quality of the ore
- (b) Use the map of East Africa below to answer question (i)

MAP OF EAST AFRICA



- (i) Name the minerals found in the areas marked **P, Q, R**
 - (ii) Name **two** areas in South Africa where diamonds are mined
 - c) Open-cast method of mining.
 - d) Benefits of trona to Kenya.
 - e) Problems facing the mining industry in Kenya.
6. Use the map of East Africa below to answer question (a) (i)



- (a) (i) Name the minerals mined in the areas marked **W, X, Y** and **Z**
 - (ii) Give **four** ways in which minerals occur
 - (b) Identify any **five** factors that influence the exploitation of minerals
 - (c) Explain any **four** problems associated with mining
 - (d) List any **four** main minerals mined in Kenya
7. (a) Give **three** main methods of mining
- (b) The following data shows the value of minerals exported from Tanzania between the years 2000-20003. Use it to answer the following questions:

Value of mineral exports in millions of Tshs.				
	2000	2001	20002	2003
Gold	680	700	500	400
Diamonds	1000	800	700	900

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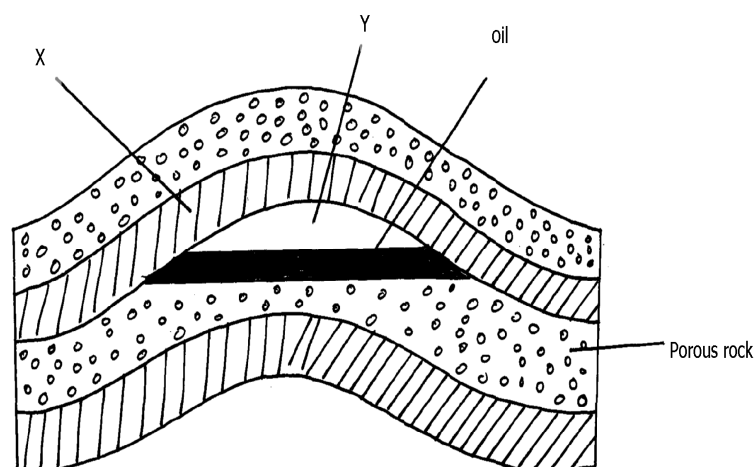
Mica	400	100	400	100
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(i) Calculate the percentage decline in the value of diamond exports between 2000 and 20001.

(c) Explain any **four** ways in which trona mining contributes to the growth of the economy in Kenya

8. (a) (i) State **three** methods used in mining
 (ii) Name the mineral found in the following areas;-
 - Magadi
 - Mwandui
 - Kasese
 (b) Describe the following factors influence mining of minerals
 (i) Technology
 (ii) Quality of ore
 (iii) Accessibility
 (c) Explain **four** problems facing the mining of Gold in South Africa
 (d) (i) List **two** leading to oil producing countries in the middle East
 (ii) State **three** uses of petroleum
 (e). State **three** problems of land dereliction

9. The diagram below shows



- (a) Identify the parts marked **X** and **Y**
- (b) Give **three** reasons why Kenya imports petroleum in crude form and not as refined by-products
10. (a) Differentiate between veins and lodes
 (b) State **three** effects of land dereliction
11. (a) (i) Give **three** methods of mining.
 (ii) Name **two** places where gold is mined in Tanzania.
 (b) Explain **four** factors which influence the exploitation of minerals.
 (c) State **four** negative effects of mining on the environment.
 (d) (i) Describe how solution mining is carried out.
 (ii) State **three** ways in which mining contributes to the economy of Kenya.
12. (a) (i) Name **three** methods of mining
 (ii) Explain **three** ways in which soda ash contributes to the economy of Kenya
 (iii) Identify **four** problems facing gold mining in South Africa

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- (b) (i) State **three** conditions necessary for the formation of petroleum
- (ii) Give **two** by-products obtained when crude oil is refined
- (c) What **three** benefits would Kenya get if the current prospecting for oil yields good results
- (d) (i) Name **two** minerals mined in the Southern Highlands of Tanzania
- (ii) Name **two** ports through which minerals from East Africa are exported through

9. INTERNAL LAND-FORMING PROCESSES

This area covers :-

- a) Earth Movements
- b) Folding
- c) Faulting
- d) Volcanicity
- e) Earth Quakes

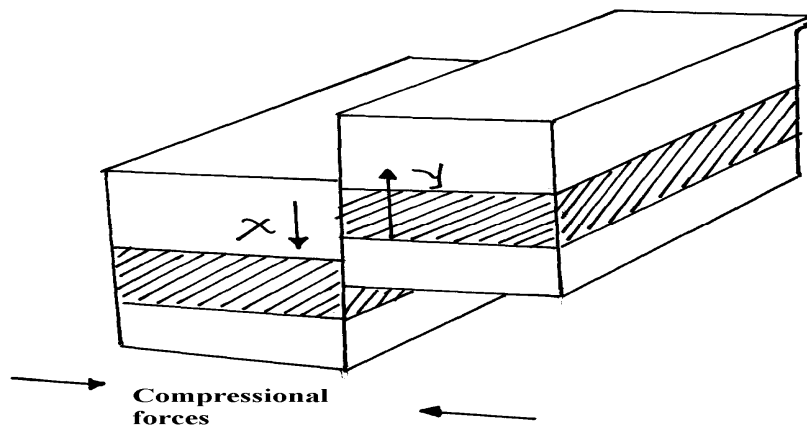
It entails:

- (i) *Definition of earth movements*
- (ii) *Explaining the causes of horizontal and vertical earth movements.*
- (iii) *Explaining theories of continental drift and plate tectonics*
- (iv) *Defining folding, faulting and vulcanicity and earthquakes.*
- (v) *Explaining the*
- (vi) *Description of types of folds, faults, and forms of vulcanicity*
- (vii) *Explaining the resultant features due to folding, faulting and vulcanicity*
- (viii) *Explain causes of earth quakes.*
- (ix) *Describing how earth quakes are measured*
- (x) *Account for the world distribution of earth quake ones, fold mountain systems and features resulting from faulting and vulcanicity*
- (xi) *Explaining the significance of the resultant features of folding, faulting, vulcanicity and the effects of earth quakes.*

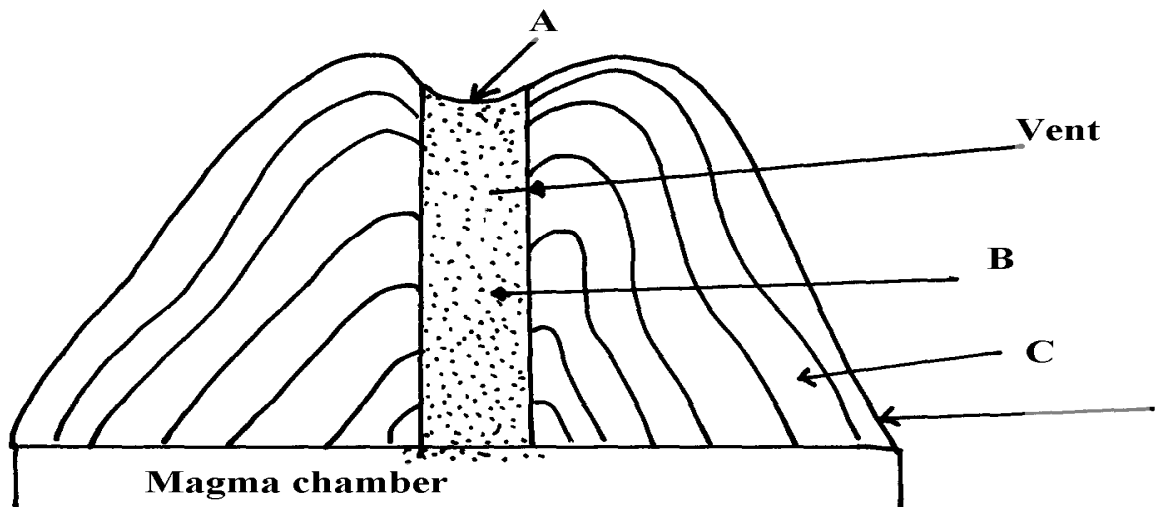
- 1. a) Composite volcano
- b) i) What are earthquakes?
- ii) Name **two** types of earthquakes waves
- iii) Explain **three** ways in which volcanic mountains positively influence human activities
- 2. a) Using well labeled diagrams, distinguish between a simple symmetrical fold and an asymmetrical fold.
- b) i) Name **two** fold mountains of the Alpine Orogeny
- ii) With the aid of well labeled diagrams, describe how a fold mountain is formed.
- c) Explain **three** positive effects of folding on the physical & human environment.
- 3. (a) State **three** ways in which the earth's crust is affected by the earthquakes
- (b) State **two** evidences of continental drift theory
- (c) Name **three** types of plate tectonic boundaries
- 4. (a) (i) What are tectonic plates
- (ii) Give any **two** examples of oceanic plates:-
- (b) Describe how the following cause earth movements:
- (i) Isostatic adjustment

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- (ii) Magma movement in the crust
 - (iii) Convectional currents in the mantle.
5.
 - (a) Give any **two** natural causes of earthquakes
 - (b) Name **three** characteristics of the rift valley lakes of Kenya
 6.
 - (a) State **three** characteristics of the rift valley lakes of Kenya
 - (b)(i) What are earth quakes?
 - (ii) Name **three** types of earth quake waves
 - c). Differentiate between extension boundaries and compression boundaries.
 7.
 - a. i) What is an earthquake
 - ii) Give **two** ways in which earthquakes can be predicted
 - b) State **two** ways in which faulting interferes with transport and communication lines.
 8. The diagram below represents a feature resulting form faulting

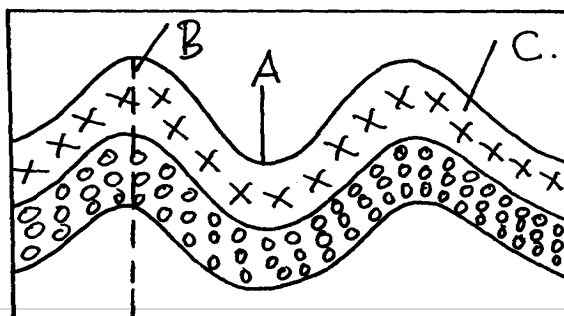


- (a) Name the feature
 - (b) Name part x and y
 - (c) Give **three** negative of an earthquake
9.
 - (a) Differentiate between hot springs and geysers
 - (b) The feature below show an extrusive landform in a volcanic area



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- (c) Using well labelled diagrams, describe how the following are formed:
- (i) An anticlinal fault
 - (ii) An overthrust fold
- (d) (i) Explain any **three** ways in which features resulting from volcanicity are a problem to people
- (ii) Describe how subsidence can lead to formation of a caldera.
10. (a) What is the plate tectonic theory ?
- (b) Name **three** types of tectonic plate boundaries .
- (c) Explain **two** evidences that support continental drift theory.
11. (a) (i) Name **two** fold mountains in Africa apart from Atlas mountains
- (ii) Differentiate between **symmetrical** and **asymmetrical** folds
- (iii) Apart from symmetrical and asymmetrical folds, name other types of folds
- (b) (i) With the aid of well labeled diagrams explain the formation of **fold** mountains
- (ii) Give **two**-examples of fold mountains in North America
12. a) Effects of the elliptical shape of the earth's orbit
- b) Mention **three** causes of the earth movements
13. a) List **two** characteristics of destructive plate boundary
- b) Outline three natural causes of earthquakes
- b) State **two** effects of earthquakes on crystal rocks
14. a) List **two** factors that determine the degree of folding in rocks
- b) State three effects of faulting on drainage systems
15. (a) State any **one** evidence that support the theory of the drifting of continents
- (b)(i) Apart from Africa, name **any two** other continents that form the Gondwanaland
- (ii) What is panthalassa?
16. a) Give **two** reasons why hardwood trees species in Kenya are in danger of extraction
- b) State **three** problems that affect forestry in Canada
17. The diagram below shows a simple fold
- (a) (i) Name the part marked **A**, **B** and **C**



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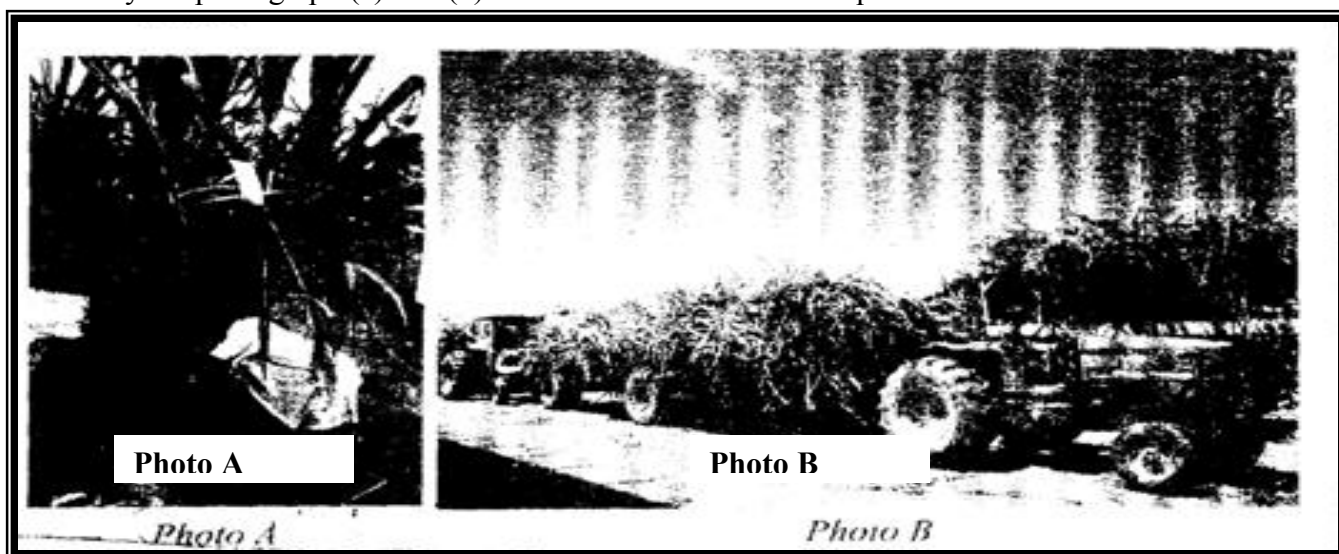
- (ii) Name **two** fold mountains outside Africa
- (iii) Give **three** landform associated with folded regions
- (b) With the aid of well labeled diagram describe the process of formation of Fold Mountain
- (c) Explain **four** ways in which folding influences human activities

10. PHOTOGRAPH WORK

The topic entails:-

- (i) *Identifying types of photographs*
- (ii) *Describing parts of a photograph*
- (iii) *Estimating the sizes of features appearing on photographs*
- (iv) *Making sketch form photographs*
- (v) *Identifying and interpreting features on photographs.*

1. Study the photograph (a) and (b) and use them to answer the questions below:-



- (a) (i) Name the type of photograph shown in the figure (b)
 - (ii) Name **three** human activities carried out in photographs (a) and (b)
 - (b) If the **two** photographs were taken from western province, name **four** districts in the province the photographs are likely to have been taken from?
 - (c) Explain **five** physical conditions that encourage the growth of crops in photograph (a)
 - (d) State **five** problems facing farmers of this crop in Kenya
2. (a) (i) State **three** characteristics of the animal breeds
- (ii) Explain why the above activity is mainly located in the highlands in Kenya

11. VEGETATION

The topic entails:

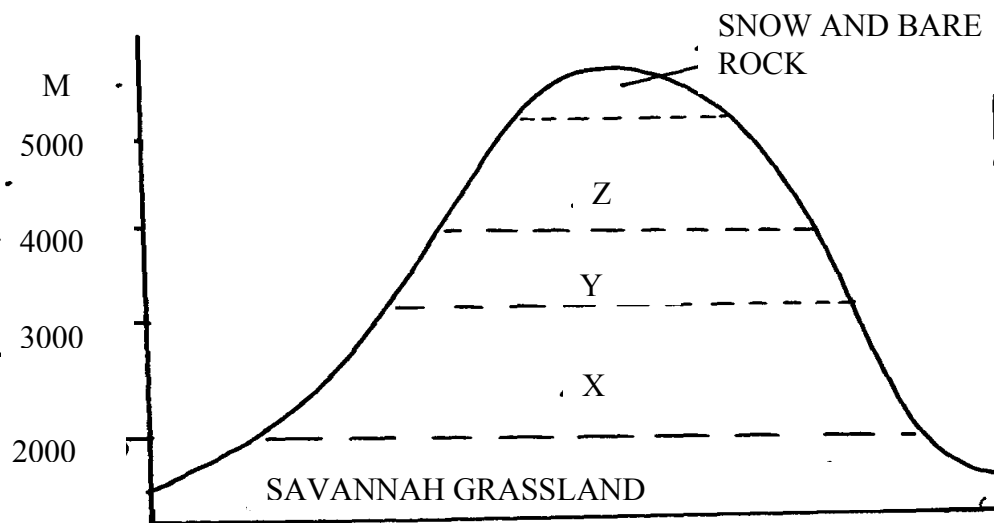
- (i) *Definition of vegetation*

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- (ii) *Discussing the factors influencing the distribution of vegetation*
- (iii) *Identifying and describing the characteristics of major vegetation regions of Kenya and the world*
- (iv) *Discussing the significance of vegetation and explain their uses within the local environment*

1.
 - a) Name **two** areas of the world with temperate grasslands.
 - b) Explain how the characteristics of the vegetation in climate **B** are adapted to the environmental conditions

2. The diagram below shows mountain vegetation zones, use it to answer the questions that follow.

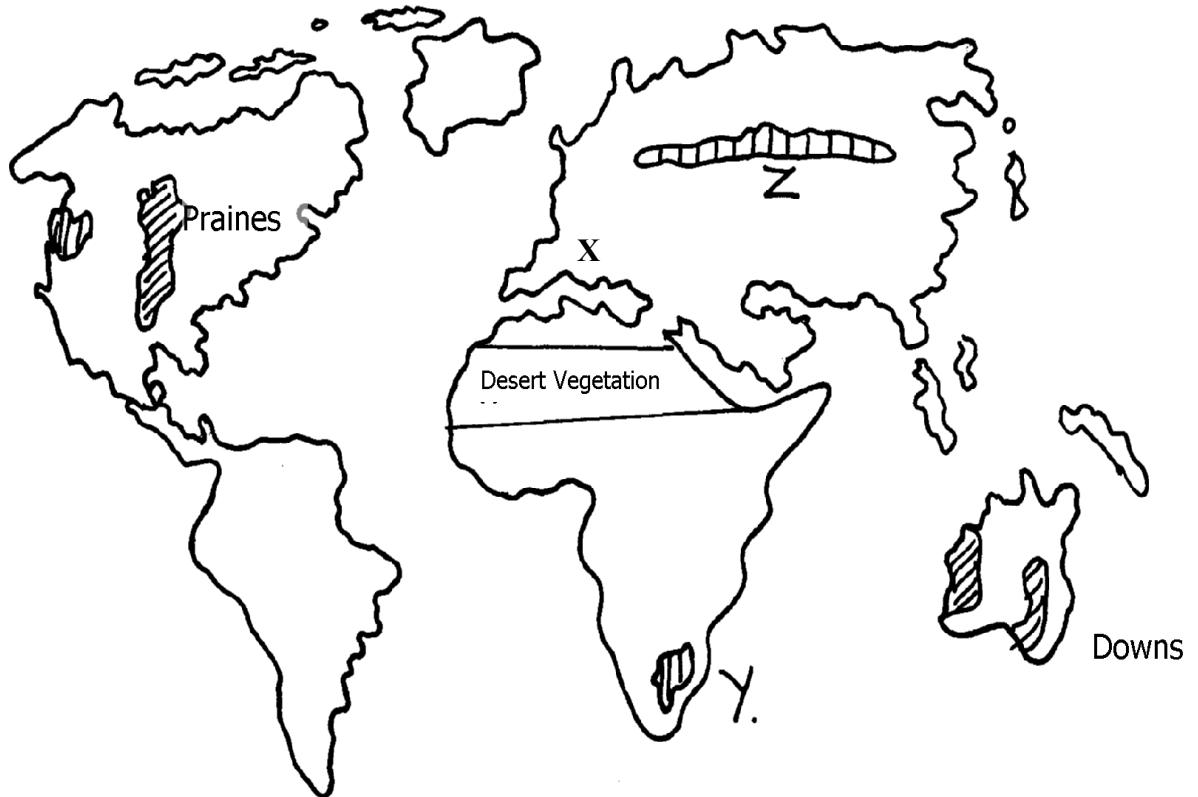


- (a)
 - (i) Name the vegetation zone marked **X,Y,Z**.
 - (ii) Give **four** characteristics of tropical savannah vegetations
 - (b) Explain how the following factors influence vegetation:
 - (i) Aspect
 - (ii) Human activities
 - (c) Name the specific countries of the world where the following grassland vegetation types are found:-
 - (i) Velds
 - (ii) Pampas
 - (iii) Downs
 - (d) Explain **three** factors that have led to a decline of natural grassland in Kenya
3.
 - (a)
 - (i) Define the term '**natural vegetation**'
 - (ii) State **three** climatic factors influencing distribution of vegetation
 - (b)
 - (i) Give **two** zones of the temperate grasslands
 - (ii) State **five** characteristics of coniferous forests

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- (c) Explain **four** ways in which vegetation of the Nyika region of Kenya has adapted to the region of Kenya has experienced in the area
- (d) Give **five** factors that have led to a decline of the natural grasslands in Kenya

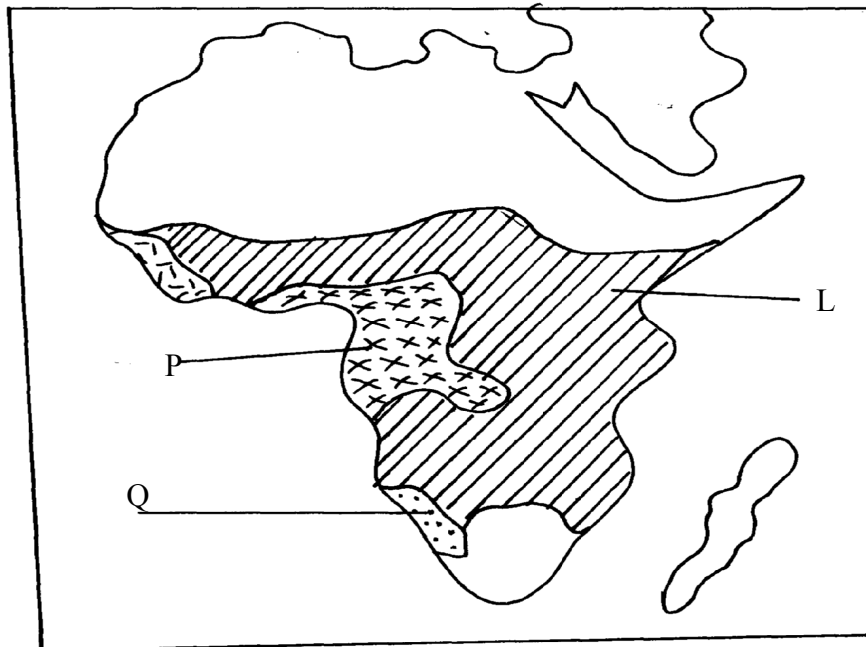
4. The map below shows major vegetation zones of the world. Use it to answer the questions that follow:



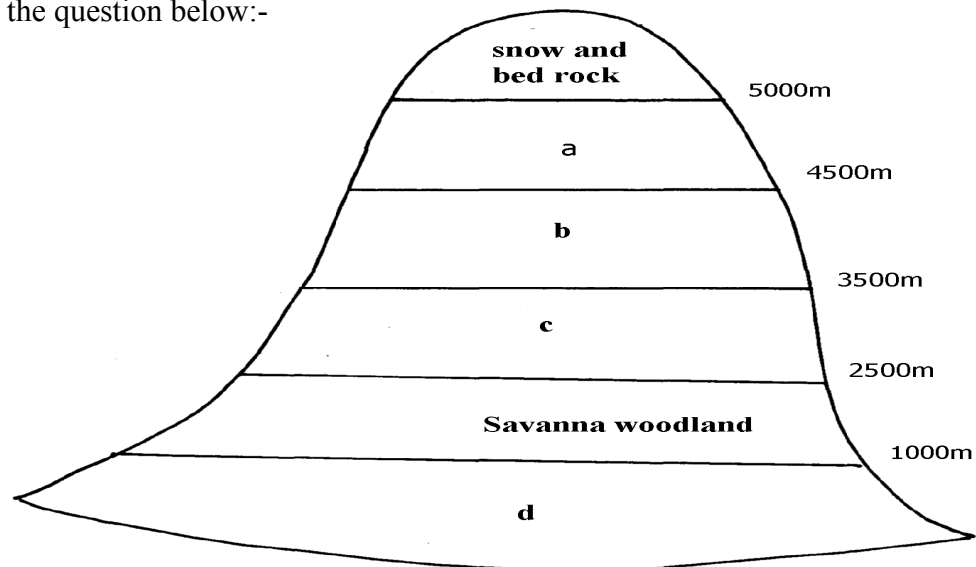
- (a) Identify the temperate grasslands marked **Y** and **Z**
- (b) Explain **four** ways in which desert vegetation are adapting to the extremely harsh conditions

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5. The map below shows the location of some vegetation regions of Africa. Use it to answer the questions that follow:



5. (a) i) Name the grassland region marked **L**
 ii) Describe the characteristics of the natural vegetation found in the shaded area **P**
 b) i) Explain **four** ways in which the vegetation found in the area marked **Q** adapts to the environmental conditions of the region
 c) i) Give **two** reasons why the mountain top has no vegetation
 ii) State **two** ways in which vegetation is of significance to the physical and human environments
6. (a) (i) Name **two** types of vegetation
 (ii) Give **two** physiographic factors influencing vegetation distribution
 (b) The diagram below shows vegetation zones of a S. Africa mountain. Use it to answer the question below:-

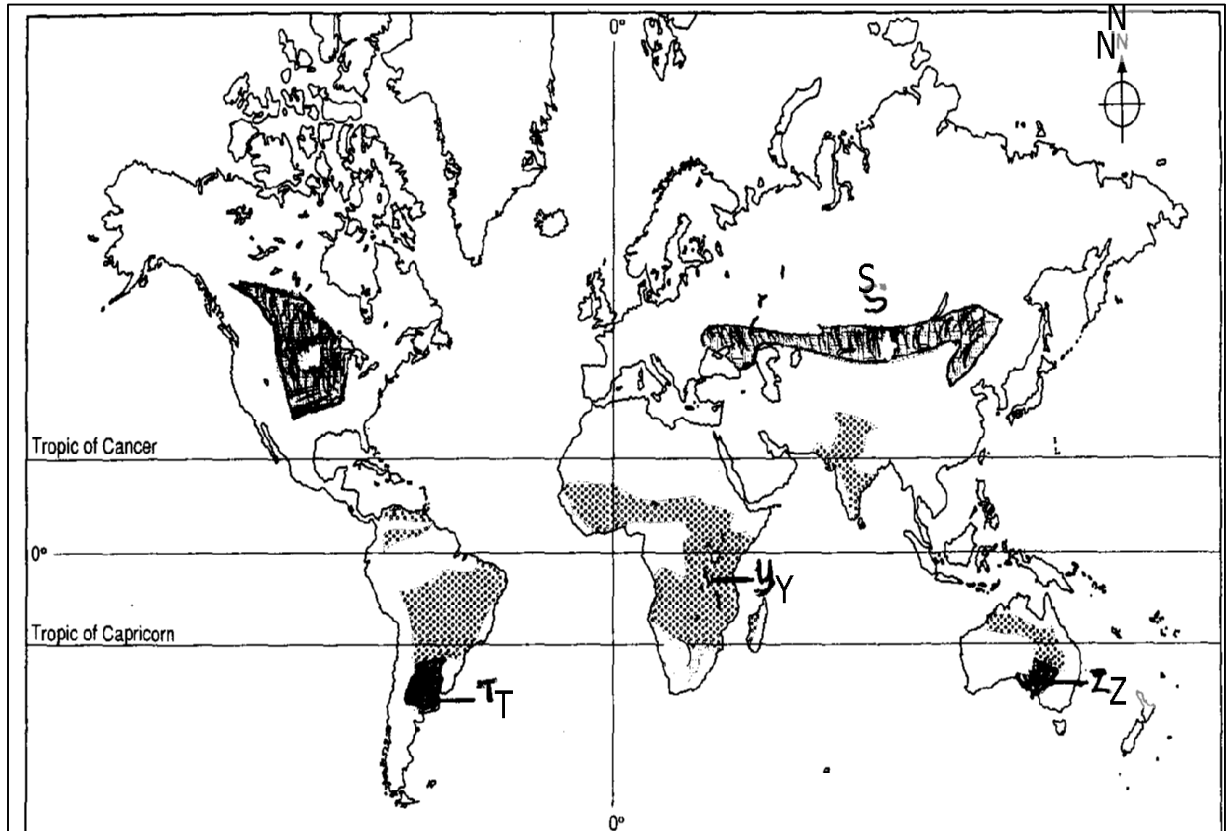


- (b) (i) Name the zones **a, b, c, d**
 (ii) Give **one** reason why there is no vegetation cover on top of the mountain

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- (c) (i) State the names given to temperate grassland vegetation in:-
 - North America
 - Argentina
 (ii) Explain **three** uses of Savanna vegetation
 (d) Explain **four** characteristics of Tropical desert vegetation

7. (a) The map below shows the grasslands of the world. Use it to answer the questions that follow:-



- (i) Name the grasslands marked S, T, Y and Z.
 (ii) State any **four** characteristics of temperate grasslands.
8. (a) (i) What is meant by derived vegetation?
 (ii) Mention **three** factors that influence the vegetation of a place.
 b) i) Give **three** natural vegetation zones on Mount Kenya
 ii) Name three temperate grasslands found in the world
 iii) Describe the characteristics of the hot desert vegetation
9. a) Explain **three** causes of the decline of the areas under forests in Kenya
 (b) Explain **four** factors that limit the exploitation of tropical rain forests in Africa.

12. FORESTRY

The topic entails:-

- (i) *Definition of forest and forestry*
- (ii) *Discussing factors influencing the distribution and types of natural forests.*
- (iii) *Discussing the significance of forests and forest products in Kenya*

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- (iv) *Identifying and explaining problems facing forestry in Kenya*
- (v) *Discussing ways and means of managing and conserving forest.*
- (vi) *Comparing and contrasting softwood forests in Kenya and Canada.*
- (vii) *Demonstrating the ability to manage and conserve forests and forest resources.*

1.
 - (a) Give **three** reasons for over-exploitation of hardwoods in Africa.
 - (b) State **four** measures taken to conserve forests in Kenya.
 - (c)
 - (i) Name **two** major lumbering maritime provinces in Eastern Canada.
 - (ii) Explain the factors that have favoured forestry in Canada.
 - (d) Explain **three** differences between softwoods in Kenya and Canada.
2.
 - (a)
 - (i) What is agro-forestry?
 - (ii) State **four** reasons why agro-forestry is being encouraged.
3.
 - (a)
 - (i) Distinguish between pure and mixed forests
 - (ii) Show how natural forests differ from planted forests in Kenya
 - (c)
 - (i) State **three** measures that are being taken in Kenya to conserve forests
 - (ii) Explain **three** factors favouring the exploitation of softwoods in Canada
4.
 - (a) Define **agro forestry**
 - (b) Outline **four** benefits of agro forestry
 - (c) Explain how the following factors influence growth of forests;
 - (i) Altitude
 - (ii) Aspect
 - (d) Explain **three** measures being undertaken to conserve forests in Kenya
 - (e) Give **four** consequences of forest depletion in Kenya
5.
 - (a)
 - (i) Distinguish between indigenous and exotic forest
 - (ii) Explain **four** ways in which natural forests differ from planted forests
 - (b) Explain **three** factors that influence the distribution of forests in Kenya
 - (c) State **three** measure that are being taken to conserve forests
6.
 - (a)
 - (i) What is **forestry**?
 - (ii) Explain **three** factors that favour the growth of natural forests on the Kenya highlands
 - (b) Explain **five** problems hindering the exploitation of tropical hardwood forests
 - (c)
 - (i) Explain **three** measures that the government of Kenya is taking to conserve forests in the country
 - (ii) State **three** factors that have led to the reduction of the area under forest in Mau forest
7.
 - (a)
 - (i) Distinguish between forestry and forest
 - (ii) Discuss the influence of the following factors on the destruction of natural forests
 - a) Climate
 - b) Human activities
 - c) Topography
8.
 - (a) Explain **three** measures which have been taken to manage forests in Kenya
 - (b) Give the differences between the soft wood forests in Kenya and

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Canada, under the following headings:

- (i) Species
 - (ii) Problems
 - (iii) Marketing
- (d) Your class intends to carry out a field study on the exotic trees of the Kenya highlands:-
- (i) Name **two** types of tree species they are likely to observe
 - (ii) Identify **three** methods you will use to record the data in the field
9. (a) Define the term **agro-forestry**
- (b) Name **three** tropical hardwoods found in Kenya
 - (c) Name **one** indigenous soft wood found in Kenya

13. EXTERNAL LAND FORMING PROCESSES

The topic entails ;-

(i) Definitions of :

- a) *Weathering*
- b) *Mass weathering*
- c) *Hydrological cycle and action of rivers*
- d) *Oceans, Seas and their Coasts*
- e) *Action of wind and water in Arid areas*
- f) *Action of water in Limestone areas*
- g) *Glaciation*
- h) *Lakes*

(ii) *Distinguishing between oceans and seas*

(iii) *Description of water movements in oceans and seas*

(iv) *Explaining the external land forming processes and the resultant features*

(v) *Identifying the sources of under ground water*

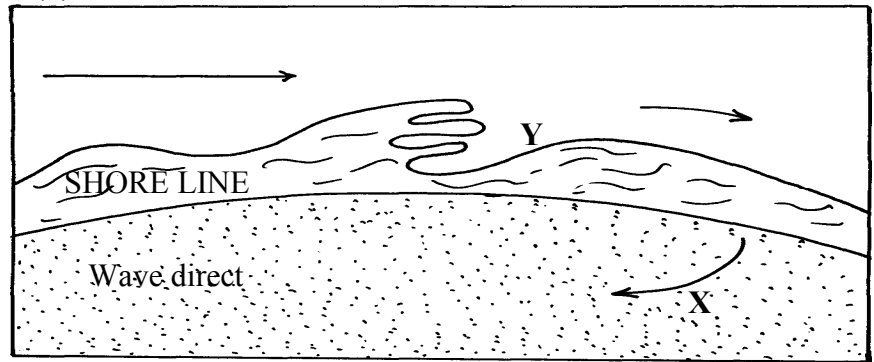
(vi) *Discussing the significance of the resultant physical features to the environment*

(vii) *Carrying out field work of land forms within the local environment*

1. (a) Name **two** rivers in Kenya West of the Rift Valley that cause frequent flooding
- (b) State **three** factors that lead to frequent flooding in the Lake region of Kenya
2. (a) i) Other than seif dunes, name two other types of sand dunes
- ii) Give **five** characteristics of seif dunes
 - iii) State **four** factors that facilitate the formation of the sand dunes
- b) Describe how a desert rock pedestal is formed
 - c) Explain **four** positive effects of desert features to human activities
3. a) i) Give **three** reasons why there are no ice sheets in Kenya
- ii) Explain **three** factors that influence the movement of ice from the place of accumulation
- b) Describe the process through which the following features are formed.
- i) an arête

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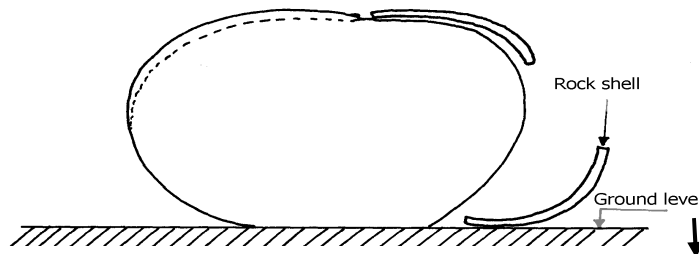
- ii) A crag and tail
 - c) Explain **four** positive effects of glaciation in lowland areas
4. a)i) Give the dates in a year during which the number of hours of darkness is equal in both the north and the south poles
 ii) Why do the lengths of days and nights vary from one part of the earth to another?
 b) State **two** effects of the rotation of the earth on its axis
5. a) What is a **lake** ?
 b) State **three** ways through which lakes are formed
6. a) i) What is **desertification**
 ii) Differentiate between **wind deflation** and **wind abrasion**
 iii) Describe how wind transports its materials through the following processes:
 I) Saltation
 II) Suspension
 III) Surface creep
 b) Describe how the following features are formed
 i) Zeugens
 ii) Oasis
7. a) State **two** ways in which glacier moves
 b) State **three** differences between a young river valley and a glaciated valley
8. The diagram below represents the action of waves on the coastline. Use it to answer questions (a) and (b)



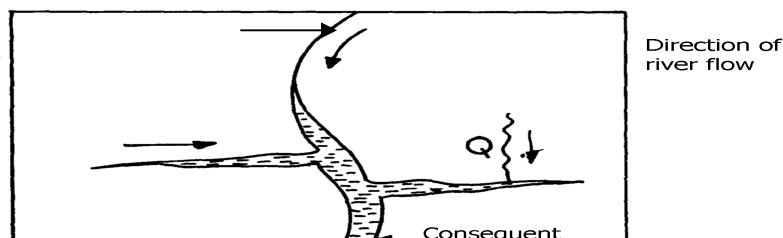
- a) Name the features marked **X** and **Y**.
 b) List **three** ways in which waves erode the coastline.
9. Name **two** sources of underground water.
10. a)i) What is chemical weathering
 ii) State **five** factors that affect the rate of chemical weathering in equatorial regions
 b) Explain the following chemical weathering process
 i) Carbonation
 ii) Hydrolysis
 c) Explain **four** economic importance of weathering to physical and human environments
11. a) Name **two** types of tides

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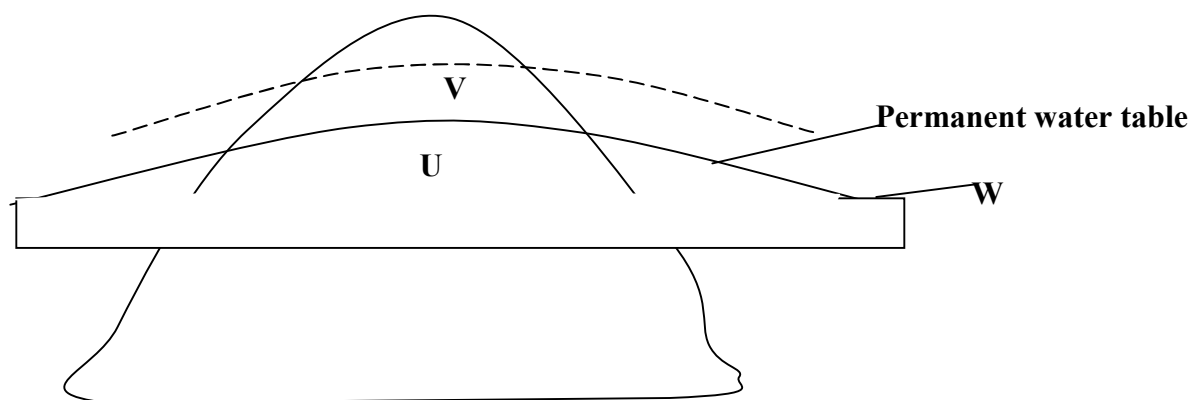
- b) Give **three** ways in which the shape of the landmasses may influence movement of ocean waters
- (c) State **four** sources of underground of water
- (d) State **five** ways in which springs occur
- (e) Account for **four** ways in which desert vegetation adopt to the climate of their environment
12. (a) Hamada and Erg
(b) Explain **three** processes of wind erosion
(c) With the aid of well labeled diagrams, describe the formation of:
i) A Barchan
ii) Rock pedestal
iii) An oasis
(d) Name **three** features resulting from water deposition in desert
13. State **three** reasons why wind erosion is predominant in desert areas.
(b) Describe **three** processes by which wind erodes.
(c) With aid of well labeled diagrams, describe the formation of the following desert features;
(i) Rock pedestal.
(ii) Zeugen.
(d) State **four** ways in which desert land forms influence man's activities.
14. (a) Differentiate between oceans and seas.
(b) Explain **three** sources of salinity in the oceans .
(c) State **three** features on the ocean floor.
15. (a)(i) What is the difference between a sea and an ocean?
ii) Explain any **three** ways in which islands are formed.
(b) (i) Describe how **Lake Victoria** was formed.
ii) Explain any **three** ways in which **Lake Victoria** has influenced the climate of surrounding areas
(c) (i) Distinguish between aridity and desertification
ii) Explain any three physical causes of aridity and desertification
16. (a) (i) What is chemical weathering?
ii) Name any **three** surface features that result from carbonation.
(b) The diagram below shows a feature resulting from weathering. Use it to answer the questions that follow:-



- (i) What is the name of the feature?
(ii) Describe how the feature is formed.
17. (a) Differentiate between a river system and a river regime.
(b) The diagram below shows a drainage pattern. Use it to answer the questions that follow:-



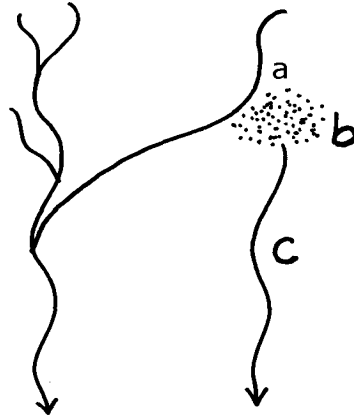
- (i) Name the drainage pattern.
 - (ii) Name the parts labelled **P** and **Q**
18. (a) Name any **two** features deposited by glacier in lowland areas.
 (b) Describe how a roche moutonee is formed.
19. (a) The diagram below shows zones of underground water. Use it to answer the questions that follow:-



- (i) Name the parts labelled **U** and **V**.
 - (ii) Name the surface water feature likely to form at the part labelled **W**.
 - (b) Explain how the nature of underground rocks influences the existence of underground water
20. (a) (i) State **two** ways in which wind erodes the surface of the earth
 (ii) Explain **three** ways thorough which wind transports its load
 (b) Using well labeled diagrams, describe how the following desert features re formed:-
 (i) Zeugen
 (ii) Rock pedestal
 (c)(i) Give the name given to rocky desert in the Sahara
 (ii) Explain **three** positive effects of desert features to both human and physical environment
21. (a) What is a lake?
 (b) (i) Name **two** types of lakes formed due to vulcanicity
 ii) explaining how lake Victoria has influence the climate of the surrounding area
 (c) (i) State **four** main reasons why lakes within the Rift valley are salty
 (ii) Name any **three** fresh water lakes in Kenya which are within the rift valley

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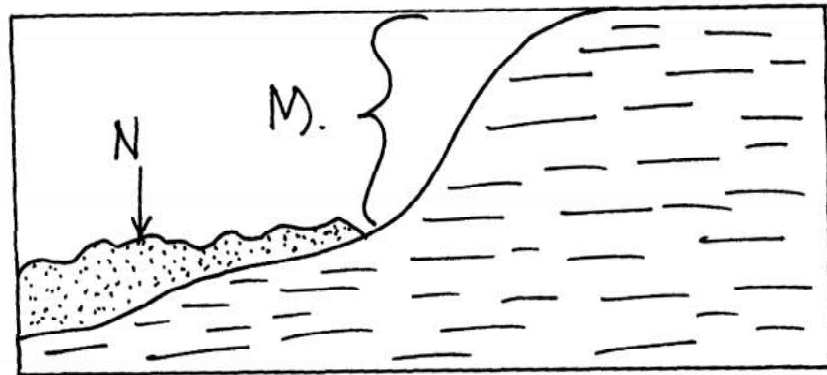
- (d) Explain any **four** economic significance of lakes to Human activities
22. (a) Name three types of tides
(b) Differentiate between oceans and seas
23. (a) (i) Differentiate between aridity and desertification
(ii) Describe **three** ways through which wind erodes the desert areas
(b) Using well labelled diagrams describe how the following desert features are formed:
(i) Rock pedestals
(ii) Yardangs
(c) State **five** uses of desert features
24. (a) State **four** factors which may cause a waterfall to form
(b) (i) Describe how river braids are formed
(ii) Name **three** features resulting from river rejuvenation
(c) Describe how a river capture occurs
(d) Give **two** reasons as to why Lake Naivasha is a fresh water lake
25. Give **three** ways through which rocks may be metamorphosed
26. (a) (i) State **two** processes through which the wind erodes the surface.
(ii) Explain **three** ways through which wind transport its load
(b) Describe how the following desert features are formed :
(i) Oasis
(ii) Zeugen
(c) Explain **four** ways in which desert land forms are of significance to human activities
27. (a) (i) Name **two** ways in which ice moves
(ii) State **two** factors which facilitate the movement of ice
(b) Identify **two** erosional features in glaciated lowlands
(c) Describe how the following features are formed:
(i) Outwash plain
(ii) Moraine-dammed lake
(d) Explain **four** ways in which a glaciated landscape is of significance to human activities
28. (a) Differentiate between mechanical and chemical weathering
(b) State **three** factors which influence weathering
29. a) Differentiate between a spring and well.
b) With aid of a well labeled diagram, show the three Zones of ground water.
c) List four candidates necessary for the formation of an artesian well.
d) Explain **three** ways in which ground water is of significant to human activities.
e) i) Define the term Karet scenery
ii) Give **two** conditions for the development of Karet landscape
iii) Outline the significance of Karet landscape to human activities.
30. (a) Use the diagram below to answer the following questions



- (i) Name the parts labelled :-
 (ii) State **four** causes of river rejuvenation
31. a i) Differentiate between denudation and mass wasting.
 ii) Explain **two** reasons why chemical weathering is dominant in humid tropics.
 b) Explain **three** factors that influence mass wasting.
 c) With aid of a diagram, explain the process of solfluxion.
 d) i) State **four** negative effects of mass wasting.

Name the:

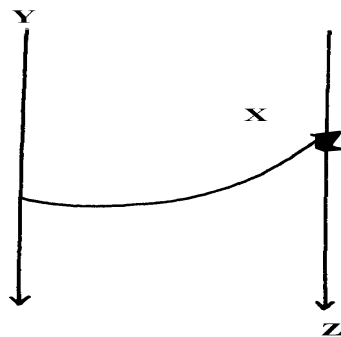
- i) Type of mass movement shown



- iii) Features M and N
 iv) State the factor that influenced the process above
- 32 a) State **three** reasons why wind is a major agent of land sculpture in the desert.
 b) Explain any **two** processes of wind erosion in deserts.
 c) With and of well labeled diagram, explain how a deflation hollow is formed.
 d) State **four** characteristics of a seif dune.
33. a) State **two** causes of coastal submergence.

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- b) Name **three** conditions necessary for the growth of coral polyps.
34. a) State **two** ways in which wind transport it's load
b) Highlight the factors that influence the formation of a desert sand dune
35. a) Name **three** sources of lake water.
b) State **three** factors that determine the size of a lake.
36. a) i) What is a **river**?
ii) Distinguish between a river confluence and a river tributary
b) Describe how a river erodes its channel by the following processes
i) Hydraulic action
ii) Abrasion
37. a) Describe the process of a river capture
b) State **five** characteristics of a flood plain
38. a) i) Name **four** features resulting from water action in deserts
ii) State **three** factors which contribute to the development of deserts
iii) Name **three** types of deserts according to the nature of their surfaces
b) i) Explain **three** factors which influence wind transportation in deserts
ii) How is an oasis formed?
39. (a) What is the difference between swash and backwash
(b) Why is wave cut platform and cliff line in Mombasa found above the sea level?
40. (a) Define the term river capture
(b) The diagram below shows a river capture, name the features marked **X, Y, Z**



41. (a) Differentiate between **accordant** and **discordant** drainage systems
(b) State **three** factors that facilitate formation of deltas
42. a) i) What is a desert
ii) Name **three** types of desert landscapes
b) i) Name **two** processes of wind erosion responsible for sculpturing desert landscapes

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- ii) Using well labeled diagrams describe how a zeugen may be formed
 - c) i) Name **four** desert landforms found in deserts associated with the action of water
 - ii) What is the significance of desert landforms
43. (a) (i) Name **three** processes of wind erosion in desert
- (ii) State **three** factors which influence the rate of wind transportation
- (b) With the help of a well labeled diagram describe the formation of the following features:-
- (i) Rock pedestal
 - (ii) Oasis
- (c) (i) List **three** external land forming processes which lead to the formation of lakes
- (ii) Explain **three** ways in which lakes influence the natural environment
- (d) (i) With the aid of a well labelled diagram, Describe how limestone pillars are formed in Karst scenery
- (ii) Explain **three** significance of resultant feature in limestone areas to human activities

14. SOILS

The topic entails:-

- (i) *Definition of soil, soil profile, soil catena, soil degradation and soil erosion*
- (ii) *Discussing the composition of soil and soil forming processes*
- (iii) *Description of properties and characteristics of soils*
- (iv) *Description of soil profile and soil catena*
- (v) *Explaining soil degeneration*
- (vi) *Classifying soils according to soil order*
- (vii) *Discussing the significance of soils*
- (viii) *Discussing ways and means of managing and conserving soils.*
- (ix) *Identifying different types of soils and their uses within the local environment*
- (x) *Demonstrating the ability to manage and conserve soils.*

1. a) Differentiate soil profile from soil catena
- b) Name three types of soil types according to soil structure
- c) i) Beside planting of trees state any other **three** measures that can be used to control desertification.
- ii) Explain how trees help in the conservation of the soil.
2. a) i) What is soil?
- ii) Differentiate between **soil catena** and **soil profile**.
- b) i) Explain how the following factors influence soil formation;
- Climate
 - Living organisms
- ii) Give **three** factors that determine the colour of the soil.
3. a) Briefly explain the factors that influence the development of soil catena.
- (b) (i) Define the term soil profile
- (ii) Draw a simple diagram of soil catena
- (iii) Give **any four** soil forming processes

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- (iv) State **four** causes of soil degeneration
4. a)i) Define the term soil
ii) Name three components of soils
b)i) Apart from living organisms state four other factors that influence soil formation
ii) Explain how living organisms influence soil formation
c) i) Distinguish between soil air and soil texture
ii) Describe the process of podzolisation
iii) State the characteristics of soils in grassland areas
5. (a) Draw a diagrammatic representation of vegetation zones on a slope common in tropical land
(b) (i) What is soil catena?
(ii) Draw a labeled diagram to show a well developed soil profile.
(iii) State **three** characteristics of the soils found in the arid regions of Kenya.
(b) Give **three** factors that determine the colour of soil.
(c) Describe how laterization occurs.
(d) Explain how the following farming practices cause soil erosion;
(i) Burning.
(ii) Continuous application of fertilizer on farm lands.
(iii) Monoculture.
6. (a) (i) Apart from latosols name any **two** other types of zonal soils
(ii) Describe how lateritic soils are formed
(iii) State any **two** characteristics of azonal soils
(b) (i) Explain any **three** ways in which rain water can lead to degeneration of soils.
(ii) Apart from growth of plants, state any **three** other ways in which soils are beneficial to people
7. (a) (i) Differentiate between a soil profile and soil catena
(ii) State **two** processes of soil formation
(b) Explain how the following factors influence soil formation:
(i) Climate
(ii) Living organisms
(iii) Topography
(c) (i) What is soil degeneration
(ii) Explain **three** ways through which vegetation naturally protects and prevent soil erosion
(iii) Give **two** sound farming methods that help conserve soil erosion
8. (a) (i) What is soil Catena?
(ii) State **three** factors which influence the development of a soil catena
(b) Explain **five** negative effects of soil erosion
(c) Describe how podzolization occurs in soils
9. (a) Name **three** components of soil
(b) Explain how the following factors influence the formation of soil :
(i) Climate
(ii) Parent material
(c) (i) Explain **three** ways in which human activities contribute to soil degeneration

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(ii) Draw a well labeled diagram to show a mature soil profile

10.
 - a) What is a **lake**?
 - b) State **three** ways through which lakes are formed
11.
 - a) Define the term “**Soil**”
 - b) Explain how the following factors influence the formation of soil
 - i) Climate
 - ii) Topography
 - c)
 - i) State **two** ways in which humus improves the quality of soil
 - ii) State **four** characteristics of desert soils
 - d) Describe how the following types of erosion occur
 - i) Sheet erosion
 - ii) Gulley erosion
 - iii) State **two** economic uses of soils
12.
 - (a) What is soil conservation?;
 - (b) State **three** methods that assist in soil conservation

15. AGRICULTURE

The topic entails:-

- (i) *Definition of Agriculture*
- (ii) *Discussion of factors influencing Agriculture*
- (iii) *Explaining types of Agriculture*
- (iv) *Location of major cash crops on the map of Kenya*
- (v) *Discussing the selected crops in respective countries*
- (vi) *Identifying and explaining similarities and differences between selected crop farming in Kenya and that of other parts of the world.*
- (vii) *Discussing Pastoral farming in Kenya*
- (viii) *Comparing and contrasting:*
 - (a) *Dairy farming in Kenya and Denmark*
 - (b) *Bee farming in Kenya and Argentina*
- (ix) *Carrying out field work on agricultural activities in the local environment*

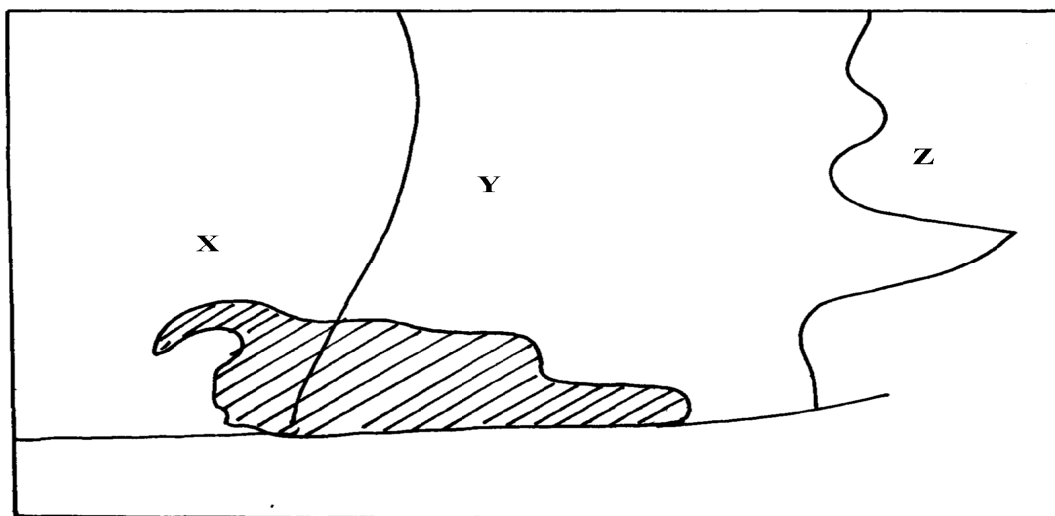
1.
 - (a)
 - (i) State the difference between land reclamation and land rehabilitation.
 - (ii) Other than tree planting, give **three** other methods employed in rehabilitation of land.
 - (b)
 - (i) State **three** objectives that led to the establishment of Mwea Irrigation Scheme in Kenya
 - ii) Explain **four** human problems that face irrigation farming in Kenya.
2.
 - (a) Describe the stages followed in the reclamation of land from the sea in Netherlands
 - (b) State **four** benefits of the delta plan to the Netherlands.
3.
 - (a)
 - (i) Name **two** districts in Rift valley where maize is grown in plantations.
 - (ii) Describe maize production from planting to harvesting.
 - (b) Outline the significance of maize growing in Kenya.

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4. Explain **three** problems facing maize farmers in Kenya.
5. (a) Name **two** exotic breeds of dairy cattle reared in Kenya.
(b) Give **three** reasons why the Kenya highlands are suitable for dairy farming.
6. (a) Name **two** horticultural crops grown in Kenya
(b) State **three** reasons why horticulture is more developed in the Netherlands than in Kenya.
7. (a) Name **two** horticultural crops that are produced in Kenya
(b) Explain **two** reasons why horticultural produce is exported by air
8. (a) (i) What is nomadic pastoralism?
(ii) State **three** characteristics of nomadic pastoralism
9. a)i) Name **three** exotic breeds of dairy cattle kept in Kenya
ii) Apart from milk name three other dairy products
b)i) State **three** factors which favour the rearing of dairy cattle in Kenya highlands
ii) Explain **four** problems facing dairy farmers in Kenya
c) i) Compare dairy farming in Kenya and Netherlands
ii) Name **two** dairy plants in Kenya
10. (a) Name **two** dairy breed found in the lowlands of Kenya
(b) State **three** problems that have hindered the dairy activities in the Kenya highland
11. a) State **three** physical factors that favour wheat farming in Kenya
b) Give **three** problems facing wheat farming in Kenya
12. (a) Name ;
(i) **Three** provinces in Kenya where wheat is grown on commercial scale
(ii) **Three** wheat producing provinces in Canada
(b) Explain **three** conditions that favour wheat farming in Kenya
(c) Explain **five** factors which enable Canada to produce more wheat than Kenya
(d) State **three** importance of wheat to the economy of Canada
13. (a) Name **three** districts in Kenya where wheat is grown on large scale
(b) Account for any **four** physical conditions needed for the growth of wheat
(c) Describe the cultivation of wheat under the following:-
(i) Processing of wheat
(ii) Marketing of wheat
(d) (i) Explain **two** problems facing wheat farmers in Kenya
(ii) Give **two** reasons why Canada produces more wheat than Kenya
(2mks)
14. (a) State **three** physical conditions that favour sugarcane cultivation in Western Kenya
(b) Give **two** reasons why we need to encourage dairy farming in Kenya
15. (a) (i) Apart from Ghana name two other countries that produces cocoa in Africa
(ii) State **three** physical conditions that favour large scale cocoa farming in Ghana

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- (b) Outline the stages cocoa goes through from harvesting to the time it is ready for export
 (c) Explain **three** problems experienced by cocoa farmers in Ghana
16. (a) (i) What is horticulture
 (ii) Horticulture is more developed in the Netherlands than in Kenya. Give **four** reasons for this
 (iii) Give **three** reasons why the growing of flowers in green house is preferred in Kenya
17. Give **four** physical factors favouring wheat –growing in Kenya
18. (a) State **two** physical conditions favoring dairy farming in the Kenya highlands
 (b) Compare dairy farming in Kenya and Denmark under the following headings:-
 (i) Feeding of the animals
 (ii) Marketing of the products
19. (a) Name **two** exotic breeds of beef cattle reared in Kenya
 (b) State **three** physical conditions that favour beef farming in Argentina
20. (a) The figure below shows wheat growing province of Canada



- (i) Name the provinces marked **X**, **Y** and **Z**
 (ii) State **two** physical factors favouring wheat farming in the named provinces above
 (b) Give **one** province where coffee is grown on large scale in Kenya
21. a) Differentiate between horticulture and market gardening
 b) State **three** factors that favour horticultural industry in Kenya
22. a) Define **nomadic pastoralism**
 b) Give **three** characteristics of pastoral farming in Kenya
23. a) Give **four** reasons why there has been a steady increase of milk production in Kenya in the recent past
 b) Give **similarities** between dairy farming in Kenya and Denmark
 c) Explain **three** benefits of dairy farming in Kenya

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24. The table below shows horticultural crop production in Kenya

Crop	Quantity in tonnes
flowers	42, 500
oranges	22, 600
tomatoes	20, 300
carrots	15, 400

- (a) (i) Draw a divided rectangle 15cm long to represent horticultural crop production in Kenya in the year 2000 using the data above.
- (a) (i) Calculate the range of the above data
(ii) What is the percentage of horticultural crop with the lowest tonnage?
- (b) (i) State **two** advantages of using divided rectangle to represent geographical information.
(ii) State **two** climatic conditions that favour horticultural crops in Kenya
- (c) Explain **four** problems facing horticultural farming in Kenya
25. (a) Name **three** main breeds of dairy cattle reared in Kenya
(b) State **three** benefits of dairy farming in Kenya
26. (a) Name **two** leading export commodities from Kenya
(b) State **three** problems facing trade in Kenya
27. (a) State any **three** roles played by livestock in the economy
(b) Identify any **two** solutions to problems facing pastoralists in the Northern part of Kenya
28. (a) List **two** economic factors that influence Agriculture
(b) (i) List **two** uses of cocoa
(ii) State **one** economic problem experienced in cocoa farming in Ghana
29. (a) Differentiate between **horticulture** and **market gardening**
(b) Name **two** main types of flowers grown in Kenya
(c) What problem do farmers in Kenya face in carrying out this activity?
30. (a) State **three** ways in which drought affect the agricultural sector in Kenya.
(b) State **three** reasons why the government of Kenya is encouraging bee keeping in the country?
31. (a) List **three** social factors that influence Agriculture.
(b) State **two** factors that favour mechanization of wheat farming in Canada.
32. (a) State **three** climatic factors that favour the growing of oil palm in Nigeria.
(b) State any **two** products obtained from oil palm.
33. (a) Define the term pollution as used in Geography.
(b) Dither their air pollution name **three** forms of pollution.
(c) State physical conditions required for the growing of tea in Kenya.

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- (d) Explain **four** problems experienced in small scale to production in Kenya.
34. (a) Name **two** exotic beef cattle breeds in Kenya.
(b) State **three** steps taken by the government to improve beef farming in Kenya.
(c) State **three** problems facing beef farming in Kenya.
(d) (i) Explain **three** physical conditions that favour beef farming in Argentina.
(ii) Explain **two** ways in which beef farming contribute to the economy of Argentina.

16. LAND RECLAMATION AND REHABILITATION

This topic entails :-

- (i) **Definition of land reclama;tion and land rehabilitation**
 - (ii) **Discussing:**
 - (a) **The factors that influence the location of the selected irrigation schemes in Kenya**
 - (b) **The significance of irrigation of farming in Kenya**
 - (c) **The problems experience in irrigation farming in Kenya**
 - (iii) **Description of the methods used in land reclamation and rehabilitation in Kenya**
 - (iv) **Comparing the methods of land reclamation in Kenya and the Netherlands**
1. (a) Apart from Mwea, name **three** other large irrigation schemes in Kenya
(b) (i) Explain **four** conditions that made Mwea-Tebere a suitable location for an irrigation scheme
(ii) Explain **four** problems facing farmers in Mwea irrigation scheme
(c) State **six** benefits which Kenya derives from irrigation farming
2. (a) A part from draining swamps, state **two** other methods used to reclaim land in Kenya.
(b) State **three** benefits that resulted from the reclamation of Yala Swamp.
3. (a) (i) Distinguish between **land reclamation** and **land rehabilitation**
(ii) Name any **three** methods of irrigation.
(iii) State **two** advantages of irrigation over natural water supplies
(b) (i) Why was Mwea Tebere irrigation scheme initiated?
(ii) Explain **four** physical conditions that favoured the establishment of Mwea Irrigation Scheme
(iii) Give **three** problems that are faced in Mwea Irrigation Scheme
(c) (i) What is a **polder**.
(ii) Describe how land is reclaimed and prepared in the Netherlands
(iii) Give any **three** benefits of the delta plan
(d) How does the above activity differ from that in Denmark?
4. Explain **three** problems caused by a large aging population
5. a) i) Apart from rice name two other crops grown under irrigation at mwea Tabere irrigation scheme
ii) Explain **four** factors that favoured establishment of Mwea Tabere irrigation scheme
b) Give **four** problems associated with irrigation farming on R.Thiba & Nyamindi
c) Explain **three** benefits of Mwea Tabere irrigation scheme
d)i) Name **two** other irrigation schemes in Kenya apart from Mwea Tabere
ii) State **three** benefits of rice farming

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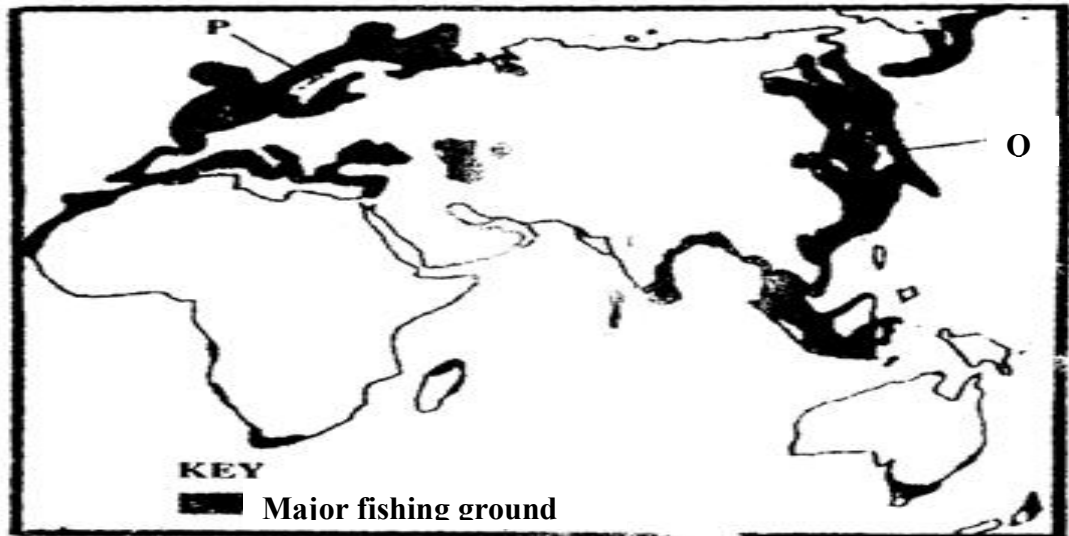
6.
 - (a) Differentiate between **land reclamation** and **land rehabilitation**
 - (b) Explain **four** methods of land reclamation in Kenya
 - (c) Explain any **two** methods used in land reclamation and rehabilitation in Netherlands
 - (d) State **four** factors which influenced the establishment of Perkerra Irrigation Scheme
7.
 - (a) Explain **two** significance of irrigation farming in Kenya
 - (b) State **three** benefits of syader see projects
8.
 - (a) Name any **three** crops grown under irrigation farming in Kenya
 - (b)
 - (i) Explain **four** conditions that made Mwea Tebere a suitable location for irrigation farming
 - (ii) Outline **two** aims of the tsetseflies control project in the Lambwe valley of Kenya
 - (iii) State **four** efforts being made to conserve water catchment areas of Kenya
9.
 - (a)
 - (i) What is a **polder**
 - (ii) Name **three** crops grown in the polders.
 - (b) Describe the stages involved in the reclamation of land from the sea in the Netherlands
 - (c) Explain **four** ways that the Netherlands benefited from the delta plan project
 - (d) State **six** problems experienced in irrigation farming in Kenya
10.
 - a) A part from Mwea name three other large irrigation schemes in Kenya
 - b) Explain **four** conditions that made Mwea a suitable location for irrigation scheme
 - c) Explain **four** ways through which tenants have benefited from Mwea irrigation scheme
 - d) Explain **three** problems faced by farmers in Mwea Tabere irrigation scheme
11.
 - (a)
 - (i) Differentiate between land reclamation and land rehabilitation
 - (ii) Name **two** methods of rehabilitating land in Kenya
 - (b)
 - (i) What is a polder
 - (ii) Name **two** crops grown in the polders
 - (c) Describe the stages of reclamation of land from the sea in the Netherlands
 - (d)
 - (i) State **three** physical factors that influence the establishment of Pekerra irrigation scheme
 - (ii) Explain **four** significance of irrigation farming in Kenya
12.
 - (a)
 - (i) Differentiate between land reclamation and land rehabilitation
 - (ii) List **three** ways in which tsetse fly menace was controlled in the Lambwe valley
 - (b)
 - (i) Name **two** areas in Kenya where swamps have been reclaimed
 - (ii) State **four** factors which influenced the establishment of Perkerra irrigation scheme
 - (c) Explain **three** problems facing irrigation farming in Kenya
 - (d) List **four** benefits of the Delta plan project in Netherlands
13.
 - (a) State **two** methods used to reclaim land in Kenya
 - (b) Outline the stages through which land is reclaimed from the sea in the Netherlands
14.
 - (a) List **four** types of wasteland that can be reclaimed
 - (b) Give **two** advantages of irrigation farming compared to rain fed farming

17. FISHING

The topic entails:

- (i) *Definition of fishing and fisheries.*
- (ii) *Explaining the factors influencing fishing*
- (iii) *Accounting for the location of the major fishing grounds of the world.*
- (iv) *Describing type and methods of fishing*
- (v) *Discussing fresh water and marine fisheries in East Africa*
- (vi) *Assessing the significance of the fishing industry in Kenya*
- (vii) *Discussing problems facing fishing in Kenya and their possible solutions*
- (viii) *Comparing and contrasting fishing activities in Kenya and Japan*
- (ix) *Explaining ways and means of managing and conserving fresh water and marine fisheries.*

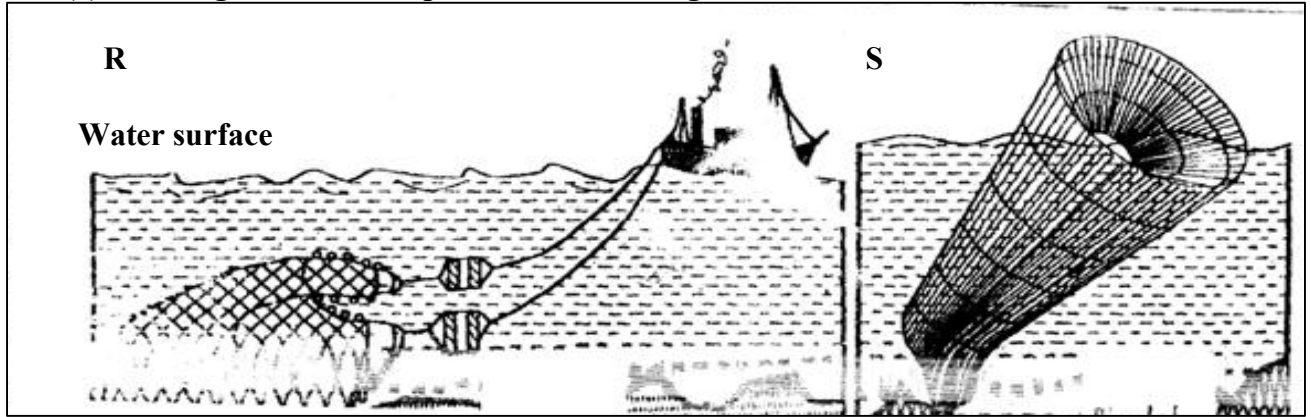
The map below shows some major fishing grounds in the world. Use it to answer questions below:



1. (a) (i) Name the countries marked **P** and **Q**
(ii) Explain **four** conditions that favour fishing in the shaded coastal water

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2. (a) The diagrams below represent some fishing methods

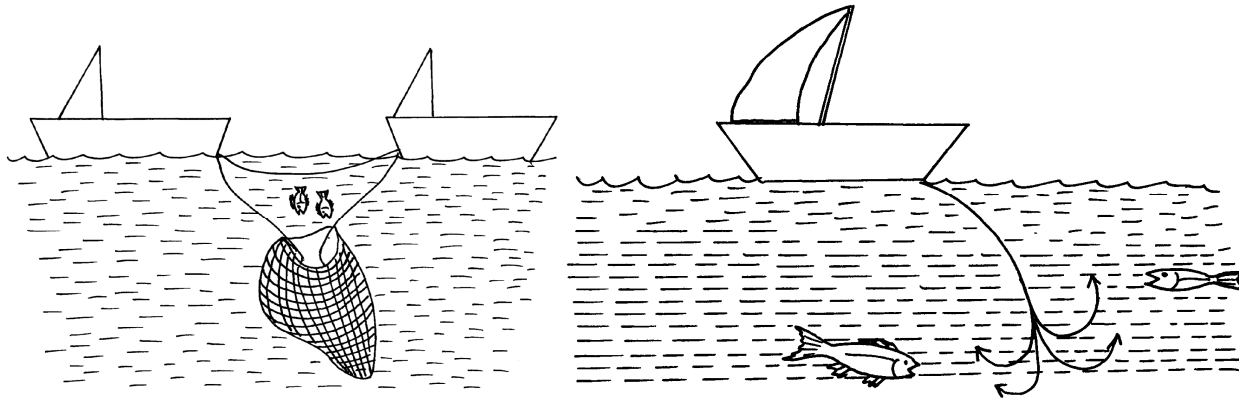


- (a) (i) Name methods **R** and **S**
 (b) Describe how each of the two methods are used in fishing
 (c) Explain **three** measures used to conserve fish in Kenya
3. (a) (i) What is fish farming?
 (ii) Explain **three** measures that have been undertaken by the government of Kenya to encourage fish farming.
 (b) Explain **four** problems which face marine fishing in Kenya.
 (c) (i) Name **three** fishing grounds in the Northern Hemisphere.
 (ii) Explain **three** physical factors that favour fishing in Japan.
4. (a) Differentiate between veins and lodes.
 (b) State the effects of dereliction
5. (a) (i) Name **two** methods of fishing.
 (ii) Name **two** types of fish caught along the Eastern Coast of Canada.
 (b) Explain how the following factors favour fishing:-
 (i) Indented Coastline
 (ii) Ocean Currents
 (c) Explain **four** ways in which fisheries in Kenya can be conserved.
6. (i) Give **three** methods used to preserve fish.
 (ii) Explain **three** problems experienced by fishermen in Lake Victoria.
7. State **four** reasons why marine fisheries in Kenya are under-developed
8. (a) (i) Name **three** types of nets used in modern fishing
 (ii) Identify the **two** main fishing grounds of the Pacific Ocean
 (b) Explain any **four** problems facing fishing in East Africa
 (c) Explain any **four** ways in which fishing is significant to Kenya
 (d) Name **four** areas where fish farmers in Kenya can obtain fingerlings
9. a) i) State the **three** categories of fish communities
 ii) Explain **four** reasons why North East Atlantic is one of the most extends
 iii) State **four** measures taken by the Kenyan government in the management and

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conservation of fisheries

10. (a) A part from trawling, name **two** other modern methods of fishing
(b) State **two** reasons why the Western Coast of Africa has high concentration of fish than the Eastern Coast.
11. (a) Name **two** commercial fishing methods commonly used in Atlantic fishing ground
(b) State **three** problems Kenyan fishermen face while fishing in Lake Victoria
12. (a) State **two** measures that have been taken to conserve fish in Kenya
(b) Give **three** reasons why Norway is a great fishing nation
13. (a) Name commercial methods of fishing shown in the diagrams below:



- (b) State **three** reasons why marine fisheries in Kenya are underdeveloped
14. (a) Differentiate between **pelagic fish** and **demersal fish**?
(b) Identify **two** problems facing the fishing industry in Kenya.

18. WILDLIFE AND TOURISM

This topic entails:

- (i) *Definition of wildlife, tourism and ecotourism*
- (ii) *Distinction between:*
 - (a) *Game reserves, National Parks, and Sanctuaries*
 - (b) *Domestic tourism and International tourism*
- (iii) *Explaining factors influencing;*
 - (a) *The distribution of wildlife in East Africa.*
 - (b) *Tourism in Kenya*
- (iv) *Location of national parks, major game reserves and sanctuaries on a map of East Africa.*
- (v) *Identification and discussion of tourist attractions in Kenya.*
- (vi) *Discussion of the significance of wildlife in East Africa.*
- (vii) *Discussion of:-*
 - (a) *Problems facing wild life in East Africa.*
 - (b) *Problems facing and associated with tourism in Kenya.*

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- (viii) *Discussion of the management and conservation of wildlife in East Africa.*
- (ix) *Discussion of the future of tourism in Kenya.*
- (x) *Comparison and contrast between tourism in Kenya and Switzerland.*

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1. Use the map of East Africa below to answer questions (a) (i)

MAP OF EAST AFRICA



- i) Name the national parks marked **P**, **Q**, and **R**
2. (a) State **two** differences between a National Park and a Game Reserve
(b) State **three** measures being taken to conserve wildlife in Kenya
3. (a) Differentiate between game reserves and game parks
(b) State **three** ways in which human activities are a threat to wildlife
4. (a) What is balance of payment?

(b) Identify **three** problems that face traders dealing with primary goods
5. (a) Why are some parts of Kenya not developed for tourism?
(b) State **two** factors that hinder domestic tourism in Kenya.
6. (a) State **three** problems facing wildlife conservation in Kenya.
(b) State **two** human factors that have made Switzerland a major tourist destination.

19. ENERGY

This topic entails:

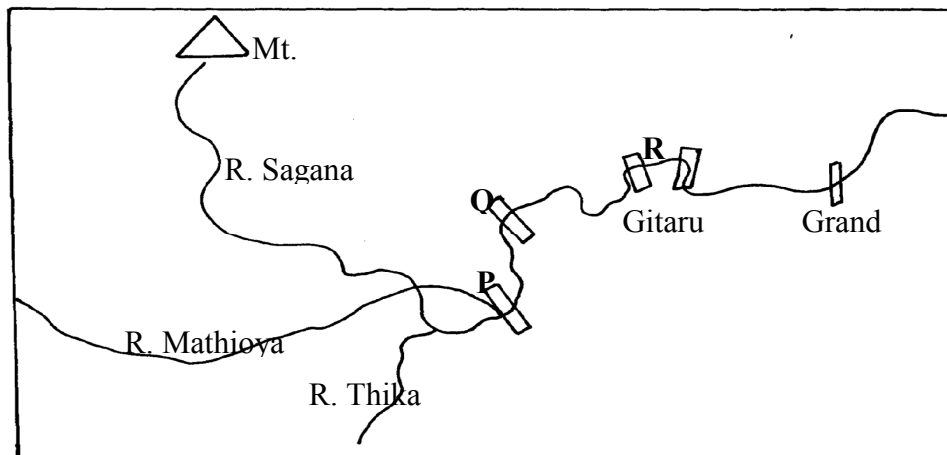
- (i) *Definition of energy*
- (ii) *Discussion of sources and types of energy.*
- (iii) *Discussion of the development of electric power projects in Kenya and Uganda.*
- (iv) *Identifying and locating other power projects in Africa*
- (v) *Explaining the significance of energy*
- (vi) *Explaining the impact of the energy crisis in the world.*
- (vii) *Discussing ways and means of managing and conserving energy.*
- (viii) *Identifying sources and uses of energy within the local environment.*

1. (a) Name **two** oil producing countries in the middle East
(b) Give **three** contributions of oil to the economies of Middle East countries

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2. (a) What is a multi-purpose project?
(b) Name **two** multi-purpose projects in Africa
3. (a) (i) Name **five** renewable sources of energy
(ii) State **three** disadvantages and three advantages of wind as source of energy
(b) (i) What is geothermal power?
(ii) Name **two** areas in Kenya which have a potential for producing geothermal power
(d) Explain **three** measures taken by the Kenyan government to conserve energy
4. a) i) What is energy crisis
ii) State **four** causes of energy crisis
b) i) Name three non- renewable sources of energy
ii) Apart from seven forks Dam project name two other hydro electric plants in Kenya
iii) Explain **four** factors that favoured the establishment of seven forks dam project
c) State **two** effects of energy use on environment

5. (a) Give **three** advantages of suing solar energy
(b) Identify the hydro-power stations marked **P**, **Q** and **R** in the diagram



- (c) Explain any **five** ways in which energy contributes to the growth of the economy
(d) (i) Explain any **three** problems associated with energy crisis
(ii) Suggest any **three** ways to minimize energy wastage
6. (a) Define the term renewable sources of energy.
(b) (i) Explain **four** physical factors influencing the generation of hydro-electric power.
(ii) Outline **three** limitations in the production of geothermal power in Kenya.
(c) (i) Explain any **four** negative effects of the energy crisis in the world.
(ii) State **four** possible methods that the government of Kenya can use to conserve energy.
7. (a) Explain the impact of the oil crisis to Kenya
(b) What measures has the Kenyan government carried out to conserve energy
8. a) What is energy crisis?
b) State **three** environmental impact of energy crisis in Kenya

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9. (a) Name **two** sources of thermal electricity
(b) Explain **four** benefits Kenya would get by striking oil in Isiolo
(c) Explain **three** measures taken by the Kenya government to manage and conserve energy
(d) Form **four** students of Nyabisawa Girls carried out a field study at Olkaria Geothermal power stations
10. (i) State **three** preparation they made prior to the study
(ii) State **three** methods of data collection they could have used
(iii) State **three** uses of Geothermal energy they learnt
11. (a) (i) Name **two** non-renewable sources of energy
(ii) Explain **four** physical factors that influence the location of hydroelectric power station
(b) (i) What is energy crisis?
(ii) State **three** causes of energy crisis
(iii) Explain **two** effects of energy crisis
12. (a) List **two** renewable fossil fuels
(b) Identify any **three** functions of hydro power reservoir other than power generation
(c) State **three** reasons why Kenya spent so little on importing maize during the year 2002
(d) (i) Name **two** provinces in Kenya where maize is grown on large scale
(ii) State any **two** uses of maize
13. (a) Name the dams marked **E, F** and **G**
(b) Differentiate between renewable and non-renewable sources of energy.
(c) Explain **three** factors that influenced the location of Owen falls dam in Uganda
(d) Explain **four** problems that hinder development of Hydro-electric power projects in Africa

20. INDUSTRY

This topic entails :-

- (i) *Definition of industry and industrialization*
- (ii) *Explaining the factors that influence the location and development of industries*
- (iii) *Describing types of industries*
- (iv) *Accounting for the distribution of industries in Kenya*
- (v) *Explaining the significance of industrialization to Kenya*
- (vi) *Discussing the problems of industrialization and possible solutions*
- (vii) *Comparing and contrasting aspects of industrialization in selected countries.*
- (viii) *Carrying out field work on an industry in the local environment.*

1. Give **five** effects of liberalization of the oil industry in Kenya.
2. (a) Define the term “**industrial inertia.**”
(b) State **three** factors that make industrialists prefer to locate their plants near already established industries.
(c) Explain **three** benefits that would result from rural electrification in Kenya
3. (a) Define the term industrial inertia

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- (b) State **three** negative effects of industrialization
4. (a) Differentiate between a factory and industry
(b) Name any **three** types of tertiary industry

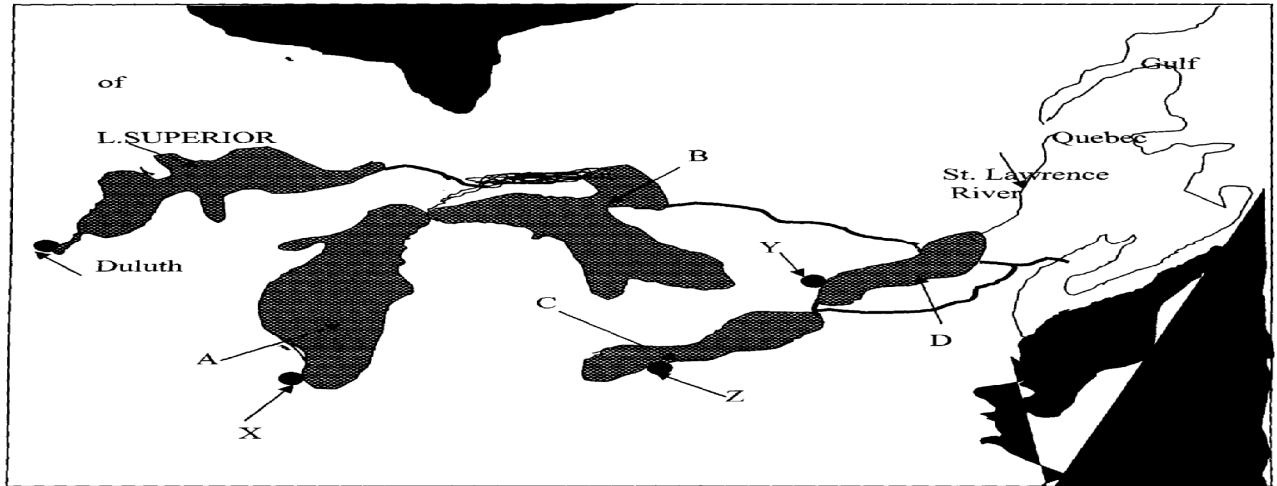
21. TRANSPORT AND COMMUNICATION

This topic entails :-

- (i) *Definition of transport and communication*
 - (ii) *Identifying modes of transport and types of communication.*
 - (iii) *Location of major lines of transport in Africa.*
 - (iv) *Outlining the role of transport and communication in economic development of Africa.*
 - (v) *Discussing problems facing transport and communication in Africa and the efforts being made to solve them.*
 - (vi) *Explanation of the role of the Great Lakes and the St. Lawrence Sea Way in the economies of USA and Canada.*
1. (a) (i) State **four** factors that influence transport and communication
(ii) State **three** problems facing transport and communication in Africa

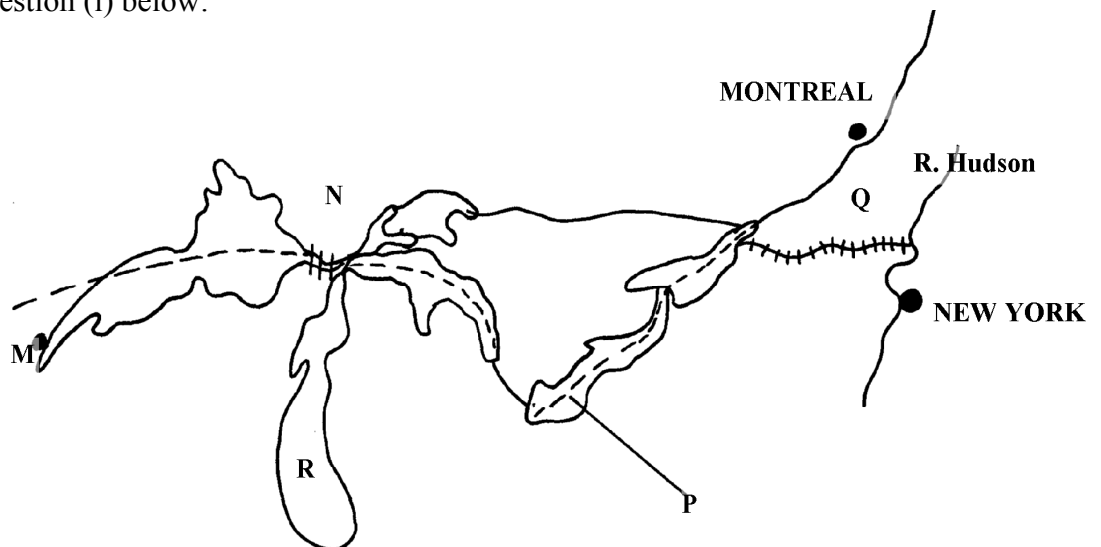
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(b) The diagram below shows the Great Lakes & St Lawrence Seaway:



- (i) Name the lakes marked **A, B, & C** .
- (ii) Why was the St. Lawrence Sea way constructed between the Great Lakes and the Pacific Ocean?
- (iii) Identify the problems that initially existed along the seaway and how they were solved
- (c) State **four** advantages of the Mombasa-Nairobi pipeline.
- (d) Discuss the role of transport and communication in development

2. (a) (i) What is transport?
- (ii) Name **three** forms of land transport commonly used in Kenya
- (b) Below is a sketch map of the Great Lakes and St. Lawrence Sea Way. Use it to answer question (i) below:



- (i) Name;
 - The Canals marked **N** and **Q**
 - The lakes marked **R** and **P**
 - The port marked **M**
- (ii) Explain **four** ways in which the Great lakes and St. Lawrence Seaway has contributed to the economies of U.S.A and Canada

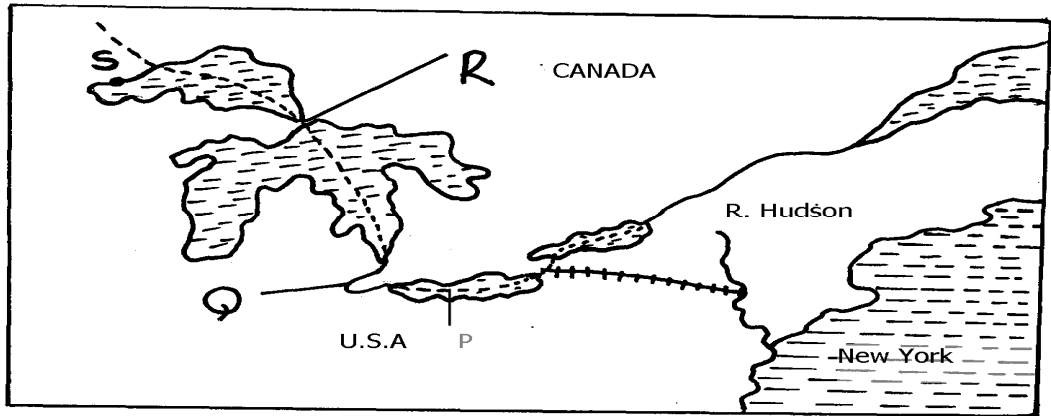
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- (c) State **four** ways in which communication has contributed to economic development of Kenya
- (d) State any **four** problems facing air transport in Africa
3. (a) Define the term **communication**
 (b) State **two** recent developments that have taken place in Kenya to improve communication
 (c) State **two** problems facing telephone as a means of communication in Kenya
4. (a) State **three** benefits that Kenya would derive from road linkage with the rest of East African countries
 (b) Give **two** major problems experienced by users of Kenyan roads
5. (a) (i) Give **two** forms of land transport
 (ii) State any **four** disadvantages of water transport
 (b) Explain any **five** ways in which transport and communication is important to the economy
 (c) Explain **three** attempted solution to African transport and communication problems
- (d) (i) Define canal transport
 (ii) Name the **two** major canals of the world
6. The table below shows the number shows the number of passengers that used railway transport in selected countries in 1966 and 1977. Use it to answer questions (a) and (b)
- | COUNTRY | PASSENGERS IN MILLIONS | |
|-----------|------------------------|---------|
| | 1966 | 1977 |
| CANADA | 4,800 | 6,000 |
| U.S.A | 27,800 | 16,600 |
| ARGENTINA | 14,100 | 12,000 |
| INDIA | 96,000 | 160,800 |
| JAPAN | 258,400 | 310,900 |
- (a) (i) Using a scale of 1cm to represent 20,000 million passengers, draw comparative bar graphs based on the data above
 (ii) State **two** advantages of using bar graphs in representing data
 (b) Calculate the percentage increase in railway passenger transport in Canada between 1966 and 1977
 (c) Explain **three** factors that hinder the development of railway links among African countries
 (d) One of the problems facing road transport is the high frequency of accidents. Explain **three** conditions of roads in Kenya that may lead to accidents
7. (a) (i) Name **three** types of communication mainly used in Kenya
 (ii) State **two** advantages of pipelines as a means of transport
- (b) (i) Why is air transport not very commonly used in Kenya?
 (ii) Explain **four** roles of transport and communications in the economic development of Africa
- (c) (i) State **three** objectives of constructing the St. Lawrence Seaway
 (ii) Name **two** canals on the St. Lawrence Seaway
8. Below is a sketch map of the great lakes and St. Lawrence sea way. Use it to answer

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question (a)

- (a) Name:- (i) The Lake marked P
(ii) River marked Q
(iii) Canal marked R
(iv) The port marked S



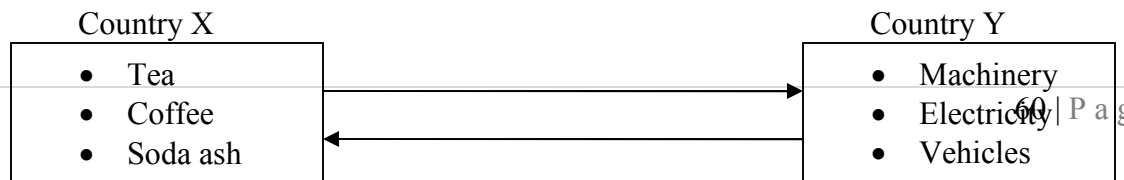
- (b) Explain **three** ways in which the great Lakes and St. Lawrence seaway has contributed to the economy of Canada and United States of America (U.S.A)
- (c) Explain the factors that hinder the development of river transport in Africa
- (d) (i) State **three** recent developments that have taken place in Kenya to improve communication
(ii) Explain **three** problems facing telephone as a means of communication in Kenya
9. (a) Define the term **containerization**
(b) State **three** advantages of containerization
10. (a) (i) Explain **three** economic importance of using mobile phones
ii) State **four** problems associated with the use of mobile phones in Kenya
b) Give **four** reasons why water transport is poorly developed in Africa
c) Explain **three** measures that have been taken to solve transport problems in Africa

22. TRADE

This topic entails :-

- (i) Identifying and defining types of trade
- (ii) Discussing factors influencing trade
- (iii) Identifying major exports and imports of Kenya
- (iv) Discussing significance of trade to Kenya
- (v) Discussing problems facing trade in Kenya
- (vi) Role played by selected trading blocks in the economies of their respective regions.

1. Use the diagram below to answer question 5a and 5b



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- a) Identify the type of trade shown above
- b) State problems country X is likely to face in the trade shown above
2. (a) Give **two** types of international trade
- (b) State **two** reasons why there is less trade between Kenya and other African countries
3. (a) State **two** problems facing trade in Kenya.
- (b) Give **three** benefits of regional trading blocs.
4. (a) Outline **two** objectives of the common market for Eastern and Southern Africa (COMESA)
- (b) State **two** efforts made by the Kenya government to enhance external trade
5. a) i) Differentiate between visible and invisible exports
- ii) Name **three** invisible exports from Kenya
- b)i) Apart from the Economic Community of West African States (ECOWAS) name two other regional trading blocks in Africa
- ii) Identify **three** member countries of ECOWAS
- iii) Explain **five** economic benefits of the Economic Community of West Africa states (ECOWAS)
6. (a) What is balance of payment?
- (b) Identify **three** problems that face traders dealing with primary goods
7. (a) Distinguish between visible and invisible export.
- (b) State **three** negative effects of a country over relying on import goods.

23. POPULATION

This topic entails :-

- (i) Definition of population
 - (ii) Explaining the factors influencing population distribution in East Africa
 - (iii) Explaining the factors influencing population growth.
 - (iv) Prescribing population structure
 - (v) Analysis of the consequences of population growth and structure
 - (vi) Comparing and contrasting population trends between Kenya and Sweden.
 - (vii) Presentation of population data using relevant statistical graphs.
1. The table below shows the population of a county in Western Europe in 1000. use it to answer part a-c

Age group	male	female
0-4	450	455
5-9	447	449

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10-14	448	450
15-19	454	458
20-24	480	472
25-29	630	632
30-34	635	639
35-39	642	671
40-44	670	638
45-49	636	568
50-54	562	641
54-59	633	639
60-64	631	634
65-69	451	452
70-74	470	468
75-79	460	459
80+	451	453

- (a) Using a scale of 1cm to 100,000 people, draw a population pyramid from the above data
 - (b) State **five** characteristics of the above population structure as shown by the pyramid
 - (c) Explain **four** problems likely to be experienced due to the population trend in the above country
 - (d) (i) What is mortality rate?
(ii) State **five** measures that have been taken in Kenya to reduce infant mortality in Kenya
2.
 - a) Name **two** primary sources of population data
 - b) Explain **four** reasons that led to rapid population growth in Kenya in the 1980's
 - c) State **four** reasons for increased infertility in Kenyan women today
 - d) Give measures taken by the government to combat child mortality
 3.
 - (a) Distinguish between **population distribution** and **population density**
 - (b) State any **three** problems associated with high population growth rate in Kenya
 4.
 - (a) (i) What is life expectancy?
(ii) Give **three** types of information which can be derived from a population pyramid.
 - (b) (i) Describe **three** ways in which population of Sweden differs from that of Kenya.
(ii) Explain **four** causes of rural-rural migration in Kenya.
 - (c) Explain **three** problems which result from the high population growth rates in the East African countries
 5.
 - (a) Define the term secondary fertility
 - (b) (i) Apart from **HIV/AIDS** give two other causes of mortality in East Africa
(ii) State **two** ways in which the spread of **HIV/AIDS** in Kenya may slow down economic development
 - c) State **five** problems facing regional trading blocks in Africa
 6.
 - a)i) What is dependency ratio?
ii) State **three** causes of a high dependency ratio in a population
 - b) i) Explain **four** factors that have led to the high population density around Lake Victoria

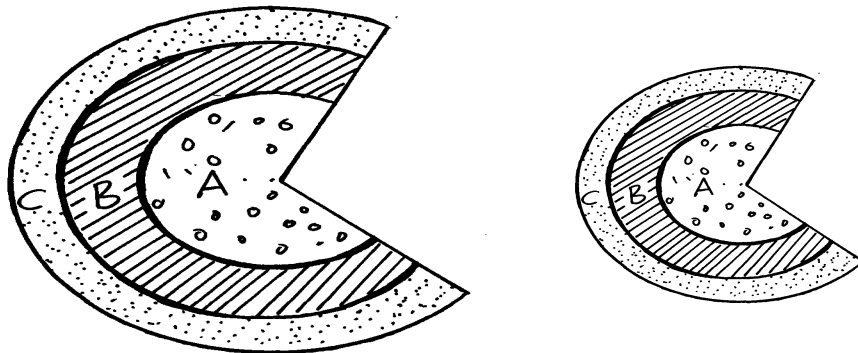
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- ii) Explain **three** problems associated with high population growth rate in Kenya
 - c) Give **three** reasons for the low birth rate in Sweden
7.
 - a) What is population census
 - b) State **three** reasons why countries conduct population census
 8.
 - (a) Differentiate between immigration and emigration
 - (b) State **three** effects of rapid population increase in Kenya
 9.
 - (a) Define the term **population explosion**
 - (b) State **three** reasons why countries conduct population census.
 - (c) Give **two** measures the Kenya government has taken to check on high population growth.

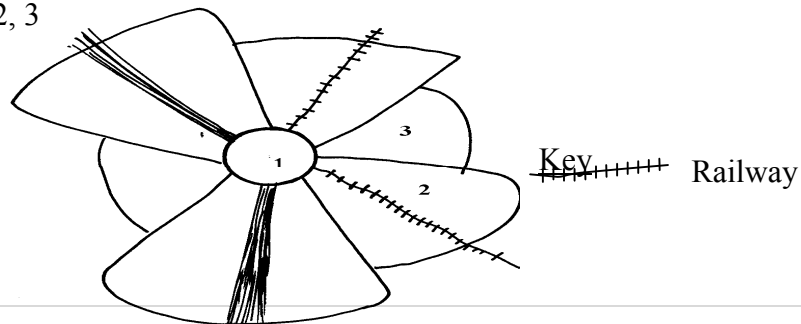
24. SETTLEMENT

This topic entails :-

- (i) *Definition of settlement and urbanization*
 - (ii) *Explaining the factors influencing settlements and settlement patterns*
 - (iii) *Accounting for the distribution and functions of selected towns in Kenya*
 - (iv) *Explaining the growth and functions of selected towns in Kenya*
 - (v) *Comparing and contrasting selected urban centres in Kenya with those of other parts of the world.*
 - (vi) *Discussing the effects of urbanization*
1. The diagram below shows part of the urban set up. Use it to answer part (a) and (b)
 - (a) Name the parts labeled **ABC**
 - (b) Give **three** characteristics of the part labeled **A**



2.
 - a) Name **two** settlement patterns common in rural areas in Kenya
 - b) The diagram below shows the internal structure of an urban centre. Name the sectors labeled 1, 2, 3

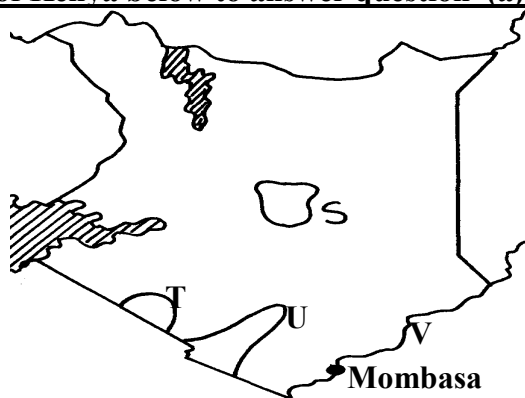


25. MANAGEMENT AND CONSERVATION OF THE ENVIRONMENT

This topic entails :-

- (i) *Definition of management and conservation*
- (ii) *Explaining why it is necessary to manage and conserve the environment*
- (iii) *Naming/identifying environment hazards*
- (iv) *Assessing the impact of selected environmental hazards and suggesting measures for combating them.*
- (v) *Discussing the measures taken in managing and conserving the environment.*

1.
 - i) Define the term **land pollution**
 - ii) Explain **four** causes of land pollution
 - iii) Explain **four** effects of land pollution
2.
 - a) Reasons why National parks are located in marginal areas.
 - (b) Explain **three** ways in which water pollution affect wildlife
 - (c) (i) State **three** causes of floods in Kenya
 - (ii) Explain **two** methods used to control flooding in Kenya
 - (d) State **six** reasons why it is important to manage and conserve environment
3.
 - (a) Give **three** causes of sound pollution
 - (b) State **two** major health effects caused by sound pollution
4.
 - (a) Apart from flooding, name three other environmental hazards experienced in Kenya
 - (b) (i) Name **two** rivers to the west of Rift valley which causes large scale flooding
 - (ii) Explain **four** problems caused by floods
 - (c) Explain **three** effects of land pollution on the environment
 - (d) (i) State **three** ways through which land pollution can be controlled
 - (ii) Give **three** effects of wind as an environmental hazard in Kenya
5. Use the map of Kenya below to answer question (a)



- (a)
 - (i) Name the wildlife conservation areas marked S, T, and U
 - (ii) Identify the Marine National Park marked V
- (b)
 - (i) Apart from Marine and Wildlife name **three** other tourist attractions along the coastal strip of Kenya
 - (ii) Give **four** reasons for wildlife conservation in Kenya

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- (iii) Explain **four** problems facing wildlife conservation in Kenya
6. (a) Give **three** reasons why it was necessary to conduct a pre-visit before the actual study (3mks)
(b) State **three** measures that they could propose to be taken to promote domestic tourism in Kenya (3mks)
7. (a) Apart from floods, name any other environmental hazard experienced in Kenya
(b) State **two** causes of desertification
(c) Give **two** reasons why it is necessary for Kenya to conserve her environment
8. (a) Differentiate between management and conservation of the environment.
(b) Give **four** reasons why we need to manage and conserve the environment.
(c) Explain **four** effects of air pollution on the environment.
(d) (i) Give **five** measures that can be put in place to combat pollution.
(ii) Identity **three** human characteristics you may learn from the garbage;
9. (a) Define the term pollution as used in Geography.
(b) Dither their air pollution name **three** forms of pollution.
10. Apart from desertification, name **two** other environmental hazards experienced in Kenya

ANSWERS SECTION I & II

1. INTRODUCTION TO GEOGRAPHY

1. - It creates awareness /understanding in the people of the environment in which they live hence the need to manage, and use resources sustainability.

2. THE EARTH AND THE SOLAR SYSTEM

1. a i) i) Tropic of cancer
ii) $66\frac{1}{2}^{\circ}$
b i) - Causes day and night
- Causes high and low tides
- Causes deflection of winds and ocean currents
- Causes time difference between longitudes
- Causes pressure difference on the earths surface
ii) Time difference = 1400 hrs – 1030hrs
= 0330 hrs
= $3\frac{1}{2}$ hrs
 $\therefore$ Difference in longitude is
 $\frac{7}{2} \times 5 = 52.5$
 $52.5 - 45$
Longitude is 7.5W
(c) - Sea breeze

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- Land breeze
 - Anabatic
 - Katabatic
2. a i) - $28^{\circ}\text{C} - 24^{\circ}\text{C} = 4^{\circ}\text{C}$
ii) - 1803mm
b) - the town experiences high temperature throughout the year ($24^{\circ}\text{C} - 28^{\circ}\text{C}$)
- the annual range of temperature is small
- rain fall throughout the year/there is no marked dry season
- the rainfall pattern has double maxima
- the wettest month is June/the driest month are December and January
- the rainfall is high i.e. 1803mm
3. a) Solstice is when the sun's overhead position is over the tropic of Cancer and Capricorn
b) - Causes seasons i.e. spring, summer, autumn & winter
- Varying lengths of day and night at different times of the year
- Changes in the position of the midday sun at different times of the year
- Changes in the position of the midday sun at different times of the year
4. a i) - Passing star theory
- Nebular cloud theory
ii) Centrifugal force
b) - Layers of the overlying rocks exert pressure hence the interior is hot
- After formation of the earth the interior cooled slowly compared to the exterior, thus the interior still retains much of its original temperature
- Radioactivity – mineral elements with the interior react with each other thus exploding to produce heat
5. a) - Passing star/ the big bang theory
- Nebula cloud theory
b) - The solar system is the group of heavenly bodies comprising the sun and the eight known planets which orbit the sun
6. a) - Mercury
- Venus
b) (i) A time zone is a group of neighbouring countries that use the same standard time
ii) - On crossing this line from east to West, a day is gained / the clock has to be adjusted backwards by 24 hours
- On crossing this line from West to East a day is lost/the clock has to be adjusted forward by 24 hours.
- c) - They are circular.
- They decrease in length Northwards and Southwards
- They are measured North and South of the equator
- Their values increase Northwards and Southwards
- They are parallel to each other
7. a) - It is the sun and the planets orbiting around it.
b) - The radio activity process taking place in the interior leads to a lot of energy production.
- This keeps the temperature in the interior very hot.
- The overlying materials exert a lot of pressure to the interior resulting to higher temperature.
- When the earth was being formed the mantle and the core cooled at a slower rate than

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- the crust.
- As a result the temperature in the interior are still hot.
8. a i) -Solar/eclipse of the sun
ii) - Q-moon
- T-umbra
b) -it cause days and night
-it causes high and low tides
-it causes the deflection of winds and oceans currents
-it causes time difference between longitudes
9. a) - Cooled at a slower rate than the outer exposed part
- Due to the process of radio- activity where atoms break releasing heat
- Weight of the overlying material that exerts pressure on the core
b) - Causes seasons
- Varying lengths of day and night
10. a i) It is the date when the sun is overhead at the equator at mid day
ii) 21st march✓
23rd September✓
iii) - causes four seasons i.e. winter, summer, autumn and spring✓
- causes varying lengths of day and night at different times of the year✓
- causes changes or altitude of the mid-day sun at different times of the year✓
11. a) - U-mohorovicic discontinuity
- V-the mantle/asthenosphere
- X-Gutenberg discontinuity
b i) - it is made up of solid rocks
- composed of two layers /sial and sima/continental crust and oceanic crust
- sial is rich in silica and a aluminum
- sima is rich in silica and magnesium
- sima rocks are like plastic/more flexible
- the top layer of the sima is made of sediments and volcanic lava
- the bottom layer of sima is made up of basalt/igneous rocks
- the sima is made up of the basaltic/igneous rocks
- sima is made up of dense rocks /2.8-3.0 gm/cc
- the sial is made up of granites/sedimentary/metamorphic rocks
- the sial rocks are rigid/brittle
ii) - the core is composed of two parts i.e. inner core and outer core
- the main mineral of the outer core are iron and nickel
- the main mineral of the inner core is iron
- the inner core has a high density i.e. 16-17gm/cc than the outer core i.e. 10.5gm/cc
- the inner core is made up of a solid rock mass
- the outer core is molten

3. WEATHER AND CLIMATE

1. -population pressure leading to clearing of forest
- climatic changes/global warming/destruction of ozone layer
- accidental fires
- poor methods of farming /overgrazing
2. a i) Aridity refers to a state of dryness or deficiency of rainfall in an area while

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desertification refers to the encroachment of desert like conditions into productive or arable land √√

ii) Weather atmospheric condition of a place over a short period of time

Climate- Average weather condition of a given place over a long period

b)- Moist warm air meets dry cold air mass

- Mountain slopes adjacent to a valley cause anabatic and katabatic winds

c) - Sea heats faster than the land during the day

- Air over the sea rises creating low pressure over the sea

- Cooler air over the land blows towards the sea

- Cool air from the land is called land breeze

3. a) - Zone of low pressure

- High temperature

- High convectional rainfall

- Trade winds converge here

- Moves with movement of the sun

(b)i) -Variation of solar output

- Volcanic eruption

- Variations in the earth's orbital characteristics

- Variation in atmospheric carbon dioxide

iii) - Increase in temperature causing rise in sea level due to melting ice

- Changes in wind speed causing erosion

- Changes in seasons leading to severe drought

- Increase in precipitation causing flooding of large areas

c) - Relief features such as high mountains influence temperature and rainfall pattern

- Presence of large water bodies that modify temperatures of the surrounding lands through breezes

- Continentality- many regions are in the interior of the continents making them receive low amounts of rains

- Winds- some areas are influenced by cold onshore winds that lower temperature of the surroundings

4. (a) - Temperature varies between 17C to 24C

- Lower slopes have higher temperatures than upper slopes

- Mean annual range of temperature is between 3C to 5C

- Rainfall is received throughout the year

- Rainfall is high 1000 -15000 mm

- Rainfall is double maxima

- Rainfall is relief- orographic

- High rain on the wind ward slopes

- Rainfall is caused by S.E trade

- Coolest months are between June and August

(b) - Fold mountains receive heavy rainfall/ snow falls give rise to many rivers for HEP, irrigation, water for domestic and industrial use

- Wind wards sides receive high rainfall which promote agriculture/ forest growth

- Leeward side have violent wind which destroy crops

- Folding leads to exposure of some minerals or bring minerals near surface for mining

- Folding hinders construction of communication

- Folding acts as barriers during wars

- Folding offer sceneries which attract tourists hence foreign exchange

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- Folds mountains make visibility poor hence hindering air transport
- 5. a) Weather is the daily condition of atmosphere taken in short period usually 24hrs while climate is the average weather condition taken for a period of 30 – 35 yrs
- b) - High rainfall/ no marked dry season/ (1500mm – 2000 mm)
- High temperatures throughout the year/ (24°– 27°)
- High humidity throughout the year with relative humidity about 80%
- Rainfall throughout the year
- Small temperature range/ 4°C
- Double maxima (March – May and Oct – Nov)
- Major winds are South East and North East
- Low pressure all the year/ Doldrums/ Equatorial low
- Rainfall mainly convectional falling in the afternoon accompanied with lightening and thunderstorm
- Thick cloud cover all the year/ thick cumulonimbus clouds
- 6. (a) i) - Warm ocean currents raise temperature of the adjacent land/ warm current increase rainfall
- Cold ocean current lower temperature of the regions/ lower rainfall/ create dryness
- ii) High areas have low temperatures/ Mt areas receive high rainfall
- Mountain sides facing sun are warmer than sides facing away from sun (Aspect)
- b i)- Rainfall of double maxima
- High temperatures throughout the year about 27°C
- Low temperature range
- Rainfall falls throughout the year
- Rainfall is high average 2000mm
- Low pressure due to overhead sun
- Rainfall is mainly convectional
- High humidity content
- 7. a) - Air should be calm so that it can remain in contact with the ground long enough to be cooled below dew point
- Day time should be warm to speed up evaporation and provide a lot of water in the air
- There should be cloudless nights
- b) - To increase precipitation in the area
- To regulate in the area
- Forest to help air purification
- 8. a) i) A fog is a mass of water droplets suspended in the lower atmosphere which limits visibility to less than a kilometer
- ii) - Air must have abundant moisture
- The night should be clear/cloudless to facilitate terrestrial radiation
- The air should be calm/gentle air currents to help the water droplets in suspension
- b) - Troposphere
- Stratosphere
- Mesopause
- 9. a) -it is a large volume of air whose temperature and humidity are fairly uniform and covers an extensive surface area
- b) -it experiences high temperature
- it is a zone of low pressure and the doldrums
- the zone migrates to the north and south of the equator with apparent movement of the overhead sun

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- it is a zone where the S.E and N.E trade winds converge
 - it is associated with convectional rain and thunderstorms
10. a) - Temperature decreases with increase in altitude.
- Rainfall increases with height upto 300m then it starts to decrease
 - Windward slopes are wetter than leeward slopes.
 - Atmospheric pressure decreases with increasing altitude.
 - Local winds are common, they blow up slope during the day and down slope during night.
 - In temp. regime slopes facing the equator are warmer than those facing the poles
- b) i)- Increases air pollution from industrial activities.
- Deforestation /uncontrolled logging.
- ii) - Widespread changes in the natural ecosystem with grasslands and deserts areas expanding as forested areas shrinks.
- Possible rise in temp. may increase evaporation rates leading increased rainfall in some areas.
 - Increased rainfall will cause flooding of low lying areas.
 - Water stored in polar glacier will melt leading to rise in sea levels and flooding of coastal lands.
 - In temperate regions, winters are likely to be wetter and summer drier.
 - Wetter & warmer conditions may increase pests & diseases thereby affecting humans, crops & livestock.
 - Change in climate is likely to cause extinction of various plants and animal species.
 - Possible drought might increase in most parts of the world.
 - Cause in change in agricultural activities & massive crop failure leading to food shortage.
- c i) - A rain gauge consists of a funnel, metallic cylindrical container and a collecting jar.
- When it rains, water from direct rain drops collects into the jar through the funnel
 - The water collected is then poured into a graduated (in mm) measuring cylinder.
 - The readings taken & recorded.
- c ii) - Thermometers /⁶six thermometers.
- Hygrometer.
11. a i) - This is climatic conditions in a restricted area due to small differences of aspect slope, vegetation and human landscapes
- ii) It is a condition where the incoming solar radiations pass through the atmosphere while the outgoing terrestrial radiation is blocked by gases/ clouds/ atmosphere making the earth retain the terrestrial radiation/ re-radiation. (This makes earth to be warmer than it would have been)/ it is a condition where the atmosphere balances the incoming and outgoing terrestrial radiation making the earth to retain optimum heat
- b) - A thermometer / maximum and minimum/ six's thermometer
- A hygrometer/ wet and dry bulb thermometers
- 12 a) -body of air/wind with uniform conditions moving in a given direction
- b) -same temperature
- air pressure-move from H.P to L.P
 - earth's movement
13. (a) Tropical convergence zone is a low pressure belt that lies between the tropics where the North East and South East trade winds converge
- b) - The forest is evergreen due to high rainfall and abundant sunshine

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- The trees grow tall to complete for sunlight
 - Trees have broad leaves to encourage water loss through evapo-transpiration / transpire excess water due to high rainfall
 - these are mainly hardwoods due to abundant sunshine
 - Trees have buttressed roots to support their great heights and large trunks in the wet soils.
 - Tree roots go deep into the ground for anchorage
 - There are many tree species due to the tropical conditions.
 - The forest has little underground because the canopies block sunlight from reaching the ground
 - The forests have numerous lianas/climbers which twine around the tall trees to reach sunlight in the dense forest.
 - Trees grow rapidly due to the high temperatures and high rainfall
14. a) It is large volume of air/ wind with similar/ uniform temperature humidity and covers over a large area and flows over along distance
- b) A- North East polar winds
- B- South West westerlies/ S. West winds
- C - North east trade winds
- D - South East trade winds
15. a i) - Global warming is the increase of average temperatures on earth as result of green house effect/ gases reflecting back to earth part of heat radiation
- ii) - The orbital position of the earth or effect of the distance of the earth from the sun
- High amount of carbon dioxide in the air from industries
 - The volcanic emption raise heat/ temperature
 - High amount of solar radiation reaching the earth due to depletion of Ozone layer
- iii) - Carbon dioxide (CO₂)
- Chlorofluorocarbons (CFS)
 - Methane
 - Nitrous oxide (N₂O)
- b)(i) - Change in the ecosystems with grass land and deserts expanding while forests shrink
- Rise in the sea level due to thawing of glaciers that leads to flooding of coastal low lands
 - Rise in global temperatures leading to increased rainfall due to increased evaporation
 - Wetter winters and dry summers in temperature regions
 - Shrinking water bodies e.g. lakes, rivers and drying up of streams
 - Reduced animal and plant species due to adaptation difficulties
 - Extinction of some species of flora and fauna
 - Low production of food due to failing soil fertility and absence of rain/ too much of rainfall in other areas
- (ii) - High rainfall evidenced by many permanent rivers
- Cool temperatures from high altitudes e.g. 2000m
16. a) Aridity is dryness/ insufficient rainfall, while desertification is the expansion/ encroachment of desert conditions
- b) - Use off defective instruments✓
- Personnel my have inadequate /limited skills✓
 - Vagaries of nature such as earth tremous✓

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- c) - painted white to prevent over heating✓
 - raised to a height of 121m to prevent contact with direct radiation from the earth's surface✓
 - it has louvers at the side to allow free air circulation✓
- 17. a)- 2-modified tropical climate of the highland
 - 3-modified equatorial climate of the lake basinb) - it receives very low rainfall of less than 250mm annually
 - the mean annual temperature are high above 29°C
 - large diurnal range of temperature
 - day temperature are high
 - humidity is low
 - sand storms are common occurrence
 - there are occasional floods caused by sporadic rain
 - the region is under the influence of dry north east trade winds
- 19. a i) X – Mediterranean
Y – Equatorial climate
- ii) - Rainfall is low/ below 250mm per year
 - Rainfall is erratic/ unreliable
 - Occasional flash floods
 - Temperatures are high throughout the year
 - Intense solar radiation
 - Diurnal range of temperature is large/ extreme hot days and cold nights
 - High rates of evaporation
 - Skies are clear/ high terrestrial radiation
 - Low humidity
 - Strong dusty winds
- b i)- Green house effect is general increase of the temperatures of the globe due to increase in the level of CO₂ and other greenhouse gases. This results into a condition where the incoming solar radiation passes through the atmosphere while the outgoing terrestrial radiation is blocked by the gases and the clouds
- ii) Increase in temperature
 - Effects
 - Rainfall increase
 - Melting of ice caps
 - Rise in sea level
 - High evaporation
 - Abnormal growth of plants
 - Change in rainfall pattern
 - Effects
 - Floodings
 - Rise in sea level
 - Drought
 - Soil erosion by water
 - Change in seasonal patterns
 - Effects
 - Severe winter/ short summer
 - Drought
 - Shifting of vegetation zones

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- Extinct of some species

Change in winds

- More frequent & more destructive
 - High waves
 - Flooding
 - Wind erosion/ wind storms
- c i) - Temperature decreases with increasing height above sea level. This is because atmospheric air at low altitude is denser than high above. Heat loss is greater at high altitude than at lower altitude
- Rise in altitude cause fall in temperature and a cooling effect that causes condensation of moisture in the air. This leads to precipitation forming at high altitude
 - Atmospheric pressure is higher at low altitude and lower at high altitude. This is because the weight of the atmospheric air at low altitude is more than at high altitude
- ii) In summer, coastlands are relatively cooler than in land areas. This is because the winds bring the cooling effect of the sea to the land, by the time winds reach inland they are heated by warm land resulting in higher temperature over the interior of the land
- In winter the onshore winds bring the warming effect of the sea into the land causing a rise in temperature. By the time these winds reach the interior of the land, they are cooled by cold land thus enhancing the low temperature
- Onshore winds cause a lot of rain in the coastal areas throughout the year. This is because the wind picks up moisture over the sea and drop it on the nearby land. The continental interiors receive less rain, mainly in summer, because the winds have dropped most of the moisture in the coastal lands
- iii) Ocean currents
- Cold ocean currents bring about the cooling effect in the temperature
 - Also bring a dry effect because the cold air is not able to rise
- Warm ocean currents
- Bring a warm effect (rise in temperature)
 - Heavy rainfall because the warm air is able to rise and condense to form rainfall

4. STATISTICAL METHODS

1. a ii) - Relatively more difficult to draw/time consuming.
 - Long bar not easy to compare.
 - Number of components to be represented is limited
 - Fluctuation in production over a period of time are not easy to see at a glance
2. (a) ii) - A better impression of totality and individual contribution.
 - Differences in quantities are easier to see.
 - Easy to read the bars as they start from a common baseline.
 - The value of each bar is easily determined.
- b) - Kenya is not self sufficient in the commodities due to the high population.
 - Frequent drought which lead to crop failure cause food shortage thus importation to supplement local produce
 - Increased cost of farm inputs leading to low production hence the need to import.
 - Occasional illegal imports/smuggling /hoarding of the commodities creates artificial shortage hence the need to import these commodities.
 - Pests and diseases reduce the amount of grain harvested thus the need for importation.

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- The requirements to maintain the trading quotas among the countries African trading blocks that Kenya imports some of the commodities in exchange of her own imports
3. a i)

DISTRICT / YEAR	1982	CT	1992	CT	2002	CT
TRANS NZOIA	24	24	26	26	40	40
KIAMBU	23	47	25	51	31	71
MERU	25	72	27	78	32	103
BUNGOMA	12	84	14	92	20	123

Milk production in 000 units in selected Districts in Kenya

4. a i) - The year 2009 recorded the highest network coverage in the country
- The year 2007 held the lowest network coverage and uncovered areas respectively since 2009
 - Safaricom network has been the leading in terms of network coverage since 2007 to 2009
 - YU network has been the least in terms of network coverage
- ii) - Suitable for absolute totals in different periods
- Gives a good visual impression of the totality
 - Easy to interpret
5. a ii) - It's easy to compare the various components within a circle.
- They are simple to construct after angles have been calculated.
 - Give clear visual impression of individual components.
 - It's easy to determine the value of each component since the size, the sector is proportional to the value it represent.
- b) $= 130,000 - 70,000 = 60,000 \sqrt{1 \text{ mk}}$
 $\% \text{ increase} = \frac{60,000}{70,000} \times 100 = 600$
 $= 85 \frac{5}{7} \% \sqrt{1 \text{ mk}}$
6. a) Gold = $\frac{26}{100} \times 360 = 93.6 = 94^0 \sqrt{1}$
 Fluorspar = $\frac{14}{100} \times 360 = 50.4 = 50^0 \sqrt{1}$
 Soda Ash = $\frac{32}{100} \times 360 = 115.2 = 115^0 \sqrt{1}$
 Zinc = $\frac{28}{100} \times 360 = 100.8 = 101^0 \sqrt{1}$
- b) - Good in showing variant types of data ✓
 - Easy to draw ✓
 - Easy to interpret ✓
 - Easy to make comparisons ✓

5. MAPS AND MAP WORK

1. a i) - Kisumu east 116/2
- Muhoroni 117/1
 - Lumbwa 117/2
 - Nyakach 116/4
 - Kericho 117/4
 - Kisii 130/2
 - Chemagel 131/1
 - Chapatarakwa 131/2
- b i) - Presence of tea plantation-tea grows under heavy amount of rainfall
 - Presence of forest cover

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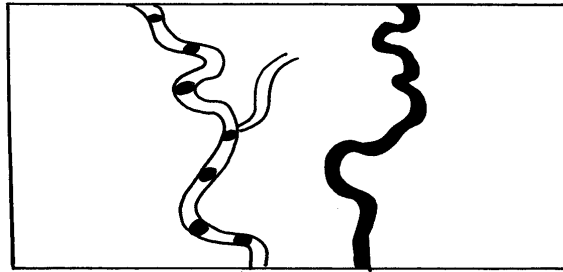
- presence of permanent river

b ii) - R. Kibol

- Scrub vegetation

- Tea plantation

c i)



-all weather road loose surface



-district boundary

d i) - the major river is R. Yurith

- river are permanent

- the river are few

ii) - Farming

- Forestry

- Mining

3. a) (i) - $0^{\circ}20'S 35^{\circ}19'E$

ii) - Power transmission line

b i) - Trigonometrical station

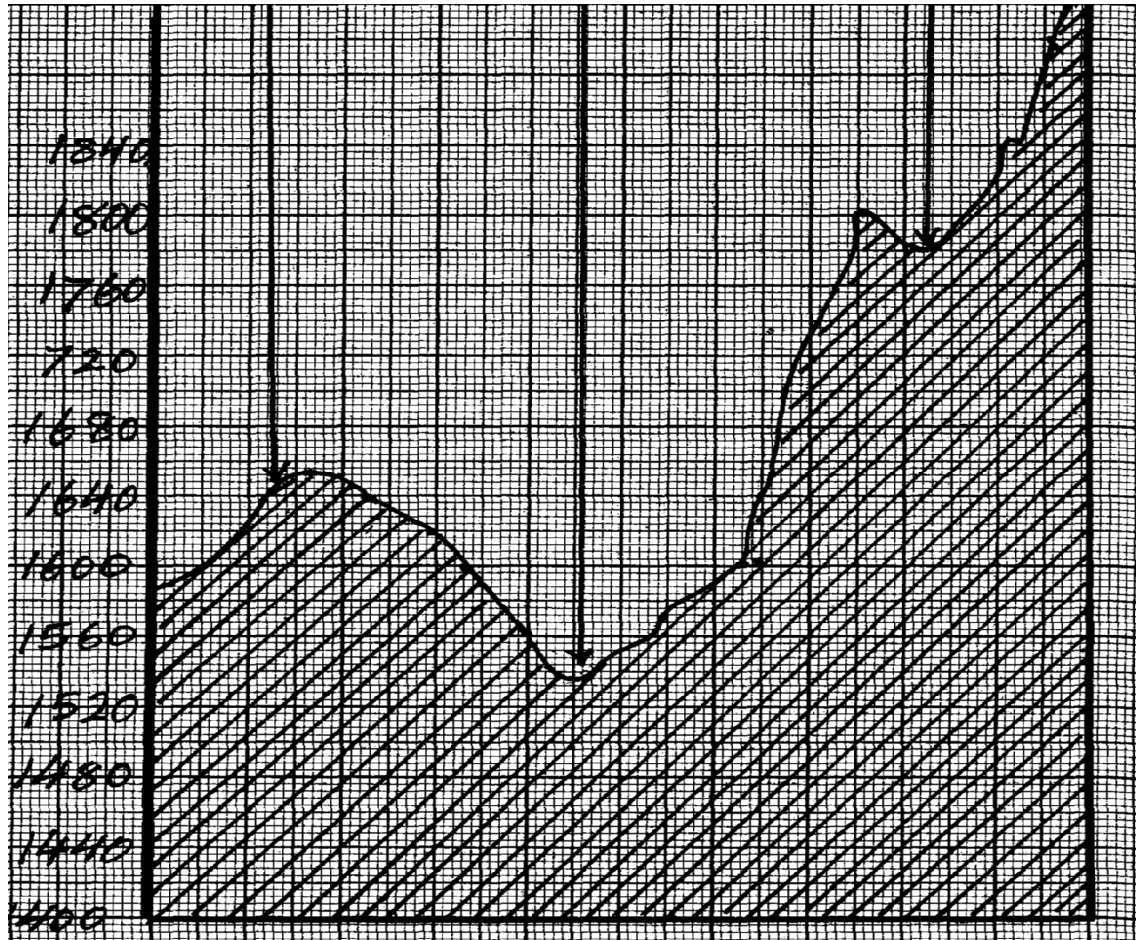
- Contours

- Hunhures e.g. along road in grid square 6867

ii) - $3.9\text{Km} \pm 0.1$ ($3.8\text{km} - 4.0\text{km}$)

iii) - $180 + 35 = 215^{\circ} \pm 1^{\circ} = (214^{\circ} - 216^{\circ})$

3. a)



ii) $VE = \frac{VS}{HS} = \frac{1}{8000} / \frac{1}{50000} = \frac{1}{8000} \times \frac{50000}{1}$

4. a i) - north west✓✓
 ii) - 000°/360°✓✓
 iii) - trigonometrical station: secondary✓
 - trigonometrical station :other✓

iv) full squares=28
 $\frac{1}{2}$ squares=26x $\frac{1}{2}$ =13
 41x1000m=41,000metres✓✓

b iii) $VE = \frac{VS}{GS}$
 $= \frac{1}{10,000}$
 $\frac{1}{50,000}$
 $= \frac{1}{10,000} \times \frac{50,000}{1} = 5✓$

c).

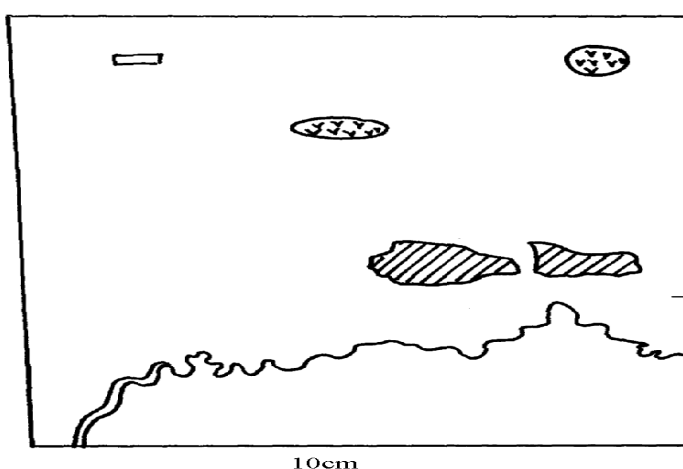
CROP	EVIDENCE
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-coffee	-coffee mill (2347)
-Grains(maize, millet, sorghum)	-mills, posho mill (4255)
-tea	-tea estate, tea nursery, tea factory.(1x2=2mks)

c.

5.
 - a) X – Savanna grass land
Y – Tropical rain forest
 - b)
 - Has very low temperature
 - Thin soils and bare rock
 - Soil is permafrost
 - c)- Mt.Kenya
6.
 - a i) - 2020m
 - ii) 15' (35°00' to 35°15')
 - iii) 36.0km² (35 to 37km²)
 - iv) 28755548
 - b i) & ii)



iii) New scale = map scale x scale factor of reduction

$$= \frac{1}{500m} \times \frac{1}{2} = \frac{1}{100000} = 1: 100000$$

- c) - The forestal area has few/no all weather roads because it is reserved for the growing of trees
 - The all weather roads avoid high hills/steep slopes because construction on/along steep slopes is difficult for example the road from Maraboi through Tamungo to Marumbasi detours to avoid steep slopes and hills.
 - The roads avoid swamps because of their water-logged conditions for example the road from Ainaboi through kabionga to Kiptula avoids the papyrus swamp
 - the roads avoid river valleys and only crosses them unavoidable
 - the roads are constructed on gentle slopes as it is easier to construct them as seen by the construction of roads on the gentler slopes to the East and no all weather roads on the ridge east of river Sondo
 - All weather roads are constructed on ridges/spurs/water sheds due to the gentleness of the slopes for example the road from Maragwa to Maraboi/the road from Ainabkoi to Kerenga
- d i) - Cool temperature as evidenced by the high relief – over 1700metres makes the area suitable for growing of tea bushes.

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- The gentle/indulating slopes as evidenced by widely spread contours allows for proper drainage of soils/mechanization making it ideal for tea farming.
 - The high rainfall as evidenced by forest vegetation/many permanent rivers originating from the area which is suitable for tea growing
 - Presence of labor as seen by the fairly dense settlement/labour lines to work in the tea farms
 - Transport is avoidable as seen by all weather roads for transportation of tea from the farms/transportation of labour
- d ii) Coffee evidence coffee mill in grid square 2347
7. a i) 1: 50,000
- 10000cm represent 1 km
- $= \frac{50,000}{100000} = 0.5$
- 1 centimeter represents 0.5 kilometers
- ii) - $35^{\circ}00' - 35^{\circ}15'E$
- iii) - $120^{\circ} + 1^{\circ}$
- b) - Wood land
- Riverine trees
 - Papyrus
 - Scattered trees
 - Scrub
- c ii) - River Sondo is the main river
- River Yurith flows S. westwards
 - Papyrus swamp present to the north and south
 - Yurith river has meanders
 - Rivers are permanent
 - Most rivers show a dendritic pattern
 - Dams in grid square 4349 and 4954
- d) - Shops and trading centre
- Presence of roads foot paths
 - Dense settlements provide market
 - Police post for security
8. a i) -Rift valley
- Nyanza
- ii) -south west direction
- iv) 420557
- b i) - 4.00km=0.1(3.9 to4.1)km
- ii) -use of place names
- grid reference
 - compass direction
- iii) -forest
- woodland
 - scrubs
 - riverine trees
 - papyrus swamp
- iv) -Tea growing-evidenced by tea factory/tea estates
- live stocks keeping-evidence by veterinary livestock centre

FREE GEOGRAPHY K.C.S.E 2012 to 2015

-cereal crops growing-evidenced by posho mill

(c)-Gentle have dense settlement due to gentle gradient which encourages settlement as evidence by spaced contours.

-Relatively flat areas have little or no settlement due to poor drainage as evidenced by papyrus swamp

-Areas with steep slopes in the western area near Kebebet have few settlements as evidence by closely spaced contours

-There has dense settlements due to fertile soil as evidenced by the growing of tea in estates

(d) -there are several permanent rivers in the area

-rivers Sondu and Yurith are the major rivers in the area

-rivers with their tributaries for dendritic drainage pattern

-there is a seasonal swamp in grid square 3656

-in the west, rivers flow north-west wards in the south they flow westwards

-there are fewer rivers in the north than in the south-

9. a i) - topographical map.

ii) - Contours

- Trigonometrical stations.

b i)

$$\begin{array}{lcl} 1.3 \text{ cm} & 1 \text{ km} \longrightarrow & 100000 \text{ cm} \\ 1 \text{ km} = 100,000 \text{ cm} & 13 \text{ cm} \longrightarrow & \frac{1 \text{ cm} \times 100000}{13} = 0.000013 \text{ km} \end{array}$$

ii) - Tea. Evidence by changoi tea factory and tea nursery.

- Coffee, evidence by coffee mill

c i) - The plains are densely settled as the land is flat/gentle land.

- The basins have been avoided as they land is water logged/flooded/swampy.

- Steep slope/escarpment have been avoided because they are unstable for construction of houses/for farming.

- There are few settlement on hilly areas because slopes are gentle.

ii) - Education/schooling evidenced by Gekonge school, Matongo school, Cheboan school, Kakibei school e.t.c

- Trading evidenced by Health centres, Dispensaries

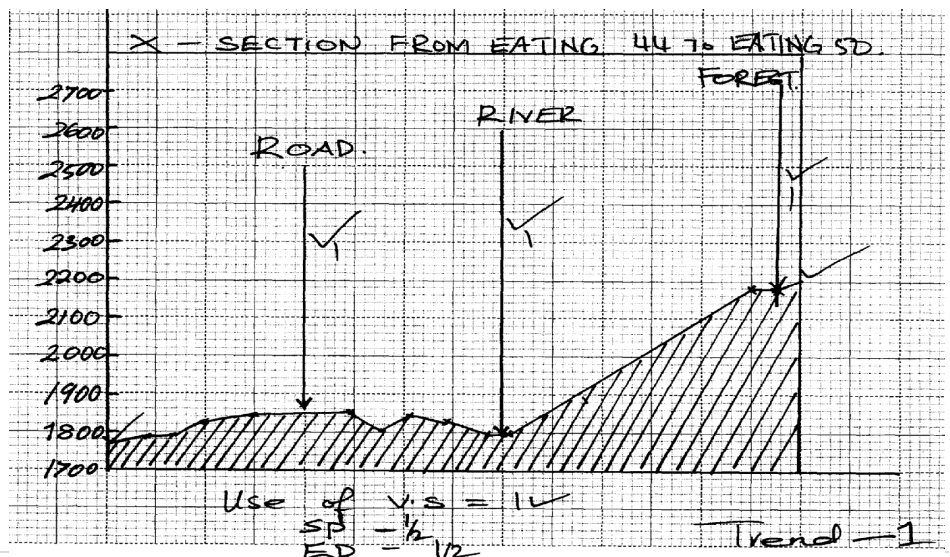
- Social work evidence by labour lines.

- Transportation evidence by murram, Dry –weather roads.

- Road

- River

- Forest



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10. a i) $35^{\circ}00' - 35^{\circ}15'$
 ii) Map scale 1:50000 i.e. 1cm represents 50000 cm

$$\frac{50000\text{cm}}{100000} = \frac{50000}{100000} \text{ km}$$

$$= 0.5\text{km} / \frac{1}{2} \text{ km}$$
 Statement scale is 1cm represents 0.5km/ $\frac{1}{2}$ km
 iii) Contours
 Trigonometrically stations
 b iii)
$$VE = \frac{VS}{HS}$$

$$= \frac{1/2000}{1/50000}$$

$$= \frac{50000}{1} \times \frac{1}{2000}$$

$$= 25$$
11. (a) - The drainage consists of rivers papyrus swamps, ponds and lakes
 - River Yurith is the main river
 - River Yurith generally flow West wards
 - The river has many meanders
 - River Yurith has two main- tributaries Itare and Kitoi
 - The area covered by the map are numerous
 - River Yurith has many tributaries that form a dendritic pattern along the course
 - The river yurith is Permanent River
 - There are many disappearing rivers
 b)
- | ECONOMIC ACTIVITY | EVIDENCE |
|-------------------|--|
| Transportation | Roads/ main tracks/ foot paths
Dry weather roads, All weather roads: (Bound surface and loose surface), Air strip |
| Trading | Shops, Coffee mill/ tea factory/ posho mill factory |
| Crop farming | Tea nursery/ tea plantation |
- c) - Employment to people work there is provided.
 - Facilitates agriculture in the regions around.
 - Processing of tea, ready for use.
 - Transportation of people evidence by all weather road e.t.c
 d)- Presence of tea that requires high rainfall.
 - Forest
 - Many permanent rivers.
 - Presence of many factories that use a lot of water.
12. a i) Topographical.
 ii) 24 54
 iii) Calculate the area enclosed by Kendu Kisii. All weather roads bound to the West of the map.

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$$7.5 \text{ km}^2 = (7.00 - 8.00 \text{ km}^2).$$

- iv) - Education - evidence school.
- Medical care – evidence hospital.
- Administration – evidence Chief's Camp, DC's, D.O's office.
- Recreation.

b) ii) Vertical scale

Horizontal scale

$$= \frac{1}{20} \times \frac{50,000}{1} = \frac{1}{5,000}$$

$$\text{V.S} = 2500$$

- c) - There are several permanent rivers like Itare and Kitoi.
- River Yurith is the main river.
- Rivers flow from North Eastern to North Western.
- There is a swamp / tree swamp around Kabiaga farm.
- Rivers form dendritic pattern
- d) - The scrub vegetation shows availability of pasture for cattle.
- The many rivers (like R. Yurith, Itare and Kitoi) in the area shows that there is adequate water for cattle.
- The high altitude of above 1800m above the sea level indicate cool conditions ideal for cattle keeping.

13. a i) 020°

ii) 556711

iii) map scale 1:50,000

1 cm rep 50,000cm

$$50,000 = \frac{50,000 \sqrt{1}}{100,000}$$

$$= 0.5$$

Statement scale is 1cm rep 0.5km/ ½ km

iv) 0° 19'S 35° 18'E

v) -contours

-trigonometrically stations/pillars

b i) -education-school/college

-recreational-club/race course

-Health-hospital

. b ii) - tea growing –tea estate/tea factory

- Trading-shop/markets

- Transportation-all weather roads/dry weather roads, murrum roads

- Quarrying-quarry

d) -the main river is tugenon

-There are many permanent rivers

-River Kipkwes and its tributan's form Dendntic drainage pattern

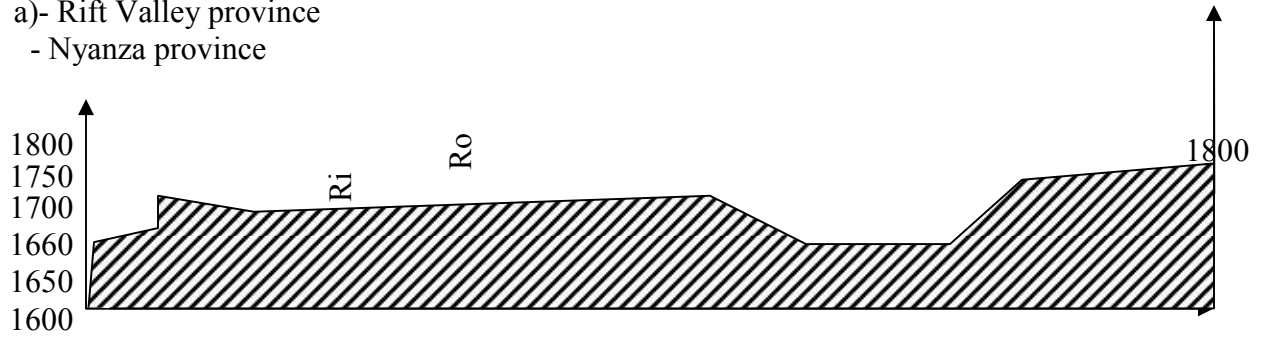
-River north of northing 69 flow generally towards the north

-River south of northing 60 flow towards west/south west

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14. a)- Rift Valley province
- Nyanza province

b i)



ii) $V.E = \frac{V.S}{H.S} = \frac{1}{50m} = \frac{1}{50000cm}$

$VE = 10 \sqrt{2}$

iv) – Topographical

15. a i) i) - 402488

ii) - 7.1 ± 0.1 Km

iii) - LUMBWA

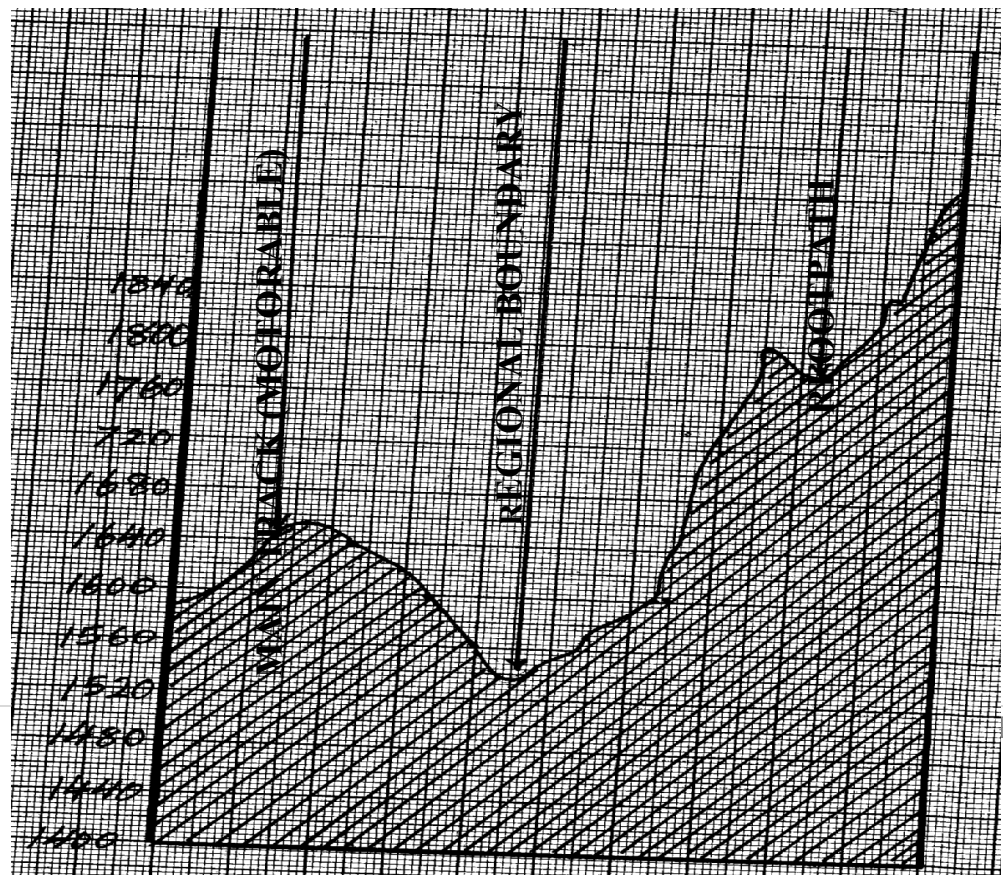
. b i) - $35^{\circ}00'E$ to $35^{\circ}15'E$

ii) - 1844 – 1847 m

- . c i) - The southern part of the area covered by the mp is well drained as evidenced by presence of many permanent rivers

- The main river is Yurith that flows South Western wards
- There is a seasonal swamp in the northern part of the area covered by the map
- There are several dams along river Kitoi
- River Sondo and its tributaries form dentric drainage pattern
- River Kitoi and Kimugung re in their youthful stages

ii)



- d i)) - Steep slopes e.g. along the escarpment have been avoided. There are more settlement on gently sloping regions
 - Forested areas e.g. to the south eastern part of the map have been avoided
 - Swampy areas have no settlement
 - Plantation forms have no settlement inside, people tend to settle around the forms
 - There are linear settlements along all weather road loose surface especially in the north eastern part of the area covered by the map
- d ii) - Presence of well developed transport network in the area ensures movement of goods and Services
 - Presence of scattered houses in the area ensures there is market provided by those in the houses
 - Tea factories and tea plantation farms is likely indication that people are employed in the area and therefore have better living standards

6. FIELD WORK

- 1. a) - To find out where farmers get their livestock/hides
 - Find out the significant of beef produce to the local areas.
- ii) - Displaying the filled questionnaires.
 - Displaying photographs taken.
 - Giving a lecture on beef farming.
 - Carry out group discussions.
 - Drawing proper sketches.
- b i) - To find out the various ways in which the power plant benefits the surrounding communities
 - To discover the amount of power generated by the plant
 - To find out the problem facing geothermal power generation in Kenya
 - To find out the power generation potential of the rift valley area
- b ii) - Formulator of objectives before actual field study
 - Write to Kengen to seek permission to visit Olkaria
 - Collect all the necessary writing materials
 - Conduct a reconnaissance to the site
 - Prepare a questionnaire for use on the day of study
 - Prepare a working schedule
- c i) - A pit is dug not too deep or shallow
 - The pit is cemented and sealed to prevent gas from escaping
 - Pipes are connected to carry the biogas to the burners
 - Pour the dung and other waste into the digester and mix with water
 - Anaerobic bacteria feed on it giving off biogas
- ii) - Saw dust
 - Agricultural wastes
 - Wood

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- Dung

- 2 . a i) - Soils are light in colour
 - Soils are sandy/ stony
 - Soils are loose in texture
 - Soils are thin
 - Soils have low moisture content
- ii) 2 columns of time & activity
 Time of departure
 Two activities indicating data collection
 Time back to school

TIME	ACTIVITY
8 a.m	Departure from school
8.30	Arrival at the field
9.00	Testing or feeling of soil texture Measuring the alkalinity of soil Carrying out experiment on water retention Observing of the soil e.t.c.
4 p.m	Going back to school

3. a i) - Savanna vegetation
 - Rainforest
 - Bamboo forests
 - Heath and moorland
- ii) - prairies
 - steppes
 - downs
 - veld
- iii) - some have thick/fleshy/succulent leaves /barks
 - some have long tap roots
 - some have no leaves /have thin/spiky/waxy/needle like leaves
 - some plants have thick/hard barks
 - some plants have thorns
 - some plants are stunted/dwarf like
 - shrubs are common
 - some plants spout during wet seasons (short time)
- b) - fire-often ,large areas of forests are destroyed by accidental and sometimes intended fire. such forests takes long to recover
- diseases causes by pests and parasites attack mainly the planted forests causing many trees to die
 - human activity /settlement /charcoal burning/logging have destroyed many forests areas of which are transformed in to farms and grasslands
 - over-exploitation leads to depletion of certain tree species such as Meru oak, camphor and Elgon teak. these trees take long to mature
 - government policy of degazetting of some forests made people free to clear many forested areas
 - prolonged droughts lead to degeneration of forests some of which take long to recover
- ci) - Conduct a previsit

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- Collect the tools and equipments needed for the study
 - Prepare a working schedule
 - Read secondary sources at information
 - Formulate the objectives and hypothesis
 - Decide on the method of data collection
 - ii) - Hygrometer (wet and dry bulb thermometer)
 - Ordinary thermometer
 - Six's thermometer (max & min thermometer)
4. a i) - Stating either activities to be carried out during the field
- Recording the data an types of vegetation
 - Collecting data on vegetation types and conditions favoring their growth.
 - Drawing the structure of plants leaves
 - By classification and characteristics of leaves of different plants.
 - Observing types of roots and stem of the different plants
- b i) - high population increase
- poor grazing methods
 - poor management of bench terraces
 - nature of the landscape
 - charcoal burning
- ii) - loss of top soil
- siltation of water reservoirs and H.E.P dams on Tana river
 - deposit of sand along river leading to sand harvesting
 - intensification of soil conversation awareness in the district
- iii)- to find out the cause of soil erosion in machakos district
- to find out the effect of soil erosion in machakos district
5. a i) - formulation of hypothesis /objective
- reconnaissance survey
 - prepare route map
 - prepare a working schedule
 - carry out secondary research(content analysis)
 - seek permission from relevant authority
 - prepare necessary stationery
 - make travel arrangement
- ii) - the residents of the area covered by the map sheet do not have access to tea products meant for export
- The tea plantation does not stretch outside the Belgut district boundary
- b i) - Factors that influence weathering
- Types and processes of weathering
 - Effects of weathering on physical and human environment
- ii) - Draw sketches of features
- Data analysis
- Data interpretation
- Discuss the findings
- 6 .a) i) - Pre – visit/ reconnaissance

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- Preparing equipments to be used
- Asking permission from relevant authorities
- Setting objectives/ hypothesis
- Preparing map of the place/ route map
- Discussion before the field day
- ii) - Climate of the area have direct influence on vegetation
 - The vegetation of the area changes with the change of climate
- b i) - Reduced amount of vegetation cover
 - Reduced mountain shows in the tropical highlands
 - Severe wind and soil erosion
 - Presence of strong winds
 - Failing of crop yield
 - Reduced size of water bodies
 - High localized temperatures
 - Low plant/ animal population densities
- i)
 - First hand information would be collected
 - Data collected would be reliable
 - It is a quick method of data collection
- . c)
 - Relief – steep slopes have no / sparse settlements evidenced in western side (KEBEWET AREA) since such slopes are not ideal for erection of houses/ gently sloping slopes have dense settlements as is in control part of near (Chemalul, Kablanga, Kapwaso) areas since they form ideal sites for erection of structures
 - Vegetation areas with thick vegetation e.g. forests, plantations have scarce settlements (Tea estates and forests in S.E). Such vegetation discourages settlement
 - Drainage – poorly drained places like mashes have no settlements since the places cannot provide firm sites for erection of houses/ well drained places have dense settlements such sites provide water for a domestic use and provide firm sites
 - Transport lines – Areas with good road network have dense settlements for easy transportation of people and goods e.g. along loose surface roads

7. a i) - Reading from secondary sources
- Carrying out reconnaissance
 - Formulation of hypotheses and statement of objectives
 - Division of class into smaller groups
 - Asking permission from relevant authorities
- ii) - Most of the crops grown are cash/ most crops grown are subsistence
- Crop farming is more dominant than livestock keeping
 - Most of the economic activities take place in areas with good transport network
- iii)

ACTIVITY	EVIDENCE
- plantation farming	- Existence of estates
- Livestock farming	- Existence of dips
- Manufacturing	- Tea factory (4647)
- Transport	- Existence of roads
- Trade	- Shops

- iv) – Topographical

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b i) INDUSTRIALIZATION

- Has led to disposal of industrial wastes into the lake contaminating /polluting the water and interfering with aquatic life
 - Agro-chemicals/industrial effluents washed into the lake has promoted weed loons
 - causing the lake to be colonized by weeds/water hyacinth

DEFORESTATION

- Has exposed soils to agents of erosion; the eroded soils have been carried and deposited in the lake siltation. This had diminished the depth of the lake.
- Deforestation in water catchment areas has interfered with the water cycle causing low rainfall and less water from feeder rivers. this has led to drying of the lakes

ii) - observation

- Taking photographs
- Taking measurements
- Collecting samples
- Interviewing/asking questions
- Reading from secondary sources

iii) - In order to write the report/summarize the data

- So as to a group/process/ analyze the data collected
- so that the various groups can report/discuss the findings from the field
- So as to display any collected samples
- So as to assess the importance of the information collected in the field
- To test samples collected from the field

. c i) -note taking

- filling in questionnaire
- tabulation
- field sketching/drawing maps
- tallying
- photographing
- tape recording
- labelling samples

ii) -bad weather-raining

- in accessibility of some area
- lack of sufficient data
- illiterate respondent
- arrogant respondents

8. a)i)- All weather road.

- Loose surface road.
- Dry weather road.

ii)- Formulate hypothesis/objectives.

- Makes a short survey/reconnaissance of the area to be studied.
- Preparation of route map.
- Carrying out literature review/secondary information.
- Organize into groups.
- Make transport arrangements.
- Prepare the necessary stationery and equipment required.

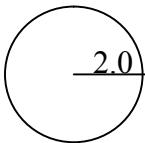
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- Prepare a working schedule.
 - Seek permission from relevant authorities.
 - b)(i) - Assemble equipment
 - Depart for the area of study
 - Report to the factory authorities
 - Embark on data collection
 - Report back to the factory authorities
 - Report back to school
 - ii) - The class will be able to study the entire course of the river
 - It will enable them to obtain detailed information on each stage of the river
 - It will save on time
 - It will enable the study to be carried out in an orderly way
 - It will encourage participation of all the members of the class/ encourage individual roles
 - It will facilitate more interaction among the group members
9. a i) - Administering questionnaire observation
- Collecting samples of rocks/ deposits
 - Drawing sketches/ diagrams/ maps
 - Measuring/ calculating
 - Taking photographs
 - Reading secondary materials/ content analysis
- ii) - The hot/ scorching sun would make it difficult to collect data
- Torrential rain/ flash floods/ dust storms would disrupt data collection
 - The rugged terrain would make it difficult to reach certain features
 - Attack by wild animals
- b i)- Measure distances/ estimation of distances/ heights
- Collect samples of plants
 - Draw sketches/ transects
 - Record/ take notes
 - Take photographs of plants/ area
 - Count plants
- ii) - By their appearance
- Their colour
 - By their leaf size/ pattern/ arrangement/ type
 - By their age
 - By the nature of their barks
 - By the texture of their leaves
10. i)- Graphs
- Pie charts
 - Preparing a written report.
 - Displaying rock samples
 - Displaying photographs.
 - Drawing maps showing distribution of area underlying desertification.
- b i) - Breaking the rocks.
- Digging the rocks.
 - Collecting samples.
 - Tasting the rocks

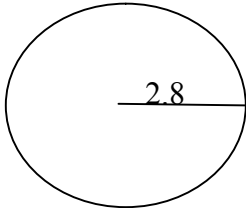
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- Observing the rocks
 - ii)- Tiredness because of rugged /steep terrain.
 - Lack of testing Material.
 - Inadequate resource materials.
 - Accidents e.g slipping/getting cut by rocks.
 - Hindrances by poor /harsh weather conditions.
 - Attack by wild animals.
 - Difficulty in carrying heavy rock samples.
 - Difficulty in climbing /ascending steep rocks.
 - 11. a i) - To determine the type of soil
 - To find out the soil colour.
 - To establish the use of the soil - ii) - Inadequate time for detailed study.
 - For further analysis in the lab.
 - Keeping for future reference in the geography room.
 - (b) i) -carried out a reconnaissance /persist
 - identified the tools to be used during the fieldwork
 - prepared a working schedule
 - asked for permission from relevant authorities
 - held discussions in groups
 - state the objectives/hypotheses
 - ii) -hostile weather due to heavy rains
 - attacks by wild animals
 - lack of resource pesos
 - some areas were inaccessible
12. a) i) -direct observation
 - administering questions
 - taking photographs
 - taking measurements
 - interviewing the local people- ii) -to identify the route
 - to prepare time schedule
 - to ask for permission from the local community
 - to identify the particular features location
- iii) -can be advised on ways of controlling flooding
 - can be advised on several uses of the river and need to conserve it
- 13. a) - long and narrow
 - some are salty
 - some are fresh
 - some have underground outlets e.g Naivasha
 - deep- b i) – Helps the researcher to decide on appropriate method of data collection
 - Helps in identifying the appropriate tools to be used during the study
 - Helps the researcher to design a working schedule
 - Helps the researcher identify problems likely to be identified

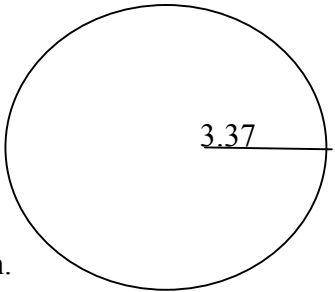
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- Helps the researcher estimate the cost to be incurred
 - Helps the researcher to familiarize with the area
 - ii)
 - Note taking
 - Filling in questionnaire
 - Mapping
 - Photography
 - c) i) - the area is too wide/extensive
 - some areas are inaccessible-steep slopes
 - harsh weather/weather changes
 - field study can take too long
 - c) ii) - report writing
 - class discussions
 - displaying of collected samples
 - asking/answering questions
 - reading more about the topic
 - analysis photographs/tape recorded work/collected samples
14. a i) $\frac{22}{105} \times 100 = 13.33$
 $= 13.33\%$
- ii) $\sqrt{\frac{2000}{165\ 000}} \quad \sqrt{\frac{2001}{318\ 000}} \quad \sqrt{\frac{2002}{455\ 000}}$
 $= 406.20 \quad = 563.914 \quad = 674.536$
 $r \ 2.0 \quad r \ 2.81 \quad r \ 3.37$
- 

2.0



2.8



3.37
- b)
 - Moderate to high rainfall/500 – 1270 mm p.a. for growth.
 - Dry spell for harvesting.
 - Warm to hot temps/15 – 20⁰c to facilitate growth/maturity.
 - Well drained volcanic soils increases yield.
 - Gentle/undulating topography to facilitate mechanization..
- c i) In Canada cultivation is highly mechanized while Kenya in Kenya its less.
- ii) In Canada wheat is mainly for export while in Kenya its mainly for local consumption.
- iii) The farms in Canada are more extensive and wide while in Kenya the cultivation is done mainly in small scale.
- d) - Used in industries to make alcohol and glue.
 - Wheat flour is food /bread/ cake.
 - The wheat stalks are livestock feed.
 - Straws are used for making papers/ straw gourds/ bedding in cow sheds.

15. a i) Firewood - $\frac{13400}{360} \times 360 = 107.20$

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45000

$$\text{Kerosene} - \frac{11200}{45000} \times 360 = 89.60$$

45000

$$\text{Charcoal} - \frac{9100}{45000} \times 360 = 72.79$$

45000

$$\text{Liquid petroleum gas} - \frac{5300}{45000} \times 360 = 42.40$$

45000

$$\text{Saw dust} - \frac{4000}{45000} \times 360 = 32.00$$

45000

$$\text{Hydro electricity} - \frac{2000}{45000} \times 360 = 15.99$$

45000

- b i) - Previsit/ reconnaissance
 - Literature review
 - Class discussion
 - Data collection instruments
 - Preparation of working schedules
 - Obtain permission
 - Transport arrangement
- b ii) - Discussion of findings
 - Write better notes/ draw better diagrams
 - Display photographs/ maps/ diagrams
- c i) - Inaccessibility of some parts the to presence of tendrils.
 - Wild animals' attack/insects sites.
 - Unfavourable weaken conditions/cold conditions.
 - Rainfall
- ii) Height of a tree measure the shadow then calculate.
 Diameter of the stem – use a tape measure.
 Trees of the same species – study leaf structures.
- d i) - Foul smell
 - Garbage may harbour smokes which are dangerous.
 - Paths may be blocked.
- ii) - Tree forming should be practiced in the area ✓
 - Agro – forestry should be practiced ✓
 - People should be encouraged to use alternative sources of energy/ energy saving jikos ✓
 - Mature trees felled should be replaced immediately ✓
 - Villagers/ people should be educated on importance of trees ✓
 - Nurseries should be established to provide seedlings ✓
 - Indigenous trees should be planted ✓

7. MINERALS AND ROCKS

1. a) - Veins and loads
 - Beds and seems
 - Weathering products
 - Alluvial/places deposits

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- b) - causes soil erosion/encourages soil erosion
- derelict landscape/make landscape ugly
- destruction of bio-diversity
- creating hiding outs of thugs/thieves
- create breeding grounds for mosquitoes if the pits are filled with water
2. a) - That is when minerals occur in crevices, crack or faults
- If they occur in small quantities in cracks or crevices then they are referred to as veins while those in large quantities are said to be in lodes
- They are formed when minerals in molten form solidify in cracks and faults or crevices
b) - tiny marine organisms called coral polyps live in colonies in the sea
- polyps extract calcium from the sea water to make their shells
- the polyps die and their hard skeletons of calcium carbonate accumulate into solid mass
- successive colonies grow
- space between the dead corals polyps are cemented by Algae to form coral rocks
3. - Mechanically formed sedimentary rocks.
- Organically formed sedimentary rocks.
- Chemically formed sedimentary rocks.
- 4 a) - Dynamic/Regional/Kinetic
- Contact
i)- They are sedimentary rocks which are formed from the hard parts of shells/skeletons of Marine. The rocks are rich in calcium.
ii)- Are sedimentary rocks which are formed from the remains of plants that were buried deep in many years. The plants are transformed into rock because of pressure of crystal rocks and resultant heat.
iii)- Coral limestone/dolomite/calcite/chalk/colite
- Sandstone
- Gypsum
- Quartzite
c i) - Particles are derived from existing rocks through process of weathering /Erosion.
- They are then transported by water/wind/ice and are deposited on land/large water bodies.
- They are deposited in layers.
- Over a period of time, they are compacted by pressure to form rocks such as alluvial, clay, sand, conglomerates.
ii) - Minerals are dissolved from land and are carried in solution into bodies such as lakes/sea.
The mixing of water of different types may result in chemical reactions followed by precipitation leading to accumulation at bottom of water body and are compressed into a rock
OR
Dissolved minerals can also be precipitated directly from water through evaporation.
This happens in very hot regions where there are constricted/shallow bays of lakes/seas, such rocks are known as evaporates.
5. a) -due to pressure changes/dynamic metamorphism
-due to intense heat/thermal metamorphism
-due to a combination of heat and pressure/thermal-dynamic metamorphism
b) -Granite

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- Diorite
- Peridotite
- Gabbro
- Syenite
- c) - The aquifer must be sandwiched between impermeable rocks so that it can retain water
 - The aquifer must outcrop in a region which is a source of water
 - The mouth of the well must be lower than the intake area to allow water to be forced out by pressure
 - The aquifer must dip from a region of water intake and the rock layers must form a broad basin

8. MINING

1. (a) i) Mining has led to exhaustion of most mineral mines in Kenya e.g. Gold in macalder
ii) Flourspar is mined in Kerio Valley✓
iii) Trona is mined on Lake Magadi through redging
 - The dredger accumulates and digs out the minerals✓
 - Inside the dredger, trona is crushed into crystals from the lake bed✓
 - It is then mixed with lake liquor and pumped to the factory on the lake shore✓
 - At the factory, the trona is mixed with water to clear it of impurities✓
 - It is then dried and send to dessicators for heating✓
 - After heating it is allowed to cool✓
 - After cooling it is then crushed into soda ash ✓b(i) - Mining results in the destruction of various species of vegetation✓
 - The destruction of plant like destroys the habitat for wildlife✓ii) - Mining leads to emission of toxic gases from underground✓
 - When it rains, such gases and chemicals are washed into rivers and lakes hence affecting aquatic life✓
 - Heavy machinery used in mining causes noise pollution✓
 - Heavy machinery and trucks used raise a lot of dust that causes air pollution
2. a i) - Open cast mining.
 - Underground mining.
 - Alluvial mining.ii) - The value of minerals.
 - The rise of the mineral deposit.
 - Methods of mining.
 - Technology.
 - Capital.
 - Market.
 - Transport cost.
 - Security.iii) - By filling the pits or the holes using the heaps of soils.
 - Through planting trees and keeping a wide range of animals thus creating a tourist centre.
 - By changing the pits or holes into a man made lake which could serve as in land fisheries or a sporting centre.

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3.
 - a) - As veins and lodes
 - As alluvial deposits
 - b) - By planting trees in the area
 - By filling up the pits with fresh soils
 - By upgrading the abandoned mine into a tourist attraction
 - By keeping a wide variety of animals in the area to restore its natural ecosystem

4.
 - a)

Angola	- Nigeria
Libya	- Chad
Sudan	
 - b)

Wax	- Bitumen/Tar/Pitch/Osphel
Sulphur	- Petro – Chemicals
Lubricants e.g grease	

5.
 - (a) (i) - method used to extract mineral and fossil fuels from the ground
 - ii) - Availability of skills and relevant modern machines are important for specialized mining operators.
 - If the skills /technology is inadequate then there would be need to bring in foreign experts mini therefore becomes dependent on foreign control.
 - High quality ores are economical to extract as they yield a large amount metal/low quality ores are rarely extracted for their metal content is very low.
 - Some rare minerals e.g uranium are exported despite their ores having low mineral content because they are important

 - b i)
 - P – Copper
 - Q – Gold
 - R – Trona
 - b ii) - Kimberley
 - Pretorca
 - Jagers Foutein
 - Koffie fontein
 - Unwanted materials /overburden is removed.
 - Excavators are used if surface materials are soft but if they are hard then explosive are used to loosen the materials.
 - Excavators are used to dig up the mineral deposits.
 - The extracted ore is loaded into lorries using excavators and transported to the processing plant.
 - Kenya earns foreign exchange from the exportation of trona. This is used to import other essential items like machinery.
 - Trona mining has created employment opportunities for many Kenyans thus improving their living standards.
 - It has led to the development of related industries e.g glass making industries in Nairobi, Mombasa e.t.c.
 - It has led to the provision of social amenities which have improved the living conditions of the people around e.g schools.
 - It has stimulated construction of transport lines e.g the Magadi – Konza railway live.
 - Has led to the growth of Magadi town.

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- Provision of water for both the domestic and industrial use within the area. This has improved the living standards of the people.
 - Government earns revenue through taxation.
 - Air pollution by dust and smoke emitted from blasting, quarrying & processing of the ores.

 - Derelict land is dangerous to both people and animals.
 - Wastage of agricultural and settlement land as the mine holes become useless.
 - Inadequate skilled personnel who are required for the industry leading to reliance on imported skilled workers.
 - Insufficient capital for the exploitation of minerals.
 - Inaccessibility of some minerals due to hilly or mountains landscape.
 - Pollution of groundwater sources as well as rivers by water leakage from processing plants
- 6 a i) W- Fluorspar X – Gold Y – Diamonds Z – Copper
- ii) - Veins/ lodes
- Beds/ seams
 - Weathering products
 - Alluvial deposits
- b) - Mode of occurrence
- Value/ cost of mining
 - Size of deposit
 - Level of technology
 - Capital availability
 - Labour supply
 - Transport
 - Government policy
 - Market availability
- (c) - Land dereliction – waste agricultural land/ makes land ugly/ limits town expansion
- Health and accident hazards – collapse of mines/ fall in open pits/ drowning in water filled pits/ toxic gases and dust
 - Unemployment/ depressed economy – after exhaustion of the minerals
 - Conflicts – cross boarder minerals
 - Loss of biodiversity – clearing vegetation for mining
 - Soil erosion – clearing of vegetation
- d) - Trona/ soda ash
- Fluor spar/ Fluorite
 - Limestone/ lime
 - Carbon dioxide
 - Gold
7. a) - Under ground/ shaft/ adit/ solution
- Alluvial/ panning/ placer/ dredging/ hydraulic
 - Open cast/ strip

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- b i) $\frac{1000 - 800}{1000} \times 100 = 20\%$
- c) - Exported to earn foreign exchange for economic development
- Creates employment opportunities reducing unemployment/ crime/ improving living standards
 - Provides raw materials for industries leading to industrialization
 - Leads to development of transport networks improving transportation trade
 - Leads to development of social amenities improving the living standards
 - Led to growth of town – magadi
 - Led to growth of tourism – revenue
 - Provides revenue through taxation for provision of facilities
8. a i) - Underground mining
- Open-cast mining
 - Alluvial/place mining
- (ii) –trona
- gold
 - copper
- b i) - Availability of technology skills and modern machines are important for specialized mining operations
- Inadequate skills, lead to importation of expatriates
 - Mining operations may up becoming dependent on foreign control
- ii) - High quality ores economical to extract as they yield a large amount of metal
- Low quality ores have low metal content and are rarely extracted
 - Rare metals e.g. uranium are exploited despite the ore having a low mineral content
- iii) - Minerals in remote areas with poor transport systems are less likely to be exploited
- Almost all ores are heavy and bulky and are therefore costly to transport. it becomes hard to exploit them in the absence of good transport systems
 - Deposits at or near ports enjoy cheap transportation compared to inland deposits
 - Deposits near the ports are likely to be more exploited
- c) - Exhaustion of the mineral because gold is non re-newable and the old mines in the rand are being depleted
- The gold grade being worked now is of poorer quality that of some years back
 - The mines are becoming deeper hence mining costs have escalated and also required new technology which is more costly
 - Inadequacy of water for processing gold due to seasonal rainfall received and increased population on the rand
 - High cost of labour because of increased demand for higher wages and competition from other factors of the economy
- (e). - Ugliness – all the natural beauty of the landscape has been lost
- Health hazard- mineral exploitation can create open pits that become breeding grounds for mosquitoes
 - Lost productivity – the soil left behind after mining may not be able to support any meaningful economic activity
9. a) x – Non porous rock.

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- y - Porous gas.
- b) - It is cheaper to transport oil in crude form.
- Oil refining creates employment opportunities to most Kenyans.
 - Some of the refined by-products are exported to land locked countries in East & Central Africa thereby earning foreign exchange.
 - Oil refining has led to establishment of industries e.g. oil refinery at Changamwe and other related industries such as fertilizer manufacturing, plastic making e.t.c.
10. a) A vein is a small crack containing minerals deposited in crystalline form while a lode is a large crack containing minerals in crystalline form
- b) - waste of Agricultural land
- Waste of industrial land
 - Lightness where land has lost its beauty
 - Health and accident hazards)
11. a i) - Shaft/underground
- Open cast mining
 - Placer/alluvial/panning/slope boring
 - Adit/drift/horizontal/hill dredging
 - Submarine mining
- ii) - Geita
- Mpanda
 - Irambal/Sekenke
 - Musoma
 - Mabuki
- b) - The value of mineral-valuable minerals e.g. gold will be mined since it earns higher profits
- Size of the deposits – should be large enough to justify mining
 - Capital- mining needs a lot of money to pay workers and purchase machinery
 - World market prices which are controlled by international bodies when prices are high more minerals will be mined
 - Transport cost- it is economical to extract ores near major industrial centres because of good transport routes
- c) - Mining leads to pollution of air/water/land/noise
- Mining leads to depletion of land
 - Mining disrupts/lowers the water table
 - Mining leads to loss of biodiversity /plants and animals
 - It leads to soil erosion/degeneration of soil
- d i) - Apply where mineral is dissolved by water
- A well vertical shaft is sunk to reach the mineral
 - Pipes are laid down through this vertical shaft
 - Superheated water is pumped into the deposits of mineral
 - Mineral dissolves in hot water and form a solution
 - Solution is pumped to the surface where it is evaporated and the mineral is extracted
- ii) - Exported to earn foreign exchange
- Generates employment opportunities
 - Has led to development of settlement

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- Has led to establishment of industries
 - Earning higher income hence better living standards
12. a i) - Alluvial mining✓
- Under ground mining✓
- Open cost mining✓
- ii) - Creation of employment opportunities which helps in reducing unemployment✓✓
- When exported it earns foreign exchange which is used in other sectors✓✓
- Development of related of related industries that helps increase the wealth of the country✓✓
- Leads to provision and improvement of social facilities✓✓
- It helps in the development of infrastructure e.g. roads✓✓
- iii) - Water shortage for power supply and processing✓
- Labour shortage and competition from other industries/ sectors✓
- Increase depth of mines makes mining expensive and risky✓
- Decreasing quality of one ✓
- b)(i)- Presence/ deposition of remains of flora and fauna fossils over a long period of time
- Presence of non- porous rocks under neath the deposits of flora and fauna✓
- Deposition of other layers of rocks/ non- porous rocks over the remains of flora and fauna✓
- Compression of the remains of fauna and flora due to folding of the layer of rocks✓
- ii) - Bitumen/ pitch/ asphalt✓
- Grease/ lubricants✓
- Resin/ petro chemicals✓
- c) - Employment opportunities✓
- Saving of foreign exchange✓
- Earning of foreign exchange from exports✓
- Industrial development✓
- Earnings would raise the standards of living/ raise the per capita income✓
- d) i) - Coal✓
- Iron ore✓
- ii) - Kilindini✓
- Dar- es- salaam✓
- Tanga ✓

9. INTERNAL LAND-FORMING PROCESSES

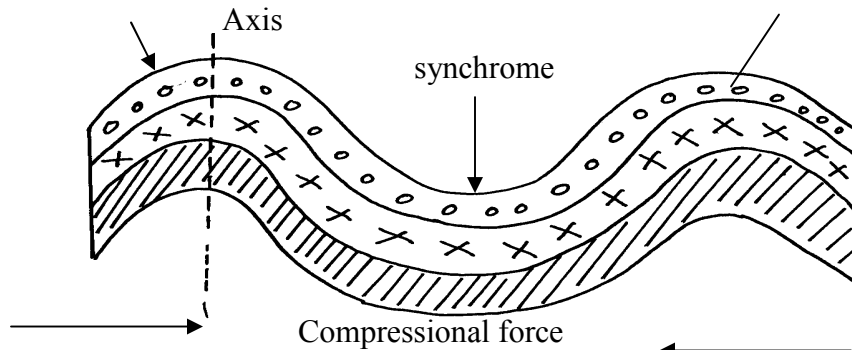
1. a) - it has vertical vent/pipe
- compose of alternating layers of ash/pyrodast and lava
- it is conical in shape /steep sided
- it has side vents
- it has conelets/parasitic cones on the sides
- at the peak, it may have a caldera /crater/plug
- b i) Earth quakes are sudden earth movement which cause vibrations/trembling within the crust✓✓
- ii) - primary/push/p-waves s
- secondary/shear waves/shake waves/s-waves
- longitudinal/L-waves/lore waves/Raleigh waves
- iii) - volcanic mountains re sources of rivers which provide water for

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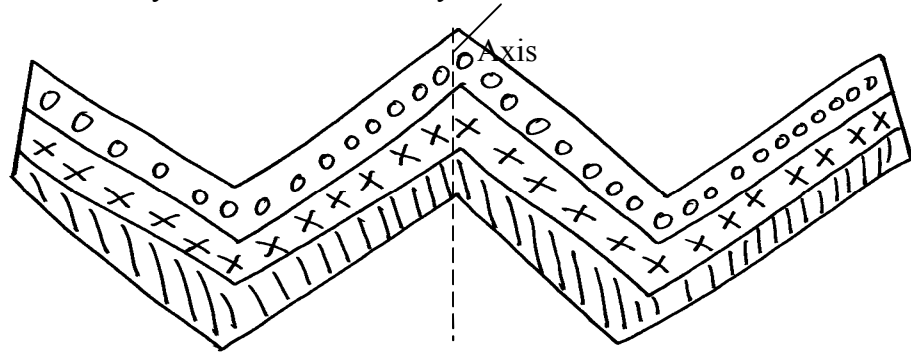
domestic/agricultural/industrial use

- volcanic mountains have forests which provide valuable timber used in building and construction industries
- volcanic mountains influence formation of relief rainfall which encourages agricultural activities
- volcanic mountains modify temperatures making them attractive to human settlements

2. a)



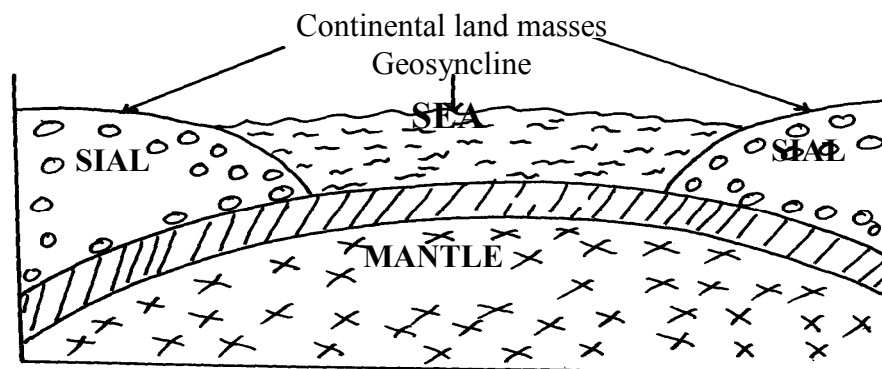
- Formed when compressional forces of equal magnitude and of moderate strength act on crustal rocks
- The rocks bend evenly and the anticline is symmetrical about its axis



- Results from unequal compressional forces acting on crustal rocks
- One of the forces is slightly stronger than the other resulting to limbs that are asymmetrical about the axis

- b i)
- Atlas mountains of Africa
 - Alps mountains of Europe
 - Himalayas mountain of Asia
 - Rockies of N. America
 - Andes of S. America

b ii) - An extensive depression called a geosynclines is formed on the surface of the earth - The geosynclines is then filled with water to form a sea

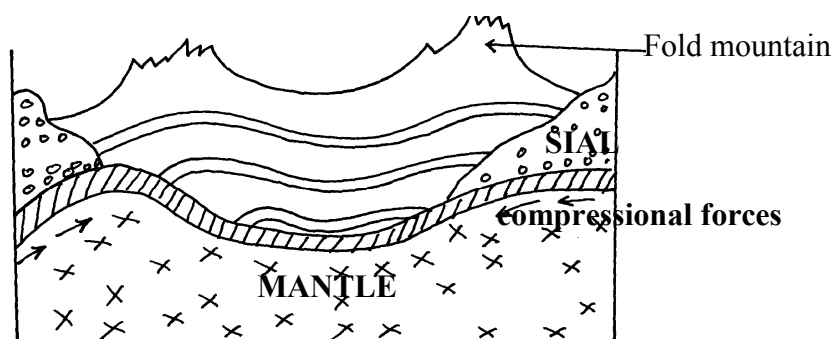


- The surrounding land masses are eroded and resultant materials deposited in the geosyncline in layers
- Accumulation and weight of sediments causes the floor of the geosyncline to subside further

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- Further subsidence of the geosynclines triggers off compressional forces drawing higher grounds close
- The layers of sediment in the geosynclines are the folded bending upwards form mountains

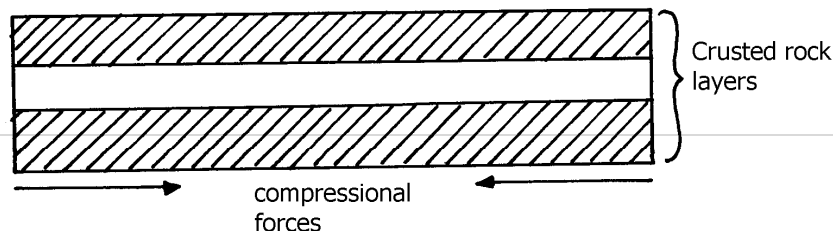
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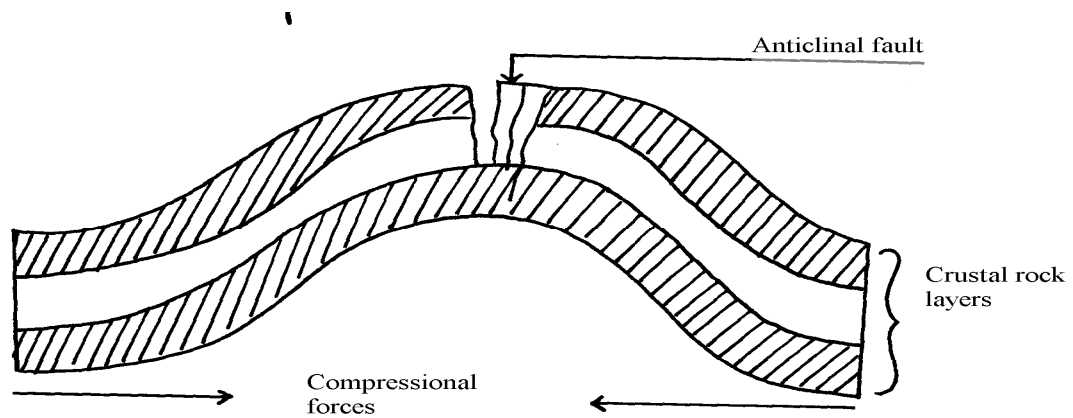


- c) - Fold mountains are often forested on their windward slopes and provide valuable timber for construction
- The windward slopes receive rainfall hence supports agriculture
 - Heavy rainfall and snow that collects in some fold mountains give rise to rivers which in turn provide water for both domestic & industrial use
 - The process of folding could bring valuable minerals to the surface cutting down the cost of mining e.g. coal in Appalachia
 - The unique mountains landscape attracts tourists thus earning the country foreign exchange
3. a) - earthquakes causes lateral and vertical displacement of rocks
- they cause raising and lowering and down warping of parts of the sea floor
 - it causes landslides/slumping
 - it leads to faulting of the crust
 - they lead to volcanic eruptions
- b) - The jigsaw fit of continental margins e.g. Southern Africa and Southern America
- Spreading of the ocean floor- rocks are younger at the oceanic trenches and younger outwards
 - Geological evidence i.e. the rock structure in some continents are similar e.g. South Africa and South America meaning that they were together
 - Palaeoclimatic evidence, similar fossils/ remains have been found in different continents
- c) - Extension boundary/ divergence
- Compression/ convergence
 - Transform fault boundary
 - Conservative boundary
4. a i) Tectonic plates are rigid blocks that form the earth's crust
- ii) - Pacific plate
- Nazca plate
- b i) This disrupts the balance between the sial and sima causing movement of continental masses
- ii) This force themselves in cracks/crevices displacing crustal rocks
- iii) These cause frictional drag/pull on the crustal rocks causing their movement
5. a) - Isostatic adjustments
- Gravitation pressure
 - Tectonic movement

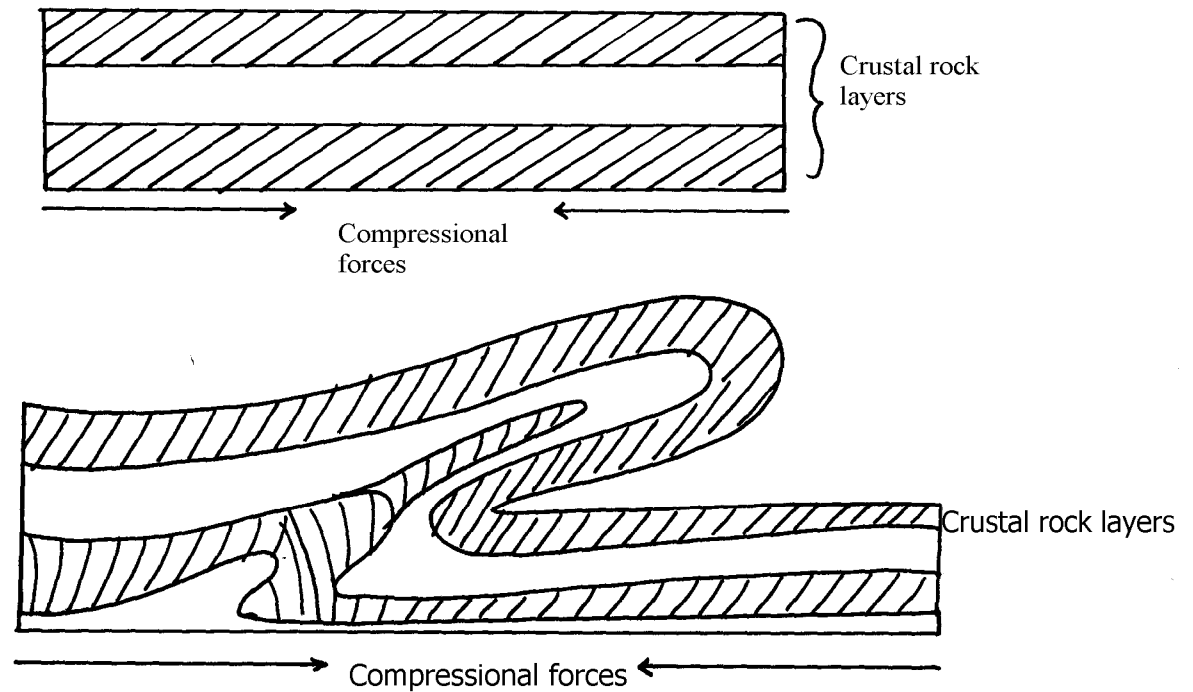
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- Magma movement (volcanic activities)
- Energy release in upper mantle
- . b) -primary (P) waves
 - secondary (S) waves
 - surface longitudinal waves
- 6. a) -they are long deep and narrow
 - most of them are salty
 - they are fault lakes
- b)i) -They are sudden earth movement which cause trembling/vibration within the earth's crust.
 - (ii) - Primary/ puse waves/p. waves.
 - Secondary/shear/S. waves
 - Longitudinal/love/L. waves
- c) Extension boundaries are zones where tectonic planets diverge or move away from each other. Space may be created and magma fill the space found in between. While compress ional boundaries are zones where tectonic plates moves towards each other. There is thus destruction of materials found at the edges
- 7. a i) - Is a sudden and rapid movement/tremor of the earth crust.
 - ii)- Use of change of velocity of seismic waves
 - Use of exotic pre-quake signals.
- b) - Faulting causes disjointing of the land which results in disruption of transport and communication lines.
 - The presence of fault scarps can cause difficulty in construction of transport and communication lines.
 - It is expensive to construct and communication lines in faulted region.
- 8. a) reverse fault
 - b) X-up throw
 - Y-down throw
- . c) - Fire outbreaks
 - destruction of transport and communication lines
 - loss if life and properties/collapse of buildings
 - Change directions of rivers flow
- 9. a) -hot springs are place where hot water or steam is emitted from the ground while geysers are jep of hot water or steam and gasses which are ejected explosively from fissure in the ground
- b) A-crates B-plug C-strato volcano
- c) i) - Crustal rock layers are subjected to compressional forces
 - This leads to formation of a massive article /up fold
 - Further compression to the same crustal rocks causes stress/tension at the crest of the anticline
 - Eventually a crack/fault and around the crust of the anticline called an anticline fault





- c ii) - Crustal rock layers are subjected to compressional forces
- Intense folding results into formation of an overfold
 - With increased pressure the over fold is compressed further to form a recumbent fold
 - When pressure is very great, a fracture occurs along the axis producing a thrust plane to form an overthrust fold

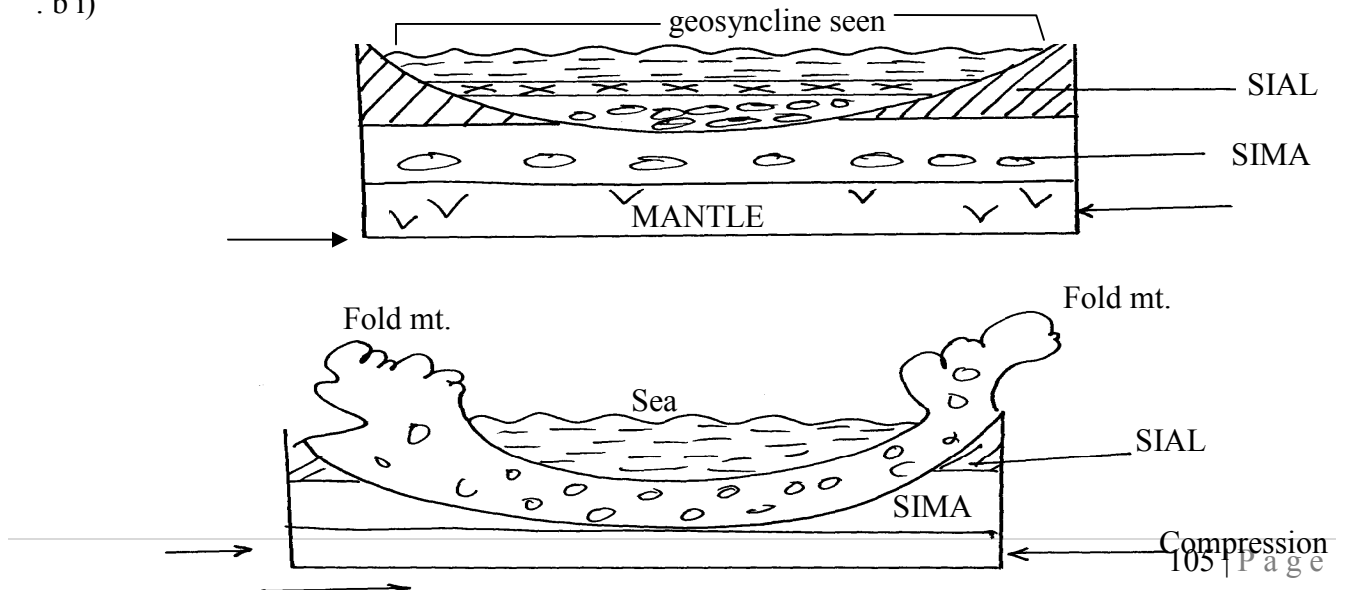


- d i) - Volcanic mountains experience low rainfall on their leeward slopes making them unsuitable for agriculture/settlement/forestry
- Volcanic mountains have steep slopes which hinder transport and communication
 - Some volcanic features like moffetes, solfataras steam jets produce poisonous which pollute the environment/are harmful to people

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- Some volcanic rock weather to form infertile solid unsuitable for agriculture/forestry
- d ii) - Outpouring of Lava forms a volcanic cone
 - When the pressure beneath stops, a hollow, void is left beneath
 - The weight of the overlying cone leads to collapse of the top of the cone into the how- /void beneath
 - This leaves a wide depression at the top of the volcano called a caldera

10. a) - It is a theory that states that the crust is divided into blocks of land (plates) that float and move towards, away or parallel
- b) - Compressional boundary
- Extension boundary
 - Transform fault boundary
- c) - Geological evidence – Rock similarities in structure, age and type of rocks found in various continents e.g. in Eastern coast of S. America and Western coast of Africa
- Jig- saw- fit – of continental coastlines. Some coastlines can fit each other when pulled together e.g. Western coast of Africa can fit with Eastern coast of South America
 - Ancient glaciations – All the four Southern continents reveal signs of a period of large scale glaciations at one time. It is evidenced by presence of glacial deposits – Tillites produced by ice sheets in Southern continents
 - Archeological evidence – Some plant and animal remains found in various continents show a striking similarity
11. a i) - Atlas
- Cape ranges
 - Akwapim
- ii) - Atlas
- Cape ranges
 - Akwapim
- iii)- Symmetrical folds have limbs dipping uniformly about the axis while asymmetrical fold one limb is steeper than the other about the axis
- b i)



- b i) - Extensive shallow depression called geosynclinal develop
 - Sediments from high areas are deposited in the geosyncline
 - Compression on either side push towards geosyncline
 - Sediments compress to form fold mountains which rise above sea level to form fold mountains
 - b ii)
 - Over fold
 - Recumbent folds
 - Nappe or overthrust fold
 - Anticlinical/ synclinal fold
 - Isoclinal
12. a) - Causes seasons
- Varying lengths of day and night time
 - Changes in the position of the mid-day sun at different times of the year
 - Causes eclipses
- b) - Movement of magma within the crust
- Gravitational force towards the center of the earth
 - Convectional currents in the mantle
 - Isostatic adjustment of sial and sima layer
13. a) - Frequent earthquakes
- Presence of faults
 - Presence of fold mountains
- b) - Tectonic movements
- Volcanicity
 - Gravitative pressure
 - Isostatic adjustment
 - Energy release in the mantle
14. a) - Nature and age of the rocks
- Intensity of compression forces
- b) - May cause a river to disappear
- Changes the rivers direction of flow
 - Forms depressions hat are filled with water forming lakes
 - Faulting across a river forms waterfalls
15. a) - The jig-saw fit✓
- Glaciatean/ancient glaciation✓
 - The coal deposits✓
 - Sea floor spreading✓
- b i) - Australia✓
- South Africa✓
- ii) this was the water body that occupied the region surrounding the pangae
16. a) - High demand for hard wood has led to over exploitation
- They take long to mature hence can not match the exploitation rate

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- Population pressure has led to cutting trees to give room for settlement and agriculture
- b) - Accidental fires which destroy the forests
- The Northern part is inaccessible (during winter) for exploitation of the forests
- Over exploitation in some areas
- Trees take long to mature due to cold climate
- Rugged landscape especially mountainous landscape makes exploitation difficult

17. (a) i) A-axis B-syncline C-limb of a fold

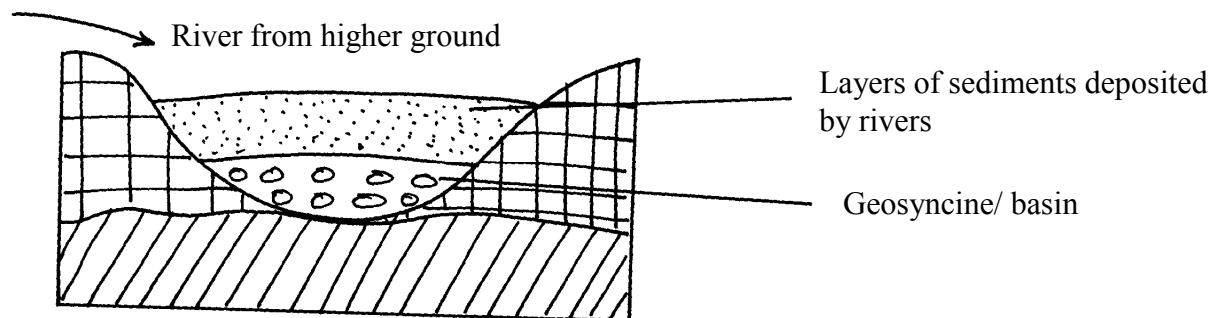
ii) - The rockets

- Andes
- Himalayas
- Alps
- Appalachians

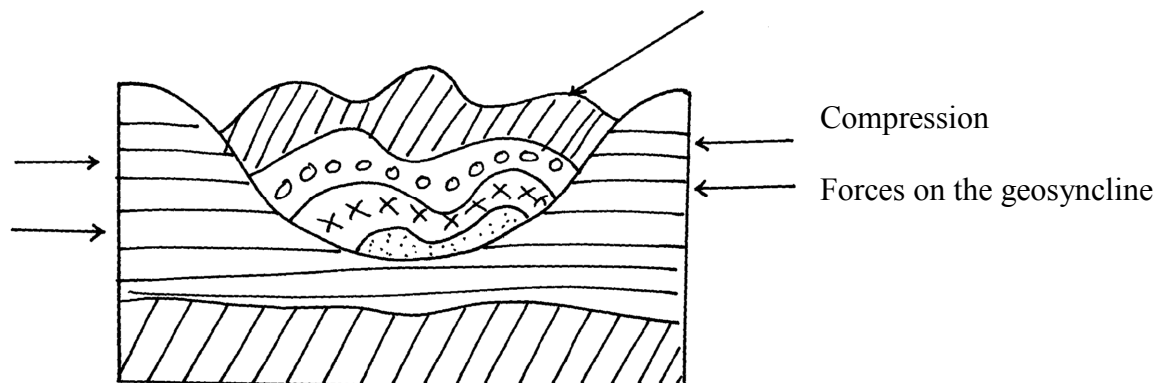
iii) - fold mountain

- cuesta
- escarpment
- plateaus
- basins

- b)
- Geosynclines are formed on the earth's surface
 - prolonged and extensive erosion occurs on the surrounding higher grounds
 - sediments are deposited in the geosyncline forming thick layers



- the weight of the sediments causes subsidence of the geosyncline leading to accumulation of more sediments
- further subsidence of the geosyncline triggers off compressional forces which draw the higher grounds closer forming fold mountains



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- c) - Fold Mountains are water catchments area. They trap rainfall which rivers which that provide water for domestic use
- fold mountains are often forested and provide valuable timber used in construction and building industry (due to high rainfall)
 - some fold mountains have valuable mineral deposits such as coal and petroleum for mining
 - fold mountains influence transport system either as barriers or as passed

10. PHOTOGRAPH WORK

1.
 - a) i) - Ground close up
 - ii) - Photograph (A) – Harvesting of sugarcane
 - Photograph B - Transportation of sugarcane
- . b) - Kakamega district
 - Busia district
 - Bungoma district
 - Mumias district
- . c) - it requires high rainfall of about 1270mm and rainfall should reduce towards harvesting
- Twice to allow for concentration of sugar
- It requires hot climate with temperatures ranging from 20°C to 27°C throughout the year
- It requires a dry sunny period for ripening and harvesting
- It requires well drained soils, with nutrients
- It is grown on fairly level gentle sloping land. This allows for the use machines during ploughing, planting and harvesting
- d) - Small farm sizes
 - Pests and diseases e.g. ratoon stunting lowering the yields
 - Corruption in factory management over production
 - Delayed payment kills the morale of the farmers
 - Low earnings from sugarcane production
2.
 - (a) i) - Well developed udder
 - Have wide hind quarters, big stomach and small chests
 - Have big milk veins
 - Short legs well set to support their heavy bodies
 - ii) - The highlands have high population that offers ready market for the product
 - They have well established infrastructure which is a major factor in dairy farming activity
 - Reliable rainfall and constant supply of water
 - The region has low temperature best for exotic breeds. Survival
 - There is humid condition
 - High quality cover of grass due to fertile soils in the region
 - The high nutrient pasture characteristic

11. VEGETATION

1.
 - a) - Pampas of Argentina
 - Veldt of S. Africa
 - Downs of Australia
 - Steppe of Asia
 - Prairies of N. America

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- b) State three characteristics of temperate grasslands.
- b) - Almost airless except along water courses
- Grass is tall and nutritious in most areas
 - Grass is shorter, tougher & less nutritious in drier areas
 - Grass withers in autumn
 - It sprouts in spring
- ii) - Some trees are umbrella shaped to reduce evaporation below the stems
- Some plants have long roots to reach the water table below
 - Some plants have thick/ fleshy/ succulent leaves to store water
 - Some plants/ grass wither in absence of rainfall but have abiding to recover quickly when rain falls
 - Some plant seeds remain dormant for long periods awaiting the start of rainfall
 - Some plants have thin leaves/ spiky/ waxy needle like leaves to reduce water loss
2. a i) X-rainforest
Y-bamboo forest
Z-heath and moorland
- ii) - it consist of a mixture of tree and grass
- in wetter areas the grass are tall and close to each other
 - in drive areas the grass are shorter and tufted
 - grass dominate the vegetation
 - trees are shorter and more scattered
 - trees are umbrella shaped
 - acacia trees are a dominant species
 - some trees i.e. bamboo have thick barks
 - river valleys have trees and thick bushes
 - during drought grass withers/trees shed their leaves
- b i) Areas on leeward slope of mountains have different vegetation from those on wind ward slope because they received different amount of rainfall✓✓
- ii) Settlement, farming and mining interferes with original vegetation leading to growth of secondary and dissertation✓✓
- c i) south Africa✓
- ii) Argentina✓
- iii) New Zealand✓
- . d) The frequent outbreak of bush fires destroys the grass retarding its regeneration✓✓
the increasing human population is encroaching into the grasslands replacing them with settlement and cultivated land✓✓
pests such as army worms /locust destroy the grass and the vegetation degenerations into a semi- desert type✓✓
wild and domestic animals overgraze and cause stunted growth of grass✓✓
3. a i) -Natural vegetation is the plant cover that exists in an area without the interference of external modifying influence especially people and their animals
- ii) -Temperature
- Precipitation(rainfall)
 - Sunlight

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- Wind
 - b i) -The prairies (of N. America)
 - The veld (of S Africa)
 - The pampas (of Argentina)
 - The downs (of Australia and New Zealand)
 - ii) -They are conical in shape
 - They have needle-shape leaves
 - In most coniferous forests a thick carpet of mosses occur
 - Forest species are generally few and large tracts of forests can consist of only one or two species of tree(occurs in pure stands)
 - Most of the species are softwoods which mature faster
 - Most species are evergreen but a few shed their leaves eg fir, pine and larch.
 - Trees have a widely spread shallow root system for utilizing moisture from the top soil since most of the time subsoil is permafrosted
 - c) -some plants have thick leaves and bark for storing water.
 - some plants produce seeds which lie dormant for a long time and germinate when the rain falls
 - some plants shed leaves during the onset of the dry season to reduce loss of water through transpiration
 - trees in the region have long roots to tap water from the water table below
 - trees in the region are umbrella shaped to provide shade that reduces the rate of evaporation around the stem
 - . d) - Frequent outbreak of bush fires destroy grass and retard its regeneration
 - frequent drought destroy the grass and so the vegetation degenerates into a semi-desert type.
 - overgrazing of domestic and wild animals causes stunted growth of grasses
 - human activities due to increased population which encroaches grasslands
 - pests such as army worms and locusts destroy the grass, thus reducing the rate of growth
4. a) X - Veldt
Z - Stepper
- b) - Some plants have a short life-cycle this enables them to survive during the short period of wetness.
- Most are salt tolerant (Halophytes) to enable them survive in the saline soils of the deserts.
 - Most plants have succulent leaves that store water to enable them survive the long drought period.
 - Some plants have fibrous stems which they use as breathing surface.
 - Most have deep roots to reach the deeper water table.
 - Some have spines/thorns which protect them from browsing animals.
 - Some are drought resistant & deciduous which enable them shed their leaves during dry season.
5. a i) L is savanna (park savanna)
- ii) - The vegetation is tropical rainforest/ equatorial forest
- The forest consists of mixed variety of tree species

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- The trees shed their leaves at different times of the year/ ever green
 - The trees are tall with large trunks
 - The trees have broad; leaves/ drip tipped leaves
 - The trees take long to mature
 - The tree species are mainly hard woods
 - The forest trees grow close together
 - The forest has numerous lianas/ climbing plants/ epiphytes
 - Some of the trees have buttress roots
 - The forest has crowns that form three distinct canopies/ layers
 - b i) - Some plants have thick/ fleshy/ succulent leaves/ barks to enable them store water
 - Some have long roots to tap the ground water
 - Some have no leaves/ have thin/ spiky/ waxy/ needle- like leaves to reduce transpiration
 - Some plants have shiny surfaces to reflect light
 - Plant seeds remain dormant awaiting the short rains
 - Some plants have thick/ hard barks to reduce transpiration
 - Some plants wilt in the absence of moisture that have a quick recovery ability
 - Some plants have thorns to protect themselves from browsing animals
 - Most plants are stunted/ dwarf like due to the harsh conditions
 - Some plants are quick sprouting to take advantage of the short lived desert rays
 - Some plants have underground bulbs that remain dormant awaiting short the rains
 - c i) - The temperatures are too low to support vegetation
 - The surface is mainly bare rock, so no soil has formed to support vegetation
 - Water is always in a frozen state, i.e. snow, so it is not available to plants
 - ii) - Vegetation is of aesthetic value as it adds beauty to the landscape
 - Vegetation binds soil together using its roots. Its aerial parts reduce the impact of raindrops on the soil. In this way, it protects the soil against erosion
 - Plant remains decay to become humus. This improves the fertility of the soil
 - Some plants are of medicinal value to people. Their leaves, barks or roots are extracted and prepared into medicines, both traditional and modern
 - Some trees are used in the manufacture of paper while others are used for the manufacture of plywood, veneer or production of timber
 - Some plants are eaten by people e.g. bamboo shoots
6. a i) - Natural vegetation
 - Derived vegetation
 - Cultivated vegetation
- ii)- altitude✓
 - aspect✓
 - terrain and drainage
- b i) A-health and moorland
B-bamboo forest✓
C-rain forest✓
D-savanna grassland
- ii) -too cold for plants to survive covered by snow✓
-rocky surface
- c i) prairies
Pampas

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- ii) - Grazing the plenty tall and short grasses are used for grazing livestock and wild animals
- Reduces soil erosion-savanna vegetation act as soil cover hence reduces soil erosion
 - Habitat for bees and wild animals-trees in savanna are habitat for bees which provide honey and home for wild animals e.g. gazelles, giraffe
 - Some shrubs and herbs growing in the savanna are used as medicine
 - Fuel-trees provide wood fuel and fuel wood for the people in the tropics
 - Grass increases humus in the soil making areas suitable for cereal farming e.g. wheat and maize
- d) - some plants have long roots to enable them tap water far deep the ground✓✓
- some plants have shallow root system which enables them to take moisture from the soil layers
 - some plants have sunken stomata and waxy leaves to reduce rate of transportation✓✓
 - some plants have succulent stems to allow them store water✓✓
 - some plants are short lived i.e. Complete their cycle within the short wet period✓✓
 - some plants e.g. tubers and corns remain dormant during dry spell until wet period stimulate
7. a i) S- Steppes
T- Pampas
Y – Savanna
Z – Downs
- ii) - the vegetation consists of a continuous cover of grass, the grass is tufted
- The grass is interspersed/mixed with bulbous/leguminous plants/shrubs
 - The grass is nutritious. Grass is tall in moist area
 - The grass is short and tough in drier areas about 15-30cm in height
 - The grass is green in spring/brownish yellowish and straw like in summer/Grass withers in autumn and die in winter/rare grass sprouts in spring/Trees are only found along rivers/ riverine trees/ trees found in hollows where rain water collects.
 - common grass varieties are gamma graze/buffalo grass
 - In drier, areas, the short grasses form bunches separated by bare soil
 - Trees found along river courses are temperate trees like poplar, willow
8. a i) - It is a plant cover existing naturally in a place but has been interfered with by man
- ii)
- Altitude
 - Aspect
 - Soil
 - Climate/ temperature/ rainfall
 - Man's activities
- b) i) - Savanna vegetation
- Rainforest
 - Bamboo forests
 - Heath and moorland
- ii) - prairies

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- steppes
- downs
- veld
- iii) - some have thick/fleshy/succulent leaves /barks
 - some have long tap roots
 - some have no leaves /have thin/spiky/waxy/needle like leaves
 - some plants have thick/hard barks
 - some plants have thorns
 - some plants are stunted/dwarf like
 - shrubs are common
 - some plants spout during wet seasons (short time)
- 9. a) - fire-often ,large areas of forests are destroyed by accidental and sometimes intended fire. such forests takes long to recover
 - diseases causes by pests and parasites attack mainly the planted forests causing many trees to die
 - human activity /settlement /charcoal burning/logging have destroyed many forests areas of which are transformed in to farms and grasslands
 - over-exploitation leads to depletion of certain tree species such as Meru oak, camphor and Elgon teak. these trees take long to mature
 - government policy of degazetting of some forests made people free to clear many forested areas
 - prolonged droughts lead to degeneration of forests some of which take long to recover
- (b) Rain forest have closely set trees with these districts canopies.
 - This crowns and closeness of the trees form a dense shade for the lower layer which makes arvesting cumbersome.
 - Most trees are tall, and have huge trunks. This makes them cumbersome to be harvested and most of them are heavy to be transported.
- High amount of rainfall.
 - This forest lies within the equatorial climatic region where high amount of rainfall is received throughout the year.
 - This means that harvesting of trees can not take place throughout the year because of the very dump condition in the forest and general disturbance caused too much rains.
- Tree species are not found in pure stands
 - Rainforest do not have pure stand and this become a problem if one need to harvest only one species meaning moving from one part of the forest to another as one harvest.
- Majority of the species are hardwood.
- These species are hard to be cut and also heavy to be transported

12. FORESTRY

- 1. a) - High demand for hardwoods
 - Population pressure which has led to increased deforestation in creation of room for farming and settlement
 - Hardwoods take too long to mature and this doesn't match the rate of felling
- b) - Agro forestry programme
 - Public awareness and education through mass media

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- Creation of forest reserve
 - Recycling of waste papers to make newsprint
 - Enacting law governing forest conservation
 - Forest research stations to conduct research on new tree species
 - Establishment of government tea zones at the edge of forests to act as buffer zones
- c i) - New found land
- Nova Scotia
 - Prince Edward Island
 - New Brunswick
- ii) - The landscape is rugged hence does not favour agriculture
- High rainfall throughout the year favours forest growth
 - Coastal location favour establishment of ports for timber exportation
 - Constant supply of H.E.P for industrial use due to many rivers with water falls
 - Plenty/abundant water supply for processing purposes e.g. bleaching of pulp
 - Ready market in Canada and U.S.A for forest products
 - Cheap means of transport or rivers like Ottawa, Fraser e.t.c
 - Low temperatures in the highlands discourage settlement hence forestry is the main use of land
- d) - In Canada they grow naturally while in Kenya they are planted
- In Canada they cover extensive areas while in Kenya they cover small areas
 - In Canada they exist in a wide variety while in Kenya the species are limited
 - In Canada they are indigenous while in Kenya they are mainly exotic
2. (a) i) - This is a collective term used to cover a variety of land uses that combine tree growing, pasture and crop production practices on the same piece of land for the purpose of increasing or improving the output of the soil.
- ii) - Remedy to deforestation.
- Source of income.
 - Environmental benefits.
 - Aesthetic saving.
 - Labour saving especially for women who spend more time fetching firewoods.
3. a i) - Pure forest is composed of trees of two or more species
- ii)
- | Natural forest | Planted forests |
|--|---|
| <ul style="list-style-type: none"> - consists of hardwood - mixed the species - random occurrence with different canopies - take long time to mature | <ul style="list-style-type: none"> - mainly softwoods - - pure woods - Planed in rows and of same height - Take 15-30years mature |
- b) - High rainfall in Kenya highlands favours tree growth
- Kenya Highlands experiences cool temperatures ideal for coniferous forests
 - High market for wood products and timber encourages forestry
 - Forestry areas are rugged and steep, discouraging settlement and agriculture
- c i) - Tree planting /re-a forestation
- People are required to seek permits if they have to cut trees
 - Forests reserves have been set aside to conserve indigenous forests

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- Forestry department carry out research to produce and distribute seedling
 - People are educated through mass media importance of the conservation
 - People are being encouraged to use of alternative source of energy/saving jikos
- ii) - Presence of cool to cold climate enhances growth
- High rainfall promotes growth of trees
 - Steep and rugged landscape discourages agriculture
 - High local and international market for forest products
 - Low population density in Canada provides more land for forestry
4. a) Agro-forestry involves cultivation of both crops and trees on the same piece of land
- . b) - Provides source of firewood and charcoal
- Source of income to farmers after selling tree products like fruits
 - Trees act as windbreakers
 - Trees create micro-climate within the farm
 - Some trees are of medicinal value
 - Trees leaves decomposes to form fertile soils
- c i) - Tree growth is limited to altitudes below 3500 because beyond this level
- the temperatures are too low for tree growth
 - Lower level of tropical mountains have rainforests, which give way to coniferous forests and then to bamboo thickets, with increase in altitude
 - In highland areas where the slope is too steep trees do not grow because the soil is thin due to heavy erosion
- c ii) - Windward slopes of mountains are generally wetter than the leeward slopes thus more forest
- The forest zone starts at a much lower altitude on the windward than on the leeward slopes
 - In the northern Hemisphere, southwards facing slopes are warmer with luxuriant forests, while North facing slopes are in the sun's shadow, cooler and with less vegetation
- . d) - People are being encouraged to plant trees and food crops in the same farms
- Regions which previously were devoid of trees are being planted with trees
 - Tree farming is practiced in many parts with the aim of raising trees for future use
 - Mature trees felled are replaced immediately
 - The reduction of wastage e.g. the use of waste paper to produce newsprint
 - People are required to seek permits if they have to cut trees. This reduces the rate of tree felling
 - The forest reserves are patrolled by guards to ensure that fires are reported promptly and also ensure unlicensed people do not cut down trees
 - Forest reserves have been set aside to conserve indigenous species
 - Forestry department carries out research to produce and distribute seedlings to ensure the extension of forests
 - People are encouraged to use alternative sources of energy or energy saving jikos
 - People are educated through mass media on the importance of trees
- e) - Desertification
- Soil erosion/ degradation
 - Extinction of some tree species
 - Loss of plants with medicinal value

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- Loss of aesthetic value
 - Decrease in wild animals
 - Decline in employment for those in forestry related areas
5. a i) Indigenous forest are those trees which have been traditionally growing in the country while exotic forest consists of trees that have been imported into the country.
- ii) - Natural forest grows in the wild while planted forests are tended by man.
- Planted forests are usually softwood while natural forest are mainly hardwood.
 - Natural forest takes a longer time to mature as compared to planted forest.
 - In natural forest, trees are scattered while planted forest are found in rows/lines.
 - Natural forest have mixed species of tree while planted forest have pure stand in a given area.
- b) - Temperature ($10^{\circ}\text{C} - 35^{\circ}\text{C}$)
- Tropical rain forest domestic high temperature areas ($21^{\circ} - 35^{\circ}\text{C}$)
 - Human activities
 - Altitude
 - Rainfall.
 - Soils
- c) - Afforestation /Reafforestation
- Legislation
 - Research.
 - Education
 - Alternative sources of energy use.
6. a i) Is the science of cultivation development and managing forests
- ii)- availability of deep fertile volcanic soils in the highlands has led to the growth of thick forests
- the high reliable rainfall throughout the year in the highland has promoted the growth of thick forests
 - their location in the low latitude region which experience high temperature has led to faster growth and early maturity for most of the trees
 - government's efforts to protect destruction of the natural forests and protect them from human settlement
- b) - there is a wide variety of tree species in a given area. This makes exploitation of valuable species difficult
- the valuable trees are scattered not in pure stands. this makes exploitation difficult and expensive.
 - The buttress roots make the falling cumbersome
 - The dense undergrowth/thick forest any quick generation of plants hinder accessibility
 - Humid climate makes working condition unfavourable
 - The forests form a habitat for dangerous wild animals which make the exploitation insecure
 - The hot humid conditions climate encourage pests and disease which are harmful to man and trees
 - The heavy rainfall throughout the year results in muddy roads-which makes exploitation difficult
 - Inadequate capital limits the use of modern techniques in the exploitation of forests
 - River transport is hindered by waterfalls and rapids. this makes transportation expensive
- c i) - carrying out public campaigns on the value of trees forests through mass media

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- e.g. Radio
- carrying out research on suitability of soils and the effects of pests diseases on forests
 - Established of training institution dealing with forestry e.g. lodiani
 - reduction of wastage in the forestry industry by recycling waste to produce other products
 - employment of forests guard and officers to guard against destruction of forests
 - creation of forest measure with the aim of protecting indigenous tree
- c ii) - frequent outbid of bush forest destroying forests
- increasing human population encroaching in to the forests
 - pests and diseases destroying some species of trees
 - frequent drought experience in the country destroying the vegetation
 - wild animals and domestic animals destroying vegetation causing stunted growth
7. a i) Forest is a continuous growth of trees and under growths covering a large tract of land while forestry is the science of developing and managing forests including cultivating them
- ii a) - Mainly temperature and precipitation
- Proper growth of tree can only take place where there is heavy precipitation throughout the year.
 - Different plants require different amount of warmth e.g. in tropical regions where rainfall is very high, big trees are common very low temperature discourage tree growth.
- b) - Human beings have destroyed any natural forests to create room for subsistence agriculture and settlement. Deforestation has been rampant while collecting wood fuel.
- At the same time human being are conserving and maintaining trees in other area
- c) - Forests do not grow on steep mountains where the surface is bare rock, the soils are thin and temperatures are low
- The windward slopes of high mountains like Kilimanjaro usually receive higher rainfall and are therefore forested. the lee ward slopes are covered with scrub vegetation
8. (a) - Research is being carried out on soil requirements for different species of trees which enables foresters to plant trees in suitable areas
- Public campaigns on the importance of forests are always being conducted through mass media
 - Alternative sources of energy are increasingly being encouraged to minimize over-dependence on wood fuel e.g. sun, wind, biomass, water, oil, kerosene, and natural gas
 - Introduction and promotion of energy saving stores to reduce the amount of wood fuel being used
 - Timber merchants and those involved in logging are being enlightened on how to improve their cutting practices through rational and selective felling and replanting to replace the ones cut.
 - Improving the infrastructure within the forested area by improving roads and providing roads and providing machinery, saw mills, tractors and seed beds
 - Control of pests and diseases which attacks trees

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- Establish of training and research institutions which specifically deal with forestry e.g. londiani forestry training college
- Government has enacted laws that enable effective management of forests e.g. forest bill 2000
- Creation of forest reserves to protect the indigenous trees and other plant species from extinction
- Posts for forest guards have been set up within the forest to protect against illegal tree felling

. c)

i)species Kenya There are both exotic and indigenous softwoods ii)problems large tracts of forests especially exotic trees are affected by pests and diseases e.g. Aphids iii)marketing most of the wood products are sold locally but some are exported to Arabian countries, USA and Europe	Canada The main species are of coniferous type (2mks) Large tracts of forests are destroyed by fires and diseases(2mks) Most of the wood products have ready markets in Canada, USA, Britain and Europe
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di) - Pines

- Cypress
- Eucalyptus

ii) - Note taking

- Field sketching
- Photographing /filming

9. a) It is the growing of trees together with crops on the same piece of land at the same time ✓

b)- Meru oak✓

- Elgon teak✓
- Mvule✓
- Elgon olive✓
- Camphor✓
- Mangrove✓

. c) - African pencil ✓

- Cedar ✓

- Podo ✓

13. EXTERNAL LAND FORMING PROCESSES

1. a) - River Nzoia✓

- River Nyando✓

b)- Heavy rainfall from the highlands ✓

- When a river contains excess water that it cannot hold in its channel ✓
- Cultivation along river banks ✓

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- Deforestation in the surrounding areas ✓
- 2. (a) i) - barchans
 - transverse/wake dunes
- ii) - they lie parallel to each other
 - they rise up to 100m/high and their length can exceed 100km/long
 - they lie parallel to the direction of the prevailing wind
 - they have a sharp knife like crest
- iii) - availability of sand
 - strong winds blowing in one direction for a long time
 - presence of obstacles e.g. rocks or vegetation
 - occasional rainfall hardening the ground causing deposition
 - presence of ground water reaching the surface
 - human activities like mining, heaping materials on the surface that acts as anchors for sand deposits
- b) - there is existence of a rock out crop of alternative layers of hard and soft rock/heterogeneous
 - the rock lies on the path of a wind carrying weathered materials
 - through abrasion, the softer parts of the rock are eroded faster than the hard parts
 - wind is more effective near the ground surface
 - the irregular shaped feature is a rock pedestal
- c) - desert features like zeugen dunes etc attract tourists
 - deflection hollow as/oasis contains water used for irrigation and other domestic uses
 - loess is deposited in water areas is fertile and used for farming
 - dwelling are curved out of loess which provide warm and cool home stead during winter and summer respectively
 - desert land forms are ideal for military training testing of weapons and space craft experiments
 - the seasonal streams can be dammed to supply water for domestic use
 - desert features provide suitable scenery for film making
- 3. a i) - Kenya experience high temperature under which ice-sheets cannot form
 - Most parts of Kenya have low attitudes
 - Kenya is found at low latitudes
- ii) - gradient of the land-ice moves faster when the slope is steep
 - temperatures-higher temperatures result in to thawing leading to faster movement of ice
 - nature of the slope-when the surface on which ice is moving is rough, it causes friction lowering the speed of the movement of ice
 - size/thickness of glacier-large masses of ice exerts pressure which leads to melting of ice underneath. this increases the speed of ice movement.
- 3. b i) - two adjacent hallows on a mountain side
 - the two hallows are filled with ice
 - the ice erode the side through plucking and deepens the hollow through abrasion
 - through erosion, the back walls of the hollows slowly recede
 - eventually the hollows(cirques)are separated by a knife-edge ridge
 - the ridge is called an arête
- b ii) - a large block of rock stands on the path of on-coming glacier

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- the ice plucks off /erode rock fragments from the upper side of the block
 - as the ice moves round and over the resistance rock, it carries the eroded materials to the lee-ward side
 - the lee side does not experience erosion
 - deposits materials on the lee-ward and together with weak rocks forms
 - flat corridors separate them from each other
 - they occur in groups
 - the width of the corridors vary from 25m to 50m
 - where the wind blows from the same direction for long, dunes become gentle on the wind ward and steep on the lee ward
 - c) - glacial till provides fertile soils for arable farming
 - ice sheets in their scouring effect reduces the land surface and depth to expose mineral seams which become easy to extract
 - out wash plains comprise of sand and gravel which are used as materials for building and construction
 - lakes formed through glaciations can be exploited for various uses such as fishing, transportation or as tourist
 - glacial features are tourist attractions
 - glacial lowlands are generally flat due to erosion and deposition and these are ideal
4. a)- 21st March - 23rd September
- ii)- because the earth is tilted on its axis
- because of the movement of the sun within the tropics
 - because of the revolution of the earth
- b) - it causes the occurrence of day and night
- it causes the deflection of winds and ocean currents
 - it causes the rising and falling of ocean tides
 - it causes the variation in time at different longitudes
 - it causes differences in the atmospheric pressure on the earth's surfaces
5. a)- An accumulation of water in a wide hollow or depression
- b) - by erosion
- by earth's movements
 - by vulcanicity/volcanic activity
 - by deposition
 - by human activity
 - falling meteorites
 - mass movements e.g. land slides
 - weathering by solution
6. a i) - Desertification is the process through which marginal lands are degraded through climatic variations and human activities/ or encroachment of arid conditions into formerly productive areas
- ii)- Wind deflation is the removal of loose unconsolidated materials by a lifting and rolling action of wind
- Wind abrasion is the process whereby wind picks loose materials and uses it to scratch and grind the earth surface

I. Wind transportation through saltation: larger fragments of rock are lifted from the

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ground in a series of hops and jumps within the wind currents

II. The fine dust are lifted clear off the ground and eventually are blown away by wind currents

III. Heavy materials are dragged along the ground by wind currents

b i) - Zeugens are formed in desert areas where alternating layers of hard and soft rocks occur

- The top layer of hard rock is jointed
- Weathering opens up the joints
- Wind abrasion erodes the joints widening and deepening them to reach the soft layer of rocks
- Deflation blows away the loose broken materials
- Furrows are formed and gradually widened as abrasion continues on the soft rock
- The hard rock forms the ridges separately the furrows

ii) - A pre- existing depression is formed through faulting

- Wind eddies remove unconsolidated materials through deflation
- As deflation continues, the depression is deepened and enlarged resulting into a wider depression
- The process of deflation is aided by weathering and abrasion
- With continued deflation, the level of water table is reached
- Water comes out of the ground and collects into the depression to form an oasis

7. a) - Plastic flowage

- Basal slip
- Extrusion flow

b)

Young river valley	Glaciated valley
Has interlocking spur	Has inundated spur
Has gently sloping cliff	Has steep rock cliffs
Has a V- shaped	Has a U shaped
Follows a winding course	Flat on the floor

8. a) X – Swash Y – Back wash

- b))
- Hydraulic action
 - Corrosion
 - Solution
 - Attrition

9.

- Rainwater
- Melt water
- Lake and sea water
- Magmatic water

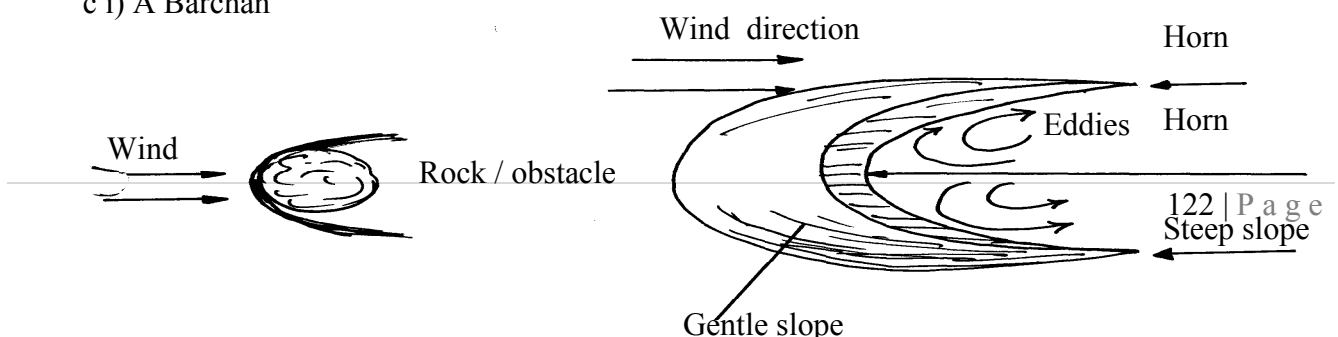
10. a i) - Chemical weathering- is the breakdown or disintegration of rocks as a result of chemical reaction or changes that cause decay of rocks

ii) - Chemical composition of rocks

- Presence of moisture/ water
- Presence of gases
- Temperature changes
- Rock permeability
- Vegetative matter

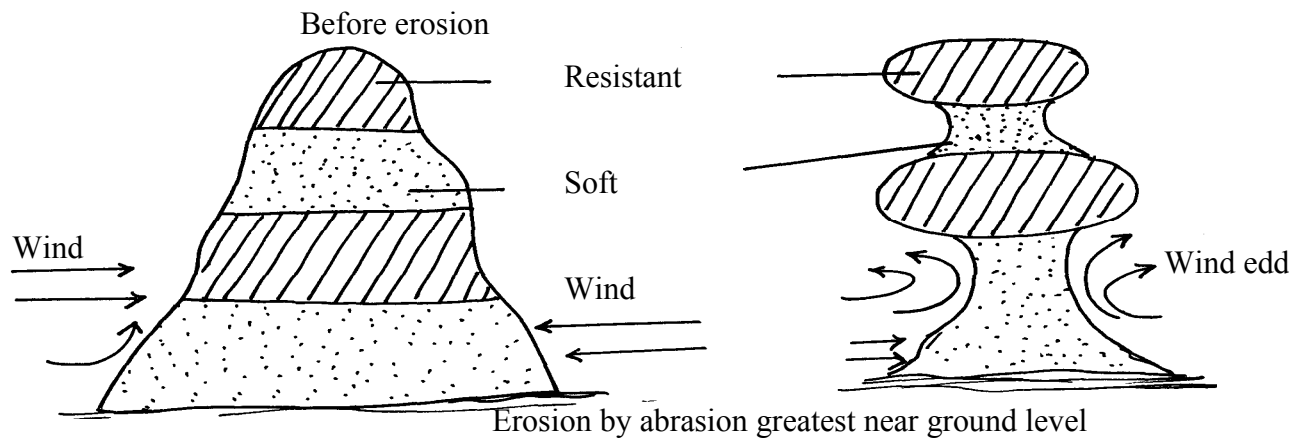
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- b i) - Water in air combines with carbon to form a weak carbonic acid
- Carbonic acid reacts with calcium carbonate to form soluble calcium bicarbonate
 - Solution is washed away causing weathering to take place
- ii) - Water enters the rocks
- Water breaks down into ions which combine with minerals
 - Rocks break down to form completely new compounds e.g. fluorspar with clay
- c) - Produces clay that is vital in brick making
- Weakens rocks making them easier to exploit i.e. mining
 - Important in soil formation
 - Weathered rocks provide scenery for tourist attraction
 - May break down large rocks hence leveling the landscape for settlement
11. a) - Spring tides
- Neap tides
 - Perigan tides
 - Apogean tides
- b) - Change direction of flow
- May force currents to flow along the coastline of the land mass
 - May split currents into two parts and flow in different directions
- c) - Rainfall that infiltrate into the ground
- Melt water from the thawing ice
 - Manmatic water that are trapped in the rock
 - Lakes and seas/ oceans that sink into the ground
- d) - When permeable rock overlie on top of an aquifer
- When well jointed rocks form a hilly country absorb water via joints which later spring out when water table meet surface
 - When dyke cuts across a layer of permeable rock
 - When a limestone or chalk locupement is under long impermeable layers
 - When urgently sloping layer of permeable rock alteranateS with layers of impermeable rock
- e) - Some plants have long roots to reach water table
- Some have small leaves to reduce transpiration
 - Some have thick stem/ barks to store water
 - Some are leaflets to minimize water loss
 - Some have leathery leaves to reduce water loss
12. a) Hamada is rocky desert while an erg is sandry desert
- b) - Deflation – wind remove dry and unconsolidated materials like dust and sand by lifting
- Abrasion – wind use materials like sand as erosive tool, where particles are hurled n rock surface causing undercutting
 - Attrition – materials carried by wind rub against each other and against rocks producing rounded land grains
- c i) A Barchan

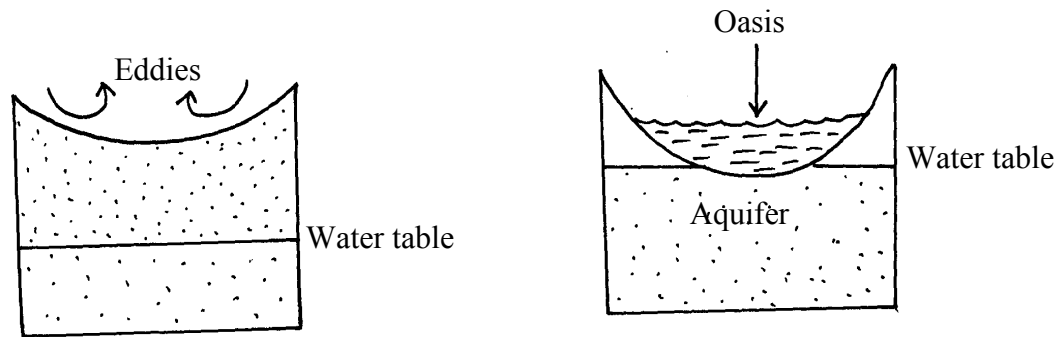


- An obstacle on path of wind e.g. rock/ bush block wind causing deposition
- Materials continue to built until windward side is smooth and leeward side steep forming a moon shaped feature called barchan

c ii)



c iii)



- Pre- existing depression/ localized fault attacked by wind eddies
 - Depression is excavated by removal of unconsolidated materials/ wind erosion by deflation
 - When water bearing rock is reached/ aquifer water oozes out to collect in depression and an oasis is formed
- d) - Alluvial fan
- Bajada
 - Salinas

13. a ii) - Presence of loose unconsolidated gravel/ stones/ soil particles that can easily

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picked by wind

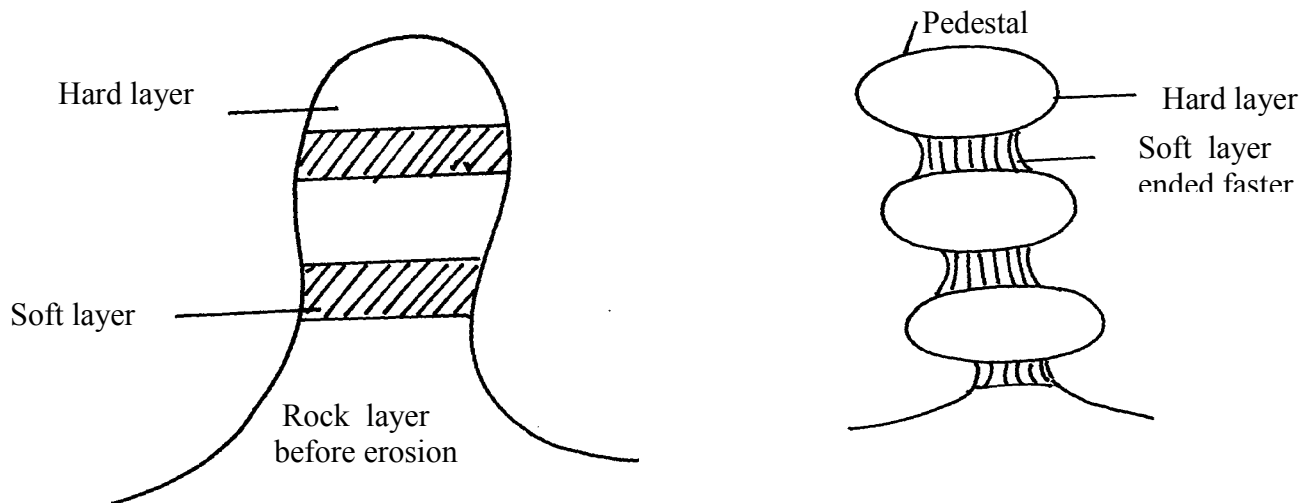
- Presence of little or no vegetation on the surface of desert
- Presence of strong prevailing winds in deserts

b) - Deflation – it is the blowing away of any dry unconsolidated materials like dust and fine particles by rolling them on the ground and lifting them up in the air

- Abrasion – materials carried by wind such as sand scratch and grind rocks and ground surface
- Attrition – particles carried by wind hit against each other and rocks and reduce in size in the process

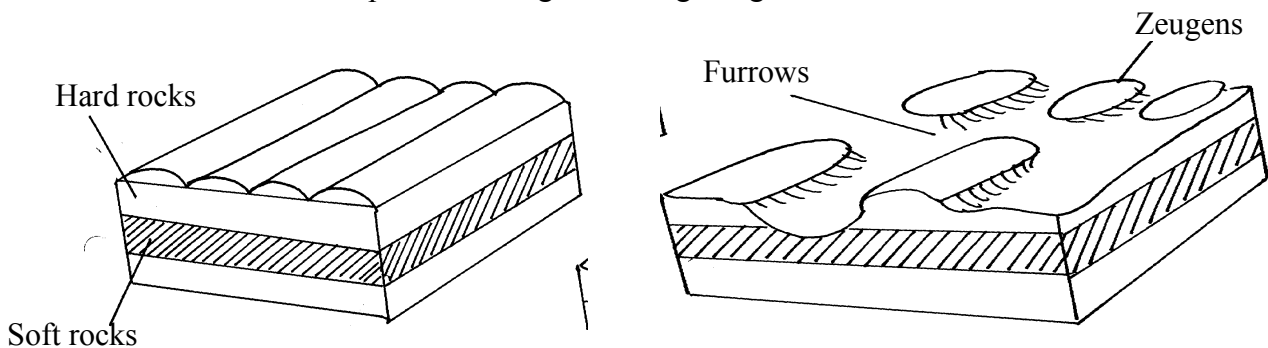
c i) - Rock outcrops with alternate layers of hard and soft layers project above the surface

- The softer rocks are eroded by abrasion faster than the hard rock. They are worn out more rapidly than resistant ones. This results in an irregular rock with protruding layers that alternate with hollows called pedestal



c ii) - Formed where hard and soft rocks lie horizontally

- Weathering attacks the soft layer leading to the formation of cracks
- Wind abrasion deepens the cracks deepening them forming furrows
- These furrows separate the ridges forming Zeugens



d) - Desert land forms e.g. Zeugeus, dunes yardang attract tourists who bring foreign exchange

- Oases provide water for domestic use/ irrigation/ livestock
- Oases attract settlements
- Loess provide fertile land for agriculture use

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- Rocky and sand surfaces can hinder communication and transportation
- 14.
 - a) - An ocean is a large/ extensive body of saline/ salty water occupying large basins between continents while a sea is a large body of saline/ salty water boarding a continent
 - b - Volcanic materials from the ocean floor add salts to the water
 - Some salt is dissolved by sea water from the underlying rocks it is in contact with
 - Some salt is added by rivers that have dissolved it from the land
 - Most of the salt is thought to have been present when water bodies were formed
 - c)
 - Ocean deeps
 - Continental shelf/ continental slope
 - Ocean ridges
 - Deep sea plains
 - Islands
- 15.
 - a i) A sea is a mass of saline water occupying depressions at the margins of continents whereas an ocean is a mass of saline water occupying depressions in between continents
 - ii) - As the coastline retreats due to marine erosion resistant rocks are isolated as islands.
 - Deposition of materials across bays, river mouths and lagoons build up barriers which project above the water level as islands.
 - Volcanic eruptions within the sea/ocean build up islands when materials pile up above the water level.
 - Accumulation of coral into reefs form coral islands
 - When upland coasts submerge valleys are drowned leaving ridges /hills above the water level as islands.
 - b i) - Earth movements led to crustal downwarping.
 - A shallow depression was created
 - The areas around the depression underwent uplifting
 - The uplifting reversed the direction of rivers such as river Kagera
 - Water from the rivers and from rain eventually filled the depression forming a lake
 - ii) - Evaporation from the lake leads to high relative humidity in the area
 - Evaporation from the lake increases moisture in the atmosphere leading to increased/high rainfall in the area
 - Evaporation from the lake leads to convectional rainfall
 - The lake encourages formation of lake breezes which have a cooling effect on the areas at the lake shores
 - Regular land and lake breezes modify the temperatures of surrounding areas keeping the diurnal range low
 - The breezes cause reversal of local winds/diversion of prevailing winds such as the Congo air mass
 - c i) Aridity is the state of land being deficient of moisture leading to scarcity/lack of vegetation. while desertification is the slow but steady encroachment of desert-like conditions onto formerly productive agricultural land
 - ii) - Continentality/distance from the sea;- Onshore winds drop most of their moisture along the coast reaching the interior as dry wind leading to aridity and desertification

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- Relief barriers/Rain shadow effect; - Rain bearing winds drop a lot of moisture on the windward side of mountains. They thus reach the leeward side of with less moisture as dry winds causing aridity and desertification
 - Cold ocean currents; where on shore winds cross over cold ocean currents, the moisture content of the winds is lowered by the chilling effect of cold currents, the winds thus drop moisture over the sea reaching the land as dry winds that cause aridity and desertification.
 - High pressure systems;- Areas with stable, high pressure systems with low humidity and descending air/diverging winds experience aridity since moist air has to rise for condensation to take place.
 - Dry wind system;- Hot and dry winds from land like the harmattan constantly blowing over some regions cause a drying effect the land resulting in development of arid conditions
 - High temperature:- Areas constantly experiencing high temperature have high rates of evaporation leading to arid conditions
 - Insufficient rainfall:- Areas that constantly receive rainfall less than 250mm annually have scanty vegetation, immature soils making the land barren
16. a i) Chemical weathering is the breakdown/decomposition insitu of rocks after reaction with other minerals, water or air
 ii) - Grikes and clints
- swallow –holes
 - Dry valleys
 - Dolines
 - urala
 - Polje
 - Gorges
- (b) i) Exfoliation dome
- ii) - In hot climate, where rocks high day temperatures cause the surface of the rocks to expand
- Low night temperatures cause the surface of the rocks to contract
 - The inner parts remain cool as the expansion and contraction only takes place on the rock surface because rocks are poor conductors of heat.
 - As the process is repeated over a long time a curved sheet of the rock/rock shall from the upper part break away from the upper part break away peeling off like onion peels.
 - *This leaves a round rock mass called an exfoliation dome
17. a) A river system refers to a river and all its tributaries while a river regime is the seasonal fluctuation/variation in the volume of water in a river
- b i) Trellis/ Trellised/ Rectilinear pattern
 b ii) P – Obsequent river Q- Secondary consequent river
18. a) - Erratics’
 - Boulder train
 - Till
 - Drumlins

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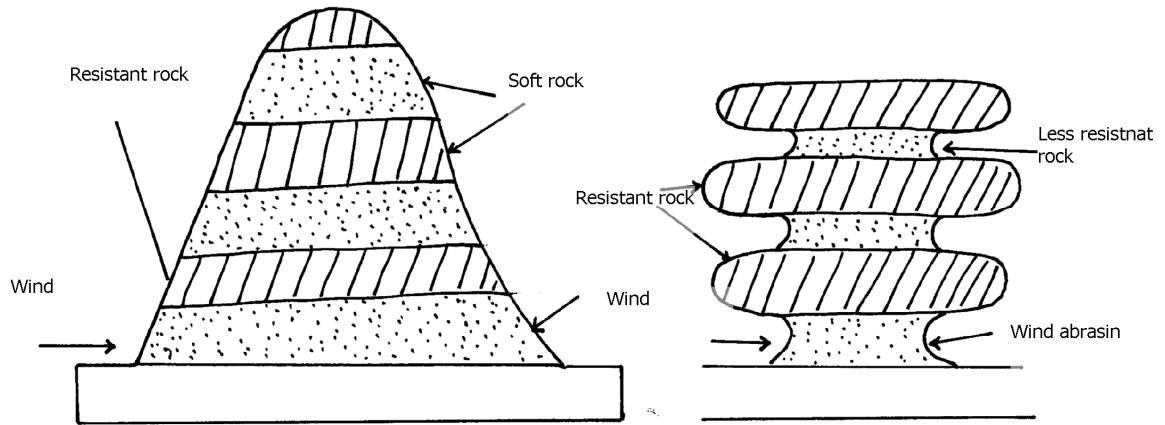
- Eskers
 - Karme
 - b) - Glacier in a low land area comes across a resistant rock
 - As the glacier passes over the resistant rock it erodes its upstream side by abrasion smoothing it
 - The downstream side is eroded by abrasion giving it a rugged nature
 - When the ice retreats a rock outcrop with a gentle smooth upstream side and a rugged steep downstream side is exposed called a roche moutonnee
19. a) i) U – Zone of permanent saturation /phreatic zone
 V- Zone of intermittent /seasonal saturation
- ii) W - Spring /river source
 Underground water
- An impermeable rock layer sandwiching an aquifer leads to accumulation of underground water
20. a i) -Deflation
 -Abrasion
- ii) – Suspension -fine particles are lifted, cleared off the ground
 saltation -larger fragments of sand particles are lifted from the ground by eddy action
 -The sand particles are moved in a series of hops/jumps within the wind current surface creep✓/traction
 -The heavy materials /small stones/pebbles are tracked along the ground.
 -these materials are rolled for short distance in a process called surface creep
- b i) -zeugen are formed in desert areas where alternating horizontal layers of hard and soft rock occur, -the top layer of hard rock is jointed/has cracks✓
 -wind abrasion opens up the joints deepening and widening them to reach the soft layer of rocks✓
 -deflation blows away the loose broken materials✓
 -furrows formed and gradually widen as abrasion continues into the soft rocks✓
 -the hard rock forms ridges separating furrows✓
 -this process creates a ridge and furrows landscape called Zeugen
- ii) -an outcrop rock eg eroded by wind from all sides
 -the rock has alternating horizontal hard and soft rocks
 -hard layers are eroded at a lower rate while soft layer is eroded faster
 -continual erosion through abrasion makes hard layer left outstanding
 -near the ground more erosion narrows the base
 -the protruding layers are the bands of resistance rock while the hollows are the areas where the soft layer was positioned
- c i) Hamada
- c ii) desert regions have very fertile alluvial soils which are used for agriculture✓✓
 -desert features such as rock pedestals yardangs and sand dunes are tourist attractions✓✓
 -salt flats are economically used for salt production especially in north Africa✓✓
 -extensive and bare desert are used as testing grounds for car and jets engines✓✓
 -desert landscape provides good site for testing military weapons e.g. nuclear bombs

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21. a) a lake is a large mass of water in a depressions
b i) -crater lakes
-lava dammed lakes
b ii) -lake Victoria modify the climate of the lake region due to convection rainfall
- Lake Victoria brings cooling effect on the surrounding area due to Lake Breeze
-convectional rainfall experienced around the lake is occasionally accompanied by thunderstorms
-land breeze and sea breeze are due to low and high pressure system✓
c i) -some lakes lacks outlets
-rock over which lake water is in contact may contain mineral salts
-some water river to pour into them
-surface run-off and rivers may dissolve a lot of salt from rocks on while they flow
-excessive exploration in areas where temperature are high
c ii) - Naivasha
- Baringo
- Turkana
. d) -some lakes provide water for both domestics and industrial uses✓✓
-some lakes forms tourist attraction site and thus earning the country foreign exchange✓✓
-some lakes provide building materials eg sand✓✓
-some lakes provide water for irrigation hence enhancing agriculture✓✓
-some lakes provide transport routes✓✓
-some lakes are sources of minerals eg magadi
-some lakes provide water for the generation of H.E.P✓✓
-some lakes are sources of fish (food) which is a source of protein for man
22. a) - spring tides
-neap tides
- Perigian tides
- Apogean tides
b) Ocean is a large extensive body of saline water occupying a basin between continents whereas, sea is a large body of saline water on margins of continents
23. a i) -acidity is the state of land being deficient in moisture leading to scanty vegetation which desertification is the slow but steady encroachment of desert like conditions into formerly productive land
ii) -abrasion-winds uses the transported materials as tools to grind/scrap the rock surfaces
-deflation-wind blows away unconsolidated materials by rolling and lifting them up the air.
-attrition-the materials carries by the wind rub against each other, breaking and becoming Smaller
b i) -a rock pillar of with differing degree of hardness lies in the path of moving wind
-the rock is acted by abrasion and weathering.
-the soft layer is worm out more rapidly than the resistant ones
-an irregular mass of rock with protruding layers is formed

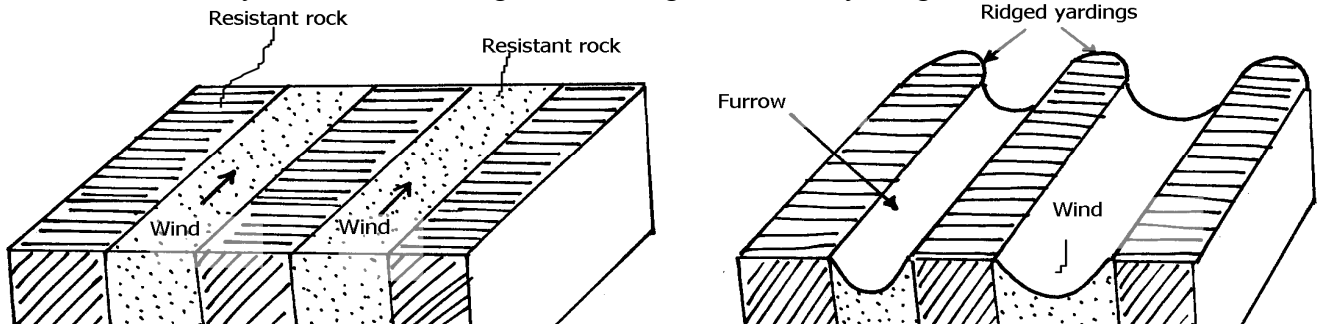
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-the base of the rock pillar is thinner because wind abrasion is more intense at lower level.



ii) -Massive rocks consisting of alternating layers of hard and soft rocks lie vertically to each other.

- these layers lie parallel to the direction of prevailing winds
- wind abrasion acts directly on the soft rocks
- wind deflation removes the worn-out parties
- large furrows between the hard layers and formed
- The hard layers are left standing as small ridges known as yardages



- c) -features like dunes, yardages are tourist attractions
- extensive and bare desert surfaces are used as testing grounds for military weapons
- salty flats are used for salt production
- the loess soils are fertile alumina deposits used for agriculture
- the loess begins of china have caved dwellings in the rocks which warmer during winter

24. a) -a difference in the resistance of the rock into which the river cuts

- a river may descend over a sharp edge of a plateau
- a river may descent over s fault scarp
- a river descends a cliff into the sea.
- a river descends a lava barrier

-a river descends a hanging valley into a flaccid trough

b i) -at the mature /old stage of a river, The river channel is wider and shallow

- the river flows sluggishly at low velocity due to due gradient.
- heavy load is deposited on the river bed
- the alluvial deposit grow into island/shoals-the river is blocked and sub-divided into small channels called distributaries

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- the water joins again later.
- the several channels intertwining each other forms river braids
- the small islands in between them are called aits /eyots /isles
- ii) -knick points
 - river terraces
 - incised meanders
 - abandoned meanders/meanders score
 - rejuvenation gorges
- c) -two rivers adjacent to each other share a common divide
 - one river is more powerful than the other due to high volume of water from a pirate river.
 - the pirate river erodes vertically and laterally faster than the weaker river.
 - the valley of the pirate river becomes deeper and wider so as the river flows at a lower level than the weaker river.
 - the stronger river extends its valley backwards by head ward erosion
 - eventually the pirate river joins the valley of the weaker river.
 - the head wards of the weaker river are diverted into the stronger river hence a river capture occurs
- d) -has underground drainage
 - rivers from its surrounding have fresh water which dilutes the salts
- 25.
 - Due to pressure changes/dynamic metamorphism.
 - Due to intense heat/thermal metamorphism.
 - Due to a combination of heat and pressure/thermal-dynamic metamorphism
- 26.
 - a i) - Deflation.
 - Abrasion.
 - Attrition
 - ii) - Suspension. The fine dust and particles are lifted off the ground. Eventually they are blown away by wind currents.
 - Saltation: Large fragments / sand particles are lifted from the ground by eddy currents. They are moved in a series of hops/jumps within the wind currents. The particles are moved along the ground surface through leaping or bouncing.
 - Surface creep/Traction: The large materials / small stones/pebbles are dragged along the ground by wind currents. These materials are rolled along to desert due to their weight
- b i) - A pre existing depression formed through faulting or otherwise is exposed to wind erosion.
 - Wind eddies remove unconsolidated materials through deflation.
 - As deflation continues, the depression is deepened and enlarged.
 - The process of deflation is aided by weathering and abrasion.
 - With continued deflation, the level of the water table is reached.
 - Water oozes out of the ground and collects into the depression to form an oasis.
- b ii) - They are formed in desert areas where alternative horizontal layers of hard rock & soft rocks occur.
 - The top layer of hard rock is jointed / has cracks weathering opens up the joints.
 - Wind abrasion erodes the joints deepening and widening them to reach the soft layer of rocks.
 - Deflation blows away the loose broken materials.
 - Furrows are formed and gradually widens as abrasion continues into the soft rock.

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- The hard/resistant rock forms ridges separating the furrows
 - c) - The loess regions have very fertile soils which are used for cultivation thus giving food to those living in that region.
 - Desert land forms like rock pedestals, yardangs and sand dunes are tourist attraction by earning foreign exchange in a region.
 - Extensive and bare desert surfaces are used for testing grounds for car and sit engine/they provide good site for testing military weapons.
 - Some deflat on hollows provide oasis which provide water for nomadic
27. a i) - Plastic flowage.
- Basal slop
 - Extrusion flow
- ii) - Step slopes/gradient.
- Summer season/high temperature.
 - Less fraction
 - Presence of thick and heavy glacier.
- b) - Ice –eroded plains
- Depression.
 - Roche mountonee
 - Crag and tail
- c i) - Large ice sheets halt their movements on gently sloping low lands. They melt at the terminus (snow line)
- The melt water from the surface and sub glacial parts of ice sheet flow beyond the terminus carrying along fine materials.
 - The melt water deposits fine materials as it flows over gentle land.
 - Pre-excusting valleys and depressions are eventually burried by these fluvio glacial deposits.
 - When ice sheet retreats it leaves behind an undulating plain of unconsolidated clay, silt, sand and gravel which is called an outwash plain
- ii) - Glacial erosion widens a valley.
- When glacier reaches its terminus it begin to melt.
 - Ice remains stagnant at the point of melting for several years.
 - Moraine is deposited across widened valley to form a ridge of terminal moraine.
 - Glacier begins to retreat towards snow field as it melts.
 - The melt water accumulates behind the ridge of terminal moraine to form a moraine dammed lake.
- d) - The warm glaciated valleys are suitable for livestock farming.
- Glacial upland areas have magnificent features like pyramid peak, an arête which encourage tourism and recreation.
 - Water falls which form at hanging valleys are exploited for generation of hydro electric power.
 - Some U-shaped valley form natural route ways and are suitable for settlement and agriculture.
 - Some fiords form deep, well, sheltered natural harbour and good fishing grounds.
 - Glaciated mountains discourage human settlement, hence growth of forests and lumbering practiced.
28. a) Mechanical weathering is the physical break up of rock material without any

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alterations in its chemical composition while chemical weathering involves changes in the chemical composition of minerals making up the rock. It's the actual decay or decomposition of rocks.

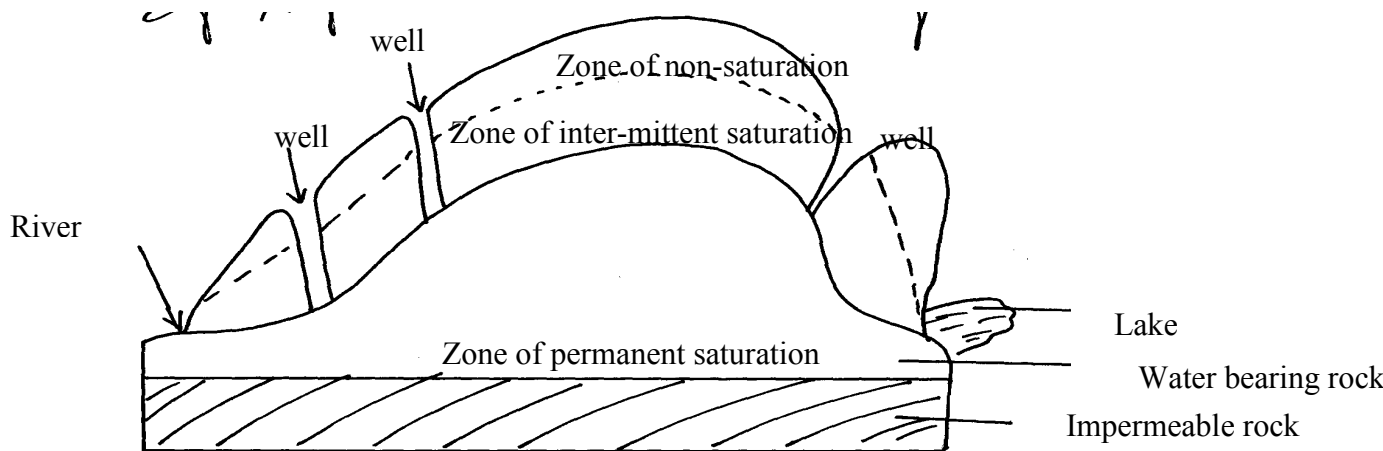
b)) - Climate.

- Topography
- Nature of the rocks.
- Plants and animals.
- Time factor.

29. a)- A spring is a point or a place where underground water flows out naturally onto the land. While a well is a dug in the ground, often with the purpose of getting water

b)- Zone of non-saturation– lies nearest to the surface. Water passes through this zone as it makes its way downwards.

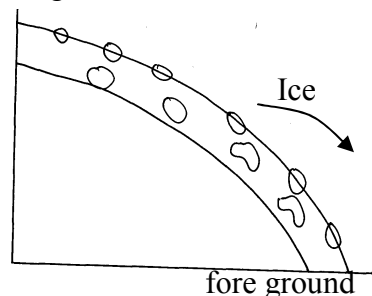
- Zone of intermittent saturation – contains water only in wet seasons or after heavy rain.
- Zone of permanent saturation – contains water both wet and dry seasons.



- c) - The aquifer must be sandwiched between impermeable rocks so that it can retain water.
 - Aquifer must outcrop in a region which is a source of water e.g Rainy area/beneath a lake.
 - Aquifer must be dip from a region of water intake and the rock layers must form a broad syncline/basin.
 - Mouth of well must be lower than the intake area. This allows water to be forced to the surface by pressure with no need of pumping it.
- d) - Acts as sources of rivers.
 - Acts as source of water- wells, springs, boreholes, oases which provide water used in homes and industries.
 - Used for irrigating agricultural land e.g Sahara desert, where dates grow near oases, Taveta- cotton and bananas are grown using water from springs.
 - Settlement – in dry areas, people tend to settle near the springs due to the availability of fresh water.
 - Provision of hot water- water from hot springs may be tapped and pumped into houses through pipes to heat up houses during cold seasons e.g Iceland.
 - Tourist attraction – Hot springs e.g Olkaria, Lake Bogoria etc, valuable mineral salt are deposited and people exploit them for economic gain.

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- e i) Is any rugged landscape whose surface rocks are limestone or dolomite and which has been acted on by carbonation and solution by rain and river water to produce features typical of limestone surfaces.
- e ii) - Rocks should be hard and well jointed.
- Hot and humid climate.
 - The surface rock and rock beneath should be thick limestone/dolomite/chalk.
 - The water –table in the limestone rocks should be deep below the surface.
- e iii) - Tourist attraction
- Discourage settlement
 - Limestone blocks are used for building houses.
 - Provide a raw material for cement manufacturing
30. a i) a. Elbow of capture
b. Wind cap
c. Misfit/small valley.
- ii) - Change in the base level.
- Drop in sea level.
 - Regional or local uplift.
 - Unequal regional subsidence of the land.
 - Increase in the river discharge.
31. a i)- Denudation refers to all external land forming process such as weathering, glacial action of water, wind and wave action which modifies the original landscape. While mass wasting is the down slope movement of weathered materials under influence of gravity
- ii) - High temperature accompanied by moisture found in the area accelerates chemical reaction thus speeding decaying of rocks.
- High rainfall and combines with carbon dioxide to form carbonic acid which makes some minerals in rock soluble.
 - Secretion of animals in such areas also contribute in dissolving of certain types of rocks.
- b) - The steeper the slope, the faster the movement.
- In bare surfaces, Mass movement is faster than in an area with vegetation cover.
 - The heavier the material, the faster the movement.
 - Human activities eg Mining, quarrying, road construction accelerate mass movement.
 - Earth movements eg vulcanicity, folding, faulting accelerate the movement of the materials.
- c) - It's a slow type of mass wasting that occurs on gently sloping land.
- Occurs in areas with alternative winter and summer.
 - During summer, melt water results to saturate soil, gravel and weather rock to move downslope as a mass over the still frozen ground on mountain region.
 - Melting of snow lubricates the movement.



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- di) - Leads to loss of life and destruction of property.
 - Destruction of transport and communication lines.
 - Accelerate soil erosion on steep slopes.
 - Leaves permanent scars on the landscape/derelict.
 - It may change the rivers course thus affecting the people and aquatic animals down stream.
 - Discourage settlement in areas prone to landslide.

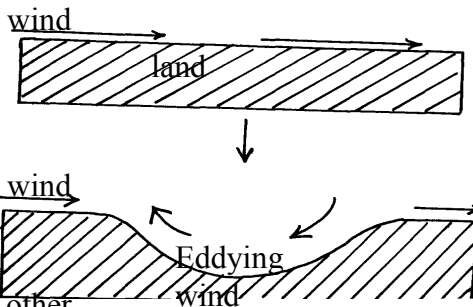
d ii) Rockfall

d iii)- M- cliff

- N-Rock debris (talus)

d iv) - Rainfall

32. a)- Wind is strong/moving at high velocity.
 - Absence of intervening obstacles/absence of vegetation cover/flat land.
 - Presence of large quantities of loose unconsolidated materials eg sand, gravels.
 b)- Attrition – Materials carried by the wind hit against rocks each and in the process are reduced in size.
 - Abrasion – Materials/load carried by wind is used to scratch and grind rocks and ground surface.
 - Deflation – materials/load carried by wind is lifted and rolled on the ground hence erodes the surface.
 c)- Dry unconsolidated materials are removed from the surfaced by wind.
 - Coarse materials /heavier are swept away by wind and erodes a pre-existing fault/crack by deflation process.
 - Heavier material move along the surface in swirling slops.
 - Continued eddying of winds leads to erosion and creation of a hollow called deflation hollow.



- d)- They lie parallel toe ach other.
 - Rise up to 100 m and about 100 km
 - Lie parallel to the direction of prevailing wind.
 - Have a sharp knife like crest.
 - Separated by flat corridors.
 - Occur in groups.
 - Where wind blows from the same direction for long, dunes become gentle on the wind-ward side and steep on leeward side.

33. a)- Melting of ice caps and sheets due to global warming and climate change.
 - Uplift of the coastal land due to tectonic forces
 b) - Warm Ocean waters (20-29)⁰c.
 - Clear and clean water, free from silt/sediments.
 - Salty water.
 - Well oxygenated water.
 - Constant washing by waves, tides and currents.

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- Plentiful supply of microscopic life for food/planktons.
- 34. a)- Suspension
 - Surface creep
 - Saltation
- . b) - Availability of sand
 - The strength and direction of wind.
 - The nature of surface.
 - Presence of intervening obstacles on path of wind movement.
 - Presence of vegetation
 - Presence of ground water reaching the surface.
- 35. a) - Rainwater
 - Rivers/streams
 - Underground water
 - Melting ice
- 35. b)- Size of the depression
 - Total annual rainfall of the catchments area and drainage basin.
 - The number of other sources from which the lake acquired its water
 - Different ways in which the lake water is lost.
 - Rate of sedimentation in the lake.
- 36. a i) - A river is a mass of water flowing in a valley
- ii) A river confluence is the point at which a tributary joins the main river while a river tributary is a smaller river which flows into a bigger one
- b i) - Water is forced into the cracks on the river banks/ water hits the banks
 - Air in the cracks is compressed
 - Compressed air creates pressure which widens the cracks
 - As the water retreats pressure in the cracks is suddenly released
 - The compression and widening of the cracks repeatedly shatters the rocks
 - The retreating water carries away the loose particles
 - The force of the moving water and the eddying effect sweep away loose materials in the river channel
- b ii)
 - River water carries sand, gravel and boulders
 - The load is used as a tool for scouring
 - The load is hurled by the river water against the banks/ dragged along the river bed
 - The load chips off the rocks on the bank and the floor (the size of the load determines the rate of erosion)
 - The load being dragged smoothens the river bed
 - Eddy currents rotate rock particles in hollows and widens them into pot holes
- a) - River capture may occur by head ward extension of the long profile
 - This happens when rivers are sharing a water shed
 - The actively eroding river gradually cuts back its slope head until it encroaches upon the divide or water shed of the other river
 - Eventually the powerful river reaches the source of the weaker river and diverts its water into its channel
 - River capture may also occur where there are two adjacent rivers

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- One of the rivers has more erosive power than the other
 - The more powerful river erodes both vertically and laterally faster than the weaker one, it flows at a lower level than the other river
 - The more powerful river erodes away the ridge that separates the two by head ward erosion
 - Eventually it encroaches into the valley of the weaker river diverting its waters into its valley
- b) - They have a gently sloping gradient/ flat surface
- They have thick alluvial deposits/ silt/ fertile soils
 - They have levees on either side of the river banks, raised river beds
 - Some flood plains have braided channels
 - Some have deferred tributaries
 - Flood plains have river bluffs at their edges
 - They have meanders/ bends
 - Some have deltas/ distributaries
 - Some have ox-bow lakes
38. a) i) Inselbergs, mesas, buttes, gorges, wadis, alluvial fans, bajadas, pediments, pediplans, playas, Salinas
- ii) - An increase in temperatures accompanied by excessive evaporation
- Prolonged period of drought or very low rainfall
 - Existence of cold ocean currents off shore which flow across the path of on shore rain bearing winds
 - Presence of high mountains which block rain- bearing winds and cause a rain shadow effect
 - Remoteness of land in the interior of a continent, far away from the direct influence of the sea
 - Location of an area in a region of anticyclones where winds are descending and diverging
 - Human activities such as deforestation, overgrazing and over drawing of underground water reserves
- iii) - Sandy desert
- Stony desert
 - Rocky desert
- b i) - Strength and speed of the wind: A strong wind can transport more and heavier load than a weak one. The strength of the wind is determined by its speed. The higher the speed the stronger the wind
- Obstacles: Presence of obstacles on the path of the wind block the movement of the wind and reduce its speed through friction. The winds capacity to transport is reduced by friction with the obstacles
 - Nature of the load: Light particles such as those of dust can be transported in suspension over long distances, while the heavier ones are transported over short distances and along the ground. Heavier peddles and stones are rolled on the ground if the wind is strong enough
- b. ii) - A pre- existing depression or localized fault is exposed to wind erosion
- Wind eddies remove the unconsolidated materials by
 - Weathering aids in breaking down the exposed rock
 - Wind abrasion excavates the depression by eroding the rock along the weak lines
 - The depression is deepened and widened as deflation continues to remove the loose materials
 - Erosion continues until the water table is reached

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- Water oozes out of the rock and collects in the depression to form an oasis
- 39. a) -swash is surge of sea water towards the beach while backwash is water that flows back to the sea from the beach
b) -fall in sea level
-rise of land near the sea
-actual reduction of water
- 40. a) Diversion of head waters of a river into a system of an adjacent more powerful river
b) X – Elbow of capture
Y- Pivotal/ powerful river
Z – misfit stream
- 41. a) - Accordant drainage system occurs with agreement with the rock structure, nature and slope of land while discordant occurs in disagreement with rock structure and nature
b) - River must have large load
- Velocity of river must be low to allow deposition in the river mouth
- River load must be deposited faster than it can be removed by currents and tides
- There should be no obstacle on the river course like a swamp to filter materials
- 42. a i) An area of land with scanty rainfall and scarce vegetation
ii) - Sandy
- Stony
- Bad lands
b i) - wind abrasion
- Wind deflation
b ii) - Rock of alternating hard and soft attacked by winds
- Cracks form on the hard surface
- Wind abrasion erodes joints deepening them
- Wind deflation blows away material
- Furrows form as soft rocks are further eroded
- Hard and more resistant rocks form ridges known as furrows
c i) - Mesas
- Buttes
- Gorges
- Salinas
- Bajadas
- Wadis
- Playa
- Pediment
- Inselbergs
c ii) - Features of tourist attraction
- Oasis are sources of water for domestic use
- Desert land forms are ideal for military training
- Loess forms fertile alluvium soils in wetter lands
- Seasonal streams can be used to supply water to surroundings

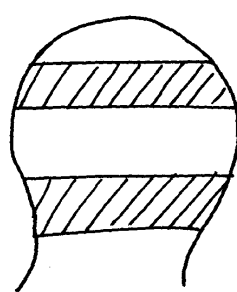
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- Sand dunes and rocky landscape pose problems to transport
- Loess in China and Europe have dug-in caves which are inhabited in winter to provide Warmth

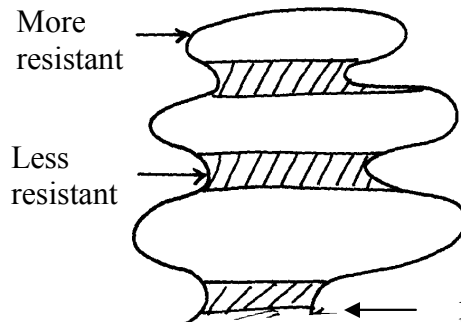
43. a i) - Abrasion✓ - Attrition✓

- a ii) - the faster the wind blows, the greater the amount of material it can carry✓
 - fine dust is easily picked up and blown away even by light wind, sand and pebbles are usually along the surface✓
 - presence of obstacles such as rock bushes on the path of the wind will slow down the wind making it drop✓
 - Deflation✓

b i) Alternate hard and soft rocks out crop laid horizontally✓ abrasion is predominant at the base level ✓soft rocks are eroded to produce furrows✓ shaped features is formed called rock pedestal

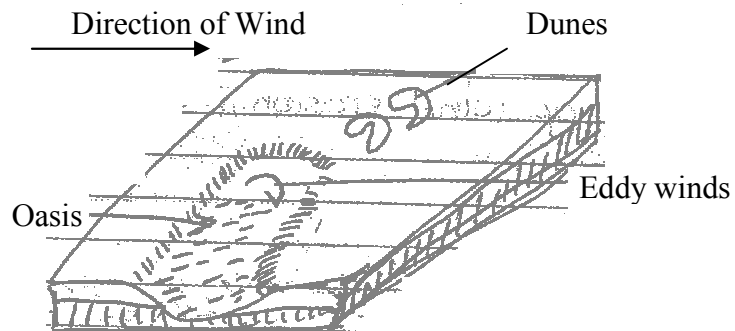


Before abrasion



Abrasion is greater at base

b ii) In desert areas wind deflation may create hollows✓.the hollows may be deepened by wind abrasion ✓process to reach the water table.✓ Water oozes out to form oasis✓



- c i) - weathering by solution lakes in limestone areas ✓
 - deposition by water/ice✓
 - erosion by wind/ice✓

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- meteorite falling✓
- human activities/damming✓
- mass movement✓
- c ii) - lakes are reservoirs acts as sources of rivers✓✓
 - support biodiversity/support flora and fauna✓✓
 - lakes enable self purification of water/air✓✓
 - modify local weather and climate✓✓
 - regulation of river from overflowing flooding✓✓
- d) - Forms when stalactite continues to grow down wards
 - Stalagmite grows upwards towards the stalactite
 - Eventually the two features meet forming a continuous column called limestone pillar
- d i) - Limestone blocks are used for building houses/ or`
 - Limestone is a raw material in cement manufacture which is used in building and construction
 - Limestone landscape is usually rugged thus discouraging settlement
 - Tourist attraction – underground and surface features are conspicuous hence attract tourists

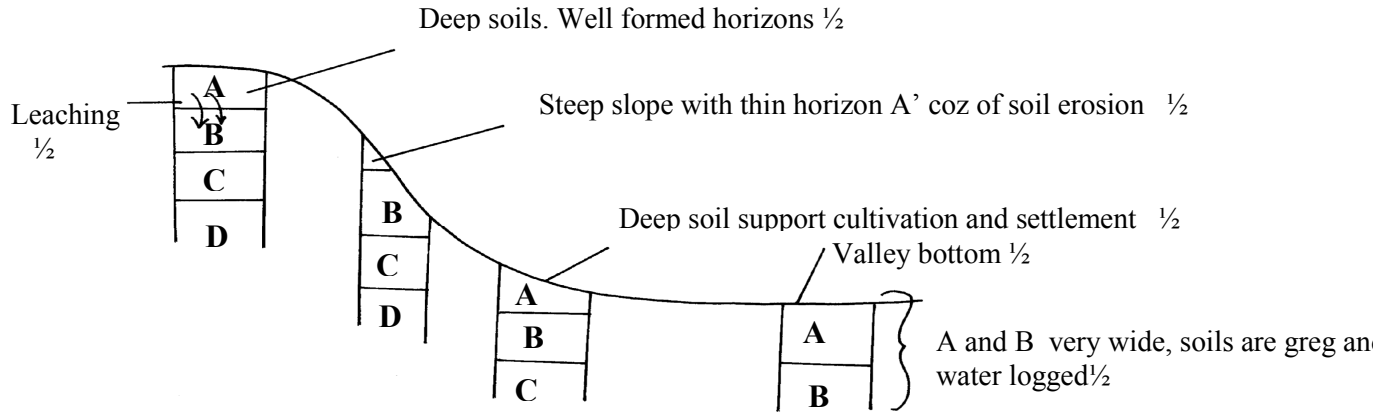
14. SOILS

1. a) - Soil profile is the vertical arrangement of various soils in layer showing the sequence of horizons from the surface to the parent materials
- Soil catena is the sequence of different soils on a slope from the top of the hill to the valley Bottom
- .
- b) - platy/plate soil structure
 - crumb soil structure
 - prismatic soil structure
 - blocky/block soil structure
 - granular soil structure
 - columnar soil structure
- c i) - Controlling overgrazing
- Avoiding bush fires
- Controlling tree cutting
- Practicing appropriate methods of cultivation e.g. planting cover crops, irrigation, mulching, terracing, contour farming
- c ii) - Leaves & branches reduce the force of rain drops which would otherwise loosen and remove soil particles
- Rate of infiltration of rain water into the soil is increased by vegetation cover thus reducing surface run off
- Tree roots which penetrate the soil help to carry surface moisture into the ground. This allows the moisture to gradually percolate deeply
- Tree roots bind the soil particles together and therefore the soil can not be easily carried away
- Trees break the force of the wind at the ground and reduces the transportation of soil by wind

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2. a i) Soil is the top layer of loose or unconsolidated rock material overlying crustal rocks and on which the plants grow
Or - Is an accumulation of rock particles minerals, organic matter, water and air found on the surface of the earth
- a ii)) Soil catena : This is the arrangement of soil on a mountain slope from the top of the valley bottom while soil profile is the vertical arrangement of various soils in layer from surface to bed rock
- b i) - - Climate
- Seasonal variation of rainfall can cause accumulated concentration of salt in the soil
 - Rainfall provides water which make it possible for rocks to disintegrate to form soil
 - Rainfall also affect the rate at which some soil forming processes can occur (leaching)
 - High temperatures increase the rate of weathering by accelerating the raise of bacterial activities which generates some of the organic matter in the soil
 - Wind, ice, water erode, transport and deposit soil particles in other areas leading to the formation of new soils
 - Living organisms
 - Living organisms add organic matter to the soil
 - Areas with thick vegetation lead to the formation of fertile humus laden soils which is quite useful in aeration
 - Bacteria help to decompose organic matter
 - Bacteria fixes nitrogen into root nodules of plants thus enriching the soil
 - Human activities can change the nature of the soil through grazing cultivation, use of fertilizers e.t.c.
- b ii) - Type of parent
- Amount of organic matter
 - Chemical composition i.e. minerals
 - Drainage of the soil or amount of water in the soil
3. a) Relief
- Slope influences the type of soil in that on steep slopes the soils that develop are thin, on gentle slopes the soils are deep and fertile. This is because the rates of erosion and weathering are quite rapid on steep slopes than gentle slopes
- Drainage
- The amount of moisture in the soil is determined by the slope. Steep slopes have well drained but thin soils
 - Hill top has well drained and mature soils
 - Valley bottoms develop soils that are poorly drained
- Mass wasting and surface run off
- This leads to development of fertile soils at the valley bottom because the top soil removed and deposited there
- Leaching
- This is the movement or washing of soil minerals from one layer to another which will also affect the type of soil developing
- b i) the vertical arrangement of soil in layers from the top the bedrock

b ii)



b iii) elluviation

illuviation

leaching

lateralization

humification

calcification

gleization

ferrilisation

b iv)- Excess irrigation

- Over stocking

4. a i) Soil is a thin layer of natural material on the earth's surface that supports plant and animal life

ii) - Inorganic matter

- Organic matter

- Soil water

- Soil air

- Soil organisms

. b i) - Parent rock

- Topography

- Climate

- Time

b ii) - Dense vegetation vital for fertile soils

- Micro-organisms like fungi fix nitrogen to the soil

- Micro- organisms aerate the soil hence improve porosity

- Activities of man i.e. grazing, use of fertilizers, construction

c i) - Soil air – gases that are available within spaces that separate soil grains

- Soil texture- size of individual soil grains

c ii) - High amount of rainfall promote chemical weathering of acid parent rock materials

- Long cold withers and short mild summers in temperate climates reduce bacterial activity

- Weathering of parent rock materials releases iron, aluminum, silica and organic acids

- Strong acid conditions enhance leaching of bases especially calcium along with iron and aluminum

- Silica that is less soluble is left back with the top layers and staining the soil beneath

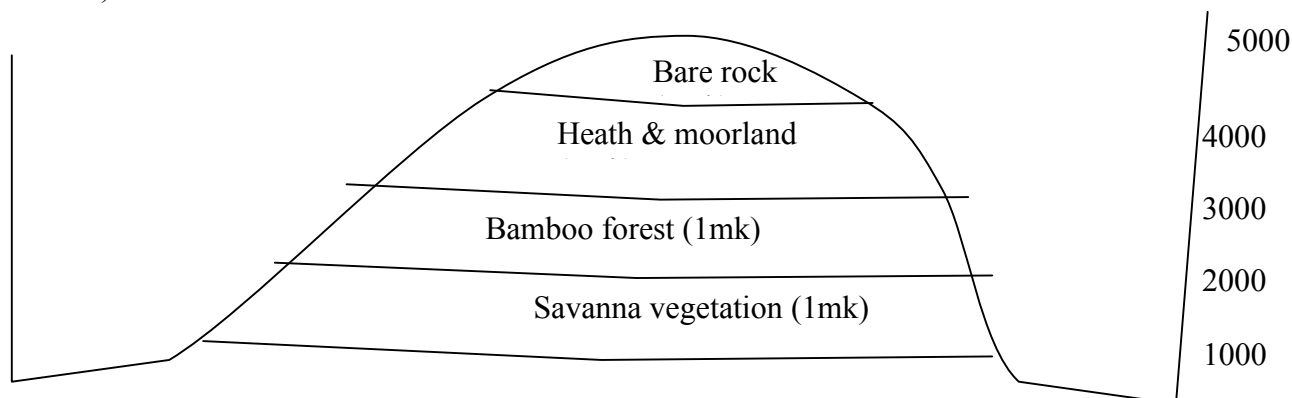
- ash- grey

c iii) - Develop mostly through calufication

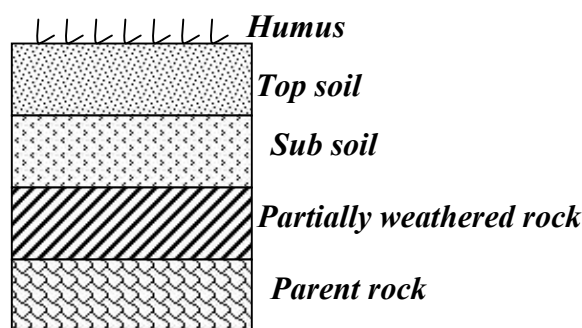
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- Have dark surface horizons
- Rich in calcium
- Have deep top layers of about 1 meter thick

5 .a)



- b i) - Soil catena is the sequence of different soils on a slope
b ii)



- a iii) - The soils are light in colour
- They are saline
 - They are sandy/ strong
 - They are loose in texture
 - They are thin
 - They have low moisture content
- b) - The type of parent rock
- The amount of organic matter/ humus
 - The chemical composition/ the degree of iron oxide/ minerals
 - The amount of water in the soil/ drainage of soil
- c) - During the wet season, mineral salts in the top layer of the soil dissolve rain water
- The dissolved minerals percolate seep downwards from the top soil to the sub soil
 - The dissolved minerals are deposited further downwards to the lower layer
 - Insoluble minerals such as iron and aluminum are left on the up layers to form a crust of late rite soils
- d i) - Burning destroys micro- organisms which are essential for formation of humus which binds the oil particles together
- Burning destroys vegetable matter that protects the soil against erosion
 - Burning destroys the nitrogen fixing bacteria making the soil less fertile
 - Burning loosens the soil making it susceptible to erosion/ leaching which drains away soluble mineral nutrients
- d ii) - This increases the aridity of the soil/ changes the PH of the soil
- The acidity destroys the micro- organisms in the soil/ bacteria/ fungi which could have helped in the formation of humus
- d iii) - Monoculture leas to exhaustion of certain minerals from the soil making it infertile

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leading to its erosion

- Monoculture leads to loosening of soils particularly there by encouraging soil erosion

6. (a) i) - Tundra soils

- Podzols
- Chernozems
- Proure soils
- Chestnut soils
- Red Desert soils

a ii) - In warm to hot humid regions silica and other bases dissolve in water are leached/percolate when it rains seep downwards from the top soil

- This leaves iron and aluminium oxides which are insoluble on upper soil horizons.
- This forms soils on upper soils on the upper soil horizons rich in iron and aluminium oxides called lateritic soils

a iii) - They are young/of recent origin

- They are immature
- They are insufficiently affected by soil forming processes
- Characteristics depend on their origin

b i) - High rainfall can lead to water logging and formation of acidic soils

- Heavy rainfall also results into surface run off that may lead to soil erosion
- Too much rain can alter the soil structure by causing crumb soil to form blocky or columnar structure
- Heavy rainfall can lead to leaching of nutrients from top soil to lower soil horizons depriving the top soil of some nutrients

- Low rainfall leads to loose soils easily blown away by the wind

- Excessive drought leads to accumulation of salts in the top soil leading to salination

b ii) - Soil is baked to make building and construction materials like bricks, tiles/whitewash

- Soils are used in making pottery/ceramics/sculptures and used by people
- Some soils contain minerals like apatite which is mined and used to make phosphatic fertilizers
- Some soils are medicinal
- Some soils are food to animals and people

7 a i) soil catena is the horizontal arrangement of soil on a mountain slope while a soil profile is the vertical arrangement of soil particles in layers or horizons

a ii) -leaching

- eluviation
- uluviation
- organic accumulation
- cheluviation
- precipitation
- organic sorting

b i) - seasonal rainfall in temperature and rainfall facilitating the rate of weathering

- the rate at which organic matter decompose rise in to increase in temperature and rainfall while dry areas are devoid of vegetation covers

b ii) - micro-organisms cause both plants and animals to decay into humus. humus are added to the upper layers of the soil

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- living organisms act on decaying organic matter through humification to form humus.
This humus consolidate forming rich soils.
- b iii)- soils formed on steep slopes generally fails to develop or mature because most of the rain water runs off along the surface ,this accelerate erosion which comes away the top layers of soil
 - on rolling and gentle slopes maximum soil development is likely to occur. This is because the rate of soil erosion is matched by the rate of soil formation due to good drainage
- c i) soil degeneration is the decline in the usefulness of soil resulting from either soil mismanagement or environment causes
- c ii) -plant roots which penetrate the soil help to carry surface moisture into the soil
 - vegetation helps to bind soil particles together
 - plant cover breaks the force of wind and therefore reduces the transportation of soil particles
 - decayed vegetation matter provide humus which binds the soil particles together
 - the leaf cover helps to reduce the force of rain drops which would loosen and remove soil particles
 - the rate of infiltration of rain water into the soil is increased by vegetation cover thus reducing surface run offs
- c iii) -crop rotation
 - contour ploughing
 - strip cropping
 - inter cropping
 - bush fallowing
 - mixed cropping
- 8. a i) -this is the sequence or arrangement of different soils down a slope
- 8. a ii) -the relief/slope/steepness of the land
 - the drainage of the area
 - the transportation of the soil debris
 - the teaching process
- 8. b) -the production topsoil is lost forever and this lowers the agricultural production of the land
 - the soil left behind after erosion is thin and cannot hold plants firmly in the ground.
 - continued loss of soil through erosion destroys vegetation cover turning the area into semi- arid/desert.
 - soil erosion causes water reservoirs to be filled up by silt adding extra cost of dredging.
 - sediments which are carried into rivers/lakes/oceans may contain industrial effluents which kill aquatic life.
 - gullies as a result of erosion expose water onto the surface leads to lowering of the water table
- 8. c) -develop best in areas which are cool and sufficiently wet
 - soluble soil constituents are moved downwards
 - it leads to severe absence of bases throughout the soil profile resulting to low PH and an accumulation of hydrogen clays
 - there is an accumulation of acid organic matter in horizon A leading to intense teaching of bases and clays
 - iron and humus are deposited in horizon B

9. a) - Living organisms.

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- Organic matter.
 - Soil minerals.
 - Soil matter.
 - Soil air.
 - b i) - Climate influence type & rate of weathering.
 - High rainfall influence leaching process runoffs resulting from high rainfall increases rate of erosion.
 - It influence rate of decomposition
 - b ii) - Influence soil profile.
 - Parent material influence soil structure.
 - It influences soil colour.
 - It influences mineral composition.
 - It influences soil texture.
 - c i) - Over cultivation exhausts the soil fertility.
 - Monoculture exhausts soil minerals.
 - Clearance of land for agriculture interrupts nutrient cycling.
 - Application of fertilizer changes the P.H.
 - Construct of roads, mining, quarrying destroy soil structure.
 - Application of fertilizers change soil PH
 - c ii) A-TOP SOIL
B- SUB-SOIL
C- PARTLY WEATHERED ROCK
D -PARENT ROCK
10. a) - A lake is an accumulation of water in a wide hollow or depression
b) - By erosion
 - By earth movements/ volcanic activity
 - By deposition
 - By human activity
 - By falling meteorites
 - By mass movements e.g. land slides
 - Weathering by solution
11. a i) Soil is the upper most surface layer of unconsolidated material which lies on the surface of the earth and in which plants grow/ soil is an accumulation of rock particles or minerals, organic matter, water and air found on the surface of the earth
b i) - Seasonal variation of rainfall can cause accumulation/ concentration of salts in soil
 - Rainfall provide water which make it possible for rocks to decay/ disintegrate to form soil
 - Rainfall can affect the rate at which some soil forming processes can occur (leaching)
 - High temperatures increase the rate of weathering/ accelerate the rate of bacterial activities which generates some of the organic matter in the soil
 - Water, ice and winds erode, transport and deposit soil particles in other areas leading to the formation of new soils (for example loess)
- b ii) - Valley bottoms/ gentle slope encourage the formation of deep and fertile soils due to deposition/ accumulation of minerals
 - Steep slopes encourage erosion of the top layer of soil thus slowing down formation of soil/ have thin soils
 - Flat plains/ flood plains are saturated with water therefore slows down soil forming processes
 - Slopes influence arrangement/ sequence of soil/ soil catena

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- Some slopes are more exposed to the sun/ rain which influence weathering of parent rocks/ soil formation (aspect)
- c i) - Humus helps to improve soil porosity by aerating the soil
 - It improves the moisture – retention capacity of the soil
 - Humus provides essential minerals to the soil
 - It improves the soil texture
- c ii)- They are thin and shallow
 - They are sandy or stony
 - They lack humus or have low organic matter content
 - They are generally saline
 - They are coarse- textured and quite porous
 - They are alkaline because of high content
 - They have low moisture content
- . d i) - This occurs on gentle slopes which are bare
 - When heavy rain falls, water spreads over a large area
 - As water moves, it removes the top layer of soil evenly over the area
- . d ii) - It occurs on steep slopes
 - Rain water cuts deep grooves/ channels/ rills on the slopes
 - The channels are widened and deepened to form gullies, through which soils are carried away
- . d iii) - Soils are sources of valuable minerals
 - Soils are used raw materials for pottery/ ceramics/ tiles/ bricks
 - Soils are used for agriculture
 - Some soils are mixed with herbs and sold for medicinal purposes e.g. clay/ sold directly for food
- 12. a) - It is a careful management/protection of soil against erosion and exhaustion
- . b) - Ploughing along the contour
 - Controls grazing
 - Stripping cropping

15. AGRICULTURE

- 1 . a i) Land reclamation is a process by which unproductive land is converted to productive land and used for crop or animal farming. Rehabilitation is the restoration of land that has been destroyed through human activities to its former usefulness/productivity
- ii) - Bush fallowing
 - Mulching
 - Planting trees
 - Filling the pits/gulleys
 - Planting cover crops
- b i) - To occupy detainee labour from Mwea detainee Camp
 - To utilize the black cotton soils in growing rice
 - To increase agricultural production
 - To settle and create employment for former detainees
- ii) - Diseases e.g. Bilharzia and malaria caused by water snails and mosquitoes, weaken and even kill the farmers hence affecting productivity
 - Stagnant water is the breeding grounds for water snails and mosquitoes
 - Fluctuation of world prices due to competition affects agricultural production
 - Poor payments for farmers produce that kill their morale

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- Monoculture practice has led to soil exhaustion resulting to use of fertilizer that increases cost of production
 - Mismanagement of irrigation bodies leading to losses of farmers killing their morale
- 2.
- a) - Construction of productive dykes/walls to enclose area to be reclaimed
 - Construction of ring canals to carry water from the reclaimed area into the sea
 - Installation of pumping stations to pump water from the enclosed area
 - Water is pumped out of the area enclosed by the dyke
 - Reeds are sown to get excess water and prevent growth of weeds
 - Drainage ditches were cut in the land and drainage pipes laid
 - More pumping stations were put up in order to drain excess water
 - Soil treated with chemical to lower salinity
 - Drained land was flushed with fresh water to remove salts from the soil
 - Pumping out of the water and crops planted or grown
 - b) - Improves distribution of fresh water for domestic use
 - Control of floods to the south west
 - Controlled seas invasion inland
 - Improved accessibility through development of roads and railways
 - Improved tourism in the area
- 3.
- a i) - Trans Nzoia
 - Nakuru
 - Uasin Gishu
 - ii) - Land is ploughed manually/mechanically
 - Sowing of seeds are done at the onset of rains
 - Two seeds are sown in holes of 2-5cm in rows of 1m apart
 - Application of fertilizers or manure is done during planting
 - Weeding and thinning is done to remove weak seedlings
 - Top dressing /nitrogenous fertilizers is applied after weeding when the crop is 30-45cm high
 - The crop is ready for harvesting in 4-8months depending on variety of seeds and altitude
 - Harvesting is done manually
 - b) - Maize is a staple food
 - Maize stalk serve as cattle feed
 - Maize is a raw material for industries
 - Stalks, leaves and cobs are important source of organic manure
 - Maize cobs and stalks are used as fuel in varied areas
 - Provide employment /income
 - Saving foreign exchange
- 4.
- Unfavourable weather conditions/ prolonged drought lead to destruction of crops hence low yields/income
 - Pests and diseases e.g. stalk rot, corn ear worm, silk worm, birds e.t.c. which attack crop reducing yields
 - Poor/low prices
 - High cost of farm inputs reduce farmers profits
 - Poor marketing strategy leading to low prices

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- Monoculture leading to soil exhaustion
- Poor storage facilities
- 5. a) - Friesian
 - Jersey
 - Guernsey
 - Alderney
 - Ayrshire
 - Sahiwal
- b) - Low temperatures ideal for exotic breeds
 - Fertile soils, which support quality pasture
 - Reliable and constant water supply for the animals
 - A high population, which provides ready market for dairy produce
 - Has humid conditions suitable for the growth of pastures
- 6. a) - French beans
 - Cabbages/kales
 - Celery
 - Spinach
 - Pawpaw
 - Mangoes
 - Avocadoes
 - Passion fruits
 - Carnations
 - Gladioli
 - Roses
 - orchids
- b) - Netherlands has a higher urban population than Kenya.
 - There is high demand for both local and foreign horticultural products in Netherlands than Kenya
 - Farmers in Netherlands have more access to the capital needed for horticultural farming than in Kenya.
 - There is more advanced and appropriate technology in Netherlands which has enhanced horticultural farming than in Kenya.
 - Netherlands unlike Kenya has a highly skilled labour for production and handling of agricultural products.
 - There is more advanced horticultural farming related to research in the Netherlands than in Kenya.
 - Netherlands unlike Kenya has well organized marketing procedures/co-operatives/auction markets which are conducive for horticultural farming
- 7. a) - Vegetables/tomatoes/onions/carrots
 - Fruits/oranges/ pineapples/ plums/ mangoes/ flowers/roses
- b) - Horticultural crops are highly perishable this necessitating faster means of transport
 - Same are light in weight which makes it easy /suitable for air export
 - There is high demand for produce thus this need to supply urgently
 - High market prices are able to pay compensate for the foreign charges
- 8. a)(i) It's a traditional way of keeping large number of animals by nomad communities

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who move from place to place in search for pasture and water.

- ii) - Frequent livestock raid by neighboring communities or amongst themselves
 - Animals are grazed communally
 - They use natural pasture for grazing in the livestock
 - It is practiced in area with low and unreliable rainfall
 - Their movement is determined by availability of water and pasture
 - Different types of animals are kept
 - Animals are kept for pride

- 9. a i) – Jersey
 - Guernsey
 - Aishire

- ii) - Ghee
 - Cheese
 - Butter

- b i) - Cool climate/ moderate temperatures which are suitable for survival of exotic breeds
 - High rainfall that favours growth of pasture and fodder
 - Deep volcanic soils which favour pasture and fodder growth

- b ii) - Attack by diseases such as East Coast Fever and pests e.g. ticks may cause death of the animal hence total loss to the farmer
 - Poor roads delay delivery of milk to collecting centers hence losses due to milk getting spoilt
 - High cost of farm inputs discourages dairy farmers
 - Mismanagement of the framers co-operatives discourages farmers due to delayed/ low or non- payment
 - Drought reduces fodder and water for the livestock causing death or low production
 - hence losses to the farmer

c i)

Kenya	Netherlands
1. Done on small scale	Done on large scale
2. Done in Kenya highlands mainly	Done in the whole country
3. Zero grazing and out door grazing on pasture	Only zero grazing on fodder

- c ii) - New K.C.C
 - Brook side

- 10. a) - Jersey
 - Ayrshire
 - Guernsey
 - Cross breed

- b) - Disease such as rindpest may lead to death of animal
 - Inadequate pasture during the dry season..
 - Failure to collect milk due to glut in production.
 - Delayed payments which discourage the farmers.
 - Poor roads which are inaccessible during rainy season.

- 11. a) - Gently sloping land
 - Deep, well drained volcanic soils

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- Moderate rainfall/ 500mm – 1270mm p.a
 - Warm conditions/ temperatures of 15C – 20C
 - Warm/ dry/ sunny spell for ripening and harvesting
 - b) - Prolonged drought which lower yields
 - Attack by pests (such as dusty brown beetle, quela birds, aphids, cereal weevils and diseases e.g. stem rust, brown leaf rust, glumme blotch
 - Exploitation by middle men
 - High costs of transport due to poor roads
 - Shortage of storage facilities
12. a i) - Rift Valley
- Central
 - Eastern.
- ii) - Manitoba
- Alberta
 - Saskatchewan
- b) - 500-12700mm RF/moderate rainfall.
- Dry sunny spell for ripening/harvesting
 - At least 3 months to temp 15⁰C – 20⁰c for maturation or growing of wheat.
 - Volcanic soils to sustain production.
 - Gentle /fairly level/undulating land enables mechanization to take place.
 - Deep /fertile/well drained soils for production of wheat.
- c) - Farmers in Canada are more mechanized while those of Kenya are less mechanized.
- Canada has more capital than Kenya which experiences financial problems.
 - Farmers in Canada are more skilled/have a long history of wheat production than in Kenya which is still developing.
 - There is advanced research in Canada providing yielding seeds/better farm inputs/control of pests & diseases/overcome limitation of whether than in Kenya where research is poorly done.
 - Farmers in Canada has more extensive tracts of land which are suitable for wheat growing than those in Kenya which are small in acreage
- d) - Canada exports wheat hence earns foreign exchange. Which is invested in other sectors of the economy.
- Wheat farming is a source of employment any Canadians who earn income raising their living standards.
 - Industrialization as many industries that deal in wheat products have been established. Leading to economic diversification.
 - The wheat farmers earn income through the sale of wheat raising their standards of living. Canada is self sufficient in food as wheat is the main cereal food consumed
13. a) - Narok
- Uasin Gishu
 - Nakuru
 - Parts of Baringo
 - Trans mara
 - Nyandarua
 - Trans Nzoia

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- Keiyo
 - Laikipia
 - b) - Gentle slope to allow mechanization
 - Altitude between 1500-2900m to reduce cases of diseases outbreak
 - Moderate rainfall between 500mm to 1,270mm to supply moisture for growth
 - Warm temperature between 15°C to 20°C to allow production
 - Warm dry sunny spell which enhances ripening of wheat and harvesting
 - c i) - Wheat is ground into tiny particles. This is done in flour mills
 - Tiny particles are sifted out as flour
 - The yellow flour is bleached
 - Bleaching gives white colour to wheat flour
 - c ii) - Wheat is sold on local markets
 - Procedures sell wheat or millers e.g. Unga limited
 - d i) - Inadequate capital limiting expansion of farms and level of mechanization
 - Pests like dirty brown beetle, aphids, quelea birds damage wheat causing low yield
 - Diseases like stem rust, leaf rust, glume blotch lead to low crop yields
 - Heavy storm/rainfall cause flattening of wheat causing rotting of wheat
 - Price fluctuation lowering productivity /profit
 - Inadequate storage facilities causing wastage
- (ii) Canada has more extensive tracks of land suitable for wheat growing while in Kenya farms are relatively small.
- Canada has a higher level of mechanization while in Kenya the level is relatively low/machines are slow
 - Farmers in Canada are more experienced due to longer history of wheat production while in Kenya the level of advancement of techniques is low.
 - Advanced scientific research in Canada enables the production of higher yielding seeds than in Kenya where the level of research is low.
 - Wheat farmers in Canada specialize in wheat production while in Kenya they practice mixed farming (any 2x1=2mks)
14. a) - High temperature between 20°C to 27°C.
- Dry sunny conditions
 - High rainfall ranging 1200-1500mm
 - Deep fertile soils
 - Well drained soils
 - Gentle sloping areas to allow mechanization
- b) - Land use conflict/competition from cash crops
- High cost of inputs which was limited mechanization
 - Impassable /muddy roads during rainy season
 - High rise of cattle diseases like East Coast fever
 - abrupt and extensive drought lowers production
 - Mismanagement of co-operatives resulting in delayed payments killing the moral of farmers
15. a i) -Nigeria
- Coted'ivoire
 - Cameroon
- ii) - heavy rainfall-1270mm-1500mm per year

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- high temperature-between 21°C and 26°C throughout the year
 - high relative humidity –over 75%throughout the year
 - deep fertile and well drained soils
 - Protection from sunshine and wind by banana plants and other shady trees
 - low altitude below 700m above sea level
- b)-The pods are harvested using long sharp knives, collected and pulled at a central place
- The pods are then split open with a sharp knife and beans are scooped out by hand
 - The beans are put in heaps on mats, covered with banana leaves and are allowed to ferment for 5 to 6 days during which the juicy pulps drain away
 - The fermented beans are washed and cleaned spread on tables covered with mat to dry in the hot sun
 - The beans are turned frequently as they dry and slowly they turn brown
 - The dry beans are put in sacks and sent to the buying centers. There the dry beans are weighted and graded ready for export.
- c) - Pest like capsid bug mealy bug destroy the crop
- The swollen shoot disease attacked the tree shoots and the black pod disease affects the pods reducing the yields
 - The strong harmattan winds break the branches and cause premature ripening of the pods
 - Fluctuation of prices in the world market discourage the farmers
 - Poor transportation facilities make it difficult for the farmer to deliver the crop in time
 - Labour is sometimes in short supply, leading to delay in harvesting
16. (a) i) horticulture is the cultivation of vegetables, fruits and flowers for commercial purpose
- ii)- there is higher local demand for horticultural crops in the Netherlands than Kenya
- The Netherlands has a long history of horticultural crop production and hence has established foreign markets than Kenya
 - Netherlands farmers are well organized making it easy to access loans whereas few Kenyans access loans
 - Netherlands has advanced technology while Kenya has low –level advancement of technology
 - Netherlands has well development transport network while in Kenya the transport network is poor
 - Netherlands has highly skilled labour for production and handling of horticulture whereas -- - Kenya has a shortage of skilled labour in this field
 - In Netherlands there is advanced research in the industry while in Kenya research is less advanced
 - Netherlands is centrally situated in Europe which gives the country an advantages of Europe market
- iii) - the plants do not suffer the effects of excessive rainfall, hailstone and strong winds
- The plants will not be affected by drought because they are constantly watered
 - The spread of pests and diseases is easily controlled
 - it is easier to control the amount of moisture that the flower require
 - uniform and constant climatic conditions for the plants are created
 - the flowers are grown throughout the year
17. - Temperature ranges from 15degrees C to 20 degrees C(moderate temperature)
- Warm dry sunny spell for ripening
 - Rainfall between 305mm and 1015mm

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- Volcanic, well drained soils
 - Gently sloping/fairly level land
18. a) - High rainfall which is well distributed throughout the year ensure abundant supply of water for the animals and pasture.
- Fertile volcanic soil ensure high quality nutritious cover of grass.
 - Cool condition/temperature averaging 18⁰c which is ideal for the survival of exotic breeds/slow breeding of pests & disease.
- b i) Dairy farming in Kenya is mainly pasture dependant while in Denmark it is mostly dependant on fodder
- ii) In Kenya, most dairy products are consumed locally while dairy products in Denmark are mostly exported to foreign markets
19. a) - Aberdeen Angus.
- Charolais
 - Shorthorn
 - Santa gertridis
 - Hereford
 - Red Angus
 - Galloway
- b) - Extensive rolling grassland which allows the cattle to graze freely.
- Fertile soils of the Andes Mountain which give rise to healthy natural grass for feeding the animals.
 - Moderate temperatures of about 24⁰c during summer and 10⁰c during winter which ensures continuous growth of grass for feeding the animals.
 - Moderate rainfall of about 1000mm which ensures growth of nutritious natural grass pasture/pampas.
 - Adequate water for the animals.
20. a i)) X- Alberta
Y -Saskatchewan
Z- Manitoba
- ii) - Warm summers with average temperatures of about 15⁰c
- Sunny late summers for ripening of wheat.
 - Extensive undulating and uninhabited low lands.
 - Rich dark – brown chernozem soils rich in potassium and phosphates/less acid soils.
- b) - Central province.
- Rift valley province.
 - Eastern province.
21. a) Horticulture is the growing of fruits, flowers and vegetables for export market while market gardening is the growing of fruits and vegetables for local market (nearest urban centre)
- b) - Hot and wet climate for growth of tropical crops and cool and wet conditions for temperature crops
- Fertile soils of volcanic origin
 - High demand for the products both locally and internationally

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- Investment by large companies providing the capital needed for horticulture production
 - Availability of technical and financial assistance from friendly countries
 - Well organized marketing system
22. a)- It refers to the extensive grazing on natural pasture involving constant or seasonal migration of people and their livestock
- b) - Pastoralists move from place to place in search of water and pasture
- Land is grazed communally, though animals may be owned by individuals
 - A large number of animals is kept resulting in overgrazing and serious incidents of soil erosion
 - There is controlled cross- breeding of animal within each herd
 - The animals are generally of poor quality due to diseases and low quality feeds
 - Animals are considered as a source of pride and not income
 - Cattle are kept as a sign of wealth, paying pride price and slaughter during cultural festival
 - Different animals are kept i.e. cattle, sheep. Donkey e.t.c.
23. a) - Improved veterinary services
- Increase milk prices
 - Good performance of co-operatives
 - Government effort through the ministry of livestock development
- b) - Breeds kept are similar
- Use of co-operatives
 - Artificial insemination is used
 - Open and zero grazing is done
- c) - Creation of employment
- Earns Kenya foreign exchange
 - Self- sufficiency in milk production
24. a i) Total tones = 100800
- 1 5cm rep 100,800 tonnes.
- Flowers : $\frac{42800 \times 15}{100,800} = 6.324 \text{ cm} = 6.3 \text{ cm}$
- Oranges: $\frac{20600 \times 15}{100,800} = 3.363 \text{ cm} = 3.4 \text{ cm}$
- Tomatoes: $\frac{20300 \times 15}{100,800} = 3.02 \text{ cm} = 3.0 \text{ cm}$
- Carrots: $\frac{15400 \times 15}{100800} = 2.29 = 2.3 \text{ cm}$

DIVIDED RECTANGLE SHOWING HORTICULTURAL CROPS IN TONES PRODUCED
IN KENYA IN THE YEAR 2000

FLOWERS	ORANGES	TOMATOES	CARROTS
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- a) ii) $42500 - 15400 = 27,100$ tonnes
 iii) $\frac{15400 \times 100}{100800}$
 $= 15.828\% / 15.3\%$
- b i) - They are easy to draw.
 - They are easy to read/interpret
 - They give clear visual impression of individual components.
 - They allow easy comparison.
 - They can be used to represent a wide range of data.
- ii) - Moderate rainfall of 800 – 1200 mm
 - Suitable for horticultural farming.
 - Cool and wet temperate crops.
 - Hot and wet climate favours the growth of tropical crops.
- c - Pests and diseases such as tomato blight attack the tomatoes and lowers the yields leading to low income for the farmers.
 - Most roads are impassable during the rainy season leads to delayed delivery of the horticultural crops to the collecting centre lowering the quality subsequently the profit to the farmers.
 - High costs of farm inputs reduces the farmer profit margins.
 - Prolonged drought/hail storms in some areas destroys the crops leading to heavy losses.
 - Stiff competition from Netherlands and Israel which reduces the international market.
 - Inadequate refrigeration facilities leads to deterioration of the quality of horticultural products and subsequently the profit to the farmers is lowered.
25. a) – Freshian - Aryshire
 - Jersey - Guernsey
 b) - Creation of employment
 - Establishment of other related industries
 - Milk is a source of protein hence healthy population
 - A source of income to farmers
 - Government collects taxes from sales of dairy products
 - Cow dung is used manure hence increased productivity
 - Dairy products are exported hence earning foreign exchange
26. a) – Tea
 - Coffee
 - Horticultural products
 b) - Kenya exports mainly agricultural commodities that fetch less revenue
 - Price fluctuations in the world market which make it difficult to plan ahead
 - Competition from cheaper commodities that are imported into the country
 - Inadequate transport and communication facilities
 - Inadequate capital to expand trading activities
 - Fees paid in form of trading license discourage traders
 - Smuggling of goods (cheaper) from neighboring countries
 - Poverty among the people which reduces their purchasing power

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27. (a) - Food supply
- Industrial raw materials
- Draught animals/ beasts of burden
- Storage of wealth
- Medium of exchange
- b) - Sinking boreholes/ harvesting rain water
- Cross breeding
- Discourage overstocking
- Extension services
- Credit availability
- Cattle dips
- Diversification e.g. bee keeping
- Sensitization on quality/ selling e.t.c.
28. a) – Operation costs
- Market expenses
- Price fluctuations
- Government policy
- b i) - Is a beverage
- Is an industrial raw material for making cocoa butter, chocolates, confectionaries cosmetics, soap and drugs.
- Cocoa husks are used as fertilizer.
- Cocoa shell is used as mulch.
- Cocoa husks may used as fuel.
- ii) - Pests such as the capsid destroy the cocoa tree.
- Diseases such as swollen shoot and black pod destroy the crop.
- Fluctuation of cocoa prices.
- Short labour supply.
- Flooded feeder roads delay collection of the crop.
- Bush fires destroy the crop.
- Strong winds e.g the harmattan cause a drying effect.
- Low cocoa prices
29. a) Horticulture is the growing of fruits, vegetables and flowers for the export market while market gardening is the intensive cultivation of fruits and vegetable mainly for urban markets
- b) -Roses
- Orchids
- Carnations
- Gladio
- c) - Inadequate capital to buy the necessary farm input
- Low production of milk from indigenous breeds
- Low managerial ability of the farmers
- Poor pasture leading to low quality breeds
- Inadequate labor supply
- Competition from imported milk and milk product by unscrupulous businessmen
- Diseases and pests
- Delayed payment to the milk producers

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- Poor transport network
 - Prolonged drought
30. a) - Loss in livestock and crops production.
- Low agricultural production.
- Low agricultural economic growth.
- Reduced production agricultural land
- Overcrowding in areas receiving adequate rainfall leading to land degradation.
- b) - Honey from bees is a major source of food as well as medicine.
- It is a major source of income in arid and semi-arid areas of Kenya.
- It provide alternative and effective use of marginal areas where crops may do well.
- Bee wax is used in making candle.
- Bees are not expensive to maintain
31. a) - Traditional diet.
- Land ownership/Inheritance.
- Religion.
- Gender roles
- Foreign influence
- b) - Gentle sloping terrain.
- Extensive pieces of land with low production.
- Adequate capital to set up large farms.
- Availability of skilled labour.
32. a) - High temperature throughout the year ($21 - 27^{\circ}$)
- Plenty of sunshine during ripening period.
- Heavy rainfall well distributed through out the year.
- b) - Cooking fat.
- Soap/both for washing and toilet soap.
- Cosmetics.
33. a) - Pollution is the contamination of the environment with substances which are harmful or poisonous to human, plants and animal life.
- b) - Land /Soil/Ground pollution.
- Water pollution
- Noise pollution.
- c) - Temperature ranging between $15^{\circ}\text{C} - 30^{\circ}\text{C}$
- Mean annual rainfall which should be between 400 – 1750 mm.
- Soil should be deep, well drained but with good water – retaining capacity.
- Altitude of between 1500 – 2200m.
- Windbreaks essential for protection of tea bushes against damaging wind.
- Shade.
- d) – Pests and diseases.
- Tea bushes are commonly attacked by different pests and disease.
- Such pests include black tea thrips, red spider mites, red crecise mite and weevils.
- Diseases include root rot.
34. a) - Aberdeen Angus
- Hereford .

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- Short horn
- Sahiwal
- b) - Establishing priority orders by creating awareness among leaders.
- Identifying methods of approach that include retorted inclusives and dept. of Agriculture.
- Educating farmers on strategies to improve their systems of livestock farming.
- Investigating the prevailing market situations and make appropriate recommendations.
- Creating Ranching schemes in arid and semi arid (ASACS) areas to improve livestock farming.
- Identifying possible financial institutions to farmers such as AFC and banks and advise farmers accordingly.
- Providing dipping facilities to improve the health of animals.
- Building dams to improve water in Reserves for the animals.
- Encouraging farmers to adopt modern methods of breeding beef cattle to improve on quantity.
- Introducing pedigree British cattle in some suitable districts or cross breeding with indigenous breeds to improve on the quality of the animals
- c-- Unreliable and inadequate rainfall leading to inadequate pastures.
- High temp. in most parts which makes it difficult to raise cattle of good quality.
- Pastoral tribes come for quantity rather than quality, land/herds of poor animals steadily ruin pasture land.
- Hard ancient rock that underlie Kenya produce poor soils prone to erosion. Natural grass is poor and not good for quality animals.
- Climate conditions of Kenya encourage spread of nagana
- d i) - Well distributed rainfall throughout the year from growth of good pasture.
- Subtropical temp. with summer temp. raising above 24⁰c lower winter temp. of about 10⁰ c favours growth of grass.
- Gently slopy Lorain provide good natural grazing landscape with good pasture.
- Fertile soils washed from foot/hills of Audeas losses that are fertile and encourage growth of pastures.
- d ii) - Provide employment in various section e.g. stockmen ranches.
- Improved feeder road to enable farmers transport cattle to the urban centres.
- Earns foreign exchange used in other section of economy (Your exports).
- By-products like hides used as raw materials to industries thus encouraging industrial growth.

16. LAND RECLAMATION AND REHABILITATION

1. a) - Perkerra
 - ahero
 - KAno
 - Bunyala
 - Bura
 - Hola
- bi) - The black cotton clay soils found in the area suitable for irrigation because they retain water
 - The gentle-sloping land enables water to reach the farm by gravity
 - Presence of rivers Thiba and Nyamidi provides regular and abundant water for irrigation
 - The un-reliable and inadequate rainfall received in the area makes it necessary for irrigation
 - More land is available for future expansion

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- Availability of labour from high population
 - ii) - Stagnant water encourages its breeding of snails and mosquitoes increasing incidences of Bilharzia and malaria
 - Quelea birds which feed on rice grain lowers crop yield
 - Siltation in canals reduces capacity to hold enough water for irrigation
 - Fluctuating water volumes reduces water for irrigation
 - Poor marketing strategies
 - Delayed payment to farmers lowers their morale
 - Shortage of capital to finance farming activities such as ploughing
 - c) - The scheme produces the bulk of Kenya's rice saving foreign exchange
 - The scheme provides employment and income to thousands of people
 - Roads have been constructed in the area to transport rice from fields to markets
 - Social amenities like schools and hospitals have been provided improving standards of living
 - Floods that used to occur during rainy season have been controlled
 - The scheme has provided land to landless people enabling them to grow food crops
- 2.
- a) - Irrigation of dry lands.
 - Clearing jungles.
 - Tsetse fly control.
 - Afforestation
 - b) - Flooding in the Yala and Nzoia plains has been controlled.
 - Hectares of land has been reclaimed for agricultural and settlement purposes.
 - The project has brought water borne diseases under control.
- 3.
- a) i) Land reclamation is bringing back the useless land into more useful while land rehabilitation is bringing the land that has been misused by man into being useful
 - ii) – Basin irrigation
 - Overhead irrigation/ drip
 - Trickle irrigation
 - Canal irrigation
 - Shadoof Archimedean screw, sakia water wheel
 - iii) – Irrigation ensures a steady and reliable supply even in arid area while rainfall may fail in a given year
 - River water used for irrigation may bring in silt which makes soil fertile and leads to more yields unlike pure rain water
 - Enables cultivation throughout the year maximizing use of land while rainfall could be seasonal
 - Water drawn for irrigation may also be used for other purposes in the farm
 - b) i) – To settle the landless people
 - The presence of rivers Thiba and Nyamindi
 - Availability of the black cotton soils – good for rice
 - To employ detainees during the days of emergency
 - Unreliable nature of rainfall in the area
 - ii) – Availability of reliable water supplies from river Thiba
 - Presence of fertile black cotton soils with high water retaining ability
 - Gently sloping land making it easy to mechanize and cheaply irrigate by gravity flow
 - Presence of high temperatures favouring rice growing
 - The unreliable nature of the rainfall made it necessary to irrigate

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- The soils were impervious thereby reducing the need to build concrete hence lowering costs
- iii) – Disease – malaria & bilharzias
 - A lot of time is spent to tend crops
 - Presence of numerous weeds
 - Mismanagement of the scheme
 - Delayed payment to the farmers
 - Few extension officers
 - Pests i.e. quela birds
 - Siltation in the canals
 - Expensive human labour
- c i) A polder is an area of low – lying reclaimed land enclosed by dykes, which protect the land against high water level that has to be maintained outside the area
- ii) – Construction of ring canals to drain water out
 - Construction of ditches within each polder which leads water to a pumping station
 - Drying of land through planting of trees/ plants
 - Desalination of the soil through chemical, flushing and planting hardy plants
 - Dividing of land into economic units
 - Laying down of good infrastructure
 - Settling of people in villages
 - Spreading of soils to improve fertility
 - Addition of fertilizers
- iii)- Control of floods of the area to the South West
 - Improved control and distribution of the regions fresh water
 - Damming has cut off salination and pollution of inland waters. This has led to the reduction of salinity of soil hence high yields
 - Islands that were isolated are now within easy reach of developed areas
 - The area is a good site for industry and a tourist resort
- . d) - Artificial insemination A.I is more widely used in Denmark than Kenya
 - There is an experience of high quality yields throughout the year
 - In Kenya, there is a practice of mixed farming while there is specialized farming in Denmark
 - Dairy product in Kenya are consumed locally whereas they are for export in Denmark
 - In Denmark it is highly mechanized while in Kenya it is low mechanized
 - In Denmark, it is evenly distributed all over the country than in Kenya where it is concentrated in highlands
 - Denmark dairy farming rely on fodder while Kenya depends mainly on grass pasture
 - Denmark has got an advanced technology in preservation of dairy products in Kenya
 - Denmark has got a well managed co-operative societies than Kenya
- 4. Heavy expenditure by the government in giving them pensions, money that could have been used to improve other sectors of the economy
 - Through economically unproductive, they require care and feeding which is expensive
 - Through unproductive economically, they require people to take care of them, hence wasting man power that could be used on productive ventures
- 5. a i) - Maize
 - Beans
 - Tomatoes

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- Vegetables
- ii) - Availability of gently slopping land that allowed irrigation water to move by natural force of Gravity
- Availability of permanent source of water from R. Thiba and Nyamindi which provided water for irrigation throughout the year
- Existence of black cotton soils (clay) which hold water on the surface longer for use in irrigation fields
- Sparse population in the region which reduced the cost of resettlement
- Stagnant irrigation water in the fields serve as breeding grounds for water bone diseases nectars e.g. bilhazia workers and mosquitoes
- Leaching of the soils due to over irrigation
- Mono culture which reduces productivity of the soil
- Interference with the life's of the people and animals who depend on the waters of these lines down stream
- . c) - The scheme has created employment which has improved the living standards of the people
- It has saved foreign exchange through production of rice that could be otherwise imported/ which is invested in other sectors of the economy
- It has facilitated urbanization e.g. Mwea town which has expanded trade opportunities
- Has increased productivity by turning less productive land. Hence adding food for consumption and sale to get income
- d i) - Perkerra irrigation scheme
- Yala swamp reclamation/ Bunyala
- Ahero irrigation scheme
- Bura
- Nola
- West Kano
-
- ii) - Is a source of carbohydrates
- Rice foliage is used as fodder
- It is a source of income through sale of rice
- 6. a) Land reclamation is the process by which wasteland is converted into farmland for growing crops and keeping animals.
- Land rehabilitation is a process of restoring land to its former productivity.
- . b) - Irrigation
- Drainage of swamps.
- Tsetse fly control and clearing of jungles
- Afforestation
- c) - Land reclamation by drawing rivers & marshlands
- Upgrading sandy and barren areas through. Use of fertile soil or application of fertilizer
- d) - To control the seasonal flood of river pekerra.
- To utilize excess water of river pekerra that used to go to waste.
- Presence of flat and gently sloping terrain which enables flow water by gravity.
- Fertile loamy soil.
- Dry condition of the area.
- 7. (a) - Improvement of standard of living.
- Settlement of the landless.
- Saving on foreign exchange.

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- Rehabilitation and development of the areas.
 - Control of environmental hazards.
 - Creation of employment.
 - Growth of urban centres.
 - Availability .
- b) - Creating of a large fresh water lake in the middle of the country lake IJssel
- The reclamation polders have attracted towns that have improved infrastructure & social amenities.
 - Has shortened the coastline distance by about 32km.
 - The protection of high tides by dykes.
 - Lake IJssel has helped in reducing the salt content of the neighbouring regions.
 - The total flooding on the coastlines has been completely reduced.
 - Reclaimed land is fertile and suitable for liable cultivation producing crops such as wheat.
 - The IJssel sea has improved the drainage .
 - Reclaimed areas are used for settlement recreation and industry
8. (a) - Rice
- Onions
 - Green grams
 - Pepper
- b i) - Presence of rivers Thiba and Nyamidi which provide water for irrigation
- Gently sloping land enabled water to reach the farms by force of gravity
 - Extensive land for future expansion of the scheme
 - Fertile soils in the area which were suitable for crop production
 - Sparse population which made it easy and cheap to resettle people
 - Low supply of rainfall received necessitated irrigation
- ii) - To eradicate tsetse flies and hence provide conducive healthy living conditions for man and animals
- To enhance the keeping of quality livestock in the area
 - To provide land for farming
 - To provide land for human settlement
- iii) - Building gabions to check the speed of water
- Agro-forestry
 - Re-afforestation
 - Gazettement of forested areas to discourage human settlement and human activities
9. a i) Is land in the Netherlands that has been reclaimed from the sea and enclosed by dykes.
- ii) - Rye - Tomatoes - Flowers
- Barley -Oat - Fodder crops
- Sugar beet - Potatoes - Wheat
- b) - Protective dykes/sea walls are constructed enclosing the part of the sea to be reclaimed.
- Ring canals are constructed.
 - Pumping stations are installed to pump out sea water from the area enclosed by the dyke.
 - Water is pumped out of the area enclosed by the dyke
 - Drainage ditches and more pumping stations are made on the land being reclaimed.
 - Drainage pipes are laid below the soil.
 - The area is divided into rectangular portions using inner dykes and ring canal.
 - The drained land is flushed with fresh water to remove salt from the soil.
 - Soils are treated with chemicals to remove salinity
 - Pumping water from the polders is a continuous process to prevent water from

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- accumulating in the reclaimed land.
10. a) Ahero Perkerra Hola/ bura
 West kano Mitunguu
 Dana katila Kibwezi
 Taveta Gezira Bunyala
- b) - The area was sparsely populated thus making it easy and cheap to resettle the people
 - The presence of river Thiba and Nyamindi which would provide water for irrigation
 - The black cotton soil in the area which was suitable for irrigation because they retain water
 - The fertile soil was suitable for crop production
 - The gentle land would allow water to reach the farm through gravity
 - The unreliable/ inadequate rainfall received in the area made it necessary for irrigation to be practiced
- c) - People who live in the area were originally nomads but now lead a settled life
 - Farmers earn income after selling rice and other crops which enables them to improve their standard of living
 - Tenants have access to credit facilities which they use to improve crop production
 - The establishment of the scheme has created employment opportunities for the people in the area
 - Roads linking the scheme to the market centers have been improved
- d) - Stagnant water encourages breeding of snails and mosquitoes which spread bilharzias and malaria respectively
 - Silting of canals/ weeds in the canals reduce the flow of water into fields. The farmers spend extra time and money dredging the canals
 - Delayed/ low payment discourage farmers
 - Diseases and pests e.g. giulea birds attack the crop which lead to low yield of rice
 - Expensive farm inputs such as fertilizer reduce the framers profit margin
 - Roads are rendered impassable during rainy seasons hence delaying delivery of rice and other crops to the market
11. a i) -Land reclamation is the process through which unproductive land is made useful for agriculture or settlement where as land rehabilitation is the process of restoring land to its former productive state.
- ii) - Afforestation.
 - Reafforestation.
 - Bush fallowing.
 - Use of grass strip and cover crops.
 - Mulching.
 -Constructing bunds (making ridges of soil across a slope)
 - Manuring.
 - Constructing of cut-off drains (digging open trenches across slopes.)
 - Controlling grazing (division of land into paddocks, allowing different sections found to regain pasture.
 - Construction of drainage trenches (digging trenches to drain off excess water)
- b i) - Is land in Netherlands that has been reclaimed from the sea and enclosed hot dykes.
- ii) - Rye - Oat - Wheat - Barley - Potatoes
 - Sugarcane - Flowers - Tomatoes - Fodder crops
- (c) - Protective dykes /sea walls are constructed enclosing the part of the sea to be reclaimed.
 - Ring canals are constructed on the interior sides of the dykes.
 - Pumping stations are installed to pump out sea water from the area enclosed by the dykes.

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- Reeds are planted to help dry out the soil and prevent weeds from growing.
 - Drainage ditches and more pumping stations are made on the land being reclaimed.
 - Drainage pipes are laid below the soil.
 - The area is divided into portions using inner dykes and ring canals.
 - The soils are treated with chemicals to lower salinity.
 - The drained land is flushed with fresh water.
 - Pumping out water from the polder is a continuous process to prevent water from accumulating in the reclaimed land.
- d i) - The presence of River Perkerra which is a tributary of Suguta river in Kerio Valley provide water for irrigation.
- Gently sloping land which permits the flow of water by gravity hence reducing costs of pumping water to the fields.
 - Presences of clay soils rich in mineral nutrients enables variety of crops to be grown.
 - Presence of extensive land which makes large scale cultivation of crops possible.
 - The area is semi-arid hence the need for irrigation farming.
- ii)- Farmers earn incomes which improve their standards of living.
- Irrigation schemes have created settlement for landless people.
 - Many people have been employed in the farms and other sectors within the irrigation schemes such as transport and processing factories.
 - It has facilitated reclamation of unproductive land, thus increasing land for farming. This has led to increase in food production.
 - Some crops grown through irrigation are exported, thus earning foreign exchange. The rice grown in irrigation schemes meets most of the domestic requirements thus saving foreign exchange that would have been used to import it.
 - It has facilitated development of infrastructure in the irrigated areas. Roads have been built to help in the transportation of inputs and commodities.
 - Social amenities such as schools and hospitals have been constructed thus improving the level of literacy and health of the people in the irrigation schemes.
 - Industries that use raw materials produced on the irrigation schemes have been developed.
 - Rice mills have been established on the rice irrigation schemes.
 - Some market centres in the irrigation schemes have grown in size due to increased trading activities e.g. Wanguru in Mwea irrigation and Margat near Perkerra irrigation scheme.
12. (a) i) Land reclamation is the practice by which less useful land is converted into more useful land While Land rehabilitation is the process of recovery/restoration of land which has been misused and destroyed through human activities
- ii) - Bush clearing
- Sterilization of the male fly
 - Use of traps
 - Spraying /use of insecticide
 - Creation of buffer zones
 - Killing or transfer of hosts
- b) (i) – Yala
- Bunyala
 - Kalusi (maragua-Euelyptus)
- ii) - The need to control seasonal floods from river Perkerra

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- The presence of river Perkerra as a source of water for irrigation
- Availability of fertile loaming soils
- The area is gently sloping
- The dry condition of the area necessitated the use of irrigation
- The need to occupy the large population of ex-detained in a productive way
- c) - The stagnant water in the plot and lands encourages the breeding of snails and mosquitoes which spread Bilharzias and malaria. this weakens the farmer causing low labour input hence low production
- Mismanagement of the irrigation projects leading to the collapse of several schemes
- Pest and diseases are attack the crops leading to low production(mention of pest here is important i.e. quelea bird)
- Poor payment for farm produce which discourages the farmers
- Poor extension services to provide for the badly needed technical advice to farmers leading to low productivity.
- Shortage of water during dry season reduced the land under cultivation and hence reduced earnings
- Frequent silting of canals which is expensive to dredge reduces the amount of water required for irrigation
- Inadequate labour which is largely required in the schemes hence burdening the farmer to hire for planting, weeding and harvesting which reduce the profits obtained.
- . d) - Floods caused by the tiles has been completely checked by construction of dams and dykes
 - The dams and dykes have completely cut off movement of salty sea water inland
 - There is plenty of fresh water for domestic and industrial use
 - The distance from Missigen to Rotter dam was shortened by 50KM opening up the area for industrialization
 - More recreational opportunities were created by the newly formed lakes becoming a tourist attraction
- 13. a) - Irrigation
 - Control of pests and diseases
 - Afforestation/reforestation
 - Control of floods
 - Control of soil erosion /construction of gabions
- b) - Protective dykes/sea wells are constructed enclosing the part of the sea to be reclaimed
 - Ring canals are constructed
 - Pumping stations are installed to pump out sea water from the area enclosed
 - Reeds are sown to both absorb excess salts
 - Soil is treated with chemical to lower salinity
 - The drained land is flashed with fresh water to remove salt from soil
- 14. a) - Swamps
 - Dry lands
 - Eroded lands
 - Pest and disease infested
- b) - Regular and reliable
 - Silt increases fertility
 - All year farming
 - Reduces year farming
 - Reduces salinity of soil

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- Dams control floods/ are used for fishing

17. FISHING

1. a i) P – Norway Q- Japan
 ii) - The areas have cool waters which have abundant planktons-main food for fish
 - The areas have shallow continental shelves which allow light to penetrate for planktons to grow
 - Convergence of warm and cool current results in upwelling of ocean waters bringing mineral for fish and plankton from sea bed to the surface
 - Cool waters encourages thriving of numerous fish species
 - Most of the coasts are indented providing secure bedding grounds for fish
 - The sheltered bays provide suitable sites for building fishing ports
 - The large population in these areas provides ready market
 - The rugged landscape limits agriculture thus people turn to alternative economic activity
 - Cool climate provides natural preservation for fish
2. a) i) R – Trawling
 S –Basket fishing
 (b) (i) Basket fishing
 - The basket channel shaped to allow easy entry for fish
 - At the mouth, there is a non-return valve which restricts the outward. The basket is held in the position with ropes/stones/ sticks downstream
 - The basket is left in that attracted position for sometime then removed for landing of fish
 (ii) Trawling
 - A bag – shaped net is attached to the ship trawler
 - The nets mouth is kept open by other boards
 - The upper part is kept a float by corks floats
 - Weights are used to keep the lower parts of the net at the sea bed
 - The trawler drags along the net
 - After sufficient fish is caught the net is hauled to the trawler
 c) - Fishing is restricted to specific reasons to allow for breeding and maturing of fish
 - Standardizing size of nets to ensure fingerlings are not caught
 - Licenses are issued to prospective fishermen to control their number and to ensure there is no over-fishing
 - Fish farming is being encouraged to ensure sufficient supply of fish.
 - There is restriction of the disposal of untreated waste into the sea
 - Artificial fertilization is carried out in special hatcheries to sustain the supply of fish/restocking of over fished waters
3. a) i) It is the rearing of fish in ponds
 ii) - The government through the fisheries department has set up fish ponds and hatcheries as demonstration farms
 - Establishment of research institutions to investigate aspects of fish farming(breeding) diseases
 - Expansion of markets through intensified campaigns aimed at encouraging more people to eat fish
 - Extension officers have been sent to the field to advise on fish farming techniques

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- The government has encouraged the establishment of co-operatives which give credit facilities to fish farmers
 - Establishment of national food policy which encourages diversification of food aid as source of protein leading to setting up of fish farms
 - b) - Inadequate storage and preservation facilities which cause heavy losses on the landed catch/leading to low catch
 - Occurrence of strong sea waves causing accidental drowning/destruction of fish vessels
 - Poor fishing equipment which confine the fishermen to a few metres of continental shelf leading to low catch
 - Sparse population in the north coast and alternative economic activities in the south coast that lead to small market for fish
 - High prices of fish which discourage many people from eating it regularly
 - Local fishermen face stiff competition from foreign fishermen mainly from Korea and Japan who have adequate and advanced facilities for deep sea fishing
 - Warm Mozambique currents and deep continental shelf which discourage the flourishing of fish
 - Regular coastline /few sheltered bays hence few sites for fish landing and breeding grounds
 - c i) - North-West Atlantic
 - North-East Atlantic
 - North – East pacific
 - North –West Pacific
 - ii) - Presence of extensive and shallow continental shelf which allow light to penetrate to the sea bed encouraging the growth of planktons used as fish food
 - Upwelling of ocean waters caused by convergence of warm Kuroshio and cold Oyashio currents bring minerals for planktons from the sea bed hence attracting fish
 - Japan has indented coastline/several bays which provide secure breeding grounds for fish
 - Sheltered bays provide suitable site for building ports/landing sites e.g. Hakkadate
 - Rugged landscape/mountains with infertile soil in some areas which does not favour agriculture hence make people resort to fishing
 - Numerous islands had exposed/provided the fishermen access to marine life
 - The area has cool waters which encourage thriving of fish
- 4.
- a) - A vein is a small crack containing minerals deposited in crystalline form while a cove is a large crack containing minerals in crystalline form ✓
 - b) - Waste of agricultural land ✓
 - Waste of industrial land ✓
 - Ugliness where the land has lost its beauty ✓
 - Health and accident hazards ✓
- 5.
- | | | |
|---|---|--|
| a i)) - Trawling
- Use of herbs
- Use of baskets | - Line fishing
- skinning
- Use of gills nets | - Use of barriers
- Harpooning
- Use of hand lanes |
| ii) – Cod
- flounder
- mackerel | - halibut
- hake | - haddock
- herring |
- (b) i) - Sheltered inlets and estuaries form ideal sites for fishing villages and ports.
 - They also offer ideal site for fish breeding because of calm water and availability of the planktons.

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- (ii) - When warm currents meet cold currents the temperature of water is regulated.
- This condition is ideal for the survival of fish.
- Convergence of these currents improve circulation of oxygen and dispersal of nutrients for the fish
- (c) (i) Provision of the necessary infrastructure such as transport routes to enable exploitation of fishing grounds located in remote areas.
- This will reduce over exploitation on a few accessible fishing grounds like L. Victoria.
- (ii) Fishing Farming
- The development of fish farms complements the fish being caught from natural waters
- (iii) Restocking of over fished grounds.
- Over fished waters are restocked using finger lings from hatcheries or obtaining them from the overpopulated fishing grounds.
- (iv) International agreements on territorial waters.
i. Countries have full sovereignty over territorial water extending for 19 km from the coast.
A further 19 km zone in which the coastal state can take action against those who break the law.
6. i) – Smoking. - Sun drying.
- Salting - Use of refrigerator.
- (ii) - The fishermen are required to have a licence from the government
This prohibits illegal fishing, over fishing and indiscriminate fishing.
- Inadequate capital.
Most of the fishermen use traditional methods of fishing which greatly reduces their catch. They lack adequate funds to enable them purchase modern fishing gear and refrigeration equipment.
--Introduction of new species.
- The introduction of new fish species in the lake has affected the breeding of indigenous fish.
- For example Nile Perch was introduced in the lake in 1960s, it now accounts for more than 80% of the fish harvests, after having caused the disappearance of more than 200 endemic fish species and yet it's not a popular species.
- Presence of weeds.
- Fishing in lake Victoria is greatly hampered by the grown of water hyacinth along the shores of the lake.
- The weed chokes the fishing boats, hence the infested areas are avoided by fishermen, yet they contain a lot of fish.
- Accidents
- Accidents affecting fishermen are common on L. Victoria.
- Have canoes and boats capsize in the lake due to the strong winds and storms prevailing in the areas.
- The fishermen loose their fishing gear and equipment, as some of them drone.
- Regional Conflicts.
- There has been boundary conflict between Kenya and Uganda over Migingo Island on L. Victoria.
- This has caused the fishermen to have fear since most of them are molested or attacked by Uganda Security groups.

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7.
 - The continental shelf is narrow
 - Poor transport connection to the fisheries
 - Local fishermen do not have adequate capital
 - The coastline is fairly straight
 - the water is too warm for fish breeding/lack of upwelling of water
 - There is low demand for fish
 - Fishermen lack modern equipment preservation facilities/storage facilities
 - Inadequate skills/inadequate research technology
 - Competition from developed countries
8.
 - a i)
 - Seine nets
 - Trawler nets
 - Gill nets
 - Drift nets
 - ii) - West N. America
 - N.E Asia
 - b)
 - Over fishing – reduced stocks
 - Pollution – killing fishing
 - Poor transport network – delay reaching market/ go bad
 - Poor market/ fish imports – reduced earnings
 - Inadequate capital – poor equipment/ preservation/ marketing
 - Fish diseases – death of fish
 - Weeds – difficulty to fish
 - Boundary conflicts – fighting/ destruction of equipment
 - c)
 - Source of income/ employment – reduce unemployment/ improve living standards
 - Tourist attraction/ sports/ marine park – forex for developing
 - Rich in nutrients – health
 - Development of industry
 - Health/ feed on mosquito- reduce malaria
 - Feeder roads – improve transport
 - Forex – country development
 - d)
 - Sagana
 - Kabaru
 - Kibos
 - Homa bay
 - Aruba
 - Haller park/ nature trail
9.
 - a i)
 - Pelagic
 - Dermersal
 - nadromous
 - ii)- Presence of numerous foods that form inlets which provide good breeding grounds for fish
 - The ruggedness of the immediate land discourage agriculture and livestock farming thus making fishing a good alternative
 - The prevailing warm Atlantic drift makes the area ice – free throughout the year thus fishing is carried out
 - The grounds have large continental shelf which are abundant in plankton
 - The dense population in western Europe provides a ready market for fish
 - The low temperature conditions in high latitudes encourages flourishing of fish

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- The highly developed technology in the region has contributed to the development of fishing
- iii) - Formulation of law against indiscriminate fishing
 - Restocking the over fished areas
 - Licensing fishermen and use of standardized nets of permitted mesh sizes
 - Prohibiting disposal of effluents into fisheries
 - Introduction of new species in the existing fisheries
 - Research of fish species their habit and migratory trends
 - Personal should be trained on how to manage fisheries
- 10 a) - Drifting method.
 - Seining method.
 - Lining method.b) - It is washed by cold ocean current which influence temperature of water favouring the survival of fish.
 - It has plenty of planktons than the Eastern coasts.
 - It has sheltered inlets/harbours which favour the thriving of fish and construction of fish harbours.
- 11. a) - Seining
 - Trawling.
 - Line fishing.b) - Privacy/sea robbery making fishermen to lose fishing equipment and their catch.
 - Water weeds such as hyacinth impede movement of vessels.
 - Storms sometimes cause accidents in the lake.
 - Territorial disputes with other countries such as Uganda.
- 12. a) - fishing is restricted to specific seasons to allow breeding licenses are issued to prospective fishermen to control their number
 - the size of nets introduced in fishing are standardized to ensure young fish are not caught- fish farming is encouraged to ensure adequate supply
 - there is restriction/control on water pollutionb) - large market both local and foreign
 - modern fishing equipment is being used in fishing
 - it has a large continental shelves/indirect coastal land- low temperature which discourages agriculture hence venturing into the sea as source of income
- 13. a) - Seining
 - Long liningb) - The Coastline is fairly straight/regular/has few indentation
 - The Continental shelf is narrow
 - The water is too warm for the breeding of fish
 - Local fishermen do not have adequate capital/have inadequate preservation facilities
 - Local fishermen face competition from developed fishing countries e.g. Japan
 - There is low demand for fish in the local market
 - Poor transport connections to fisheries resources
- 14. a) Pelagic fish are the fish that mainly live near the surface or at the shallow depths or lakes and seas, while demersal fish are those which live at or close to the bottom of water bodies

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- b) - Over fishing – Some fish species in the fresh water lakes have been over- harvested to the extent that natural replacement has been limited
- Pollution – Some industries release their wastes directly into rivers, lakes and oceans. The wastes are hazardous and make the water bodies inhabitable for fish
 - Inadequate capital – Most fishermen use traditional methods which reduces their catch
 - Inadequate transport – Due to the poor maintained roads the fishermen don't reach the market on time
 - Introduction of new species – This introduction has affected the breeding of the indigenous fish
 - Presence of weeds – Weeds like hyacinth traps the fishing boats hence the areas are avoided

18. WILDLIFE AND TOURISM

1. i) P – Sibiloi
Q - Serengeti
R - Murchison falls

a) National park	Game reserve
• Its fenced off • Managed by the central government	• There's no fence ground it • Managed by the local authority of the area they are located

- b) - Establishment of game parks
- Outlawing illegal hunting and trade in game trophies
 - Setting up of wildlife protection agencies
 - Public education campaigns
 - Protecting the endangered species
 - Animal translocation programmes
 - Encouraging the formation of wildlife clubs and societies in institutions
2. a) Game reserves are areas where wildlife is protected but some limited human activity may be allowed while game park are gazetted regions where settlements hunting and cutting of plants is no allowed whatsoever
- b) - poaching by illegal hunters has reduce the population of animals
- pollution caused by industrial and domestic refuse has greatly affected marine life
 - soil erosion due to improper farming methods has led to degradation of wildlife habitat
3. a) Balance of payment is the difference between visible exports and imports and also invisible exports and inputs in the value of trade taking place between two countries.
- b) - They are perishable.
- Some are bulky making transportation difficult.
 - Are of low value
 - They are exhaustible/supply may fluctuate negatively depending on season.
 - Inadequate capital/ for expansion since a large capital outlay is needed.
4. a) - Remoteness caused by poor means of transport
- Insecurity in the areas
 - Human wildlife conflict where man kills wildlife
 - Hostile climate e.g. drought which leads to death of wildlife
 - Settlement in the people in places meant for game parks and reserves
- b) - Negative attitudes towards local tourism limits the number of people who engage in tourism

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- Inadequate local campaigns and advertisement of tourist attractions/ special packages lead to low public awareness
 - Familiarity with the tourist attractions among the local people makes them fail to appreciate their beauty and value
 - Insecurity from poachers in national parks and game reserves scare people away from visiting them
 - The high cost of accommodation in the game lodges discourages local tourism
 - The roads leading to tourist sites are poorly maintained, discouraging people from visiting such sites
5. a) - Poaching which has led to decline in the number of some animal species.
- Encroachment into conserved land by human settlement.
 - Drought in the National Parks and game Reserves which sometimes cause death of wild animals
 - Overgrazing in the National parks and game reserves cause soil erosion.
 - Inadequate capital has made it difficult to adequately conserve wildlife.
 - Over-exploitation of water resources e.g. fish, seaweeds which pose danger of extinction
 - Migratory behaviour of wild animals which makes it difficult to eradicate /control occurrence of pest and diseases.
- b) - Political neutrality which removes any travel restrictions to the country
- Fluency in many languages that enable easy communication accessible
 - Hospitality of the Swiss that encourage tourists.
 - Trained personnel who provide excellent services.
 - Effective systems which make it easy for transaction to be made

19. ENERGY

1. a) - Saudi Arabia ✓
- Iraq ✓
 - Iran ✓
 - Kuwait ✓
 - United Arab Emirates/ Abu Dhabi/ Bahrain/ Bahrain Island ✓
 - Qatar ✓
- b) - Earning foreign exchange ✓
- Improvement of hospitals and schools ✓
 - Development of manufacturing industries e.g. petrol chemicals ✓
 - Creates employment ✓
 - Development of towns and cities ✓
 - Incomes/ profits have made the countries have high per capita income ✓
 - Revenues and royalties earned from oil have enabled those countries to invest abroad thus increasing their wealth ✓
2. a) - A multi purpose project is a project that serves more than one purpose e.g. HEP, irrigation, fishing ✓
- b) - Seven forks project ✓
- Kariba project ✓
 - Aswan high dam project ✓

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- Akosombo dam project✓
- Cobra Basa project✓
- 3. a i) The sun, wind, water, wood, waves and tides, geothermal steam, biomass and animal
 - ii) - It is always available as long as it is blowing
 - It is cheap source as we do not pay for it
 - It is clean source which does not pollute environment
 - It can be produced on small scale for local consumers
 - The land occupied by windmills can also be simultaneously used for cultivation of crops
- b) i) Geothermal power is electricity which is generated from the earth's internal heat that reaches the surface through geysers and hot springs
 - ii) - It is relatively cheap to produce
- Generation of electricity is continuous because steam is continuously being produced naturally
- The cost of operating of geothermal plant is relatively low
- Geothermal steam is renewable source
- It is a clean source of energy which does not pollute the environment
- it helps reduce over-dependence on exhaustible sources such as fossil fuels
- .c) - Government bureaucracy and political interference
 - Inadequate capital for investment
 - Lack of skilled labour
 - Inadequate technology
- d) - People are being encouraged to use energy-saving devices such as the energy-saving Stoves
 - the government is encouraging the use of alternative sources of energy such as wind, the sun and biomass.
 - there is development of wood fuel programme through a forestation and reforestation involving planting of quick-growing trees.
 - The government uses consumer prices to discourage unnecessary uses of oil as fuel
 - People are being encouraged to switch off electrified gadgets when they are not being used
 - People are also encouraged to have proper maintenance of motor vehicles to reduce the amount of fuel consumed
- 4. a i) - Refers to the chain of negative reactions emanating from a fuel shortage of essential energy
 - ii) - Over reliance on one source of energy (oil)
 - High increase in oil prices by oil producing countries
 - Hoarding of oil to create artificial shortage leading to skyrocketing of prices
 - Depletion of fossil fuels
 - War which hinders mining and supply of oil to the world market
- b i) – Coal
 - Uranium
 - Petroleum
- ii) - Turkwel power project
 - Sondu Miriri project
- iii) - Steep gradient that allowed fast flow of water (water falls) to turn turbines to produce Electricity

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- Permanent source of water from RTana which provided plenty and regular supply of water to turn turbines and provide electricity throughout the year
 - Sparse population in the region which reduced the cost of resettlement
 - Impermeable soils in the region which reduce water loss from the reservoir through seepage
 - Presence of hard basement rock which provided firm foundation for construction of dams
5. a) - Inexhaustible
- Versatile- many uses
 - Reduces dependence on oil
 - Cheap/ free access everywhere
 - Minimum maintenance cost
 - Environment friendly
 - Can be stored
- . b) P- Masinga
Q – Kamburu
R – R. Kindaruma
- . c) - Industrial growth – creation of employment/ goods
- Transport – employment/ enables trade
 - Agriculture e.g. tractors e.t.c. – food production
 - Water supply – industrial use
 - Health e.g. X- rays – healthy work force
- . d i) - Reduced industrial production – shortages/ loss of employment
- Increased prices/ fares – limits purchase/ travel
 - Domestic problems e.g. darkness/ cooking e.t.c. reduced standards of living
- . ii) - Put off gadgets not in use
- Keep gadgets in good working condition
 - Use of public transport
 - Increase capacity of public vehicles
 - Better roads – avoid traffic jams e.t.c.
 - Use fewer lighting bulbs
 - Minimise outdoor advertising i.e. bill boards
 - Energy saving jikos/ stoves
 - Use low capacity cars
- 6 . a) - Renewable sources of energy are those sources that are continually being replaced or regenerated.
- They are considered inexhaustible
- b i) - Presence of steep gradient of where there is a waterfall through out the year to make the operations economical
- Hard basement rock to provide a firm foundation for the construction of a dam and accommodate weight of the reservoir
 - Presence of a deep valley-where there is a deep valley or river gorge, it provides space for the reservoir therefore saves on the costs of constructing a dam.
 - Presence of non-porous rocks to prevent seepage of water from the reservoir
- ii) - Adequate capital to develop more plants in Olkaria and other potential areas
- Low level of technology in exploitation of geothermal

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- Economical imbalance between the cost of setting up a station and the subsequent power outlay hence need for more stations.
- Potential areas are scattered and located in remote sparsely populated areas which are not potential markets for electricity produced
- High transmission costs of the potential markets
- c i) - It leads to an increase in prices of imports and other locally manufactured goods.
 - Third world countries spend huge sums of money on importation of crude oil
 - Affects balance of trade since earnings from exports will be lower than imports.
 - Results into inflation since the government passes the costs to consumers
 - Oil crisis lowers GDP of third world countries since there will be more expenditure than investments
 - It causes recession making an economy unable to create more jobs
 - Causes a rise in living standards of people
 - Increase production costs in other industries
- c .ii) - Reducing the consumption rate of energy by encouraging people to put off electricity when not in use
 - Rationing of power by the distributing company
 - Encouraging the use of public vehicles, proper maintenance of vehicles to cut down on the amount of petroleum used/consumed
 - Encouraging the use of alternative sources of energy like biogas and solar energy that can be renewed hence saving on non-renewable energy sources.
 - Ensuring the development of energy saving devices e.g. jikos which use wood/charcoal
 - Development of industrial machines which can use coal that is cheaper as an alternative form of energy
 - Reduction of taxes on gas and solar panels to cut down on the use of wood fuel
- 7. (a) - Leads to increase in prices of imports which result in material shortage lowering of the gross domestic products (GDP) and creates a recession hence economy cannot be able to create more jobs
 - Leads to inflation i.e. the rise of prices of various commodities leading to reduced consumption of goods and services hence drop in profits
 - Increase in price of other forms of energy due to higher transport and production cost
 - Decline in agricultural production due to decrease in and under maize and wheat and reduced tourist activities due to the hike in oil prices that result in higher transport costs. air fares rise steeping making it expensive for tourists to travel
 - Depletion of foreign exchange reserves due to the increase in the oil import bill, the country spends a lot of its revenue on purchase of unrefined oil
 - Environmental degradation due to search for alternative sources of energy hence demand for harcoal and wood fuel causing deforestation and soil erosion
- b) - To conserve energy means the efficient utilization of energy to avoid wastage
 - Encouraging he use of solar for heating and lighting
 - Encouraging Kenyans to put off electricity gadgets when they are not being used
 - Proper maintenance of motor vehicles
 - Encouraging the use of public transport instead of personal cars
 - Encourage the use of energy saving jikos /stores to reduce wood fuel used
- 8. a) Is usually portrayed as a question of price and supply uncertainties and the rapid

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depletion of fossil fuels

- b) - Destruction of forest as alternatives source of energy
 - Promote erosion due to deforestation
 - Destabilize ecological balance
 - Facilitate desertification due to deforestation
- 9. a)
 - Oils.
 - Natural gas.
 - Coal.
 - b) - Kenya would save foreign exchange it used to spend on oil inputs and channel the funds to projects.
 - More industries would be established because industrial fuel would be cheaper/increase in investments. .
 - More job opportunities would be created in the oil sector and other related industries thereby improving standards of living of Kenyans. .
 - Transport costs would reduce leading to cheaper commodities in the market. .
 - Kenya would earn more foreign exchange through oil exports and these earnings used to develop other sectors of the economy. .
 - Infrastructure of social amenities improved leading to better standards of living.
 - c) - The government is encouraging the use of alternative sources of energy such as ,
Geothermal biomass, wind & sun.
 - The government is developing modified programmes through afforestation, reafforestation & planting of fast-growing trees.
 - Controlling importation of vehicles with high engine capacity.
 - People are encouraged to use energy saving jikos.
 - A lot of emphasize is being put on protecting the existing forest by resettling people]who have settled on forest land.
 - Encouraging the use of public transport by improving it so as to reduce the number of private cars on the roads.
- 10. i) - Conducting reconnaissance
 - Preparing relevant tools & equipment.
 - Preparing working schedule.
 - Forming groups.
 - Content analysis/doing more research.ii) - Direct observation.
 - Interviewing.
 - Administering Questionnaire.
 - Photographing/filming.iii) - Lighting.
 - Heating.
 - Powering machines.
- 11. a i) - Coal
 - Natural gas
 - Uranium
 - Petroleum products/oilii) - a large and constant volume of water harvested from large river with large volume of

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water through out the year

- a deep narrow valley/gorge to minimize the construction cost.
- Sparse population in order to minimize relocation/resettlement cost
- A fall water /water fall-water falling from a high point to supply force required to rotate the turbines
- A hard basement to reduce the amount of water that will seep into the ground and provide storing foundation for the dam
- Adequate capital for dam construction, transmission of power and compensation displaced people

b i) Energy crisis is the price and supply uncertainties that are usually accompanied by the rapid depletion of fossil fuels

ii) - overdependence on oil and its products

- depletion of wood fuel
- exhaustion and deepening of coal mines
- artificial shortages cause when some countries decide to conserve their resources
- wastage and misuse of energy
- wars/disagreements within oil producing countries

iii) - every crisis has led to petroleum price adjustment causing a general inflation and the prices of various commodities.

- Some industries in the economy are affected to a points where they lay off some of their staff
- The price increase usually leads reduced consumption of goods and services leading to a drop in profits

12. a) - Petroleum

- Gas

b) - Fishing

- Irrigation
- Industrial/ domestic use
- Water ways
- Tourism
- Regulate river flow
- Water storage

c) - There could have been bumper harvest of maize locally.

- There might have been sufficient food for all.

- The country may not have experienced famine to necessitate emergency importation.

d i) - Rift Valley

- Western
- Central
- Coast.

d ii) - Maize stalk is used as cattle feeds.

- Cobs /stalk is used as fuel.
- Grain is used as human food.
- Industrial material – oil.

13. a) E – Owen falls Dam in Uganda

F – Akosombo Dam in Ghana

G – Kariba dam in Zambia/ Zimbabwe

b) Renewable sources of energy are those that have the capacity to be regenerated/ reused

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e.g. sun, wind, water while non-renewable sources of energy are those that can be exhausted if not well managed such as coal, petroleum and natural gas

- c) - Availability of large volume of water from the River Nile
 - Presence of a natural water fall, the Owen falls; to turn the turbines
 - Presence of a hard basement rock to support the weight of the reservoir
 - Availability of a ready market for the power from the surrounding high population
- d) - High seasonal fluctuations in the flows of the river regime especially in times of drought
 - Inadequate capital since H.E.P require high capital outlay which many African countries lack and they rely heavily on foreign aid
 - Poor maintenance of machinery at power production stations caused by inadequate capital to buy spare parts for the machines
 - Siltation of reservoirs caused by poor farming methods upstream and lack of machinery to remove the silt from the reservoirs
 - Small markets for HEP hinder production due to high poverty level of the population found in the counties
 - The big rivers such as Niger, Nile, Volta, Zambezi and Orange have their waters reduced by evaporation because they pass across dry regions

20. INDUSTRY

- 1. - Created variation in prices of petroleum products
 - Led to mushrooming of many petroleum products dealers
 - Encouraged importation of refined petroleum products
 - Caused frequent price adjustments by oil dealers
 - Created widespread / availability of petroleum products
- 2. a) Industrial inertia is the tendency of an industry to maintain its location and activities after factors of its initial location have ceased
 - b) - Presence of an already established transport system
 - Availability of skilled labour
 - Utilization of out-puts of other plants as raw material
- c) - it would encourage setting up of industries in the areas thus stimulation decentralization of induction
 - it would reduce the cutting down of trees and electricity would be available for domestic use.
 - It would attract/improve social amenities in rural area reducing the need for people to move to urban areas
 - Most people would invest in the rural area which would led to higher standards of living
 - It would encourage development of horticultural farming to have ideal storage for perishable products
- 3. a) -it is the inability of an industry to move from an area through original location factors here changed/exhausted/are no longer existing
 - b) - it has resulted in pollution leading to environmental deterioration
- it has led to rural-urban migration depriving the source area able bodies people
- it has led to uneven economic development
- other economic activities e.g. agriculture production have been underrated in favour of industrial activities
- due to technological advancement some people have been rendered jobless
- 4. a) An industry is an enterprise or a commercial profit making undertaking while a factory

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refers to the structures or buildings which may house an industry

4. b)
- Transport
 - Communication
 - Banking
 - Insurance
 - Trade

21. TRANSPORT AND COMMUNICATION

1. a i) – Demand
- Availability of infrastructure
 - Topography/ terrain
 - Capital availability
 - Political factor
 - Historical factors
 - Climate
- ii) – Difficult terrain – This makes it difficult to develop transport and communication
i.e the great rift valley, the rugged scarps inselberg and numerous rocks out crops
- Harsh environments – e.g. Sahara desert and many isolated others
 - The harsh desert climate associated with low population. Such makes it uneconomical to construct the transport and communication facilities
 - Non – navigable rivers
- This is due to
1. Fluctuating water levels
 2. Presence of rock outcrops
 3. Rugged terrain i.e. many rivers have rapids waterfalls
- (b) i) A – L. Michigan
B – L. Huron
C – L. Erie
- ii) - To provide cheap water transport for bulky commodities like coal and iron ore to industrial centres around the great lakes
- To shorten the route to external markets especially for wheat from Canadian prairies and manufactured goods from the Great lakes
 - To expand manufacturing industries along the entire route, provide employment and develop towns
 - Increase agricultural production and tourism
 - Open up this part of America for international trade
 - To ease strain on American railways
 - To increase HEP production along parts of the Seaway e.g. at Niagara waterfall and at dams constructed along the route
- iii) – The long sault and Lachine rapids between Ontario and Montreal hindered navigation. Waterfalls e.g. Niagara was preserved for HEP and tourist attraction. This was solved by construction of alternative route i.e. Welland canal
- Shallow ports e.g. Montreal that could not handle large ships – they were dredged and deepened
 - Problems of trans- shipment along the seaway causing delay. Seaway has been widened to accommodate large ships and these can now sail through the water way without stopping

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- c) - Reduces deterioration of Nairobi – Mombassa highway due to reduction of tankers
- Has reduced cases of accidents especially on Nairobi – Mombasa highway
- Very fast and convenient means of transporting such highly risky products since the products are firmly sealed in the pipe
- Less labour used after installations i.e. cuts down on labour costs
- It cuts down on occurrences of fuel spillage and pollution causes by derailment or accidents on trail or roads
- Different kinds of petroleum products for various industrial and domestic uses can be conveyed one pipeline
- Ensures constant supply of petroleum
- d) - It has promoted international cooperation created by no distance
- has led to expansion of trade for aircraft goods especially those with perishable goods to international market without delay
- Has provided many people with employment opportunities e.g. pilots, engineers e.t.c.
- has resulted in loss of life especially when passenger plane crashes people die in large numbers
- It has resulted into environmental pollution through discharge of wastes and fuel into the space
- Has led increased terrorism through plane jacking
2. a i) Transport is the movement of goods and people from one place to another
ii) Road, pipe land, railway
b i) N – 500 canal
- Q – New York state Barge canal.
Lakes marked O – Lake Michigan P - Lake Erie
Port marked M – Duluth.
- ii) - Accessibility to raw materials which has led to extensive industrial development in the area.
- Cheap means of transport in import and exports has encouraged interval and international trade.
- Provision of H.E.P from the dams f or both domestic and individual use.
- Growth of port and towns like Duluth etc. have become points for various economic activities.
- The sea way is a tourist attraction which generate income in the region.
- Employment opportunities have been created in the transport industry thus raising the living standard of the local people.
- The lakes and the dams are a source of water for both domestic and industrial use.
- The two countries earn some revenue from toll charfes levied on ships that use the route
- c) - Communication enhances trade
- Many people are employed in the communication sector.
- It promotes international and domestic tour ism.
- Governments disseminate information through the media.
- Communication promotes international understanding which lead to increased trade thus boosting economic development.
- d) - African governments do not have adequate capital for modern transport equipment/system.
- Stiff competition from well established air lines in the developed countries.
- High insurance cost of planes discourages investment in air transport.

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- Mismanagement of African national air lines incur huge losses.
- Some regions in Africa are not served by direct flights and thus increases the cost of travelling to passengers.
- Small/low volume of passengers and cargo makes it expensive for some airlines to operate.
- Insecurity in some parts of Africa is also discouraging the use of air transport.
- 3. a) Communication is the exchange of information, ideas a, messages or opinions among people.
- b) - Liberalization of airwaves/licensing of more private radio stations/TV stations.
- Introduction of the fiber optic cable to enhance communication via internet /e-mail fax (telefax)
- Licensing many companies to provide cell phone services.
- Liberalization of the press.
- Expansion of telephone facilities.
- Liberalization of postal services.
- High competition from faster and more efficient electronic mail.
- High cost of installation.
- Vandalism of telephone equipment.
- Mismanagement
- Poor reception /overlapping of telephone line.
- 3. c) - High competition from faster and more efficient electronic mail.
- High cost of installation.
- Vandalism of telephone equipment.
- Mismanagement
- Poor reception /overlapping of telephone line.
- 4. a) - Trade would grow and develop in Kenya
- There will be more diplomatic ties with rest of East African states.
- More Kenyans will get employment in other East African countries
- Kenya would earn more revenue paid y the users form East African states
- b) - Inadequate road signs make the drivers to cause accidents
- Slippery roads lead to accidents
- pot-holed roads surface cause vehicles to breakdown
- Traffic jams, especially in urban centres cause delays
- Narrow roads/bridges makes the drivers to strain and cause accidents
- Inadequate drainage system of roads especially in towns cause accidents
- 5. a i) – Road
- Railway
- 5. a ii) - Fluctuations/ interruptions e.g. freezing
- Obstacles – usually unseen
- Limited network
- Incase of accident loses are certain and enormous
- Relatively slow
- Expensive – ports/ habours/ facilities
- May be in uneconomical areas
- Contribute to water pollution
- b) - Influence of choice of location of economic activities – moving of raw materials/ finished Products
- Expands markets size – possible to move goods and transfer information
- Employment creation – reduce unemployment/ improve living standards

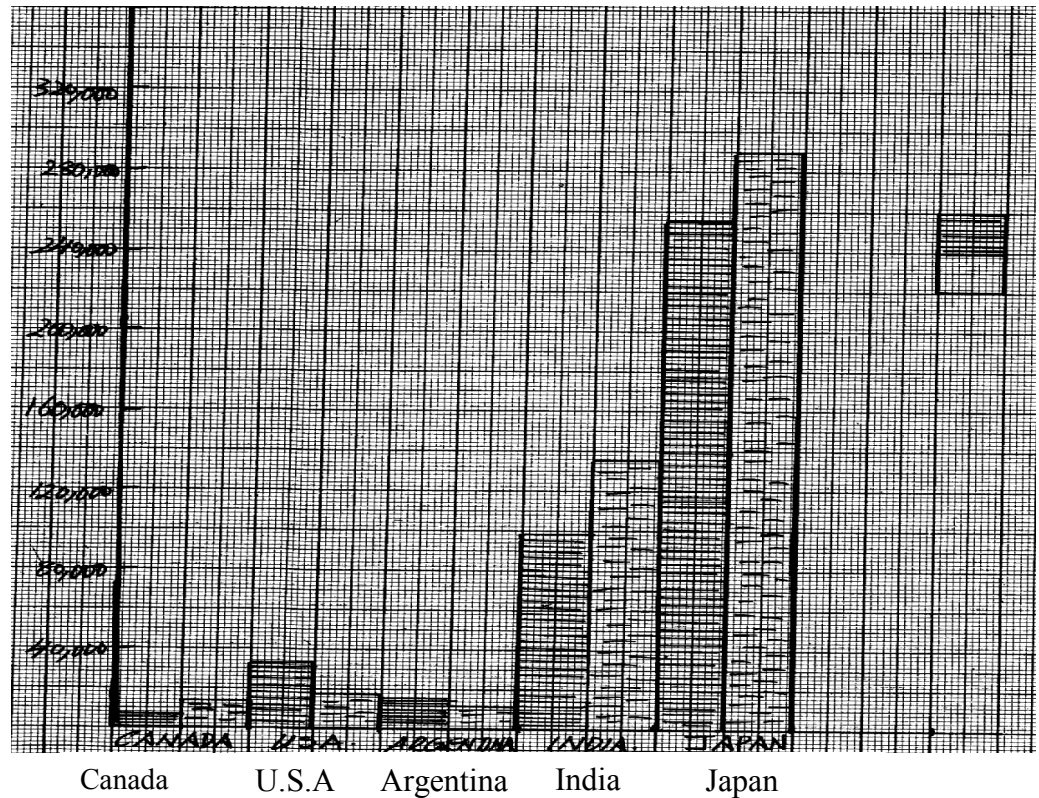
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- Information/ movement hence tourism hence forex
- Revenue through tax – development
- Dissemination of information brings about change
- Leads to urbanization
- . c) - Creation of trans Africa high ways to increase connectivity
- Creation of international railways to increase connectivity
- Regional economic integration to increase trade hence the need for networks
- Establishment of natural networks and transport/ communication companies to increase local transport/ communication
- d i) - It is an artificial water way
- ii) - Suez
- Panama

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6. a i)

No. of passengers in millions



Key
1966
1977

- ii) - Easy to construct/draw
 - Easy to compare
 - Easy to read
 - Give clear visual impression
 - Depict data more accurately

$$6. b) \frac{6000-4800}{4800} = \frac{1200 \times 100}{4800} \\ = 25\%$$

6. c) - African countries have railways of different gauges which make it difficult for them to join
 - The countries were colonized by different European powers who constructed railways to transport raw materials from the interior to the ports within their own colonies
 - Political difference /different political ideology /political stability among African countries hinder efforts to construct railway line to link them
 - African countries produce similar goods hence there is limited trade between them and this does not warrant them to construct railway lines
 - Railways are expensive to construct/inadequate capital and hence expensive to construct
 - Parts of Africa are unproductive so it would be uneconomical to construct railway lines
 - Variation in train has hindered the development of various lines
6. d) - narrow roads where heavy traffic limit ease of movement and overtaking
 - the pot-holes sections of the roads may cause tyre burst/vehicle breakdown /may make drivers who are avoiding potholes crash vehicles
 - The narrow bridges may cause vehicles to crash
 - Sub-standard surface may cause vehicles to skid/overtake

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- Dusty roads may reduce visibility leading to accidents
 - 185 - muddy roads during rainy seasons may cause vehicles to collide
- 7. a i)
 - News papers and magazines
 - Radio and television
 - Telephone/mobile phones
 - Internet
 - Carrier service
- ii)
 - Provide a reliable continuous flow of the commodity being transported
 - It is relatively cheap/inexpensive to operate
 - It is a safe mode of transporting oil, gas and water
 - Are free from accidents/accidents are minimal
 - It does not pollute the environment unless there's a leakage
- bi)
 - Airports are very expensive to build
 - Buying and maintaining the aircraft is a very expensive venture
 - In case of accidents there are usually few survivors and the losses are very high
 - It is very expensive to travel by air
 - The transport of bulky goods on planes tedious is and expensive
 - The government requires a lot of land to put up airports
 - The cost of insuring plane is very high
 - Running airlines requires skills and personnel
- ii) - Creation of employment opportunities as they are industries. This leads to raising the standard of living of the people.
 - Promotes international and domestic tourism hence earns foreign exchange to the countries concerned
 - Expansion of new trade areas and markets hence boosts industrial, commercial and agricultural activities hence development
 - Leads to increased mobility of labour thus reducing cases of labour shortages in industries and other sectors
 - It has promoted international understanding as people are able to interact and learn about each other
 - It generates income for the government and the people of the country. The income earned boosts the economic development of the country
 - Growth and development of different settlements is influenced by the availability of transport networks. As towns develop other facilities expand hence economic development
- c i) - To regulate the difference water levels along the sea way through dredging
 - To smoothen the river channels by removing the existing rock outcrops, rapids and small
 - To regulate the flow of St. Lawrence River through the construction of dams and locks
 - To promote trade and industrialization in the two countries
 - To remove the silt that was brought about by depositing between lakes Erie and Huron
- ii) - Soo canals
 - Welland canal
 - Trent canal
 - New York state barge canal
- 8. a i) (i) Lake marked P – L. Erie
 - ii) River marked Q – Detroit river.
 - iii) Canal marked R – Soo Canal
 - iv) Port marked S – Duluth.
- b) - Provides easy means of transport for both imports and exports, thus encouraging

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- internal and external trade.
- It has led to growth of parts and towns along its course.
- Due to accessibility to raw materials, there has been industrial development in the area.
- The dams along the route provide hydro-electric power for industrial and domestic use.
- The sea way is a tourist attraction which generates income in the region.
- The sea way has created employment opportunities in the transport and industrial sector raising the standards of living of the people in the area.
- Tariffs charged earns the country incomes.
- c) - Most rivers in Africa pass through different climatic regions with alternating wet and dry seasons. This makes the volume of water fluctuate from season to season making it difficult to use the rivers.
- Many rivers in Africa have rapids and water falls which hinder the movements of vessels.
- Some rivers in Africa have floating vegetation that makes it difficult for vessels to move through them.
- Many rivers in Africa are either short, too shallow or too swift making navigation on them difficult.
- Rivers are affected by siltation at their mouths and along their courses. This reduces its depth required for the vessels to move.
- African countries have insufficient capital to use in the development of water ways, ports and for purchasing vessels.
- African countries have inadequate technology. This hinders the process of developing river transport.
- Some rivers in Africa flow across different countries. This calls for negotiation between countries concerned. This hinders the development of river transport.
- Most of the rivers pass through unproductive areas making it uneconomical to develop
- d i) - Liberalization of airwaves/licensing of more private radio stations (T.V. stations).
- Introduction of E- mail /internet/ Fax. e.t.c
- Introduction of mobile phones/cell phones/pagers.
- Expansion of telecommunication facilities to new areas.
- Liberalization of postal services.
- ii) -Development of other faster and more efficient means of communication e.g. electronic mail has led to reduces use of telephones.
- High costs of installation and maintenance of telephone lines limit the number of the subscribers.
- Vandalism of the telephone equipment renders most of the telephone services unavailable to would be users.
- Mismanagement in the organization that provides telephone services has made it difficult to expand the services to many areas of the country.
- Poor reception/overlapping of the telephone lines limit the use of the facility/description by natural hazards.
- Lack of modernization of telephone in some areas causes delay and discourages the use of telephones.
- 9. (a) Is the parking of goods in large standardized box – like structure which once filled in a factory /exporter can be transported
- b) - Saves space in ships as their dimension is more or less uniform.
- It speeds up the process of handling goods hence saves time.
- It is relatively cheap as it requires little labour.\reduce loss of goods as they are sealed

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at the exporter premises and delivered to the importer without the seal being broken.

- Reduces loss of goods through breakages and spoilage.
 - Checking of goods is fairly easy.
10. (a) (i) - Facilitates faster business transactions due to fast means of transport hence increasing volume of trade
- Through M- Pesa and M- Kesho it has facilitated money transfer for business hence increasing trade transactions and investment
 - It has created employment which improves the living standards of the people
 - The company pays taxes which are used to improve other sectors of the economy
- ii) - Facilitates theft cases/ aids stealing
- It is expensive to acquire the device and to maintain it
 - Health associated risks e.g. hearing impairment and cancer
 - Can easily be lost
 - Lack of network coverage in some areas
 - Noise pollution over ringing
- b) - Most rivers are seasonal
- Many major rivers are unnavigable due to presence of water weeds, cataracts and water falls
 - Meanders on some rivers make the distance longer than land routes
 - Many rivers flow through empty or inhospitable lands with little economic value
 - Siltation especially near the river mouth makes the channels shallow
- c) - Construction of super high ways where by each country is responsible for construction of the stretch within her national boundaries to improve linkage of countries
- Establishment of regional trade blocs that spearhead the development and management of transport network among member states ensures good transport network in Africa
 - Negotiations among leaders of various African nations to provide security to avoid high way robbery

22. TRADE

1. a) - Bilateral trade
- b) - Low earnings due export of agricultural products and raw materials which have low value
Heavy expenditure due to importation of manufactured goods which are expensive
- Unpredictable production since agriculture she relies on for export depends on climate which is unreliable
2. a) - Bilateral
- Multilateral
- b) - Similar products
- Limited transport/ communication links
 - Colonial patterns of trade
 - Limited manufactured goods
 - Prohibitive tariffs to protect local industries
3. (a) - Scarcity of goods
- Inadequate capital supply
 - High costs of getting trading permits/licences
 - Lowly developed roads/transport network
 - Trade barriers

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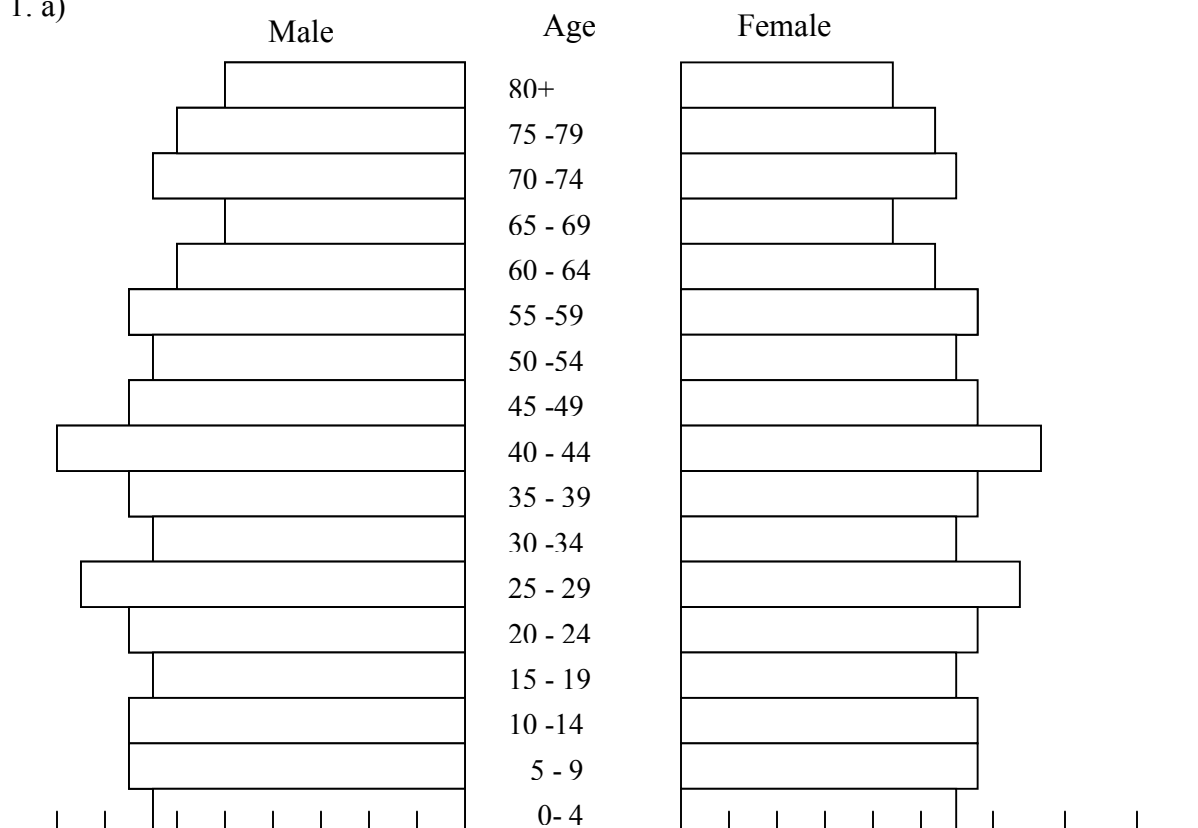
- High cases of poverty among the people
- Inadequate market buildings
- Insecurity
- Smuggling of essential goods
- b) - They have helped crate cooperation among member states
- Member states have a longer market for their goods
- The expanded market ha promoted industrial development
- Goods have been made cheaper for people in the region through reduction of tariffs
- Employment opportunities have been created through industrial development
- Member states are able to invest in joint development projects like railway construction.
- Trade in the regions has boosted agricultural development
- The common market has made people of the member state enjoy a variety of commodities
- There is reduced reliance on goods and services from other parts of the world
- 4. a) - To eliminate taxes on goods produced within the member countries.
- To enable the member states to increase use of their raw materials.
- To enable people in the region to interact and exchange ideas freely.
- To reduce unnecessary competition among member states.
- To promote transport & communication between the countries.
- To create a common market for the goods produced in the member countries.
- To establish a common bank COMESA bank to aid transaction.
- b) - Improving infrastructures.
- Joining trading blocs such as COMESA, EAC.
- Creating Export Processing Zones. (EPZ) to promote the volume of exports.
- 5. a i) Visible exports are tangible goods sent o other countries for sale while invisible export are transactions between countries which lead to monetary returns
- ii) -Tourism
- Financial services
- Transport services
- Loans/ grants/ Aids
- b i) - COMESA
- Southern African Development Community (SADC)
- ii) Berlin Niger Nigeria Ghana
- Burkina faso Mauritania Liberia
- Mali Guinea Senegal Togo Cape town
- Cote de voire Gambia Guinea Bisau Cameroon
- iii)- Encouraged the development of industries
- Phased out all customs and tariffs on goods originating within West Africa
- Improvement of tele- communication
- Exchange of technology
- Campaigned for the sale of petroleum from Nigeria to member states at reduced prices
- 6. a) Balance of payment is the difference between visible exports and imports and also i nvisible exports and inputs in the value of trade taking place between two countries.
- b) - They are perishable.
- Some are bulky making transportation difficult.

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- Are of low value
 - They are exhaustible/supply may fluctuate negatively depending on season.
 - Inadequate capital/ for expansion since a large capital outlay is needed.
7. a) Visible exports are tangible goods sent to other countries for sale while invisible exports are transactions between countries which lead to monetary returns like interest and dividends on the foreign investments
- b) - Hinder growth of home infant industries due to stiff competition/ slows industrialization
- Creates a state of dependency on developed countries/ slows exploitation of national resources
 - Hinder diversification of the economy
 - The country experiences unfavorable balance of trade/ retards economic development
 - Whenever there is poor relationship with the trading partner the country experiences shortage of the import goods
 - The country spends her foreign exchange reserve on imports/ faces devaluation of her local currency

23. POPULATION

1. a)



b) - Has a high life expectancy.

- Has a large working population.
- Has a low dependency ration.
- Has a low fertility rate.
- Has low death rate

c) - Inadequate manpower making labour expensive.

- Rural depopulation due to increased urbanization/leading to labour shortage.
- High old age dependency ratio due to high life expectancy.
- Under utilization of social amenities due to low birth rates.

d i) – mortality rate is the number of deaths in a population of 100 people per year.

d ii) - Improving medical facilities and immunizing children to control disease.

- Educating parent on child care during pre natal period.
- Educating parents to have planned families.
- Encouraging parents the benefits of breast feeding and balanced diet.

2. a) - Population census

- Sample surveys

b) - Improved nutrition and medical care which lowered mortality and increased fertility hence leading births and longevity

- Increase in early marriages which increased reproductive life span hence increase in birth rate
- Low level of family planning due to low awareness leading to large families hence rapid population growth

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- Many people were still entrenched in cultural beliefs which favour having large families for security investment hence rapid population growth
 - c) - High literacy level and awareness on need to have small families
 - Most of the women go to school and spent more time in schools or colleges thus has reduced indulgence into sex or early marriages by most women
 - High abortion rate which damages the reproductive system
 - Misuse of family planning gadgets and drugs which damage the reproductive system
 - d) - Early vaccination against polio, measles e.t.c.
 - Provision of free mosquito nets to expectant mothers
 - Free health care to all infants in government hospitals
 - Training of traditional aids to birth delivery on modern methods of birth delivery
 - Expansion of health facilities to all rural areas to increase access to health care
 - Employing nutritionists to educate mothers on better nutritional practices to evade child mortality
3. a) - Population distribution is the way people are spread out on the land whereas population density is the number of people unit area of land.
- b) - A high rate of unemployment.
- High crime rate as people seek ways of supporting themselves.
 - High demand for social amenities.
 - The dependency ratio decreases.
 - High demand for food sometimes leading to food shortages.
 - Strain on natural resources and scarcity of land.
4. a i) Life expectancy is the average age to which the people of a country expect to live/the average age at which people die
- ii) - The composition by sex
- The size of population
 - Proportion of dependency ratio
 - Different age groups
 - The proportional males to females
- b i) - Population growth rate is high in Kenya and low in Sweden
- Kenya's population has a large number of young people below 20 years of age while Sweden has a high medium age population
 - Death rate is high in Kenya and low in Sweden
 - The fertility rate is high in Kenya and low in Sweden
- ii) - Natural hazards e.g. floods force people to migrate to other areas for safety
- Pastoralists migrate from one rural areas to another in search of water and food for their livestock
 - Land disputes make people move and settle elsewhere
 - Pressure on land makes people to move and buy land elsewhere for settlement
 - Insecurity in some areas forces people to move
 - Settlement schemes attract people to settle in them
- c) - Causes high dependency for social basic needs
- Leads to high rates of unemployment
 - Leads to food shortages
 - Leads to land fragmentation and this decreases agricultural production
 - Leads to high rates of crime

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5. a) It is the fertility achieved after the abstinence that is continued, when the initial fertility was broken.
- b i) - Natural calamities
- Low nutritional standards/famine /lack of food
 - Conflicts
 - Other epidemics/diseases.
 - Inadequate/poor medical facilities
 - Road carriage.
- ii) - The sickness leads to absenteeism from work/reduced productivity.
- Money spent in treating the sick could be used for other economic activities.
 - Deaths resulting from the disease lead to loss of economically productive population.
 - Care-takers at family level use moral time caring for the sick/orphans instead of engaging in economic activities/high dependency ratio.
- c) - Production of similar goods
- Failure to remit annual subscriptions by members
 - Different levels of industrialization
 - Poverty among the population in the regions
 - Poor transport and communication linkages
 - Desire to [protect local industries by member countries
 - Lack of common currency
6. a i) Is the number of unresourceful people between zero to fourteen years and above sixty five years per every 1000 resourceful people
- ii) - High birth rate
- Low death rate
 - Scarcity of employment opportunities
- (b) i) -Presence of large towns e.g. Kisumu
- High rainfall
 - Fertile land
 - High fertility rate
- ii) - High employment rate
- Pressure on social amenities e.g. schools
 - Rural- urban migration hence overcrowdings
 - Too many dependants
 - Pressure on land
 - Food shortages
- c)- Acceptance of family planning methods
- Career advancement among Swedish women
 - Decrease in infant mortality
 - Advanced living standards that ensure only a small family can be catered for
7. a) Is the enumeration of the people in a given area and the compilation of demographic, social and economic information of the population being enumerated at a given time
- b) - To determine the composition of the population
- To know the trends and levels of mortality and fertility
 - To plan for provision of basic facilities

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- To aid in creating new administrative units
 - To estimate the dependency ratio
 - To know the literacy level
 - To know labour supply and predict any unemployment problems
8. a) - Emigration is the movement of people out of their country and settling in another while Immigration is the movement of people from another country into a country.
- b) - Higher dependency ratio leading to low investment.
- Causes land fragmentation leading to food shortage.
 - Shortage of water in urban centres.
 - Leads to excessive tree felling for energy requirements and to pave way for settlement and agriculture.
 - Congestion of social facilities such as schools and hospitals.
 - Leads to increased unemployment.
9. a) Population explosion is the abnormal increase in people in a region, thereby overstretching the available resources
- b) - To determine the composition of the population
- To know the trends and levels of mortality and fertility
 - To plan for provision of basic facilities
 - To aid in making decisions regarding the creation of new administrative
 - To estimate dependency ratio
 - To know the literacy level
 - To know labour supply and predict any unemployment problems
- c) - Introduction of the national family planning programme through creation of national and development (NCPD)
- Introduction of adult education programme to check illiteracy and teach on importance of family planning
 - Creation of public awareness through mass media i.e. radios, TVs
 - Encouraging men and women to opt for voluntary sterilization and discouraging early marriages through legal action and education
 - Organizing family life seminars and public baraza's

24. SETTLEMENT

1. a) A – Central Business District
B – Transitional zone.
C - Zone of industry and working class.
- b) - Has full buildings/sky scrappers.
- Population density very high at low at night.
 - Has high land value.
 - Mainly have officers and shops.
 - Is the focal point of a town
2. a) - nucleated
- dispersed/ scattered
- b) 1- Central Business District
2- Industrial Zone
3- Low class residential zone

25. MANAGEMENT AND CONSERVATION OF THE ENVIRONMENT

- 1
 - i) Land pollution Is the contamination of the land through improper disposal of waste
 - ii) - Spraying of crops with chemicals contaminates the land by affecting soil composition
 - Careless dumping of domestic waste and industrial waste contaminates land
 - Burning of vegetation during land
 - Preparation leads to death of micro- organisms making the land unsuitable for plants to survive
 - Poor disposal of non- biodegradable material like plastic and polythene papers contaminates the land
 - iii) - Bad Odour leading to air pollution
 - Accumulation of domestic and industrial wastes on the road side blocks the drainage system
 - Mounds of wastes are a breeding ground for flies, mosquitoes and rodents which can cause outbreak of diseases
 - Injuries can result from broken bottles, rusted metals and iron
2.
 - a) ii) - Its best economic use of land because Agriculture does not do well in those areas.
 - Population is low and this reduces conflicts between man and animals.
 - It requires extensive land that is only available in those areas.
 - It reduces the conflict of resettling people.
 - b) - Polluted water cause death of aquatic animals.
 - High concentration of organic and inorganic nutrients in water causes electrification in lakes.
 - Water pollution causes diseases to animals
 - c i) - When a river contains excess waters which it cannot hold within its channel.
 - When the level of the sea or lake rises due to increased rainfall.
 - Exceptional heavy rainfall like El nino results in excess water on land.
 - When a dam built across a valley breaks, the water in the reservoir floods lower land down stream.
 - When massive earthquake occurs on the seabed – causes tsunami which causes flooding in coastal areas.
 - c ii) - Construction of dykes, dams and levees. These can be constructed across rivers to control the speed of water/amount of water flowing downstream.
 - Reforestation on slopes and river banks reduces surface run off and erosion / increases the rate of water infiltration in the ground.
 - Improving and diverting river channels.
 - Shallow rivers can be dredged to increase their depth. /Rivers with meanders can be straightened to enable water flow swiftly./ Tributaries can be drenched to reduce the volume of water getting into the main river.
 - d) - To sustain human life.
 - To protect endangered species.
 - For sustainable utilization of resources.
 - For aesthetic value.
 - For future generations/posterity.
 - To curb global warming.
3. a) - Hooting and roaring off engines by motor vehicles

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- Reverting of machines in rolling mills and welding
 - Banging of metal containers by Jua kali artisans
 - High pitched music played in vehicles, dance and disco halls, homes and religious gathering
 - Aircrafts as they take off or land in airports
 - Blasts from mines and celebrations e.g. Dawali
- b) - Rise in blood pressure and tension of muscles/physical stress
- Affects the nervous system causing neurosis and irritation
 - Damage of eardrums causing deafness
- a) – Desertification
- Lightning
 - Windstorms
 - Pest and diseases
 - Pollution
4. (a) - Earth quakes
- b i) - Nyando
- Nzoia
 - Yala
 - Kuja/Gucha
- ii) - the stagnant water becomes breeding ground for vectors that cause water related diseases.
- Floods cause loss of property/lives
 - Floods cause soil water logging which lower crop production
 - Floods wash away crops leading to food shortages/famine
 - Floods wash away bridges/roads/telephone lines/air field disrupting transport and communication
 - People are dispatched by floods are made homeless
- c)- the garbage man result to foul smell/air pollution which is hazardous to human health
- when it rains the dumped waste garbage is washed to rivers causing water pollution
 - garbage can be a breeding ground for rodents /flies/cockroaches which can cause diseases outbreak e.g. plague
 - accumulation of garbage leads to blockage of roads/drainage systems
 - garbage heaps are eye sore as they made the environment ugly
- d i) - burning waste materials
- digging pits throwing rubbish
 - minimizing use of harmful chemicals/use of organic manure
 - creating public awareness on the dangers of land pollution and how to control it
 - recycling of waste materials
 - government legislation against dumping
 - setting up proper garbage collections programmes
- ii) - strong winds destroy trees
- wind blow off roofs of houses
 - winds cause strong see storms and lead to boats capsizing/communication lines are destroyed/destruction of transport lines

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- winds cause soil erosion
 - winds spread air-borne disease
 - winds spread bush fires
5. a i) S – Mt. Kenya National Park
T- Amboseli National Park
U – Tsavo National Park
ii) Malindi Marine park
- bi) - Pie-historic sites e.g. Gedi ruins
- Sandy beaches
 - Caves, cliffs/coral reefs (coastal landforms)
 - Warm, sunny weather
 - People's culture
- ii) - Ensures maximum use of less productive land e.g. arid and semi-arid regions
- Source of foreign exchange through payment of fees at entry points/gates to parks/reserves
 - Creation of employment opportunities as guides in parks, lodges e.t.c.
 - Protection and conservation of endangered species e.g. rhinos, elephants e.t.c. for future generation/prosperity
 - Enhances research/promotes education in plants and animal species
 - For aesthetic purpose/recreation
 - Has stimulated the need to build roads and airstrips connecting parks with urban areas/opening up remote areas.
 - Promoted diversification of the economy from over-relying on agriculture to tourism
- iii) - Illegal hunting/poaching of wildlife/game leads to extinction of some animal species
- Overstocking of wild animals leading to destruction of the natural environment/overgrazing
 - Frequent drought leading to loss of animals through starvation and death
 - Human wildlife conflict leading to destruction of crops and death of people, through high costs of compensation
 - Inadequate capital limits government conservation effort especially to construct game parks
 - Pollution of the environment leading to loss of wildlife
 - Fire outbreaks which destroy vegetation/animals
 - Pests and diseases e.g feline immune deficiency virus that threatens lion population
6. (a) - To get in contact with the park management and seek permission for the visit
- To identify and engage the services of a tour guide
 - To identify the methods to be used in data collection
 - To formulate the relevant objectives and hypotheses
 - To assemble the appropriate equipment for the study
 - To help in estimating the cost of the study
 - To prepare an appropriate work schedule
 - To determine the appropriate routes to be followed
 - To identify possible problems that may be experienced and ways of avoiding them
- (b) - Offering incentives like reduced tariff rates by hotels during the low season for local people
- Lowering the entrance fees for local people into national parks and game reserves
 - Encouraging employers to offer incentive holidays to their employees
 - Promote domestic tourism through the media

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- Encourage young people to tour their country by establishing wildlife clubs in schools

7.
 - a) - Desertification.
 - Lightning
 - Wind storm
 - Pests and diseases
 - Pollution
 - b) - Deforestation
 - Poor agricultural activities e.g overgrazing, monoculture, overcropping etc
 - Increase in population.
 - Bush fire
 - Global warming
 - c) - To ensure proper utilization of resources without damage.
 - To sustain human life since it wholly dependent on the resources for survival.
 - To preserve the aesthetic value such as landscape and vegetation of her environment.
 - To protect the endangered species of plants and animals.
8. a) - Management of environment refer to the effective planning and control of the processes that could harm the environment while.
 - Conservation of environment refers to the protection and presentation of natural resources from destruction wastage or loss.
- b) - For sustenance of human life.
 - To protect endangered species.
 - For aesthetic value.
 - To interlize natural resources
 - For future generations.
 - To preserve cultural heritage.
- c) - Gases from factories corrode roofs of houses.
 - Some poisonous gases lead to earth to form acid rain which is harmful to life.
 - Smog and smoke reduce visibility which way lead to road/air accidents.
- d) (i) - Control soil erosion
 - Regular inspection of factories
 - Setting up recycling plants.
 - Treatment of sewage.
 - Managing garbage collect and disposal well.
 - Use of unleaded fuel
 - Proper legislation.
- (ii) - Age of the population
 - Social – economic status of the population.
 - Culture of the people.
9. a)- Pollution is the contamination of the environment with substances which are harmful or
 - poisonous to human, plants and animal life.
- b) - Land /Soil/Ground pollution.
 - Water pollution
 - Noise pollution.
10. a) - Flooding

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- Lightning
- Pests/diseases
- Hailstorms
- Dusts storms
- Landslides
- Pollution
- Soil erosion
- fire

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FORM ONE WORK

CHAPTER 1

INTRODUCTION TO GEOGRAPHY

1. What is practical geography? (2mks)
2. Name two branches of geography (2mks)
3. Name any three study areas in human geography (3mks)
4. Define the term environment (2mks)
5. Explain three reasons for importance of studying geography (3mks)
6. Name five human features (3mks)
7. What is habitat (2mks)
8. List six disciplines related to geography (6mks)
9. Differentiate between each of the following:
 - Democracy and population geography
 - Economics and economic geography (4mks)
10. Name two Greek words from which term geography originate (2mks)

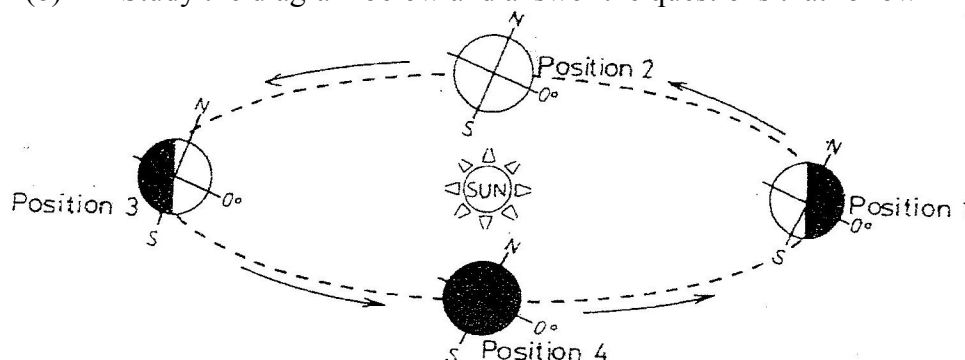
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CHAPTER 2

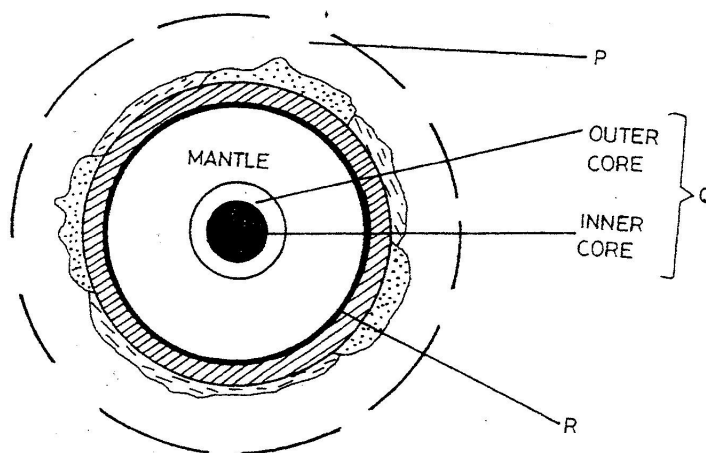
THE EARTH AND THE SOLAR SYSTEM.

PAST KCSE QUESTIONS ON THE TOPIC

1. (a) State two effects of the rotation of the earth (2mks)
- (b) Study the diagram below and answer the questions that follow

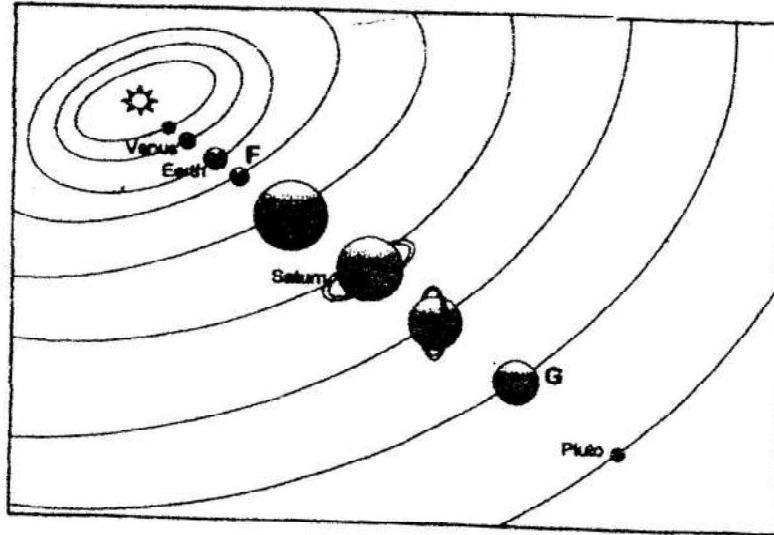


- (i) Which movement of the earth is represented by the diagram? (1mk)
 - (ii) Give two effects of the movement represented by the diagram (2mks)
2. The diagram below represents the structure of the earth. Use it to answer question

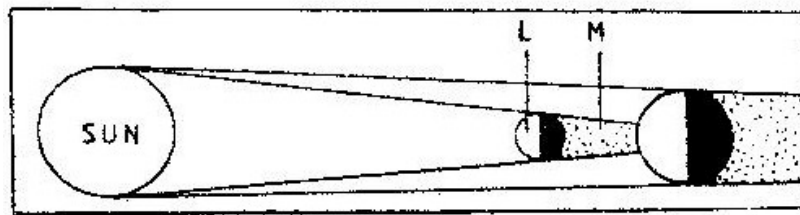


- (a) Name
 - (i) The parts marked P and Q (2mks)
 - (ii) The discontinuity marked R (1mk)
- (b) State three characteristics of the mantle (3mks)

3. The diagram below shows the composition of the solar system



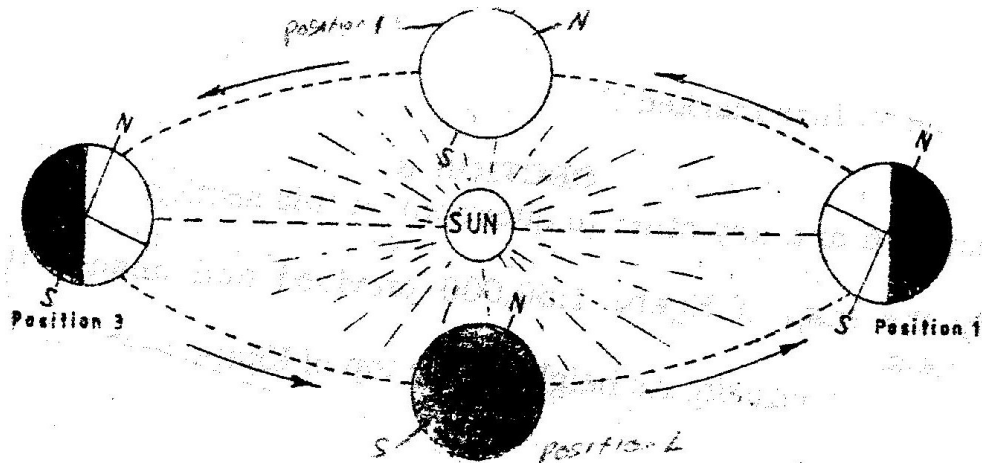
- (a) Name the planets marked F and G (2mks)
- (b) State three effects of the rotation of the earth on its axis (3mks)
4. a) What is the solar system?
- b) Use the diagram below to answer the questions that follow.



- i) What type of eclipse is represented by the diagram?
- ii) Name the features marked L and M
5. (a) (i) Give the two dates in a year during which the number of hours of darkness is equal in both the north and south poles.
- (ii) Why do the lengths of days and nights vary from one part of the earth to another?

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- (b) The diagram below shows the revolution of the earth around the sun. Use it to answer the questions that follow



- (i) If the earth takes 366 days to make a complete revolution during a leap year, how long will it take to move from position 1 to position 4?
 - (ii) What season is experienced in the southern hemisphere when the earth is in Position 1?
6. Define the following,
 - i. Solar system
 - ii. Galaxy
 - iii. Star
 - iv. Asteroids (6mks)
7. Differentiate between the following
 - (a) Latitude and longitude
 - (b) Dateline and international dateline
 - (c) Meteors and Meteorite. (6mks)
8. State three differences between solar eclipse and lunar eclipse. (2mks)
9. State four factors that support life on planet earth. (4mks)
10.
 - (a) List four effects of earth rotation. (4mks)
 - (b) At Nairobi on longitude 37°E local time is 1 p.m. What time would it be at Sarissa on longitude 41°E ? (4mks)
11.
 - (a) Define equinox. (2mks)
 - (b) State characteristics of summer solstice. (4mks)

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12. The earth is inclined to the ecliptic plane at an angle of..... and the axis is also inclined at an angle to perpendicular line. (4mks)
13. Fill in the table from (a) - (f) (10mks)

Property s/Layer	Major constituent	Thickness	Density	Temperature
Outer crust	(a)	iii. 16-24 kms	(b)	
Inner crust	ii. Magnesium	S (c)	2.8-30 gms/cc	
Asthenosphere	i. Iron	2900 kms	(d)	5000 C
Centrosphere	ii. Nickle	(e)		(f)

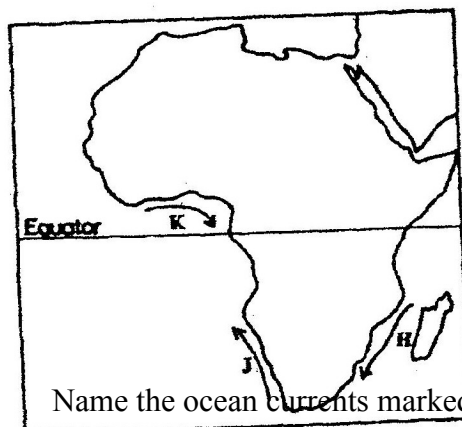
14. State three weaknesses of the passing star theory. (6mks)
15. Differentiate between hydrosphere and atmosphere. (4mks)
16. Planet ... 1... is seventh planet from the sun and is greenish in colour. Planet ...2... takes shortest time to revolve round the sun about 88 earth day. Planet ...3... and ...4... are referred to as twin planets. Planet ...5... takes about 11.86 earth years to revolve round the sun. All the planets have satellite orbiting round them except planet ...6... and ...7... (7mks)
17. Explain reasons for flattening and bulging of earth. (4mks)
18. State characteristics of winter solstice (4mks)
19. Differentiate between summer solstice and winter solstice. (4mks)
20. (a) What is an eclipse? (2mks)
21. Apart from planets name other heavenly bodies.
22. What is a longitude? (2mks)
23. State the effects of the elliptical shape of the earth's orbit. (6mks)
24. If the local time in Nairobi on longitude 37°E time is 10 p.m. What will the time be at Buchanan Liberia on longitude 10°W.? (4mks)
- (a) What is the effect of International Date Line on crossing the line? (4mks)
- (b) What is the angle of inclination of the earth axis from its orbit? (2mks)
- (c) Give four proofs that the earth is spherical in shape. (8mks)

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CHAPTER 3

WEATHER

1. (a) How does a sea breeze occur? (2 mks)
- (b) Use the map of Africa below to answer questions (b) (i)



- (i) Name the ocean currents marked H, J, and K (3 mks)
 - (ii) State two effects of a warm ocean current on the adjacent coastlands (2 mks)
2. (a) Name two theories of the origin of the earth (2 mks)
 - (b) Name four layers of the earth's atmosphere (4 mks)
 3. (a) State two conditions that are necessary for the formation of fog.
 - (b) The diagram below shows some types of clouds. Use it to answer the questions that follow.



- (i) Name the clouds marked R
 - (ii) Give two weather conditions associated with cumulonimbus clouds
4. a) the tables below represent rainfall and temperature of stations X and Y.
Use them to answer questions (a) and (b)

MONTHS	J	F	M	A	M	J	J	A	S	O	N	D
TEMPERATURE IN ⁰ c	30	31	31	31	30	29	29	28	28	29	29	30

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RAINFALL IN MM	250	250	325	300	213	25	25	25	100	275	380	200
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MONTHS	J	F	M	A	M	J	J	A	S	O	N	O
TEMPERATURE IN °C	21	20	20	17	15	13	12	13	15	16	18	20
RAINFALL IN MM	12	12	15	50	90	110	87	87	50	35	20	15

- a) (i) For each of the two stations calculate the mean annual temperature.
- X -
- Y -
- (ii) Calculate the annual rainfall for station Y
- (iii) On the graph paper provided, draw a bar graph to represent rainfall for station x. Use vertical scale of 1cm to represent 50mm
- b) Describe the climatic characteristics of station Y.
5. a) The table below shows climatic data of a station in Kenya.
- Use it to answer question (a)

Month	Jan	Feb	Mar	April	May	June	Jul	Aug	Sep	Oct	Nov	Dec
Temp in °C	28.9	29.7	30.3	29.9	29.7	29.2	28.4	28.7	29.6	30.1	29.2	28.7
Rainfall in mm	9.0	8.0	21.0	49.0	25.0	9.0	20.0	10.0	4.0	10.0	17.0	11.0

- i) What is the annual range of temperature at the station?
- ii) Calculate the total rainfall for the station.
- b) State three factors that influence climate.
6. (a) Name two elements of weather that can be recorded at a school weather station
- (b) Give three reasons why the recording of data at a school weather station may be inaccurate
7. (a) Describe a suitable site where you would locate a weather station in your

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School

(2 mks)

- (b) Give reasons why a Stevenson's screen is:
- (i) Painted White (2 mks)
- (ii) Has louvers (2 mks)
8. Define relative humidity. (2 mks)
9. (a) Identify four characteristics of convectional rainfall. (4mks)
- (b) State the difference between radiation fog and advection fog. (4mks)
10. (a) Briefly describe how the six thermometers operate. (5mks)
- (b) Three ways in which clouds are classified. (3mks)
11. (a) Give three precautions to be taken when citing a weather station. (3mks)
- (b) State three factors determining the amount of solar radiation reaching the earth's surface. (3mks)
12. Define the following terms:
- (i) Climate
- (ii) Relative humidity
- (iii) Weather forecasting
- (iv) Absolute humidity
- (v) Weather lore (5mks)
13. State the advantages of studying weather through field work. (5mks)
14. (a) Describe how you would use the following apparatus during a field study.
Rainfall, maximum and minimum thermometers. (3mks)
- (b) Identify and explain the formation of the type of rainfall found in the Lake Region or Kenya. (8mks)
- (c) Briefly write down two problems associated with the type rainfall above. (4mks)
15. (a) What is weather forecasting? (2mks)
- (b) List four problems of weather forecasting. (4mks)
- (c) State four ways in which weather forecasting is important to the human activities. (4mks)
16. (a) Explain three ways in which clouds influence weather. (3mks)
- (b) Use the data below to answer questions that follow.

Month of the year	J	F	M	A	M	J	J	A	S	O	N	D
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Temp in °C	25	26	26	24	23	22	21	21	22	22	22	22
Rainfall in mm	42	40	73	171	90	89	163	160	71	68	64	42

- (i) Calculate mean annual temperature
 - (ii) Calculate annual rainfall
 - (iii) Calculate annual range of temperature.
 - (iv) Calculate the mean annual rainfall
 - (v) Which is the wettest month? (10 mks)
17.
 - (a) Define 3 air mass. (2mks)
 - (b) Name types of air masses. (3mks)
 - (c) A mass of air at 15°C can hold 20gm/cm³ of moisture. The same air at the same temperature has 6gm/cm³ of moisture. What is its relative humidity? (4mks)
 18. Name two instruments placed in the Stevenson Screen. (2mks)
 19. Why does sea breeze flow at night time? (3mks)

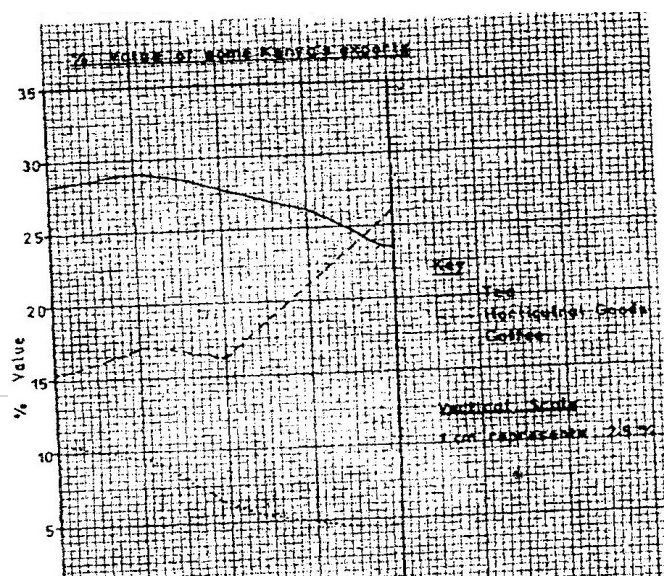
CHAPTER 4

STATISTICAL METHODS

1. The table below shows petroleum production in thousand barrels per day for countries in the Middle East in April 2006. Use it to answer question (a)

Country	Production in '000" barrels
Iran	3800
Kuwait	2550
Qatar	800
Saudi Arabia	9600
United Arab Emirates	2500
Iraq	1900

- a) (i) What is the difference in production between the highest and the lowest producer (1mk)
- (ii) What is the total amount of petroleum produced in April 2006 in the region? (1mk)
- b) State three conditions that are necessary for the formation of petroleum (3mks)
2. The graph below shows percentage value of some export commodities from Kenya between 1999 and 2003. Use it to answer questions (a) and (b)



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- (a) (i) What was the percentage value of the tea exported in the year 2000? (2mks)
- (ii) What was the difference in the percentage values of the horticultural products and coffee exports in 1999? (2mks)
- (iii) Describe the trend of the value of coffee exports from 1999 to 2003 (3mks)
- (iv) Explain three factors which may have led to the increased export earnings from horticultural produce in Kenya between years 1999 and 2003 (6mks)
- (v) Give three advantages of using simple line graphs to represent data. (3mks)
- (b) State four reasons why Kenya's agricultural export earnings are generally low (4mks)
- (c) State five reasons why the common market for Eastern and southern Africa (5mks)
3. (a) Define the following terms
- Statistics
 - Statistical data
 - Statistical methods (6mks)
- (b) State two types of statistical data. (2mks)
- (c) Write down two types of questionnaires. (2mks)
4. (a) What factors must be considered in selecting methods of data collection. (3mks)
- (b) Differentiate between discrete data and continuous data giving relevant examples. (4mks)
5. (a) What is sampling (1mk)
- (b) State 3 types of sampling. (3mks)
6. (a) Name two main methods used in analyzing statistical data. (2mks)
- (b) What is the significance of statistics in geography? (5mks)

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7. (i) Name two types of graphs that you have learnt about. (2mks)
- (ii) What are the advantages of using graphs named above in representing statistical data?
Give advantages. (4mks)
8. (i) What is a questionnaire?
- (ii) State four advantages of using questionnaires in collection of statistical data.
(4mks)
- (iii) Explain oral interview method. (2mks)
9. Explain the following methods of data recording.
- Tabulation
 - Photographing
 - Tape recording
 - Tallying
10. What is data? (2mks)
11. Marks 72, 60, 65, 70, 65, 80, 65, 70, 80, 84, 63, 75, 63, 71, 74
Use the data above to find out mean and mode. (4mks)
12. With the help of data above explain how median is obtained. (3mks)

CHAPTER 5

FIELD WORK

PAST KCSE QUESTIONS ON THE TOPIC

1. State two ways in which information collected during the field study would be useful to the local community. 2mks
2. Your class is required to carry out a field study of a river. What would be the advantage of dividing the class into groups according to the stages of the long profile 3 of a river?
3. What would be the disadvantages of c using secondary data in this kind of a field study?
4. You intend to carry out field study on population in the local open air market,
 - (i) State three reasons why it would be necessary for you to visit the market before actual field study.
 - (ii) Give two methods you would use to collect information on pollution. ,
 - (iii) State three follow up activities necessary for the study. -51
5. You are supposed to carry out a field study on the uses of vegetation in the area around your school.
 - (a) State three reasons why it would be necessary to visit the area *before* the day of the study.
 - (b) Give four uses of vegetation you are likely to identify during the study.
(3mks)
 - (c) Why is it necessary to sample part of the forest for the study?
6. List three types of fieldwork. (3mks)
7. Explain the importance of field work. (5mks)
8. Outline the procedure for carrying out field study. (5mks)
9. List some topics in physical geography on which you can carry out a field study.
(4mks)
10. State five ways in which you would prepare for field study to a weather. (5mks)
11. What is the importance of carrying samples from the field to the school? (4mks)
12. Formulate five suitable objectives for field study on a visit to a forest. (5mks)
13. Discuss types of hypothesis. (2mks)
14. List five methods of data presentation. (5mks)
15. Explain five problems one would encounter on field study in a forest? (5mks)
16. Why is reconnaissance important?

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CHAPTER 6

MAP WORK

1. Study the map of Taita Hills (1:50,000) sheet 189/4 provided and answer the following questions
 - (a)
 - (i) What is the bearing of the peak of Mwatunga hill in grid square 3214 from the water tank in grid square 2619? (2mks)
 - (ii) What is the length in kilometers of the section of the Mwatate – Voi railway line in the south – eastern part of the map? (2mks)
 - (b) Draw a rectangle measuring 16cm by 12 cm to represent the area enclosed by the Eastings 24 and 40 and Northings 20 and 30 (1 mk)

On the rectangle, mark and name the following features:
 - Mgange hills (1 mk)
 - A rock out crop (1 mk)
 - All weather road, bound surface (1 mk)
 - River Ruhia (1 mk)
 - Ronge forest (1 mk)
 - (c) Using evidence from the map, explain three factors that have favoured the establishment of the Teita sisal Estates in the Southern part of the area covered by the map (6 mks)
2. Study the map of Nyahururu, 1: 50,000 (sheet 105/4) provided and answer the following questions
 - (a)
 - (i) Give the six figure grid of the junction where the road to Ndaragwa (D 388) meets with the road to Nyeri & Nanyuki (B5) (2mks)
 - (ii) Calculate the bearing of point X from point Y (2mks)
 - (iii) Name three physical features found along the line XY (3mks)
 - (b)
 - (i) Draw a square 12 cm by 12 cm to represent the area enclosed by the Easting 10 and northing 10 to the North- eastern part of the map (1mk)

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- (ii) On the square, mark and label
- The main river (1mk)
 - All weather loose surface road (1mk)
 - A forest (1mk)
- (c) Citing evidence from the map, explain two
- (i) Physical factors that may have influenced the location of Nyahururu town (4mks)
- (ii) Factors that favour saw milling in the area covered by the map (4mks)
3. Study the map of Taita Hills (150: 50,000 sheet 189\4) to answer the following questions.
- a) What is the approximate height of the hill at the grid square 3926. (2mks)
 - b) Measure the length of all weather 6 to roads (bound surface) from Wundanyi to southern edge of the area covered by the map. (2mks)
 - c) Citing evidence from the map describe the relief of the area shown. (5mks)
 - d) State differences between a map and a plan. (2mks)
 - e) Explain two importance of scale in maps. (2mks)
4. Study the map of Kisumu East (1:50,000) and answer the following questions.
- (a) (i) What is the bearing of the trigonometrical station at grid reference 081980 from the rock antelope at grid reference 071992. (2mks)
 - (ii) Measure the length of the all weather road (bound surface) 1321, from, the junction at grid reference 974911 to the edge of the map, grid reference 947967. (2mks)
 - (b) (i) Describe the relief of the area covered by the map.
 - (ii) Explain how relief has influenced the settlement in the area covered by the map. (8mks)
 - (c) Citing evidence give three economic activities carried out in the area covered by the map.
 - (d) Students from the school at Masago (grid square 0681) carried out field study of the course of river Ombeyi.

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- (i) State three findings they are likely to have come up with. (3mks)
- (ii) Give three advantages of studying rivers through field work

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CHAPTER 7

ROCKS AND MINERALS

1. (a) Describe the following characteristics of minerals
 - (i) Colour (2mks)
 - (ii) Cleavage (2mks)
 - (iii) Hardness (2mks)
- (b) (i) Give two types of igneous rocks (2mks)
- (ii) Explain three conditions necessary for the growth of coral polyps (6mks)
- (c) State four uses of rocks (4mks)
- (d) You are planning to carry out a field study on the rocks within your school environment
 - (i) Give two secondary sources of information you would use to prepare for the field study (2mks)
 - (ii) State why you would need the following items during the field study:
 - A fork jembe (1mk)
 - A polythene bag (1mk)
 - (iii) Suppose during the field study you collected marble, sandstone and granite, classify each of these samples according to its mode of formation (3mks)
2. (a) State two characteristics of sedimentary rocks (2mks)
- (b) Give two examples of chemically formed sedimentary rocks (2mks)
3. a) Name the type of rocks which results from the metamorphism of:
 - (i) Granite
 - (ii) Clay (2mks)
- b) Give two reasons why sedimentary rocks are widespread in the coastal plain of Kenya. (2mks)
4. (a) (i) What is a rock? (2mks)
- (ii) Describe three ways through which sedimentary rocks are formed
 - Mechanically formed
 - Organically formed
 - Chemically formed (6mks)

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- (b) Describe two process through which sedimentary rocks changer into metamorphic rocks
- (c) Give an example of each of the following types of igneous rocks
- (i) Plutonic rocks (1mks)
 - (ii) Hypabyssal rocks (1mks)
 - (iii) Volcanic rocks (1mks)
- (d) Suppose you were to carry out a field study of rocks within the vicinity of your school
- (i) Name three secondary sources of information you would use to prepare for the field study (3mks)
 - (ii) State four activities you would carry during the filed study (3mks)
 - (iii) State three problems you are likely to experience during the field study (3mks)
5. (a) Differentiate between plutonic rocks and volcanic rocks
- (b) Describe how lava plateau is formed
- (c) (i) Name three volcanic features found in the rift valley of Kenya
- (ii) Explain four negative effects of vulcanicity in Kenya
- (d) You intend to carry out a field study of a volcanic landscape
- (i) State four reasons why it is necessary to conduct a reconnaissance of the area of study.
 - (ii) During your field work, you intend to study volcanic rocks, state why you would need the following items
6. (a) State two main conditions that influence the characteristics of igneous rocks. (2mks)
- (b) Write down three characteristics of sedimentary rocks. (3mks)
- (c) Name two examples of organic sedimentary rocks and where found in Kenya. (2mks)
- (d) Name four examples of metamorphic rocks and state the original rock from which each was formed. (4mks)
- (e) Describe the importance of rocks to human activities. (5mks)
7. (a) State with examples three classes of mechanically formed sedimentary rocks. (6mks)
- (b) Differentiate between regional metamorphism and contact metamorphism.

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(4mks)

8. (a) List two examples of extrusive igneous rocks. (2mks)
(b) Differentiate between extrusive and intrusive rocks giving an example in each case. (2mks)
9. What is a rock? (2mks)
10. What is a mineral? (2mks)
11. Describe changes that occur in sedimentary rocks when they are subjected to high heat and pressure. (4mks)
12. Describe calcareous rocks. (2mks)
13. Describe carbonaceous rocks. (2mks)
14. Give examples of chemically formed sedimentary rocks. (2mks)
15. How are coral rock formed? (3mks)
16. How do rocks become metamorphic? (3mks)

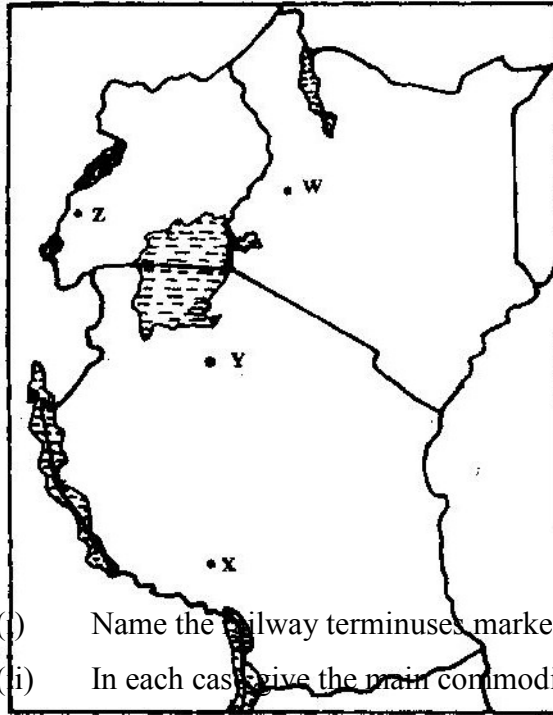
CHAPTER 8**MINING**

1. The table below shows petroleum production in thousand barrels per day for countries in the Middle East in April 2006. Use it to answer question (a)

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Kuwait	2550
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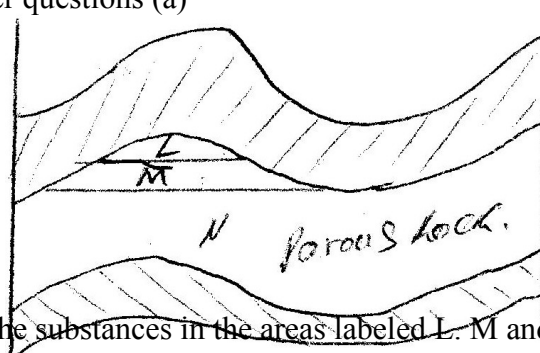
- a) (i) What is the difference in production between the highest and the lowest producer (1mk)
- (ii) What is the total amount of petroleum produced in April 2006 in the region? (1mk)
- b) State three conditions that are necessary for the formation of petroleum (3mks)
2. Use the map of East Africa below to answer questions (s).

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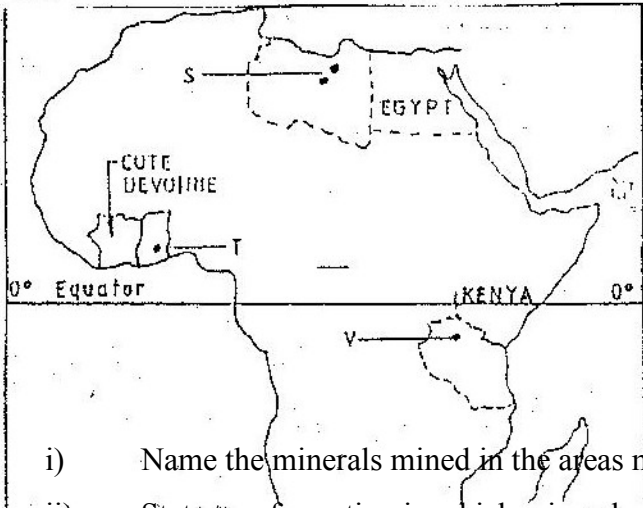
- a) (i) Name the railway terminuses marked P, Q R (3mks)
- (ii) In each case give the main commodity transported by the railway lines marked S and T. (2mks)
- b) (i) State four reasons why road network is more widespread than railways in East Africa. (4mks)
- (ii) One of the problems facing road transport is the high frequency of accidents. Explain four conditions of roads in Kenya that may lead to accidents. (8mks)
- c) i) Name three physical regions through which River Tana passes (3mks)
- ii) Explain three effects of land pollution can be controlled
- d) State four ways through which land pollution can be controlled (4mks)
3. The diagram below show the occurrence of petroleum in the earth's crust.

Use it to answer questions (a)



- a) Name the substances in the areas labeled L, M and N (3mks)

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- b) Give two by-products obtained when crude oil is refined (2mks)
4. Use the map of Africa to answer question (a) (i)
- 
- i) Name the minerals mined in the areas marked S, T and V.
- ii) State two formation in which mineral ores occur.
- b) Explain four problems, which Zambia experiences in the exportation of copper.
- c) Explain three ways in which coal contributes to the economy of Zimbabwe.
- d) Describe three negative effects of open cast mining on the environment.
5. a) Explain how deep shaft mining is done (2mks)
- b) Disadvantages of using the above method (2mks)
6. Explain four effects of land dereliction on the environment. (4mks)
7. Describe how panning mining is carried out. (3mks)
8. Identify four problems facing gold mining in South Africa. (4mks)
9. (a) In what ways has Kenya benefited from the mining of soda ash in Lake Magadi? (2mks)
- (b) What are the negative effects of mining on the environment? (4mks)
10. (a) Explain what is meant by placer mining. (2mks)
- (b) Name three mining methods.
11. Describe the occurrence and exploitation of Trona in Kenya till it is ready for marketing.
12. Name seven significances of minerals in Kenya. (7mks)
13. Explain diamond and gold in South Africa under following headings:
- Occurrence

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- Extraction
 - Benefits to the economy
 - Problems
- uses of soda ash.

(10mks)14. Name five
(5mks)

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FORM TWO WORK

CHAPTER 1

INTERNAL LAND FORMING PROCESSES – EARTH MOVEMENTS.

1. (a) Name the two types of earth movements that occur within the earth's crust (2mks)
 (b) Describe the origin of the continents according to the Theory of continental Drift (3mks)
2. Explain what you understand by each of the following:
 - (i) Earth movements.
 - (ii) Internal land forming processes.
 - (iii) External land forming processes. (6mks)
3. Explain four evidences put forward to proof continental drift theory. (8mks)
4. Explain plate tectonic theory. (4mks)

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CHAPTER 2

INTERNAL LAND FORMING PROCESS – FOLDING

1. (a) In your answer booklet, draw a diagram to show a simple fold and on it mark and name,
 - (i) An anticline. 1 mk
 - (ii) A limb. 1 mk
 - (iii) A syncline 1 mk
- (b) Name two fold mountains in Africa. 2 mks
2. (a) Name one fold mountain in;
 - (i) Asia
 - (ii) North America
 - (iii) South America
- (b) (i) Apart from Fold Mountains, name three other features resulting from folding.
- (ii) With the aid of a labelled diagram, describe the formation of an overthrust fold.
- (c) Explain four effects of Fold Mountains on human activities.
- (d) (i) How would students in your school prepare themselves for study of landforms in your district,
- (ii) State two advantages of studying landforms through field work.
3. Define orogenesis. 2 mks
4. What is folding? 2 mks
5. Explain the meaning of compressional boundaries. 2 mks
6. Differentiate between limb and axis in relation to folding. 4 mks
7. Differentiate between foreland and back land. 4 mks
8. Fill in the table provided details on age, period and features formed in each named orogenies.

Orogeny	Years (age)	Period	Mountains/features built
Charnian	1	Pre-cambrian period	2
Caledonian	Old 440 million years ago		-Akwapim Hills of Ghana - Scottish highlands

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Hercynian	3	Upper Carbon ferrous period	- Cape ranges - Appalachian mountains - Ural mountains
Alpine	Youngest 70 million years ago		4

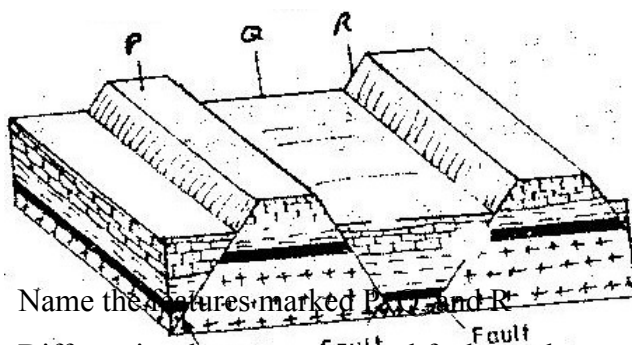
6 mks

9. Explain formation of Fold Mountains by contraction theory.

CHAPTER 3

INTERNAL LAND FORMING PROCESSES – FAULTING

1. a) A part from the Rift Valley name two other relief features that were formed as result of faulting. (2mks)
b) With the aid of a well labeled diagram, describe how a Rift Valley is formed by tensional forces. (8mks)
2. The diagram below represents features produced by faulting.
Use it to answer questions that follow.



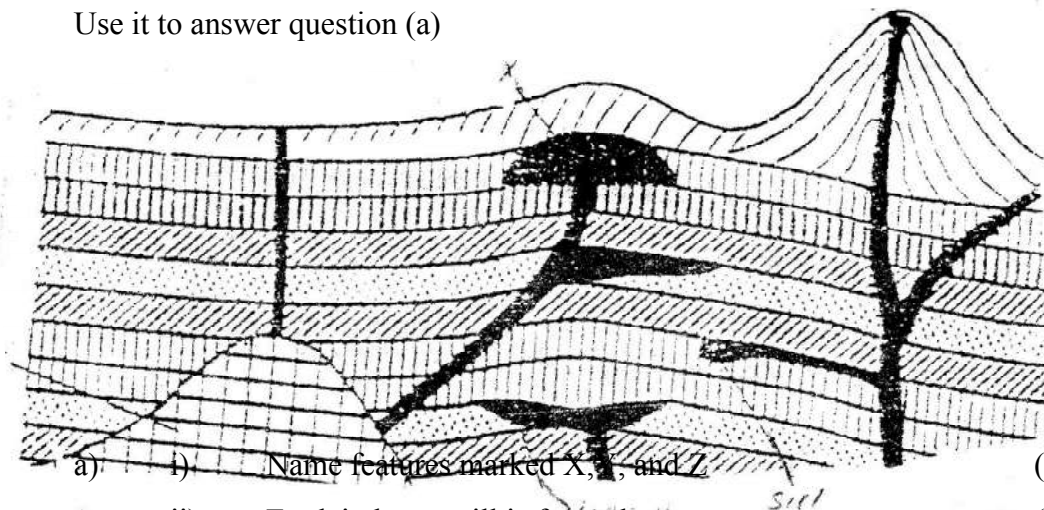
- a) Name the features marked P, G and R.
- b) Differentiate between a normal fault and a reverse fault.
3. State ways in which faulting influences drainage. (3mks)
4. Name two examples of Horst Mountains in East Africa. (2mks)
5. Explain two ways in which features resulting from faulting are of economic importance (4mks)
6. (a) (i) With aid of diagrams outline formation of rift valley by tension theory. (5mks)
(b) Students are planning to carry out field study of an area affected by faulting.
(i) State four importance of having a pre-visit of the area.
(ii) Give three disadvantages of using observation to study such an area.
7. Name three types of fault. (3mks)
8. Explain how compressional forces lead to formation of rift valley. (5mks)
9. Give two of escarpments in East Africa. (2mks)
10. Explain ways in which features resulting from faulting are of importance. (8mks)
11. Describe formation of fault steps with aid of diagrams. (6mks)

CHAPTER 4

INTERNAL LAND FORMING PROCESSES – VULCANICITY

1. The diagram below shows some intrusive features formed by vulcanicity.

Use it to answer question (a)



- a)
 - i) Name features marked X, Y, and Z. (3mks)
 - ii) Explain how a sill is formed. (4mks)
 - b) Describe the characteristics of a composite volcano. (4mks)
 - c) Explain **four** ways in which volcanic mountains positively influence human activities. (8mks)
 - d) Students carried a field study on volcanic rocks
 - i) Give **four** reasons why it is necessary to collect rock samples during such a field study. (4mks)
 - ii) State **two** problems they are likely to have experienced during the field study. (4mks)
- 2.
- (a) Differentiate between plutonic rocks and volcanic rocks
 - (b) Describe how lava plateau is formed
 - (c)
 - (i) Name three volcanic features found in the rift valley of Kenya
 - (ii) Explain four negative effects of vulcanicity in Kenya
 - (d) You intend to carry out a field study of a volcanic landscape
 - (i) State four reasons why it is necessary to conduct a reconnaissance of the area of study.
 - (ii) During your field work, you intend to study volcanic rocks, state why you would need the following items
3. Name three volcanic features found in the Rift Valley of Kenya. (3mks)
4. Explain four negative effects of vulcanicity in Kenya. (8mks)

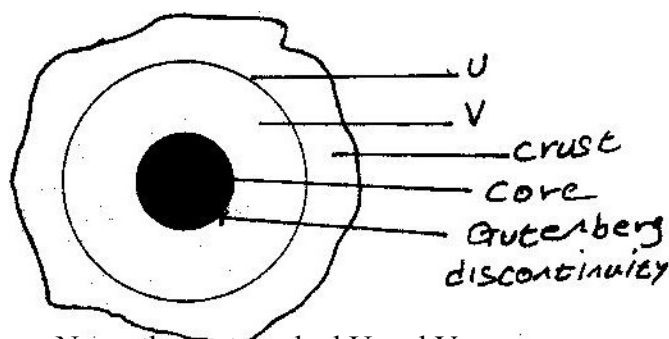
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5. Describe how lava plateau is formed. (5mks)
6. Differentiate between sill and dyke. (4mks)
7. What is vulcanicity. (2mks)
8. Describe how Crater Lake is formed. (5mks)
9. Describe how Mt. Kenya was formed.

CHAPTER 5

INTERNAL LAND FORMING PROCESSES – EARTHQUAKES.

1. (a) Name two scales used to measure the intensity of an earthquake (2mks)
(b) Give three causes of earthquakes (3mks)
2. The diagram below represents the internal structure of the earth. Use it to answer question(a)



- (a) Name the part marked U and V.
 - (b) Describe the deposition of:
 - i. The crust
 - ii. The core
 - (c) (i) What are earthquakes
(ii) Name two types of earthquakes.
(iii) State the five ways in which the earth's crust is affected by earthquakes.
 - (d) You intend to carry out a field study of an area recently affected by intense earthquake.
 - i. Give two sources of information that you would use in preparation for the study.
 - ii. Explain two factors that would make it difficult for you to collect accurate data during the field study.
3. (a) State three causes of earthquakes
(b) Give two effects of earthquakes in built up areas
 4. State the major causes of earthquakes (2mks)
 5. Explain how intensity of earthquake is measured. (2mks)
 6. List major effects of earthquakes where they occur. (4mks)
 7. Distinguish between seismograph and seismogram (4mks)
 8. Differentiate between intensity and magnitude of earthquake (4mks)

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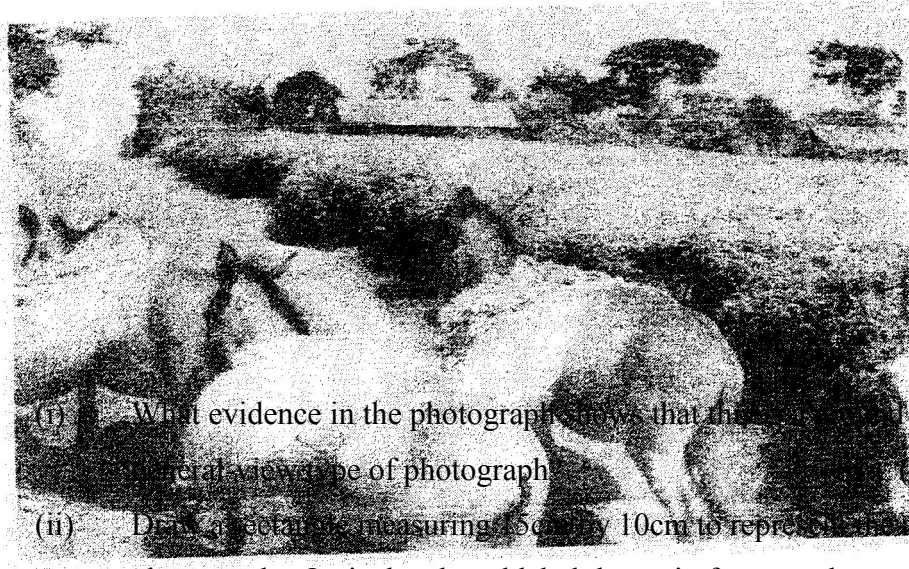
9. Students from your school intend to carry out a field study of an area recently affected by intense earthquake.
- (i) Give two sources of information that you would use in preparation for the study.
 - (ii) Explain two factors that would make it difficult for you to collect accurate data during the field study.

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CHAPTER 6

PHOTOGRAPH WORK.

1. The photograph provided shows a tea growing area in Kenya. Use it to answer questions (a) and (b)



- a) (i) What evidence in the photograph shows that this is a small scale tea growing area? (2mks)
- (ii) Draw a rectangle measuring 15cm by 10cm to represent the area of the photograph. On it sketch and label the main features shown on the photograph. (5mks)
- (iii) Identify two features from the photograph that show that this is a small scale tea farm. (2mks)
- b) Describe the stages involved in the cultivation of tea from land preparation to the stage shown on the photograph.
- c) (i) Name two districts in the Eastern province where tea is grown. (2mks)
- (ii) Explain four ways in which the Kenya Tea development agency (KTDA) assists small scale tea farmers in Kenya (8mks)
2. What is the type of photograph shown?



3. Name each of the following:

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- (i) Crop under cultivation (2mks)
 - (ii) Type of farming (2mks)
 - (iii) Other groups of crops in this type of farming. (2mks)
 - (iv) The province in Kenya where this photograph was taken. (2mks)
- 4. Name five problems facing this type of farming in Kenya. 5mks
 - 5. Name two major export destinations (countries) for farm produce shown. (2mks)
 - 6. Name three physical features at the background of the photograph. (3mks)
 - 7. Draw a rectangle of 12cm by 7cm to represent the area covered by the photograph.



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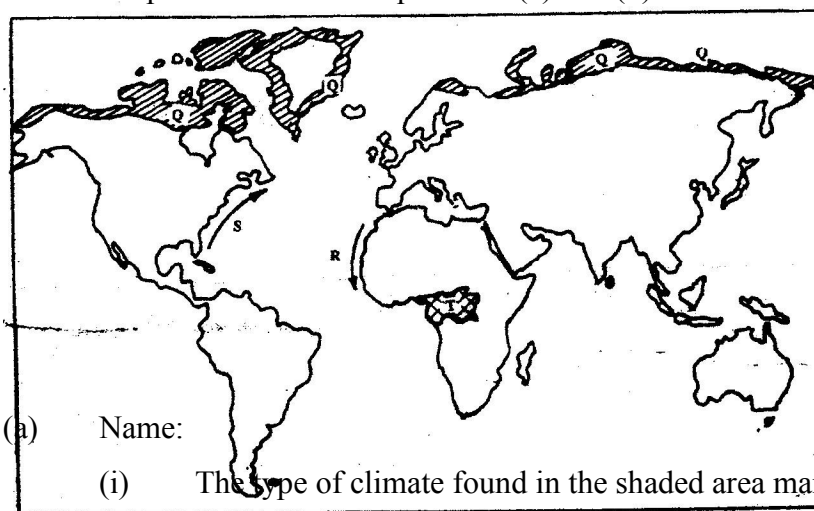
CHAPTER 7

CLIMATE.

1. (a) (i) What is climate? (2mks)
- (ii) Explain two effects of climate change on the physical environment
2. The table below represents rainfall and temperature figures for a town in Africa. Use it to answer the questions that follow

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temp (⁰ C)	27	28	28	28	27	25	25	24	25	26	27	26
Rainfall (mm)	25	38	99	140	277	439	277	69	142	201	71	25

- (a) Calculate the annual range of temperature for the town (2mk)
- (b) Calculate the total annual rainfall for the town. (2mks)
- (c) State two characteristics of the climate experienced in the town. (2mks)
3. Use the map below to answer questions (a) and (b)



- (a) Name:
 - (i) The type of climate found in the shaded area marked Q (1mk)
 - (ii) The ocean current marked R and S (2mks)
- (b) Describe the characteristics of the type of climate found in the shaded area marked T (8mks)
- (c) Explain how the following factors influence climate
 - (i) Altitude (4mks)
 - (ii) Distance from the sea (4mks)
- (d) (i) Describe a suitable site where you would locate a weather station

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in your School (2mks)

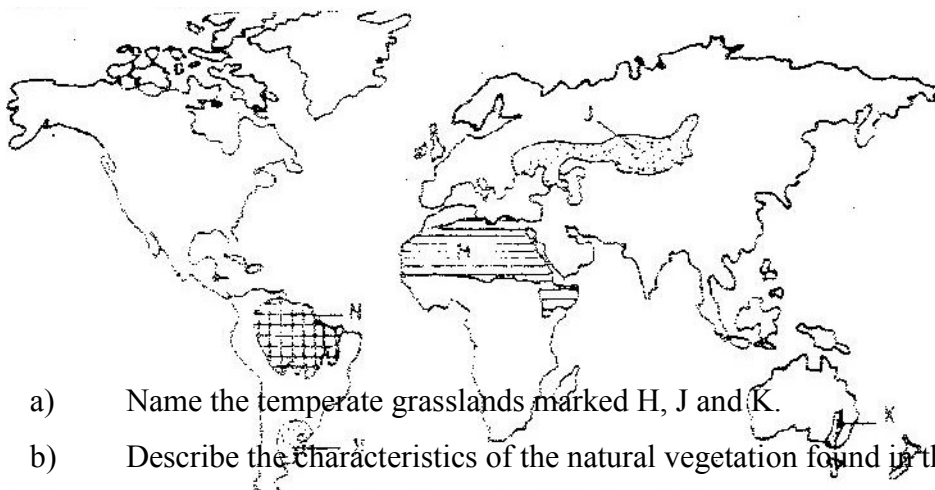
- (ii) Give reasons why a Stevenson's screen is:
- Painted White (2mks)
 - Has louvers (2mks)
4. Describe the characteristics of natural vegetation associated with equatorial climate (4mks)
5. Give five characteristics of hot desert climate (5mks)
6. How does the following factors influence climate?
- (i) Wind/air masses.
 - (ii) Latitude.
7. Explain characteristics of climatic conditions experienced in the Kenyan highlands. (8mks)
8. Explain four ways in which mountains influence climate. (8mks)
9. What is greenhouse effect? (2mks)
10. How do human activities influence climate change? (6mks)
11. How does clearance of vegetation cause climate change? (3mks)
12. Define climate. (2mks)
13. What is isothermal layer? (2mks)

CHAPTER 8

VEGETATION

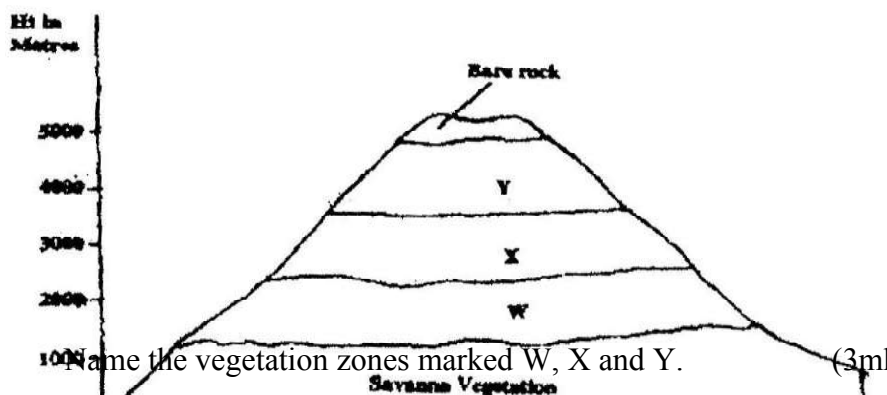
1. The map below shows some vegetation regions of the world.

Use it to answer questions (a) to (c).



- a) Name the temperate grasslands marked H, J and K.
- b) Describe the characteristics of the natural vegetation found in the shaded area marked N.
- c)
 - i) Explain four ways in which the vegetation found in the area marked M adapts to the environment conditions of the region.
 - d) You are required to carry out a field study of the vegetation within the local environment:
 - i) A part from identifying the different types of plants, state three other activities you will carry out during the field study.
 - ii) How will you identify the different types of plants?
2. The diagram below represents zones of natural vegetation on a mountain in Africa. Use it to answer question (a) (i) and (ii)

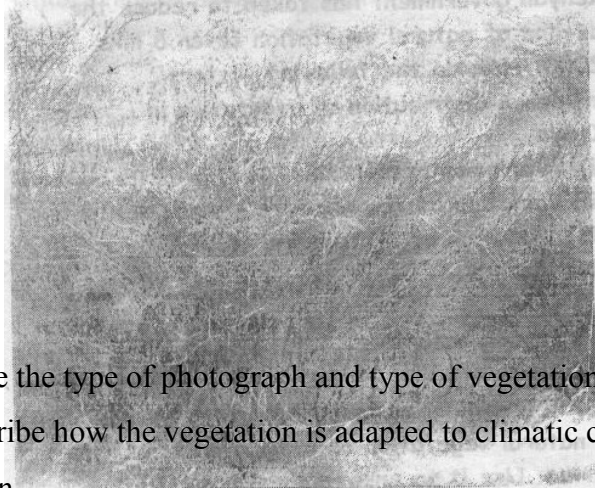
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- a) (i) Name the vegetation zones marked W, X and Y. (3mks)
- (ii) Describe the characteristics of the savanna vegetation. (6mks)
- (iii) Name the temperate grasslands found in the following countries:
- Canada (1mk)
 - Russia (1mk)
 - Australia (1mk)
- b) Explain three causes of the decline of the areas under forest in Kenya. (6mks)
- c) You are supposed to carry out a field study on the uses of vegetation in the area your school.
- (i) State three reasons why it would be necessary to visit the area before the day of the study. (3mks)
- (ii) Give four uses of you are likely to identify during the study (4mks)
3. (a) What is a natural vegetation? (2mks)
- (b) State three characteristics of Mediterranean vegetation (3mks)
4. Explain three measures that the Kenyan government has taken to reduce the decline of natural vegetation cover. (6mks)
5. How do the following factors influence distribution of vegetation in Kenya.
- (i) Variation in rainfall
 - (ii) Variation of temperature
 - (iii) Variation of altitude/relief.
 - (iv) Aspect
 - (v) Soil
 - (vi) Human activities (10mks)

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6. State two reasons why mountain tops have no vegetation. (2mks)
7. Define vegetation. (2mks)
8. Name areas where coniferous forests are found. (3mks)
9. State characteristics of temperate grassland. (5mks)
10. Distinguish between secondary vegetation and planted vegetation. (4mks)
11. State two ways in which vegetation is significant to human and physical environment. (5mks)



- (a) Name the type of photograph and type of vegetation. (2mks)
- (b) Describe how the vegetation is adapted to climatic conditions of the region. (3mks)

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CHAPTER 9

FORESTRY

1. a) i) What is forestry? (2mks)
 ii) Explain three factors that favour the growth of natural forests on the slopes of Mt. Kenya. (6mks)
 iii) State five factors that have led to the reduction of the area under forest on the slopes of Mt Kenya. (5mks)
- b) Explain four measures that the government of Kenya is taking to conserve forests in the country. (8mks)
- c) Give the differences in the exploitation of softwood forests in Kenya and Canada under the following sub-headings;
 i) Period of harvesting; (2mks)
 ii) Transportation (2mks)
2. Explain factors favouring forestry in Canada
3. List two species of indigenous hardwood forest trees in Kenya. (2mks)
4. Name two industries associated with forestry. (2mks)
5. Explain three measures being undertaken to conserve forests in Kenya. (6mks)
6. Explain four problems that are being experienced in exploitation of hardwood forests in Kenya. (8mks)
7. Explain three reasons why only a small part of Kenya is forested. (6mks)
8. Outline three consequences of forest depletion in Kenya. (3mks)
9. State the factors favouring forestry in Kenya. (8mks)
10. Name four secondary products of forests. (4mks)
11. List eight major indigenous tree species in West Africa. (8mks)
12. (a) Define agro-forestry. (1mk)
 (b) Outline four benefits of agro-forestry (4mks)

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FORM THREE WORK

CHAPTER 1

EXTERNAL LAND FORMING PROCESSES – WEATHERING.

1. (a) (i) What is the difference between weathering and mass wasting?
 (ii) Apart from plants, give three other factors that influence the rate of
 Weathering (3mks)
 (iii) Explain two ways in which plants cause weathering (4mks)
- (b) (i) List two types of mass wasting other than soil creep (2mks)
 (ii) Explain three factors that cause soil creep. (6mks)
- (c) Explain four effects of mass wasting on the environment. (8mks)
2. Give two processes involved in each of the following types of weathering
- (a) Physical weathering (2mks)
- (b) Chemical weathering (2mks)
3. (a) What is mechanical weathering? (2mks)
- (b) How is an exfoliation dome formed? (5mks)
4. Describe five processes involved in chemical weathering? (3mks)
5. Name physical weathering processes that take place in the arid areas.
6. List factors that determine rate of weathering.
7. Define the term denudation
- 8.



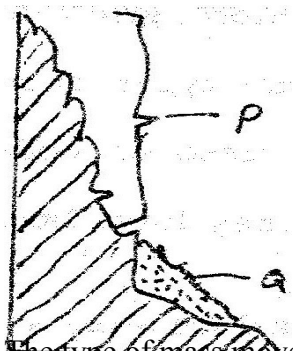
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- a) Name the above type of weathering.
- b) Describe the process shown by the photograph

CHAPTER 2

EXTERNAL LAND FORMING PROCESSES – MASS MOVEMENT.

1. a) State two conditions which may influence the occurrence of landslides
- b) Using the diagram (in question paper), name

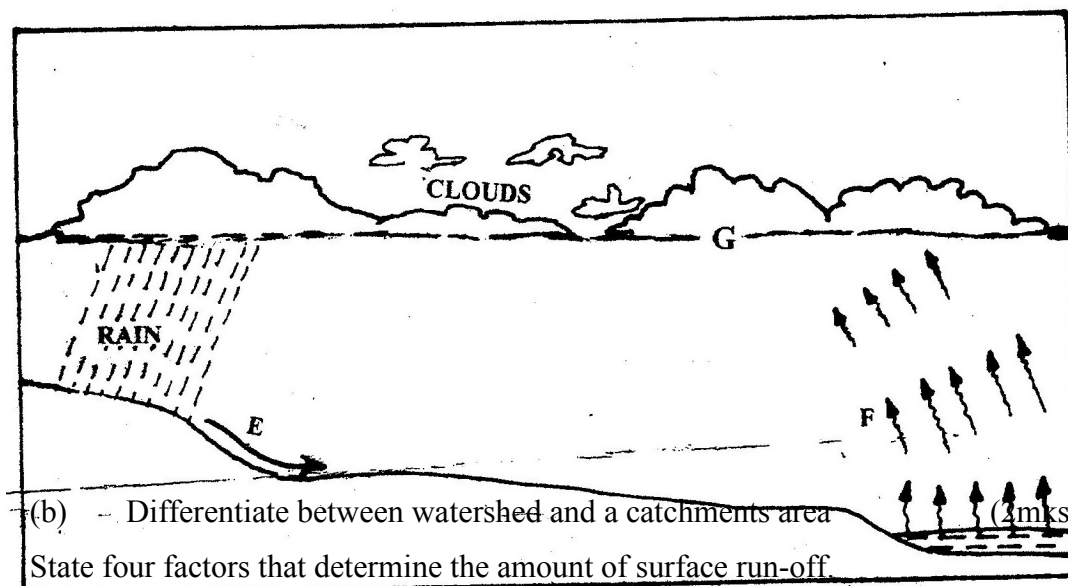


- (i) The type of mass movement shown
 - (ii) The features marked P and Q (2mks)
2. Explain five ways in which soil creep occurs. (10mks)
 3. Describes the effects of soil creep. (6mks)
 4. Define the following:
 - a) Mass wasting.
 - b) Mass movement. (2mks)
 5. Name and explain three process of slow mass movement. (4mks)
 6. Explain the factors that are responsible for rapid mass wasting.
 7. List the evidences of soil creep. (4mks)

CHAPTER 3

THE HYDROLOGICAL CYCLE

1. (a) The diagram below shows the hydrological cycle. Name the stages marked E, F, and G (3mks)



- (b) - Differentiate between watershed and a catchments area (2mks)
2. State four factors that determine the amount of surface run-off (4mks)
3. What is hydrological cycle? (2mks)
4. Explain factors that influence percolation of water. (8mks)
5. (a) What is cryosphere? (2mks)
- (b) Name the significance of hydrological cycle. (4mks)
6. List other forms of precipitation other than rainfall that may also form major inputs into the system. (4mks)
7. What factors can influence surface run off or overland flow. (5mks)
8. In what ways can we sustain the process of the hydrological cycle? (8mks)

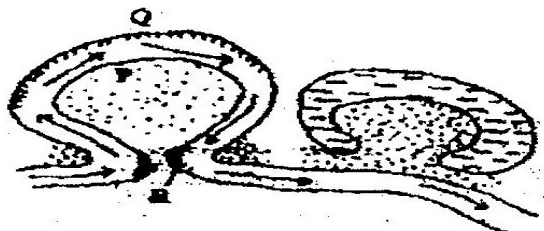
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CHAPTER 4

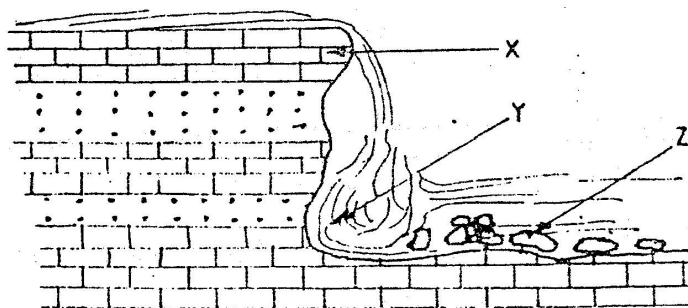
ACTION OF RIVERS

1. (a) Name two types of the coastal deltas (2mks)
- (b) State two conditions that lead to deposition of silt at the mouth of a river (2mks)

2. The diagram below shows river Mander. Use it to answer question (a)



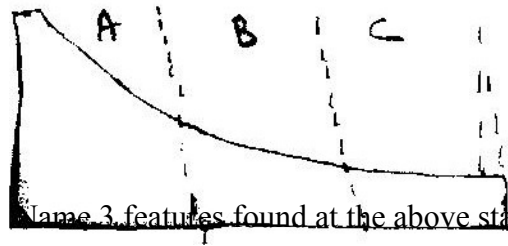
- (a) (i) Name the process that take place at each of the points marked P and Q. (2mks)
- (ii) Name the feature formed at the point marked R (1mk)
- (iii) Describe how an Ox- bow lake is formed (5mks)
- (b) State five characteristics of a flood plain (5mks)
- (c) Explain three causes of river rejuvenation (6mks)
- (e) Your class is required to carry out a field study of a river
 - (i) What would be the advantages of dividing the class into groups according to the stages of the long profile of a river? (4mks)
 - (ii) What would be the disadvantage of using secondary data in this kind of a field study? (2mks)
3. (a) State two factors which influence the occurrence of surface run- off
- (b) The diagram below shows a waterfall. Name the feature marked X, Y and Z



4. Describe three ways in which rivers transports its load.
5. Describe the following drainage patterns

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- (i) Dedritic.
 - (ii) Trellis.
 - (iii) Centipetal.
6. a) State two factors that influence the rate of erosion by the river in its upper course.
- b) (i) Define river rejuvenation
- Name two features that result from river rejuvenation
7. Explain the following:
- (a) River basin
 - (b) Watershed
 - (c) Catchment area
 - (d) River regime (8mks)
8. With examples from Africa, explain the differences between the following river features:
- (a) Inland delta and alluvial fan.
 - (b) Estuarine delta and an estuary.
 - (c) Bluff and river cliff.
 - (d) Levees and river bank.
 - (e) River valley and river channel.
 - (f) Paired terrace and unpaired terrace.
 - (g) Drainage pattern and drainage system,
 - (h) Misfit river and deferred river,
 - (i) Antecedent drainage and superimposed drainage. (18mks)
9. Describe how a river erodes its channel through the following processes
- (i) Abrasion
 - (ii) Hydraulic (4mks)
10. (a) (i) In which stage is the river at 'A'



- (ii) (3mks)
- (b) (i) In which stage is the river at 'B'

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(ii) Which are the characteristics of the river at stage B?

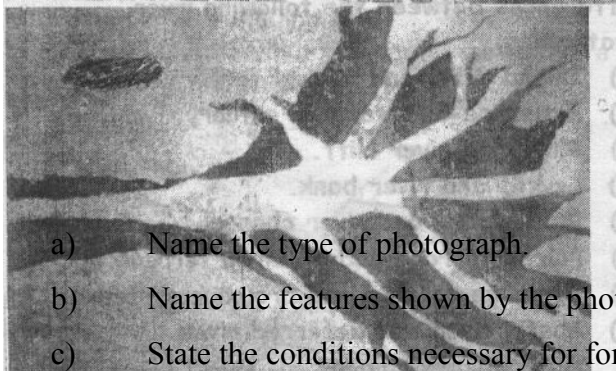
(iii) Describe the characteristics of the river at the above stage C.

(4mks)

(c) In which stage is the river at C.

11. Explain the significance of rivers to man.

(10mks)12.



a) Name the type of photograph.

(1mk)

b) Name the features shown by the photograph. I and II.

(2mks)

c) State the conditions necessary for formation of these features.

(3mks)

CHAPTER 5

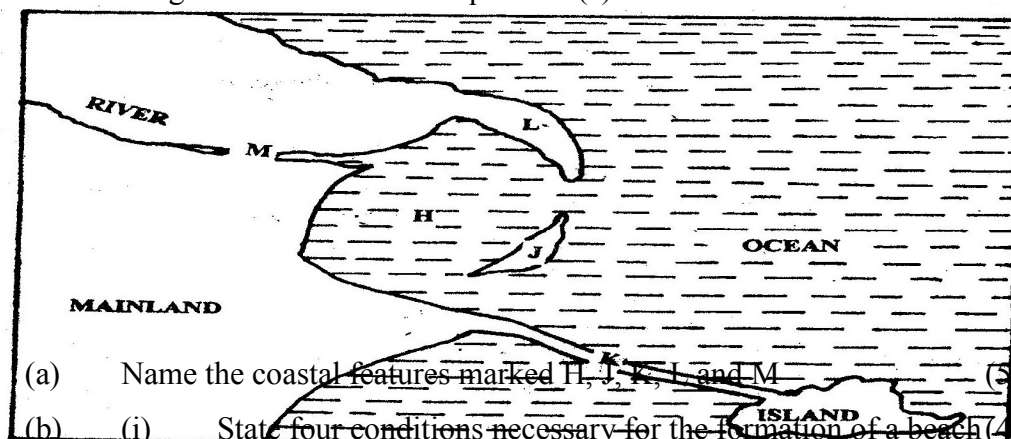
LAKES.

1. Give three processes that lead to formation of lakes.
2. Describe how Lake Victoria was formed.
3. Explain how Lake Victoria influences the climate of the surrounding areas.
4. What is a lake?
5. State three ways in which lakes are formed.
6. Explain how each of the following lakes were formed :
 - (a) Victoria
 - (b) Tanganyika
 - (c) Chala
 - (d) Sare
 - (e) Kivu (15mks)
7. State the differences between the lakes on the eastern and western areas of East African Rift Valley. (6mks)
8. With reference to specific lakes in East Africa, explain the significance of lakes in the region. (8mks)

CHAPTER 6

OCEANS, SEAS AND THEIR COASTS.

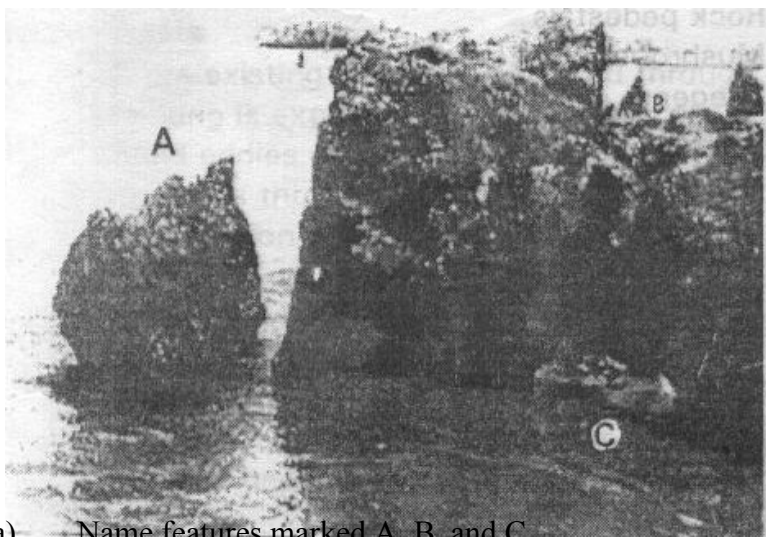
1. Use the diagram below to answer question (a)



- (a) Name the coastal features marked H, J, L, and M. (3mks)
- (b) (i) State four conditions necessary for the formation of a beach. (4mks)
- (ii) Describe three processes involved in marine erosion. (6mks)
- (c) You are planning to carry out a field study on the depositional features along the coast of Kenya
- (i) State five objectives you would formulate for your study. (5mks)
- (ii) Give five methods you would use to record the information collected. (5mks)
2. (a) Name two types of submerged coasts. (2mks)
- (b) Explain how the following factors determine effectiveness of wave erosion along the coast.
- (i) Nature of the material transported by waves
- (ii) Nature of the coastal rocks. (4mks)
3. State two causes of submerged coasts. (2mks)
4. Name two features that result from submergence of coasts. (2mks)
5. Define term coastline. (2mks)
6. What are destructive waves? (2mks)
7. Name three resultant features of wave erosion. (3mks)
8. Describe formation of cliff. (5mks)
9. Describe formation of a wave-cut platform. (5mks)
10. Name three types of coast. (3mks)
11. (a) Describe formation of coral coast. (5mks)

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- (b) Explain the significance of coral coast to Kenya.
12. Distinguish between shingle beaches and sand beaches. (6mks)
13. Name three types of submerged coasts. (3mks)
14. Name two types of movements of ocean water. (2mks)
- 15.

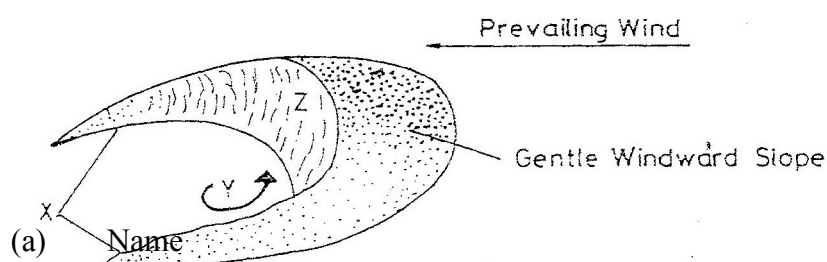


- (a) Name features marked A, B, and C.
- (b) Describe the formation of feature marked C.

CHAPTER 7

ACTION OF WIND AND WATER IN ARID AREAS.

1. The diagram below represents a barchan. Use it to answer questions (a)



- (a)
- (i) The feature marked X (1mk)
 - (ii) The air current marked Y (1mk)
 - (iii) The slope marked Z (1mk)
- (b) Give two ways in which wind transports its load (2mks)
2. a) (i) Two process through which wind erodes the surface
- (ii) Three ways through which wind transports its load
- b) (i) How an oasis is formed
- (ii) How zeugens are formed
- c) You are supposed to carry out a field study of a semi-arid area in Kenya.
- (i) Two ways of preparing for the Field study
 - (ii) Information that would be collected through observation of the arid area
 - (iii) Measures to be recommended for controlling desertification.
3. Explain the process of abrasion. (2mks)
4. Name four features of wind erosion. (4mks)
5. Describe formation of zeugens, (4mks)
6. List features of wind deposition. (4mks)
7. Describe formation of wadis. (5mks)
8. Differentiate between suspension and saltation. (4mks)
9. Name four types of desert surface (4mks)
10. Identify and describe the processes of wind erosion. (6mks)
11. (a) Explain how wind transports its load.
- (b) State the factors influencing wind transportation. (3mks)
12. Explain the formation of the following features:
- (a) Bajadas.

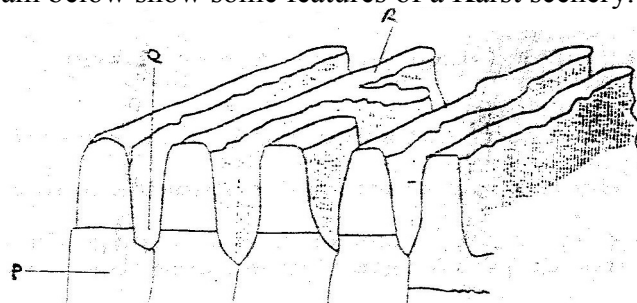
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- (b) Pediments. (6mks)
13. Students carried out field study on desert landforms.
- (i) State two type of information they collected through observation.
 - (ii) Which measures would they have recommended to control desertification?

CHAPTER 8

UNDERGROUND WATER

1. The diagram below show some features of a Karst scenery. Use it to answer questions (a)

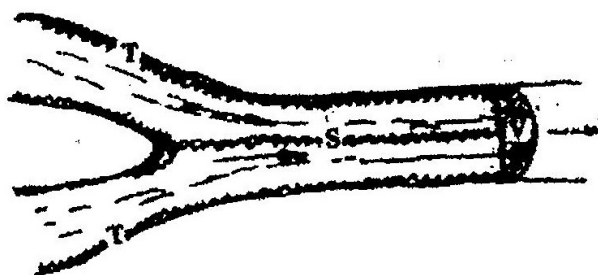


- a) Name the features marked P, Q, and R. (5mks)
- b) Describe carbonation as a process of Chemical weathering (3mks)
2. State three conditions necessary for the development of Karst scenery, (3mks)
3. Give two reasons why there are few settlements in a Karst landscape. (4mks)
4. Explain factors influencing formation of springs. (8mks)
5. Distinguish between the following.
- (i) Effluent streams and influent streams. (4mks)
- (ii) Artesian basins and artesian well. (4mks)
6. Name three surface features of Karst landscape. (3mks)
7. What are stalactites? (2mks)
8. Explain the significance of limestone regions. (8mks)

CHAPTER 9

GLACIATION

1. (a) (i) What is an ice sheet? (2mks)
- (ii) Give two reasons why there are no ice sheets in Kenya (2mks)
- (iii) Explain three factors that influence the movement of the ice from the place where it has accumulated (6mks)
- (b) Describe how an arête is formed (4mks)
- (c) The diagram below shows types of moraines in a valley glacier



- (i) Name the type of moraines marked S, T and V (3mks)
- (iii) Explain four positive effects of glaciation in lowland areas. (8mks)
2. a) (i) What is a glacier? (2mks)
- (ii) Distinguish between valley glaciers and ice sheets (4mks)
3. The diagram below shows a glaciated upland area



- (a) Name the feature marked P, Q, and R (2mks)
- (b) How is a U- shaped valley formed? (5mks)
4. a) Describe how pyramidal peak is formed. (6mks)
- b) Explain the significance of upland glaciated features to human activities. (6mks)
- c) Students from a school near Mt. Kenya were planning to carry out a field study on the glaciated features on the top of the mountain.
- (i) Give the reason why it would be difficult to undertake the field study on the glaciated features on the mountain. (4mks)

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(ii) Describe how students would use a photograph of Mt. Kenya to identify the
glaciated features on the mountains. (3mks)

5. Differentiate between snout and snow niche. (4mks)
6. Name three glaciers on Mt. Kenya. (3mks)
7. Describe the formation of a glacial trough. (3mks)
8. What is ice cap? (2mks)
9. Name three resulting features of glacial erosion on Mt. Kenya (3mks)
10. What is a nivation hollow? (2mks)

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CHAPTER 10

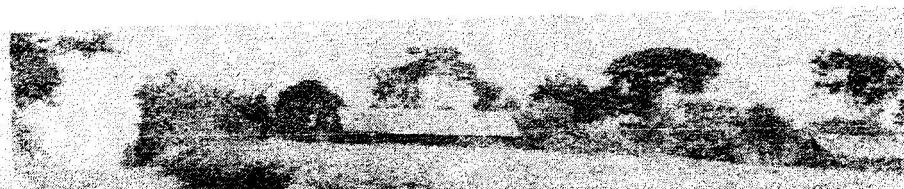
SOIL

1.
 - a)
 - (i) What is soil catena? (5mks)
 - (ii) Draw a labeled diagram to show a well developed soil profile. (3mks)
 - (iii) State three characteristics of the soils found in the arid regions of Kenya. (6mks)
 - b) Give three factors that determine the colour of soil. (2mks)
 - c) Describe how laterization occurs. (2mks)
 - d) Explain how the following farming practices cause soil erosion. (2mks)
 - (i) Burning (2mks)
 - (ii) Continuous application of fertilizer on farm lands. (2mks)
 - (iii) Monocultures. (2mks)
2.
 - (a) Name two types of soil according to texture. (2mks)
 - (b) State two ways in which humus improves the quality of soil. (2mks)
3. What is soil? (2mks)
4. Identify classification of soil according to order. (3mks)
5. Describe formation of soil through decomposition of organic matter. (3mks)
6. How does salination occur? (3mks)
7. What do you understand by zonal order soil?
8. List four soil conservation and management practices. (4mks)
9. What do you understand by podzolisation? (2mks)

CHAPTER 11

AGRICULTURE.

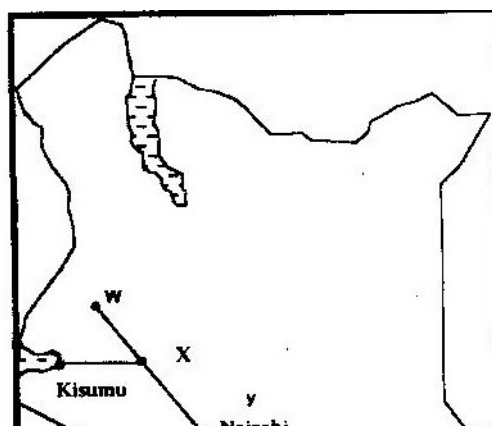
1.
 - a) State two climatic conditions that favour the growing of oil palm in Nigeria. (2mks)
 - b) Give two problems experienced in the marketing of palm oil in Nigeria. (2mks)
2. The photograph provided shows a tea growing area in Kenya. Use it to answer questions (a) and (b)



- a) (i) What evidence in the photograph shows that this is a ground
general-view type of photograph? (2mks)
- (ii) Draw a rectangle measuring 15cm by 10cm to represent the area of the
photograph. On it sketch and label the main features shown on the photograph.
(5mks)
- (iii) Identify two features from the photograph that show that this is a small scale tea
farm. (2mks)
- b) Describe the stages involved in the cultivation of tea from land preparation to the stage
shown on the photograph.
- c) (i) Name two districts in the Eastern province where tea is grown.
(2mks)
- (ii) Explain four ways in which the Kenya Tea development agency
(KTDA) assists small scale tea farmers in Kenya (8mks)
3. (a) State three physical conditions that are necessary for the growing of cocoa
(3mks)

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- (b) Give three economic problems experienced in cocoa farming in Ghana
(3mks)
4. a) Give three physical factors that favour coffee growing in Kenya highlands.
b) State two problems facing coffee farming in Kenya
5. a) i) Name two provinces in Kenya where wheat is grown on large scale
(2mks)
ii) Explain four physical conditions that favour wheat growing in Kenya
(8mks)
- b) Compare wheat farming in Canada and / Kenya under the following
i) Storage (2mks)
ii) Transportation (2mks)
iii) Marketing (2mks)
- c) i) Explain three climate problems that affect wheat farming in Canada
(6mks)
ii) Give three uses of wheat (2mks)
- d) Name two districts in Kenya where wheat is grown on commercial scale.
(2mks)
- e) Name two wheat producing provinces in Canada (2mks)
- f) Explain five factors which enable Canada to produce more wheat than Kenya.
(5mks)
6. a) State five physical conditions required for the growing of tea in Kenya
(5mks)
b) Explain four problems experienced in small scale tea farming in Kenya
(8mks)
7. The map below shows some major tea growing areas in Kenya.



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- a) Name the areas marked W, X and Y. (3mks)
- b) Give two reasons why there was an increase in tea production over the given period. (2mks)
- c) Describe the stages through which tea is processed from picking to the time it is ready for marketing. (5mks)
- 8. Name major cocoa growing areas in Ghana. (3mks)
- 9. List suitable conditions for cultivation of cocoa. (4mks)
- 10. Name types of commercially cultivated coffee. (3mks)
- 11. Explain ways in which Brazilian government responds to problems facing coffee industry. (6mks)
- 12. Name four uses of maize. (4mks)
- 13. List four problems facing maize farmers. (4mks)
- 14. Outline stages in industrial processing of cocoa. (5mks)

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CHAPTER 12

AGRICULTURE – LIVESTOCK.

1.
 - a) Name two exotic breeds of dairy cattle reared in Kenya. (2mks)
 - b) State three physical conditions that favour dairy farming in Denmark (8mks)
2.
 - a) Explain four ways in which the government of Kenya assist nomadic pastoralist to improve the quality of their livestock
 - b) Explain three factors that favour beef farming in Argentina.
 - c) State three environmental conditions which favour commercial beef farming in Kenya. (3mks)
 - d) Name two exotic breeds of cattle reared in commercial ranches in Kenya. (2mks)
3. Mention three problems facing beef farming in Kenya. (3mks)
4. State five human factors that have favoured beef farming in Argentina. (5mks)
5. State differences in dairy farming in Kenya and in Denmark. (6mks)
6. What effort is Kenyan government making to improve dairy farming? (5mks)
7. What is nomadic herding? (2mks)
8. State five features of nomadic herding. (5mks)
9. Explain two efforts Kenyan government has made to improve beef farming. (4mks)
10. Explain four physical conditional that favour dairy farming in Kenya. (8mks)
9. The table below shows data on average milk yield in kg per cow in Denmark.

Year	1990	1991	1992	1993	1994	1995
Yields (Kg)	5243	6693	7398	7610	7792	7946

- (a)
 - (i) Draw a divided rectangle 15cm long to represent milk yield in Denmark.
 - (ii) State two advantages of using divided rectangles.
- (b)
 - (i) Explain three factors that have favoured dairy farming in Denmark. (6mks)
 - (ii) State 3 problems facing dairy farmers in Kenya.

- (c) Explain why beef farming is more developed in Argentina than in Kenya.

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FORM FOUR WORK

CHAPTER 1

LAND RECLAMATION AND REHABILITATION.

1. (a) Give two methods used to reclaim land in Kenya. (2mks)
(b) Outline the stages through which land is reclaimed from the sea in the Netherlands. (5mks)
2. (a) Difference between land reclamation and rehabilitation. (4mks)
(b) Five ways through which land is being reclaimed. (5mks)
(c) Describe of polderization process in Netherlands. (4mks)
(d) State three benefits that resulted from the reclamation of the Yala Swamp. (3mks)
3. (a) State the objectives of Mwea irrigation scheme. (4mks)
(b) State the conditions that favoured establishment of Mwea irrigation scheme. (4mks)
(c) Explain problems faced by rice farmers in Mwea irrigation scheme. (5mks)
(d) State benefits of Perkerra irrigation scheme. (5mks)
(e) Explain factors that limits Perkerra irrigation scheme. (8mks)
4. Differentiate between horticulture and market gardening. 4 mks
5. Outline three problems facing horticulture farming in Kenya. 3 mks
6. Name five irrigation schemes in Kenya. 5 mks
7. State two factors that favour the occurrence of tsetse flies in Olambwe Valley.
8. State measures that were taken by Kenyan government to control Tsetse flies in Olumbwe Valley. 5 mks
9. Give three conditions that made Mwea suitable for establishment of irrigation scheme. 3 mks
10. Explain two effects of tsetse flies on humans and livestock. 4 mks
11. (a) Differentiate between land reclamation and land rehabilitation. 4 mks
(b) State the reasons for the establishment of Mwea-Tebere irrigation project.
(c) Explain five problems facings Mwea -Tebere irrigation scheme. 5 mks
12. State the problems facing Perkerra irrigation sceme. 5 mks
13. (a) Outline the benefits which Kenya derives from irrigation farming. 5 mks

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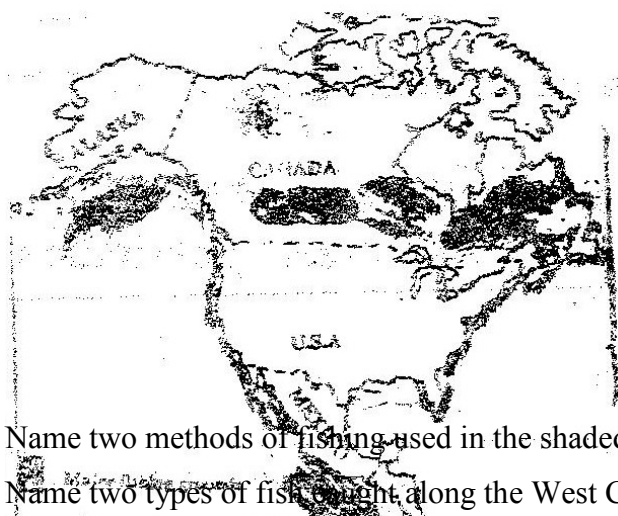
- (b) What are the problems experienced in irrigation farming in Kenya. 5 mks
14. (a) Apart from irrigation, list other methods of land reclamation in Kenya.
(b) Briefly describe the methods listed in 12 (a). 5 mks
15. State the benefits of Yala-Bunyala project. 5 mks
16. (a) What is a polder. 5 mks
(b) Describe the stages involved in ^{1f}reclamation of a polder. 5 mks
(c) State five benefits of land reclamation in Netherlands.
17. State the main differences between the methods of land reclamation in Kenya and Netherlands.
6 mks

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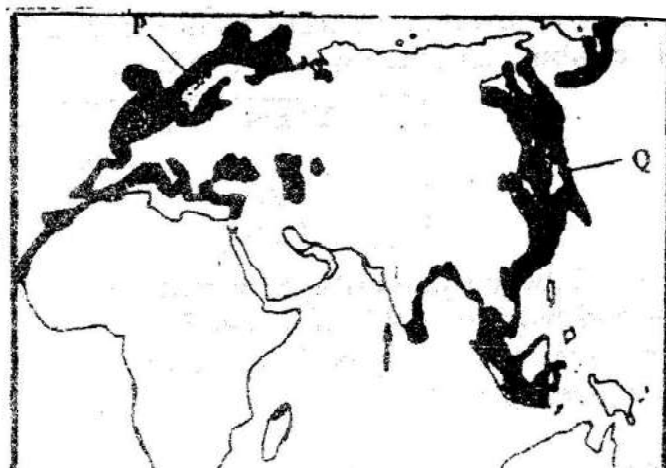
CHAPTER 2

FISHING

- 1 (a) State three measures that have been taken to conserve fisheries in Kenya. 3 mks
- (b) Give four reasons why Norway is a great fishing nation. 4 mks
- (c) Two traditional/ subsistence methods of fishing. 2 mks
- (d) Three measures the government of Kenya is undertaking TO encourage fish culture. 3 mks
2. Use the map of North America to answer question (a).



- (a) Name two methods of fishing used in the shaded areas. 2 mks
- (b) Name two types of fish caught along the West Coast of Canada. 2 mks
- (c) Explain how the following factors favour fishing in the shaded areas,
- (i) Indented coastline
- (ii) Ocean currents 4 mks
- (d) Give three methods used to preserve fish. 3 mks
- (e) Explain three problems experienced by fishermen while fishing in Lake Victoria. 3 mks
3. The map below shows some major fishing grounds in the world. Use it to Answer the questions below.



- (a) Name the countries marked P and Q. 2 mks
- (b) Explain four conditions that favour fishing in the shaded coastal waters. 8 mks
4. (a) The diagrams below represents some fishing methods.
- Diagram R shows a boat on the water surface with a net being pulled from the water. Diagram S shows a large, funnel-shaped net being pulled from the water. The diagrams are labeled 'R' and 'S' respectively. The water surface is labeled 'Water surface' and the river bed is labeled 'River bed'. The shaded area is labeled 'Shaded'.
- (i) Name R and S. 2 mks
- (ii) Describe how the above methods are used in fishing. 4 mks
5. (a) Differentiate between fishing and fisheries. 4 mks
- (b) Identify the physical and human factors influencing fishing. 5mks
6. (a) Draw an outline map of the world and locate the major fishing grounds. 2 mks
- (b) Account for the location of fishing grounds located in 6(a). 4 mks
7. (a) Discuss the main types of fishing. 3 mks
- (b) List the traditional methods of fishing. 3 mks
- (c) Describe how the following methods of fishing are carried out:
- (i) Drifting
- (ii) Trawling 4 mks
8. (a) (i) Apart from Lake Victoria, name other fresh water fisheries in Uganda and Tanzania. 4 mks
- (ii) Explain the factors favouring fishing on Lake Victoria. 6 mks
- (b) Explain why marine fisheries are underdeveloped in East Africa. 5 mks
9. (a) State the significance of fishing to the economy of Kenya. 5 mks
- (b) (i) Describe the problems facing fishing in Kenya. 5 mks

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(ii) Identify the solutions in b(i) above. 5 mks

10. Draw a table showing the similarities and differences between fishing in Kenya and Japan.

6 mks

11. (a) Define:

(i) Management of fisheries.

(ii) Conservation of fisheries. 4 mks

(b) State the measures which can be undertaken to manage and conserve fisheries.

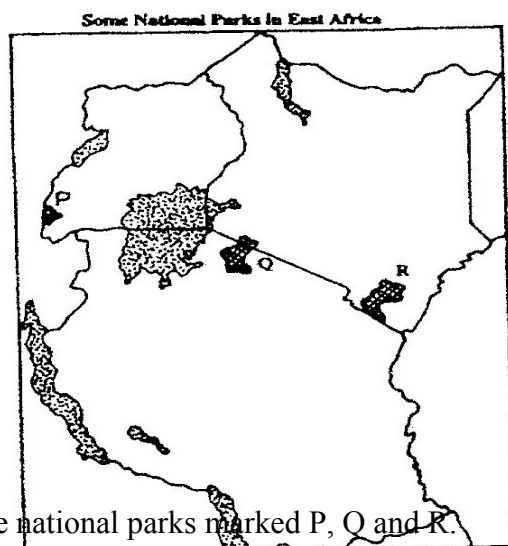
4 mks

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CHAPTER 3

WILDLIFE.

1. Use the map of East Africa below to answer question (a) (i)



- Name the national parks marked P, Q and R. 3 mks
2. Give FOUR reasons why wildlife conservation is encouraged in Kenya. 4 mks
3. State three reasons why National Parks have been established in Kenya. 3 mks
4. (a) Differentiate between game reserves and game parks. 4 mks
- (b) State four steps taken by Kenyan government to promote wildlife resources. 4 mks
5. Explain three ways in which human activities are a threat to wildlife. 6 mks
6. What is wildlife? 2 mks
7. State five reasons for the need to conserve wildlife? 5 mks
8. State three types of wildlife one may find at I, Nakuru. 3 mks
9. Explain four problems the government of Kenya faces in efforts to conserve wildlife? 8 mks
10. State human factors that affect existence of wildlife. 5 mks
11. State ways in which government of Kenya can strengthen anti poaching unit.
12. (a) Define:
- (i) Wildlife
- (ii) Tourism 4 mks
- (b) Distinguish between Same Reserves, National Parks and Sanctuaries.
- (c) State five significance of wildlife in East Africa. 5 mks
13. (a) Discuss the problems facing wildlife in East Africa. 5 mks

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- (b) Explain the measures taken to manage and conserve wildlife in East Africa

6 mks

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CHAPTER 4

TOURISM.

1. (a) Explain the differences between the tourist attractions in East Africa and in Switzerland under the following subheadings:
 - (i) Climate.
 - (ii) Culture 4 mks
- (b) Explain five benefits that Kenya derives from tourism. 10 mks
- (c) Explain four measures that Kenya should take in order to attract more tourists. 8 mks
2. (a) Apart from historic sites, name two tourist attractions along the Coastal strip of Kenya. 2 mks
- (b) Give -three reasons why it is necessary to preserve historical sites. 3 mks
3. (a) Name two game reserves in Kenya. 2 mks
- (b) Define of domestic tourism 2 mks
- (c) How has the recent negative travel advisories affected Kenya's economy? 4 mks
4. (a) Explain four measures the Kenyan government has taken to attract more tourists. 8 mks
- (b) Explain three factors that have led to development of tourism in Switzerland.
5. (a) Differentiate between the following
 - (i) Ecotourism
 - (ii) Domestic tourism 4 mks
- (b) Explain similarities between tourism in Kenya and in Switzerland. 8 mks
6. Explain why tourism is 'invisible export'. 2 mks
7. Explain four positive effects of tourism. 8 mks
8. Explain four ways in which tourism in Kenya differ from that Switzerland. 8 mks
9. Why are some parts of Kenya not developed for tourism? 3 mks
10. What are the problems facing tourism in Kenya? 5 mks
11. State five efforts being made to improve tourism industry in Kenya. 5 mks
12. What is domestic tourism? 2 mks
13. Name tourist attractions found in Rift Valley province of Kenya. 5 mks
14. What factors hinder development of domestic tourism in Kenya? 5 mks

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15. Name two historical attractions along Kenyan coast. 2 mks
16. (a) Define:
- (i) Eco-tourism
 - (ii) Domestic tourism
 - (iii) International tourism 6 mks
- (b) (i) State five tourist attractions in Kenya. 5 mks
- (ii) Explain five factors influencing tourism in Kenya. 10 mks
17. (a) State six factors influencing tourism in Switzerland. 6 mks
- (b) Explain five problems facing tourism in Kenya. 10 mks
18. Compare and contrast tourism in Kenya and Switzerland. 8 mks
19. The table below shows the number of tourists who visited Kenya.

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Year	1999	2000	2001	2002	2003
No. Of Tourists	1.53 m	1.64 m	1.65 m	1.77 m	1.54 m

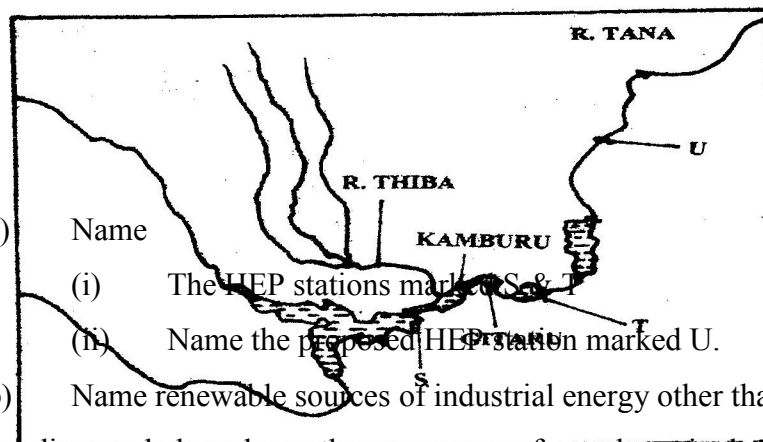
- (a) Using a radius of 3cm draw a pie-chart to represent the above data.
- (b) List three advantages of using pie-charts to represent statistical data.

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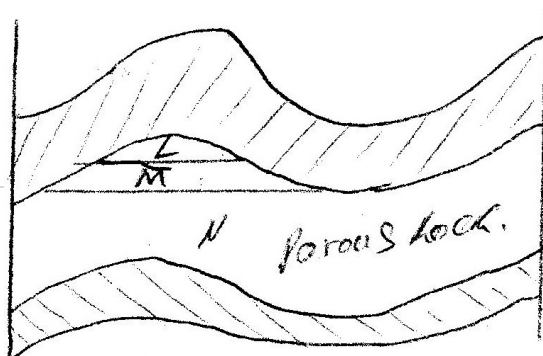
CHAPTER 5

ENERGY.

1. (a) Apart from providing power, state three other benefits of the dams along River Tana. 3 mks
 (b) State two problems that affect hydroelectric power production along river Tana. 2 mks
 (c) State two factors that hinder the expansion of geothermal power. 2 mks
2. (a) Name two non-renewable sources of energy. 2 mks
 (b) Explain four physical factors that influence the location of a hydroelectric power station. 8 mks
3. (a) Explain three benefits that would result from rural electrification in Kenya. 6 mks
 (b) In what three ways did the power shortage resulting from the drought of the years 1999 and 2000 affect this industrial sector in Kenya? 5 mks
4. (a) Give three reasons that make tropical countries to have the potential to develop HEP. 3 mks
 (b) Give two reasons why tropical countries are not sufficient in HEP production. 2 mks
5. Use the diagram to answer questions below.



- (a) Name 2 mks
 (i) The HEP stations marked S & T. 1 mk
 (ii) Name the proposed HEP station marked U. 2 mks
- (b) Name renewable sources of industrial energy other than water
6. The diagram below shows the occurrence of petroleum in the earth's crust. Use it to answer question (a).



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- (a) Name the substances labeled I, M and N. 3 mks
- (b) Give two by-products obtained when crude oil is refined. 2 mks
- (c) Explain efforts by Kenyan government to reduce cost on importation of petroleum. 6 mks
7. Name two main disadvantages of using coal as a source of energy. 2 mks
8. Suggest four solutions to Kenya's energy deficit. 4 mks
9. What are the causes of energy crises? 4 mks
10. List uses of nuclear energy. 4 mks
11. Name uses of wind energy. 2 mks
12. Name two non- renewable sources of energy. 2 mks
13. Why has usage of coal as a source of energy declined? 4 mks
14. Give four reasons why Kenya has been unable to exploit high geothermal power potential. 4 mks
15. Name the various methods that can be used to conserve energy. 4 mks
16. Define biomass. 2 mks
17. (a) Define energy. 2 mks
- (b) List three renewable; sources of energy. 3 mks
- (c) State the disadvantages of coal as a source of energy. 4 mks
18. (a) What is the name of power projects along river Tana. 4 mks
- (b) Apart from hydroelectric power production, state the other benefits of the dams along the Tana. 4 mks
- (c) Identify the problems facing hydroelectric power production along the Tana. 4 mks
19. (a) Apart from the Tana, name three other hydroelectric power projects in Kenya. 3 mks
- (b) State the benefits of the Owen Falls Dam in Uganda. 4 mks
- (c) State the factors limiting the expansion of geothermal power production in Kenya. 4 mks
20. Apart from hydro-electric power production; state the other benefits of the dams. 4 mks

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21. State the significance of energy. 5 mks
22. (a) What is energy crisis? 2 mks
(b) State the causes of energy crisis. 2 mks
23. Explain the problems Kenya faces due to overdependence on petroleum. 4 mks
24. (a) Differentiate between management and conservation of energy. 2 mks
(b) Describe the measures of management and conservation of energy. 10 mks

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CHAPTER 6

INDUSTRY

1. (a) State two reasons why some industries are located near the sources of raw.
2 mks
(b) Give three characteristics of the cottage industry in India. 3 mks
2. (a) State characteristics of jua kali industries in Kenya. 5 mks
(b) State problems facing Jua kali industries in Kenya. 5 mks
(c) Differentiate between manufacturing and tertiary industries. 4 mks (d) State three factors that led to the growth of iron and steel industry in the:
Ruhr region of Germany. 3 mks
3. (a) Name three agricultural non-food manufacturing industries in Kenya.
3 mks
(b) Explain three causes of the decline in the textile industry in Kenya. 3 mks
(c) Describe the measures that should be taken to control the following.
(i) Water pollution
(ii) Rural urban migration. 2mks
(d) Explain four factors which have favoured the development of the electronics industry in Japan. 8 mks
4. (a) What is industrialization? 2 mks
(b) Name a town in Kenya where each one of the following industries is located,
(i) Oil refinery
(ii) Paper manufacturing
(iii) Motor vehicle assembly. 3 mks
(c) Give five factors why the development of the Jua kali industry is encouraged In Kenya.
5 mks
(d) Name two non-food agricultural industries in Thika. 2 mks
5. State benefits of exploiting soda ash in Kenya. 5 mks
6. State two major steps currently -undertaken by the government to promote industrial growth and development. 2 mks
7. What is industrialization? 2mks
8. State three reasons why the jua kali industry is encouraged in Kenya. 3 mks
9. (a) What is an industry. (2mks)

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- (b) State two ways in which each of the factors affect the location and development of industries.
- (i) Raw materials
 - (ii) Transport
 - (iii) Markets 6 mks
- (c) Explain why power is not considered as a major industrial locational factor in the modern world. 2 mks
10. (a) Define:
- (i) Primary industry
 - (ii) Secondary industry
 - (iii) Quaternary industry 6 mks
- (b) (i) What is meant by Jua Kali industry in Kenya? 2 mks
- (ii) State five economic benefits of the Jua Kali industry in Kenya. 5 mks
- (c) Explain the significance of industrialization to Kenya. 6 mks
11. (a) State two main industries found in the towns below:
- (i) Thika
 - (ii) Athi River
 - (iii) Kisumu
 - (iv) Nanyuki
 - (v) Nakuru 10 mks
- (b) Explain the problems associated with industrialization. 6 mks
12. State five reasons why the Ruhr region in Germany is the leading industrial centre in Europe. 5 mks
13. (a) (i) Mention four industrial regions in Japan. 4 mks
- (ii) List five factors that have aided car manufacture and electronic industries in Japan. 5 mks
- (b) Explain the reasons which contributed to the growth and development of the cottage industry in India. 8 mks

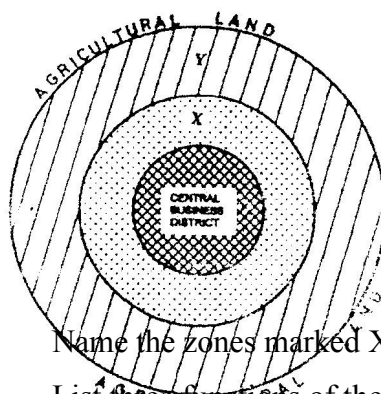
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CHAPTER 7

SETTLEMENT.

1. The diagram below represents the functional zones of urban centre. Use it to answer question

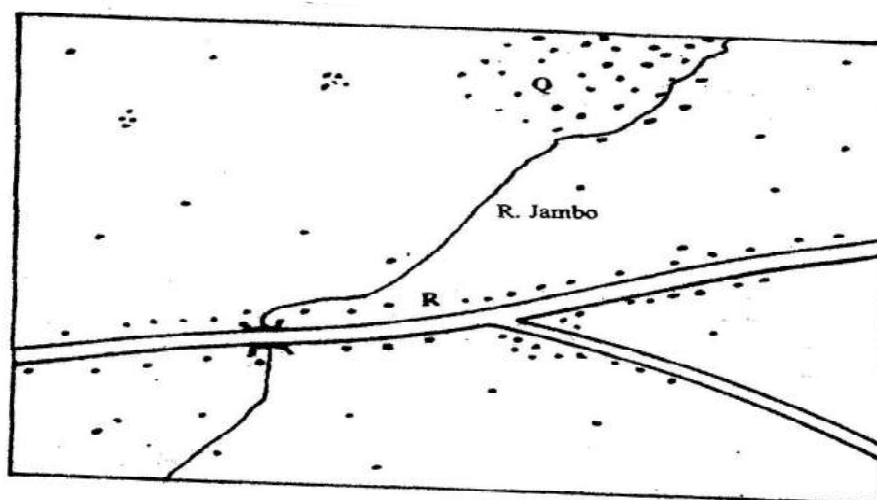
(a)



- (a) (i) Name the zones marked X and Y. 2 mks
(ii) List three functions of the Central Business District. 3 mks

2. (a) Name two types of human settlements. 2 mks

Use the sketch below to answer question (b)



KEY	
	Settlements
	Road
	River

- (b) Settlement patterns marked Q A R. 2 mks
3. (a) Name two types of rural settlement patterns. 2 mks
(b) Apart from urban-rural migration, name two other types of migration. 2 mks
- (c) State three factors that may lead to urban-rural migration. 3 mks
4. State three factors which led to the development of Mombasa into a major sea port in the region. 3 mks
5. (a) What is urbanization? 2 mks
(b) Give two differences in the functions of New York and Nairobi cities.

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4 mks

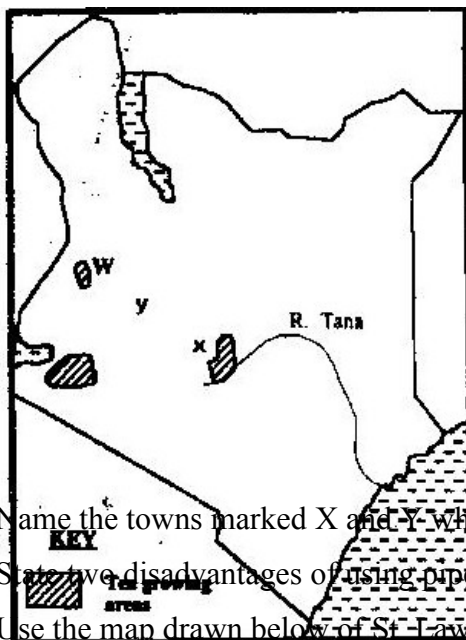
6. Apart from pollution, explain four problems experienced in urban centres. 8 mks
7. Explain similarities and differences between Nairobi and New York cities. 8 mks
8. State two main functions of rural settlements. 2 mks
9. Explain four problems that are experienced in Nairobi as one of the major urban centres in Kenya. 4 mks
10. List three factors that have led to rapid growth of Mombasa town. 3 mks
11. State ways through which the Kenyan government is using to solve problems of Nairobi city. 5 mks
12.
 - (a) Differentiate between the terms settlement and urbanization. 4 mks
 - (b) Identify factors influencing settlement. 4 mks
 - (c) State the factors influencing patterns of settlement. 5 mks
13.
 - (a) By use of relevant examples, explain the distribution of urban centres in East Africa. 5 mks
 - (b) Explain 4 human factors which may lead to development of towns.
14.
 - (a) Explain factors leading to the growth of Kisumu as a Lake Port. 8 mks
 - (b) What are the functions of Thika town? 5 mks
15. Explain the problems facing New York City. 6 mks
16.
 - (a) Outline the main differences between the cities of New York and Nairobi. 4 mks
 - (b) Compare the port of Mombasa to the port of Rotterdam. 8 mks
17. Discuss the effects of urbanization. 4 mks

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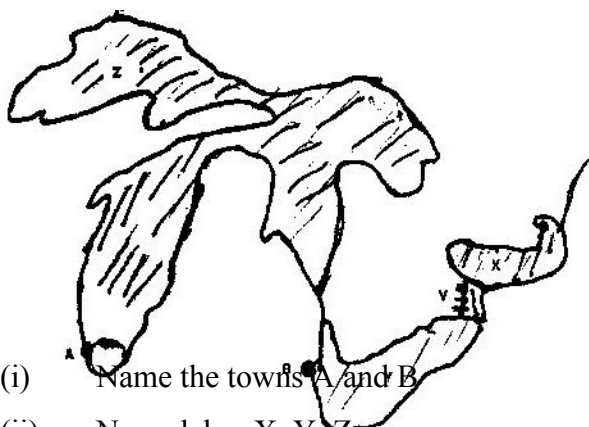
CHAPTER 8

TRANSPORT AND COMMUNICATION.

1. State the causes of the decline in the use of letter writing as a means of communication in Kenya. 5 mks
2. (a) Give three advantages of railway over road transport. 3 mks
(b) The map below shows the extent of the oil pipeline in Kenya. Use it to answer question (b).

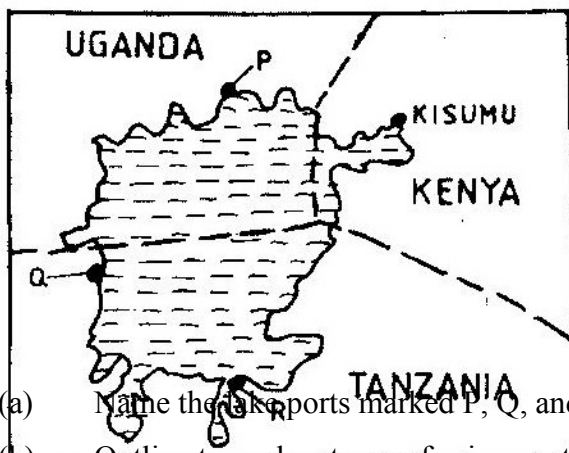


- Name the towns marked X and Y which are served by the pipeline. 2 mks
- (c) State two disadvantages of using pipelines as means of transporting oil.
 - (d) Use the map drawn below of St. Lawrence sea way to answer the questions that follow.



- (i) Name the towns A and B.
 - (ii) Name lakes X, Y, Z.
 - (iii) Name canal marked V.
3. Use the sketch map of Lake Victoria below to answer question (a).

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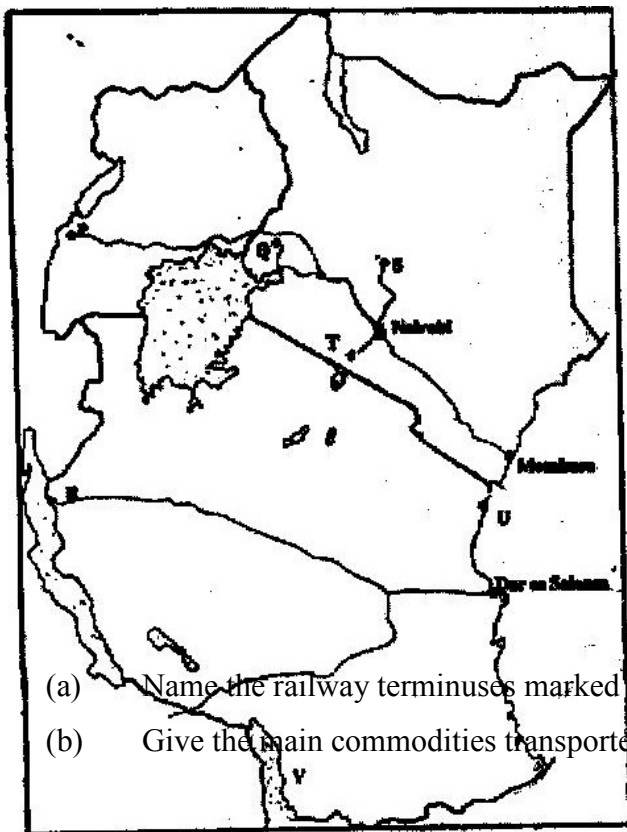
- (a) Name the lake ports marked P, Q, and R. 3 mks
- (b) Outline two advantages of using containers while handling goods at the port of Kisumu.

2 mks

4. (a) Give three reasons why road transport is used more than air transport in East Africa. 3 mks
- (b) In what three ways does Kenya benefit from air links with the rest of the world?

3 mks

5. Use the map of East Africa below to answer question (a).



- (a) Name the railway terminuses marked P, Q and R. 3 mks
- (b) Give the main commodities transported by the railway lines marked S and T. 2 mks
- (c) Name the port marked U and the lake marked V. 2 mks

2 mks

2 mks

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- (d) State four reasons why road network is more widespread than railway network in East Africa. 4 mks
- (e) One of the problems facing road transport is the high frequency of accidents. Explain four conditions of roads in Kenya that may lead to accident. 8 mks
6. Give four benefits of the efforts the government is data in streamlining pubic transport sector. 4 mks
7. Outline two major problems affecting the development of trans- African highway in Africa. 2 mks
8. State two major problems hindering river transport in Kenya. 2 mks
9. Suggest three benefits of the proposed Southern bypasses to be constructed in Kenya. 3 mks
10. (a) Define containerization.
(b) Outline three merits of using containerization as a method of transportation. 3mks
11. (a) State three problems facing railway transport in Kenya. 3 mks
(b) Identify importance of railway transport in a country. 4 mks
12. Differentiate between 'transport' and communication. 4 mks
13. Apart from cell phones, mention other two modern methods of communication. 2 mks
14. State advantages of using cell phones communication. 3 mks
15. State reasons why St. Lawrence sea way was set up by the government of USA. 4 mks
16. (a) Define:
(i) Transport
(ii) Communication 4 mks
(b) Apart from water transport, list the other modes of transport. 2 mks
(c) Name the two types of waterways used in transportation. 2 mks
17. (a) Give three reasons why river transportation in Africa is poorly developed. 3 mks
(b) Name the major ocean routes of the world. 3 mks
(c) State the advantages of water transport. 6 mks

18.	(a)	Account for the poor rail linkages <i>j</i> , between the African countries.	2 mks
	(b)	Name three trans-continental rail lines in Africa.	3 mks
	(c)	What are the advantages and disadvantages of transporting goods by rail?	6 mks
19.	(a)	Why is railway transport less used in Africa?	4 mks
	(b)	Name two railway systems in Africa.	2 mks
	(c)	Discuss the advantages and disadvantages of railway transport.	4 mks
20.		State the advantages of road transport.	4 mks
21.	(a)	What is containerization?	2 mks
	(b)	Discuss advantages and disadvantages of containerization.	6 mks
	(c)	State advantages and disadvantages of air transport.	6 mks
22.		Identify the main types of communication.	4 mks
23.	(a)	Name two trans-African highways	2 mks
	(b)	What are the benefits of trans-African highways?	2 mks
	(c)	Explain the problems facing the trans-African highways.	4 mks
24.		Discuss the role of transport and communication in economic development of Africa.	4 mks
25.	(a)	State the problems facing transport and communication in Africa.	4 mks
	(b)	Outline the efforts being made to solve these problems.	4 mks
26.	(a)	(i) Identify the obstacles that face navigation along the St Lawrence seaways.	3 mks
		(ii) State ways in which navigation on the seaway was improved.	4 mks
	(b)	Explain the benefits of the St. Lawrence seaway on the economies of the United States of America and Canada.	6 mks

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CHAPTER 9

TRADE.

1. (a) State five reasons why the common market for Eastern and Southern Africa was formed. 5mks
2. (a) What is international trade? 5mks
(b) Name major imports from Europe to Kenya 2 mks
(c) List factors that influence the import and export of goods in Kenya. 4 mks
(d) Explain ways through which Kenya will benefit from the renewed East African co-operation. 6 mks
(e) Explain negative effects of international trade. 6 mks
3. (a) Explain four measures which Kenya may take to reduce the unfavourable Balance of trade. 8 mks
(b) Explain four benefits that Kenya derives from international trade. 8 mks
4. (a) What is Trade? 2 mks
(b) Identify the two types of internal trade. 2 mks
5. State the factors influencing trade. 4 mks
6. (a) Differentiate between visible and invisible exports. 4 mks
(b) Draw a table showing the major visible export and imports of Kenya. 8 mks
7. Explain the significance of trade to Kenya. 5 mks
8. State the future on international trade in Kenya. 3mks
9. (a) What is a regional trading bloc? 2 mks
(b) Apart from the European Union, name three other regional trading blocs. 3 mks
(c) Outline the role played by the European Union in the economy of Europe. 4 mks
10. Explain why the Kenya's exports are more to the outside world than her neighbors. 6 mks

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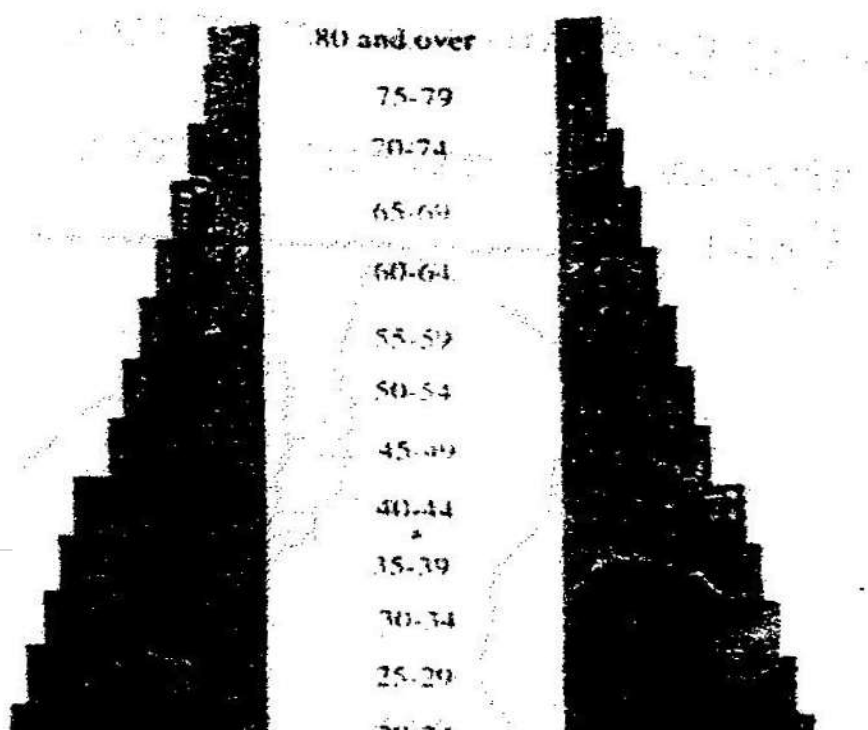
CHAPTER 10

POPULATION

1. (a) State four reasons why the northern parts of Kenya are sparsely populated. 4 mks
- (b) Give two primary sources of population data 2 mks
- (c) What information can be derived from a population pyramid? 4 mks
- (d) Describe three ways in which the population of Kenya differ from those of Sweden. 6 mks
- (e) Explain four causes of rural-urban migration in Kenya. 8 mks
2. Explain three problems which result from the high population growth rate in the East African countries. 6 mks

2005:

3. List three factors that have contributed to a decline in infant mortality in Kenya. 3 mks
4. Explain two reasons why Thika District has a higher population than Murang'a district. 4 mks
5. (a) State the reasons for carrying out population census. 5 mks
- (b) How the following factors lead to population increase in Kenya.
(i) Early marriages
(ii) Improved medical facilities
(iii) Cultural beliefs. 6 mks
6. The pyramid below represents the population of country X. Use it to answer question (a).



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- (a) Describe the characteristics of the population represented by the pyramid.
4 mks
- (b) Explain three problems which may result from a high population growth rate.
6 mks
- (c) Describe THREE measures that have been taken in Kenya to reduce infant mortality.
6 mks
- (d) Explain four factors that have led to a high population density in Lake Victoria basin.
8 mks
7. a) Define the term population. 2 mks
- (b) Explain factors influencing population distribution. 6 mks
8. (a) Explain factors influencing population growth. 8 mks
- (b) Describe the main features of population structure of a developing country.
4 mks
9. Explain the factors leading to high fertility levels in a population. 6 mks
10. Compare and contrast population trends between Kenya and Sweden. 8 mks
11. The table below shows population distribution in Kenya by province in 1999. Use it to answer the questions below.

Province	Population	Area in Sq Kms
Nairobi	2,143,254	696
Central	3,724,159	13,220
Rift Valley	6,987,036	182,539
Western	3,358,776	8,264
Nyanza	4,392,264	12,547
Coast	2,487,264	82,816
Eastern	4,634,779	153,473
N. Eastern	962,143	128,124

Table 9.3 *Population of Kenya by Provinces*

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- (a) Calculate the population density of each province. 2 mks
- (b) Give reasons why there is a high population density in Central Province of Kenya.
3 mks

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CHAPTER 11

MANAGEMENT AND CONSERVATION OF THE ENVIRONMENT.

1. (a) Why is it necessary to conserve water? 3 mks
(b) How does terracing help in water conservation? 2 mks
2. (a) Apart from desertification, name two other environmental hazards experienced in Kenya. 2 mks
3. (a) Name three physical regions through which river Tana passes. 3 mks
(b) Name two rivers in Kenya to the West of the Rift Valley which causes large scale flooding. 2 mks
(c) Explain four problems caused by floods. 8 mks
4. (a) Define the term pollution. 2 mks
(b) Explain three effects of land pollution on the environment. 6 mks
(c) State four ways through which Sand pollution can be controlled. 8 mks
5. (a) State ways in which drought affects the agricultural sector in Kenya. 4 mks
(b) What is soil conservation? 2 mks
(c) State three farming methods that assist in soil conservation. 3 mks
6. State three factors contributing to occurrence of floods 3 mks
7. State five negative effects of floods. 5 mks
8. State measures currently undertaken by Kenyan government to control natural disasters. 5 mks
9. Explain five economic importance of controlling floods. 10 mks
10. Name five natural hazards. 5 mks
11. Name three human induced hazards. 3 mks
12. Define desertification. 2 mks
13. Name causes of desertification. 4 mks
14. List effects of desertification. 4 mks
15. Explain five types of environment. 10 mks
16. (a) Define:
 - (i) Environment
 - (ii) Management of the environment
 - (iii) Conservation of the environment6 mks

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- (b) Why is it necessary to manage and conserve the environment? 4 mks
17. (a) Mention six environmental hazards. 6 mks
- (b) Explain:
- (i) The causes of floods.
- (ii) The effects of floods. 4 mks
18. (a) (i) What is lightning? 2 mks
- (ii) What causes lightning? 2 mks
- (b) (i) What are the effects of lightning? 2 mks
- (ii) How can lightning be controlled? 4 mks
19. (a) (i) Define pollution. 2 mks
- (ii) Name the four types of pollution.
- (b) (i) What is air pollution? 2 mks
- (ii) State five causes of air pollution. 5 mks
- (iii) How can pollution be controlled? 5 mks
20. (a) (i) Name four non -governmental organizations involved in the management and conservation of environment in Kenya. 4 mks
- (b) Explain how legislation in Kenya is used to manage and conserve the environment. 8 mks

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PAPER 1 (A)

2 HOURS 45 MINUTES

Answer all questions in this section.

1. (a) Define Geography. 2 mks
(b) Explain the relationship between Geography and Biology. 2 mks
2. (a) Differentiate between mass wasting and mass movement. 2 mks
(b) Outline four factors that cause soil creep to occur. 4 mks
3. (a) Name two fold mountains formed during Alpine Orogeny. 2 mks
(b) State three theories which explain the formation of fold mountains. 3 mks
4 Explain what you understand by each of the following;
(a) Vertical movement.
(b) Isostatic adjustment 4 mks
5. (a) Give three conditions necessary for the formation of Karst scenery. 3 mks
(b) List three zones of saturation below the earth surface. 3 mks

SECTION B

Answer question 6 and any other two questions

6. Study the map of Taita Hills (1: 50 000) sheet 189\4 provided and answer the following questions.
 - (a) What is the grid square of:
 - (i) Water reservoir at Mwasere
 - (ii) Water tank at Kirutai 4 mks
 - (b) What is the magnetic variation as of January 1991? 2 mks
 - (c) What is the direction of flow of River Goshi? 2 kms
 - (d) Explain four ways in which relief influences settlements in the area covered by the map. (8mks)
 - (e) Name three types of natural vegetation in the area covered by the map. 3 mks
 - (f) Students of the school at Zare carried out field study around the school,
 - (i) List four preparations they made. 4mks

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- (ii) State two methods they would use to collect data. 2 mks
7. (a) Define the term Lake, 2 mks.
- (b) Explain three reasons why some Rift Valley lakes are saline. 6 mks
- (c) (i) Describe how caldera lake is formed. 5 mks
- (ii) Give three examples of caldera lakes in Africa. 3 mks
- (d) Explain four ways in which lakes modify the climate of the surrounding areas. 8 mks
8. (a) (i) Define faulting. 2 mks
- (ii) Distinguish between reverse and tear faults. 2 mks
- (b) Explain formation of each of the following;
- (i) Fault scarp
- (ii) Tilt block 4 mks
- (c) Explain four economic significances of faulting to human activities 8mks (d) (i) State three advantages of oral interviews. 3 mks.
9. (a) (i) Differentiate between zero lapse rate and environmental lapse rate. 4 mks
- (ii) Name two types of fronts in air masses. 2 mks
- (iii) State two effects on climate from urbanization. 2 mks
- (b) (i) Name one type of hot climates. 1 mk
- (ii) Account for the characteristics of rainfall experienced in Equatorial climate. 8 mks
- (c) (i) State four causes of the recent global climatic change. 4 mks
- (ii) State six major effects of climate change. 6 mks
10. (a) (i) State three characteristics of arid areas. 3 mks
- (ii) Give two factors that promote wind deposition in arid areas 2 mks
- (b) (i) Identify two processes of wind erosion. 2 mks
- (ii) Describe how the barchan is formed. 4 mks
- (c) Differentiate between a rock pedestal and a mushroom block. 4 mks
- (d) Suppose you are asked to carry out a field study on the action of water in a desert.
- (i) State two objectives for your study 2 mks
- (ii) Name two water depositional features you are likely to observe. 2 mks

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(iii) Explain three significance of arid landforms.

6 mks

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- b) Discuss the formation following Lakes.
- i) Lake Victoria
 - ii) Lake Tanganyika 8 mks
- c) Explain how human activities have negative impact on lakes. 8 mks
- d) i) List three positive effects of lakes to humans. 3 mks
- ii) Name four follow up activities in fieldwork. 4 mks
8. a) i) Differentiate between mineral and rock. 4 mks
- ii) State characteristics of sedimentary rocks. 3 mks
- b) Classify rocks according to form and origin giving two examples in each type. 9 mks
- c) Explain formation of following examples of rocks. 6 mks
- i) Tuff
 - ii) Coral rock
- d) State use of equipment listed below in fieldwork. 3 mks
- (i) Geological hammer
 - (ii) Lenses
 - (iii) Hydrochloric acid
9. a) i) Name three hot deserts. 3mks
- ii) State two characteristics of arid lands. 2 mks
- b) Describe formation of following features.
- i) Rock pedestals
 - ii) Yardang 8 mks
- c) Name three resultant features of action of water in deserts. 3 mks.
- d) i) Name three processes of wind transport. 3 mks.
- ii) State advantages of using secondary sources of data. 2 mks
- iii) List four evidences of desertification. 4 mks.
10. a) i) What is a fault 2 mks.
- ii) Name the parts of a fault? 4 mks
- b) i) Name two resultant features of faulting. 2 mks
- ii) Explain formation of Rift valley by theory of anticlinal arching. 3 mks
- c) i) Distinguish between basic lava and acidic lava. 6mks

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- ii) State four characteristics of composite volcano. 4 mks
- iii) State four positive influences of volcanicity. 4 mks

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KCSE MODEL PAPER 2 (A)

2 HOURS 45 MINUTES

SECTION A

Answer all questions in this section

1. (a) Name two exotic beef cattle reared in Kenya. 2 mks
(b) State three physical conditions favouring beef farming in Argentina 3 mks
2. (a) Name two products which can be transported using pipelines. 2 mks
(b) Give three advantages of using pipelines as a means of transport. 3mks
3. a) What is land reclamation? 2 mks
b) State three physical factors which influenced the establishment of the Perkerra irrigation scheme. 3 mks
4. a) Name three cities in Kenya. 3 mks.
b) State three problems facing the growth of Mombasa city. 3 mks
5. a) Name two environmental hazards other than floods 2mks
b) Give three problems resulting from flooding in Lake Victoria basin. 3mks

SECTION B

Answer question 6 and any other two questions

6. a) (i) What is statistics? 2 mks.
ii) Name three ways in which data analysis is done. 3 mks
iii) Name two advantages of a wind rose. 3 mks
b) State four physical conditions necessary for the growing of sugar cane.
c) Describe the commercial production of sugar cane from land preparation to harvesting. 8 mks
d) Explain three problems facing sugar cane farmers in Kenya. 6 mks
7. a) Name three national parks in Uganda 3 mks
b) (i) Explain two differences between a national park and a game reserve. 4 mks
ii) Name three tourist attractions found in the Rift Valley Province of Kenya. 3 mks
c) i) State three physical conditions that influence distribution of wildlife in East Africa. 3 mks

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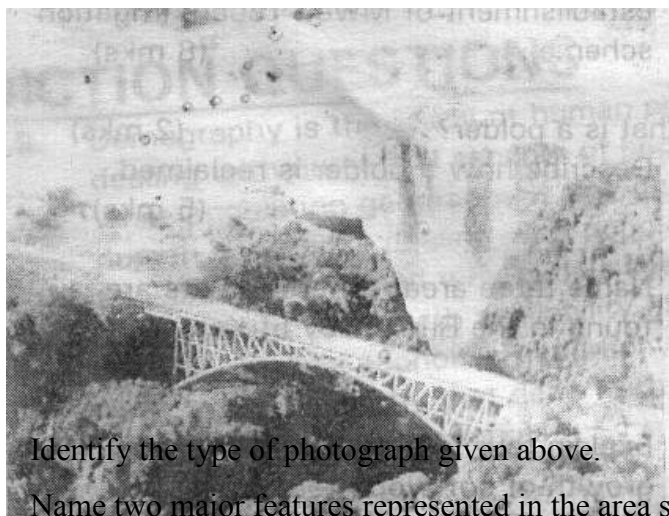
- ii) List four problems experienced by the Kenya Government in its effort to conserve wildlife. 4 mks
- d) Explain four factors which have made Switzerland a major tourist attraction in Europe. 8 mks
8. (a) i) What is agroforestry? 2 mks
ii) Give three reasons why agroforestry is being encouraged in Kenya. 3 mks
- (b) i) Name three types of coniferous trees found in Kenya. 3 mks
ii) State five characteristics of coniferous forest. 5 mks
- (c) Compare forestry in Kenya and Canada under the following sub-headings.
i) Tree harvesting.
ii) Transportation of logs
iii) Marketing
- (d) Explain three problems that affect forestry in Canada. 6 mks
9. a) i) Name two non-renewable sources of energy other than coal. 2 mks
ii) Give three disadvantages of using coal as source of energy. 3 mks b) i) Name two Geothermal potential areas in Kenya. 2 mks
ii) Give three reasons why Kenya has not been able to fully exploit her geothermal potential. 3 mks
- c) Explain four conditions that favour the location of hydroelectric power station. 4 mks
- d) i) What is energy crisis? 2 mks
ii) Explain three effects of energy crisis in Kenya. 3 mks
10. a) i) Differentiate between pelagic and demersal fishing. 2 mks
ii) Name any two examples of demersal fish. 2 mks
- b) State four reasons why fresh water fishing is more important than marine water fishing in Kenya. 4 mks
- c) i) What is fish farming? 2 mks
ii) State four ways through which fish farming contributes to the economy of Kenya. 4 mks
- d) State three problems facing fish farming in Kenya. 3 mks
- e) Explain four conditions that favour fishing in Japan. 4 mks

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SECTION B

Answer question 6 and any other two questions from this section.

6. Use the photograph below to answer questions (a) i and (ii)



- a) i) Identify the type of photograph given above. (1mk)
- ii) Name two major features represented in the area shown on the photograph. (2mks)
- b) Identify four road conditions that trigger mad accidents. (4mks)
- c) Explain four limitations of road transport. (8mks)
- d) Explain four problems faced by African states in efforts to improve roads. (8mks)
7. (a) Give five reasons why wildlife conservation is encouraged in Kenya. (5mks)
- b) Explain what you understand by the following terms:
- i) Domestic tourism.
- ii) Eco-tourism (4mks)
- (c) Explain three factors that have hindered the development of domestic tourism in Kenya.
- (d) Explain four factors that have made Switzerland a major tourism destination in Europe.
8. (a) What do you understand by the following terms:
- i) Land reclamation.
- ii) Land rehabilitation. (4mks)
- b) i) State any four methods used in land reclamation and rehabilitation in Kenya. (4mks)
- ii) Apart from Mwea Tebere, name two other large scale irrigation schemes in Kenya. (2mks)

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- iii) Explain four factors that led to the establishment of Mwea Tebere irrigation scheme. (8mks)
- (c) i) What is a polder? (2mks)
- ii) Describe how a polder is reclaimed. (5mks)
9. a) i) Name three areas where forests are found in the Rift Valley of Kenya. (3mks)
- b) Explain four factors that favour the growth of softwood forests in Swaziland. (4mks)
- c) i) Explain four problems experienced in commercial exploitation of the equatorial forest in Africa. (4mks)
- ii) Give three species of trees found in Gabon. (3mks)
- d) State three measures being taken to conserve forests in Kenya. (3mks)
10. (a) i) Name two conditions that are necessary for the formation of petroleum. (2mks)
- ii) Give two reasons why Kenya imports her oil in crude form. (2mks)
- (b) i) State two advantages of geothermal power. (2mks)
- ii) State four factors that hinder the expansion of geothermal power production in Kenya. (4mks)
- (c) Explain four ways by which the government of Kenya should apply to conserve her energy. (4mks)
- (d) i) What is multi-purpose dam? (2mks)
- ii) State three reasons why Akosombo dam was established. (3mks) iii) List three benefits of establishing the Akosombo dam. (3mks)

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ANSWERS TO TOPICAL QUESTIONS

PAST KCSE AND PREDICTION QUESTIONS

FORM ONE

CHAPTER 1

INTRODUCTION TO GEOGRAPHY

1. Collective term for methodologies of fieldwork, maps and map work and photograph interpretation used in study of geography.
2.
 - Physical geography
 - Human geography
3.
 - Climate
 - Rocks and minerals
 - Earth and the solar system
 - Soil
4.
 - External conditions surrounding of an organism.
5.
 - Provides knowledge that promote conservation of resources
 - Admission in careers that generate income e.g. Geologist.
 - Learn skills in time management useful in personal activities
6.
 - Transport lines e.g. roads.
 - Settlement structures e.g. houses
 - Drainage features e.g. boreholes, water dams
 - Industries e.g. coffee mills
 - Farms e.g. tea estates
7. Part of physical conditions that provide home in which certain organisms live.
8.
 - History
 - Biology meteorology
 - Demography
 - Sociology
 - Agriculture
 - Economics
 - Physics
 - Chemistry
 - Geology

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- Medicine
- 9. • Demography is the study of human population dealing with numerical aspects of population while population geography is branch of human geography dealing with population explaining where and why people live.
- Economics deals with availability of resources while economic geography is branch of geography that deals with location and distribution of resources.
- 10. • Geo, graphein

CHAPTER 2

THE EARTH AND THE SOLAR SYSTEM

1. (a)
 - Causes day and night/apparent movement of sun from east to west.
 - Causes differences in time at different longitudes.
 - Causes deflection of winds\ ocean currents.
 - Causes rising and falling of sea tides.
 - Causes variation in atmosphere pressure on the earth surface.(b)
 - Earth revolution
 - Causes changes in position of midday sun at different times of the year.
 - Causes varying length of the days and nights in northern and southern hemisphere.
 - Causes changes in seasons; spring, summer, autumn and winter. • Causes lunar eclipse
2. (a) (i) P - Atmosphere
(ii) Q - Barysphere/centrosphere/core
(iii) R -Mohorovicic discontinuity/moho dicontinuity
(b)
 - Divided into two-upper and lower mantle
 - Mantles' main constituent minerals are ferro-magnesium and silicate.
 - Mantle is about 2, 900 km thick.
 - Upper mantle has low temperature than lower mantle.
 - Mantle has temperatures of about 1000°C.
 - Mantle is made up heavier rocks than rocks of earth crust.
 - Upper mantle is made up of an elastic solid/semi-molten
 - Inner mantle is made up an elastic solid/semi molten basic rocks/ viscous liquid.
3. (a) The planets marked F and G is
 - Mars
 - Neptune
4. (a) Solar system refers to the composition of the sun, the planets and other heavenly bodies related to the sun.
(b) i) Solar eclipse
(ii) L - Moon
M- Shadow

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5. (a) i) 21st March and 23rd September
ii) Due to revolution of the earth.
- (b) i) 274.5 days
ii) Summer season
6. (i) Solar system - Organization made up of the sun with the nine planets orbiting around it and heavenly bodies,
(ii) Galaxy - Group/cluster of stars in the universe.
(iii) Star - Hot mass of glowing gases that transmit light to outer bodies.
(iv) Asteroid - Small planet-like objects orbiting around the sun between the planets of Mars and Jupiter.
7. • Latitude is the distance north or south of equator measured as an angle from the earth's centre while longitude is the distance of the earth's surface measured east or west of prime meridian and expressed as an angle.
Latitude is imaginary line running from East to West showing how far North or South a place is from Equator.
- Dateline is line 180° at which a day is lost or gained while international dateline is zigzag line along longitude 180° deviating land surfaces and at which day is lost or gained.
- Glowing objects that quickly cross the sky before they burn up and disappear while meteorites are those meteors that pass through the atmosphere brightly but do not burn up.
8. • In solar eclipse moon lie between sun and earth while in lunar eclipse earth lie between moon and sun.
- In solar eclipse shadow of moon is cast on earth while in lunar eclipse shadow of earth is cast on moon.
- Solar eclipse occurs during the day while lunar eclipse occurs during the night.
- Lunar eclipse is caused by earth's revolution while solar eclipse is caused by revolution of moon.
9. • Presence of water that support life.
- Presence of atmosphere with adequate O₂ and CO₂ levels that support life of animals and plants respectively.
- Enough heat and light due to earth's favourable distance from the sun.

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- Proportional gravitational force that allow objects to be upright on the earth's surface.
10. (a) • Causes deflection of the winds
- Causes time difference between Meridians.
- Causes variation in speed of air masses.
- Causes rising and falling of ocean currents.
- Causes variation in atmospheric pressure.
- | | |
|------------------------|--------------|
| G.M | East |
| 34°E | 41°E |
| 1 p.m. | ? |
| 1° | = 4 minutes |
| 4 x 4 | = 16 minutes |
| Local time = 1.16 p.m. | |
11. (a) Periods 21st March and 23rd September when the sun is overhead at midday along the equator.
- (b) • Sun is overhead at mid-day along the tropic of cancer/Capricorn.
- The Arctic Circle experiences 24 hrs of daylight.
- Days are longer than nights.
- Temperatures are high in the region experiencing summer solstice.
- 24 hour sunshine within the circles.
12. 66 ½⁰, 23 ½⁰
13. (a) Silica, aluminium
- (b) 2.7 gms/cc
- (c) 6 -10 kms
- (d) 3.0-3.3 gms/cc
- (e) 3470 kms
- (f) 5,500 cc
14. • Chances of another star approaching the sun are minimal.
- High temperature materials drawn from the sun would disperse rather than condense.
- It does not explain where the sun and the star came from.
15. Hydrosphere is part of the earth surface covered by water masses e.g. oceans, seas, rivers and swamps while atmosphere refers to thin layer of gases surrounding the earth and held by earth's gravitational pull.

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16. (a)
 1. Uranus
 2. Venus
 3. Earth
 4. Venus
 5. Jupiter
 6. Venus
 7. Mercury
17.
 - Earth rotates on its own axis to make a complete turn; and its poles rotate of this axis and pulled towards each other (centripetal forces)
 - Equator covers a long distance and therefore rotates faster, with more speed causing a flinging force (centrifugal force).
18.
 - Low temperatures
 - Longer night times than day times at latitudes beyond equator
 - The sun is overhead Tropic of Capricorn on 22nd December and its winter Solstice in the Northern Hemisphere.
 - On 21st June the sun is directly overhead the Tropic of Cancer and its winter solstice is in the Southern hemisphere.
 - Sun is not visible at cycles and there is darkness for 24 hrs.
19.
 - Winter solstice occurs on 22nd December and 21st June when the sun is overhead at mid-day along tropic of Capricorn and Cancer respectively. At the Arctic Circle and Antarctic circles the sun is visible for only a few minutes when it appears above the southern/ Northern horizon.
 - Summer solstice occurs on 21st June and 22nd December when the sun is overhead in the tropic of Cancer and Capricorn respectively. The sun rises higher in the sky and is visible for 24 hrs at the arctic and Antarctic cycles.
20. (a) It is a shadow that is formed when rays of the sun are blocked from reaching the earth or the moon.
- (b)
 - Comets
 - Asteroids
 - Meteorites
 - Satellites
21.
 - It is an imaginary line running from North to South that shows how far

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east and West a place is from the Greenwich prime meridian.

- It refers to angular distance east or west of the Greenwich prime meridian.
22. • It causes aphelion whereby the earth is sometimes in its farthest position from the sun.
- It causes perihelion whereby the sun is sometime at its closest position to the sun.
 - It influences the occurrence of spring and neap tides.
 - It changes cycle of equinoxes and solstices
 - It influences occurrence of seasons
23. $1^{\circ} - 4 \text{ minutes}$
 $47^{\circ} \times 4 = 188 \text{ minutes}$
 $188 \text{ minutes} = 3 \text{ hrs } 8 \text{ minutes}$
Buchanan is 3 hrs 8 minutes behind Nairobi.
Time is 6.52 a.m.
24. (a) On crossing this longitude while going to the West, a day is gained and while crossing to the East a day is lost.
- (b) $66\frac{1}{2}^{\circ}$
- (c) Photographs taken from the outer space or satellites show the curvature of the earth.
- During eclipse of the moon, the earth casts a spherical-shaped shadow on the moon.
 - Circumnavigation of the earth along a straight path will bring one back.
 - Earth's horizon
 - Other planets are curved and earth is one of planets.
 - Different rising and setting times in different places.
 - An approaching ship.

CHAPTER 3

WEATHER

1. (a) During the day the land heats faster than the sea.
 - The air over the land rises
 - Cooler air from the sea blows towards the land to replace the rising air
 - The cool air from the sea is called sea breeze(b) (i) H- Mozambique, J - Benguera
(ii) Raising temperature Causes rainfall
2.
 - Troposphere
 - Stratosphere
 - Mesosphere
 - Ionosphere
3. (a)
 - Air must have abundant moisture.
 - A cloudless night to facilitate terrestrial radiation.
 - Air should be calm to remain in contact with the ground in order to be cooled.
 - There should be gentle air currents to hold water droplets in suspension.
 - The air must be cooled below dew point.(b) (i) R - cumulus
4. (a) (i) X - 3°C
Y - 9°C
(b) (i) 583 mm
(ii)
 - Sea make water is heated intensely by solar radiation.
 - Heating is intense in the afternoon
 - Warm moisture laden air rises and condenses at higher altitude.
 - Condensed water vapour forms cumulonimbus clouds.
 - Clouds eventually give rain accompanied by thunderstorm.
5. (a) (i) $30.3 - 28.4 = 1.9^{\circ}\text{C}$
(ii) $9.0 + 8.0 + 21.0 + 49.0 + 25.0 + 9.0 + 20.0 + 10.0 + 4.0 + 10.0 + 17.0 + 11.0$
 $= 1930 \text{ mm}$
(b)
 - Altitude - High altitude areas have low temperature and low pressure. Temperature varies with height because air is heated from below.
 - Winds transfer heat from one place to another causing changes in temperature.

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- Latitude influences climate such that areas near equator are warmer.
 - Aspect influences climate as south facing slopes in the northern hemisphere are warmer than north facing slopes in the same.
 - ITCZ- zone of low pressure which migrates North and South equator affects rainfall.
6. (a)
- Sunshine
 - Rainfall
 - Wind
 - Cloud cover
 - Air pressure
 - Humidity
7. a)
- Open area free of shade by trees and buildings.
 - Gentle land free of flooding
 - Area with wide view of surroundings.
 - Away from concrete surfaces.
- b) Reasons why Stevenson screen is;
- Painted white - can reflect direct heat from the sun.
 - Louvred on sides - To allow free flow of air and regulate temperature.
8. Relative humidity refers to the ratio between water vapour actually present in the air and its capacity to hold water vapour at a given temperature.
9. (a)
- It's heavy and torrential/falls in large drops.
 - Usually accompanied by lightning and thunderstorms
 - Falls mainly in the late afternoon
 - It's highly localized and lasts for a short while (15-20 mins)
- (b) Radiation fog forms when air in contact with the ground is cooled through terrestrial radiation while advection fog forms when warm moist air is cooled as it passes over cool surface e.g. land/sea.
10. (a)
- When the temperature rises, the alcohol in the left hand column expands and pushed the mercury column. The mercury in turn pushes the mercury in the right hand column and steel metal index up.
 - The maximum temperature is shown by the end of the index pushed by the mercury.

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- When the temperature falls, alcohol in the left hand column contracts and pulls the index along the tube. When the temperature rises, the alcohol expands leaving behind the index. Then the minimum temperature is read.
- (b)
 - According to the altitude of their bases.
 - Their appearance/structure
 - Their formation
- 11. (a)
 - It should be in an open place with free flow of air.
 - Away from barrier e.g. trees
 - Should be on a fairly level ground.
 - The site should be free from flooding
 - The site should provide a wide view of the surrounding landscape and the sky.
- (b)
 - Intensity of the sun's radiation in space the average distance from the sun.
 - The transparency of the atmosphere
 - Position of the earth in its orbit
 - The area and nature of the surface on which the rays fall.
- 12. Climate It's the average weather condition of a given place over a period or time usually (30-35 years)
- Relative humidity Refers to the ratio between water vapour actually present in the air and its maximum capacity to hold water vapour at a given temperature.
- Weather forecasting it's the prediction of the weather situation for a given place within a given period of time e.g. hour, a day, a week.
- Absolute humidity It is the total amount of water vapour that a given volume of air can hold.
- Weather lore Refers to a body of traditional facts and beliefs relating to weather e.g. a halo around the moon, croaking of frogs, a rainbow, migration of birds
- 13.
 - The students are able to relate what they have learnt in class to the real environment hence making geography real and interesting.
 - It breaks the class monotony.
 - It enables learners to develop skills of observation measurement, recording and analyzing data.
 - It improves the visual memory through observation.

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14. (a)
- Rain gauge
 - The rain gauge is kept in an open space in the weather station from above. Its raised to avoid splashes from entering into the gauge.
 - The water collected is emptied into the measuring cylinder every 24hrs.
 - Take readings on the measuring cylinder.
 - This cylinder is graduated in mm and the level the water emptied reaches gives us the reading amount of rainfall for the day.
 - Record the readings and interpret.
 - A maximum and minimum thermometer
 - When the temperature rises, alcohol in the left hand column expands and pushes the mercury column and maximum temperature is read.
 - When the temperature rises, alcohol in the left hand column contracts and pulls the index along the tube and the minimum temperature is read from the upper end of the index.
 - After recording the reading, the thermometer is reset using a magnet.
 - Interpret the readings.
- (b) (i) Convectional rainfall
- Its formation**
- The intense heating from the sun results into warm air rising in form of convectional currents.
 - The rising air reaches the high atmosphere and moisture in it condenses. Forms clouds and falls rain.
 - It falls in the late afternoon accompanied lighting and thunderstorms.

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- (c) **Problems**
- Lightening and thunderstorms which are destructive to life and property.
 - The torrential/large drops which are harmful to the crops and other vegetation.
 - The hailstones also are destructive to the crop leaves.
15. (a) • Weather forecasting
- Weather forecasting is the prediction of weather conditions
- (b) (i) Problems of weather forecasting
- Inaccurate data
 - Defective instruments
 - Personnel with limited skills
 - Vagaries of nature such as earthquakes
- (ii) • Determines times for sea and air travel.
- Determine time when sporting activities take place.
 - Determines the fishing activities and habits in the area.
 - Help determine suitable clothing for the day.
 - Help plan farmers calendar of activities.
 - Help plan suitable housing.
16. (a) How clouds influence weather.
- Clouds determine the amount of solar radiation reaching the earth's surface and the amount leaving the earth's surface. This determines temperature conditions.
 - Day temperatures are moderated by clouds.
 - Areas of thick rain clouds have high rainfall.
- (b) (i) Mean temperature $^{276}_{12} = 23^{\circ}\text{C}$
- (ii) Annual rainfall 1073 mm
- (iii) Annual range of temperature 5°C
- (iv) Mean rainfall - 1073 mm
- (v) Wettest month - April
- NB. MUST SHOW WORKING!
17. (a) Large volume of air with uniform temperature and humidity and flow over considerable distance
- (b) • Equatorial air mass
- Tropical air masses

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- Polar air masses
- Arctic and Antarctic air masses

(c) $15^\circ - 20\text{g/cm}^2$

$6\text{g/cm}^3 = ?$

$$\text{RH} = \frac{A.H}{\text{Max}} \times 100\% = \frac{6}{20} \times 100 = 30\%$$

Max 20

18. (a) A thermometer/ maximum/ minimum/ six thermometer Hygrometer/wet and dry bulb thermometer.
- 19.
- At night, land loses heat faster than sea.
 - Air upon land becomes cooler and heavier than that upon the sea.
 - The relatively warmer air upon the sea is lighter and therefore it rises while the cooler heavier air at the land flows towards the sea to replace the warm rising air.

CHAPTER 4

STATISTICAL METHODS

1. (a) (i) $9600 - 800 = 8,800,000$ Barrels
(ii) 21,150,000 Barrels.
(iii) $21,150,000 \div 30 = 705,000$ Barrels
2. Graph
 - (i) 29.3% (29 - 29.5%) 29%
 - (ii) 4.75%/4.8%/4.9%
(4.75% - 4.9%)
 - (iii) Describe the trend of the value of coffee exports from years 1999 to 2003.
 - The value was generally declining over the five year period.
 - The value was highest in 1999.
 - The decline between 1999 and 2000 was minimal/gradual.
 - The highest drop was between 2000 and 2001
 - There was a minimal drop between 2002 and 2003.
 - The decline between 2001 and 2002 was minimal/gradual.
 - The value was lowest in 2003.
 - (iv) Explain three factors which may have led to the increased export earnings from horticultural produce in Kenya between years 1999 and 2003.
 - Improved technology which leads to advanced crop husbandry/increase the volume of fresh horticultural products.
 - Aggressive promotion of trade abroad leading to a wide/ready market in foreign countries.
 - Improved ways of packaging have made the produce more competitive/attractive,
 - Improved infrastructure/air/road transport have helped in the quick means of transportation of fresh produce to the market.
 - The declining benefits from traditional agricultural exports leading to the expansion of the areas under horticultural crops.
 - The government has encouraged the formation of organizations that are assisting horticultural farmers.
 - (v) Give three advantages of using simple line graphs to represent data.

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- Give clear visual impression.
 - Easy to construct.
 - Easy to interpret.
 - Can be used to represent a wide variety of variables.
 - Appropriate for comparison.
- (b) Reasons why Kenya's agricultural export earning generally are low
- Kenya sells most of her agricultural products in their raw form and they are priced lowly.
 - International prices keep fluctuating from year to year.
 - Prices of some commodities are externally determined.
 - There is competition from other producing countries/from other similar products.
 - Some products are inferior in quality.
 - There are fixed quarters for some agricultural products.
 - Decline in quantities of some agricultural exports.
3. (a) Statistics It refers to the art or science that is concerned with the interpretation of numeric information.
- Statistical data Refers to the information collected and arranged in a systematic manner.
- Statistical methods Refers to the techniques used in collecting, recording, analyzing and presenting data.
- (b) Primary data and Secondary data
- (c) Closed-ended (rigid) - Open-ended
4. (a)
 - The method should be inexpensive.
 - Should be time saving
 - Should give accurate data
 - Most applicable method
- (b)
 - Discrete data refers to the non-continuous data over time given in whole numbers only e.g.
 - Total population in a nation.
 - Monthly rainfall totals.
 - No. Of livestock per district

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- Continuous data can be given in any value including decimals e.g. 1.8km.
5. Sampling refers to the process by which a representative portion of the whole phenomena under study is analyzed and generalized/ generalization is made.

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Types of sampling

Systematic sampling Stratified sampling Random sampling

6.
 - (a)
 - Calculation of percentages
 - Measuring of Central tendency (mean, median and mode)
 - Frequency distribution
 - (b)
 - Predicting for future trends.
 - Showing changes through time
 - Establishing Geographical relationships
 - For economic planning
 - For explaining geographical phenomena.
 - Useful for making comparisons.
7.
 - (i)
 - Simple line graph
 - A combined line and bar graph
 - Simple bar graph
 - (ii)
 - The simple bar graph
 - Prominent values stick out well
 - Bars are appealing to the eye
 - Easy to draw, read and interpret the data represented
8.
 - (i) A set of pre-questions which are related to the topic of study.
 - (ii)
 - Its a source of first hand information
 - The researcher can ask for clarification from the respondent.
 - Similar questions are used for all respondents and comparison can easily be made.
 - When posted, rigid questionnaires reduce fieldwork expenses.
 - (iii) Interview involves collection of information by asking questions directly and recording the answers given. In this method the researcher established contact with the respondent and agree on time for face to face interview with the respondent. Interviews can also be carried out on telephone with the interviewee.
9. Methods of data recording
 - Tabulation** This is recording of data by arranging facts of figures in form of table or list.
 - Photographing** This is done by use of a camera to record geographical

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information.

Tape recording This can be done when one is collecting data through an oral interview where one uses tape recording device to record conversation.

Tallying Used when the data is collected through counting. One counts and puts a vertical strike for every item counted, on the fifth count one puts a diagonal crossing the four strokes.

10. Data refers to facts and figures collected from the field.

11. The mean is

$$72 + 60 + 65 + 70 + 65 + 80 + 65 + 70 + 80 + 84 + 63 + 75 + 63 + 71 + 74$$

$$= \frac{1057}{12} = 70.47$$

12

12. This is got by arranging the data in an ascending order as follows: 60, 63, 63, 65, 65, 65, 70, 70, 71, 72, 74, 75, 80, 80, 84. The middle number is the median: 70

CHAPTER 5

FIELD WORK

1.
 - The information on rainfall can be used by farmers to plan their calendar of activities.
 - The information on humidity can be used in improving storage of produce e.g. Cereals.
 - The findings can be used to plan suitable time for drying farm produce.
2. Advantages of dividing class into groups.
 - The class will be able to study the entire course of the river.
 - Would enable them to obtain information on each stage of the river.
 - Would save on time.
 - Would enable studies to be carried out in an orderly way.
 - Would encourage participation of all members of class/ encourage individual \ roles.
 - Would facilitate more interaction among the group members.
3. Disadvantages of using secondary data.
 - Recorded data could be out of date.
 - Condition under which data was collected may have changed.
 - Obtaining records on the particular river may be difficult.
4. (i)
 - To design appropriate research method.
 - To prepare the working schedule.
 - To be able to identify relevant equipment for data collection.
 - To identify suitable areas for study/ to familiarize with people who will provide information
 - To seek permission from owners of the land.(ii)
 - Interviewing
 - Taking photographs/video recording
 - Measuring the extent of polluted area
 - Administering questionnaires
 - Tape recording.(iii)
 - Analyzing data
 - Writing report
 - Giving relevant advice to the stake holders

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- Discussing the findings
- Displaying photography, sketches from the study area.
- 5. (a)
 - To get permission from the relevant authority.
 - To be able to formulate objectives/hypothesis
 - To be able to prepare a working schedule/decide on appropriate methods of data collection.
 - To determine the respondents/resource persons.
 - To determine methods of data collection required.
 - To access the problem likely to be experienced in the area.
- (b)
 - Use as fodder.
 - Use for providing fruits/roots/vegetables as food.
 - Providing wood fuel.
 - Controlling soil erosion
 - Use of ornaments/beauty, aesthetics.
- (c)
 - In order to do a detailed study
 - To reduce cost of study
 - To save time
 - The whole forest is too large to cover within a day.
 - It would be less boring to study a small area.
 - Some parts may be inaccessible
 - To reduce bias
- 6.
 - Types of field work
 - Field excursions
 - Field study
 - Field research
- 7.
 - Importance of field work ~
 - Breaks classrooms monotony.
 - Make study of geography real.
 - Helps learner to acquire skills.
 - Encourages students to appreciate the environment.
 - Enables learners to get first hand information from the field.
 - Improves visual memory through observation.

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- Enhances what has been learnt in class.
- 8.
 - Procedure of carrying out fieldwork
 - Identify type of study
 - Statement of objectives
 - Formulate hypothesis
 - Prepare necessary materials and tools.
 - Conduct actual field study.
- 9.
 - Soils
 - Rocks
 - Vegetation
 - Landforms
 - Drainage
- 10.
 - Fieldwork preparation
 - Seek permission from relevant authorities
 - Conduct pre-visit
 - Hold class discussions
 - Determine methods of data collection and recording.
 - Prepare objectives and hypothesis
 - Dividing into groups.
 - Read secondary materials
 - Prepare work schedule.
- 11.
 - Importance of carrying samples
 - For laboratory analysis
 - There is no adequate time to analyze samples in the field
 - Lack of adequate skill to analyze the samples hence need to expert opinion.
 - For future reference.
 - To expose more students to their findings.
- 12. Suitable objectives
 - To know the type of forest.
 - To determine the factors which have favoured growth of forest?
 - To know the type of trees found in the forest.
 - To find out the problems faced by the forest.

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- To find out the economic significance of the forest.
13. Types of hypothesis
- Positive hypothesis This is stated in positive form.
 - Null hypothesis Stated in negative form.
- 14.
- Writing reports
 - Drawing tables, graphs and charts.
 - Drawing maps
 - Displaying photographs.
 - Displaying photographs.
 - Displaying samples.
15. Problems during fieldwork
- Inaccessibility in some areas due to rugged topography.
 - Attack by wild animals
 - Harsh weather conditions
 - Tiredness
 - Accidents
 - Language barrier
- 16.
- To help in identifying methods of data collection
 - To help in formulating hypotheses/objectives
 - To help in assembling equipments
 - To help in estimating the cost of the study
 - To seek permission for the visit
 - To determine appropriate routes
 - To identify likely problems
 - To identify location of data.

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CHAPTER 6

MAPS AND MAP WORK

1. (a) (i) 14km (13.9-14.1 km)
- (ii) 134° (133°-135°)
- (b)
 - Availability of transport evidenced by railway\road to transport sisal.
 - Large tracts of land with scanty settlements allow expansion of far due to availability of land.
 - The area receives low rainfall as evidenced by presence of scrub which discourages growing of other cash crops.
 - Gentle sloping land evidenced by spaced contours ideal for mechanization.
 - Supply of labour evidenced by nucleated settlement around Mwatunge.
- (c) (i)
 - Escarpments/steep slopes have few or no settlements.
 - There are no settlements in the forested areas.
 - Most settlements are found near\along roads\motorable tracks.
 - South East area is sparsely settled.
 - There are many settlements around shops.
 - There are few settlements along rivers.
 - Sisal estate has no settlements.
- (ii) Citing evidence from the map, give two economic activities carried out in the area covered by the map other than sisal farming.

ACTIVITY	EVIDENCE
Cattle keeping	Cattle dips/scrub
Trading	Shops
Transport	Railway\road
Crop growing	Sisal plantation

2. a) Ans; 114031
- b) (i) Ans; 317° (316°- 318°)
- (ii)



(iii)

- c)
 - Rivers
 - River valleys
 - Scarp slope\escarpment\scarp face
 - Gentle slope
 - Seasonal swamp
 - Woodland
- d) i) Two types of physical factors that influenced the location of Nyahururu town.
 - Availability of water from the nearby rivers for domestic and industrial uses.
 - The high altitude (over 2,300m) which makes the area experience cool climate ideal for settlement.
 - Gentle sloping terrain ideal for settlement\road construction shown by contours wide apart.
 - Presence of Thomson Falls which are a tourists' attraction and encourage settlement by construction of tourist houses.
 - Availability of building stones from nearby quarry for construction of houses.
- e) Describe drainage of the area covered by the map.
 - The area has many rivers\ high density of rivers.
 - The water courses are generally permanent.
 - Some rivers end in swamps.

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- There is a pond 020130.
 - Main rivers includes Iguameti, Nyahururu.
 - There are rapids at Thompsons falls.
 - There is parallel drainage pattern along the escarpment.
 - There are papyrus swamps and seasonal swamps.
 - Rivers have denclintic drainage pattern.
3. a) 873m
- b) Ans- 15 km $\pm$ 0.2 KM. (1 5km-1 5.2km)
- c) Ans. -0 30"
- d) **Citing evidence from the map describe the relief of the area shown.**
- Land rises from about 600m to about 2,200m above sea level.
 - The highest point is Vuria peak at 2208 m \lowest point is about 600m.
 - The landscape is generally hilly\numerous hills\mountains\ many ridges eg. Mugange hills ridges\Muraru ridge\ many spurs.
 - There is a highland mass in the Mugange Wundanyi area which slopes downwards in all directions. ^J
 - There is river valley of river Voi which has pronounced meanders.
 - Some areas are plain e.g. Lower valley Kisushi.
 - South-Eastern part is gently sloping.
 - Central, Western and Northern side has steep slopes.

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e) State differences between a map and a plan.

- The scale of a plan is large while the scale of map may be large or small.
- Plan is drawn to show specific information while map shows general information.
- Plan shows many details of specific feature while map shows few details of many features.

f) Explain two importance of scale in maps.

- Express relationships between objects on map and in real life enabling one to estimate or measure size.
- Control arrangement of symbols avoiding overcrowding.
- Controls space as large scale gives more details than the small scale maps.

4. (a) (i) $139^{\circ} \pm 1^{\circ}$ (138 - 140°)

(ii) $7.2 \text{ km} \pm 0.1$ (7.1 - 7.3 km)

(b) (i) Relief of the area covered by the map.

- The highest area is Nandi escarpment/187m above sea level.
- The lowest area is to the south West which is about 1140 m above sea level.
- The east is a plain\Kanu plain/plateau
- North western part is hilly with some steep slopes
- To the North Eastern is the Nandi escarpment
- The northern part is dissected by rivers
- The South-west is a basin occupied by a lake.
- There are numerous river valleys with steep sides in the highlands and are broad in the lowlands.

(ii) Influence of relief on settlements.

- The steep slopes/escarpment have been avoided because they are unsuitable for construction of houses/for farming.
- There are a few settlements on the hilly areas because the slopes are gentler.
- The plains are densely settled as the land is flat/gently sloping.
- The basin are avoided as the land is water logged/flooded/swampy.

(c) Economic activity Evidence

- | | |
|-------------|-----------|
| - Quarrying | - Quarry |
| - Trading | - Markets |

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- Transportation - Roads/railway/main tracks/foot paths
- Processing - Sisal factory
- Manufacturing - Ginnery/flour mills

(d) (i)

- The river has many meanders
- The river has tributaries/confluences
- The river disappears into a swamp
- The river has a wide flood plain
- The river is at its old age stage (ii) Advantages of studying rivers through field work.
- It enables students to relate what is learnt in classroom to what is in the field.
- Students are able to measure and calculate the velocity of a river and its size.
- Students are able to count the number of tributaries.
- Students are able to gauge the impact of the river on the area.
- They are to find out the uses of the river.
- It allows students to acquire appropriate attitude towards environment.
- It breaks classroom monotony for students and teachers.
- It allows students to use their observation skills to make conclusion.

CHAPTER 7

ROCKS AND MINERALS

1. (a) (i) **Colour** Distinct appearance by colour used to identify specific minerals eg. Gold is yellow.
- (ii) **Cleavage** Tendency of mineral to break in certain direction.
Some minerals break along planes on which atomic bonds are relatively weak.
- (iii) **Hardness** Ability to resist scratching. Various minerals have varying degree of hardness eg. Talc is softest while Diamond is hardest.
- (b) (i) **Hyperbbyssal rocks**
 - Volcanic rocks\extrusive igneous rock.
 - Plutonic rocks/intrusive igneous rocks.
- (ii)
 - The water should be salty
 - Water should be clear free from silt.
 - Sea water should be warm with temperatures between 20°C to 29°C
 - Shallow water with depth not exceeding 60m.
 - Polyps must be in submerged condition.
 - Water should be well oxygenated.
- (c)
 - Some unique rocks e.g. crying stone of Kakamega present spectacular scenery for tourist attraction which helps earn the country some foreign exchange.
 - Rocks are parent material for soil formation exploited in agricultural activities.
 - Valuable rocks and minerals such as gemstones and diamond are exploited to generate income.
 - Rocks provide building and construction materials e.g. marble, ballast and sand used in construction of houses.
 - Rocks are useful as raw materials in construction industry e.g. The coral rocks and coral limestone are used in manufacture of cement.
- (d)
 - A folk jembe- excavating rocks for closer examination.
 - A polythene bag -for carrying rocks samples for subsequent studies.

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2. (a) • The rocks are formed from sediments of preexisting rocks.
- Rock sediments are arranged in layers.
- Processes involved act at ordinary temperatures
- Sediments are non-crystalline
- Some sediments contain fossils
- Sediments are compressed, hardened and consolidated by cementing material to form sedimentary rock.
- (b) Give two examples of chemically formed sedimentary rocks.
- Trona, gypsum, flint, rock salt
3. (a) In each case name the type of rock which results from the metamorphism of:
- (i) Granite
- (ii) Clay
- Granite → Gness
- Clay → slate
4. (a) (i) Rocks are naturally occurring agglomerations of mineral particles forming part of the earth crust.
- (ii)
- (a). Mechanically formed sedimentary rocks formed from deposition of sediments of other rocks in layers.
- (b). Organically formed - formed from remains of dead plants and animals which are laid down to layers.
- (c). Chemically formed - formed from mineral particles dissolved from tend and deposited in layers into water bodies.
- (b) • Weight of averlying layers cause change in grain arrangement in dynamic metamorphism.
- Heat of magma get into contact with sedimentary rocks causing grains to crystallize or form new minerals.
- During mountain building rocks are compressed and heat generated in thermodynamic metamorphism causing changes in structure and recrystallization of minerals.
- (c) i) Granite, diorite and peridotite

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- ii) Dolerite, porphyrite and diabase.
- iii) Basalt, obsidian and pumice.
- (d) i) Secondary sources
 - Text books/pamphlets/journals/ periodicals/ magazines/ news papers/handouts.
 - Photographs/pictures/video tapes/slides/films
 - Maps/geological maps
 - Tape recorded information
- ii) Activities during the field study
 - Drawing of sketches
 - Observation
 - Collecting rock samples
 - Making notes
 - Taking photographs
 - Asking/answering question.
 - Studying geological maps -
 - Labelling samples
 - Breaking rocks
 - Digging to access rocks
 - Filling in the table.
 - Filling in questionnaires
 - Tape recording
- iii) Likely problems
 - Inability to identify the rocks
 - Inability to access the rocks
 - Accidents/slipping
 - Difficulties in climbing/descending steep rocks
 - Hindrance by poor weather conditions/rainy/sunny
 - Attack by wild animals.
- 5. (a) i) Plutonic rocks are igneous rocks which form beneath earth surface when magma cool slowly forming large crystals\course grained/course textured.
- ii) Volcanic rocks are igneous rocks formed on the earth surface when lava cool rapidly forming small crystals fine grained/textured.

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6. (a) Conditions influencing characteristics of igneous rocks
- Mineral composition
 - Mode of formation
- (b) Characteristic of sedimentary rocks
- Arranged in layers/strata
 - Non- crystalline
 - Have bedding planes
 - Contain fossils
- (c) Limestone, chalk, coral reefs, ironstone, diatomite, coal.
- (d)
- | | |
|---------------|-------------------|
| Original rock | Metamorphic rocks |
| Limestone | Marble |
| Sandstone | Slate |
| Coal | Graphite |
| Clay/shale | Stale/schist |
| Mudstone | Slate |
| Augite | Hornblend |
| Granite | Gneiss |
- (e)
- Some rocks forms uniqueness features which attracts tourists and helps to earn foreign exchange.
Rocks are parent material for soil exploited for agriculture.
 - Valuable rocks and minerals are exploited to generate income.
 - Provides building and construction materials e.g. sand.
 - Source of raw materials for cement industry.
7. (a) Mechanically formed sedimentary rocks.
- (i) Arenaceous - Sandstone and grit
- (ii) Argillaceous - Shale, claystone, siltstone, loess, mudstone
- (iii) Rudaceous - Conglomerate, breccia and boulder clay.
- (b) Contact metamorphism is due to heat from magma which leads to changes in appearance and character while regional metamorphism is due to heat and pressure which creates changes in rock structure and minerals.
8. (a) Basalt obsidian, Pumice, tuff, rhyolite, andesite.

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- (b) Intrusive igneous rocks are rocks formed when magma cools and solidifies below the earth's surface while extrusive are formed on the surface of the earth when lava has solidified.

Extrusive rocks - Basalt, obsidian

Intrusive - Granite, gabbro, diorite, peridotite, dolerite, porphyrite, diabase.

9. A rock is an aggregate of mineral particles forming part of the earth's crust,
10. A mineral occurring inorganic substances with definite chemical composition and physical properties.
11. Heat and pressure - causes re crystallization of minerals. This creates new minerals. It also alters the structure of the minerals particles.
12. Calcareous rocks are formed from shells and skeletons of marine creatures. The shells skeletons accumulate in layers and are compressed to form hard compact mass.
13. Carbonaceous rocks are formed from remains of plants which are buried by overlying materials compacting them into hard mass.
14. Coral rocks results from accumulation of skeletons of coral polyps. The skeletons accumulates in layers to form hard compact mass (coral rocks).
15.
 - By being subjected to
 - Pressure - dynamic metamorphism
 - Heat - contact/thermal metamorphism
 - Pressure and heat - thermal -dynamic metamorphism.

CHAPTER 8**MINING**

1. Conditions that are necessary for the formation of petroleum.
 - Presence/deposition of remains of flora and fauna fossils over a long period of time.
 - Presence of non porous rocks underneath the deposits of flora and fauna
 - Deposition of other layers of rocks/ non -porous rocks over the remains of flora and fauna.
 - Compression of remain of flora and fauna due to folding of the layer of rocks.
2. (a) (i) Minerals mined in area marked

W	-	Fluorspar
X	-	Gold
Y	-	Diamonds
Z	-	Copper

 - (ii)
 - Alluvial mining
 - Underground mining
 - Open-cast mining
 - (iii) Sea ports through which some minerals mined in East Africa are expected through. -Mombasa, Dar-es-Salaam.
- (b) Factors that influence exploitation of minerals.
 - Modes of occurrence
 - Economic value of the mineral/quality of the minerals/cost of mining.
 - Size of the mineral
 - Level of technology
 - Availability of capital
 - Labour supply
 - Availability of transport facilities
 - Government policy/political influence
 - Availability of market
- (c) Significance of soda-ash mining
 - Creation of employment opportunities.
 - Development of infrastructure.

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- Development of related industries.
- Improvement of social facilities.
- Earns Kenya foreign exchange.
- (d)
 - Planted trees
 - Creating a park to attract tourists
 - Introducing aqua culture
 - Landscaping for settlement /farming
 - Refilling
- 3. (a)
 - Gas
 - Oil\petroleum
 - Water

(b)
 - Wax
 - Bitumen\pitch\asphalt
 - Grease lubricants
 - Resin\petrol-chemicals
- 4. Use the map of Africa below to answer the questions below.

(a) Name the minerals mined in the areas marked S, T and V.
S - Oil/Petroleum
T - Bauxite/Gold
V - Diamond

(b) State two formations in which mineral ores occur.

 - Some minerals occur as evaporates.
 - Others occur as veins/lodes.
 - Some minerals occur as alluvial deposits.
 - Some occur as weathered products.
 - Some minerals are found in seam

(c) Explain four problems which Zambia experiences in the exportation of copper.

 - Zambia is landlocked/ has no coastline hence copper has to pass through other countries to reach the seaport.
 - The distance from Zambia to the coast is long which makes transportation of copper expensive.

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- Political instability in the neighbouring countries makes it insecure to transport copper through them to the coast
 - Congestion at the seaports causes delays in loading and off-loading of copper
 - Loss of copper through theft while on transit deprives Zambia of the part of the expected revenue.
 - Copper is bulky thus it can only be transported by rail which is slow.
- (d)
- Describe three negative effects of open cast mining on the environment
 - The land is left with gaping quarries which are ugly interfere with the natural beauty of the landscape.
 - The heaps of rock waste hinder any other forms of land use/create a landscape that is expensive to rehabilitate/barren landscape.
 - The dust produced during the mining pollutes the atmosphere/is a health hazard.
 - Open cast mining causes shortage of land as it hinders settlement/leads to displacement/hinders agriculture.
 - Large scale blasting of rocks leads to instability of the basement rocks.
 - Water collects in the hollows left by open cast mines creating ponds which become habitats for disease causing organisms
 - It interferes with the natural vegetation which is cleared before extraction of the mineral begins/takes time to regenerate.
5. (a)
- A vertical shaft is dug to reach the mineral Tunnels are then dug horizontally.
 - The roof of the tunnel must be supported to prevent it from collapsing.
 - The mineral is then removed by blasting using explosives.
 - Its then brought to the surface using a lift.
 - The mineral is then transported to the factory for processing.
- (b)
- It is an expensive method
 - Accidents due to collapsing of mines.
6. (a)
- Ugliness of the land.
 - The open pits are health hazards once filled with water.
 - The land losses productivity.
 - Accidents are likely to occur especially children falling into pits.

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7.
 - It involves digging out sand from river beds and swirling it around with water in a shallow pan.
 - This pan is tilted such that lighter sand is washed over leaving the mineral behind.
8.
 - Water shortage for power supply and processing
 - Labour shortage
 - Increased depth of mines
 - Increased cost of mining
 - Decreasing availability of ore
9.
 - (a)
 - It earns the country foreign exchange
 - It generate jobs to Kenyans
 - Leads to uplifting of living standards
 - Development of settlements e.g. Magadi town.
 - Development of transport system
 - (b)
 - Causes scenery ugliness
 - Pollution of dust, noise and overburden
 - Dereliction of land
 - Loss of land productivity which can lead to desertification.
10.
 - (a)
 - It involves mixing the alluvial deposits with water in a container. The mixture is rotated until light particles (sand, mud) are washed off
 - Leaving minerals particles such as gold behind. This is called panning,
 - (b) Three mining methods
 - Open-cast mining
 - Underground mining
 - Alluvial mining
11.
 - Occurrence It forms when rain water seeps down through volcanic rocks which contain soda ash. The water is heated by underground hot rocks. This forces the water to move upwards into Lake Magadi.
 - Exploitation Its extracted using the dredger which floats on the lake. Pumps the mixture of trona and water to the factory on the shores through a pipeline. At the factory impurities are removed. This is heated and turned to soda ash ready for packaging and export.
12.
 - Earns foreign exchange

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- Generates employment opportunities
- Development of settlements
- Lead to development of industry
- Earning higher income hence better living standards.
- Land dereliction
- Pollution
- Loss of biodiversity
- Soil degradation
- Enhancing mass wasting

13. **Gold** is found in quartz or redistributed sediments. Gold is extracted using the deep shaft method. Gold ore is crushed, dissolved and then precipitated to obtain the gold. Gold is used in making of jewellery and ornaments, in chemical industries, in density and for medals. Gold earns export revenue, has led to infrastructure development and provide employment. Rising costs of production, high labour costs and low quality gold are some of problems facing gold mining.

Diamonds are associated with volcanic activities. They occur in igneous rocks in pipes. The ore is blasted crushed and washed. It is then passed through filtering screen and then through a special solution. Diamonds are used in the jewellery industry, for polishing, for drilling, and for cutting instruments. Diamonds; earn foreign exchange, have led to development of towns, have created employment and contributed to development of infrastructure. The problems facing diamond mining to include exhaustion, unstable world market prices, high costs of processing and inadequate labour.

- 14.
- Uses of soda ash
 - Glass manufacture
 - Paper making
 - In oil refinery
 - In textile industry
 - In soap manufacture

FORM TWO

CHAPTER 1

EARTHS MOVEMENT

1.
 - (a)
 - Vertical movement/ epierogenic
 - Horizontal movement/ lateral/orogenic
 - (b) Describe the origin of continents according to the theory of continental drift.
 - Theory first explained by F.B Tylor (91908), J.B Backer (1911) and adopted by German meteorologist Alfred Wegner in 1912.
 - Says the present distribution of continents was due to disruption of super continent known as Pangea. Pangea was surrounded by a large super water body - Panthalasa.
 - Pangea broke first into two continents i.e. Godwanaland and Laurasia.
 - The two continents were separated by a long narrow ocean known as Tythys.
 - Godwanaland broke further and drifted into southern contents of present Africa, South America e.t.c. while Laurasia drifted to present Eurasia, North America e.t.c.
2.
 - Earth movement are movements which occur within the crustal rocks due to tectonic-forces.
 - Internal land forming processed are those processes operating inside the earth. They are also known as endgenetic.
 - External land forming processes operates on the surfaces of the earth. Also known as exogenetic.
3. Evidences of continental drift
 - Geometrical fitting of Western coastline of Africa and Eastern Coastline of America.
 - Similar plant and animal remains in different continents.
 - Similarities in rock structures along coast of West Africa and Eastern South America.
 - Evidence of glaciation in Southern continents which are formed from glaciated regions.
 - Presence of coal in mild and high latitude regions coal is usually formed in tropical areas with dense vegetation.
 - Similarities of fold mountains found in S. Africa and Argentina. Both in age and East - West trend.
4.
 - Plate tectonic theory suggests the earth is made of rigid blocks (plates)

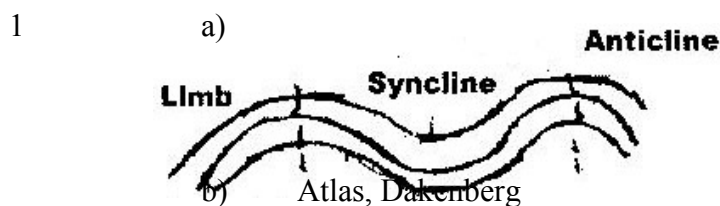
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floating of molten material.

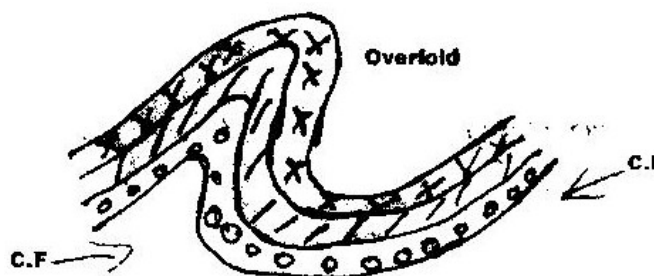
- The plates are mobile.
- They move towards each other away from each other or parallel past each other.

CHAPTER 2

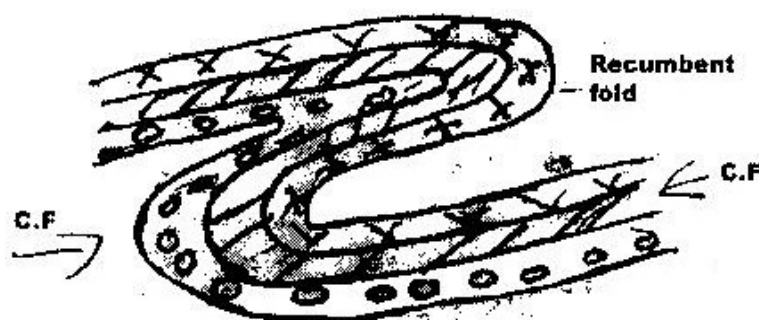
INTERNAL LAND FORMING PROCESSES - FOLDING



2. a) Fold mountains in
- Asia – the Himalayas
 - North America – Rocky Mountains and the Appalachians
 - South America – The Andes Mountains
- b) i) Rolling plains, ridge and valley landscape, intermountain plateau, inter-mountain basin, escarpments.
- ii) Formation of over thrust fold.
- Layers of rock of the earth's crust are subjected to compression forces.
 - Intense compression results in formation of overfold

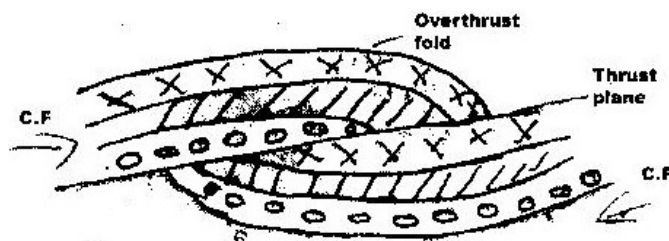


- With increased pressure the overfold results in the formation of recumbent fold



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- When pressure is very great a fracture occurs along the axis in the recumbent fold producing thrust plane.
- The upper part of recumbent fold slides forward over the lower part along the plane resulting to the formation of an over thrust fold.



- c) Effects of fold mountains to human activities.
- windward side of fold mountains receive heavy precipitation which
 - Enhance agricultural activities / forestry.
 - Rivers which originate from fold mountains provide water which is used
 - For generation of HEP/irrigation/domestic and industrial purposes.
 - Some fold mountains have exposed minerals deposits which are exploited.
 - Fold mountains are important tourist attractions/snow capped mountains encourage sporting activities.
 - Fold mountains may act as barriers to transport and communication.
 - Topographic nature of the landscape may encourage/discourage agriculture/settlement.
- d) i) formulate study objectives/hypothesis.
- Identifying methods of data collection/representation.
 - Planning a schedule of activities
 - Carrying out reconnaissance survey.
 - Seeking permission from relevant authorities.
 - Identifying/collecting/sorting out relevant equipments/tools for study.
 - Drawing a route map
 - Assembling relevant stationery
 - Reading relevant information/literature review.
 - Dividing themselves in groups
 - Hold class discussions
- ii) Gives first hand information on different types of land forms.

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- Application of knowledge gained to real life situations.
 - Development of various skills/ application of skills learnt.
 - Help in familiarizing with the environment.
 - Reduces monotony and boredom in the classroom.
 - Provides in – depth/ broader learning
 - Enables one to appreciate landforms
 - Enhances visual memory of landforms better than the theory.
3. Orogenesis is the process through which Fold Mountains are build.
 4. folding is the process of crustal distortion which causes the rocks to bend upwards or downwards.
 5. Compression boundaries – is one where plates move towards each other holding or connecting line in a fold which rock layers dip or rise from opposite directions.
 6. **Limb** – layers of rock on either side of the axis while axis is the central line in a fold which rock layers dip or rise from opposite direction.
 7. Foreland – is static block of land that is pushed in formation of geosyncline fold while backland is block of land where forces originate that cause sediment in the geosyncline to wrinkle.

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8. Fill in the table.

Orogeny	Years (age)	Period	Mountains/features built.
Charnian	Oldest 600 million years ago	Pre- Cambrian period	-deccan plateau of India -Laurentin shield of North Americ
Caledonian	Old 440 million years ago	Silurian period	-Akwapim Hills of Ghana -Scottish highlands
Hercynian	Young 350 million years ago	Upper Carbon Ferrous Period	-Cape ranges -Appalachaian mountains -Ural mountains
Alpine	Youngest 70 million years ago	Palaeocene period	-Himalayas-Asia -Rockies – USA -Andes- S. America

9. Contraction theory.

According to the after earth had formed, the surface rocks of the crust cooled faster than those in the interior. As the interior continues the cool, the surface rocks wrinkled to fit on the contracting interior leading to formation of Fold mountains.

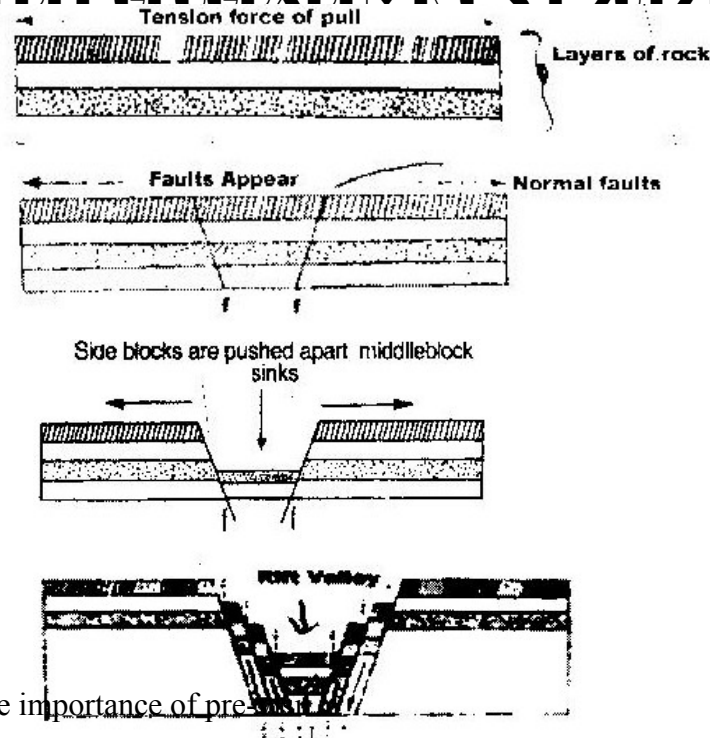
CHAPTER 3

INTERNAL LAND FORMING PROCESSES - FAULTING

1. a) Fault scarp
 Tilt block
 Block/horst mountain
 b)
 - Faulting disrupts lines of transport and communication
 - Some features like Rift Valley form barriers which make establishment of transport and communication expensive.
 - Faulting cause sinking of land which leads to destruction of property such as buildings and crops.
 - Leads to formation of depressions which are filled with water to form lakes
 - Unique features are formed which attracts tourists.
 - Faulting exposes minerals making exploitation easier.
 - Makes rivers to have waterfalls.
2. a) P- Horst
 Q- Rift valley
 R- Escarpment
 b) Normal fault is fault resulting from tension in inclined place with inclination of fault plane and direction of downthrow on same side while reversed fault is fault that results from compression forces where the one block is pushed upwards in relation to another forming up throw.
3. - Vertical faulting, across a river may cause waterfall/river rejuvenation.
 - Rift faulting in an enclosed area may lead to formation of a depression which can be filled with water for form a lake.
 - Some rivers flow along fault lines/fault guided drainage
 - Uplifting of landscape which may cause reversal of direction of river flow
 - Rivers may disappear to the ground through a fault line.
4. Pare, Usambara, Ruwenzori, Nyandarua and Mau Ranges.
5. - Block/ horst mountains are a source of rivers which provide water for industrial/ agriculture/domestic use.

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- Rift Valley formation has led to exposure of minerals such as diatomite, soda ash which are mined on rift valley.
 - Mountains formed are barrier to moisture carrying wind leading to orographic rainfall which favours agriculture and settlement.
 - Some Rift valley lakes are important fishing grounds/mining sites/provide water for irrigation
 - Faulted features provide beautiful scenery which promotes tourism.
6. a) Layers of rocks are subjected to tensional forces.
- Two normal fault develop
- Middle block subsides between two side block
- Middle part form depression called rift valley enclosed by escarpment



- b) i) The importance of pre-study visit
- To enable them to draw up study objectives hypothesis.
 - To familiarize themselves with themselves with area of study/identify problems.
 - To enable them prepare a work schedule plan of activities.
 - To enable them identify/sort out relevant tools/ equipments for the study.
 - To identify suitable methods of data collection.
 - To seek permission from the occupants of the site of study.
 - To enable them prepare financial requirements.
- ii) Disadvantages of direct observation of the area.
- It is expensive
 - It is time consuming
 - It is tiresome
 - It is limited to only direct sources/primary sources of data.
 - It only suitable to sighted people
 - Some features may be hidden out of view.
 -

7.

- Normal fault
- Reverse fault

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- Tear fault
- Anticlinal fault
- Thrust fault

8.

- Compression force tends to push rocks together.
- Lines of weakness develop and lead to formation of parallel reversed faults on the crustal rocks.
- Further compression thrust side blocks over middle block leaving it to form rift valley floor.

9.

- Nyandarua Range
- Lake Manyara escarpment.
- Nyando escarpment

10.

- Rift valley provides a spectacular scenery that attracts tourists earning foreign
- Mining of soda ash in rift valley generate export earnings
- Rift valley floor has fertile soils suitable for farming
- Rift valley lakes are suitable for fishing grounds.

CHAPTER 4

INTERNAL LAND FORMING PROCESSES- VULCANICITY

1. a) i) X-Laccolith
 Y- Batholith
 Z- Dyke

 ii)
 - Magma is forced along horizontal lines of weakness/ bedding planes of rock strata.
 - Intrusion of magma cool and solidify horizontally along bedding planes.
- b)
 - It has a vertical vent
 - It is made up of alternating layers of ash and pyroclasts
 - Conical in shape with steep sides
 - May have crater on its peak or a plug
 - It has sidevents
 - Has parasitic cones/conelelets
- c)
 - Volcanic ash and basalt on pouring provide fertile soils exploited for agriculture
 - Occurrence of minerals such as fluorspar in Kerio Valley and Diatomite in Shinyanga are exploited to generate foreign exchange.
 - Volcanic mountains act as water catchments areas from which major rivers and springs originate. The drainage features provide water for industrial and domestic use.
 - Geoghermic areas which owe existence due to volcanic activities are tapped to generate electricity.
 - Volcanic mountains form beautiful sceneries that attract tourists.
- 2 a)
 - Formed when magma reaches the surface of the earth / or from lava through multiple vents/fissures.
 - The lava is ultrabasic / extremely fluid of low viscosity.
 - Lava flows over large area of distance and spread widely before cooling covering valleys and low lying hills
 - Lava cools slowly forming an extensive plateau.

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- Plateau may form through a series of eruptions which results in thick layers.

b)

- Some volcanic features create barriers making construction of transport and communication.
- The rugged nature of volcanic landscapes makes settlement and agriculture difficult
- Volcanic mountains create rain shadow effect which result too aridity.
- Recent volcanic lava flows have poorly developed soils which are unsuitable for agriculture.
- Volcanic eruption may produce poisonous gases which pollute the environment posing danger to life.

c) Ash and cinder cones

- Fumaroles/solfatara
- Hot spring/geysers/stream jets
- Crater caldera
- Volcanic cones
- Lava plateau

3.

- Hot springs/geysers/steam jets
- Craters/calderas/crater lakes
- Volcanic mountains
- Ash and cinder cones
- Fumaroles/solfatara/Muffette

4. Explain four negative effects of vulcanicity in Kenya

- Some volcanic features create barriers making construction of transport and communication lines difficult.
- Rugged nature of volcanic landscape make settlement and agriculture difficult
- Volcanic eruption may produce poisonous gases which pollute environment thus posing danger to life.
- Volcanic eruption may produce poisonous gases which pollute environment thus posing danger to life.
- Volcanic mountains create rain shadow effect which result in aridity on leeward side.

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- Recent volcanic lava flows have poorly developed soils which are unsuitable for agriculture.
- 5.
- Forms from lava when magma reaches surface of earth through fissures.
 - Lava is ultra basic or extremely fluid or low viscosity.
 - Lava cools slowly forming extensive plateau
 - Plateau may further continue to form through series of eruption that follow
 - Examples: Kaoutu, Yatta, Laikipia
6. A sill is near horizontal /tabular sheet of igneous rock formed from solidified magma between bedding plane while a dyke is a sheet of intrusive rock which cut near vertical/discordantly across bedding plane.
7. Vulcanicity is the process of eruption/escaping of magma/through which gaseous, liquid/molten and solid materials are intruded in to the earth's crust or are extruded onto the surface.
- 8.
- Crater lake
 - Outpouring of lava forms a volcanic cone.
 - The vent may be sealed when lava solidifies in it.
 - This leads to building of pressure below the plug.
 - This leads to explosion of cone leaving depression on top.
 - Water from rain or underground source accumulates in the depression.
 - Examples: L. paradise on Mt. Marsabit, crater lake in Central Island of L. Turkana, L. Sonachi on South west of Lake Naivasha, Lake Simbi Nyamia in Nyanza.
9. Formation of Mt. Kenya
- Due to earth movements, the rocks of the crust were disturbed leading to formation of a vent.
 - The underlying molten rock escaped through a central vent to the surface /volcanic eruption occurred
 - There were violent eruptions which ejected acidic lava that cooled and solidified.
 - These lava piled in layers around the vent.
 - The resultant mountain was cone shaped
 - Over the years, eruption ceased and the volcano became extinct.

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- Erosion set in exposing the plug and produced the jagged peaks of the mountain.

CHAPTER 5

INTERNAL LAND FORMING PROCESSES - EARTHQUAKES

1. a)
 - The Mercalli scale
 - Rossi foren scaleb)
 - Collision of tectonic plates
 - Energy release in the mantle
 - Violent volcanic eruptions
 - Nuclear explosions
 - Gravititive pressure
 - Magma movement within the crust
2. c)
 - (i)
 - Earthquake is sudden movements or tremors of the earth crust.
 - (ii)
 - Primary waves
 - Seconday waves
 - Rayleigh waves
 - Love waves
 - (iii)
 - Rocks of the earth crust are displaced laterally.
 - Earthquake causes uplift or subsidence of land.
 - Earthquakes can rigger off landslides on the surface of the earth crust
 - Earthquakes can lead to faulting of the crustal rocks.
 - Earthquakes can lead to volcanic eruptions
3. a)
 - Collision of tectonic plates
 - Energy release in the mantle
 - Violent volcanic eruptions
 - Nuclear explosions

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- Gravititive pressure
 - Magma movement within the crust
- b)
- Effects of earthquakes in built up areas
 - Loss of life (human, animal and plant)
 - Disruption of transport and communication lines.
 - Outbreak of fires
 - Avalanches and landslides may occur covering build up areas
 - Tsunamis may drown coastal settlements

4.

- Magma movement within earth crust
- Isostatic adjustment resulting in breakage of rocks
- Sudden explosions e.g Nuclear testing.

5.

- The strength of an earthquake is measured by its intensity and magnitude. Intensity measures how hard the earthquake hits the ground.
- Intensity is measured using Mercalli scale. Magnitude measures amount of energy given off. It is measured using Richter scale.

6.

- Damage to property
- Loss of human life
- Can cause landslide
- Disruption of infrastructure.

7.

- Seismograph is an instrument which detects and records seismic waves of earthquakes while seismogram is a graph-like record on which earthquake impulses are recorded.

8.

- Intensity measures how hard earthquake hits the ground. It looks at the effects while magnitude measure amount of energy released during an earthquake.

9. i)

- Written materials/books/magazines/newspapers.

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- Maps
- Photographs/films/videos/slides (Visual aids)
- Resource persons
- Electronic media-radio, TV (Audio aids)

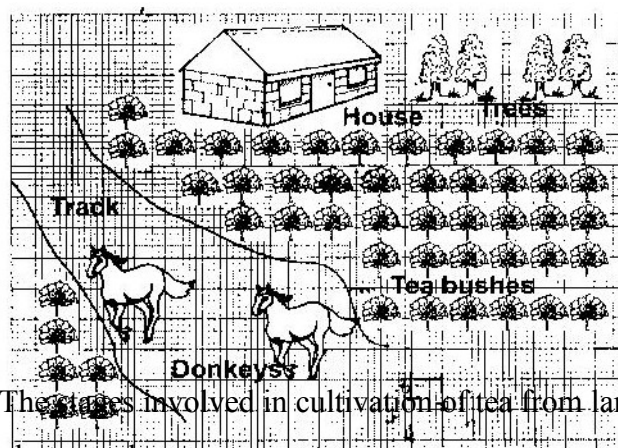
ii)

- Inaccessibility of the area due to massive destruction/restriction.
- Lack of informers because people may have been evacuated
- The rubble may obscure the evidence of the amount of damage.

CHAPTER 6

PHOTOGRAPHIC WORK

1. The photograph below provided shows a tea growing area in Kenya. Use it to answer question (a) and (b)
 - i) What is the evidence in the photograph that this is a ground general photograph.
 - It FREEs on all/ many objects.
 - It does not FREE on particular object.
 - The objects become progressively smaller towards the background.
 - Photograph captures the general appearance of the area.
 - ii) Draw a rectangle measuring 15cm by 10 cm to represent the area of the photograph on the sketch and label the main features shown on the photograph.



- iii) The stages involved in cultivation of tea from land preparation to the stage shown in the photograph.
 - The land is cleared of vegetation
 - The land is ploughed /tilled.
 - Seedlings are planted in nursery and allowed to grow to 20 cm.
 - Seedlings are planted in rows which are about 1.5 metres apart.
 - The plants are weeded and manure / mulching applied regularly.
 - Once the bushes start growing. The tips of branches are pruned regularly to encourage plant to form more branches.
 - The crop is harvested every two weeks once it attains maturity.
 - After harvesting, the green tea leaves are transported to the factory within 24hrs.
 - iv) Identify two features from the photograph that shows that this is a small scale tea farm.
 - The simple houses
 - Mode of transport by use of donkeys

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- Untrimmed edges of tea bushes
 - Dry maize stalks near the houses.
2. Ground general photograph.

3.

- Flowers
- Horticulture/horticultural farming
- Fruits/vegetations
- Rift valley province
- Fruits/Vegetables

4.

- Inaccessible feeder roads
- Inadequate refrigeration facilities
- Improper marketing system

5.

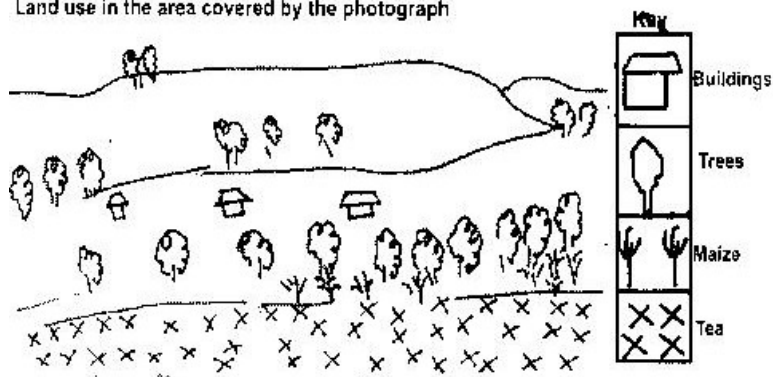
- Netherlands/Switzerland
- Germany
- Britain
- France

6.

- Escarpment
- Hills/mountains
- Valleys

7.

Land use in the area covered by the photograph



CHAPTER 7

CLIMATE

1. a) Climate is the average weather conditions of atmosphere of a place for a long time usually 30 to 35 years.
b)
 - Disruption of natural ecosystems due to ecological changes that affect existence of some organisms and resources.
 - Abnormal fast growth of plants due to increased carbon dioxide
 - Flooding from rising sea due to melting or antarctic ice.
 - Increased rainfall to between 7% and 11% annually due to increased rate of evaporation.
 - Severe draughts due to reduced rainfall.
2. a) $28^{\circ}\text{C} - 24^{\circ}\text{C} = 4^{\circ}$
b) -1803 mm
c)
 - The town experiences high temperatures throughout the year.
 - The annual range of temperature is small about 4°C .
 - The rainfall pattern has double maxima.
 - The wettest month is June/the driest months are December and January.
 - The total annual rainfall is high 1803mm.
3. a) (i) Polar climate
(ii) R - Canary current
S - Gulf stream current
b)
 - High temperatures all year about 27°C
 - Low diurnal range of temperature of approximately 6°C
 - High rainfall of between 1500mm and 2000mm throughout the year.
 - High humidity due to high rainfall and high evaporation
 - Major winds are S. east and North East trade winds
 - Thick cloud covers throughout the year.
 - Rainfall is mainly convectional accompanied by thunderstorms
 - Long hours of sunshine.

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- Low atmospheric pressure.

c)

(i) How altitude influences climate.

- Temperature decreases with increase in height above sea level. This is because atmospheric air is denser at low altitude than high altitude.
- There is greater heat loss at high altitude due to few obstacles to interfere with outgoing terrestrial radiation hence low temperatures.
- Air pressure is higher in the lowlands due to greater weight of air above.

ii) Distance from the sea

- During summer cooler winds from the sea are onshore and modify temperature of the coastal land.
- During winter the sea water which is relatively warmer than land brings warming effect to the coastal land therefore modifying temperature.
- Onshore winds carry a lot of moisture from the sea bringing rainfall to coastal lands.

4.

- Characteristics of natural vegetation associated with equatorial climate.
- The natural vegetation is equatorial rain forest vegetation.
- Growth of this vegetation is due to high rainfall well distributed throughout the year.
- Forests consist of tall trees with straight and smooth stems. The trees reach great height of 40 metres because of competition for sunlight.
- The trees form canopy.
- Forest is characterized by close growth of trees of different species.
- Trees have broad leaves drip-tip in shape.
- Many trees have buttress root system mainly for support.
- Forest is evergreen

5.

- Five characteristics of hot desert climate
- Low annual rainfall less than 250mm/dry climate.
- Occasional flash floods/sporadic rains
- Clear sky/clear sunny days/high terrestrial radiations.
- High temperature during the day.

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- Relatively low temperatures during the night/high diurnal range.
 - Strong winds
 - Low humidity
 - High evaporation
 - Unreliable rainfall
6. How following factors influence climate:
- i) Wind/air masses
 - Warm winds bring warming influence in the cool lands leading to warming effects
 - Areas under influence of dry winds have little or no rainfall while areas under moist winds are usually wet.
 - ii) Latitude
 - Areas near equator are hotter than areas far away from equator. This is due to concentration of sun rays per unit area at the equator.
 - Amount of solar radiation and temperatures decreases polewards
7. Climatic conditions experienced in the Kenyan highlands.
- Region receives rainfall throughout the year
 - Total rainfall ranges from 1000mm to 1500mm
 - The region has double maxima in east and single maximum in west
 - Rainfall is higher on the windward slopes than on leeward slopes
 - Rainfall is higher on the windward slopes.
 - Rainfall is caused mainly by S.E trade winds.
 - Average temperature ranges between 17⁰C to 24⁰C.
 - Area receives mainly relief rainfall.
- 8.
- Because of aspect slopes facing the sun are warmer than slopes facing the opposite direction.
 - The mountains cause anabatic winds which have a cooling effect on hill side during the day. Valley bottoms katabatic winds bring effect during the night
 - Reduction of air pressure with increase in altitude
 - Occurrence of relief rainfall on windward side as an influence of relief.
9. Green house effect is a condition where incoming solar radiation passes through

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the atmosphere while outgoing terrestrial radiation is blocked by gases and clouds in the atmosphere. This makes earth to retain much of terrestrial radiation therefore becoming warmer.

10. Climate change due to human activities
 - Burning of fossil fuels
 - Forest and grassland fires
 - Industrial and agricultural development.
11. Clearance of vegetation reduces disposal of carbon dioxide from the atmosphere by photosynthesis. Due to inadequate vegetation that would otherwise utilize CO₂ in photosynthesis, there is build up of excess carbon dioxide gases in the atmosphere leading to global warming.
12. Climate- is the average weather conditions of a place or region which have been observed for long period of time usually 30-35 years.
13. Isothermal layer – is a layer within atmosphere within which temperature remains constant despite increasing height.

CHAPTER 8

VEGETATION

1. d) (i) Measure distances/estimation of heights of plants
 - Collect sample of plants
 - Draw sketches/transects
 - Record/take notes
 - Take photographs of plants/area
 - Count plants
 - Feeling the leaves
 - Conduct interview
- (ii) How to identify different types of plants
 - By appearance
 - Their colour
 - By their leave size/patterns/type
 - By their age
 - By the nature of their bark
 - By texture of their leaves
 - By their system of the roots
 - The type of fruits.
2. a) (i) W- Rainforest
X- Bamboo
Y-Health and moorland
- (ii)
 - Savanna vegetation consists of trees and grass
 - Wetter areas/near forests the vegetation consists of tall trees similar to those found in forests and woodlands
 - Wetter areas have tall thick grass.
 - Gradually away from the forest, the trees become fewer and shorter
 - Grass is shorter in drier areas
 - In drier areas the trees are short and more scattered.

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- Some trees are deciduous type
- Most trees are umbrella shaped
- Most common trees are acacia and other thorny trees.
- Where the rainfall is lowest grass is tufted and coarse/trees scrub
- There are scattered baobab trees and other drought resistant trees.
- Along river valleys there is riverine vegetation and thick bush.

(iii) Canada-Prairies

Russia- Steppe

Australia-Downs

b)

- Fire- Often large areas of forests are destroyed by fires and take long to recover.
- Diseases and pests attack mainly the planted forests causing many trees to die.
- Human activities/settlements/charcoal burning/logging have destroyed many forest areas.
- Over exploitation leads to depletion of certain tree species such as Meru oak, Campor and Elgon teak. These trees take long to manure.
- Government policy of degazetting of some forests made people free to clear many forested areas.
- Prolonged drought leads to degeneration of forest some of which take long to recover.

3. a) Natural vegetation is the plant cover which is growing wildly on its own.

b)

- The vegetation is adapted to long, hot dry summers.
- Some plants are evergreen
- Grasses dry up during summer and germinate during winter.
- Woody scrub is common in very dry areas.
- Some plants have small, spiny leaves while others have thick skinned or leathery leaves.
- Some plants have long roots.
- Some plants have thick barks
- Some plants have large and fleshy bulbous roots.
- Some trees are deciduous.

4.

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- Campaigns against indiscriminate cutting down of trees/educating people/ reducing overgrazing.
 - Establishment of vegetation/forest reserves
 - Restriction on cutting down of trees
 - Development of energy saving technology to reduce high consumption of wood fuel
 - Use of alternative sources of energy
 - Encouraging the planting of more trees to reduce reliance on existing ones
 - Establish Nyayo tea zones to act as buffer zone.
5. (i) Variation in rainfall
- (ii) Variation of temperature
- (iii) Variation of altitude/relief.
- (iv) Aspect
- (v) Soil
- (vi) Human activities

Variation of rainfall

Areas that receive high rainfall are forested while those receiving low rainfall have grassland vegetation.

Variation of altitude/relief

Vegetation varies with height above sea level (e.g. montane in high altitude) as altitude influence climate and soil.

Aspect

Areas on leeward slopes of Mountains have different vegetation from thick growth of vegetation in the windward side because they receive different amounts of sunshine and rainfall.

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Soil

Sandy soil/swamp soil/saline soil influence growth of different types of vegetation. Vegetation on slopes is determined by soil catena.

Drainage

Vegetation is as luxuriant along water courses/along coastal flats because surface water supply is reliable/waterlogged areas support swamp vegetation.

Human activities

Settlement/mining/ farming interferes with the original vegetation leading to growth of secondary / derived vegetation/desertification.

Wild animals

Destroy vegetation leading to secondary type/desertification. They aid in seed dispersal.

6. State two reasons why mountain top have no vegetation.
 - Temperatures are too low to support plant growth.
 - There is no soil to support plant growth/bare rock.
 - Water is in frozen state.
7. Vegetation refers to plant life on earth surface.
8. Areas where coniferous forests are found.
 - Cool temperate continental climate/Siberian type.
 - Cool temperate eastern margin- Laurentian type.
 - West coast of Canada.
 - Scandinavian region
9. Characteristics of temperate grasslands
 - Trees are scarce except along water courses.
 - In moist areas the grass is tall.
 - Where it is drier there is short tough grass
 - Grass withers in autumn and dries up in winter but sprouts during spring.
 - Presence of scattered trees
 - Common trees are acacia
10. Secondary vegetation comprises natural processes colony on a place which is in the process of receiving due to interference by man while planted vegetation comprises of plants grown in a place by people e.g. agro forestry.
11. Two significance of vegetation to physical and human environment.

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- Vegetation is of aesthetic value as it adds beauty to landscape.
- Vegetations roots binds soil together protecting soil against erosion
- Plant decay to form humus adding fertility to soil.
 - a)
 - Ground close-up
 - Acacia vegetation
 - b)
 - Thorny like leaves to reduce rate of water loss.
 - Have long tap root to tap underground water
 - Plant seeds remain dormant awaiting short rains.

CHAPTER 9

FORESTRY

1. a)i) Is science of planting caring and using trees/forests and their associated resources or the practice of managing and using trees/ forests and their associated resources.
 - ii)
 - The area receives high rainfall/1000 mm-2200mm throughout the year which encourages continuous growth of trees.
 - The area has deep fertile volcanic soil that allow the roots for penetration deep into the ground support trees.
 - The area is a gazetted reserve prohibiting cultivation and settlement hence allowing growth of trees
 - The steep slopes discourage settlement thus forests thrive.
 - iii)
 - The government policy of degazetement has allowed illegal cultivation and settlement in forest areas.
 - Increased population of elephants that destroy trees
 - Prolonged droughts have caused drying up of some trees.
 - Plant diseases and pests destroy parts of forests.
 - Over exploitation of certain species of trees
- b)
 - Legislations aimed of curbing encouraging public to participate in conservation of forests.
 - Legislations aimed of curbing encroachment in forested areas.
 - Encouragement in forested areas
 - Encouraging public to participate in conservation of forests.
 - Setting up of buffer zones to hinder human encroachment into forested areas (Nyayo Tea Zones)
 - Creation of forest reserves
 - Encouraging agro-forestry

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- Setting up presidential commission with the mandate of rehabilitation destroyed forested areas.
- c) Give the difference in exploitation of softwoods forests in Kenya and Canada under following sub-headings.
 - i) Period of harvesting
 - ii) Transportation.

Period of harvesting

- In Kenya harvesting is done throughout the year while in Canada harvesting is in winter and early spring.

Transportation

- In Kenya transportation is mainly by road transport while in Canada transport is mainly by water transport.

- To ensure continuous supply of wood fuel, timber and raw materials for paper industries.
- To protect soil from being carried away (erosion)
- To protect water catchment areas
- To create room for conservation of wildlife
- To create employment opportunities
- To reduce importation of forest products thus saving foreign exchange
- To promote scenic beauty.

2.

- The low temperatures limit other land use activities making forestry a good alternative.
- The thin infertile soils due to keep glacial erosion limit agriculture and favours forests
- Adequate precipitation throughout the year
- Many rivers with waterfalls provide HEP used in the timber related industries
- River provides cheap transport and water needed in the processing purposes.
- An extensive ready market for forest products in Canada and U.S.A.

3.

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- Meru Oak
- Elgon teak
- Cmpbor

4.

- Furniture making
- Wood carving industry
- Construction

5.

- Mature trees felled are replaced immediately.
- Tree farming is practiced in many parts with the aim of raising trees for future use.
- Regions which previously were devoid of trees are being planted with trees
- People are now being encouraged to plant trees and food crops in the same farms.
- The reduction of wastage e.g. the use of waste paper to produce newsprint.
- The reduction of wastage e.g. the use of waste paper to produce newsprint
- People are required to seek permits if they have to cut trees. This reduces the rate of trees felling/unlicensed people do not cut down trees.
- The forest reserves are patrolled by guards to ensure that fires are reported promptly and also to ensure unlicensed people do not cut down trees.
- Forest reserves have been set aside to conserve indigenous species.
- Forestry department of the ministry of natural resources carry out research to produce and distribute seedlings to ensure the extension of forests.
- People are being educated through mass media on the importance of trees
- People are being encouraged to use alternative sources of energy/energy saving jikos.

6.

- There is a wide variety of trees species in a given area. This makes the exploitation difficult and expensive
- The buttress roots make the felling cumbersome
- The dense underground/thick forests and quick generation of plants hinder accessibility, exploitation difficult
- The heavy rainfall throughout the year results in muddy roads which makes transportation difficult.

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- Inadequate capital limits the use of modern techniques in the exploitation of the forests.
- The demand within the region/markets are discouraging exploitation/expensive to transport/difficult to transport.
- River transport is hindered by waterfalls/rapids. This makes transport expensive.

7.

- Aridity that has reduced area under forests
- High population leading to clearing of forests for settlement
- Over exploiting of forests resources e.g timber fuel.
- Government policy of resettling people in forested areas.
- Forest fires which have contributed to destruction of forests.
- Pests and diseases which destroys trees.

8.

- Desertification
- Soil erosion/degradation
- Extinction of some tree species
- Decline in employment for those in forestry related areas
- Loss of plants with medicinal value
- Loss of aesthetic value
- Decrease in wild animals

9. Factors favoring forestry in Kenya

- Cool climate especially in the Kenya highlands.
- Fertile soils-Volcanic soils within the Kenya highlands and Rift valley.
- High precipitation ranging between 1000-2000 which favours growth of trees.
- Rugged landscape in some areas which discourages settlements and agriculture leaving growing of trees as the only alternative
- Varied altitude which favours growth of different types of trees due to varied temperatures and rainfall.
- High demand of forest products
- Creation of forests reserves which enables forests to develop without interference from humans

10. Secondary products of forests-Ply woods, Vanner, fibre board, paper, wood, carvings wine, tannins

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11. Mahogany, ebony, ironwood, rosewood, green heart, goarea, mahure, supele, duhuma, African walnut, camphor, palms.
12.
 - a) Agro-forestry involves cultivation of both crops and trees.
 - b)
 - Provides sources of firewood and charcoal
 - Source of income to farmers after selling tree products like fruits
 - Trees act as wind breakers
 - Trees create micro climate within the farm
 - Some trees are of medicinal value
 - Trees leaves decomposes to form fertile soils

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FORM THREE ANSWERS

CHAPTER 1

WEATHERING

- 1 a)(i) Weathering is disintegration/breaking down and decomposition of rocks in situ due to exposure on the surface while mass wasting is down slope movement of weathered material under influence of gravity.
- (ii)
- Climate changes
 - Relief/topography
 - Nature of rock
- (iii)
- Pressure of expanding roots in cracks cause rocks to disintegrate.
 - Plants like algae release organic acids on rock causing its decay.
 - Lichens, mosses maintains rock moisture which facilitates chemical weathering.
- 2 a)
- Exfoliation
 - Block disintegration
 - Slaking
 - Crystal growth
- b)
- Hydrolysis
 - Oxidation
 - Solution
 - Carbonation
3. a) Is the physical break up or disintegration of rock material without any alteration in the chemical composition?
- b)
- During the day suns' heat in arid areas cause surface layers of rock to expand.
 - At night the low temperatures result in cooling and contraction of outer layers
 - This occur repeatedly
 - Rocks with poor heat conductivity do not transmit heat to the inner rock.

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- Expansion and contraction causes stresses within outer parts of the rock,
- Eventually shell of outer rock layer peel off from main rock mass.
- This leaves behind a rounded-off mass known as exfoliation dome.

4. Five processes;

Hydrolysis

This is the process where chemical reactions takes place between hydrogen ions in water and minerals in a rock.

Hydration

This is the process where some minerals in a rock take up water and expand causing stress in the rock. This causes the rock to fracture.

Solution

This is the process where some rock materials dissolves in water and are washed away in solution. This leaves behind a weak rock.

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Carbonation

This is the process by which rainwater with carbonic acid dissolves calcium carbonate in rocks.

This leaves behind a weak rock.

Oxidation

This is the process in which oxygen in the air reacts with iron compounds in the rocks. The ferrous state of iron minerals changes to ferric state which weakens the rock.

5. Pressure release/unloading

Exfoliation

Block disintegration

Crystal growth

Slaking

6. Climate changes

Relief/topography

Nature of rock

Vegetation cover plants

Action of humans and animals

7. Denudation is the wearing away/sculpturing of land surface by processes of weathering/mass wasting/transport and erosion.

8. a) Block disintegration

b)

- A well jointed rock is subjected to intense heating during the day and cooling during the night.
- The rock minerals expand due to heating and contract as a result cooling during the night.
- The rock minerals expand due to heating and contract as a result of cooling
- The joints enlarge due to alternating expansion and contraction of the rock mass.
- When this occurs repeatedly the rock mass eventually break into blocks along joints hence the name block disintegration.

CHAPTER 2

MASS WASTING

1. a)
 - Amount of precipitation and extent of saturation.
 - The gradient of the slope
 - Human activities such as mining.
 - Occurrence of earthquakes and faultingb)
 - i) Rock fall
 - ii) P-cliff face/steep slope/scarp slope
q-Talus/rock debris/boulders
2.
 - Due to temperature changes soil particles expand and contract hence shift position down slope.
 - Moisture/rain water cause soil to become wet and compact. On drying the particles loosen and shift position down slope.
 - Moisture acts as lubricant to soil particles causing their movement down slope.
 - Removal of soil on the downhill side makes the rest of soil to shift down slope.
 - Human activities/action of borrowing animals may cause the removal of soil on lower part of slope. This triggers soil particles on the upper part of the slope to shift down slope.
 - Freezing of soil water expands the space between soil particles. Once water thaws particles fall by gravity shifting position down slope.
 - External forces e.g moving a trigger effect which causes downwards movement of soil particles.
3.
 - Soil creep pushes posts and fences from their original positions
 - Displacement of soil particles down slope leaving steep upper slopes bare
 - Burry roads and railways making repair expensive
 - Causes slope retreat
 - Leads to formation of terrace
 - Leads to formation of terrace
 - Leads to formation of deep fertile soils down slope which favours agriculture
4. **Mass wasting:**

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This is the down slope movement of weathered materials under the influence of gravity.

Mass movement:

This is the down slope movement of weathered materials after lubrication by water.

5. Soil creep:

Movement of fine soil down a gentle slope. It is the slowest movement and quite hard to notice.

Talus creep:

Slow movement of angular waste of rock of various sizes down a cliff, hill, scarp and mountain side.

Solifluction: gravitational flow of surface materials saturated with water.

6.

- Very steep slope these accelerate the rate of movement.
- Very high rainfall which makes the materials to be extremely fluid and the ground to be generally unstable.
- Tectonic movement such as earthquakes and faulting.
- Human activities e.g mining or removal of soils at the base of slopes making the upper layers unstable.

7. Evidence of soil creep includes:

- Joint blocks of distinctive rocks types are dislodged from the outcrop
- Edges of strata seem to bend in the down hill direction.
- Fences posts and telephone poles lean downwards and even shift measurably out of line.
- Retaining walls of road cuts lead and break outward under pressure of soil creep from above.
- Accumulation of deep soil at the base of slopes while the upper slopes are left bare.
- Bare and exposed steep upper slopes due to soil displacement
- Slope retreat
- Bent tree trunks.

CHAPTER 3

THE HYDROLOGICAL CYCLE

1. a) E - Surface run-off
 F - Evaporation
 G - Condensation
 b) A watershed is a ridge line boundary line separating drainage basins or river systems while a catchment area wetland which a river draws its water.
2.
 - Heavy rainfall /high intensity of rainfall/
 - Low rate of evaporation
 - Sloping ground/steep slopes
 - Presence of vegetation/bare surfaces
 - Saturated soil surfaces.
3. Is the endless circulation of water from earth's surface to the atmosphere as moisture or water vapour and back to the surface of the earth as rain or snow with source of energy being the sun.
4. **Type, amount and duration of rainfall** e.g light showers in long duration reaching the ground facilitate infiltration as apposed to heavy rain in showt duration
Nature of slope- Level land hold water on ground long enough to infiltrate than on steep slope where run-off is accelerated.
Level of soil saturation-Infiltration is greater in areas with lower water table and lower soil water than in higher water table with high amount of soil water
Soil type- Coarse grained soil allow greater infiltration than fine grained compact soil
5. a) Refer to all water held in form of ice in storage on the earth surface.
 b)
 - Provide underground water
 - Ecological balance
 - Formation of clouds
 - Oxygen and carbon dioxide cycles
 - Occurrence of leaching which is soil forming process.
6. Dew, mist, fog snow, snow, frost
7. **Type, amount and duration of rainfall** e.g. light shower reaching the ground gently will facilitate infiltration as opposed to heavy storm which encourage run off.

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The duration the rainfall takes also determine the rate at which it infiltrates i.e. longer shower enhance infiltrates while short heavy storm encourages run off.

Nature of the slope: flat land holds water onto the earth surface longer hence encourages infiltration but when the land is sloppy surface, run off is accelerated.

Amount of water already in the soil e.g the lower the water table, the lower the chances of run off; but the higher the water table, and amount of water in the soil, the higher the surface run off.

Vegetation cover: Where vegetation is thick, the rate of run off would be reduced because the rain drop impact will be reduced. A lot water will be retained through interception hence lower rates of surface run off.

Soil type: Course-grained open textured sandy soil have higher infiltration rate than fine grained compact soil. This reduces run off. Likewise, deep uniformly permeable soil have extensive infiltration hence lessens the chances of overland flow.

Environmental factors e.g where the rate of evaporation exceeds the rate of infiltration, surface runoff is minimized (if other factors are treated constant) but where the rate of evaporation is lower overland flow will be higher.

Human activities: This can control runoff deliberately by construction of reservoirs acting like natural water bodies.

In urban areas, surface sealing by concrete and bitumen accelerates surface runoff.

Agricultural practices can also modify runoff through contour farming and levee construction which will store water and increase both infiltration and evaporation at the expenses of runoff.

8.

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- Reducing the rate of deforestation and increasing afforestation and reforestation programmes so as to facilitate transpiration process
- Activities which can cause global warming such as releasing chlorofluorocarbons in the atmosphere should be reduced so that water held in the cryosphere can still be maintained and used as storage in future
- Avoiding excessive harvesting of sand on river beds.

CHAPTER 4

ACTION OF RIVERS

1. a)
 - Birds foot delta.
 - Estualine delta
 - Aruate delta
 - Cuspate deltab)
 - River must have a large load of sediments
 - absence of filters like lakes or swamps in the river's course
 - Rate of river deposition should be faster than rate of erosion by tides and waves.
2. a)
 - (i) P- Deposition
Q- Erosion
 - (ii) R- Bluff/cliffs
 - (iii)
 - In a stream, water flows in meandering motion.
 - The motion sweeps the current to outer bank where lateral erosion/undercutting occurs and deposition occurs in the inner bank.
 - This leads to formation and enlargement of a meander
 - Subsequent cutting through the neck of the meander by the stream establishes a new channel
 - The abandoned cut-off meander channel forms a crescent shaped lake-ox-bow lakeb)
 - Gentle sloping/flat surface
 - Has levees
 - Has deltas/distributaries
 - Has meanders
 - Land is fertile-alluvial deposits
 - Has marshes/swamps
 - Has braided channels

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- Presence of ox-bow lakes
3. a)
- Increased discharge because of increased rainfall increases erosive activity of the river
 - A fall of sea level leads to river renewing its head ward erosion. Starting from the new base level.
 - Uplift of land causes a stream to increase its erosive power
 - Changing rock resistance from hard to soft along the river course reactivates rivers erosive power.
- b) X- Resistant rock
Y- Plunge pool
Z- Rock boulders/rock pebbles.
- 4.
- Some particles are carried in suspension because they are light and can be maintained within turbulence of water.
 - Some load is dissolved in water and carried in solution.
 - Some particles are heavy and are momentarily lifted by turbulence of the water and then dropped onto the river bed by **saltation process**.
 - The heavy particles are rolled along the river bed through process known as **traction**.
5. i) Dedritic pattern resembles a tree trunk and its branches. The tributaries join the main river at acute angles.
- Trellis pattern comprises a main river with tributaries joining at right angles. The minor tributaries also join the main tributaries at right angles.
- ii) Trellis pattern comprises a main river with tributaries joining at right angles. The minor tributaries joining at right angles, the minor tributaries also join the main tributaries at right angles.
- iii) Central petal pattern comprises rivers which flow into a common inland basin or depression such as a lake or swamp.
6. a)
- Gradient of the river channel which determine river energy.
 - The nature of bed rock whereby soft rocks would be more eroded.
 - Volume of the water which affects river ability to erode.
- b)

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- (i) Define river rejuvenation
Refers to the revival of rivers' erosive energy.
- (ii)
- Incised meanders
 - River terraces
 - Knick points
 - Rejuvenation gorges
7. a) River basin: The entire system of the river including tributaries and Distributaries
- b) Water shed: A boundary line separating neighbouring river basins. It's also called river divide.
- c) Catchment area: the entire area from which the river draws its water.
- d) river regime: this is the seasonal variation of the amount of water in the river channel or the variation of the quantity of water (discharge) passing down a stream.
8. a) Inland delta is formed at any part of the river course before it reaches its mouth e.g Niger delta in Mali and Okavango delta in Botswana, while an alluvial fan is deposit composed of silt, sand, gravel and boulders found at a point where a river enters a plain from high land e.g. Ewaso Narok fan on river Ewaso Narok in Kenya, kilombo alluvial fan in Tanzania and lumu alluvial fan in Uganda.
- b) Estuarine delta: develops at the mouth of a drowned river e.g Zambezi and volta river deltas
An estuary is a mouth of the river which drains its water in the ocean in one complete channel e.g. river Zaire in democratic republic of Congo
- c) A bluff is a blunt, low embankment on the bank of the river as a result of recession of interlocking spurs due to lateral erosion.
A river cliff is an overhanging undercut river bluff (river bank) due to lateral erosion.
- d) Levees and river banks

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Levees are temporal raised river banks made of sand deposits on the flood plains between which a river flows. A river bank is an elevated land alongside a river which may either be vertical or sloping towards the river.

e) River valley and river channel

A river valley includes both the cross and long profile of a river including all the associated features from its source to its mouth, while a river channel is the narrow groove or furrow through which a river flows.

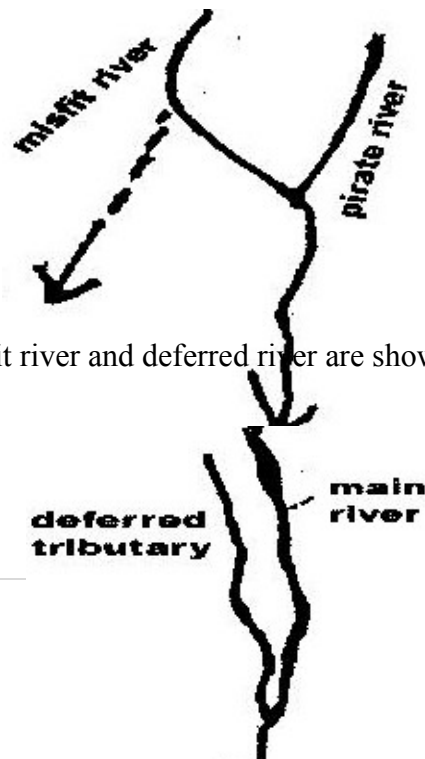
f) Paired terraces and unpaired terraces. Both are associated with river rejuvenation (renewed erosion of river to cut new channels). Paired terraces have valley levels on the opposite side with equal elevation while in the case of unpaired terraces benches on the opposite sides do not match.

g) Drainage pattern and drainage system. A drainage pattern is the layout or pattern made by the river and its tributaries on the landscape whilst a drainage system develops where rivers either flow in accordance or discordance to the existing rock structure and slope.

h) A misfit river and deferred river

A misfit river is one whose water has been pirated by another powerful river flowing adjacent and on a lower ground.

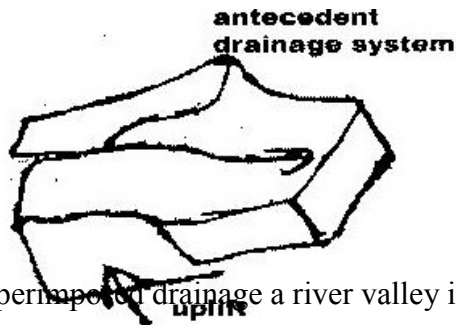
A misfit river is also referred to as an under fit or beheaded stream. A deferred river is a tributary that flow for long distances parallel to the main stream before joining it due to the presence of the levees.



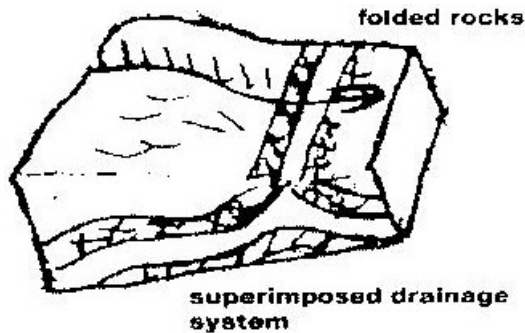
A misfit river and deferred river are shown in the diagram below.

- i) Antecedent drainage and superimposed drainage.

Antecedent drainage is a drainage system where a river maintains its course while the surrounding land is uplifted.



In superimposed drainage a river valley is developed on the former cover of rock which have been exposed and have a completely different structures



9.

- **Abrasion**

This is the wearing down of river bed and sides by use of the load carried by the river. The load is hurled by water against the banks and dragged along the bed acting as a scouring tool.

- **Hydraulic action**

High force moving water removes loose materials e.g. gravel. The water scoops out rock particles.

10.

- a)
- The river is in youth stage
 - The main function of the river at this stage is erosion
 - The main features of the river includes:

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- Water fall

- Rapids

- Gorges

- Potholes

- Interlocking spurs

b) i) The river is in mature stage

ii)

- Main function of the river are erosion and deposition though lateral erosion is more dominant than vertical erosion.

- Main features of the river includes.

- Wider valleys with an open v-cross section

- Gentler river gradient, wider valley floor

- River bend begins to appear

- Concave banks stand out as river cliff while

- Convex banks become slip off slopes

- Interlocking spurs change to bluff due to gradual removal by lateral erosion

c) i) The river is in the old stage

ii)

- The main function of the river is deposition

- The main features of a river at this stage include;

- Meanders and ox-bow lakes, natural levees and deferred tributaries, Incised meanders and terraces, Braided channels, Flood plains, Deltas and tributaries, Very gentle gradient, Shallow broad and flat valley.

11.

- River provides water for irrigation this has enhanced food production of the local people in the areas where the river passes.
- River provides route ways e.g. the navigable rivers. This has facilitated easy transport and communication between any area served by the river
- River provides sand which is a building material. People can put up permanent structures and this has changed their lifestyles

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- Rivers are sources of alluvial soil. These deposits of river soils form rich agricultural land e.g. along the Tana river valley. This has increased food production hence better living standards.
 - Rivers form natural boundaries between communities, districts provinces or countries. Such boundaries have helped in settling disputes related to land at the borders
 - During floods, rivers destroy properties and human life. This has led to migration of people to higher ground and inadequate development of physical infrastructures such as permanent housing, roads e.t.c.
 - Rivers are tourist attraction features. The revenue earned through tourism where there are rivers is a source of income to the local inhabitants, and hence the revenue raises their standard of living
 - Rivers provide fishing ground. Fish which is a rich source of protein is used to supplement other food resources such as meat. This ensures a healthy population devoid of diseases such as kwashiorkor.
 - Communication barrier some river form barrier between communities making communication difficult e.g communities making communication difficult e.g. communication between Lamu and Tana river district. The local therefore have to use alternative longer routes which is expensive and time consuming.
 - Water borne diseases: In rivers where water is almost stationary many water borne diseases are a problem. These diseases may include river blindness, malaria and bilharzias. The treatment of this diseases is difficult and very expensive
 - Rivers provide site for hydro-electric power generation. The lifestyle of locals in other benefits which comes along with power generation. This may include trade, urbanization; better road and communication network this has led to improved living standard.
 - Rivers provides port facilities. The presence of ports also offers employment opportunities to peoples within the region.
 - This has enhanced the living standard in the region e.g port of Mombasa.
 - Rivers supply water for both domestic and industrial use. The construction of industries in the area is also a major source of employment to locals hence improvement of standards of living.
12. a) Vertical aerial photograph
- b) I. Arcuate delta
II. Birds foot delta
- c) - Presence of large load/ample supply of load.

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- Absence of strong waves/Currents in the sea/lake.
- Decrease in the velocity/speed of river
- Presence of gentle gradient
- A shallow shore at the river mouth.

CHAPTER 5

LAKES

1.

- Erosion
- Volcanic
- Tectonic movement
- Deposition

2.

- Earth movements caused crustal warping
- A basin was formed
- Uplift of land masses/back-tilting of plateau around the depression diverted the flow of rivers into the depression by reversed drainage.
- Deposition into the depression resulted into further down warping
- Water from the rivers accumulated in the depression forming a lake.
- Water from the rivers accumulated in the depression forming a lake.

3.

- The lake breezes have a cooling effect on the surrounding areas
- The high rate of evaporation from the lake leads to formation of convectional rainfall in the area.
- Moistures from the lake leads to an increase in the amount of rainfall received in the area
- Evaporation from the lake leads to increase in the relative humidity of the lake region.

4.

- Lakes formed due to earth movements such as faulting and down warping.
- Lakes formed due to vulcanicity such as when water occupies crater
- Lakes formed due to glaciations e.g tarns.

5. a) Formation of lake Victoria:

Lake Victoria was formed as a result of warping and tilting of the earth crust. During the drainage evolution process, the western part of the depression was uplifted to form a Ruwenzori and middle part down warped Rivers like Yala, Nzoia, Mara, and Kuja were cutoff and started pouring their water into this depression to form lake Victoria.

b) Formation of lake Tanganyika: lake Tanganyika was formed as a result of

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faulting. During the formation of the rift valley some parts of the valley were deeply faulted to form long narrow hollows. When water accumulated in this hollow, Lake Tanganyika, which is faulted or Rift Valley Lake was formed.

- c) formation of lake Chala: Lake Chala was formed when water accumulated in the crater of a volcano.
- d) Formation of lake Sare is a lagoon that formed at the shore of Ugowe Bay in Siaya District. Action of long shore drift deposited materials across the by which cutoff part of Lake Victoria to isolate the water mass to form a lagoon new lake Sare.
- e) Formation of Lake Kivu; Lake Kivu is a lava dammed lake. It was formed during the eruption of Virunga volcanoes which blocked the west rift. One of the tributaries of river Zaire was dammed to form Lake Kivu.

6.

Eastern	Generally fresh
• Generally alkaline • Have no outlets except Naivasha and Baringo • Short narrow and deep	• Generally fresh • Have outlets • Long and deep

7. Significances of lakes in East Africa

- Sources of fishes e.g. Lake Victoria turkana
- Hydro electric power production e.g. Owenfalls on Lake Victoria
- Irrigation purposes e.g. Lake Naivasha
- Transport e.g. Lake Victoria source of water for both domestic and industrial use e.g. Lake Victoria
- Sources of rivers e.g. Lake Victoria is source of river Nile.
- Tourist attraction e.g. Lake Nakuru and Elementaita
- Source of minerals e.g. soda ash at Lake Magadi
- Building materials e.g. sand from Lake Victoria
- Modifying climatic condition of the environment e.g. Lake Victoria
- Recreation purposes e.g. Lake Turkana
- Habitat for destructive animals e.g. crocodiles and snakes in lake Shakaababo
- Destruction of properties and displacement of people e.g. Masinga dam due to flooding.

CHAPTER 6

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OCEANS, SEAS AND THEIR COASTS

1. a) H-Lagoon
J-Island
K-Tombolo
L-Spit

b)(i)

- A gently sloping shore.
- The shore should be shallow
- The breaking waves should have a strong swash and a weak backwash.
- Waves should carry a large load of materials to be deposited.

(ii) Processes involved in marine erosion.

Hydraulic action

- Breaking waves/swash hits against cliff shattering the rock
- The force of breaking waves compress air into the cracks/joints in the cliff face. This enlarges the cracks and part of the rock break off.

Corrosion/Abrasion

- The rock fragments carried by the waves are used as a tool to erode the cliff as the wave break at the cliff face.
- The material/ load carried by the backwash erodes the sea floor.

Attrition

- The searing down of particles/loads as they continuously hit against each other and against the cliff

Solution/corrosion

- The solvent and chemical action weakens and removes the minerals found in the cliff and sea-floor where there are limestone rocks.

c) (i) Objectives to formulate for the study.

- To assess/find out the importance of depositional features.
- To identify different types of depositional features.
- To find out how the features were formed
- To establish how features are distributed along the coast
- To establish how features are distributed along the coast
- To find out the materials that make up each of the depositional features

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- To find out how constructive wave break at the shore.
- ii) Methods to use to record the information collected
 - Photographing/video taking/filming
 - Tape recording
 - Taking notes
 - Sketching/drawing
 - Filling tables Tallying
- 2. a) Submerged highland coasts
Submerged lowland coast
- b) i) Hard rocks carried by waves increase the erosive power of the waves as they hit against the Coast.
ii) A coast made of soft rocks wears away easily when subjected to sea waves.
- 3. Rise in sea level/eustatic change in base level/positive eustatic change.
Depression of coastlands/submergence of coastlands.
- 4. Fjords/fjord
Rias/creeks
Islands
Estuaries
- 5. Coastline is the line reached by the highest storm water and demarcated by a cliff.
- 6. Destructive waves are waves which have strong backwash and weak swash leading to enhanced erosion and less deposition.
- 7. Erosion features include:
 - Cliffs -Blowholes
 - Caves -Arch
 - Geos -Stacks
 - Stump
- 8.
 - Cliffs are formed by action of destructive waves
 - These waves start by cutting a small notch or hollow on the rock face called a notch.
 - As soil erosion continues a notch is enlarged.
 - The upper section collapses due to its own weight forming a cliff.
- 9.

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- A steeply sloping coast is subjected to sea waves
- Due to wave attacks a notch is formed.
- When the upper side of the notch collapses a cliff develops
- Continued undercutting of the cliff makes the cliff to collapse and to retreat inland
- As cliff retreats it leaves behind a rocky floor which slopes gently towards the sea (wave-cut platform)

10.

- Emerged coast
- Submerged coast
- Coral coast

11. a)

- Formed from tiny marine organisms called coral polyps
- Coral polyps live in colonies
- They extract from sea water, calcium and use it to build protective shells
- When they die their skeletons pile together and are cemented together by calcareous algae to form a ridge like rock parallel to the shore called coral reef.
- Coral reefs include fringing reefs barrier reefs and atolls.

b)

- Rias have been used to develop harbours
- Most resultant land forms are tourists attractions sites
- Some Rias are habitat for marine life which promotes fishing industry
- Coral rock is a raw material for cement industry
- Most of landforms have promoted education and research.

12.

- Shingle beaches are beaches made up of unsorted particles of shells, mud, stones and sand particles of various sizes. Such kind of beaches allow backwash to infiltrate into the beach.
- Sand beaches-these are beaches made up purely of sand. But since sand is compact they don't allow easy infiltration of water into the ground.

13.

- Ria coast
- Fjord coasts

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- Dalmatian coast
- 14.
- Horizontal movement
 - Vertical movement
- 15
- a) A- Stack
B- Arch
C- Cave
 - b) Abrasion and wave action attacks the pre-existing lines of weakness at the base of headland leading to formation of a hollow.
The hollow is enlarged to form a tunnel like chamber known as cave.

CHAPTER 7

ACTION OF WIND AND WATER IN ARID AREAS

- 1.
- a) (i) X- horns
(ii) Y- Eddie currents
(iii) Z- Steep leeward slope
 - b) Traction
Suspension
Surface Creep
- 2.
- a) (i) Processes through which wind erodes the surface.
 - Deflation
 - Abrasion
 - Attrition
 - ii) Ways through which wind transports its load
 - Suspension**
 - The fine dust particles are lifted and suspended in the air
 - Eventually they are blown away by wind currents
 - Saltation**

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- Larger fragments/sand particles are lifted from the ground by eddy actions
- They are moved in a series of hops/jumps by wind currents

Surface creep/attraction

- The heavy materials /small stones/pebbles are dragged along the ground by wind currents

b) (i) How oasis is formed

- A pre-existing depression formed through faulting or otherwise is exposed to wind erosion
- Wind eddies remove unconsolidated materials through deflation
- As deflation continues, the depression is deepened and enlarged.
- The process of deflation is aided by weathering
- With continued deflation, the level of the water table is reached.
- Water oozes out of the ground collects into the depression to form a lake known as oasis.

ii) Zeugens

- Zeugens are formed in desert areas where alternating horizontal layers of hard and soft rocks occur
- The top layer of hard rock is jointed/has cracks.
- Weathering opens up the joints
- Wind abrasion erodes the joints deepening them to reach the soft layer of rocks
- Abrasion continues, farrows are formed and gradually widened
- The hard/resistant rock forms ridges separating the fallows
- This process creates a ridge-furrow landscape

c) (i) Ways through which students would prepare for field study

- Reading from the relevant written materials.
- Assembling relevant tools/equipments/materials for the study.
- Formulating hypothesis/objectives
- Grouping /appointing group leaders
- Planning a schedule of activities
- Carrying out reconnaissance
- Studying /drawing a route map

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- Identifying methods of data collected
- ii) Information that would be collected through observation of the arid area.
 - Sparse vegetation/large patches of bare soil
 - Sparse settlements
 - Presence of drought resistant crops
 - Stunted trees/tufts of grass
 - Dust storms/land storms
 - Evidence of wind erosion/deposition
- iii) Measures to be recommended for controlling desertification.
 - Planting of trees
 - Controlling overgrazing
 - Avoiding bush fires
 - Controlling tree cutting
 - Practicing appropriate methods of cultivation/planting cover crops
 - Irrigation/mulching/terracing/strip cropping/contour farming
 - Gabion construction
- 3. Abrasion- Wind picks loose weathered, material and transports them. During the course of transportation the material scrubs other rock surfaces it comes into contact with.
- 4.
 - Zeugen
 - Rock pedestal
 - Yardangs
 - Deflation hollows
 - Mushroom blocks
- 5.
 - Occurs in alternating soft and hard layers
 - The hard layer is underlain by soft layers
 - Weathering breaks the hard cap in the well joined rock.
 - Wind abrasion deepens and widens the joints to produce a landscape of furrows and ridges
 - The ridges are called zeugens
- 6.

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- Barchans
 - Seif dunes
 - Transverse and wake dunes
 - Loess
 - Drass
7. - Sheet floods develop on gently sloping surfaces surrounding upland areas
- On steep sided and undulating landscape flash floods cut out rills which are then enlarged to form gullies.
- Continues erosion of gullies enlarges them to form a steep sided rocky valley or ravine known as wadi.
8. Differentiate between suspension and saltation
- Suspension**
- Fine particles are carried within the turbulence of wind while in
- Saltation**
- Saltation- Medium size particles are tolled along the group and when they collide they bounce off into the air and cause other particles to be lifted in the air.
9. Name four types of desert surface
- Erg desert
 - Reg desert
 - Hamada
 - Badlands
10. i) **Abrasion**- materials carried by wind scours/grinds against the desert surfaces leading to removal rock particles.
- ii) **Deflation**- strong winds blows away dry unconsolidated materials.
- iii) **Attrition**- Heavy materials carried by wind hits against each other leading to reduction of size facilitating their transportation.
11. a) **Suspension** – fine and light materials are picked by the wind and carried within the air turbulence.
- Saltation** – medium sized materials are transported through a series of hops and jumps along the surface.
- b) - Wind speed and force
- Nature of the load

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- Presence/absence of obstacles

- Weather changes

12. **Bajadas-** Bajada starts with formation of alluvial fans when alluvial cones coalesce along the edge of a depression

The margin leads to formation of gently surface.

Due to erosion, land mass waste a high area recedes

The retreating mountain leave a gentle sloping rock known as pediment.

13. i)

- Sparse vegetation/large patches of bare soil

- Sparse settlements

- Presence of drought resistant crops

- Stunted trees/tuff grass

- Dust storms/sand storms

- Evidence of wind erosion and deposition features.

ii)

- Planting trees

- Controlling tree cutting

- Practicing appropriate farming methods-strip, cropping, mulching, gabion construction.

CHAPTER 8

UNDERGROUND WATER

1. a) P – Clint

Q - Grike

R - Jointed limestone

- b) Rain water dissolves carbon dioxide in the atmosphere to form weak carbonic acid.

Acid rainwater falls on jointed limestone rocks.

The reaction forms calcium bicarbonate which is soluble and easily disintegrated.

2. - The surface rock and the rock beneath should be thick limestone or dolomite
- The rock should be hard and well jointed

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- The climate should be warm or hot
- Rainfall should be moderate to high
- The water table in the rocks should be deep below surface.

3.

- The rock has thin soils which discourage agriculture
- Inadequate surface water for domestic use
- Surface is rugged hindering construction of houses and infrastructure
- Presence of underground caves which may collapse.

4.

- Influence formation of spring.
- Occurrence of permeable rock on top of an aquifer: A spring will develop at the point where the two meet.
- Existence of a well joined rock; water is absorbed through the joints and eventually spring out where the water table meets the surface.
- Dyke cutting across a layer of permeable rock:
The water on the upper slope of the dyke is impounded which causes the water table to rise and causes springs where the water table meets the surface.

5

- a) Effluent streams are underground streams in limestone areas which are fed by a water table above their level while influent streams are underground streams which are fed by the water table lying beneath their bottom.
- b) Artesian basin is the status in the earth's crust in which one or more aquifers are sandwiched between impermeable rock strata from a permanent water holding facility while an artesian well is a well dug in the basins to tap water.

6.

- Grikes
- Clints
- Sink holes
- Dolins
- Uvalas

7.

Stalactites are finger like underground masses of calcite hanging vertically from the roof of a limestone cave.

8.

- Limestone regions are very good for grazing purpose especially for sheep because soil is thin and the surface is dry.

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- The Karst landscape is characterized with intermitted streams or absence of streams leading to scarcity of water supply in these regions.
- Limestone is extracted for use in industries e.g Building industry, iron and steel industry.
- The features such as gorges, caves and burns form good tourists attraction.

CHAPTER 9

GLACIATION

1.
 - a) Is large continuous mass of ice which covers vast areas of lowland
 - b)
 - Initially ice collects in shallow hollows on the mountain sides
 - The hollows are enlarged by the plucking action of the ice to form cirques
 - More ice accumulate in the hollows leading to further erosion
 - The cirques recede until a knife edged rock called arête separates them.
 - c)
 - (i) S- Medial moraine
T-Lateral moraine
V-terminal moraine
 - (ii)
 - Alluvial fans and outwash plain have fertile soils exploited for agriculture.
 - Fjords coastline provide good fishing grounds because they are deep and sheltered.
 - Lakes and rivers form channels for development of route ways
 - Rivers provide water for domestic and industrial uses.
 - Some features attract tourists earning foreign exchange
2.
 - a)(i) Is a mass of ice of limited width which moves outwards from a central area of ice accumulation.
 - (ii) Valley glaciated mountain while ice sheets are large expanses of ice covering large areas of permafrost land.
3.
 - a) P- Pyramidal peak/horn
Q- Arete
R- Hanging valley
 - b)
 - A pre-existing U-Shaped valley is filled with ice/glacier.
 - The glacier erodes the valley by abrasion or plucking
 - The end of spurs are truncated/cut
 - Ice melts away leaving behind a U-shaped valley
4.
 - a) Formation of pyramidal peak
 - Ice exerts pressure on the hollows

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- Plucking actions of ice enlarges the hollow allowing more ice to collect in them.
- Freeze-thaw action leads to expansion of cracks/hollows making them large basins.
- Moving ice plucks off loose rock materials from the basin enlarging them further.
- Nivation eats into the back wall of basins making them recede into the mountains side
- Steep-sided knife edged ridges are formed separating the basins.
- Three or more these ridges/arêtes converge at the mountain top forming a jagged peak known as pyramidal peak/horn.
- b) Significance of upland glaciated features to human activities.
 - The upland glaciated valleys are suitable for livestock farming
 - Glacial upland areas forms magnificent features that encourage recreation and tourism.
 - Glaciated mountains encourage the growth of forests hence lumbering is practiced.
 - Waterfalls formed in glaciated uplands provide suitable sites for hydro-electric power production.
 - U-shaped valleys form a natural route way
 - Flooded coastline form deep well sheltered natural harbours/good fishing grounds.
- c) i) Why it is difficult to carry out field study of glaciated feature.
 - Climbing the mountain is difficult due to rugged terrain.
 - Features are found far from the schools/settlements
 - Time may be inadequate
 - Poor weather conditions i.e. rainfall and low temperatures
 - Thick forest and dangerous animals which makes it difficult to access such areas.
 - It is difficult to conduct a previsit
 - Avalanches
- ii) How students would use the photograph of Mt. Kenya to identify

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the glaciated features.

- By dividing it into parts
 - By observing and identifying the features in each part of the photograph
 - By recording the features observed
 - By drawing sketches of the features observed
 - By labeling the features observed.
5. A snout is the lower part of glacier where it begins to melt while a snow niche is a smaller niche mass laying on a steeply sloping hollow, gulley or bench in the high mountains.
6. - Lewis glacier (between Lenana and Nelion)
- Tyndall Glacier (North of Lenana peak)
7. It is caused by Glacier as it erodes vertically and laterally. The sub-glacial moraines scrub the floor while the lateral moraine scrubs the walls. The trough is thus broad, flat bottomed, and steep sided with a U-shaped cross-section.
8. It is a permanent cover of ice on the land surface extending to a small area of land
9. Pyramidical peaks e.g Lenana, Batian, Nelion Cirques e.g Teleki tarn, Hobley Gorges, Aretes
10. It is a shallow pre-glacial depression that has progressively enlarged. A patch of snow produces alternate freezing and thawing on rocks around the margins which then cause them to rot and disintegrate. Melt water helps to remove the resulting debris thus forming nivation hollow.

CHAPTER 10

SOIL

1.
 - a)(i) This is the arrangement of soil layers along a mountain slope from top to bottom.
 - (iii) - Soils are thin/shallow
 - Have low organic content
 - Soils have low moisture content
 - Soils are rich in calcium/alkaline
 - c)
 - In savanna areas there is alternating wet and dry season
 - During wet season, mineral salts in the horizon A dissolve in the percolating rainwater.
 - The dissolved minerals are precipitated/deposited in the lower layer. This process is called eluviation.
 - Insoluble minerals such as iron and aluminium accumulate in A horizon/top layer to form a crust known as laetrile
 - During dry season illuviation occurs. Soluble minerals are dissolved in the capillary water and moves upwards to horizon A.
 - Evaporation occurs on the horizon A.
 - Minerals are precipitated near/on the surface to form crust.
2.
 - a) Loamy
Clay
Silty
Sandy
Gravel
 - b) -Humus help to improve soil porosity
-Humus improves the moisture retention capacity in the soil
-Humus improves soil texture
-Humus provides essential minerals in the soil
3.
 - Soil accumulation of rock particles, minerals, organic matter, water and air found on the surface of the earth.

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- Is the superficial layer of loose/unconsolidated rock material overlying on crustal rocks and on which plants grow.

4.

- Zonal
- Intrazonal
- Azonal

5.

Take place through to stages namely mineralization and humification **mineralization** is the biological and chemical breakdown of dead plant tissue by soil micro-organisms to produce simple soluble organic substances. This is the initial stage of decomposition.

Humification is the second stage of decomposition where the dead plant material which had been mineralized are regrouped into large molecules to form humus.

6.

- Leaching process common in arid areas.
- Soils are leached upward through capillarity
- In some areas this leads to formation of thin salty crust on the surface

7.

Zonal order soil is a type of soil classification which group together soils which have undergone long time of soil formation process under good drainage conditions.

8.

- Planting cover crops
- Ploughing along contours
- Practicing crop rotation
- Controlled grazing
- Agro forestry
- Mulching
- Adding fertilizer

9.

- Is a severe leaching process
- Is limited to middle and high latitude areas

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- The climatic conditions are too cold that bacteria action is inhibited while moisture is sufficient enough to allow large green plant to grow.
- Humic acid produced from a abundant leaf mould and humus, leach the upper soil strongly of bases, colloids and oxides of iron and aluminum leaving composed largely ash gray as soil horizon of silica.

CHAPTER 11

AGRICULTURE –CROP FARMING

1. a)
 - High temperatures throughout the year temperature range of 20⁰C to 30⁰C.
 - High rainfall that is evenly distributed throughout the year 1500mm to 2100mm,
 - High relative humidity of 80% to 90%
 - Plenty of sunshine during the ripening season.b)
 - Competition from other oil vegetables
 - Poor road network
 - Production of low quality oil
 - Reduction of low quality oil
 - Reduced production that has lowered the amount of oil exported.
2. b) Describe the stages involved in cultivation of tea from land preparation to the stage shown in the photograph.
 - The land is cleared of vegetation
 - The land is ploughed/tilled
 - Seedlings are planted in nursery and allowed to grow to 20cm
 - Seedlings are planted in rows which are about 1.5 metres apart.
 - The plant are weeded and manured/ mulching applied regularly.
 - Once the bushes start growing. The tips of branches are pruned regularly to encourage plant to form more branches.
 - The crop is harvested every two weeks once it attains maturity
 - After harvesting. The green tea leaves are transported to the factory within 24hrs.c) (i) Name two districts in Eastern province where tea is grown
 - Embu

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-Meru North

-Meru South

-Meru Central

(ii)

- Organizes farmer education days/ provides extension services for the farmers at a low prices
- Buying farm input in bulk and sells to farmers at low prices.
- Providing credit facilities to the farmers to enable them purchase farm inputs
- Collecting the green leaves and delivers the factory on behalf of farmers
- Establishing factories where the green tea leaves are processed
- Undertaking the marketing of tea on behalf of the farmer.
- Improves feeder roads to ease transportation of green leaves
- Conducts researches on tea crop varieties/diseases/pests in order to produce high yield tea/ better quality tea.

3. a) -High temperatures/24.30⁰C.
-High rainfall/1,200-1,500mm.
-Well distributed rainfall throughout the year.
-Deep, well drained, fertile soil/volcanic/light clays
-High relative humidity
-Shade from strong sun rays for seedlings/young plants
-Shelter from strong hamattan wind.
-Undulating lowlands/below 750m above sea level.
- b) -Fluctuation of prices in the world market
-Competition from other land uses.
-Inadequate labour during harvesting
-High production costs
-Competition from other beverages like coffee, flowers, fruits, vegetables
4. a) -Temperature ranging from 14⁰C – 26⁰C
-High rainfall 1100mm - 2030mm per year.
-Well distributed rainfall throughout the year
-Gently sloping landscape.
-Deep, fertile, well drained soils

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- b) -The crop is attacked by pests and diseases
-Fluctuation of coffee prices in the world market
-Poor infrastructure
-Mismanagement of coffee co-operatives
-Inadequate rainfall
-Inaccessibility of credit/inadequate capital
-Poor marketing strategies
-Low payments
5. a) (i) -Central
-Rift Valley
-Eastern
- (ii)
- Moderate rainfall/500mm to 1270mm to enhance the growth of wheat.
 - Temperatures ranging from 15⁰C to 20⁰ C/warm conditions to facilitate growth/maturity of wheat.
 - A warm /dry /sunny spell for ripening and harvesting.
 - Fertile volcanic soils to sustain high production
 - Gentle sloping/undulating landscape to allow proper drainage/mechanized cultivation.
- b) i) Storage
- In both Canada and Kenya wheat is stored in grain silos.
 - In Canada wheat on transit is stored in huge grain elevators/special car boxes while in Kenya it is stored in sacks.
- ii) Transport
- In Canada wheat is transported by railways (CPR and CNR), roads and waterways while in Kenya it is transported by roads and railways.
- iii) Market
- In Canada wheat is for both local and export markets while in Kenya wheat is for local market
 - Canada has a larger and reliable local market than Kenya.

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- Kenya wheat sold through NCPD or directly to the millers, Canada sold by government/individuals.
- c) i) Climate problems that affect wheat farming in Canada.
 - Low rainfall/Unreliable rainfall which leads to crop failure
 - Low temperature/long and cold winters which limit outdoor activities/delays cultivation of wheat.
 - Frost which destroys wheat leading to low yield.
 - Hailstones which destroys wheat leading to low yield
 - Strong winds causes soil erosion especially affects ploughing resulting to loss of fertile soils
- ii)
 - Uses of wheat
 - Used as animal feed
 - Used as human food
 - Used for brewing/distilling alcohol
 - Used for making adhesives/glues
 - Used for paper and straw boards.
- d)

-Nakuru	-Samburu
-Uasin Gishu	-Trans Mara
-Narok	-Nyandarua
-Trans nzoia	-Keiyo
-Laikipia	-Mt.Elgon
- e)
 - Alberta
 - Saskatchewan
 - Manitoba
- f)
 - Wheat growing in Canada is more mechanized leading to higher production than in Kenya
 - More capital is available in Canada enabling farmers to sustain production
 - Farmers in Canada are more experienced due to long history of wheat production than in Kenya

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- Advanced scientific research in Canada enables the production of higher yielding seeds/better farm inputs/control of pests and diseases/overcome limitations of weather
- Wheat farmers in Canada specialize in wheat production while in Kenya, farmers practice mixed farming
- In Canada, there are more extensive tracts of land suitable for wheat growing than in Kenya.

6 a)

- Cool/warm climate/ 10°C to 18°C throughout the year.
- High rainfall/100-2000 mm per year
- Well distributed rainfall throughout the year
- Areas that are frost-free
- Deep, light and well-drained soils
- Gently sloping/undulating land
- Acidic/volcanic soils
- High altitude/1200 mm -2300 mm

b)

- Delayed payments/low payments that lower the morale of the farmers.
- Mismanagement /embezzlement of funds thus farmers are discouraged
- Poor feeder roads in the tea growing areas lead to delays in collection/delivery of the green leaf hence wastage.
- Adverse weather conditions such as long droughts/hail storms lead to destruction of the crop.
- Fluctuation of prices in the world market makes it difficult for the farmers to plan ahead.
- High production costs due to high prices of farm inputs leads to lower yields since most farmers cannot afford to buy them
- Pests/ Fungal diseases destroy/reduce yields
- Inadequate /unreliable transport facilities delay the collection/delivery of green leaf reducing the quality.

7.

a) W- Kapenguris

X- Kericho /Kisii

Y- Meru/Embu/Nyeri/Kirinyaga/Mt Kenya region

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b)

- Expansion of tea growing areas and the establishment of the Nyayo tea zones.
- Increase in the number of small-scale tea farms in the country
- Improved marketing strategies through KTDA
- Expansion/increase in the number of tea factories.

c)

- When the bushes are ready only the two top leaves and a bud/flush are picked
- The green leaves are transported in airy baskets to a collecting centre for weighing.
- The weighed leaves are transported by lorries fitted with bags to the processing factories. The leaves are weighed again at the factory.
- The tea leaves are spread out on long wire trays.
- The leaves are then dried by blasts of warm air from beneath the trays.
- The dry leaves are passed through a set of rollers to chop them/the leaves are crushed.
- The leaves are placed in containers for fermenting reducing tanning acid and changing the above to grey-brown.
- The leaves are passed through a conveyor belt which takes them to a tunnel which is a temperature of 100°C for roasting after which they turn black.
- The leaves are sifted for grading /lasted for classification.
- The graded tea is packed in tea sets for export and small packages for local market.

8. Cocoa growing areas

- Kumasi
- Tokoradi
- Accra

9.

- High rainfall 1200-1500 mm per year
- High temperature throughout the year $24-27^{\circ}\text{C}$
- Deep soils which are well drained
- Shelter from strong sunrays
- Plenty of labour force during harvesting
- High relative humidity 70-90%

10. Types of commercially cultivated coffee.

- Arabica

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- Robusta
- Liberia

11. Ways through which Brazilian government responds to problems facing coffee industry.

- The government has established an institute for the permanent defence of coffee. The institute manipulates the amount of coffee released to the international market thus creating artificial shortages consequently maintaining high prices.
- The government lobbies for higher quota in the international market
- The government has been encouraging crop diversification/mixed farming by introducing annual crops such as sugar cane and soya beans in the coffee growing areas to reduce over dependence on coffee.
- When the prices are low the government buys coffee from farmers and stores it thus stabilizing the prices for the farmers.
- To solve the problem of overproduction the government prohibits planting of more coffee.

12. Use of maize

- Animal feed
- Industrial use in manufacturing of alcohol
- Vegetable oil, rayon, elastics, paper and wall boards.
- Fuel (stalks, cobs, especially in rural areas)
- Manure.

13. Problems facing maize farmers

- Pests and diseases (stalk borer, rodents)
- Unreliable rainfall
- Poor storage facilities leading to loss of harvest (rodents/Weevils)
- Exploitation by middlemen
- Rising production costs e.g expensive fertilizer, chemicals, certified seeds
- Fluctuating market seeds

14. Processing of cocoa

- Cocoa pods are split open with a machete
- Beans embedded in the pulp are removed by hand

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- Beans are fermented for a week by heaping them into a single heap and covering them with banana leaves
- Beans are washed after fermenting cleaned and sun dried.
- They are cleaned roasted and husks removed to produce cocoa nuts
- Cocoa nibs are ground into powder as cocoa butter is separated
- The powder is mixed with milk.

CHAPTER 12

AGRICULTURE –LIVESTOCK

1. b)
 - The landscape/gentle sloping land which is suitable for grazing.
 - The climate has warm and sunny summers that allow outdoor grazing
 - The climate has warm and sunny summer that allow outdoor grazing
 - There is cool climate suitable for pasture growing
 - There moderate rainfall/ rainfall suitable for grass/fodder crops
 - Soils are fertile to support high quality pasture.
- 2 a)
 - The government has set up demonstration ranches to educate the pastoralists on better ways of keeping livestock.
 - Cattle dips have been constructed to control pests
 - Extension services are provided to give advice to the pastoralists
 - Boreholes and dams have been constructed to provide water for the livestock.
 - Road s have been constructed to enable the pastoralists to transport their produce to markets
 - Through formal education the pastoralists have leaned the advantages of keeping manageable sizes of herds
 - The government encourages group ranching to enable the pastoralists to view livestock keeping as a commercial undertaking.
- b)
 - Replacement of coarse grass with alfalfa/corn has improved the quality of pastures for the beef cattle
 - Cross breeding of the traditional cattle with higher quality breeds/Hereford, Aberdeen Angus, Short horn has improved the quality of the yields.
 - The maritime climate of the area makes grazing of cattle possible throughout the year.
 - Availability of water supplied using wind pups ensures constant supply of water cattle.
 - Availability of vast lands/pampas suitable for cattle grazing encourages beef ranching.
 - Availability of market both local and external encourages the farmers to expand the beef industry sustains the industry .

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- Availability of refrigeration facilities enables beef to reach far off markets in good condition.

c)

- Presence of undulating landscape
- Adequate water supply
- Large tracks of land/expensive tracks of land
- Adequate pasture for the animals
- Moderate rainfall which supports growth of natural pasture

d)

- Aberdeen Angus
- Hereford
- Charolais
- Red Angus
- Short horn
- Santa Getaways

3.

Pests and diseases e.g foot and mouth, nagana.

Poor pastures due to soil infertility

Low and unreliable rainfall thus inadequate pasture drought.

Low quality animals due to high temperatures and pastoral preference for large quantity of animals rather than quality

Inaccessibility which limits transportation of beef products to market.

4.

- Availability of refrigeration facilities
- Well –organized and fairly mechanized beef ranches
- Availability of well developed transport networks
- Supplementing natural grass with artificial pasture e.g. alfalfa
- Availability of local and external market.

5.

Denmark	Kenya
i) Breeds are exotic	i) Most breeds are indigenous
ii) Machinery is supplied to the farmers	ii) Little or no use of machinery by farmers
iii) Cool climate provides fodder throughout the year	iii) Shortage of fodder during the dry spell
iv) Most of the products are for export	iv) Most of the products are local consumption

6.

- Introduction of high quality breeds through cross breeding.
- Introduction of various animal feeds to promote more milk production
- Better prices offered to farmers create an incentive to farmers to produce more milk
- Control and elimination of disease through veterinary services
- Development of dairy co-operatives
- Better management techniques like zero grazing

7.

- It's the extensive grazing on natural pasture involving constant or seasonal migration of nomads and their flock.
- Cattle are kept as a sign of wealth.
- Uncontrolled breeding
- Lack of organized land tenure (community owned)
- Diseases incidences are common
- Animals are of low value
- Marketing systems are inefficient
- Involves seasonal movements
- Many kinds of animals are grazed/kept

8.

- Introduction of pedigree British cattle/cross breeds them with indigenous breeds
- Educating and encouraging farmers to adopt modern methods of rearing breeds

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- Providing water by building dams and reservoirs ploughing and resourcing pasture land with special strain of drought resistant (more nourishing grasses)
- Funding research in animal disease control and management, educational programmes and drug supply
- Providing extension officers to give the farmers necessary advice
- Decontrolling the price of meat products

9.

- Low temperature of about 18⁰C for survival of exotic breeds
- Cool conditions hence few diseases hence greater survivor for the animal feed
- Plenty of nutritious grass
- Fertile soils which ensure high quality of grass throughout the year
- Constant water supply throughout the year.

FORM FOUR WORK

CHAPTER 1

LAND RECLAMATION AND REHABILITATION.

1. a)
 - Irrigating dry land.
 - Draining of swamps
 - Adding manure to infertile soils
 - Introducing drought resistant crops
 - Planting of trees/Afforestation
 - Tse-tse fly control
 - Part of the low lying land covered by sea water is enclosed using strong walls/ ring dykes
 - Ring canals are constructed to lead water to pumping station.
 - The water is pumped out using windmills/diesel pumps/electricity pumps.
 - Ditches are then dug to drain the excess water from the enclosed land
 - Chemicals are added to the soil to reduce salinity /fresh water is pumped into the enclosed land to reduce salinity.
 - Oats, rye and sugar beets are planted to improve the PH of the soil and reduce the land salinity further.
 - The land is dry and (ready for use)
- 2 a)
 - Land reclamation is a process by which unproductive land such as deserts, marsh or swamp is converted into a land fit for cultivation.
 - Land rehabilitation is the restoration of land that has been ruined through man's negligence to its former reputation or good condition.b)
 - Control of soil erosion
 - Afforestation/ reforestation/agro-forestry
 - Irrigation
 - Draining of swamps and flood prone areas
 - Introduction of drought resistant crops
 - Control of pests e.g. tsetse flies
 - Use of manure to improve soils

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c)

- Construction of the ring dykes and ring canals
- Construction of the ring dykes and ring canals
- Construction of ditches within each polder leading water to a pumping station
- Water is pumped out into the canal
- Land is allowed to dry
- Desalination to improve the soil through flushing with fresh water and planting hardy plants
- Dividing the polder land into economic units
- Infrastructure is laid out
- People are settled in villages.
- People are settled in villages
- Farming activities began with spreading of soil to improve fertility
- Addition of fertilizers to the soil

d)

- Floods were controlled
- Pests/water borne diseases were controlled
- There was an increase in the land for agriculture
- Better farming methods were introduced
- There was an increased agricultural output/new crop introduced
- There was an increase in the employment opportunities.

3.

b)

- The area was sparsely populated hence less displacement of people
- Low rainfall received in the area made it suitable for irrigation
- The area had fertile soil suitable for rice farming
- The black cotton soils in the area have a high water retention, a condition required for rice cultivation
- Presence of rivers Nyamindi and Thiba which provides irrigation water.

c)

- Stagnant pools of water have led to water borne diseases e.g bilharzias and malaria which weakens farmers hence lowering their productivity.
- Siltation of canals which calls for regular dredging which is expensive.

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- High rates of evaporation has led to salinisation of soils.
- Pest like quelea destroy crop and reduce yield thereby reducing farmers income
- Mismanagement of funds has led to delayed payments to farmers.

d)

- It has helped resettle landless people
- The scheme has opened productivity in former wasteland
- Tenants generate income from horticultural farming which has helped raise their living standards
- Creation of employment for the local people.
- 2001
- There is insufficient amount of water in River Perkerra and this limits expansion of the scheme.
- The harsh climate and high temperatures in the area hinders production of some variety of crops.
- The scheme is located in remote areas with sparse population limiting market for the produce
- Financial constraints which affects farmer activities.

4. Horticulture is the intensive cultivation of vegetables, fruits and flowers while market gardening is the intensive cultivation of vegetables and fruit for the nearest urban centre.

5.

- Price fluctuation due to over production
- High degree of perishability of the products
- Pests and diseases destroy crops and reduced yield.
- Lack of adequate capital to run the farms

6.

- Mwea Tebere irrigation scheme mitunguu/Ishaaira scheme
- Kibwezi scheme
- Kibirigwi scheme
- Taveta scheme
- Daula scheme
- Bura/Hola Gabole scheme

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- Perkerra scheme
 - Perkerra scheme
 - Bunyala scheme
 - A hero scheme
7. Hot and dump climate providing conducive physical conditions for the tsetsefly
Bush vegetation in the area provides environment preferred by the tsetse fly.
- 8.
- Clearing of the bush
 - Spraying of the bush area from flying aircraft.
 - Sterilizing the male fly by curing it to some chemicals substances placed at strategic place
 - Killing the host animal
 - Creation of consolidated zones
 - By using traps
- 9.
- Sparse population making it easy and cheap to resettle people
 - Land availability due to low population tributaries of R. Tana (Thiba, Nyamidi)
 - Black cotton soils which retain water longer
 - Gently sloping topography enabling gravity flow of water
 - Low, unreliable rainfall received in the area make irrigation necessary.
- 10.
- It causes sleeping sickness in humans
 - It causes trypanosomiasis in cattle
- 11 a) Land reclamation is the process by which unproductive land e.g a desert or a swamp is converted into a useful agricultural land, while land rehabilitation is the restoration or bringing back to its former good condition a land that has been wasted through human negligence.
- b)
- To settle thousands of hitherto landless people in central Kenya
 - To provide some form of work or political detainees during the state of emergency
 - To increase agricultural production aimed at attaining self-sufficiency in food production.
 - To harness the rainy season

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c)

- Stagnant pools of water have led to waterborne diseases e.g bilhazia and malaria weakens the farmers hence lowering their productivity.
- Siltation of canals which calls for regular dredging which is expensive
- High rates of evaporation which have led to salination of the soils.
- Pests e.g. quela birds attacks the crop thus lowering the yields and farmers income
- Mismanagement of funds has led to delayed payment to the farmers thus lowering their morale.

12.

- There is insufficient amount of water in River Perkerra and this limits the expansion of the scheme.
- The harsh climate and high temperatures in the area hinders the production of some crops.
- The scheme is located in a remote area with sparse population and poor transport and communication network. This limits the market for products.

13. a)

- The schemes have helped in boosting food production thus enhancing self sufficiency.
- Through irrigation farming, many people are employed
- It has helped to resettle landless people
- Through irrigation, farmers have earned income. Through the direct sales of their produce thus raising their standards of living
- It has opened up remote areas for development by promoting the development of infrastructure and social amenities.

b)

- Siltation of canals
- High rates of evaporation
- Fluctuating regimes of rivers
- Inadequate capital
- Closing up of canals by water weeds
- Presence of waterborne diseases
- Presence of pests and diseases
- Delayed payments

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- Low pricing of the crops
14. a)
- Draining in wet and low lying areas
 - Planting of vegetation
 - Tsetse fly control
 - Planting of drought resistant crops in marginal lands
- b)
- Draining wet and low lying areas
 - Used to reclaim swampy and marshy lands
 - Ditches and canals are dug to drain away excess water
 - Trees with high water absorption capacity are planted
 - River channels are straightened to improve the flow of water.

Planting of vegetation

- Reafforestation and Afforestation programmes are undertaken.
- Reafforestation is the planting of trees where they have been cut while Afforestation is the planting of trees where none existed.

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Tsetse fly control

This is done by:

- Bush clearing
- Sterilizing the male fly
- Creation of consolidated zones
- Spraying of tsetse fly infested areas
- Killing of the host animals

Planting of drought resistant crops in marginal areas

- Marginal areas are transition zones between high rainfall and very low rainfall areas. They receive low rainfall. They receive low rainfall.
- In these areas, research has been done in the drought resistant crop varieties e.g sorghum, millet and cassava
- These tend to withstand long spells of no rainfall and mature within the short cycle of rain.
- The marginal areas include Kitui, Turkana, Busia, Baringo etc

15

- It has led to the introduction of farming through irrigation
- It has helped in the control of floods in the area
- It has increased the amount of agricultural land
- Contributed to the improvement of transport systems
- It has increased the amount of agricultural land
- Contributed to the improvement of transport systems
- It has created employment opportunities
- Over 800 hectares of land have been rehabilitated and are now being utilized for agriculture
- It has assisted in the control of floods
- The project has enhanced agricultural production

16. a) A low lying land reclaimed from the sea and enclosed by dykes in the Netherlands.

b) Construction of ring dykes and canals

- Construction of ring dykes and canals
- Construction of ditches within each polder which leads water into a pumping station
- Water is pumped out into the canals

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- The land is allowed to dry
- The land is allowed to dry
- The soil is improved through desalinization by flushing in with fresh water, planting of hardy plants and additions of soil.
- The polder land is then divided into economic units
- Infrastructure is laid out and people are settled in villages
- Farming activities then commence.
- c) Creation of fertile agricultural land
 - Control of floods
 - Contributed to urbanization
 - Production of horticultural products for export has earned foreign exchange
 - Provision of fresh water through Lake Yssel.
 - Improvement of road transport between
 - North Holland and Friesland.

17.

- Methods employed in Kenya are less capital intensive as compared to those Netherlands which is capital intensive.
- More varied methods of reclamation are employed in Kenya while in Netherlands mainly one method is used
- Reclamation is mainly due to insufficient moisture in Kenya while in Netherlands it is due to excess water.

CHAPTER 2

FISHING

1. a)
 - The over fished areas are being restocked
 - There are laws enacted against indiscriminate fishing types of nets/seasons for fishing/areas free for fishing.
 - Special hatcheries have been set up for artificial fertilization of eggs (pisciculture)
 - Disposal of effluent into fisheries is prohibited
 - Research is carried out to expand and improve fisheries
 - Institutions have been set up to train personnel to manage fisheries
- b)
 - It has an extensive continental shelf
 - Its waters are rich in plankton
 - It has a long forded coastline which provides sheltered waters ideal for fishing/breeding ground for fish.
 - Cold climate/rugged terrain drove people to fishing.
 - Norway has a long tradition in sailing and fishing
 - It has a well developed ship building industry which provides fishing vessels.
 - Its cool climate makes preservation of fish easy/it has advanced technology in fish preservation.
- c)
 - Harrowing
 - spearing
 - Poisoning
 - Using of rafts
 - Single lining
- d)
 - Setting up fish ponds and hatcheries demonstration farms
 - Provision of extension officers to advice farmers
 - Encouraging farmers to set up fish ponds
 - Provision of extension officers to advice farmers
 - Encouraging farmers to set up fish ponds

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- Provision of technical and financial assistance to fish farmers by fisheries department
- Government's food policy encourages eating of fish by communities that never ate fish.

2

a)

- Trawling
- Purse seining/serning
- Drifting
- Lining

b)

- Salmon
- Mackerel
- Herring
- Haddock
- Flounder
- Sardine

c) **Indented coastline**

- These provide secure breeding grounds for fish because the bays are sheltered from sea waves
- The sheltered bays provide suitable sites for building fishing ports/fishing landing site.

Ocean currents

- The meeting of cool and warm currents causes upwelling of ocean water which brings plankton/fish food to the surface and for oxygenated waters.
- Cool waters provide ideal water temperatures for survival of numerous species of fish.

d)

- Freezing
- Sun drying
- Salting
- Smocking
- Canning

e)

- The occurrence of strong winds leads to high waves causing accidental drowning/destruction of fishing vessels and nets.

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- Most fishermen have poor fishing equipment/motor boats engines which are ineffective.
This leads to low catch delayed landing
- Fishermen lack appropriate storage/preservation facilities which cause heavy losses of the catch/low catch.
- Presence of floating vegetation/water hyacinth which entangle and tear the fishing nets which is a loss to the fishermen
- Parts of the lake shores are swampy/marshy which makes the landing of the catch difficult.
- Insecurity/theft of fish and fishing equipment discourages the fishermen.

3.

a) P-Norway

Q- Japan

b)

- The areas have cool waters which have abundant supply of planktons which is the main food of fish.
- The areas have shallow continental shelves which allow light to penetrate to the sea bed encouraging the growth of micro-organisms used as food by fish.
- The areas experience convergence of warm and cool currents which result in upwelling of ocean water thus bringing minerals for fish and planktons from the sea bed to the surface.
- Cool waters experienced in most of these coastal areas encourage thriving of numerous fish species
- Most of the coasts are indented/have numerous sheltered bays which provide secure breeding grounds for fish.
- The sheltered bays provide suitable sites for building fishing ports/fish landing sites
- The large population in these areas provide a ready market which promotes the fish industry.
- The rugged landscape in some of the areas limits agricultural activities thus people turn to fishing as an alternative economic activity.
- Cool to cold climate provides natural preservation of fish.
- Advanced technology enhances fishing
- Cold climate limits agriculture

4

a)

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i) R- Trawling

S - Basket fishing

ii) **Basket fishing**

- The basket is funnel shaped to slow easy entry for fish.
- At the mouth there is no –return valve which restricts the outward/escape movement of fish once inside the basket
- The basket is lowered in water with the mouth facing the direction from which the water is flowing.
- A bait is put in the basket.
- The basket is held in position with ropes/stones/sticks to prevent it from being swept downstream
- The basket is left in that position for sometime/overnight for landing.

Trawling

- A gag-shaped net is attached to a ship/trawler
- The nets mouth is kept open by other boards/head beams
- The upper part of the net is net kept afloat by corks/floats.
- Weighs are used to keep the lower parts of the net at the seabed.
- The trawler drags along the net
- After sufficient fish is caught, the net is hauled to the trawler.

5.

a)

- Fishing are all activities involved in harvesting of aquatic life in the seas and inland waters world over
- Fisheries refer to all the water resources which form the habitat from which fish is harvested. They are numerous such as seas, oceans, rivers, ponds, lakes among others.

b)

- Presence if planktons which fish feeds on.
- Cool climate favouring the growth of fish food and for preservation of the catch.
- Indented coastlines for breeding of fish as well as development of fishing ports and villages
- Environmental influences such as numerous islands
- Large population to offer ready market and labour.

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- Advanced fishing technology for high volume of catch as well as processing of catch.
6. a) World's major fishing grounds
- High latitudes offers cool climate ideal for flourishing of planktons and preservation of fish.
 - They are washed by warm and cool ocean currents and indented coasts offering ideal breeding ground for fish.
 - Environmental factors limiting meaningful agricultural activities hence fishing is the alternative occupation
 - A wide continental shelf for a rich fishery.
7. a)
- Pelagic fishing; fishing used to catch fish which swims near the water surface
 - Dimersal fishing: Fishing done close to the shores in shallow and sheltered waters along the coast and stretches of rivers
 - Inshore fishing: Fishing done close to the shores in shallow and sheltered waters along the coast and stretches of rivers
 - Fresh water fishing: Fishing done in streams, rivers, lakes and ponds having fresh waters.
- b)
- Traditional methods of fishing include;
 - Harpooning
 - Wooven baskets
 - Traps
 - Handlines
 - Herbs
- c)
- Drifting method**
- Tennis like nets are hanged vertically in water
 - The nets are kept open by floats on upper side and weights on the lower side and weights on the lower side.
 - On both sides , the edge of the net is anchored on the seabed.
 - It is left in the position overnight
 - When fish swims on it. They are entangled by their gills.

d)

Trawling method

- A cone shaped bag is attached to a slow moving ship
- The bag is dragged over the seabed
- Its upper part is kept open by floats (otter boards) and lower part kept open by weights.
- When sufficient catch is realized, the net is hauled back to the trawler.

8. a) Uganda

- i) Lakes Kyoga, Albert, George and Katwe. Rivers Nile, Kafu, Smiliki, Katonga and Kagera.

Tanzania

Lakes Rukwa and Malawi

Malagasi swamp

Rivers Ruvuma, Rufiji, Rungwa and Great Ruaha.

ii)

- The lake has many landing grounds
- Dense population around the lake offer cheap labour and ready market
- Lake has many fish species of commercial and domestic value
- High demand for fish around the lake region.

b)

- Inadequate market for marine fish.
- Inadequate capital to purchase the required fishing gear.
- Narrow continental shelf hence poor fishery.
- Competition from other foreign countries
- Rough sea wave limiting deep sea fishing

9. a) Source of employment

- Tourist attraction- sport fishing
- Source of food rich in protein
- Facilitate development of industries
- Medical value
- Income generation

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- Source of foreign exchange
- Development of settlement
- Improvement of infrastructure

b)i)

- Inadequate capital to purchase modern equipment
- Low level of technology to undertake commercial fishing
- Inadequate transport network leading to high perish ability of fish.
- Limited internal market
- Presence of aquatic weeds hindering fishing activities
- Pollution of fisheries

ii)

- Formation of co-operative societies to advance loans and market fish.
- Ban on indiscriminate fishing in order to harvest only fish required sizes
- Restocking of overfished waters with fast maturing fish
- Legislation to control any form of pollution on the fisheries
- Improvement of infrastructure to the fish landing grounds
- Construction of cold storage facilities to preserve fish
- Establishment of research centers to research on fast maturing fish species
- Increased surveillance of fisheries to curb illegal gear and overfishing.

10. Similarities

- In both countries, fishing co-operative societies have been formed
- Fisheries in both centres face the problem of overfishing and pollution of fisheries.
- In both countries, fish farming is done
- Similar methods of preservation and processing is employed in both cases

11. a)

- i) Management of fisheries refers to all measures undertaken for better and useful exploitation and rehabilitation of fisheries
- ii) Conservation involves the proper utilization of fisheries to ensure little or no wastage.

b)

- International agreement on fisheries
- Forbidding indiscriminate fishing

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- Carrying out research
- Restocking overfished waters
- Control of pollution
- Artificial fertilization

CHAPTER 3**WILDLIFE**

1. i)

- Name the national parks marked P, Q and R.
- P - Ruwenzori/Queen Elizabeth
- Q - Serengeti
- R - Tsavo (East/West)

ii)

Differences	
Kenya	Japan
1. Traditional methods of fishing are used	1. Advanced fishing technology is applied
2. Expansion of fishing industry is limited by insufficient capital	2. Expansion of fishing is favoured by easy access to capital
3. Limited fishing grounds with regular coastlines	Many fishing grounds with indented coastlines.
4. Fishing is done on a small scale	Fishing is done on large scale
5. Few species of fish of commercial value	5. Many fish species of commercial value
6. Both marine and fresh water fishing is practiced	6. Only marine fishing is practiced
7. fish research is not quite extensive	7. Advanced research in fishing is carried out

2.

- To protect the endangered animal and plant species
- To promote tourism
- To promote foreign exchange
- To keep them for posterity
- To sustain the raw materials for supply of herbal drugs
- For education and research purposes
- For aesthetic value.

3.

- To preserve the natural beauty

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- To conserve wildlife/plants and animals
- To promote tourism/provide for recreation
- To create an environment for education and research.

4. a)

Game reserves are areas where wildlife is protected but some limited human activity is protected but some limited human activity may be allowed while game parks are gazetted regions where settlement hunting and cutting of plants is not allowed whatsoever.

b) State four steps taken by Kenyan government to promote wildlife resources

- The establishment of anti-poaching unit has helped curb poaching.
- Encouragement of ecotourism has helped reduce wildlife human conflict and negative impact on wildlife habitat
- Isolation and treatment of sick animals/extending veterinary services to wild animals
- Ban on hunting and trading in game trophy
- Construction of electric fencing around game parks and reserves in order to reduce wildlife human conflict.

5.

- Poaching wildlife by illegal hunters has reduced the population of the animals
- Pollution caused by industrial and domestic refuse has greatly affected marine life
- Soil erosion due to improper farming methods has led degradation of wildlife habitat.
- Cutting down trees in forest to pave way for settlements and agriculture.

6. Refers to animals and plants in their natural habitat

7.

- Reasons for the need to conserve wildlife
- Wildlife earns foreign exchange through the tourist industry.
- While conserving wildlife in its natural habitat, forest has been preserved and this has helped protect water catchment areas and soil.
- Commercial exploiting of wildlife population through consumptive methods for meat or other products earns income
- Tourism which is associated with wildlife generates employment

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- Scientific benefits include biological studies which have led to some conservation and management of wildlife and environment problems being solved
- Marginal land in tropical lands where most of the parks and game reserves are located cannot adequately support domestic animals, crops and forestry due to unreliable rainfall has been utilized.

8.

- Birds e.g flamingo
- Trees e.g Acacia
- Animals such as zebra, buffalo, rhino.

9. **Poaching-it threatens the survival of some animal species**

- Destruction of habitats due to population pressure
- Over exploitation of eater resources e.g sea weeds thus posing the danger of extinction.
- Interference with animals' way of life due to frequent visits hence trampling on vegetation by vehicles. Animals are also harassed by the noise from the vehicles.
- Destruction of crops and property by wildlife.
- Destruction of crops and property by wildlife.
- Inadequate capital to assist in conservation measures
- Drought since most national parks and reserve are in arid and semi arid areas
- Overstocking of some wild animals leads to destruction of environment through overgrazing
- Fire outbreaks destroy wildlife
- Pollution of the environment leads to death of wild animals
- Illegal hunting of wild game threatens the conservation efforts
- Rapid human population growth leads to encroachment of game parks and reserves.
- Inadequate capital limits government conservation efforts.
- Frequent droughts experienced in some of the national parks and reserves leads to loss of animals through starvation and death.

10.

- Pollution of their habitats through dumping of untreated wastes leading to their
- deaths.
- Over exploitation of water resources through over fishing leading to extinction of some species.

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- Poaching : man kills for their trophies or food thus makes them timid since they have been taken from their natural habitats
- Man clears their habitats like forests and therefore threatens their existence.

11.

- Should buy them fire arms and ammunicions in order to protect themselves and the animals from poachers
- Should equip them with two-way radios so that they are able to communicate effectively with their colleagues in cases of any impending danger
- Should be provided with vehicles so that they can travel easily throughout the national parks and also carry any injured animals to safety.
- Should be provided with camping equipments so that they can take care of any part of the park away from their houses/homes

12. a) i)

Wildlife refers to all untamed plants (flora) and animals (fauna) found in their natural habitat.

ii)

Tourism is the act of traveling and visiting places within ones country for pleasure and recreation

b)

- Game reserves are areas where wildlife is protected but some limited human activity may be allowed while national parks are gazetted regions where settlement, hunting and cutting of plants is not allowed whatsoever.
- Sanctuaries are special set aside to give protection to specific plants or species of animals which are threatened with extinction.

c)

- Forms the basis of the tourist industry
- It has offered employment to many people
- It has also encouraged education and research in the animal species
- Conservation and management of wildlife and environment
- Wildlife has provided game meat for local consumption and export
- Forested areas are sources of rivers which in turn has been harnessed for hydro-electric power projects, irrigation and fishing.

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- Some of the trees have medical value and their products have been used for the manufacture of drugs
- Wildlife further has aesthetic value which enhances scenic beauty of the environment.

13.

a)

- The encroachment of wildlife habitat as the population increases. There is need for more land for settlement and farming
- Poaching wildlife by illegal game hunter has reduced the population of the animals.
- Pollution caused by both industrial and domestic refuse has greatly affected marine life
- There has been inadequate funding to the organizations managing the wildlife department
- The changing climate trends have led to droughts leading to death of wild game in some areas.
- The soil erosion has combined with the changing
- Conditions to the natural habitat of wildlife.

b)

- The establishment of anti-poaching unit has also been established to curb poaching.
- Encouragement of eco-tourism where the communities living within the parks and reserves are involved in the management of the parks and in return a percentage of the proceeds is used in community projects.
- Intensifying of the training offered to the personnel handling wildlife
- The construction of electric fence around game parks and reserves.
- The isolation and treatment of sick animals
- The ban on hunting and trading in game products especially ivory has helped in reducing cases of poaching

CHAPTER 4

TOURISM

1 a) (i) **Climate**

While the climate of East Africa is warm and sunny most of the year, encouraging sun bathing, in Switzerland there are cold winters which enable winters which enable winter sports and hot summers that expose beautiful sceneries.

ii) **Culture**

In east Africa there are varied/diversity of African cultures while Switzerland tourists are attracted to the Swiss hospitality than European languages.

b)

- Development of tourist facilities provide employment opportunities thus reducing unemployment and raising the standards living
- Exchange revenue which is used to develop other sectors
- Tourists provide a ready market for trade items such as handcrafts and other curios
- tourists in hotels and lodges has stimulated the growth of agriculture and other related industries
- The need for improved transport and communication has led the promotion of infrastructure to tourist sites which also benefits the local people
- Establishment of national parks and museums as tourist attractions enabled Kenya to protect/preserve its rich cultural heritage.
- Tourism encourages cultural exchange which promote international understanding.

c)

- Development of infrastructures/roads/airports/communications to all tourist-sites which make them easily accessible
- Improved securities to ensure the safety of the tourists is guaranteed.
- Advertising the country more aggressively in order to make it more attractive/improve the image or the country a broad.
- Establishing a diversity of tourist attractions/emphasis to avoid depend the traditional attractions and reduce competition with others
- Establishing/modernizing tourist facilities in areas such as western Kenya where such facilities are inadequate.

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- Intensify domestic marketing to reduce reliance on foreign to improve/train more tourist personnel to sensitize citizen on the need hospitable.
- 2
- a)
 - Sandy beaches
 - Marine life /wildlife
 - People's culture
 - Coastal land forms e.g caves/cliffs
 - b)
 - They are a tourist attraction
 - for education purposes/research purposes
 - For aesthetic/beauty of land
 - For posterity/for future generation
 - For preservation of culture
- 3.
- a)
 - South Kitui
 - Shimba Hills
 - Buffalo Springs
 - Maasai Mara
 - b) A type of tourism where people visit certain attraction sites within their country for pleasure and recreation.
 - c)
 - Low capacity at hotels
 - Reduction on employment when hotels close down
 - Affected transport industry
 - Less market for agricultural goods e.g pork and eggs
 - Loss of foreign exchange
 - Scaring away potential investors in the sector
- 4.
- a)
 - Improving transportation to tourists' sites to make them accessible
 - Building hotels to increase bed capacity
 - Improving air links with other countries so as to facilitate direct movement of tourists to Kenya

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- Preserving wildlife/eradicating poaching/maintenance of tourist sites
- Improving security in the parks
- Promoting cultural heritage so as to attract more tourists
- Advertising abroad-opening tourist offices abroad
- Providing package tours
- Charging low rates during low tourists
- Advertising abroad-opening tourists offices abroad
- Providing package tours
- Charging low rates during low tourist seasons

b)

- Switzerland has a well developed transport network e.g electrified railways/cable cars
- The country enjoys political stability and neutrality which enables people from different parts of the world to visit Switzerland anytime
- The Swiss are known for their hospitality/tourists feel at home.
- The Swiss speak more than three international languages i.e Italian, German, French and English which enables them to communicate with visitors from all over world
- Switzerland has well developed financial institutions and international bank
- The well developed hotels offer excellent services/all inclusive hotels/package tours attract many tourists because it is cheaper.

5. a) (i)

Ecotourism

Ecotourism is practice of involving tourists and community living around the tourists attractions in order to minimize negative environmental influences and to maximize economic gain.

ii)

Domestic tourism

Domestic tourism is whereby local people visit places of interest within their country for pleasure or recreation

iii)

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International tourism

International tourism involves tours by people to foreign countries

b)

- Both countries are endowed with beautiful sceneries consisting of snow capped mountains.
- In both countries tourists are the main foreign exchange earner.
- Both countries enjoy sight of wildlife of tourist attraction.
- Climate plays a vital tourist attraction in both countries.

6. It is produced and consumed within the country though it is an export item which brings foreign exchange.

7 **Employment** –Directly and indirectly e.g tour guides, drivers, hotel employees, game rangers etc.

Development of infrastructure- Tourism has encouraged the development of roads and air transport

Direct income – This comes from payment made when entering national parks and game reserves

Training facilities have expanded in Kenya to cater for tourism industry.

8. i)

Kenya has no winter season while Switzerland has double season i.e winters and summers which enable tourists to visit the country.

ii)

Kenya has few glaciated mountain while Switzerland has more glaciated mountain scenery and glaciers of alphas

iii)

Kenya has game parks whereas Switzerland has zoos

iv)

Kenyan cultures are more varied than those of Switzerland

v)

Kenya's beaches are used throughout the year where in Switzerland they are seasonal.

9.

- Remoteness due to poor means of transport
- Hostile climate e.g drought which leads to death of wildlife

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- Settlement of people in places meant for game parks and reserves
- Human wildlife conflict where man kills wildlife.
- Insecurity in some areas

10.

- International terrorism that has scared tourists
- Inadequate capital to provide the necessary facilities for tourists
- Decreased tourist attractions especially the flora and fauna due to encroachment into conserved land by human beings
- Hostility from local residents
- Poaching which reduces some animals species
- Poor means of transport and communication

11.

- Kenya has established parks and game reserves to conserve wildlife
- High class hotels have been built along the coast, the major towns and in parks
- Infrastructure facilities such as roads have been improved in most of the parts
- More qualified personnel are being trained for the tourist industry in Utalii
- Tourist promotion offices have been opened in major cities of Western Europe, North America, Japan and many African countries
- Domestic tourism have been encouraged
- Efforts have been made to keep a stable political climate. This has led to an increase in the number of tourists.
- Improvement of cultural activities
- Creation of the ministry of tourism
- Preservation of historical sites
- Overseas advertisements.

12. It's where people visit places of attraction in their own country for leisure and recreation.

13.

- Hot springs/geysers/geothermal
- Birds/flamingoes
- Varied relief features
- Mining sites
- Sports fishing

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- Wild animals
- Vegetation

14.

- Negative attitude towards local tourism limits the number of people who engage in tourism
- Inadequate local campaigns and advertisement of tourist attractions/ special packages lead to low public awareness
- Familiarity with the tourist attractions among the local people makes them fail to appreciate their beauty and value.
- Insecurity from poachers in national parks and game reserves scare people away from visiting them
- The high cost of accommodation in the game lodges discourages local tourism
- The roads leading to tourist sites are poorly maintained. This discourages people from visiting such sites.

15. Gede ruins, Fort Jesus, Lamu museum

16

a)

- Eco-tourism is the practice of involving the community living around the tourist attractions in the management and conservation of the tourist attractions in the management and conservation of the tourist attraction sites and the surrounding environment for economic gain
- Domestic tourism is whereby the local people visit places of interest within their country for pleasure and recreation
- International tourism involves organized tours by people from other countries of world.

b) i) Wildlife

- Beautiful scenery e.g. snow capped mountain
- Excellent beaches
- Warm sunny climate
- Culture and tradition

ii)

The relative peace and political stability enjoyed in the country since independence
International terrorism where Kenya has been a victim which has led to decline in the
number of visitors

Insecurity in the countries parks and major urban areas

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Improved airlines between Kenya and other countries which have contributed to growth of international tourism

The establishment of hotels of international standards has led to increase in beds in hotel rooms.

The economic recession in the world often reduces the number of tourists to Kenya because the spending power of tourists is reduced.

17. a)

- Climatic conditions
- Beautiful mountains
- Improved infrastructure
- Peace and stability
- A highly skilled manpower

b)

Climatic change which have affected the wildlife habitat

Decline in wildlife because of illegal hunting of wild game and harvesting of trees

Erosion of moral values due to the youth copying some of the tourists antisocial behaviour

Incidences of banditry activities which have scared tourist away from the tourist attraction spots

18.

- Both countries are endowed with beautiful scenery consisting of snow capped mountains
- Both countries have magnificent features e.g. the Rift Valley Kenya and glaciated valleys in Switzerland.
- Both countries have their tourism industries backed by the government
- In Kenya the tourists enjoy the sight of animals in national parks while in Switzerland the animals are put in zoos. Kenya enjoying sunny periods throughout the year but Switzerland has sunny periods only in summer and very cold periods in winter.
- Kenya have a varied culture while Switzerland have limited culture.

19. a)1999

Total number of tourists – 8.13 million

1999

$$\frac{1.53}{8.13} \times 360^0 = 67.7^0$$

8.13

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1999

$$\frac{1.77}{8.13} \times 360^\circ = 67.6^\circ$$

8.13

2001

$$\frac{1.65}{8.13} \times 360^\circ = 73.1^\circ$$

8.13

2002

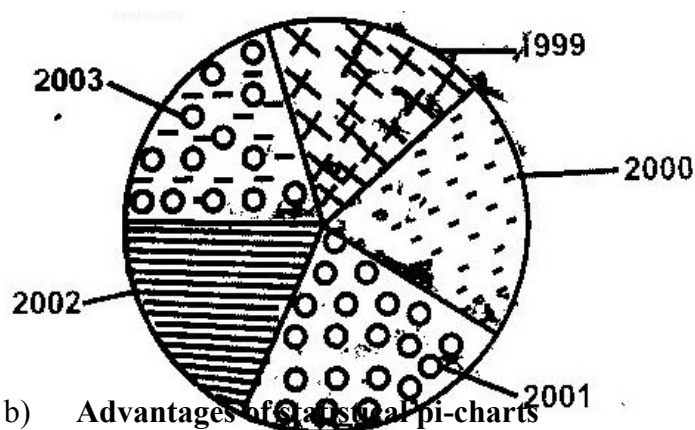
$$\frac{1.77}{8.13} \times 360^\circ = 78.4^\circ$$

8.13

2003

$$\frac{1.54}{8.13} \times 360^\circ = 68.2^\circ$$

8.13



b) Advantages of statistical pi-charts

- Gives good/clear visual impression
- They are easy to construct
- Easier to compare information represented

CHAPTER 5

ENERGY

1. a)
 - Provision of water for domestic use
 - Provision of water for irrigation
 - The dams serve as bridges across the river
 - The dams and the reservoirs are fresh water fisheries
 - The reservoirs have modified the local climate.
- b)
 - Changes in the river regime
 - Silting or reservoirs
 - Poor maintenance of machinery at the power houses
 - Inadequate capital to purchase spare parts.
- c)
 - Limited number of suitable sites
 - Inadequate capital for investment
 - Scarcity of skilled labour.
 - Uranium
 - Coal/peat
2. b)
 - Presence of large volume of water from a river/lake /large catchment area to provide water to drive turbines
 - Regular/constant supply of water to ensure continuous generation of power
 - Hard basement rock to provide a firm foundation for the construction of a dam
 - Presence of rapids/water falls/nick points to provide a massive hydraulic force/head for power generation
 - Presence of a deep valley/a river gorge to reduce the cost of the construction of the dam
 - Non-porous rock to prevent seepage.
3. a)
 - It would encourage setting up of industries in the rural areas thus stimulating
 - decentralization of industries.
 - It would attract /improve social amenities in rural areas reducing the need for

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- people to move to urban centres
- More people would invest in the rural areas which would lead to higher standards of living
- It would encourage development of horticultural farming/to have ideal storage of perishable products.

b)

- It leads to closure of some industries
- It led to unemployment/redundancy/early retirement of workers
- It led to an increase in the cost of production of goods
- It led to an increase in the cost of electricity
- It led to power rationing

4. a)

- High volume of water e.g. River Tana, Nile, Niger, etc.
- Regular flow of water throughout the year
- Several falls and rapids provide good sites
- Presence of hard basement rocks

b)

- Inadequate technological resources
- Lack of adequate capital
- Lack of sufficient skilled manpower

5. a) i)

S – Masinga

T- Kindarum

ii) U- Mutonga

b)

- Wind
- Wood fuel
- Solar
- Geothermal/underground stream

6. a) L- Gas

M-Oil/petroleum

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N- Water

b)

- Wax
- Bitumen/tar
- Sulphur
- Lubricants
- Resin/petro-chemicals

c)

- Proper maintenance and manufacture of fuel efficient vehicles.
- Improvement of public transport system encouraging people to walk, use public means of transport/use bicycles
- Making domestic appliances like refrigeration and cookers more energy efficient
- Switching off electrical gadgets when not in use
- Development of energy saving jikon and other techniques and technologies
- Developing alternative sources of energy other than petroleum.

7.

- It's highly pollutant
- Non-renewable
- Dirty
- Bulky to transport
- Limited in usage

8. **Expansion and establishing more power plants**

- Increasing importation of bulky HEP from Uganda and SA to add to the national grid
- Encouraging and streamlining public transport so as to attract more commuters other than use of private vehicles reduce number of vehicles on the roads.
- Improvement expansion of road network to ensure easy flow of traffic to reduce fuel consumption
- Contacting foreign investors to explore oil in Kenya
- Encouraging use of bicycle by lowering or eliminating tariffs

9. **What are the causes of energy crisis?**

- Increase in oil prices
- Depletion of wood fuel in developing countries

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- Exhausting of coal mines
- Economical and political embargoes
- Over development of oil and its products
- Artificial shortages may be created
- Waste and misuse of energy

10.

- Uses of nuclear energy
- Generation of electricity
- Production of heat
- Making atomic weapons

11.

- Uses of wind energy
- Used to turn propellers and rotors that run machines e.g windmills
- Pumping water
- Grinding grains
- Generation of electricity

12.

- Coal
- Nuclear
- Peat
- Petroleum and natural gas

13.

- Why has coal as a source of fuel/energy declined? Explanation:-
- Has low calorific value. It's dirty compared to other sources of energy/pollution
- It is bulky and cumbersome to transport
- Development of other sources of energy e.g. like oil, solar, nuclear and HEP
- Coal seams are exhaustible
- Inadequate capital
- Inadequate technological know how
- Over-dependency on other sources of energy e.g. HEP and petroleum
- Low demand for power
- Inadequate skilled personnel

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- 14.
- Switching off power source when not in use
 - Use of energy saving means e.g. use of public as opposed to private vehicles
 - Exploiting the renewable sources of energy e.g. solar, HEP
 - Aforestation, reforestation, agro forestry
 - Buy product which consume less energy e.g. do away with fuel guzzling vehicles
15. i) Source of energy derived from organic matter
16. a) Energy is a resource of fuel used to operate machinery
- b)
- Hydro-electric power
 - Geothermal power
 - Wind energy
 - Solar energy
 - Sea energy/wave energy/Tidal
 - Biomass energy
- c)
- Coal is bulky making it difficult and costly to transport
 - It contribute a lot to air pollution through soot and smoke
 - It has low caloric value
 - It causes a lot of environmental degradation during mining by leaving
 - Ugly cars on the land scape
- 17 a) Seven Folks Scheme
- b)
- They have provide water for domestic use and irrigation
 - They have served as a bridge across the river.
 - They are tourist attractions
 - They have acted as fresh water fisheries
 - They have led to micro-climate along the area
- c)
- changes in river regimes
 - Inadequate funds for maintenance
 - Silting of the reservoirs

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- Inadequate skills and technology
- Mismanagement of funds

18 a)

- Turkwell Gorge Dam on River Turkwell.
- Gogo Falls on River Kuja
- Sondu-miriu project on river Miriu

b)

- It has promoted the growth of industries within the region
- It has earned the country revenue through the exportation of electricity
- It has earned the country revenue through the exportation of electricity
- It has acted as a bridge across the lake
- It has acted as a bridge across the lake
- It is a tourist attraction

c)

- Limited number of suitable sites
- Inadequate capital for investment
- Inadequate technology
- Scarcity of skilled labour

19.

- They have enhanced fishing
- Provision of water transport
- Promoted agriculture through irrigation
- Some have enhanced flood control
- Creation of micro-climate

20. It is used in the domestic sector for various purpose e.g. cooking, lighting, heating etc. It is the engine that runs the industrial sector.

21 a)

- Energy crisis is the price and supply uncertainties they are usually accompanies by the rapid depletion of fossil fuels
- b)
- Overdependence on oil and its products

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- Depletion of fossil fuel in the developing countries
- Economic and political embargoes against the leading producers
- Exhaustion and deepening of coal mines
- Artificial shortages caused when some countries decide to conserve their resources e.g. U.S.A
- Wastage and misuse of energy

22.

- The country does not produce crude oil hence relies on importation thus ignoring other sectors of the economy
- Oil imports affects the countries' balance of payments
- When oil prices are high, the cost of manufactured goods and services increases causing inflation in the country
- The Oil producing and Exporting Countries dictate the prices without consulting the consumer countries such as Kenya. This necessitates higher taxation to increase revenue for importing oil
- Frequent shortage of petroleum products leads to destruction of forests.

23.

- a) Management of energy implies the need and careful use of energy resources while conservation of energy means the efficient use of energy to avoid wastage.

b)

Proper maintenance and manufacture of fuel efficient vehicles.

Improvement of the public transport system encouraging people to walk/use bicycles

Making domestic appliances like refrigerators, televisions and cookers more energy efficient and switching off.

Electrical gadgets when not in use

Development of energy saving techniques and technology

Development and use of alternative sources of energy other than petroleum.

CHAPTER 6

INDUSTRY

1. a)

The raw materials may be too bulky and thus expensive to transport

Some raw materials are perishable so they have to be processed before transportation

Processing reduces transport costs

b)

- Give three characteristics of the cottage industry in India
- They rely on simple equipment/machines
- They are labour intensive
- They are owned by families/individuals
- They produce mainly for local markets
- They are widespread in the country/located in homes
- Labour is provided by individuals/members
- They need little capital to start or operate

2 a)

- Are operated by individuals/small groups
- Are pursued as part time or full time occupation. The industry:
- Require low capital investment
- Use simple equipment
- Use local or recycles raw materials
- Use basic/simple skills in craft
- Are operated in the open
- Are widespread

b)

- Difficulty in getting raw materials
- Inadequate capital
- Competition from locally well established industries/competition from imported products
- Adequate marketing skills
- Inadequate security

c)

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Manufacturing is a process of changing raw materials into a finished product/commodity ready for use while tertiary industries provide services and facilities for use by other industries.

d)

- Availability of coal and iron ore as raw materials.
- Cheap water transport on River Rhine.
- Availability of capital.
- Abundant sources of power such as coal and HEP
- Ready market from Central and Western Europe
- Availability of skilled labour
- Industrial interdependence
- Availability of water.

3

a)

- Tobacco processing
- Soap manufacturing
- Footwear making
- Vegetable chemical products making/pyrethrum processing
- Leather tanning
- Rope making
- Textile manufacturing
- Saw milling/pulp and paper making

b)

- large scale importation of second hand clothes has reduced demand for locally produced textile products/second hand clothes are cheaper than the locally produced raw clothes
- There has been a decline in the production cotton which has led to limited supply of raw materials for the textile industry.
- Mismanagement of textile factories has led to closures of such operations.
- Liberalization of the economy has encouraged business people to import textile from other countries instead of selling locally produced ones.
- Belief that imported garments are superior to locally produced ones has reduced demand for local garments.

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- Decline in the economy has discouraged investors who would set up textile industries in Kenya.

c) i)

- Water pollution
- Treating the industrial waste to reduce the negative impact particularly industrial effluents/enforcing environmental laws on the use of improved technology for industry efficiency.
- Recycling wastes in order to reduce the industrial waste turn over

ii)

- Rural –urban migration
- Improve agriculture to create employment in rural areas
- Encourage growth of jua kali industries in rural areas to promote self employment
- Decentralize industries so as to control the number of people moving to industrial centers in search of employment.

d)

- Availability of adequate financial resources which have helped in the setting up and expansion of the industry.
- Presence of large population which provides a large domestic market for electronic goods/availability of large external markets
- Government policy on industrialization has led to rapid development of electronic industries.
- The highly developed sources of power encourage growth of electronic industries
- Advanced technology/research has promoted efficiently methods of production/high quality goods which are competitive in the world market.
- The numerous sea ports ease the importation raw materials and exportation of finished electronic products

4.

a)

It is the process of change from primary to secondary and tertiary production.

b)

- Oil refining-Mombasa
- Paper manufacturing –Webuye
- Motor vehicle assembly –Nairobi, Mombasa

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c)

- It requires less capital to establish since it is made up of small-scale units
- It creates employment for the growing labour force
- It produces mainly for the local market thus the country saves foreign exchange
- It requires less expensive machinery since production is manual
- It facilitates decentralization of industries since it spreads easily/thus checking rural urban migration.
- It produces relatively cheap products that are affordable by many/improving the quality of living
- It uses locally available/recycled raw materials thus reducing the cost of imports/conserves the environment
- It uses locally available /recycled raw materials thus reducing the cost of imports/conserves the environment
- It imitates the products that are already in the market thus spreading technological skills
- It operates at grass roots level thus uses locally available skills
- It empowers the people to initiate projects thus reducing reliance/dependence on the government, donors etc

d)

- Leather tannin
- Tobacco treatment
- Textiles

5.

- Development of industry e.g. Magadi Soda company
- Source of foreign exchange from export of soda ash
- Creation of employment opportunities e.g in mining, industry
- Opening up of previously remote parts of Kajiado through construction of tarmac roads and railway line
- Cheap source of mineral e.g common salt

6.

- Encouraging foreign investors through organizing international investment conference
- Improvement in transport system e.g tarmac roads easing congestion at the port of Mombasa

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- Making electricity supply reliable and affordable.
- Establishment of export processing zones
- Exploiting market for Kenya's industrial goods e.g. under AGOA
- Fighting corruption
- Easing licence issuing procedures.
- Joining economic organizations e.g EAC and COMESA

7. It is the process of change from primary to secondary and tertiary production or the setting up of more industries and expansion of already existing industries.

8

- It creates employment for the growing labour force
- Requires less capital to establish since it's made up of small units
- Produces mainly for the local market thus country
- Local production saves foreign exchange or foreign exchange where exported
- Does not require expensive machinery since production is labour intensive
- Facilitates decentralization of industries hence reducing rural-urban migration
- Produces relatively cheap products that are affordable by many
- Uses locally available raw materials e.g. scrap metal, plastics hence conserve the environment
- Uses locally available skills thus encourages people to initiate projects and reduces reliance on the government and foreign procedures
- Raises the peoples living standards

9 a)

An industry is an establishment set up to process and transform simple and ordinary raw materials to a more complex materials of great value. Industrialization is the process and the speed at which a country plans and establish a country plans and establishes industries.

b) i)

- Its availability
- Its in exhaustibility
- Its bulkiness

ii)

- Industries dealing with perishable goods are close to the market
- Industries whose products are bulky hence increasing transport costs, are located near the market.

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- c) Modern fuel and power can be transported for long distance for example, electricity can be transferred by high tension wires, petroleum by oil tankers and through pipelines. Natural gas is also transportable by road, air and water.
- 10 a) i)
Primary industries; these are the simplest forms of industries which first extract raw materials from their sources. They are also referred to as extractive industries
- ii)
Secondary industries: These are industries which depend on the primary industries for raw material
- iii)
 - It makes use of locally available materials
 - It produces cheap consumer goods which substitutes the expensive imported goods
 - They are found in nearly every corner of the country thus improving the economy of the rural areas
 - They also assist in solving unemployment problems
 - It has stimulated development where they have been established
 - It has encouraged the utilization of local resources
 - It has enhanced the production of agricultural products
 - It has led to the growth and development of urban centers
11. a) Significance of industrialization to Kenya.
- i) Provision of employment. People are employed in different categories of industry e.g transport
- ii) Reduces reliance on imported goods thereby saving foreign exchange
- iii) Development of settlements e.g. Magadi town
- iv) Improvement of infrastructure e.g road and railway network connecting Magadi and the Mombasa port
- v) Earning of foreign exchange which is used to develop other sectors of economy.
- vi) Promotion of relationship between Kenya and other countries through trade relations involving industrial goods.
- vii) Helps in promoting rural areas where industries are established.
12. a) i) Thick -Fruit canning

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- Motor vehicle
- ii) Athi river
 - Cement making
 - Meat canning
- iii) Kisumu
 - Fish processing
 - Textile manufacture
- iv) nanyuki
 - Textile manufacture
 - Garment making
- v) Nakuru
 - Soft drink making
 - Floor milling

b)

- It has resulted in pollution leading to environmental deterioration
- It has led to rural-urban migration depriving the areas of able bodied people
- It has led to uneven economic development creating two district regions; the developed rural areas
- Other economic activities
- In some instances it has led to loss of life
- Due to technological advancement, some people have been rendered jobless.

13. - Availability of raw materials i.e iron ore and limestone.
- Availability of coal which provides power to iron and steel industries (used for smelting iron ore)
- Accessibility-The area has direct links with the rest of Europe thorough air, road and water. This allows transportation of raw materials and finished goods
- Availability of capital which is used to invest in industries
- Availability of ready market within the region and other parts of the world.

14. a)

- i) Tokyo - Yokohama industries zone
- Osaka Kobe- Industries zone
 - Okazaki- Yokaichi industries zone
 - Yawata- Kokura-Moji industries zone

ii)

- Availability of capital
- Availability of market

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- Production of simple and unsophisticated cars and electronic gadgets
- Development of hydroelectric power
- Availability of abundant water
- Presence of skilled manpower
- Advancement in technology
- Japan's geographical location.
- Inadequate agricultural land.

b)

- Availability of labour which is skilled
- High demand for products from cottage industries due to high population
- Need to earn living to uplift their living standards which motivated Indians to establish cottage industries.
- Availability of localized raw materials
- Availability of simple machines which are affordable

CHAPTER 7

SETTLEMENT

1. a)(i) X- Industrial zone/lower class housing
Y- Suburb area/high class residential area
ii) -Trading
-Administration
-Commerce/banking/insurance
-Location or light industries
2. a) -Rural
-Urban
b) Q- Nucleated
R- Linear
3. a) -Nucleated
-Scattered/ dispersed
-Linear

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- b) -Urban-urban
-Rural-rural
-Rural-urban
- c) -Retirement/from formal employment in urban areas
-Lack of jobs in urban centres
-Insecurity in urban centres
-The strategy of District FREE for rural development
-Provision of infrastructure facilities in the rural areas.
4. -It has a deep-sheltered harbour/natural/harbour
-It is located at a strategic point on the East African Coast/enter port/gateway.
-It is well linked to the interior by railway, road and air
-Early settlement/Early trade
5. a) It is the process whereby an increasing proportion of the total population in a country settles/concentrates in town/the process which towns or cities grow.

b)

New York	Nairobi
-It is seaport	-It is an inland port
-It is a state capital	-It is a national capital
-It is an international commercial/financial centre.	-It is a national commercial centre

- 6.
- Insecurity due to high rate of criminal activities
 - Unemployment resulting from too many job seekers competing for limited job opportunities.
 - Inadequate housing facilities caused by inadequate resources for putting up housing estates.
 - Limited land for expansion caused by rapid population growth in urban centers
 - Traffic congestion due to increase and concentration of activities in the C.D.
 - Inadequate social amenities e.g schools, hospitals to cater for the rapid population growth in urban centers
- 7.
- Nairobi is situated on low lying plane interior of continent while New York is sited on island along the downwards mouth of River Hudson

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- Expansion of Nairobi is horizontal extending into adjacent rural areas. New York is extending vertically due to limited horizontal space.
 - New York is characterized by many high building called skyscrapers while in Nairobi skyscrapers are few.
8. Two main functions of rural settlements
- a) Production of agricultural goods
 - b) Mining activities
 - c) Forestry/lumbering
 - d) Fishing
- 9.
- a) Large unemployed population which is idle and encourages Criminal/immoral practice e.g. parking boys and girls, baggers
 - b) Rapid growth of population which has led to an acute shortage of housing/slums
 - c) Traffic congestion during rush hours leading to loss of time
 - d) Heaps of uncollected garbage which are health hazard as they can lead to Epidemic/land pollution
 - e) Perennial water shortages due to increased number of consumers
 - f) poor sewage system in some parts which is a health hazard for poor Drainage system
 - g) Rapid growth of population has led to inadequate provision of health and Education services and facilities/social facilities.
 - h) Pollution of air/water/sound from industries and vehicles causing a health Hazard.
- 10.
- a) sheltered harbor making it an international sea port
 - b) Central location making it local point of routes/strategic sites
 - c) Large hinterland rich in agriculture and minerals.
 - d) Development of many industries e.g tourism, oil, refinery, etc has attracted many people due to employment
 - e) Development of commercial activities attracting many people.
 - f) Administrative centre attracting many people
 - g) Construction of the railway and other lines of transport to interior Contributing to rapid growth of the town.

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11. a) It has deployed traffic police to control traffic/installation of traffic lights/construction of fly over construction of tunnels to control traffic congestion.
- b) Its encouraging self –employment like jua kali sectors to solve Unemployment problems in Nairobi.
- c) encouraging use of easily available raw materials for building low cost house/funding high rise on multi-stored flats/expanding urban area boundaries.
- d) Location of industries away from residential areas establishing proper waste/garbage collection system to reduce pollution in the city.
- e) Increased police surveillance to reduce high crime rate in the city/strengthening the security intelligence to reduce high crime rate in the city/strengthening the security intelligence unit.
- f) The government has encouraged independent bodies to come up with different ways to sensitize the public on ways to take care of cleanliness, drug abuse, vandalizing equipment and public policing.
12. a) Settlements is a place on the earth surface occupied by people.
Urbanization is the process in which an individual or proportion of a country's population is transformed from rural based agricultural life to urban based non-agricultural life style.
- b) Availability of water
- Natural calamities
 - Presence of building materials
 - Relief
 - Defence
 - Transport
 - Shelter
13. a) Kenya
- Major urban centres are found in the Rift Valley and associated highland
 - Others are in the lake basin and the coastal region
 - They include: Kampala, Mbarara etc.
- Uganda
- Major urban centres are found in the southern part of the country around Lake Victoria

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- They include: Kampala, Mbarara etc

Tanzania

- Urban centres are found along the coastal belt e.g Tanga and Mtwara.
- Around Lake Victoria e.g Mwanza and Musoma
- Others are found in mineral rich areas of Mwandui and Shinyanga.

b)

- Population increase put stress on existing infrastructural facilities forcing urban centres to expand their territories
- Industrial development attracts large population who seek for employment hence leading to emergence of industrial towns.
- Administration functions leads people to settle close to administrative services. Such places grow into urban centres.
- Mining activities attracts subsidiary economic activities in an area. These attract large settlement leading to growth of urban centres.
- Tourism attracts infrastructure, other investments and large population. This leads to the expansion of urban boundaries hence urban growth.

14. a)

- Railway terminus reaching the shores of lake Victoria made it handle a lot of exports and imports from Uganda leading to its growth
- The early Asian settlers who came during the construction of railway line to Uganda established commercial activities in Kisumu which further contributed to its growth.
- Administration function i.e for long time, Kisumu served as provincial headquarters of Nyanza, Western and Kericho districts. It was also used by early railway engineer.
- High population from the densely populated districts of Vihiga, Kakamega, Kisii and Siaya ensures steady supply of labour for industries and the provision of market to the manufactured goods.
- Road linkages i.e it is served with road linkages to the surrounding districts of Kakamega, Kericho, Busia and Kisii which has led to increased trade hence leading to its growth
- Development of pier which is strategically located at the shores of Lake Victoria has led to trade between Tanzania and Uganda. This has provided a trading advantage.
- Rich hinterland i.e the town is surrounded by agricultural hinterland of Vihiga, Kakamega, Kisii and Kericho which provide food for residents and raw materials to industries.

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b)

- Industrial centre
- Administrative centre
- Tourist centre
- Education centre
- Communication centre
- Commercial centre

15.

- Housing problems due to ever-increasing population in the city that has led to development of slums like Harlen,
- Brooklyn and Bronx
- Limited space for expansion because of its site has led to emergence of skyscrapers
- Traffic congestion during rush hours leads to waste of time and fuel
- Social evils such as crime rates lawlessness and prostitution due to lack of employment
- Pollution due to heavy concentration of industries, motor vehicles and domestic refuse leading to air, water and noise pollution
- Loneliness due strenuous life and joblessness leading to homicidal tendencies
- Climatic problems in which heat concentration makes the city residents feel uncomfortable. At times during winter, blizzards are common.

16. a)

- Nairobi is sited on a low lying plain while New York is sited on an island of a
- long the drowned mouth of River Hudson
- Nairobi expansion is horizontal emerging into adjacent rural areas. New York on the other hand is expanding vertically into the air due to limited space.
- Mode of transport within Nairobi city is motor vehicles and in some cases railways while in New York a part from motor vehicles and in some cases railways while in New York a part from motor vehicles and railways, rivers and canals are being used.
- New York is characterized by very high buildings known as skyscrapers while in Nairobi skyscrapers are very few.
- The climate in Nairobi is warm and sunny throughout the year, New York city experiences summers and winters

b)

- Port of Mombasa serves as a developing and agricultural hinterland whereas Rotterdam serves developing and highly industrialized hinterland
- Exports through the port of Mombasa are mainly agricultural products and minerals, while exports through the port of Rotterdam are manufactured goods and chemicals
- Mombasa's imports are mainly manufactured goods while Rotterdam's imports are agricultural commodities
- Both ports are ice free throughout the year

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17. Positive effects

- Urban centres lead to the development of infrastructure to facilitate movement of
- People and goods.
- Creation as it leads to development of industries
- Provision of market as population in urban areas leads to high demand of goods
- Leads to national integration as it encourages interaction of people.

Negative effects

- Inadequate housing due to uncontrolled population influx
- Inadequate public amenities as the existing ones are overstrained
- Unemployment as the number of job seekers exceeds employment opportunities.

CHAPTER 8

TRANSPORT AND COMMUNICATION

1.

- The high cost of postage
- Competition from cheaper and faster means of communication
- The delay in the delivery of letters
- Loss of letters
- Tampering with letters

2. a) Give three advantages of railway over road transport

- Railways can carry more bulky goods over long distances at once.
- Railways are cheaper than road
- Railways are less susceptible to traffic jams.
- Once built, railways do not require frequent relaying unlike roads, which are frequent relaying unlike roads, which are frequently resurfaced.
- Vehicles get stuck on dry weather roads during wet seasons while trains cannot get stuck.
- Railways are more efficient because they operate on a rigid timetable

b) X- Nakuru

Y- Eldoret

c)

- They are expensive to maintain
- They are not flexible

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- They do not serve intermediate locations
- They can cause excessive loss of in case leakages.

d) i) A- Chacago B- Detroit

ii) X- L. Ontanrio

Y- Ere

Z- Superior

iii) Welland canal

3. a)

P – Jinja

Q – Bukoba

R – Mwanza

b)

- Containers guard against destruction of goods
- They make loading and off-loading easy
- They guard against theft of goods
- It saves time when loading and off-loading

4. a)

- Motor vehicles are cheaper to buy and maintain than aircrafts
- Road transport is more flexible than air transport
- Construction of roads is cheaper than that of airports
- Fares/freight charges on roads are lower than that of air transport
- Skills required to operate air crafts are higher and rare than those required to
- Operate motor vehicles.

b)

- It encourages the growth of tourist industry
- It promotes horticultural exchange between Kenya and other countries.
- It encourages international cooperation/facilitates international emergency services
- Earns foreign exchange from landing fees

5. a)

P- Kasese

Q- Butere

R- Kigoma

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b) S- Maize /Wheat/ cattle

T-Soda ash

c) U- Tanga

V-Malawi/Nyasa

d)

- It is cheaper to construct and maintain
- Roads are flexible and provide door to door services
- Roads can be used by a wide range of transport
- The roads are faster to use
- There is greater demand for road transport than railway transport
- The roads are faster to use
- There is greater demand for road transport than railway transport
- Roads can be constructed at varied terrain

e)

- **Narrow roads where heavy traffic limit use of movement and overtaking**
- The pot-holed sections of the roads may cause tyre busts/vehicle break downs/may make drivers who are avoiding potholes crash the vehicles.
- The narrow bridges may cause vehicles to rear off the roads.
- The narrow bridges may cause vehicles to crash
- Sub-standard surfaces may cause vehicles to skid and overturn
- Blurred/missing road signs may make drivers lose control of vehicles to skid and overturn.
- Blurred/missing road signs may make drivers lose control of vehicles
- Unavailability of pedestrian paths/sidewalks may cause pedestrians to walk on the roads.

6.

- Reduction in road accidents through use of speed governors
- Source of government revenue through payment of income tax
- Improves comfort in public transport
- Increased profitability due to reduced insurance premiums

7.

- Dense forest covering large areas

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- Political instability and political differences between some African Countries
- Presence of large and many rivers that make road construction difficult and expensive to bridge.
- Insufficient capital by individual governments to construct and maintain parts of the highway in their countries.
- some flow in the areas lack substantial resources
- Most rivers are narrow and shallow
- Some rivers fluctuate in the volume of water
- Some have waterfalls, rapids and cataracts
- Presence of floating vegetation along some rivers

8.

- Ease congestion within Nairobi
- Promote both local and international trade
- Ease transportation of bulky goods
- Opening of rural urban areas around Nairobi

9.

a) **Define containerization**

Means packing of a commodity in large box like structures which are once filled in a factory can be transported by railway or lorry to the dock and into and off the ship by cranes.

b)

- Speeds up the process of handling goods
- Saves spaces in a ship as their dimensions are uniform
- Minimizes loss of goods through theft
- Relatively cheap as it requires little labour-checking the content on entity is quite easy.

10.

a) **State three problems facing rail transport in Kenya**

- Poor maintenance of rail transport in Kenya
- Poor maintenance of rails and wagons leading to accidents
- -Inadequate capital to purchase new Wagons.
- -does not offer door to door services –not flexible
- Faces stiff competition from other forms of transport
- Its initial objective was to serve foreign markets and not local markets hence most sections passes through low density areas

b) **Identify the advantages of rail transport in a country**

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- Enables transport of goods over long distances at cheap rates
- There 's economy of spaces as many trains can be scheduled to use the same rail
- -they have minimal pollution
- Its effective in moving a large number of people and goods within a short time
- Has fewer accidents
- Requires less lab our because many wagons are pulled by one driver
- -Convenient to clients because they follow a time schedule-can plan his movement.

11 Differentiate between transport and communication.

Transport is the act of moving, carrying and conveying items and people from one place to another while communication is the transmission of words and messages from one part to another

12. Apart from cell phone state two other modern means of communication

- Telephone
- Telex
- Telegram
- Email/Internet

13. State the advantages of using cell phone in communication.

- Are portable
- Are quick/fast
- There is immediate feedback
- Messages can be sent over long distances within a short time.
- Message can be stored

14.

- To remove rock shoals, rapids and several small islands in river channel which hampered navigation
- To deepen the river channel which hampered navigation
- To deepen the river and regulate flow of the river
- To construct locks along the route to regulate flow of water as well as movement of vessels
- To construct locks along the route to regulate flow of water as well as movement of vessels
- To generate H.E.P.

15. a)

- i) Transport is the carriage of goods and people from one place to another

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- ii) Communication is the transmission of information from one point to another
- b)
- Air transport
 - Road transport
 - Railway transport
 - Pipeline transport
- c)
- Inland waterways
 - Oceanic waterways
16. a)
- Presence of floating vegetation makes it difficult for vessels to sail on rivers
 - Silting at river mouths hinder the development of ports and makes the river
 - Channel shallow.
 - Some of the rivers are too swift in their courses.
- b)
- The north Atlantic sea route
 - The Mediterranean Asiatic sea route
 - The European-Eastern South America sea route
 - The panama Canal sea route
 - The Cape of Good Hope, sea route
 - The North pacific sea route
- c)
- It offers the cheapest form of transport for large bulky goods
 - It is route free/uses the natural routes e.g rivers, seas ect
 - Accidents are minimal
 - Water transport experiences no traffic congestion
 - It is able to accommodate vessels of any dimension
17. a)

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- The African countries were administered by different colonial governments who constructed rail lines only within the areas of their jurisdiction
- Many African countries have political differences which lead to mistrust and hostility thus working against any effort to undertake railway construction jointly.
- African countries have railways of different gauges which makes it difficult for them to be connected.
- The little interstate trade among African countries doesn't warrant construction of railways to transport bulky goods
- African countries lack sufficient capital to establish railways

b)

- Trans Cameroon railway
- Tanzania – Zambia railway
- Trans DRC railway
- Kenya Uganda railway

c) **Advantages**

- It is cheap when transporting bulky goods
- It is all-weather e.g. it is not affected by weather changes
- The train follows a regular timetable hence helps people to plan their movement

d) **Disadvantages**

- It is a slow mode of transport
- It is not flexible
- It is not economical on short distance

18. a)

- Motor vehicles are cheaper to buy and maintain
- Fares and freight charges on roads are lower
- Skills required to operate motor vehicles are readily available
- Construction of roads is cheaper
- Road transport is more flexible
- It is relatively cheaper over short distance

19.

This is the packing of goods in large standard sized box-like structures which are sealed at the factory or by the exporter and transported by road, railway/water to the importer while sealed.

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(a) Advantage

- It reduces the loss of goods through theft
- It lowers the insurance premiums due to reduced risks
- It speeds the process of handling goods
- Breakages of goods is minimized

(b) Disadvantages

- It is expensive to adopt as it requires special docks, ships etc
- It may lead to redundancy of labour at the port as machines require human labour.

(c) Advantages

- Provide the fastest means of transport
- Relatively independent of physical barriers
- Accidents are limited

(d) Disadvantages

- Highly vulnerable to poor weather.
- Accidents though rare are fatal.
- Airports are expensive to construct.

20. -Telegraph -Fax
 -Telephone -Internet
 -Television -E-mail
 -Radio

21. a) -The Great North road from Cape town (South Africa) to Cairo (Egypt).
 -The Trans-African highway from Mombasa (Kenya) to Lagos (Nigeria)

b)

- They have provided employment in the transport sector
- They have provided employment in the transport sector
- They have enhanced international trade
- They are source of government revenue through toll station revenue
- The highways have contributed to urbanization
- They have opened up remote areas for development

c)

- Low level of inter-state trade among the African countries.

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- Hostile environment in some sections through which the highways pass e.g deserts, thick/dense vegetation
- Rugged landscape caused by mountainous and hilly terrain.
- Political differences amongst some of the countries through which the highways pass.

22.

- It promotes industrial development by facilitating the movement of raw materials to the manufacturing centres and finished goods to the markets.
- It has promoted both local and international trade
- Promotion of tourism by opening up potential sites
- It has promoted both local and international trade
- Promotion of tourism by opening up potential sites
- It has opened up remote areas for exploitation e.g. through mining agriculture etc
- It has enhanced interaction among people leading to exchange of technology.

23. a)

- Inadequate manpower of the development of new networks.
- Insufficient funds for the development of new infrastructure
- Civil wars leading to destruction of existing network
- Most of the continents rivers are not navigable
- Colonial heritage which led to the development of externally oriented transport and communication facilities

b)

- Construction of trans-African highways
- Construction of international railways
- Establishment of regional economic blocs to improve transport and communication lines
- Seeking donors to help finance the construction of infrastructure

24. a)

- Frequent fog and mist at the mouth of the St. Lawrence
- Different water levels
- Rocky islands within the river channel
- Shallow and narrow sections of the water ways
- Freezing during the winter season

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- Presence of rapids and winter season.
 - Presence of rapids and waterfalls
- b)
- Construction of canals
 - Dredging of river channels
 - Blasting to remove the rocky islands
 - Use of fog lights and radars to avoid accidents due to fog.
- c)
- Development of urban centres and ports e.g. Toronto, Port Arthur, Buffalo and Detroit
 - It has led to the production of hydroelectric power for both domestic and industrial development by offering cheap transport for raw materials and finished goods
 - It has led to an increase in the volume of trade within the region especially due to proximity to Europe.

CHAPTER 9

TRADE

1. a)
 - To encourage member countries to reduce duties charged on goods among the countries from COMESA member states.
 - To promote trade among member states.
 - To acquire greater economic strength/higher bargaining power with trading blocs of the world.
 - To establish a larger market for the goods produced in the region.
 - To remove trade barriers among member states.
 - To create regional specialization in order to improve the quality of goods.
 - To create political cooperation among s member states.
 - To create monetary and financial cooperation among member states
2. a) International trade is the exchange of goods and services between different countries.
- b)
 - Machinery
 - Capital equipment
 - Textiles
 - Pharmaceutical products
 - Fertilizers
 - Automobiles
- c)
 - Governments policy/government legislation
 - demand for goods both locally and outside Kenya
 - Variation of natural resources/goods/quality of goods
 - Availability of transport and communication
 - the purchasing power
 - the level of industrialization
 - Tariffs imposed on Kenya exports
- d)
 - There will be improved access to raw materials for industrial development
 - The expanded market will attract new investments from local and foreign sources which will lead to expansion of industries/more earnings.

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- There will be improved negotiating powers in the international arena
- There will be improved transport links between Kenya, Uganda and Tanzania which will facilitate faster movement of goods and people.
- There will be mutual political understanding between Kenya and its neighbors.

e)

- Overspecialization/overdependence on a particular items is risky in case of a fall in the prices in the world market.
- Imported items may become a threat to the local industries leading to closure of some of items
- Some imported goods e.g expired goods e.g. expired goods sub-standard goods may have adverse effects on the citizens/economy
- If a country depends on another, it may sometimes have to tolerate some undesirable gestures from such countries.
- There may be over exploitation of natural resources leading to their depletion e.g. minerals
- Over-emphasis or export oriented product at the expense of other sectors of the economy.

3. a)

- Encouraging the development of jua kali industries which do not require importation of heavy machinery.
- Restricting the importation of luxury items through taxation.
- Establishing/importation of luxury items through taxation.
- Establishing/import substitution industries to cut down on importation of some commodities.
- Developing alternative sources of energy in order to reduce importation of some commodities
- Encouraging the production of high quality manufactured goods for exports in order to earn higher income
- Diversifying the agricultural export base to enable the country to have a variety of exports
- Opening new markets to avoid dependence on the trading partners.

b)

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- The imported industrial inputs led to growth of manufacturing industries in the country.
 - Demand for Kenya's exports has led to the expansion of the industries that produce those goods
 - Kenya earns foreign exchange which enables it to import goods from other countries
 - Transport and communication network in Kenya has been improved to facilitate the movement of trade goods/modernization of the facilities for handling goods at the port of Mombasa.
 - Taxation of commodities and services rendered has generated revenue for the country.
 - Employment opportunities have been created in the manufacturing and service industries that handle imports and exports
 - Trade has enhanced exchange of technology between Kenya and the trading partners.
 - Trade encourages specialization which leads to production of high quality goods in some industries in Kenya, thus enabling the country to earn higher income.
4. a) Trade is the exchange of goods and services for mutual benefit
- b) - Wholesale trade
- Retail trade
5. - The availability of capital for investment
- Presence of well developed transport and communication system
- Demand for the trade items and the presence of an adequate source of supply
- The absence or presence of trade barriers
- The creation of common markets
6. a) Visible exports are tangible goods sent out of the country for sale while invisible exports are the payments received for services rendered outside the country.

Visible exports	Visible imports
Coffee	Cereals
Tea	Crude oil
Soda	Pharmaceuticals
Fluorspar	Industrial machinery
Horticulture products	Iron and Steel
Petroleum products	Motor vehicles

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7.

- Through trade, the country is able to obtain goods which it cannot produce
- Trade enables Kenya to dispose surplus production
- International trade enabled Kenya to earn foreign exchange
- Through trade, the country has been able to generate many job opportunities
- Trade offers market for industrial goods and also enables industrial goods and also enables industries to obtain raw materials and other inputs. This has promoted industrial development in Kenya
- Trading centres have attracted dense settlements. This has turn promoted urbanization.

8.

- With increased cordial relationship with her neighbors, trade will continue to prosper.
- Through COMESA, trade with the rest of the region will further improve
- The country's balance of trade will continue being unfavorable unless the quality of the exports is improved on and import substitution industries set up
- With the signing of multi-lateral and bi-lateral agreements, and through the Lome convention, the country will gain access to the lucrative markets in Western Europe.

9. a) A group of countries usually sharing a common history and within the same geographical region who come together for economic benefits

b)

- Common market for Eastern and Southern Africa (COMESA).
- The southern Africa Development Community (SADC)
- The Economic Community through the removal of customs among member state.
- It has promoted the free movement of persons, services and currencies hence increasing the volume of trade.
- Has established financial institutions e.g. the European Investment Bank
- This has financed economic projects.

10.

- The neighboring countries produce similar agricultural goods to Kenya thus reducing trade.
- Trade tariffs have discouraged free trade with her neighbors
- In the past there has been restriction of movement which has limited the degree of movement of people and goods within the region.

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- The development countries offer manufactured goods and industrial machinery hence they are better trading partners.

CHAPTER 10

POPULATION

1. a)
 - The area receives low unreliable rainfall.
 - Most parts have thin undeveloped soils/sandy soils unsuitable for agriculture
 - -the area has scanty vegetation that cannot support livestock
 - There is inadequate supply of surface water
 - Some areas have rugged terrain unsuitable for settlement
 - -some areas have ragged terrain unsuitable for settlement
 - -Some areas are insecure and therefore avoided
 - The area experiences high temperatures unsuitable for settlement
- b)
 - National census/head count
 - Sample survey/sample count
 - Vital statistics/registration of birth/death/Marriages
- c)
 - The size of the population
 - The different age cohorts (groups)
 - The proportion of males to females
 - The composition by sex
 - The proportion of the youthful/working/ageing/dependency ratio
 - The population of Kenya has large number of young people below 20 years of age while Sweden has an ageing population
 - Kenya's population has a lower life expectancy while Sweden has a high life expectancy
 - The population birth rate in Kenya and low in Sweden it is low
 - The death rate is high in Kenyan and low in Sweden
 - The population growth rate is high in Kenya and low/negative in Sweden.
 - A high percentage of the population in Kenya lives in the rural areas while in Sweden most people live in urban centres.
- e)
 - Pressure on land makes people to buy/look for land elsewhere and move to settle there.

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- People move from one rural area to another/to plantations in search of employment
- Insecurity in some rural areas make people move to safer places
- Settling up of government development projects cause displacement of people who are settled elsewhere such projects attract settlements in those areas (settlement schemes).
- Natural hazards force people to migrate to other areas for safety
- Pastoral communities migrate from one rural area to another in search of pasture/water for their livestock.
- Changes in land tenure system/land dispute cause people to move and settle elsewhere.

2.

- The rate of population growth is higher than the rate at which job opportunities are generated leading to high unemployment
- The high demand for social amenities caused by the high population growth rate leads to congestion in schools/hospitals/housing/transport facilities
- The large number of youthful population creates a high depending ratio which causes slow economic growth.
- The high demand for agricultural land cause land fragmentation/landlessness
- The large number of poor people/leads to temptation to commit crime/high crime rate.

3.

- Improve diet
- Improved medical facilities
- Improved education for mothers/patients

4.

- Nearness to Nairobi – many people live in Thika town and work in Nairobi because house rents are lower. This increases the population of the District.
- Thika town is an industrial centre and attracts large population of workers unlike Murang'a where there are few industries that are rural based.-Thika town is a large commercial centre compares to Murang'a town.
- Thika attracts many people who operate different businesses. This increases the population of the district.

5 a)

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- For planning purposes
 - To help in the distribution of resources
 - To make estimates of population growth
 - To identify the rates of deaths and births
- b)
- i) Early marriage
- People who marry early are likely to get more children because they have a long period during which they can get children
- ii)
- Improved medical facilities
 - Both the child, mothers and the general population have better chances of survival because of the available medical facilities/
 - The country is able to control the spread of diseases and has ability to cure diseases. This leads to higher survival rates.
- iii) Culture beliefs
- Some cultures encourage large families. In almost all cultures, there is a tendency of people preferring male children. This may lead those who are not getting male children to have a large family as they hope to get a boy.
6. a)
- The number of male and female population is almost equal at all levels
 - From 0-14 years, the population is low
 - From 15-44, the population is high
 - The ageing population is low
 - The population has a low birth rate
 - The population has a low birth rate
 - The population has low death rate
- b)
- There is likely to be a high unemployment rate because job opportunities do not increase at a rate that can cope with the increasing number of job seekers.

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- The standard of living is likely to be low because the government is not able to promote adequate social amenities.
- It may lead to a high dependency ratio which will slow down the economic growth
- Strain on natural resources/scarcity of land which would lead to landlessness and fragmentation.
- There would be low food production hence food shortages.

c)

- Improving medical facilities such as immunization of children to control diseases. This has created a healthy environment for child survival.
- Providing more education opportunities for parents ensures better care for their children e.g. in providing balanced diet
- Introduction of family planning programmes has led to emergence of manageable sizes of families which promote higher chances of child survival
- Carrying out research on infant related diseases to come up with ways of controlling them ensures higher chances of survival.

d)

- Presence of large towns with industries has attracted large of numbers of job seekers
- High rainfall which influences production of a wide variety of crops hence sufficient food.
- Fertile land which attracts settlements
- High fertility rate leads to a high natural increase
- The fairly level land encourages agriculture and settlement
- Increased commercial activities/trade attract a large number of populations
- Early settlement in the reason encouraged growth of towns which formed a FREE migration
- Developed communication has enhanced movement in the area.

7. a) Population is the sum total number of people living in a given area either a home, village, town, rural area or even the whole world.

b) i) Physical factors:

- **Relief:** Areas with rugged relief, valleys and steep slopes are avoided while gentler slopes are densely populated.

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- **Climate:** High rainfall and cool temperatures encourages high population while, low and unreliable rainfall with high temperatures leads to low population.
- **Soils:** Fertile soils, which are well drained encourages population concentration, while poorly drained soils which are less fertile are sparsely populated.
- **Drainage:** Poorly drained areas discourages population while well drained areas encourages high population
- **Vegetation:** Densely vegetated areas, woodlands and savannas are sparsely populated.

ii) **Human factors**

- Economic activities: areas of industrial concentration, mining centres, fishing ports are all responsible for high population concentration
- Security: Areas where security is poor discourages population while areas with good security attracts population.
- Government policy: Creation of land for settlement encourages population while settlement of forests and game reserves discourages population

8. a)

- High fertility encourages high births. This encourages high population growth where fertility is low; there are low hence low population growth.
- Increase in mortality rates leads to low population growth rate while mortality decrease leads to high population growth.
- Immigration results into population increase in the receiving area while emigration results into population decrease in the area of origin

b)

- High birth rates
- Low life expectancy
- High dependency ratio
- High death rates

9.

- Cultural factors e.g. polygamy and early age. Improved diet resulting to better nutrition leads to healthy people who become parents at an early age
- Religious influences whereby certain churches advocate for natural family control method which are less effective.
- Disapproval of artificial family planning practices leads to high fertility levels.

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10.

Kenya	Sweden
a) Higher population	Lower population
b) Population is predominantly youthful	Population is predominantly ageing
c) Lower life expectancy	Higher life expectancy
d) Population growth is higher	Population growth rate is lower
e) majority of population is rural based	Majority of population is urban based
f) High birth rate	Low birth rate
g) Dependency ratio is high	Dependency ratio is low

11. Population density in Kenya

Region	Density
Nairobi	3079 p/km ²
Central	281.7 p/km ²
Rift Valley	38.3 p/km ²
Western	406.4 p/km ²
Nyanza	350 p/km ²
Eastern	30 p/km ²
North Eastern	7.5 p/km ²

12.

- Suitable climate: Central Province experiences high rainfall totals per annum. This encourages various agricultural activities hence attracts high population densities.
- Fertile soil: The red volcanic soils which favours variety of agricultural activities have attracted a high population.
- Industries: the establishment of manufacturing industries in the towns of Central Province like Thika have attracted many people who seek for employment.

CHAPTER 11

MANAGEMENT AND CONSERVATION OF THE ENVIRONMENT

- 1 a)
 - To ensure that there is supply for present and future generations
 - To maintain the hydrological balance
 - To maintain the eco-systemb) By reducing surface run-off which ensures that rain water seeps slowly into the ground.
2. a)
 - Lighting
 - Strong winds
 - Dust storms
 - Pest/diseases
 - Floods
 - Pollution
 - Soil erosion
 - Fires
 - Land slidesb)
 - Population pressure/clearing of forests for farming/settlement
 - Climatic changes
 - Accidental fires
 - Poor methods of farming/overgrazing/Overstocking/poor irrigation methods
3. a)
 - Central highlands
 - The Nyika plateau
 - Coastal lowlands
 - Nyando
 - Nzoia
 - Kuja/Guchac)
 - The stagnant water becomes breeding ground for vectors that cause water related diseases.

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- Floods cause loss of property/lives
 - Floods cause soil water logging which lowers crop production
 - Floods wash away crops leading to food shortages/famine
 - Floods wash away bridges/roads/telephone lines/air fields disrupting transport and communication
 - People are displaced by floods/are made homeless
4. a) The presence in the environment of contaminants which are injurious to human, land, plant animals life.
- b)
- The garbage may result to foul smell/air pollution
 - When it rains, the dumped waste/garbage is washed to rivers causing water pollution.
 - Garbage can be a breeding ground for rodents/flies/cockroaches which can cause disease outbreak e.g plague.
 - Accumulation of garbage leads to blockage of roads/drainage systems
 - Garbage heaps are an eye sore as they make the environment ugly.
- c)
- Burning waste materials
 - Digging pits for throwing rubbish
 - Minimizing use of harmful chemicals/Use of organic manure
 - Creating public awareness on the dangers of land pollution and how to control it.
 - Recycling of waste materials
 - Government legislation against dumping
 - Settling up proper garbage collection programmes
- 5 a)
- Leads to shortage of pasture
 - Leads to crop failure
 - Leads to shortage of water for livestock and irrigation
 - Leads to shortage of agricultural raw materials for agro-based industries
 - Leads to reduction in export of agricultural commodities/reduction in the farmers' income.
- b) What is soil conservation?
- It is careful management/protection of soil against erosion/exhaustion

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- c) State three farming methods that assist in soil conservation
- Ploughing along the contour
 - Controlled grazing
 - Strip cropping
 - Making terraces
 - Digging cut off drains/trenches
 - Planting cover crops
 - Mixed cropping
 - Agro-forestry
- 6.
- Occurrence of heavy rainfall leading to high volume of water than river banks can hold e.g. Nzoia and Nyando
 - Siltation of lakes, river channels and canals on the lands due to soil erosion upstream thus reducing carrying capacity triggering floods.
 - Nature of land- plain areas are prone to floods as drainage tends to be poor
 - Deforestation-Clearing of vegetation on catchment areas along the river (river line vegetables) entrance run off and reduce percolation.
- 7.
- Stagnant grounds are breeding grounds for carrier diseases e.g. malaria, typhoid, bilharzias
 - Loss of life
 - Loss of property which take time to regain
 - Causes soil water-logging thus lowers crop production
 - Food shortages or famine where farmers calendar is disrupted or crops washed away
 - Traumatized people or make people panic
 - Loss of confidence and interest in undertaking development and economic activities.
 - Disrupts clean water supply to major towns where floods wash away water pipes.
- 8.
- Construction of dams to reduce volume and velocity of river discharge downstream
 - Construction of dykes or artificial levees which help to restrict outflow of rivers or canals e.g. along river Nyando and Nzoia.

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- Conservation of vegetables on catchment areas and along the course of the river meander thus smooth flow of rivers
- Engaging government and non-governmental organizations ICRAF in creating awareness on importance of activities e.g. agro-forestry, water management and conservation which enhance infiltration
- Clearing deepening and widening of rivers and canals to facilitate easy flow of water into lakes and dams.

9.

- Excess water stored in dams can gainfully be used to generate HEP
- Excess water can also be used for commercial farming through irrigation
- Creation of environment e.g. irrigation schemes
- Promotion of fishing activities in dams and lakes due to reduced sedimentation and pollution
- Poverty would also go down due to reduced incidence of loss live, crops, homes and livestock
- Flood plains are highly rated for agricultural due to fertile soil deposits.

10.

- Floods
- Droughts
- Earthquakes
- Diseases Pests
- Lighting

11.

- Fires
- Soil erosion
- Pollution
- Desertification

12.

It's spread of arid conditions/deserts into formerly productive areas

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13.

- Climatic changes
- Wanton destruction of vegetation
- Overstocking resulting on overgrazing
- Cultivation of marginal lands
- Poor cultivation methods
- Population pressure leading to opening up of virgin land and clearing forest

14.

- Food shortage
- Reduction of livestock due to lack of water and pasture-dwindling of agricultural and potential land
- Low standards of living of farmers
- Stagnation of Agro-based economy

15. i) **Natural environment**

Includes things that are availed by nature e.g. landscape and air

ii) **Geographical environment**

Refers to environment factors whose relationships are considered in terms of spatial location

iii) **Physical environment**

Includes all phenomena apart from man and the things he creates.

iv) **Non human environment**

Includes all those things that are not in a social system, whether man made or not

v) **Cultural environment**

Includes all aspects of human culture found within a given environment for his survival e.g. taboos, totems, traditional beliefs, etc.

16. a) i) Environmental refers to all those external conditions that surrounds a living organism. It comprises of the atmosphere, water, land, vegetation and animals.
- ii) Management of the environment involves all the measures aimed at better and useful exploitation and rehabilitation of natural resources.
- iii) Conservation of environment involves the proper utilization of resources that ensures little or no wastage.

b)

- In order to sustain life
- For recreation purpose
- For protect wildlife
- For aesthetic value.
- Lightning
- Windstorms
- Pests and diseases
- Pollution
- Droughts
- Earthquakes
- Windstorms

c) i)

- Heavy downpour which increases the river volume leading to the river bursting its banks
- Impermeability of the soils which reduces the rate at which rain water percolates into the ground
- The rise of lake levels that causes the lake water to rise hence flooding the adjacent land
- The occurrence of tsunamis which produces strong sea waves leading to sea water spilling onto the land
- Deforestation that results into land being exposed to run-off causing flash floods.
- Poor urban drainage that clogs the drainage system.

ii)

- It leads to loss of life
- It results in the destruction of property
- People have been displaced
- Flooded areas are associated with waterborne diseases
- It also disrupts the infrastructure.

17. a) i)

It is a bright sudden flash of natural electricity produced during a thunderstorm

ii)

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It is caused by a formation of the clouds whose ions are positively and negatively charged thus releasing an electrical charge.

- b) i)
- Claims many lives
 - destroys property
- ii)
- By installing lightning arresters
 - By public awareness campaigns
 - By conducting research
18. a) i) Pollution is the addition of harmful substances into the environment
- ii)
- Air pollution
 - water pollution
 - Land pollution
 - Noise pollution
 - Radiation pollution
- b) i) Air pollution is the additional of harmful substances into the atmosphere making it to be contaminated.
- Discharge of smoke and gases by industries
 - Emission of fumes from motor vehicles
 - Engagement in agricultural activities that leads to such pollution.
 - Discharge of smoke from domestic heating using charcoal, firewood and coal.
 - Smoke emitted from air crafts
 - Mining and quarrying
 - Poor disposal of domestic waste
- iii)
- Ensuring that industrial effluents and fumes are properly treated
 - Encouraging the use of biodegradable materials
 - Legislation to regulate the amount of noise in social places
 - Increase in the farm manure as an alternative to organic fertilizer

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- Regulating by law the setting up of industries which are likely to pollute the environment

19. a)i)

- The international Centre for Research in Agro forestry (ICRAF)
- The Swedish International Development Agency (SIDA).
- The United Nations Environmental Programme (UNEP)

b)

- Several laws have been enacted to combat environmental degradation like the Water Act which gives guidelines on the safety and welfare of employees and the working conditions in factories.
- The Wildlife Conservation and Management Act whose aim is to protect the wild game and their natural habitats.

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MODEL PAPER 1 A

312/1 PAPER 1

SECTION A

1. a) The study of distribution and interrelationship of natural and human phenomena on the earth's surface.
b)
 - Biology is the scientific study of organisms focusing on their anatomy and physiology and behaviour while Geography is interested in the distribution of organisms and the factors influencing this distribution, i.e. Biogeography.
 - Biology applies geographical information to explain factors which determine the distribution of species on the earth's surface
2. a) Mass wasting is the down slope movement of rock materials under force of gravity. Mass movement is the down slope movement of rock materials due to lubrication of water.
b)
 - Alternating heating and cooling
 - Freezing of soil particles
 - Removal of soil down slope
 - Rainwater
 - Ploughing down slope
3. a)
 - Himalayas
 - Atlas in Africa
 - Alps in Europeb) State three theories which explain the formation of fold mountains.
 - The contraction theory
 - The convectional theory
 - Continental drift
 - Plate tectonic theory
4. a) Vertical movements along a radius from the earth's center to surface or

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from the surface towards the earth's center.

- b)
- Isostatic adjustment occurs when sialic (continental crust) layer which is
 - less dense floats on the dense simatic (oceanic crust) layer
 - Erosion on the continents reduces pressure on continental crust (sial)
 - leading to isostatic uplift, while deposition of sediments on sea crust(sima) adds pressure which results to isostatic sinking.
 - The continental crust (sial) and oceanic crust (sima) constantly adjust to one another in process known as isostatic adjustment.
5. a)
- The rock below should be limestone/chalk/Dolomite/Soluble
 - The rock should be well jointed to allow water percolation
 - The area must have high rainfall/moderate rainfall to facilitate solution
 - The water table must be deep to allow water percolation.

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SECTION B

6. a) The grid squares are;
- i) 3926
 - ii) 4225
- b) $0^{\circ} 30'$
- c) North West to South east
- d)
- The steep slopes/escarpments have been avoided because they are unsuitable for construction of houses /for farming
 - The plains/gentle sloping areas are densely settled as the land is flat/gentle sloping suitable for construction of houses/farming
 - The basins are avoided as land is waterlogged/flooded/swampy therefore not suitable for settlements
 - Some river valleys are avoided because of flooding .
 - Very high areas have few settlements because the areas experience low temperatures
- e)
- Woodland
 - Forests
 - Scrub
 - Scattered trees
 - Papyrus
 - Thicket
- f) i)
- Carry out reconnaissance
 - Read from reference books
 - Prepare questionnaire
 - Assemble necessary tools and equipment
 - Prepare sketch map of study area
- ii)
- Oral interview method
 - Questionnaire method

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- Observation method
7. a) This is a body of water which collects or accumulates in hollow/basin/depression on the earth's surface
- b)
- High evaporation due to high temperatures in the rift valley. This results to concentration and accumulation of dissolved salt in the lakes
 - Washing of mineral rich soils by surface run-off and rivers into the lakes.
 - The lakes have no streams flowing out which causes the accumulation of minerals in the lakes.
 - Lake bed is made of salty bedrock.
- c) i)
- It occurs after an eruption
 - Outpouring of lava forms a volcanic cone vent, which is sealed
 - Lava cools and solidifies in it
 - This leads to piling of pressure below the plug
 - The pressure leads to a violent eruption that blows off the top of the cone
 - This leaves a wide depression at the top of the cone called a caldera
 - This depression is later filled either melt water, ground water or rainwater to form a caldera lake.
- ii)
- L. Shala in Ethiopia. Largest in Africa.
 - L. Ngozi in Tanzania
 - L. Nyos in Cameroon.
- d)
- Evaporation from lakes leads to formation of convectional rainfall.
 - The lake breezes have a cooling effect hence lower the temperature of surrounding areas
 - Regular land and sea breezes help to moderate the temperature
 - Evaporation from lakes leads to increase in the relative humidity of the surrounding areas.
8. a) i) Faulting is the formation of cracks or lines of weakness in the earth's crust due to tensional and compressional forces acting on crustal rocks.

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- ii) Reverse faults are formed when compressional forces push a block of a land on one side of a fault upwards in relation to another while a tear fault is formed where two each other which causes a faults to form two adjacent lands sliding past each other
- b) i) Fault Scarp
A fault scarp is formed when there is an upward and downward displacement on either side of a single fault line- the exposed part is the fault plane
- ii) Tilt Blocks
During faulting vertical tectonic forces may push the blocks of land upwards, the process some parts may be raised more than others due to uneven forces forming an uplifted tilted block.
- c)
- Faulting may result to formation of rift valley lakes which are used for fishing/irrigation/transportation/mining.
 - Faulting results to attractive sceneries such as rift valley, geysers which attract tourist which earns foreign exchange.
 - Hot springs and geysers resulting from faulting can be utilized for geothermal electricity.
 - The windward side of block mountains receive high rainfall useful for Agriculture/settlement/forestry
 - Faulting exposes minerals such as diatomite which a source of income.
- d) i)
- Gives first hand information
 - One can seek clarification
 - Its useful method in collecting information from people who cannot read or write
 - Further discussions can be initiated
 - One can gauge accuracy of the responses
9. a) i)
- Zero lapse rate is where temperatures remain constant with increase in height
 - Environmental Lapse rate is the decrease in temperature with altitude.
- ii)
- Polar front/cold front

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- Intertropical front/ITCZ/Warm front

iii)

- Smoke from factories form fog/increase the temperatures
- Presence of tall buildings/planned roads change the direction of winds
- Smoke dust particles act as nuclei increasing precipitation.

b) i)

- Tropical Desert climate
- Savanna climate
- Tropical maritime climate
- Tropical monsoon climate

ii)

- The areas experience high rainfall/1500-2000mm all year round due to ITCZ where moistened wind converge leading to high rainfall
- Double maxima rainfall/two rainfall seasons due to sun's being overhead twice a year.
- The rainfall is mainly convectional due to high temperatures accompanies by continuous evaporation/Eva transpiration
- High grounds receive relief/orographic rainfall due to high altitude effect
- The rainfall is accompanies is accompanied by thunderstorms due to cumulo nimbus clouds/rapid convectional currents
- Rainfall mainly occurs in the afternoon due to intense heat/evaporation
(4x2= 6mks)

c) i)

- Appearance of sunspot/dark area of the surface of the sun causing higher/lower temperatures on the surface f the earth
- Higher solar radiation received during perihelion period leading to higher surface temperatures

ii)

- Global warming encouraging farming in cold areas/poles
- Ecosystem is affected where grassland/desert areas/deforestation will increase
- Expansion of ocean water will lead to sea level rising leading to floods.
- Rise in temperatures/evaporation will increase rainfall in some areas

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- Temperature land winters will be wetter/summers drier.
- Increase in pests/diseases due to wetter and warmer climatic conditions
- Extinction of some plant/animal species.

10. a) i)

- The areas receive very low rainfall less than 250mm per annum
- Rainfall is irregular/sporadic
- The areas have high rate of evaporation
- The areas have high diurnal/daily temperature range
- The areas have little or no vegetation cover.
- The areas are sandy
- The areas are rocky

ii)

- Reduced wind spread/velocity
- Presence of obstacles/vegetation

b) i)

- Abrasion
- Deflation

ii)

- Barchan form when sand deposition occurs around an Obstacle/Rock /bush
- Continued deposition leads to enlargement of the mound of sand
- Wind pushes sand forward along the edges to form horns
- Further push by the wind causes the horns to grow longer thus giving the barchan a crescent shape
- The windward side is gentle due to gradual deposition.
- The leeward side is steepened by the eddies.
- They lie at right angles to the prevailing wind.
- They form either singly or in groups.

(Any 4 well explained x 1 = 4 marks)

(c)

ROCK PEDESTAL	MUSHROOM BLOCK

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It is formed where the rock is heterogenous	It is found where the rock is homogenous.
It is irregular in shape and thin at base	It is broad at the top and narrow at the bottom.

(Any 2 well explained x 2 = 4 marks)

- (d) (i) - To find out how water erodes the desert surface.
- To find out how water transports its load.
- To find out factors influencing water erosion and deposition.
- To find out the features resulting from the action of water.
- To find out the significance of resultant features.
- (ii) - Alluvial fans
- Bajadas
- (iii) - Some of them e.g. rock pedestals, Zeugens etc are unique hence a tourist attraction. This may earn a country foreign exchange.
- Some of them offer suitable sites for film making.
- Some areas have cases which provide water for domestic use/livestock/irrigation/attract human settlement.
- Some areas have loess which is fertile/suitable for crop growing.
- They offer suitable sites for testing military weapons/for military training.
- Hot sun in arid areas provides a large potential for solar energy which can be used for domestic purposes.
- Arid areas may have dug - in caves which may provide human shelter.
- Sand - dunes may cover transport routes thus hindering transport/wadis may be too wide thus making construction of bridges expensive.

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K.C.S.E MODEL PAPER 1B

SECTION A

1. a) - Iron
 - Nickel
 b)
 - Analysis of volcanic materials that have erupted
 - Examine mines and bore holes matter and profiles
 - Study earthquake/seismic waves.
 - Analyse rocks of meteorites and moon

- 2 a) A theory that state that the earth crust is divided into block of land (plates) that float and move towards, away, or parallel past each other. Eg. African block etc.
 b)
 - Geological evidence -Rocks similar in Eastern coast of South America and Western Coast of Africa.
 - Jig-Saw-fit-some coasts can fit each other e.g. Western Coast of Africa and Eastern South America.
 - Similarity in climate e.g. Southern, continents share tropical climate and lack winter.
 - Fossil remains are similar at Coasts that fit each other.
 - Paleo magnetism. .

3. a)
 - Temperature decreases with increase in height above sea level
 - Is most dense layer
 - Contain weather forming elements
 - Life supporting zone
 b)
 - Far from obstacles e g houses, trees which may provide shade.
 - Placed upright on the ground.
 - On flat ground / free from flooding
 - Cylinder kept at height of 30cm above the ground

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4. a)
- Alternating cooling and heating shift soil particles
 - Moisture in soil and its loss compact and loose soil particles
 - Earth quake triggers off soil creep
 - Freezing and thawing
- b)
- Afforestation and reforestation
 - Proper agricultural practices e.g. terracing, planting cover crops
 - Regulating livestock and controlled grazing
 - Building gabions, dams and cutoffs.
5. (a)
- Rock must be chalk, limestone, and dolomite
 - Climate be high temperatures and high rainfall
 - Deep water table
 - Thick vegetation to release and increase amount of carbon dioxide
- b) Name two sources of under ground water. (2mks)
- Water trapped in rocks (Manmatic water)
 - Lake and sea water percolating into the ground
 - Melt water sinking to the ground
 - Rain water percolating into the ground

SECTION B

6. a) (i) Ans. 5.4 +0.1 km

(ii)

$$\begin{aligned}\text{Gradient} &= \frac{\text{Vertical rise} = \text{VI}}{\text{Horizontal equivalence (HE)}} \\ \frac{360}{4600} &= \frac{1}{13} \quad \quad \quad 1 \text{ in } 13\end{aligned}$$

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b) (i)

- Rivers
- Seasonal swamps
- Water reservoirs
- Dams
- Waterfalls

(ii)

- Scrub vegetation
- Seasonal swamps
- Water reservoir
- Seasonal streams.

(c)

- Western part of the area has a continuous steep side of a scarp. This is evidenced by close contours.
- Southern part is plain land evidenced by wide apart contours.
- Northern part of the map has numerous hills evidenced by concentric contours.
- There is presence of numerous river valleys.
- Rugged relief (steep slope / Escarpment) on the western part discourages settlement
- Gentle relief on the central and Eastern region encourage settlement -Forests have little or no settlements
- Swamps not settled

(d)(i)

- Good transport/ railway /roads
- Gentle relief/wide contours
- Availability of water for domestic use from rivers
- Expansion room/sparse neighborhood
- Railway roads
- Contours apart
- Rivers
- Sparse neighborhood

(ii)

- Education center - school

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- Health center – hospital
 - Recreation center -club, hotel
- 7.
- a) Accumulation of water on a large inland basin (depression)
 - b) Formation of lakes formation of the following lakes.
 - (i) Lake Victoria
 - (ii) Lake Tanganyika
 - (i) Formed by down warping and reverse of rivers waters. Forces within the crust during rift valley formation. (Western rift valley and Eastern rift valley) made land between them to sink. Rivers from 'Kenyan side' moving westwards were cut. Rivers from 'Ugandan side' like Kagera and Kitonga reversed their westwards movement, and flowed backwards. Their waters filled created depression
 - (ii)
 - Lake Tanganyika
 - Is a rift valley lake
 - Formed through faulting
 - During rift valley formation areas between minor faults in rift valley sunk deeper than rift valley level
 - Water collected in the depression between minor faults. The lakes are usually long and narrow
 - Lake Tanganyika was formed in this way
 - Both surface runoff and ground water filled the depression
 - (c)(i) Explain how human activities have negative impact on lakes.
 - Industrial disposal of waste pollute lakes
 - Oil spillage / leakage affect aquatic life in water
 - Insecticides and pesticides from agricultural farms find way to lakes polluting it.
 - Over exploitation of water has shrunk lakes.
 - d) i) List three positive effects of lakes to humans.
 - Fish source
 - Transport by water

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- Domestic and industrial use
 - Hydro- electric power production (e.g. Lake Masinga)
- ii) Name four follow up activities in fieldwork.
- Group discussion
 - Report writing
 - Group leaders read reports to class
 - Displaying samples/pictures.
8. a)(i) Differentiate between mineral and rock.
- Mineral- Inorganic substance occurring naturally beneath the earth's surface
 - Rock- Naturally occurring materials made of one or more aggregate of mineral particles forming part of the earth's crust.
- ii) State characteristics of sedimentary rocks.
- Form from sediments of already existing rocks.
 - Laid down in layers / strata.
 - Rocks are non- crystal line.
 - Some contain fossils.
 - Are stratified
- b) Classify rocks according to form and origin giving two examples in each type.
- Igneous e.g. Granite, Gabbros
- Sedimentary e.g. Iron stone, Clay
- Metamorphic e.g. Gneiss, Graphite
- c) Explain formation of following examples of rocks.
- (i) Tuff (ii) Coral rock
- (i) Tuff
- During violent volcanic eruption
 - Magma burned to ash /dust like substance
 - Thrown to space in volcanic explosion
 - Finally dust falls to surface

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- Particles pile on each other/get segmented
- On consolidation of particles, rock called tuff is formed.
- (ii) Coral rock
 - Skeleton of dead microscopic / sea organisms / sea polyps are buried under - sea waters.
 - The skeletons pile to each other
 - Get compacted in mass.
 - A organically formed sedimentary rock called coral forms.
- d) (i) State use of equipment listed below in fieldwork.
 - (a) Geographical hammer - Check rock hardness
 - (b) Split the rock to get samples Lenses -magnify and identify rock particles / crystals
 - (c) Dilute hydrochloric acids – test chemical composition of rocks e.g. carbonate compounds.
- 9. a)(i) Name three hot desert
 - Arabian desert
 - Sahara desert
 - Great Australian
 - Namib
- (ii) State two characteristic of arid lands.
 - Lack of sufficient.
 - High evaporation
 - Little or no vegetation.
- b) Describe formation of following features.
 - i) Rock pedestals ii) Yardang
- (i) Rock pedestals
 - An outcrop rock is eroded by wind from all sides
 - The rock has horizontal alternating hard and soft layers.
 - Hard rock is eroded at a slower rate, while soft rock layer is eroded faster. - Continued erosion through abrasion, make the softer layer to be highly

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eroded and hard layer is left outstanding

- Near the ground more erosion narrow it because it has more load.
- A feature with thick head and narrow base formed is called rock pedestal.

ii) Yardang

- Occur where alternating parallel vertical bands and soft rocks lie parallel to the prevailing wind.
- Wind abrasion erodes the weaker rock more than hard rock.
- Weaker rock forms furrows / trough and the hard rock form ridges called yardangs.

c) Name three resultant features of action of water in deserts,

- Inselbergs
- Masha + buttes
- Wadis
- Alluvial fans
- Bajada

d)(i) Name three processes of wind transport.

- Traction.
- Surface creep
- Saltation

ii) State advantages of using secondary sources of data.

- Not expensive since publication may already be available
- Skilled people have written information / books
- Such information may not be found in primary sources.

iii) List four evidences of desertification.

- Increasing temperatures.
- High evaporation.
- Low and unreliable rainfall.
- Reduction in vegetation densities.
- Drying of surface water lowering of water table.

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- Poor and reducing crop yield.

10. a) i) What is a fault

Fracture / crack on the earth's crust

ii) Name the parts of a fault?

- Upthrow
- Fault scarp / escarpment
- Throw
- Heave

b) Name two resultant features of faulting.

- Fault blocks.
- Tilt blocks.
- Fault steps.

ii) Explain formation of Rift valley by anticlinal arching.

- Sedimentary young rocks experience upthrust force at the anticline
- Minor and major faults form at the fold crest
- Bigger arch is formed
- More upthrust cause gaping at crest
- Either side of the faults raise more than central block
- Central block is left at a lower level forming the rift valley.

c)i) Basic lava

- Fluid or less viscous
- Travel long before solidifying / cooling.
- Silica Content between 45% - 55%.

Acidic lava

- Does not cover long distance from vent
- Silica content 65%.
- Lava is viscous.

ii)

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- Has a crater / caldera.
- Pipe.
- Parasitic cones.
- Alternating layers ash, lava pyroclasts.
- iii) State four positive influences of volcanicity.
 - Features attract tourists thus earning foreign exchange.
 - Lava forms fertile volcanic soil good for agriculture.
 - Geothermal energy can be tapped from geysers.
 - Volcanic rocks are good building materials.

MODEL PAPER 312/2 PAPER 2a

SECTION A

1. a) - Aberdeen Angus
 - Hereford
 - Charolais
 - Red Angus
 - Shorthorn
 - Galloway
 - Santa Gertrudis

 b) Adequate water for the animals
 - Extensive rolling grassland which allows the cattle to graze freely.
 - Fertile soils of the Andes mountains which give rise to healthy natural grass for feeding the animals
 - Moderate temperatures of about 24°C during summer and 10°C and during winter which ensures continuous growth of grass throughout the year
 - Moderate rainfall 1000mm which ensures growth of nutritious natural pastures/pampas.

2. a) Gas - Oil/petroleum
 - Water

 b) It allows continuous flow and supply of the commodities.
 - It is not affected by bad weather conditions
 - It is reliable and convenient especially on highly inflammable products
 - Low operating costs/cheap to maintain
 - It is free from pollution
 - It helps to reduce the number of tankers/reduces congestion in roads/reduces accident on the roads.

3. a) It is the process by which unproductive land is converted into useful land fit for cultivation.
 - The pressure of river Perkerra which ensures a constant supply of water
 - Gently sloping land which allows the natural flow of water by gravity
 - Availability of fertile/loamy/alluvial soils rich in minerals which ensures

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growth of variety of crops

- Availability of extensive land for expansion
- The area receives unreliable rainfall/low rainfall/ semi arid conditions which necessitated irrigation

4. a) - Nairobi
- Mombasa
- Kisumu
- b) State three problems facing the growth of Mombasa city.
- Acute shortage of water
 - Pressure on available social amenities
 - Inadequate housing facilities
 - Narrows streets/congestion/traffic jams
 - Limited space for expansion
 - High rate of unemployment
5. a)
- Pests and diseases
 - Drought
 - Lighting
 - Volcanic eruptions
 - Pollution
 - Earthquake
 - Windstorm
- b)
- Loss of life and property
 - Displacement of people
 - Floods wash away crops
 - Floods cause soil water logging lowering crop production
 - Flooding results to waterborne diseases which can cause death

SECTION B

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6. a)(i) Organized facts and numerical figure collected for a purpose.
- (ii)
- Measures of central tendency
 - Percentages
 - Measures of dispersion.
- iii)
- Easy to construct
 - Easy to interpret
 - Clear visual impression - easy to compare frequency.
- b)
- Deep soils.
 - Well drained soils.
 - Alluvial/drainage 'black cotton soils.
 - High temperatures (20°-27°C -throughout the year.
 - High rainfall/(1, 200-1, 500mm)
 - Undulating land
- (c)
- Land is cleared.
 - Land is ploughed using machines.
 - Furrows are made in the field.
 - Cutting of cane from the old plants is done.
 - The cane cutting/setts are dipped into insecticides before planting.
 - The setts are buried/planted in furrows and covered with soil.
 - Fertilizer is applied regularly
 - Weeding/spraying with herbicides is done severally.
 - The crop matures after 18 months/ 18-24 months.
 - The cane is harvested using pangas/matchete.
 - The cane is piled in heaps in the field.
 - The cane is loaded into tractors/lorries and transported to the factories.
- d)
- Pests e.g. White scale, white grub and termites and diseases e.g. Ratoon stunting, smut, yellow wilt, leaf spot etc. destroy the cane stagnating the growth thus lowering the

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yields. Burning the cane by arsonists/fire outbreaks destroys the cane lowering the yields.

- Delay in harvesting of sugarcane reducing the quality and tonnage of the harvest making the farmers to earn less. Flooding of the local market with cheap imported sugar leads to insufficient market for local producers lowering their income.
- Delayed payments/low payments discourage the farmer.
- Mismanagement of sugar factories demoralizes the farmers.
- High costs of farm inputs greatly reduce the farmers profit margin.

7. a)

- Bwidi
- Semliki
- Kidepo Valley
- Murchuson falls
- Queen Elizabeth
- Mt Elgon

b)i)

- A national park is established exclusively for wildlife while a game reserve can accommodate both wild and domestic animals.
- A national park is set up and controlled* by the central government whereas a game reserve is set up and managed by a local authority where it is located.

(ii)

- Wild animals.
- Lakes.
- Birds/flamingoes.
- Vegetation.
- Hot springs/geysers/fumaroles.
- Varied relief (any feature).
- People's culture.
- Mining sites.
- Pre-historic sites.
- Sport tourism

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c)i)

- Climatic
- Vegetation.
- Soil type.
- Relief/terrain
- Drainage

(ii)

- Illegal hunting/poaching.
- Overstocking of some wild animals.
- Frequent drought.
- Human-animal conflict.
- Inadequate capital.
- Pollution of environment.
- Fire outbreaks

d)

- Favourable climate with warm sunny summers, which allows sunbathing and cold winters which encourage winter sports e.g. Skiing.
- Varied sceneries i.e. Snow capped mountains, cascading waterfalls provide varied tourist attractions.
- Central position of Switzerland within Europe makes the country easily accessible from other European countries.
- Diversity of languages spoken in Switzerland makes it possible for tourist to communicate.
- Well developed transport network to tourist sites provide easy accessibility.
- Availability of health resorts/spas.
- Inherent hospitality encourages more tourists to visit the country.
- Well developed financial institutions banks promote easy transactions hence encouraging tourism.
- Switzerland is the headquarters of several international agencies leading to influx of delegates.

8. a)i) It is growing of trees and crops in same area and at the same time.

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(ii)

- To ensure continuous supply of woodfuel/timber/herbal medicine/raw material for paper making.
- To protect soil from erosion.
- To protect water catchment areas/create microclimate/ maintain hydrological cycle.
- To create scenic beauty.
- To expand the habitat for wildlife/conservation of wildlife.
- To create employment opportunities.
- To reduce importation of forests products/save foreign exchange.

b)(i)

- Pine
- Cedar
- Cypress
- Fir

(ii)

- The trees are conical in shape.
- Trees occur in pure stands.
- Trees grow tall and straight (30-40m)
- Trees have needle like leaves.
- Trees bear cones.
- The forests are evergreen.
- The forests have no undergrowth.
- The trees have thick barks.
- Trees have shallow roots.

c)(i) Tree harvesting.

- In Canada harvesting is done through clear cutting while in Kenya it is selective logging.
- In Canada logging is done in winter while in Kenya cutting takes place throughout the year.
- In both countries commercial logging is mechanized.

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- ii) Transportation of logs
In Canada logs are transported using melt, water/rivers while in Kenya transportation is by roads.
- iii) Marketing
 - In Canada the forest products are mostly for export to USA and European markets while in Kenya they are consumed locally.
 - In both countries the forest products have a ready market locally.
- d)
 - Accidental fires which consume large tracts of forests reduce the area under forest.
 - Cold climate lead to trees to take too long to mature which delays harvesting time.
 - Rugged landscape especially in mountainous area hinder smooth exploitation of forests/Northern parts are inaccessible in winter because of cold climate.
 - Over-exploration of some areas has created a shortage in some of the tree species which take a long time to mature when re-planted.
- 9. a)(i)
 - Petroleum/oil
 - Natural gas.
 - Uranium
 - It is a major pollutant.
 - It is bulky/heavy/costly to transport.
 - It has low calorific value.
 - It is dirty/sooty.
 - It is exhaustible.
- b)i)
 - Ol Karia.
 - Lake Bogoria.
 - Eburru.
 - Lake Magadi
 - South of Lake Turkana.

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- Menengai crater.
- Around L. Baringo.
- Area between Mt. Longonot and Suswa.

(ii)

- Limited number of suitable sites
- Inadequate capital for investment.
- Scarcity of skilled labour/inadequate technology.
- Pressure from steam in some sites is not powerful enough to turn the turbines.

c)

- Pressure of large volume of water from a river/lake to provide water to drive the turbines.
- Regular/constant supply of water to ensure continuous generation of water.
- Hard basement rock to provide a firm foundation for construction of a dam/power house.
- Presence of rapids/ waterfalls/ knick points to provide a massive hydraulic
- Presence of a deep valley/a river gorge to save on the cost of constructing a dam/to provide a space for a reservoir.

d)(i) It is a situation where demand for a given fuel exceeds supply/acute shortage of energy in the world caused by increase in oil prices.

(ii)

- Petroleum becomes too expensive for the importers, affecting their balance of trade.
- The government passes extra cost to the consumers, this may result to inflation. The country undergoes through a period of economic recession leading to diminished job opportunities and closure of factories.
- Prices of other forms of energy rise due to higher transport and production costs.
- Agricultural production drops and the economy in general declines.
- Increased deforestation creates environmental problems as demand for fuel wood and charcoal increases.

10. a)(i) Differentiate between pelagic and demersal fish.

- Pelagic fish are those fish that live near the surface or at shallow depths of seas while demersal fish live at or close to the bottom of the sea.

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- Pelagic live and move in shoals while demersal live and move singly.
- (ii)
- Cod
 - Halibut
 - Plaice
 - Skate
 - Dogfish
 - Haddock
 - Hake
 - Flounder
 - Catfish
- b)
- There are numerous inland fishing ground e.g. Rivers/lakes.
 - Fresh water fish is more popular/high demand.
 - Kenya has a regular coastline with narrow continental shelf which is unsuitable for fish breeding.
 - Marine water is too warm for fish breeding.
 - Fresh water fishing requires simple fishing equipment/technology.
 - Fresh water fishing requires little capital.
- c)(i) It is the rearing of fish confines in ponds.
- (ii)
- Provides another alternative source of protein.
 - Saves foreign exchange.
 - Creates job opportunities.
 - Is a sources of income/earns foreign exchange.
 - Encourage development of infrastructure
 - Provides industrial development.
 - It leads to better use of derelict land by creating fish ponds e.g. around Bamburi
- d)
- Limited/shortage of market.
 - Inadequate capital
 - Presence predators

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- Inadequate skills/technology.
 - Poor transport network.
- e)
- Japan has cool waters which have abundant supply of planktons the main food for fish.
 - Japan has shallow continental shelf which allow light to penetrate to the sea bed encouraging growth of microorganisms used as food by fish.
 - The country experiences convergence of warm and cool currents which results in upwelling of ocean waters thus bring minerals for fish.
 - It has cool waters which encourage thriving of numerous fish species.
 - It has indented coast/numerous sheltered bays which provide secure breeding grounds for fish.
 - Japan has a large population which provides a ready market.
 - It has advanced technology which enhances fishing.
 - Cool to cold climates provides natural preservation for fish.

MODEL PAPER 2B

SECTION A

1. a) State three advantages of using solar energy in Kenya.
 - It is a cheap source of energy as it is absolutely free.
 - Solar energy is available in all parts of the world, however the duration and intensity differs.
 - It is easy to tap and requires minimized maintenance costs e.g. Solar panels lasts 20-30 years before they can be replaced.
 - Solar energy is a clean and environmentally friendly form of energy.
 - Solar energy can be stored in batteries and used later in the night where there is no sunlight.
 - The development of solar power lessens the dependence on oil or coal especially for those countries which do not have fossil fuels.
 - It is versatile in its use to power diverse items ranging from simple light bulbs to solar cars and satellite.
 - It is an inexhaustible source of energy; solar energy is available as long as the sun continues to rise
- b) State two factors hindering the development of solar energy
 - It is difficult to store solar energy and can only be stored in small portable batteries. The cost of producing the equipment required for large scale solar energy production is high.
 - The sophisticated technology required to produce the equipment for solar energy is lacking.
2. a) State three ways through which Kenya has benefited from international airports.
 - Boosting industries like tourism and in food processing.
 - Encourages precision and advancement in technology.
 - Can now easily export its perishables.
 - Has offered employment opportunities to Kenyans at different levels.
 - It has promoted international understanding and harmony.

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- b) Suggest two possible solutions to Africa's transport and communication problems.
- Construction of Trans-African highways to improve the quality and volume of international road traffic in Africa.
 - Construction of international railways.
 - Encouraging regional economic cooperation.
 - Establishment of national airways.
 - Establishment of ground satellites.
3. a) Distinguish between population distribution and population density. –
- Population distribution refers to the way people are spread out on the land while population density describes the concentration of people in a specific area.
- b) State three problems created by a decline of population in a developed country.
- Inadequate manpower/labour is expensive.
 - Underutilization of social facilities such as schools/ under utilization of resources.
 - High old age dependency ratio.
 - Underutilization of public utilities such as schools, health facilities, roads etc.
4. a) State three physical factors which influence the location of settlements.
- Availability of water supply/good drainage.
 - Availability of land/space.
 - Nature of relief.
 - Suitability of climate or good rainfall.
 - Absence of pest and diseases or health environment or fertile soils.
- b) Which are the three functional zones of an ideal urban centres.
- Central/commercial business district.
 - Residential zone.
 - Manufacturing/industrial zone.
5. a) Types of fish reared in ponds
- Tilapia

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- Trout
 - Mud fish
 - Carp
- b) State three measures that have been undertaken by the government of Kenya to encourage fish farming.
- Establishment of research institutions.
 - Setting up of ponds and hatcheries as demonstration farms.
 - Advising farmers on fish farming techniques.
 - Establishment of cooperatives which give credit facilities to fish farmers. --
- Diversification of food and source of protein leading to setting up of fish farms.

SECTION B

6. Use the photograph provided to answer question b
- a) i) Identify the type of photograph given. Ground photograph 1 mark
- ii) - Gorge
- Bridge
- River
- Road
- b) - Narrow roads.
- Potholes
- Sharp bends
- Unmarked roads
- Slippery road surfaces.
- c)
- Roads cannot carry heavy and bulky goods
 - Roads are vulnerable to adverse weather conditions eg. Heavy rain.
 - Road transport is more prone to accidents.
 - Traffic congestion make the transport slow especially during rush hours. (d)
 - Inadequate skilled manpower required for construction of road network.
 - Inadequate capital for development of transport system.

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- Political differences among African states have hindered development of inter-state linkages.
 - Rugged terrain \ thick vegetation makes it difficult to construct roads.
7. a) Give five reasons why wildlife -conservation is encouraged in Kenya.
- To protect the endangered animals/plant species/regeneration.
 - To promote tourism.
 - To generate foreign exchange/revenue.
 - To keep them for posterity/ future generation
 - To sustain the raw materials for supply of drugs.
 - For education and/research purposes
 - For aesthetic value/beauty/recreation.
- b) Explain what you understand by the following terms.
- i) Domestic tourism.
- Is the visit of citizens of a country to places of interest within the country
- ii) Eco-tourism
- It is a word derived from ecology and tourism. It is a tourism approach which involves the community around the tourist attraction in conservation and management of tourist site and the surrounding environment.
- c) Explain four factors that have hindered the development of domestic tourism in Kenya.
- The roads leading to the tourist sites are poorly maintained. This discourages people from visiting such sites.
 - Inadequate local campaign/advertisement of tourist attractions/special packages lead to low public awareness.
 - Familiarity with the tourist attractions among the local people makes them fail to appreciate their beauty and value.
 - Negative attitude towards local tourism limits the number of people who engage in tourism.
 - Insecurity from gangsters/poachers in the national parks and game reserves scare people away from visiting them.

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- The high cost of accommodation in the game lodges discourages local tourism/the high cost of hiring tourist vehicles discourages people from touring/low income
- d) Explain four factors that have made Switzerland a major tourist destination in Europe.
 - Favourable climate, with warm sunny summer which allow for swimming and sunbathing/the cold winters which encourage winter sports such as skiing.
 - The varied scenery consisting of snow capped mountains, cascading waterfalls and glaciated landscape provides varied tourist attractions which are lacking in other parts of Europe.
 - The central position of Switzerland within Europe makes the country easily accessible from the other European countries.
 - Political neutrality of Switzerland removes any travel restrictions to the country as a tourist destination. Diversity of languages spoken in Switzerland makes it possible for tourist to communicate and move around with ease.
 - Well developed transport network to tourist sites provides easy accessibility.
 - Advanced training in the tourist management enables Switzerland to provide the necessary service to tourists thus attracting more tourists to the country.
- 8. a) What do you understand by the terms
 - i) Land reclamation.

This is the practice by which less useful land is converted into more useful lands
 - ii) Land rehabilitation.

This is the process of recovery of land which has been misused and destroyed through human activities such as quarrying, overgrazing, deforestation, charcoal burning or over cultivation.
- b) i) State any four methods used in land reclamation and rehabilitation in Kenya.
 - Drainage of swamps.
 - Irrigation dry land.
 - Application of manure.
 - Rehabilitation of eroded landscape.
 - Filling up of quarries.

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- Control of pests.
- Afforestation and reafforestation.
- Agro-forestry.
- Planting of drought resistant crops.
- Control of floods.
- ii) Name two large scale irrigation schemes in Kenya.
 - Perkerra.
 - Bura.
- iii) Explain four factors that led to the establishment of Mwea Tebere irrigation scheme.
 - Presence of perennial rivers of Thiba, Murubara and Nyamindi from Mt. Kenya provide adequate water throughout the year.
 - The Mwea plains have black cotton soils which are suitable for rice farming as they are capable of retaining water.
 - The climate of Mwea is not suitable for agriculture. The rainfall is low, poorly distributed and unreliable. The rainfall varies from 635 mm to 1270 mm p. a. falling in April to May and October to November.
 - During the years of emergency in 1950's many people were detained and the government need projects to keep the detainee busy hence it provided work to the detainees.
 - The land is gently sloping making it possible for irrigation farming.
 - The Mwea plains were sparsely populated having been previously used as a communal grazing land.
 - The presence of loamy soils made it possible to cultivate other crops to support the families settled.
 - Number of landless people from Central province who lost their land to white settlers so the government needed to settle them hence they used Mwea Tebere project as most appropriate site to settle the landless.
- c) i) What is a polder.

This is land reclaimed from the sea and enclosed by dykes
- ii) Describe how a polder is reclaimed

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- Each polder is surrounded by a ring canal and a high dyke to protect it against the higher water level.
 - The ring canal carries excess water to the sea or to a water reservoir.
 - Barges are used to bring boulder clay to constructing two outer walls of the dykes.
 - Sand is pumped from lake or sea bed to fill the space between two outer walls of the dyke.
 - The boulder clay walls are reinforced from their outer surface by use of blocks of basalt.
 - Water from the polder is then pumped out or drained through ditches and canals into the ring canal until the former lake or sea bed is exposed.
 - The exposed land is studied in detail to understand the nature of soil e.g. Acidity, fertility, depth and composition.
 - Reeds are then grown to drain the excess water and use up the polder. -Drainage pipes are then laid and ditches are dug to dry up the polder.
 - Experimental crops are planted and buildings put up for five years before the land is open for general use.
9. a) Areas in Kenya where forests are found in Rift Valley
- Uasin Gishu.
 - Timboroa.
 - Burnt Forest.
 - Mau Summit.
 - Tinderet.
 - Turbo
- b) Four factors that favour growth of softwood forest in Swaziland.
- Swaziland receives high rainfall/adequate /reliable/sufficient which favours the growth of softwoods.
 - Swaziland experiences cool temperatures due to high altitude hence favouring the growth of softwood.
 - The ruggedness of the landscape in Swaziland hinders the development of agriculture making the forestry the major land use.

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- It is the government policy to increase the areas under soft wood forest/afforestation.
 - This is major programme of reafforestation which encourages planting of softwood.
 - There is high demand for wood products in South Africa.
- c)i) Explain four problems experienced in commercial exploitation of the equatorial forest in Africa.
- Poor roads hinders transportation of logs especially due to bad weather.
 - Logs are heavy and bulky hence making it difficult to transport.
 - Some trees species have big buttresses which makes tree felling difficult.
 - Inadequate capital limits the use of modern facilities like powered saws.
 - Thick undergrowth which makes development of roads impossible.
 - Tree species grow in a mixture (no pure stands) in case of natural forests which makes tree extraction very difficult.
 - Inadequate skilled or expertise personnel as managers and lumberers.
- ii) Three species of trees found in Gabon.
- Mahogany.
 - Okuome.
 - Iron wood.
 - Rose wood.
 - Iroko.
 - Ebony.
 - Sapele.
 - Green heart.
 - Obeche.
 - Teak.
 - Camphor
- d) Measures taken to conserve forest in Kenya.
- Reafforestation.
 - Afforestation.
 - Agroforestry.
 - Soil conservation

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- Use of energy saving jikos.
 - Creation of awareness on forest
 - Management through chief barazas.
 - Setting tree planting day.
 - Employing forest guards.
 - Creation of forest reserves.
10. a) i) Conditions necessary for formation of petroleum.
- Presence of sedimentary rocks.
 - Presence of organic remains/fossils.
 - Presences of non-porous rocks/cap rock.
 - Presence of pressure to compress organic remains.
 - Presence of porous rocks.
 - Time taken for sedimentation and pressure to compress.
- (ii) Reasons why Kenya import her oil in crude form
- To create more employment opportunities.
 - To get by-products used in making other materials e.g. Polythene bags, plastic.
 - Tar a by-product is used for making roads.
 - To earn foreign exchange through export (earning the country foreign exchange).
 - It is cheaper.
 - To diversify her source of energy?
 - To reduce the cost of transport.
- b) i) Two advantages of geothermal power.
- It is provided naturally.
 - It is not exhaustible.
- ii) Four factors that hinder expansion of geothermal production in Kenya.
- Inadequate capital.
 - Low level of technology.
 - Inaccessibility of areas where the geysers and hot springs are found.
 - They are limited to few areas.

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- c) Four ways by which the government of Kenya should apply to conserve her energy.
- Limiting the use of fuel guzzler vehicles.
 - Encourage people to walk on short distances.
 - Using power rationing criteria where applicable.
 - Educating the mass on the need of conserving energy for future use.
 - Encouraging afforestation and reafforestation.
 - Encouraging use of alternative sources of energy e.g. Solar.
 - Use of energy saving jikos.
- d) i) A multi purpose dam
- A reservoir that serves more than one purpose i.e. Providing electricity, irrigation, macro-climate, tourism etc
- (ii) Reasons for establishment of Akosombo Dam.
- To provide water for irrigation.
 - To provide electricity.
 - To control floods.
 - To create employment opportunities.
 - To improve transport of the surrounding area.
- iii) Benefits of establishing Akosombo Dam.
- Lighting the area.
 - Provided water for irrigation.
 - Attracted tourists.
 - Created employment opportunities.
 - It has helped to regulate the flow of river Volta.
- It has modified the climate of the surrounding region
- It has improved transport in the surrounding region.

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PAPER 2

Agriculture livestock

1. Name two exotic breeds of dairy cattle reared in Kenya
2. State three physical conditions that favour dairy farming in Denmark.

Agriculture crop farming

3. State two climatic conditions that favour the growth of palm oil in Nigeria.
4. State two problems experienced in the marketing of palm oil in Nigeria
5. Explain for ways in which the Kenyan governments assists tea farmers.
6. Name two districts in Eastern province where tea is grown

Mining

7. State three conditions that are necessary for the formation of petroleum.

Industry

8. Name three agricultural food processing industries in Kenya
9. Explain how the following factors have favoured the development of industries in Thika town.
 - i) Proximity to Nairobi
 - ii) Availability of water
 - iii) The hinterland

Field work

13. You intend to carry out field study on population in the local open air market.
 - i) State three reasons why it would be necessary for you to visit the market before actual field study.
 - ii) Give two methods you would use to collect information
 - ii) Give two follow-up Activities you would carry out after the field study.

Photographs

14. The photograph below provided shows a tea growing area in Kenya use it to answer question (a) and (b)



- a)
 - i) What is the evidence in the photograph that this is a ground general photograph.
 - ii) Draw a rectangle measuring 15cm by 10cm to represent the area of the photograph on the sketch and label the main features shown on the photograph
 - b) Identify two features from the photograph that shows that this is a small scale tea farm
15. Describe the stages involved in cultivation of tea from land preparation to the stage shown in the photograph.

Forestry

- 16. What is forestry?
- 17. Explain the factors that favour the growth of natural forests on the mount Kenya.
- 18. State five factors that have led to reduction of area under forests on mount Kenya
- 19. Give the difference in exploitation of softwoods forests in Kenya and Canada under following sub-headings.
 - i) Period of harvesting
 - ii) Transportation.

Transport

- 20. Name three international airports in Kenya.
- 21. Give four advantages of air transport over road transport
- 22. Explain measures improve road transport
- 23. Explain why there are few rail links among African Countries
- 24. Give four reasons why there is limited use of the river transport in Africa

Environmental conservation

- 25. State two causes of water pollution
- 26. Give two effects of water pollution on environment
- 27. Explain four methods used to control floods in Kenya

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28. Explain how the following soil conservation methods are used to improve the quality of soil
- i) Contour ploughing
 - ii) Mulching
 - iii) Crop rotation

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PAPER 2

QUESTIONS AND ANSWERS

Agriculture livestock

1. Name two exotic breeds of dairy cattle reared in Kenya
 - Friesian/Holstein
 - Arshire
 - Guernsey
 - Alderney
 - Brown swiss/swiss brown
2. State three physical conditions that favour dairy farming in Denmark
 - The land scape/gentle sloping land
 - The climate has warm and sunny summers that allow outdoor grazing.
 - There is cool climate suitable for pasture growing
 - There is moderate rainfall/rainfall raining of grass/fodder crops
 - Soils are fertile enough to support high quality pasture.

Agriculture crop farming

3. State two climate conditions that favour the growth of palm oil in Nigeria
 - High temperature range of 20⁰C to 30⁰C
 - High rainfall that is evenly distributed throughout the year/ 1500mm to 2100mm.
 - High relative humidity of 80% to 90%
 - Plenty of sunshine during the ripening season
4. State two problems experienced in the marketing of palm oil in Nigeria
 - Competition from other oil vegetables
 - Poor road network
 - Production of low quality oil
 - Reduced production that has lowered the amount of oil exported.
5. Explain for ways in which the Kenyan governments assist tea farmers
 - It organizes farmer education days/provides extension services for the farmers at low prices.
 - It buys farm input in bulk and sells to farmers at low process

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- It provide credit facilities to the farmers to enable them purchase farm inputs
 - It collects the green leaves and delivers to the factory on behalf of farmers
 - It establish factories where the green tea leaves are processed
 - It undertakes the marketing of tea on behalf of the farmer
 - Improves feeder roads to ease transportation of green leaves
 - Conducts researches on tea crop varieties/diseases/pests in order to produce high yield tea/better quality tea.
6. Name two districts in Eastern province where tea is grown
- Embu
 - Meru North
 - Meru south
 - Meru Central

Mining

7. State three conditions that are necessary for the formation of petroleum.
- Presence/deposition of remains of flora and fauna fossils over along period of time.
 - Presence of non porous rocks underneath the deposits of flora and fauna
 - Deposition of other layers of rocks underneath the deposits of flora and fauna
 - Compression of remain of flora and fauna due to folding of the layers of rocks

Industry

8. Name three agricultural food processing industries in Kenya.
- Tea Processing
 - Milk processing
 - Sugar refining
 - Fruit canning/processing
 - Grain milling
 - Vegetable canning /processing
 - Oil processing

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9. Explain how the following factors have favoured the development of industries in Thika town

i) Proximity to Nairobi

ii) Availability

iii) The hinterland

i)

- Nairobi provides some inputs required by the industries in Thika there is industries interdependence among the industries in Nairobi and Thika.
- The rail and road connection between Nairobi and Thika provide cheaper movement of goods and services for the industries in Thika
- Nairobi provide ready market for manufactured goods from Thika.

ii) Availability of water.

- River chania which passes through Thika town provide fresh water for industrial use especially coffee processing and fruit canning industries
- Water for use in the industries in available through the year since river chania is permanent.

iii) Hinterland

Thika town has rich agricultural hinterland is densely populated hence provides cheaper labour for the industries/ready market for industries.

- 10.

- It has created employment opportunities hence raising the standards of living of people/reducing problem of unemployment
- It has led to development of transport and communication network thus facilitating development of other sectors of the economy.
- It has facilities development of social amenities in the area where industries are located .s
- It has led to increased agricultural production since some industries use agricultural raw materials.
- It has led to acquisition of management and technical skills which are also used in other sectors of economy reducing/enhance the expansion of industries.
- It has led to improved balance of trade since there is added value to the export products
- It has led to improved balance of trade since there is added value to the export products

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- It has led to growth and expansion of settlements and urban centres as labour migrates to the industrial centres.

11.

- Nairobi
- Mombasa
- Thika

12.

- The country has adequate capital to invest in industry
- Advanced technology has led to efficient methods of production of high quality cars which are competitive in the world market.
- Japan provides dual saving vehicles leading to high demand for them in the world.
- Japan has highly skilled industrious workforce which enhances efficient production
- Japan has many sea ports which makes the importation of raw materials and exporting cars possible.
- The government policy/peace and stability encourage industrialization led to rapid development of industries.
- Japan has highly development hydro-electric power projects which provide power projects which provide power needed for industries.
- The presence of high population provide large local market.
- Japan's terrain is rugged unsuitable for development of agricultural and thus industries provide an alternative source of income to be used for buying food and other commodities
- Strategic position of Japan in relation to other countries encourages trade thus promoting production of vehicles/Japan as accessible from all directions.

Field work

13. You intend to carry out field study on population in the local open air market
- i) State three reasons why it would be necessary for you to visit the market before actual field study.
- To get permission from the relevant authority
 - To be able to formulate objectives/hypothesis

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- To be able to prepare a working schedule /decide on appropriate methods of data collection
- To determine the respondents/resource persons
- To access the problem likely to be experienced in the area
- ii) Give two methods you would use to collect information on pollution
 - Interviewing
 - Taking photographs
 - Measuring the extent of polluted area
 - Administering questionnaires
 - Tape photographs
 - Measuring the extent of polluted area
 - Administering questionnaire
 - Tape recording / video recording.
- iii) Give the following up activities you would carry out after the field study
 - Analyzing data
 - Writing report
 - Giving relevant advice to the stake holders
 - Discussing the findings
 - Displaying photography/sketches from the study area.

Photographs

14. The photograph below provided shows a tea growing area in Kenya use it to answer question (a) and (b).
- a) (i) What is the evidence in the photograph that this is a ground general photograph that this is a ground general photograph.
- It FREEs of all /Many objects
 - The objects become progressively smaller towards the background
 - Photograph captures the general appearance of the area
- ii) Draw a rectangle measuring 15cm by 10cm to represent the area of the photograph on the sketch and label the main features shown on the photograph.



- b) Identify two features from the photograph that shows that this is a small scale tea farm
- The simple houses
 - Mode of transport by use of donkeys
 - Untrimmed edges of tea bushes
 - Dry maize stalks hear the houses.
15. Describe the stages involved in cultivation of tea from land preraration to the stage shown in the photograph
- The land is cleared of vegetation
 - The land is ploughed/tilled
 - Seedlings are planted in nursery and allowed to grow to 20cm
 - Seedlings are planted in rows which are about 1.5 metres apart.
 - The plants are weeded and manure/mulching applied regularly.
 - Once the bushes start growing. The tips of branches are oruried regularly to encourage plant to form more branches
 - The crop is harvested every two weeks once it attains maturity.
 - After harvesting. The green tea leaves are transported to the factory within 24hrs.

Forestry

16. What is forestry?
- Is science of planting caring and using trees/forests and their associated resources.
 - The practice of managing and using trees/forests and their associated resources.
- 17 Explain the factors that favour the growth of natural forests on the slopes of Mt. Kenya.
- The area receives high rainfall /1000mm -2200mm throughout the year which encourages continuous growth of trees.

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- The area has deep fertile volcanic soil that allow the roots for penetration deep into the ground support trees.
 - The area is a gazetted reserve prohibiting cultivation and settlement hence allowing growth of trees.
 - The steep slopes discourage settlement thus forests thrive.
18. State five factors that have led to reduction of area under forests on mount Kenya.
- The government policy of degazetement has allowed illegal cultivation and settlement in forest areas
 - Increased population of elephants that destroy trees
 - Prolonged drought has caused drying up some trees.
 - Plant diseases and pests destroy some trees forests
 - Over exploitation of certain species of trees.
19. Give the difference in exploitation of softwoods forests in Kenya and Canada under following sub-headings.
- i) Period of harvesting
 - ii) Transportation
- Period of harvesting
- In Kenya harvesting is throughout the year while in Canada harvesting is in
 - Winter and early spring.
 - In Kenya transportation is mainly road transport while in Canada transport is mainly water transport.
- Transport
20. Name three international airports in Kenya.
- Nairobi- Jomo Kenyatta international airport
 - Mombasa – Moi international airport
 - Eldoret international airport.
21. Give four advantages of air transport over road transport
- Air transport is faster
 - Air transport is efficient method of transporting perishable goods/ valuable items.
 - Air transport is not affected by traffic jams
 - Helicopters are flexible and can land in remote areas.

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- There are few accidents.
 - It is more comfortable
22. Explain measures that should be taken to improve road transport.
- Construction of by-passes /tunnels/flyovers to reduce congestion in large Towns
 - Construction of highways/dual carriageways to accommodate more traffic
 - To educate road users on road safety precautions/discipline on roads to ease traffic on roads.
 - To enforce traffic rules to regulate traffic flow
23. Explain why there are few rail links among African Countries
- Most of the existing rail lines were constructed by colonialists who had no interest in linking the colonies
 - o The rail lines were constructed in different gauges making it difficult for the countries to link them up.
 - o Political differences discourage attempts to link the lines
 - o Inadequate capital limits the construction of new lines and for maintenance of railways.
 - o Large areas of the continents are economically unproductive thus it would be uneconomical to link them by railway
 - o Difficult terrain/thick forests make it expensive to construct railway lines.
 - o Limited trade links due to production of similar commodities fail to justify construction of railway links
 - o Competition from preference of other means of transport lead to neglect of railways.
24. Give four reasons why there is limited use of the river transport in Africa
- Some rivers have seasonal regime/varying volume
 - Some rivers have seasonal regime/varying volume
 - Some rivers have shallow water/silted river mouths
 - Some rivers have floating vegetation that choke the course
 - Some rivers have narrow channels unsuitable for sailing vessels.

Environmental conservation

25. State two causes of water pollution
- Oil leaks from ships/trucks

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- Industrial effluents when discharged into rivers/lakes\
 - Washing away (into rivers and lakes) of chemicals, fertilizers, pesticides/insecticides
 - Dumping of soil waste into water courses
 - Disposing of raw sewerage into rivers / lakes
26. Give two effects of water pollution on environment
- It may cause death of aquatic life
 - It may destroy beaches
 - It leads to spread of water borne diseases
27. Explain four methods used to control floods in Kenya
- Dykes are constructed on raised banks / levels to increase their heights and prevent water from over flowing.
 - o Dredging of river channels to deepen/widen them to, take it possible for them to accommodate excess water
 - o Dams are built across the rivers to control the amount of water discharged by the river
 - o Draining/redirecting a river/straightening of a river to control its wild flow
 - o Planting of trees in catchments are to reduce surface run- off and increase infiltration.
28. Explain how the following soil conservation methods are used to improve the quality of soil.
- i) Contour ploughing
 - ii) Mulching
 - iii) Crop rotation

Ans

- i) **Contour ploughing**
 - It helps trap water to reduce the speed of water thus preventing the formation of gullies and removal of topsoil from slope
 - It helps reduce the speed of water thus preventing removal of top soil.
- ii) **Mulching**
 - It enhances the retention of water in the soil by preventing it from direct sunlight wind
 - It increases the rate of infiltration by holding the rainwater and releasing it gradually
- iii) **Crop rotation**

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- Since different crops utilize different minerals rotation helps in balancing / replacing the material content in the soil.

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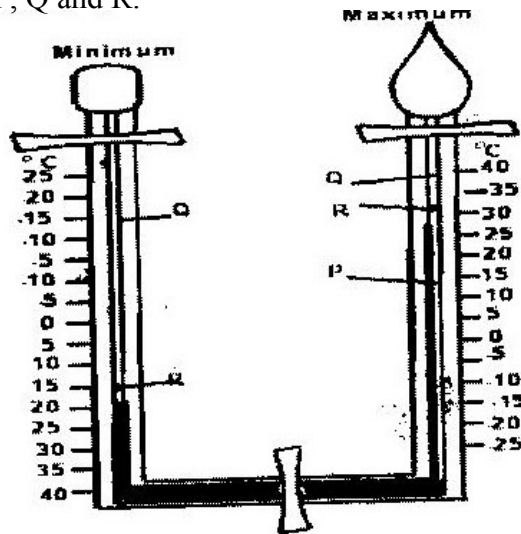
2008 KCSE GEOGRAPHY QUESTIONS

PAPER 1 312/1

SECTION A

Answer all the questions in this section

1. a) Give three reasons why it is necessary to study the plate tectonics theory. (3mks)
- b) Name two types of tectonic plate boundaries (2mks)
2. a) The diagram below shows a Six's Thermometer. Name the parts marked P, Q and R. (3mks)



- b) The table below shows temperature readings at a weather station for one week

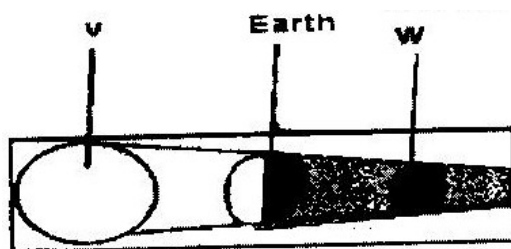
Temp/Day	Mon	Tue	Wed	Thur	Fri	Sat	Sun
Max. °C	28	27	28	29	29	29	26
Min. °C	18	18	20	16	22	21	19

Calculate the following:

- i) The diurnal range of temperature for Tuesday: (1mk)
- ii) The mean temperature for Saturday. (1mk)
3. a) Give two examples of non-metallic minerals. (2mks)

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- b) Why is industrial diamond used in shaping hard stones and metals? (1mk)
4. a) Apart from water vapour, name two other substances that are suspended in the atmosphere.
- b) i) Give two factors that are considered when classifying clouds. (2mks)
ii) Name two types of clouds that give rise to rainfall in the tropical regions. (2mks)
5. a) The diagram below shows an eclipse. Name the features marked V and W. (2mks)



- b) State four proofs that the shape of the earth is spherical. (4mks)

SECTION B

Answer question 6 and any other Two questions from this section

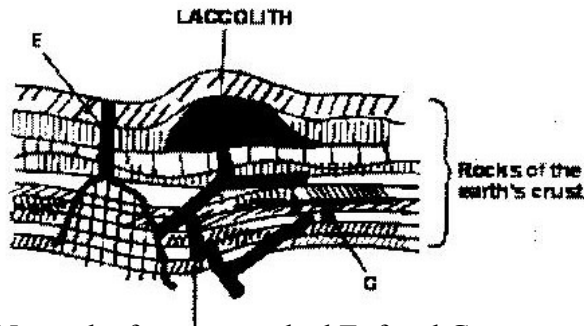
6. a) Study the map of Kericho 1:50,000 provided and answer the following questions.
- i) Give the longitudinal extent of the area covered by the map. (1mk)
- ii) Convert the scale of the map into a statement scale. (2mks)
- iii) What is the approximate height of the hill in the grid square 6770? (2mks)
- iv) Calculate the area of Kericho Municipality. Give your answer in square kilometers. (2mks)
- b) i) Give three types of natural vegetation found to the west of Easting 53 (3mks)
- ii) What is the bearing of the trigonometrical station at grid reference 554668 from the factory at grid reference 610626? (2mks)
- iii) Identify three forms of land transport found to the north of

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Northing 68 and west of Easting 53?

(3mks)

- c) Describe the distribution of settlements in the area covered by the map. (4mks)
- d) Citing evidence from the map, explain three factors that favour the establishment of tea estates in the area covered by the map. (6mks)
7. a) Differentiate between magma and lava. (2mks)
- b) The diagram below shows some intrusive volcanic features.



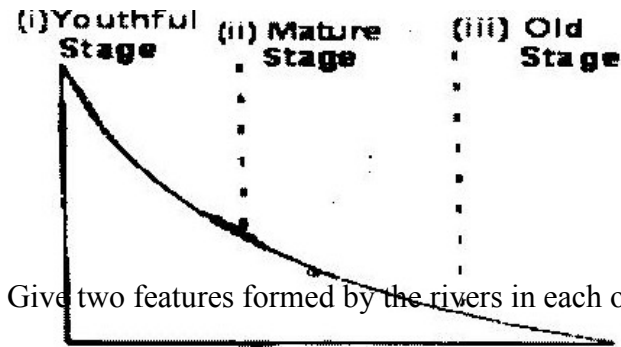
Name the features marked E, f and G.

(3mks)

- c) Describe how the following features are formed and for each give an example from Kenya:
- i) A crater (3mks)
- ii) A geyser (5mks)
- iii) A lava plateau (4mks)
- d) Explain four ways in which volcanic features influence human activities. (8mks)
8. a) i) Name two sources of rivers. (2mks)

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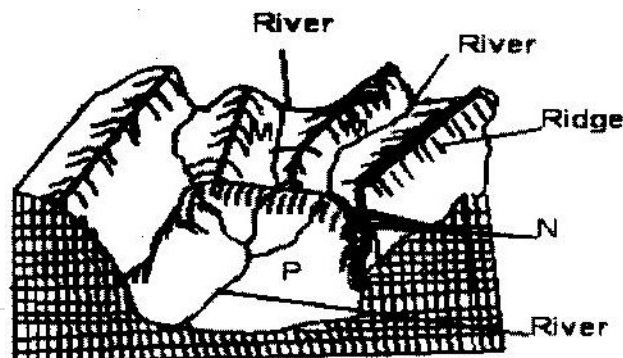
- ii) The diagram below shows the three stages of the long profile of a river.



Give two features formed by the rivers in each of the three stages.

(6mks)

- b) Describe the processes by which a river transports its load. (6mks)
- c) Describe each of the following drainage patterns;
- State two methods you would use to collect data. (3mks)
 - State three advantages of studying the work of rivers through fieldwork. (3mks)
9. a) i) Describe how ice is formed on a high mountain. (3mks)
- ii) Apart from a valley glacier, name two types of ice masses found on Mountains in East Africa.
- b) Explain how the movement of a valley glacier is influenced by the following factors:
- Temperature (2mks)
 - Width of a glacier channel. (2mks)
- c) Describe the distinctive characteristics of the following features resulting from glacial erosion:
- A corrie (3mks)
 - A pyramidal peak (3mks)
 - a fiord (fjord) (3mks)
- d) i) The diagram below shows a glaciated upland area



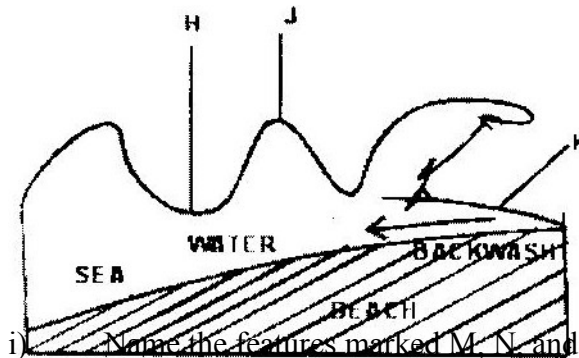
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Name the features marked M, N, and P.

- ii) Describe the process through which a crag and tail is formed

(4mks)

10. a) The diagram below shows a breaking sea wave.



- ii) Describe the process through which a crag and tail is formed.

(4mks)

- b) Describe three processes of wave erosion along the coast. (2mks)
- c) Explain how the following factors influence wave deposition:
- i) Gradient of the shore (4mks)
- ii) Depth of the sea (4mks)
- e) Using well labeled diagrams, describe how a bay bar is formed. (6mks)

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GEOGRAPHY PAPER 2

SECTION A

Answer all the question in this section.

1. State four characteristics of shifting cultivation. (4mks)
2. Give the difference between softwood forests in Kenya and Canada under the following sub-headings.
 - a) Distribution of softwood forests (2mks)
 - b) Transportation of the logs. (2mks)
3.
 - a) Differentiate between land reclamation and land rehabilitation. (2mks)
 - b) State two ways in which each of the following problems experienced at the Mwea irrigation Scheme can be solved;
 - i) Low prices of rice (2mks)
 - ii) Fluctuating water levels in the irrigation canals. (2mks)
4. Give five reasons why it is necessary to conserve wildlife in Kenya (5mks)
5.
 - a) State two economic benefits of the common Market for Eastern and Southern Africa (COMESA) to the member countries. (2mks)
 - b) Give four factors that limit trade among countries of Eastern Africa.

SECTION B

Answer question 6 and any other two questions from this section.

6. The photograph below show cattle rearing in an area in Kenya. Use it to answer question (a)



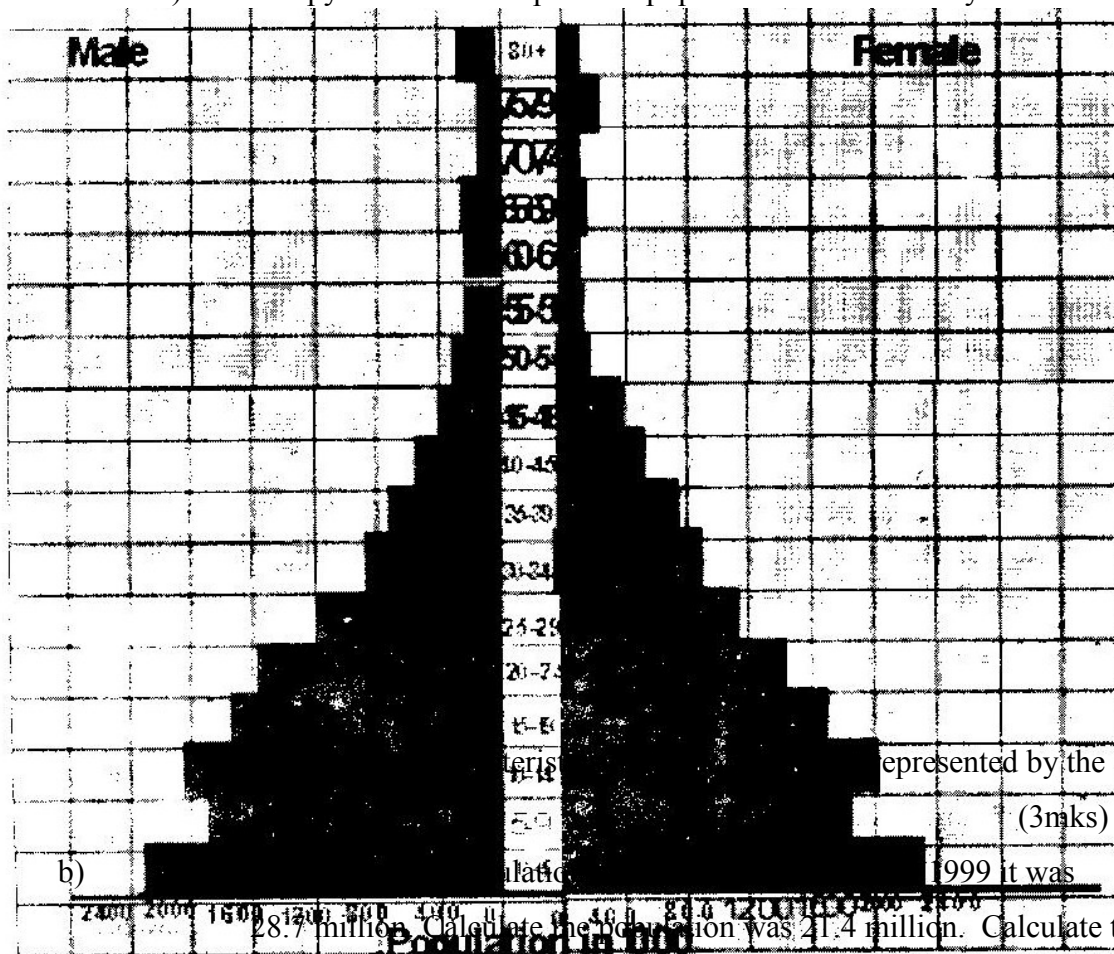
- a)
 - i) (1mk)
 - ii) Describe the features shown on the photograph. (3mks)
 - iii) What three indicators show that the area was experiencing drought

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When the photograph was taken. (3mks)

- b) Discuss nomadic pastoralist in Kenya under the following sub-headings;
- i) The cattle breeds kept (2mks)
 - ii) The pattern of movement (2mks)
 - iii) Marketing of the animals. (3mks)
- c) i) give three reasons why nomadic pastoralists keep large herds of animals. (3mks)
- ii) Explain four measures taken by the government of Kenya to improve beef cattle farming (8mks)
7. a) i) Give two documents from where information on population data is obtained (2mks)

- ii) The pyramid below represents population structure Kenya.



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- ii) Explain two factors which may have led to the large population increase between 1989 and 1999. (4mks)
- c) Explain three consequences of high population growth rate. (6mks)
- d) Explain four physical factors that influence population distribution in East Africa. (8mks)

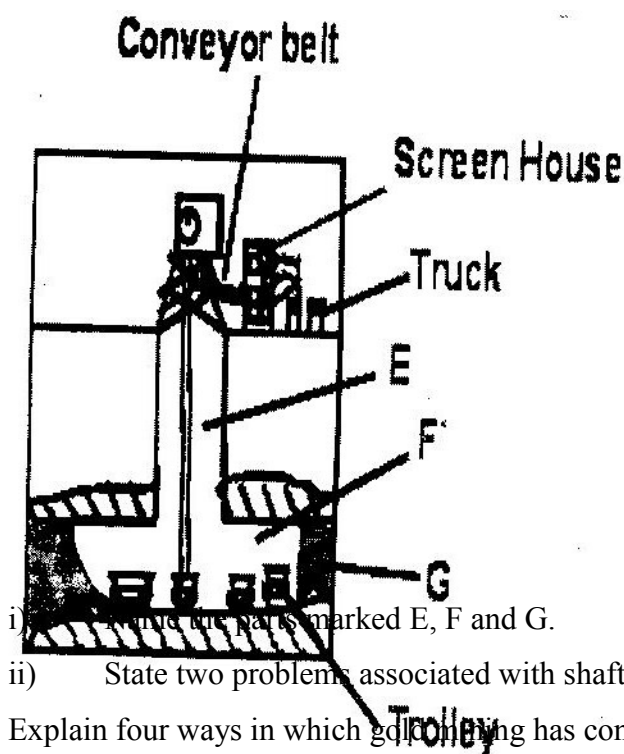
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8. a) The table below shows the quantity of minerals produced in Kenya in tones between years 2001 and 2005. Use it to answer questions (a) (i) and (ii).

Mineral/Year	2001	2002	2003	2004	2005
Soda ash	297,789	304, 110	352, 560	353, 835	360, 161
Fluorspar	11,885	85,015	80, 201	117, 986	26, 595
Salt	5,664	18,848	21,199	31,139	26,595
Others	6,093	7,000	4,971	6,315	8,972

Source: Economic Survey 2006

- i) Calculate the average annual production of soda ash over the 5 year 5 years period. (2mks)
- ii) Calculate the total mineral production for the year 2003. (1mk)
- b) The diagram below shows shaft mining.

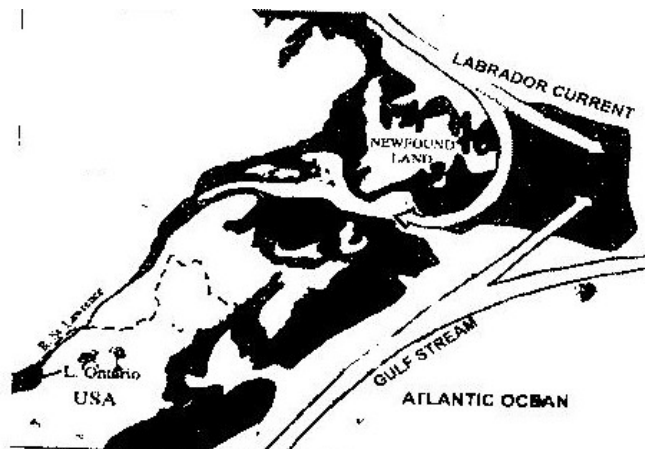


- i) Name the parts marked E, F and G. (3mks)
- ii) State two problems associated with shaft mining. (2mks)
- c) Explain four ways in which gold mining has contributed to the economy of South Africa. (8mks)
- d) Explain three negative effective effects of mining on the environment (6mks)
9. a) State three physical conditions that favour large scale sugarcane faming in Kenya. (10mks)

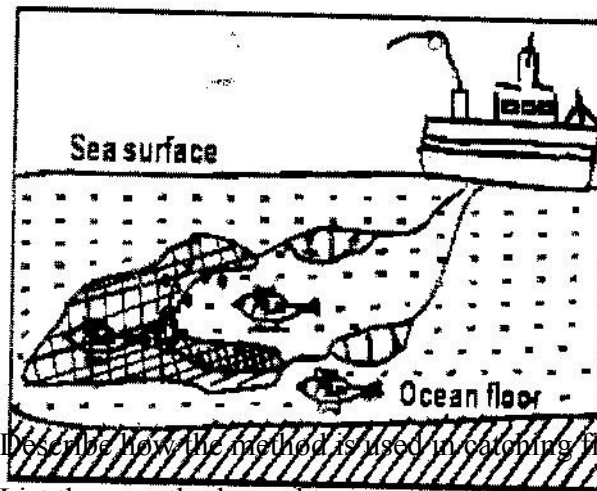
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- b) Describe the cultivation of sugarcane farming in Kenya. (10mks)
 - c) Explain five problems facing sugarcane farming in Kenya. (10 mks)
 - d) Your class visited a sugar factory for a field study on sugar processing.
 - i) Outline four stages if sugar processing that the class may have observed. (4mks)
 - ii) Name two by-products of sugar that the class may have identified during the study. (2mks)
10. Use the map of North-West Atlantic below to answer questions (a) and (b).

NORTH –WEST ATLANTIC FISHING GROUNDS



- a)
 - i) Name the country
 - ii) Explain how the two ocean currents shown on the map influence fishing in the area shaded on the map other than ocean currents. (6mks)
- b) Explain three factors that favour fishing in the area shaded on the map other than ocean currents (6mks)
- c) Explain why in East Africa, fresh water fishing water in the area shaded on the map other than ocean currents. (6mks)
- d) The diagram below shows a fishing method



- i) Describe how the method is used in catching fish. (5mks)
- ii) List three methods used to preserve fish. (3mks)

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PAPER 1

SECTION A

1. a) Give three reasons why it is necessary to study the plate tectonics theory.
- It explains the current position of the continents
 - It enables one to understand the creation of the structural land forms
 - It helps one to understand how the earth maintains balance/isostasy
 - It explains the cause of earth quakes/volcanicity
- b)
- Divergence/extension/constructive
 - Convergence/compression/destructive
 - Transform /conservative.
2. a) P- Mercury
Q- alcohol/oil of Cleo salt
R- Metal index.
- b) i) The diurnal range of temperature for Tuesday; $27-18=90^{\circ}\text{C}$
ii) The mean temperature for Saturday
- $$\frac{29 + 21}{2} = 25^{\circ}\text{C}$$
3. a) -Coal
-Petroleum
-Diamond
-carbon
- b) Because it is the hardest mineral/ it does not break easily
4. a) -Dust particles
-Pollen grains
-Gases
-Salt particles/sodium chloride
-Smoke
- b) i) -Their height
-Their shape/form
-Appearance
- Any 2x1 mks

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- ii) -Cumulonimbus
-Cumulus
-Nimbostratus
5. a) V - The sun
W - The moon
- b)
- The gradual emergence of a ship approaching the shore.
 - Circumnavigation of the earth along a straight path leads one to the starting point from the opposite direction.
 - The different times during which the sun rises and sets in different parts of the world.
 - The appearance of the middle pole to be relatively higher than other poles placed along a straight line on a level ground at equal distances. (curved)
 - The circular shape of the earth seen on photographs taken from satellites
 - The circular shadow cast by the earth during a lunar eclipse
 - The earth is a planet and all planets are sphere. Any 4x1 mks

SECTION B

6. a) i) $-35^{\circ} 15'$ to $35^{\circ} 25' / 10'$
ii) Map scale 1: $\frac{50,000}{100,000}$ km
 $= 0.5$ km
Statement scale is 1cm represents 0.5 km / $\frac{1}{2}$ km
iii) Just over 2120 m and below 2140m
iv) $10.5 - 11.0 \text{ km}^2$
- b) i)
- Scrub
 - Woodland
 - Scattered trees
 - Thicket
 - Papyrus/papyrus swamp vegetation Any 3x1 = 3mks
- ii) 305°C ($304^{\circ} - 306^{\circ}$) / N 55° ($55^{\circ} - 56^{\circ}$)
iii) All weather loose surface

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- iv) Dry weather road
- v) Motorable tracks/main track
- vi) Foot paths /other tracks.

c)

- There are few settlements/labour lines within the tea estates and forested areas
- To the north and west of Kericho-Lumbwa road, the settlements form a dispersed pattern
- To the north of Tugenon river, there are few or no settlements
- There are nucleated settlements in the market/shopping centres/labour lines/villages
- Some areas with the steep slopes and river valleys have a few or no settlements
- Kericho town is the main settlement are/forms a large cluster of settlement
- Few settlement in the tea estate.

d)

- The high relief modifies temperatures making the area suitable for the growing of the area suitable for the growing of tea bushes.
- The relatively undulating slopes allow proper drainage of soils making it ideal for tea farming/allows mechanization
- Presence of forests/many permanent rivers show that the area receives high rainfall which is suitable for tea growing.
- The area has fairly dense settlements which indicates availability of labour needed in tea farming.
- The area is well served by all weather roads which are needed for the transportation of tea from the farms to the factory /transportation of labour

Any 3x2=6mks

7. a) Magma is the molten rock material which originates from the interior of earth, cools while below the earths surface (and has large crystals) while lava is the molten rock materials that has reached the surface. (Has solidified and has small crystals. (2mks)
- b) E- Dyke
F- Lapolith
G- Sill

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- c) i) A crater
- Eruption of lava through a central vent causes building up of a cone.
 - The lava in the vent cools and contracts.
 - The cool lava withdraws into the vent leaving a shallow depression of the cone
 - Gas explosions may blow away surface rocks causing a crater
- Examples
- Mt Longonot
- Menengai
- Mt/suswa
- Mt Marsabit
- ii) A geyser
- Rainwater percolates down through cracks in the rocks.
 - The water gets into contacts with hot igneous rocks
 - The water gets into contacts with hot igneous rocks
 - The water is super heated and gases/steam form
 - Pressure builds up in the cracks.
 - The pressure causes steam and water to be ejected explosively as jet to the surface intermittently.
 - The water and steam are emitted intermittently as pressure level changes.
- Example – Lake Bogoria
- iii) A lava plateau
- It is formed when magma reaches the surface of the earth through a series of vents/fissures
 - The lava is extremely fluid/ultra-basic
 - The lava spreads evenly over a large area
 - The lava cools slowly and solidifies
- Example
- Yatta plateau
- Uasin Gishu plateau
- Laikipia plateau

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d)

- Volcanic highlands/mountains are sources of rivers which provide water for domestic/agriculture/industrial use.
- Volcanic rocks weather down to form fertile volcanic soils which support agriculture
- Volcanic rocks are important building materials in the construction industry
- Volcanic features are tourist attractions which promote tourism.
- Volcanic mountains/highlands influence formation of relief rainfall which encourages agricultural activities.
- Volcanic highlands influence formation of relief rainfall which encourages agricultural activities
- Volcanic highlands/mountains modify temperatures making them making them attractive to human settlements
- Volcanic features such as steam jets and geysers provide suitable sites for geothermal power generation.

8.

a)

i)

- Lake /swamp
- Melting ice/snow
- Springs
- Surface run off.

ii)

Youthful stages

- Rapids/water fall/cascades
- V-shaped valleys/canyons
- Potholes
- Plunge pools
- Interlocking spurs

Mature stage

- Meanders
- River cliff/bluff
- Wide v shaped valley
- Slip –off- siopes
- Alluvial fans

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- Braids

Old stage

- Meanders
- Ox-bow lakes
- Braided channel/isels
- Flood plain
- Levees
- Devees
- Deferred tributaries
- River terraces
- Distributaries/deltas
- Bluffs
- Meanders scar

b) Traction process

The large and heavy loads of the river are rolled /dragged along the river by the force of the moving water and gravity.

Saltation

Some large fragments that cannot remain suspended in the water are momentarily lifted and dropped by water turbulence. The series of hops move the load down the river.

Suspension

Light insoluble materials such as sand and silt grains are carried and maintained within the water by river turbulence and transported downstream.

Solution process

The soluble minerals /materials are dissolved in river water and carried away

c) i) Dendritic patterns

- It develops in areas where rocks have uniform structures.
- The direction of flow is influenced by the slope of the land
- The tributaries converge on the main river forming a shape like that of a tree and its branches.

ii) Trellis pattern

- The pattern develops where soft and hard rocks alternate vertically

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- The tributaries join the main river at acute angles.
 - The consequent streams flow to the opposite direction of the main river
 - The main river and its tributaries form a rectilinear pattern.
- d) i)
- Observation/digging up the deposits to expose the inner layers
 - Collecting samples
 - Taking photographs
 - Interviewing the people around the river.
- ii)
- It enables one to collect first hand information
 - It promotes development of practical skills
 - It promotes application of acquired knowledge
 - One is able to develop skills of data analysis.
9. a) i)
- Due to low temperatures, water vapour freezes and forms snow
 - Snow falls and accumulates on the mountain top/higher slopes
 - Snow continues piling and new layers exert pressure on the lower layer
 - Lower layers become compressed/compacted as air is expelled from the spaces by snow particles
 - The compacted layers are ice.
- ii)
- Ice caps
 - Cirque glaciers
- b) i) **Temperature**
- Glaciers move faster in summer/when the temperatures are higher because the ice melts due to the warm conditions whereas in winter/ when temperatures are low, ice movement is slow due to cold conditions.
 - The temperature of the bottom of the valley glacier rises with Pressure. Thereby thawing and enabling its movement down slope.
- ii) **Width of a glacier channel**
- When the channel is wide ice movement is slow that is because ice

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spreads out forming a thin layers there is less pressure to cause thawing that would facilitate ice movement/vice versa.

- c) i) **A corrie**
 - Is a deep rock basin
 - Has steep sides
 - Is arm-chair in shape/semi circular
 - Has a high back wall
 - Has a reverse slope on the lower side
- ii) **Fiords**
 - Has steep walls
 - Is a narrow sea inlet
 - Is a U-shaped
 - Has hanging valleys
 - Has deep water shallower seawards/deeper landward
- d) i) M- Hanging valley
N-Water valley
P-U-shaped valley / glacial trough
- ii)
 - A large block of rock stands on the path of oncoming glacier
 - The moving ice plucks off/erodes weak rock fragments from the upper side of the rock
 - As the ice moves round and over the resistant rock it carries the eroded materials to the lee side
 - The lee side does not experience erosion.
 - Eroded materials are deposited materials increase on the lee side.
 - With time the moving ice smoothens the side of the on coming ice deposited materials increase on the lee side
 - The resistant rock is the crag while the materials deposited on the leeward to form the tail.
- 10. a) i) H-Trough
J-Crest
K-Swash

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- ii) A backwash is the return flow of water down the beach to the sea after a wave is broken.

b) **Abrasion/corrosion**

- Rock fragments carried by waves are used as a tool to grind against the cliff. As waves break rock fragments carried by the back wash erodes the sea.

Solution/corrosion

- The solvent and chemical action of the sea water dissolves and removes the minerals that are found in the cliff/sea floor especially where there are limestone rocks.

Hydraulic action

- The swash/breaking waves hit against the cliffs shattering the rocks. The breaking waves compress air into the cracks/joints in the cliff face. This widens the cracks and parts of the rocks may break off.

Attrition

- Particles that are carried by waves are constantly colliding against each other and wears them into smaller sizes

c) i) **Gradient of the shore**

A shore with a gentle gradient reduces the velocity/speed of the flow of the backwash causing the waves to deposit the load on the shore. Where the shore is steep, the velocity/speed of flow of the backwash will cause the materials to be moved from the shore back into the sea. (There will be little or no deposition)

Gentle gradient 2 marks

Steep gradient 2 marks

ii) **Depth of the sea**

Shallow water causes waves to break thus encouraging deposition.

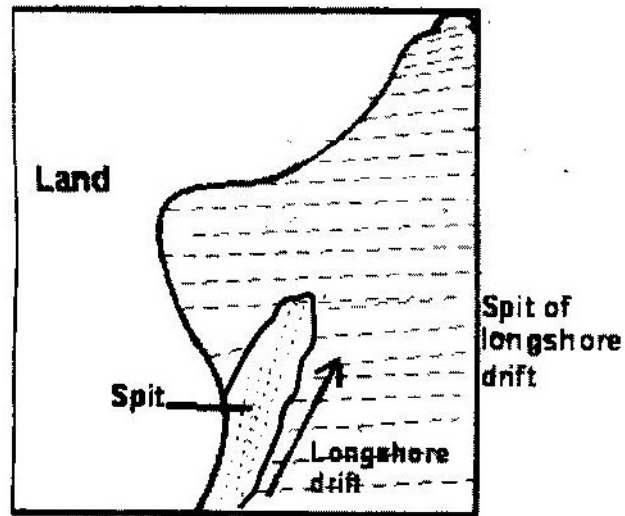
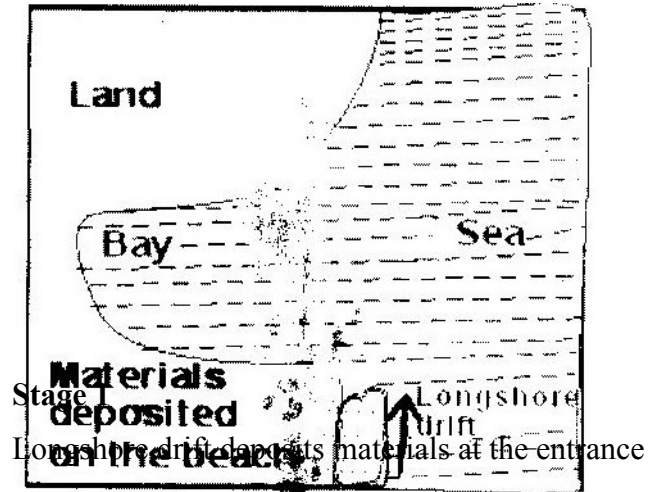
Where the sea is deep, there is less deposition because the sea bed is not in contact with the waves carrying deposits.

Shallow water 2 marks

Deep water 2 marks

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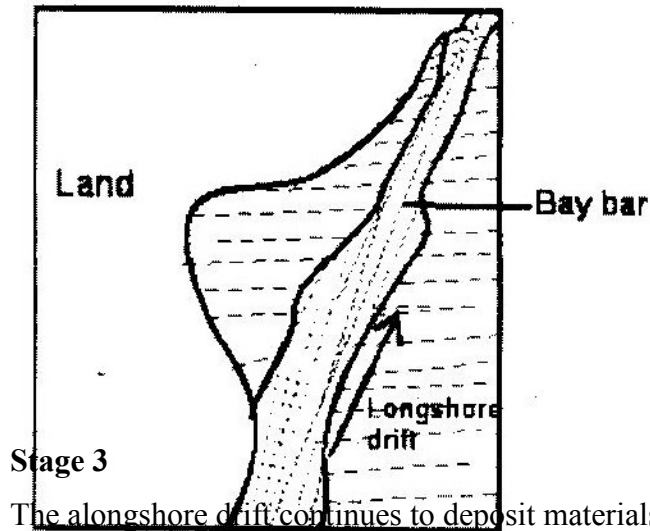
- d) Using well labeled diagrams, describe how a bay bar is formed



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Stage 2

A spit forms at the entrance of the bay



The alongshore drift continues to deposit materials and the spit extends towards the other end of the lagoon. Eventually the spit reached the other end thus forming a bay bar.

Diagrams 3 marks

Text 3 marks

PAPER 2 ANSWERS

SECTION A

1.
 - Vegetation is cleared by slashing and burning
 - There is the use of little or no manure/use of ash
 - The land is communally owned
 - The yields decline after a certain period of continuous use. The land is abandoned when the yields decline
 - Both the settlements and plots are temporary
 - Farming depends mainly on family labour
 - The farmers use simple implements
 - It is mainly for subsistence
 - Plots are small and scattered
2. a)
 - In Kenya, softwood forests are found mainly in the highlands while in Canada they are found both in highland and lowland areas.
 - In Kenya, softwood forests cover a small percentage of the total land area while in Canada they cover large tracts of land.

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- b)
 - In Kenya, logs are transported by road/trucks while in Canada transport is mainly by water by rivers.
 - In Kenya, transportation is expensive while in Canada it is cheap.
- 3. a) Land reclamation is the process of converting less productive land into a more productive state for agricultural or settlement purposes while land rehabilitation is the process of restoring degraded/impoverished/damaged land back to a useful state.
- b) i)
 - Diversifying the crops produced in the scheme.
 - Improving the quality of the rice produced through research
 - The government should restrict the importation of rice to reduce competition.
 - Improve the marketing strategies to enable farmers to source for market outside Kenyaii)
 - Continuous dredging of canals/deepening of canals
 - Construction of dams to store water for use during dry season.
 - Government to enforce laws on proper land use in the catchment areas of the rivers that supply water to the scheme.
- 4.
 - To maintain the genetic diversity/genetic pool
 - To preserve wildlife for future generation/posterity.
 - To protect the endangered species of plants and animals
 - To ensure sustainable utilization of species
 - To attract tourists/to earn foreign exchange.
 - To use wildlife for research/for education.
 - To maintain aesthetics for recreation
 - To provide materials for medicinal extracts.
- 5. a)
 - It has created a large market for goods produced in members countries
 - It has resulted in the availability of a variety of goods
 - It has led to the establishment of common tariff
 - The removal of visa requirements has made it easier for traders to move

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across borders within the region.

- b)
 - Membership to different trading blocks by different countries.
 - Lack of a common currency.
 - Underdeveloped infrastructure/poor transport network
 - Restriction of movement of people and goods/high taxes rates
 - Political instability
 - Political differences
- 6. a) i) Ground /ground general view
 - ii)
 - On the foreground there is bare ground/some short vegetation cattle browsing /grazing.
 - In the middle ground there is a herdsman and some cattle on the move/raising dust.
 - There is a road in the middle ground
 - At the background there are some patches of grass/some trees/shrubs/thickets
 - Some parts of the background are bare surface
 - There are in the middle
 - Clear skies in the background
 - There is a fence in the middle ground
 - iii)
 - The cloudless sky
 - The malnourished/thin animals
 - Dust raised by moving animals
 - Bare ground/sparse vegetation /patches of vegetation/little vegetation
 - Patches of dry /brown grass
- b) i) The pastoralists keep mainly indigenous breeds such as Zebu and Boran.
 - ii)
 - Their movement is seasonal.
 - During the dry season the pastoralists migrate with their livestock

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to the highlands where there is pasture and water

- During the wet season they move to the plains since pasture is available.

iii)

- Some cattle are sold to slaughter houses/to individuals.
- Some pastoralists sell their livestock through community groups/ranches.
- Some livestock are sold to the livestock are sold to the livestock

marketing Department

- Some pastoralists sell their animals to Kenya Meat Commission.

c) i)

- It is a form of insurance against natural calamities /diseases / drought.
- Animals are kept as a sign of wealth/prestige/social status.
- Animals are kept for use to pay dowry.
- Animals are used as a source of food/milk/meat/blood
- Animals are a source of income

ii)

- It encourages research /the cross-breeding of traditional cattle breeds with exotic ones. This improves the quality of the animals/cross breeds are more resistant to diseases than pure exotic breeds
- It strengthens community education to teach beef cattle farmers better livestock managements
- It has constructed roads to make services accessible to farmers/make transportation of animals to markets easier
- It encourages the replacement of the coarse grass with nutritious pasture to improve the quality of animals.
- It has sunk bore holes/dug wells/constructed dams to provide water for the animals.
- It has revived Kenya Meat Commission (KMC), a government parastatal that buys animals from farmers for slaughter.

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7. a) i)
- National census report
 - Text books
 - Magazines
 - Periodicals/Journals
 - Statistical abstracts
- ii)
- The number of male and female is almost equal to all ages.
 - The dependency ratio is high
 - The number of infants from age 0-4 is high/the population has a high birth rate.
 - The middle age/working population is low
 - The number of youth aged 5-19 is high
- b) i)
- $$\frac{-28.7 - 21.4}{21.410} = 7.3$$
- $$\frac{7.3}{21.410} \times 100 = 3.4\%$$
- ii) **Early marriages.**
- Many people in Kenya get married early and this allows them a longer period of fertility resulting in many children being born.

Improved medical care:

This leads to higher chances of survival for both the mothers and infants as well as the general population, thus increasing the survival rates.

Improved Diet:

This results into better health for the entire population hence reducing the mortality rate.

Cultural beliefs:

Some cultures encourage large families due to the preference of one gender to other/some /some cultures/religions discourage the use of contraceptive/family planning leading to couples getting many children.

Migration:

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Due to political instability in neighboring countries e.g Sudan, Ethiopia there was an increase of refugees hence high population increase.

- c)
- It leads to high dependency ration resulting into little savings by the working group /low investments/low living standards
 - There is likely to be a high unemployment rate since job opportunities may not increase at a rate that can cope with the increasing number of job seekers /may increase the rate of crime
 - It increases demand for food which may lead to food shortage
 - It increases demand for agricultural land causing land fragmentation/landlessness/destruction of forests.
 - Expenditure while meeting demands for the large population reduces revenue that would be used for development of income generating projects hence slow economic growth.

d) **Climate:**

The cool and wet/hot & wet/high & reliable rainfall areas are densely populated because they are suitable for farming/hot dry areas have sparse population because they are unsuitable for farming

Relief:

Mountains and hilly areas have low temperatures/are rugged and this discourages settlement/development of infrastructure/agricultural activities.

Plains and gently sloping areas are usually densely populated because they are suitable for settlement and other economic activities. Flat areas depression are sparsely populated because of poor drainage, which causes swampy conditions/flooding.

Soil:

Fertile soils are suitable for agriculture thus attracting large population/areas that have poor soils have sparse population.

Pests and diseases:

Areas that are infested with pests and disease-carrying vectors discourage settlement since the conditions are unhealthy

Drainage:

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Low-lying areas that are prone to periodic flooding and water logging have sparse population because they are unsuitable for agriculture and other economic activities/well drained areas attract settlement

Vegetation:

Forested areas/savannah woodland have wild animals, disease vectors and discourage human settlement and other economic activities

8. i) Soda ash average production for 5 years
 $= 1668446 \div 5$
 $= 333689.2$ Tonnes
- ii) Mineral production for the year 2003
 $= 45369.2$ Tonnes
- iii)
- It is a raw material for making glass
 - It is used in king detergents
 - It is used in some chemical industries /petroleum refining
 - It is used as a water softener/water treatment
 - It is in desulphurising steel
 - It is used in paper industries.
- b) i) E- main shaft/vertical shaft
F- Tunnel/horizontal shaft/Gallery
G- Mineral ore
- ii)
- Sometimes, mines get flooded with sub/terrain water.
 - There are occasional emission of poisonous gases in the mines.
 - The dust produced causes respiratory diseases
 - Sometimes tunnels collapse causing deaths of miners.
- c)
- Gold is highly prices, thus it earns foreign exchanger which used to improve other sectors of the economy.
 - Gold provides raw materials for industries that make jewellery and other highly valued items thus promoting industrial expansion.
 - Gold as a medium of exchange in the world is used in South Africa as a

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means of paying international debts.

- Gold mining industry has generated employment opportunities, which raises the standard of living of the people /earn more income
- Gold mining has led to development of towns in the Rand and the Orange Free State creating a large demand for agricultural products.
- Mining of gold has led to the expansion of infrastructure such as transport and communication/provision of social amenities.
- Gold mining has led to the development of industrial mining skills that are useful in other sectors of the economy.

d)

- The dumping of rock waste had led to the loss of biodiversity/destruction of natural vegetation
- Dereliction of land due to dumping of waste materials is an eye sore/destroys the natural beauty of the land
- Dereliction of land due to dumping of waste materials is an eye sore/destroys the natural beauty of the land
- Pollution of the areas by noise/blasts smoke and water pools are all health hazards.
- Mining disrupts the water table which may lead to shortage of water.
- Mining takes up land that would have been used for agriculture thus interfering with food production.
- Mining displaces human settlements thus disrupting people and necessitating expensive resettlement processes.

9

a)

- Well drained fertile soils /Black cotton soils
- Gently sloping/undulating landscape
- High rainfall 1200 to 1500 mm well distributed throughout the year.
- Moderate high temperature/20⁰C- 28⁰C
- Long periods of sunlight.

b)

- The land is cleared of its natural vegetation
- It is ploughed using either tractors or ox-drawn ploughs

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- Harrowing is done to loosen the large lumps of soil.
 - Shallow furrows are dug at intervals of 1.2 and 1.8 metres apart.
 - Cutting/seed cane are planted in the furrows
 - Top dressing/nitrogen fertilizers are applied
 - Weeding is done regularly/herbicides are applied
 - After 18 months the cane is ready for harvesting
 - The cane is cut/harvested using pangas
 - The harvested cane is loaded into Lorries for transportation to the factory.
- c)
- Pests such as termites and white grub/diseases such as ratoon stunting and smut attack the plants and lowers the yields leading to low income for the farmers
 - Accidental fires/fires set by arsonists destroy the cane resulting in heavy losses to the farmers.
 - Flooding of market by cheap imported/sugar results in unfair competition causing delay in payments to the farmers.
 - Delays in harvesting reduce the quality tonnage of the cane reducing the farmer's earnings.
 - Closure of some factories such as (Ramisi and Miwani) has deprived farmers of the source of income/annual closures of factories for servicing of machines disrupts the farmers' calendar of activities.
 - Poor feeder roads in some areas leads to delayed delivery of the cane to the factory lowering the quality and subsequently the profit to the farmers.
 - Prolonged droughts in some areas destroys the crop leading to heavy losses.
 - High cost of farm input reduces the farmer' profit margins
 - Mismanagement of factories and cooperatives leads to delayed payments thus discouraging the farmers
- d) i)
- Weighing of the cane
 - Chopping of the cane
 - Crushing of the cane
 - Boiling

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- Filtering
- Grading
- Packing/bagging of sugar

ii)

- Molasses
- Bagasse
- Wax
- Aconitic acid
- Filter cake/mud

10. a) i)

- Canada

ii)

- The convergence of the warm and cold currents causes upwelling of ocean water which bring minerals for planktons to the surface attracting large number of fish to the area
- The convergence of warm and cold currents modifies the temperature of the ocean water making the area ideal for fishing throughout the year
- The cool waters favour survival of a wide variety of fish species which makes the area an important fishing ground

b)

- The area has a broad shallow continental shelf which provides suitable conditions for the growth of plankton used by fish as food
- The region experiences low temperatures that are favourable for the survival of fish/for preservation/storage of fish.
- Due to the ruggedness of the land bordering the shaded area/the short growing season. Many people therefore concentrate on fishing as an alternative economic activity.
- The hinterland is densely populated thus providing ready market for the fish.
- Advanced technology has resulted in highly developed ship building/fishing vessels are equipped with modern preservation facilities, thus making it possible for fishermen to carryout large scale fishing.

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- The indented coastline provides ideal fish breeding sites/sheltered bays are ideal for setting up fishing villages and ports.

c)

- There numerous inland fishing grounds such as lakes and rivers which are accessible to many people.
- There is low demand for sea fish compared to fresh water fish making fresh water fishing more preferable.
- The narrow continental shelf along the coast of East Africa limits the growth of plankton thus limiting the breeding of fish/limiting the variety of edible fish.
- The stiff competition if the open sea from the industrialized countries whose fishermen use modern fishing equipment discourages local fisherman
- The limited technology and inadequate capital make it difficult to develop marine fishing.

d) i)

- A bag-shaped net is attached to a trawler/ship
- The net is cast into the water by the trawler
- The nets' mouth is kept open by other boards/head beam
- The upper part of the net is kept afloat by corks/floats.
- Weights are used to keep the lower part of the net at the sea bed
- The trawler drags the net along the sea bed.
- After sufficient fish has been caught, the net hauled to the trawler to empty the fish,

ii)

- Canning
- Freezing
- Smoking
- Salting
- Sun-drying
- Frying